SMITHIAN FOLD THEORY V3

After Turing: The Fold Machine

An Exact, Parameter-Free and Machine-Closed Derivation of Classical Computational Science from Smithian Fold Theory

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An Exact, Parameter-Free and Machine-Closed Derivation of Classical Computational Science from Smithian Fold Theory

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in the same concept DOI lineage. The correction changes no scientific claim,
evidence classification, mathematical value, source identity or machine record.
Lean verification integration. *The current SFT ordered census passed the*
independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims,
898,902 candidate records and decisions, 11,108 controls, seventeen branches
and no reported issue. Lean natively proves the two-class operational root and
its unique survivor and constructs proof-bearing acceptance-gate certificates.
The remaining claim content is checked as the complete registered artifact
graph; that distinction is retained throughout this successor.

Abstract

This successor presents the complete current `computation` paper surface: 375 live model-admitted claims, 96,000 generated candidates, 375 unique survivors and 1,500 passed controls. It integrates 6 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

The reconstruction spans formal computation; computability; computational complexity; algorithms and mathematical data structures; semantics and programming theory; concurrent and distributed computation; cryptography and computational security; learning and intelligence theory; and scientific computation. It derives a native SFT tape machine, exact languages and automata, rewriting and recursion, model equivalence, universality, halting and incompleteness boundaries, exact resource laws, deterministic-support randomized computation, semantics-preserving compilation, consensus and impossibility boundaries, explicit adversary laws, traceable learning and exact scientific calculation. Its native famous-problem results remain precisely typed: `BB_F(k)=k`, `P_F=NP_F`, and the depth-`k` full-edge circuit lower bound hold inside the admitted Fold grammar and are not silently exported to conventional

external grammars.

The proof domain imports no conventional machine, axiom, free or fitted parameter, semantic numerical zero, negative proof magnitude, irrational or imaginary proof scalar, floating proof quantity, completed infinity, ontic randomness, cryptographic hardness assumption, pretrained model, benchmark answer or application result. Displayed `0` is structural absence and schedule support is deterministic. Quantum operations, physical hardware, software platforms and the later Unison AI, Fold Chess, Fold Go and Fold Protein rebuilds remain explicit downstream handoffs.

Version 1.4 preserves the complete version 1.3 foundation and its public scientific argument, then executes the full-field roadmap claim by claim. No protected engine or verification-authority source was modified. Every later extension must enter as a new registered question and pass the same public protocol.

Keywords: Smithian Fold Theory; complete-field classical computation; Fold machine; computability; halting; complexity; algorithms; semantics; distributed computation; cryptography; learning theory; scientific computing; superdeterminism; computational proof; open science.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned `PASS` for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 375 current claim(s) in its declared branch ownership boundary; 6 later claim section(s) are integrated in the successor supplement.
6. **Newly integrated results.** `SFT-COMP-AIX-CAUSAL-ATTENTION-TRANSFORMER-001` - Exact SFT causal attention-transformer assembly; `SFT-COMP-AIX-EXACT-TEACHER-LEARNING-002` - Exact source-bound teacher-observation learning; `SFT-COMP-OAI26-PERMANENT-FORMULA-001` - SFT reconstruction of the permanent rational-formula lower bound; `SFT-COMP-OAI26-GAPCVP400-002` - SFT reconstruction of GapCVP400 NP-hardness; `SFT-COMP-OAI26-PERMANENT-FORMULA-VALIDITY-001` - SFT disproof of source-artifact validity: PermanentFormulaLowerBound.permanent_rational_formula_logarithmic_lower_bound; `SFT-COMP-OAI26-GAPCVP400-VALIDITY-002` - SFT disproof of source-artifact validity: GapCVP.Comparator.gapCVP400IsNPHard.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

1. Central scientific claim and exact boundary

> *Within the frozen SFT V3 Classical Computation census dated 29 July 2026, all 369 obligations in all 12 registered families are individually admitted through the untouched engine, independently reconstructed, adversely controlled, observed under their declared custody boundary and exactly reconciled. The current closed count is 369, the current open count is structural absence, and completion remains open to lawful versioned extension.*

The frozen census identity is `sha256:411959c9505e6e0e9ec57e03923572f249cb5402996dca17911b3e1616e81b6ea`. The final reconciliation identity is `sha256:84e436aa27ed392c4c2a5cab77833952a027f43e878151627e958e64be3ee30e`. The branch contains

94,464 completely decided candidate structures, 369 unique survivors and 1,476 passing controls. Every new full-field registration has an empty axiom list and an empty free-parameter list; the preserved foundational registrations prohibit imported answer-producing premises and retain their admitted upstream dependency routes. Every dependency chain reaches `SFT-ROOT-THERE-IS-NO-NOTHING` through actual receipts rather than narrative citation.

Closure means the exact generated kernels named in the census are complete. The native Fold Busy-Beaver, Fold-P/Fold-NP and circuit lower-bound theorems retain their explicit grammar boundaries. They do not decide conventional Turing-machine Busy Beaver tables, conventional P versus NP, arbitrary imported circuit models, practical cryptographic security or physical device performance. Quantum operations belong to the Quantum Computation branch; application rebuilds and platform engineering remain downstream. This is dated completion, not a permanent prohibition on new questions, counterexamples or stronger lawful certificates.

2. Headline findings and results at a glance

Result	Exact executed result	Scientific consequence and boundary
Complete-field Classical Computation	**369/369 obligations** across 12 families; **94,464** generated candidates; 369 unique survivors; **1,476** passed controls; 369 current receipts.	The version 1.3 roadmap is now an executed, independently replayable corpus rather than future intent.
Native Fold Busy Beaver	For every admitted positive finite description depth `k`, the maximal halting transition count is exactly `BB_F(k)=k`.	Closing populations through depth 14, attaining witnesses and the successor certificate establish the native theorem; no imported Turing-machine grammar is claimed.
Native Fold complexity equality	Exact evaluation emits the unique lawful trace checked by the verifier at the same admitted description-depth resource, so `P_F=NP_F`.	Both containments are executed over complete generated supports and tampered certificates halt; the theorem is not exported to conventional P versus NP.
Arbitrary admitted Fold-circuit lower bound	A depth-`k` Fold circuit needs all `k` forced transition edges. At depth 14, all 16,384 edge subsets were enumerated and only the full support survived.	The lower and attaining upper witnesses coincide inside the admitted Fold circuit grammar.
Superdeterministic randomized computation	Every registered schedule is a deterministic trace; apparent randomness is complete schedule support plus observation-relative uncertainty.	Randomized search and learning do not import an uncaused transition or ontic probability.
Formal limits	The self-description plus held-complement construction defeats a total internal halting decider; proof enumeration retains an explicit incompleteness boundary.	Finite generated processes remain exactly executable as terminal or recurrent without mislabelling bounded evidence as unrestricted proof.
Semantics, distribution and security	Exact binding, substitution, evaluation, types, correctness, compilation, causality, consensus, consistency, adversary and transcript laws are separately admitted.	Every theorem retains its machine, resource, fault, network and adversary boundary; finite demonstrations are not advertised as practical security guarantees.
Learning and scientific computation	Hypothesis support, evaluation custody, generalization, planning, reinforcement, stability, convergence, discretization, simulation and inverse reconstruction retain full traces.	A benchmark, pretrained oracle or result-only simulation cannot substitute for the premise-to-result and measurement route.
Dated validation and handoff	Twelve validation laws replay all 351 predecessor receipts; six handoff laws bring the branch to 369/369 with one-owner boundaries.	The Grand Lock is a dated evidence reconciliation, while lawful extensions remain open and quantum, domain measurement and engineering ownership stay explicit.

These results were not selected by their historical names. Their carriers, alternatives and transition laws were registered and sealed before correspondence. Historical terminology enters only after the SFT result exists, allowing comparison without importing the historical model as a premise.

Current status, evidence language and reader map

Status field	Current position
Branch and completion boundary	Classical Computation: 369/369 frozen obligations; 369 live claims. This is dated, versioned completion and remains open to lawful extension.
Formal status	Every live claim represented here has a current model-admitted receipt. Formal admission does not by itself imply empirical confirmation.
Empirical status	Claim-specific. Blind, non-blind, development-observed, holdout, adverse, unresolved, unavailable and formal-only distinctions remain exactly as recorded in the claim ledger and evidence packages.
Chronology	Claim-specific registration, seal, custody and observation order governs. Later evidence does not retroactively become an earlier prediction.
Publication status	Version 1.5.1 is published open access at 10.5281/zenodo.21761653 in the existing Zenodo concept DOI lineage.
Existing record lineage	Current record DOI `10.5281/zenodo.21627721`; concept DOI `10.5281/zenodo.21518310`. Publication is permitted only through the existing record's new-version route.
What is not claimed	Permanent closure of science, universal empirical confirmation, transfer of another branch's ownership or permission to replace an adverse or unresolved record.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the specified formal, computational or empirical test; its kind must be named.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications. None silently substitutes for another.
Foundational closure	Completion of the registered foundation only. It does not imply field-wide closure.
Field-wide closure	Completion of the frozen or registered dated field census named in this paper.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may be added by a later version without rewriting existing receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relationships; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The cited repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. Those files remain authoritative where prose abbreviates their display.

Editorial change control

This version may improve expression, order, typography and navigation. It may not change scientific meaning, scope, chronology, evidence class, ownership, claim status or machine identity. Any apparent conflict between authoritative records must be traced and either resolved by explicit supersession or recorded for Maria Smith; prose alone cannot manufacture agreement.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorise. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigour.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Computational Science, this constitution refuses both imported machine authority and benchmark theatre. A Turing machine, complexity class, randomized schedule, cryptographic adversary, learning model or simulation may enter only after its native Fold carrier and resource boundary have been forced. A terminal program output or opaque benchmark score cannot replace the transition trace that makes the computation scientific.

Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through https://discord.gg/ucwGryVxGr, or inspect the public project at https://github.com/MettaMazza.

Evidence-boundary inventory identity:
`sha256:cebee4dc4c4b66fa6c73b587dd93ca56a9ea7b5c86593fda179967ee70fb49e8`.

3. Derivational constitution

Every dependency terminates at the single operational root theorem, *there is no nothing*. Admitted proof objects are structural One, exact positive parts, generated finite traces, Fold labels, words, relations, maps, graphs, information ledgers and composition records. Semantic numerical zero, negative quantity, irrational or imaginary values, floating proof arithmetic, a completed infinity and an ungenerated continuum are excluded. Empty One is a form; held complementary labels replace signed proof values.

Python booleans, indices, lengths, empty containers and process return codes are quarantined host mechanics. They execute and count evidence but carry no SFT mathematical authority. All registrations have empty axiom and free-parameter tuples. No application, benchmark, pretrained model or earlier SFT result selected a survivor.

4. The native Fold machine

The native machine begins with exact generated configurations, two held Fold labels and a source-bound transition relation. Blank support is empty One, not numerical zero. A word is a complete sequence of held labels at canonical positions. Reading is an observation relation; writing is exact held-label substitution; movement is a counted transition across generated positions. A configuration retains its word, held position, process state and proof trace. Every step records its premise, action, successor, resource use and lost or retained distinctions.

Universality is derived only after languages, automata, rewriting, recursion, binding, abstract machines, circuits, processes and composition have receipts. The universal process interprets every description inside the frozen machine grammar. Conventional Turing, Church and circuit models are comparison encodings and cannot select the machine law.

5. Superdeterminism, branching and randomized computation

Randomized computation closes without an uncaused random premise. The machine retains complete registered schedule support and executes one deterministic trace per held schedule. Apparent uncertainty is an observation-relative distinction ledger over that support. Exact success parts summarise the branch partition after execution. If a physical source supplies schedules, its distribution is measured at the one-way empirical boundary and cannot flow backward into the algorithm law.

6. Halting, incompleteness and famous limits

The universal description grammar supports self-description and a held complement operation. Assuming a total internal halting decider permits construction of a process that rejects exactly when the decider predicts acceptance and accepts exactly when it predicts rejection on its own description. Neither held verdict is a fixed point. The contradiction closes the total-decision boundary without treating nontermination as numerical zero. The same missing distinction appears in effective self-verification: a complete internal proof enumerator cannot both retain consistency and supply every demanded self-referential verdict.

7. Security, learning and scientific evidence

Security claims expose the message, key, observation, adversary and resource supports. Information-theoretic security means an observation closes the relevant source distinctions over complete support. Computational security means the registered adversary grammar and resource boundary fail to reopen them. Learning claims similarly retain complete hypothesis alternatives, exact evidence compatibility and sealed evaluation. A blind opaque score cannot replace the premise-to-result trace. Scientific simulations prove consequences of a registered model; only separate, sealed, target-custodied measurements can test a natural law.

8. Dependency order and complete execution census

Order	Sub-branch	Claim	Candidates	Closure
1	Formal Computation	`SFT-COMP-FORM-STATE-TRANSITION-001`	256	depth-independent
2	Formal Computation	`SFT-COMP-FORM-LANGUAGE-GRAMMAR-001`	256	depth-independent
3	Formal Computation	`SFT-COMP-FORM-AUTOMATON-001`	256	depth-independent
4	Formal Computation	`SFT-COMP-FORM-REWRITING-001`	256	depth-independent
5	Formal Computation	`SFT-COMP-FORM-RECURSIVE-FUNCTION-001`	256	depth-independent
6	Formal Computation	`SFT-COMP-FORM-LAMBDA-CALCULUS-001`	256	depth-independent
7	Formal Computation	`SFT-COMP-FORM-ABSTRACT-MACHINE-001`	256	depth-independent
8	Formal Computation	`SFT-COMP-FORM-CIRCUIT-001`	256	depth-independent
9	Formal Computation	`SFT-COMP-FORM-OPERATIONAL-PROCESS-001`	256	depth-independent
10	Formal Computation	`SFT-COMP-FORM-COMPOSITION-001`	256	depth-independent
11	Formal Computation	`SFT-COMP-FORM-MODEL-EQUIVALENCE-001`	256	depth-independent
12	Formal Computation	`SFT-COMP-FORM-UNIVERSALITY-001`	256	depth-independent
13	Computability	`SFT-COMP-CBL-RECOGNITION-DECISION-001`	256	depth-independent
14	Computability	`SFT-COMP-CBL-HALTING-001`	256	depth-independent
15	Computability	`SFT-COMP-CBL-ENUMERATION-001`	256	depth-independent
16	Computability	`SFT-COMP-CBL-REDUCTION-001`	256	depth-independent
17	Computability	`SFT-COMP-CBL-DEGREES-001`	256	depth-independent
18	Computability	`SFT-COMP-CBL-UNIVERSAL-MACHINE-001`	256	depth-independent
19	Computability	`SFT-COMP-CBL-UNDECIDABILITY-001`	256	depth-independent
20	Computability	`SFT-COMP-CBL-INCOMPLETENESS-001`	256	depth-independent
21	Computability	`SFT-COMP-CBL-RELATIVE-ORACLE-001`	256	depth-independent
22	Computability	`SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001`	256	depth-independent
23	Computational Complexity	`SFT-COMP-CPLX-INPUT-SIZE-001`	256	depth-independent
24	Computational Complexity	`SFT-COMP-CPLX-TIME-SPACE-001`	256	depth-independent
25	Computational Complexity	`SFT-COMP-CPLX-CIRCUIT-RESOURCE-001`	256	depth-independent
26	Computational Complexity	`SFT-COMP-CPLX-COMMUNICATION-QUERY-001`	256	depth-independent
27	Computational Complexity	`SFT-COMP-CPLX-RANDOMNESS-001`	256	depth-independent
28	Computational Complexity	`SFT-COMP-CPLX-REVERSIBILITY-COST-001`	256	depth-independent

Order	Sub-branch	Claim	Candidates	Closure
29	Computational Complexity	SFT-COMP-CPLX-PARALLEL-001	256	depth-independent
30	Computational Complexity	SFT-COMP-CPLX-BOUNDS-001	256	depth-independent
31	Computational Complexity	SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001	256	depth-independent
32	Computational Complexity	SFT-COMP-CPLX-AVERAGE-WORST-001	256	depth-independent
33	Computational Complexity	SFT-COMP-CPLX-APPROXIMATION-001	256	depth-independent
34	Computational Complexity	SFT-COMP-CPLX-PARAMETERIZED-001	256	depth-independent
35	Computational Complexity	SFT-COMP-CPLX-DESCRIPTIVE-001	256	depth-independent
36	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-SEARCH-ORDER-001	256	depth-independent
37	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-ARITHMETIC-001	256	depth-independent
38	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-STRINGS-SEQUENCES-001	256	depth-independent
39	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-TREES-GRAPHS-001	256	depth-independent
40	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001	256	depth-independent
41	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001	256	depth-independent
42	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-OPTIMIZATION-001	256	depth-independent
43	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-RANDOMIZED-001	256	depth-independent
44	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-PARALLEL-001	256	depth-independent
45	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-DISTRIBUTED-001	256	depth-independent
46	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-ONLINE-STREAMING-001	256	depth-independent
47	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-NUMERICAL-001	256	depth-independent
48	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-SYMBOLIC-001	256	depth-independent
49	Algorithms and Mathematical Data Structures	SFT-COMP-ALG-APPROXIMATE-001	256	depth-independent
50	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-SYNTAX-001	256	depth-independent
51	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-BINDING-SUBSTITUTION-001	256	depth-independent

Order	Sub-branch	Claim	Candidates	Closure
52	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-EVALUATION-001	256	depth-independent
53	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001	256	depth-independent
54	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-TYPE-001	256	depth-independent
55	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001	256	depth-independent
56	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-TERMINATION-001	256	depth-independent
57	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-CORRECTNESS-001	256	depth-independent
58	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-SPECIFICATION-001	256	depth-independent
59	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-TRANSFORMATION-001	256	depth-independent
60	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-COMPILATION-001	256	depth-independent
61	Semantics and Mathematical Programming Theory	SFT-COMP-SEM-VERIFICATION-001	256	depth-independent
62	Concurrent and Distributed Computation	SFT-COMP-DIST-CONCURRENCY-001	256	depth-independent
63	Concurrent and Distributed Computation	SFT-COMP-DIST-CAUSALITY-001	256	depth-independent
64	Concurrent and Distributed Computation	SFT-COMP-DIST-SYNCHRONIZATION-001	256	depth-independent
65	Concurrent and Distributed Computation	SFT-COMP-DIST-COMMUNICATION-001	256	depth-independent
66	Concurrent and Distributed Computation	SFT-COMP-DIST-PARTIAL-ORDER-001	256	depth-independent
67	Concurrent and Distributed Computation	SFT-COMP-DIST-CONSENSUS-001	256	depth-independent
68	Concurrent and Distributed Computation	SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001	256	depth-independent
69	Concurrent and Distributed Computation	SFT-COMP-DIST-REPLICATION-CONSISTENCY-001	256	depth-independent
70	Concurrent and Distributed Computation	SFT-COMP-DIST-FAULT-MODEL-001	256	depth-independent
71	Concurrent and Distributed Computation	SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001	256	depth-independent
72	Concurrent and Distributed Computation	SFT-COMP-DIST-LOCALITY-001	256	depth-independent

Order	Sub-branch	Claim	Candidates	Closure
73	Concurrent and Distributed Computation	SFT-COMP-DIST-NETWORK-COMPUTATION-001	256	depth-independent
74	Cryptography and Computational Security	SFT-COMP-SEC-ONE-WAYNESS-001	256	depth-independent
75	Cryptography and Computational Security	SFT-COMP-SEC-SECRECY-001	256	depth-independent
76	Cryptography and Computational Security	SFT-COMP-SEC-INTEGRITY-001	256	depth-independent
77	Cryptography and Computational Security	SFT-COMP-SEC-AUTHENTICATION-001	256	depth-independent
78	Cryptography and Computational Security	SFT-COMP-SEC-HASHING-001	256	depth-independent
79	Cryptography and Computational Security	SFT-COMP-SEC-COMMITMENT-001	256	depth-independent
80	Cryptography and Computational Security	SFT-COMP-SEC-SIGNATURE-001	256	depth-independent
81	Cryptography and Computational Security	SFT-COMP-SEC-PROOF-KNOWLEDGE-001	256	depth-independent
82	Cryptography and Computational Security	SFT-COMP-SEC-ZERO-KNOWLEDGE-001	256	depth-independent
83	Cryptography and Computational Security	SFT-COMP-SEC-MULTIPARTY-001	256	depth-independent
84	Cryptography and Computational Security	SFT-COMP-SEC-ADVERSARIAL-001	256	depth-independent
85	Cryptography and Computational Security	SFT-COMP-SEC-SECURITY-DEFINITION-001	256	depth-independent
86	Cryptography and Computational Security	SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001	256	depth-independent
87	Learning and Intelligence Theory	SFT-COMP-LEARN-INFERENCE-001	256	depth-independent
88	Learning and Intelligence Theory	SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001	256	depth-independent
89	Learning and Intelligence Theory	SFT-COMP-LEARN-REPRESENTATION-001	256	depth-independent
90	Learning and Intelligence Theory	SFT-COMP-LEARN-GENERALIZATION-001	256	depth-independent
91	Learning and Intelligence Theory	SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001	256	depth-independent
92	Learning and Intelligence Theory	SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001	256	depth-independent
93	Learning and Intelligence Theory	SFT-COMP-LEARN-INDUCTION-001	256	depth-independent
94	Learning and Intelligence Theory	SFT-COMP-LEARN-SEARCH-PLANNING-001	256	depth-independent
95	Learning and Intelligence Theory	SFT-COMP-LEARN-REINFORCEMENT-001	256	depth-independent
96	Learning and Intelligence Theory	SFT-COMP-LEARN-MULTIAGENT-001	256	depth-independent

Order	Sub-branch	Claim	Candidates	Closure
97	Learning and Intelligence Theory	SFT-COMP-LEARN-ADAPTATION-001	256	depth-independent
98	Learning and Intelligence Theory	SFT-COMP-LEARN-LEARNING-LIMITS-001	256	depth-independent
99	Learning and Intelligence Theory	SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001	256	depth-independent
100	Learning and Intelligence Theory	SFT-COMP-LEARN-CLASSICAL-LEARNING-001	256	depth-independent
101	Scientific Computation	SFT-COMP-SCI-EXACT-APPROXIMATE-001	256	depth-independent
102	Scientific Computation	SFT-COMP-SCI-STABILITY-001	256	depth-independent
103	Scientific Computation	SFT-COMP-SCI-ERROR-PROPAGATION-001	256	depth-independent
104	Scientific Computation	SFT-COMP-SCI-DISCRETIZATION-001	256	depth-independent
105	Scientific Computation	SFT-COMP-SCI-CONVERGENCE-001	256	depth-independent
106	Scientific Computation	SFT-COMP-SCI-SYMBOLIC-001	256	depth-independent
107	Scientific Computation	SFT-COMP-SCI-SIMULATION-001	256	depth-independent
108	Scientific Computation	SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001	256	depth-independent
109	Scientific Computation	SFT-COMP-SCI-INVERSE-PROBLEM-001	256	depth-independent
110	Scientific Computation	SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001	256	depth-independent
111	Scientific Computation	SFT-COMP-SCI-HIGH-DIMENSIONAL-001	256	depth-independent
112	Scientific Computation	SFT-COMP-SCI-MANY-BODY-001	256	depth-independent
113	Scientific Computation	SFT-COMP-SCI-MATHEMATICAL-MODELLING-001	256	depth-independent
114	Computability	SFT-COMP-CBL-NATIVE-BUSY-BEAVER-002	256	depth-independent
115	Computational Complexity	SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002	256	depth-independent
116	Computational Complexity	SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002	256	depth-independent

The branch total is **29,696 candidates**, **116 survivors**, **464 adverse controls** and **116 independent validations**.

Family results, complete claim inventory and claim-level audit records

The dependency-ordered ledger below is the complete human-readable Classical Computation claim inventory for this version. Each claim section exposes its claim identity, family, formal and empirical status, exact statement, dependency route, carrier and boundary, candidate grammar and count, unique survivor, elimination logic, falsification condition, controls, source and custody records, chronology, scientific meaning and receipt identities. Where a generic clause is referenced, its full unchanged wording is retained in the shared claim-record appendix. The machine packages remain authoritative for complete candidates, decisions and executable traces.

9. Derivation 1: Exact state and transition law

Claim: `SFT-COMP-FORM-STATE-TRANSITION-001`

9.1 Scientific necessity and exact theorem

Exact state and transition law is required to derive generated configurations and change without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced generated configurations and change kernel retains complete state carrier; held action labels; source-bound transition rows; initial and terminal classes; retained execution traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

9.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete state carrier; held action labels; source-bound transition rows; initial and terminal classes; retained execution traces.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for generated configurations and change.

> *All generated finite generated configurations and change structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Does not yet define a language, machine model or resource bound.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

9.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced generated configurations and change kernel retains complete state carrier; held action labels; source-bound transition rows; initial and terminal classes; retained execution traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

9.4 Operational laws and consequences

- Carrier law: complete state carrier; held action labels; source-bound transition rows; initial and terminal classes; retained execution traces.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

9.5 Depth-independent closure

Base: The structural One supplies one canonical generated configurations and change form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated configuration-and-change form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

9.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-STATE-TRANSITION-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-STATE-TRANSITION-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-STATE-TRANSITION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not yet define a language, machine model or resource bound.	`PASS`

9.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: state, transition system.

Does not yet define a language, machine model or resource bound.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Does not yet define a language, machine model or resource bound.

9.8 Exact evidence identities

- Source manifest: `sha256:d6baac855792700e19eea8e212398cc9d04f90d5282e644b593777276c3aa6ec`
- Complete-census certificate: `sha256:a9e5bca9c8d3f9281489c4f6ddb6dfa34b58fa0915aff9d33a77eec0d33495f0`
- Derivation seal: `sha256:52d77e7c18c95f893f56bae7f7710e36ce0ab176b01d1f8b868be0ad6e0385f0`
- Independent implementation: `sha256:c2917eb384de1fd43f363ae486fd3db8c6670b6199a9f0de76519dc719d78e62`
- Independent certificate: `sha256:36071490b1ec6ae387124089b784f55bfdc63f7664cc91f3e45d45298b673299`
- External validation: `sha256:8df5268b3c4793c2dc19ad1082c591e928ab99d7aca1469d9d9870749143a0f9`
- Model-admitted engine receipt: `sha256:6ff1fdc0b1d041805addd6ef9350dee1fe38a41082072c9f7888d4a22975228f`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-STATE-TRANSITION-001-6ff1fdc0b1d04180.json`

10. Derivation 2: Fold formal-language and grammar law

Claim: `SFT-COMP-FORM-LANGUAGE-GRAMMAR-001`

10.1 Scientific necessity and exact theorem

Fold formal-language and grammar law is required to derive generated symbol words and formation without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced generated symbol words and formation kernel retains complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-STATE-TRANSITION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

10.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for generated symbol words and formation.

> *All generated finite generated symbol words and formation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Closes generated finite languages and depth-independent word succession, not a completed infinite language object.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

10.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.

Eliminated candidates	Exact first-failure reason
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced generated symbol words and formation kernel retains complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

10.4 Operational laws and consequences

- Carrier law: complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

10.5 Depth-independent closure

Base: The structural One supplies one canonical generated symbol words and formation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated symbol-word-and-formation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

10.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-LANGUAGE-GRAMMAR-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-LANGUAGE-GRAMMAR-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-LANGUAGE-GRAMMAR-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closes generated finite languages and depth-independent word succession, not a completed infinite language object.	`PASS`

10.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: formal language, grammar.

Closes generated finite languages and depth-independent word succession, not a completed infinite language object.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Closes generated finite languages and depth-independent word succession, not a completed infinite language object.

10.8 Exact evidence identities

- Source manifest: `sha256:e7e49951c44319178d53b75d6dbc14f7fcdf81dea5b263bf7e4215c3f358094e`
- Complete-census certificate: `sha256:009322561ae741ea9b0f0f7bc97850497effbfa14b253d5eb1ffa19ce738e68f`
- Derivation seal: `sha256:688db340733ac157427caf7da72b4bb5f97b86627b4a080ce828a3d5bd9adb4e`
- Independent implementation: `sha256:70802d179f08a4a5a6b2f178c5bd11fa87d64689159b14694def28731b79e5c8`
- Independent certificate: `sha256:8a40d9a22d068e806f0c2d49a9af923f6fa29101600069cb311d9f82d40d8cfd`
- External validation: `sha256:9ace6f1cbeddd3086e42f5b9dce75bf37c34fa29abb5ed8b97fd5bf9d476ac67`
- Model-admitted engine receipt: `sha256:9247a863bf580966b3db1091ef6024a0f4562506076136dce33acda2a6039d14`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-LANGUAGE-GRAMMAR-001-9247a863bf580966.json`

11. Derivation 3: Fold finite automaton law

Claim: `SFT-COMP-FORM-AUTOMATON-001`

11.1 Scientific necessity and exact theorem

Fold finite automaton law is required to derive finite recognition processes without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced finite recognition processes kernel retains finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-LANGUAGE-GRAMMAR-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

11.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for finite recognition processes.

> *All finite generated recognition-process structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Does not yet assert equivalence with every other model or universality.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

11.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.

Eliminated candidates	Exact first-failure reason
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced finite recognition processes kernel retains finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

11.4 Operational laws and consequences

- Carrier law: finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

11.5 Depth-independent closure

Base: The structural One supplies one canonical finite recognition processes form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite recognition processes form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

11.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-AUTOMATON-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-AUTOMATON-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-AUTOMATON-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not yet assert equivalence with every other model or universality.	`PASS`

11.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: finite automaton, recognizer.

Does not yet assert equivalence with every other model or universality.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Does not yet assert equivalence with every other model or universality.

11.8 Exact evidence identities

- Source manifest: `sha256:2875e2c5834ff96fc547959b332f1a1f7a7e84c85bea5e3873518751be60cc6e`
- Complete-census certificate: `sha256:59bade0640ecca5d80d75d3a8b0a37298a314db962cfb92dcbeb82b6873ee981`
- Derivation seal: `sha256:de24c317c1f621d5ec62ecb401e94a4fccfba4de95dabad9ab144a71748f731c`
- Independent implementation: `sha256:0e08a4bae652b4772475643b97ad32e52bf14f55de8fefa0ff3218b5ef5e22e4`
- Independent certificate: `sha256:c6d7e75b9dbd097a50865aa3fb476bc057d2a5a2b933b069a694f021a4cc7d27`
- External validation: `sha256:17f43954c36004983c2299b4c7468c5e5816b764b5c8735af7b0275b942a66d2`
- Model-admitted engine receipt: `sha256:a28d36b12b07be8aa4f32af257b75d6da331b0cebfea99fda3d6240e9ccaf109`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-AUTOMATON-001-a28d36b12b07be8a.json`

12. Derivation 4: Fold rewriting-system law

Claim: `SFT-COMP-FORM-REWRITING-001`

12.1 Scientific necessity and exact theorem

Fold rewriting-system law is required to derive held-position replacement without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced held-position replacement kernel retains generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-AUTOMATON-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

12.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for held-position replacement.

> *All generated finite held-position replacement structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Only registered finite rules and generated words are admitted.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

12.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.

Eliminated candidates	Exact first-failure reason
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced held-position replacement kernel retains generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

12.4 Operational laws and consequences

- Carrier law: generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

12.5 Depth-independent closure

Base: The structural One supplies one canonical held-position replacement form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated held-position replacement form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

12.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-REWRITING-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-REWRITING-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-REWRITING-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only registered finite rules and generated words are admitted.	`PASS`

12.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: rewriting system, term rewriting.

Only registered finite rules and generated words are admitted.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Only registered finite rules and generated words are admitted.

12.8 Exact evidence identities

- Source manifest: `sha256:4efb2a905b4b82b88562af8bc58df30b63eaa7e37fc6d2026698945df864753d`
- Complete-census certificate: `sha256:ea26ea052050e13e4329e7e31cca7d1afdc3560dd5a87c6621a4ea7f58e381ca`
- Derivation seal: `sha256:7f1279c1b5b5d6b2e31d821616cdc40fbb8e89e782ad571f2cd65486a9569727`
- Independent implementation: `sha256:1d03336bec2e9efb5b8a8338de71b228a4a517b8fee08c2146539375583d26b8`
- Independent certificate: `sha256:f5faf737b0c4341fe3c50b95d147053ef79cdb85a9ad4971d22b3304c7bb3466`
- External validation: `sha256:13b91a821e5ab3ea07326aed719fde2054ecd40561f9787f4d5d3b1218d24ef1`
- Model-admitted engine receipt: `sha256:b1f6c54c1f8be568e0dbb809dc19f1f882840e53c693aa208db0c2410860defe`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-REWRITING-001-b1f6c54c1f8be568.json`

13. Derivation 5: Fold recursion law

Claim: `SFT-COMP-FORM-RECURSIVE-FUNCTION-001`

13.1 Scientific necessity and exact theorem

Fold recursion law is required to derive structural finite recursion without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced structural finite recursion kernel retains structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result; every operation is source-bound, trace-complete,*

boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-REWRITING-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

13.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for structural finite recursion.

All generated finite structural finite recursion structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Closes recursion over every generated finite successor trace, not completed transfinite recursion.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

13.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.

Eliminated candidates	Exact first-failure reason
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced structural finite recursion kernel retains structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

13.4 Operational laws and consequences

- Carrier law: structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

13.5 Depth-independent closure

Base: The structural One supplies one canonical structural finite recursion form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated structural finite recursion form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

13.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-RECURSIVE-FUNCTION-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-RECURSIVE-FUNCTION-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-RECURSIVE-FUNCTION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closes recursion over every generated finite successor trace, not completed transfinite recursion.	`PASS`

13.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: recursive function, primitive recursion.

Closes recursion over every generated finite successor trace, not completed transfinite recursion.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Closes recursion over every generated finite successor trace, not completed transfinite recursion.

13.8 Exact evidence identities

- Source manifest: `sha256:348ad460880a7aa65b3209f59bb02bdcbeaa373fa7b536e7ccaefea2bad37d18`
- Complete-census certificate: `sha256:549d897e9cba6810b09f251e85e82e2d398e63bd5b8bab2cc820dcbf015ffe97`
- Derivation seal: `sha256:f0d1440d842607d5258c5accdcb83682ab3eaf0a58c3dceac901a2c9c730723d`
- Independent implementation: `sha256:f0103e97dcbd277c3c09a157d04752e90b18f719eac050beb7d45f706b36c486`
- Independent certificate: `sha256:9dc55f7ee3aa968c744de817d3bdef595ab14824948958aeb1d57417152fa061`
- External validation: `sha256:24365aa8a72572993447d2cafcf0abb11e9491ece06a19c1441ae242ceecc411`
- Model-admitted engine receipt: `sha256:291230ba67c75760c319e58fe0ae120c6c300a83e39d5f51c70de936c73e580c`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-RECURSIVE-FUNCTION-001-291230ba67c75760.json`

14. Derivation 6: Fold binding and self-application calculus law

Claim: `SFT-COMP-FORM-LAMBDA-CALCULUS-001`

14.1 Scientific necessity and exact theorem

Fold binding and self-application calculus law is required to derive binding, substitution and reduction without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced binding, substitution and reduction kernel retains generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-RECURSIVE-FUNCTION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

14.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for binding, substitution and reduction.

> *All generated finite binding, substitution and reduction structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No conventional lambda calculus is assumed; the result is the generated Fold term kernel.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

14.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced binding, substitution and reduction kernel retains generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

14.4 Operational laws and consequences

- Carrier law: generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

14.5 Depth-independent closure

Base: The structural One supplies one canonical binding, substitution and reduction form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated binding, substitution and reduction form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

14.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-LAMBDA-CALCULUS-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-LAMBDA-CALCULUS-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-LAMBDA-CALCULUS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No conventional lambda calculus is assumed; the result is the generated Fold term kernel.	`PASS`

14.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: lambda calculus, substitution.

No conventional lambda calculus is assumed; the result is the generated Fold term kernel.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No conventional lambda calculus is assumed; the result is the generated Fold term kernel.

14.8 Exact evidence identities

- Source manifest: `sha256:52bb83eb5794290940d213fa2d1cc6cdc62db73ee5dfb6eccf81583fbca292c4`
- Complete-census certificate: `sha256:dfff75c37a0aa841000744b9aeb2b111230f40159b8ebe14a1f419338ab33a87`
- Derivation seal: `sha256:2a809ddc1fcc1b0a016920d46311c76dcd0cd465af22930ae3b823a0ba7e14e3`
- Independent implementation: `sha256:d745cfdbf195dec5c6646f1dfdc6039f85e6170cde0ce4f36f97627314a62ed6`
- Independent certificate: `sha256:f89a69bbf74e6f348b37e64f672de3968639ca340f9d50e9c18b7a29a4a36c06`
- External validation: `sha256:6a8daaa4987a604def6e709ad88ba6c6c5a18065e3d9bc8ce61e12f537d000d6`
- Model-admitted engine receipt: `sha256:c5df90aeea7c0f8459bb98cdfa9343167723cc7b320af2c9fec60f7f5f5bee63`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-LAMBDA-CALCULUS-001-c5df90aeea7c0f84.json`

15. Derivation 7: Fold abstract-machine law

Claim: `SFT-COMP-FORM-ABSTRACT-MACHINE-001`

15.1 Scientific necessity and exact theorem

Fold abstract-machine law is required to derive finite executable machine descriptions without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced finite executable machine descriptions kernel retains description word; configuration carrier; transition kernel; generated store or tape; initial configuration; complete execution and halt trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-LAMBDA-CALCULUS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

15.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: description word; configuration carrier; transition kernel; generated store or tape; initial configuration; complete execution and halt trace.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for finite executable machine descriptions.

> *All finite generated executable-machine-description structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Machines outside the frozen description grammar remain outside the theorem.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

15.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced finite executable machine descriptions kernel retains description word; configuration carrier; transition kernel; generated store or tape; initial configuration; complete execution and halt trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

15.4 Operational laws and consequences

- Carrier law: description word; configuration carrier; transition kernel; generated store or tape; initial configuration; complete execution and halt trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

15.5 Depth-independent closure

Base: The structural One supplies one canonical finite executable machine descriptions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite executable machine descriptions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

15.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-ABSTRACT-MACHINE-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-ABSTRACT-MACHINE-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-ABSTRACT-MACHINE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Machines outside the frozen description grammar remain outside the theorem.	`PASS`

15.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: abstract machine, Turing machine.

Machines outside the frozen description grammar remain outside the theorem.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- Machines outside the frozen description grammar remain outside the theorem.

15.8 Exact evidence identities

- Source manifest: `sha256:64c1a785e68928851b39460f04755c042d9c7cf57ea547b9f90e1604d55a00b4`
- Complete-census certificate: `sha256:45abdaafd0f0e6e537991e14ab4dda06923162c64e70075289ee5b3a4a320976`
- Derivation seal: `sha256:1bf45a77a8defe269a94cc9471ce829b8b518f30acb09bcaab1a18d351cc508f`
- Independent implementation: `sha256:b0c8e900076f04ce4e88db72f4300ddec7b44e871c574fda6828f04743b6437a`
- Independent certificate: `sha256:3ce91feb3e76ea25d296bfcc777100313354b171985af741b72cbbe204dc2113`
- External validation: `sha256:0062195e4657ede78fadd3877e80e38bd845ffef5f981c6a577d68b4c6e04815`
- Model-admitted engine receipt: `sha256:2b2471771a06a549143d109bd88b40339b476c36ab65b278fd86a7b54adbdd1b`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-ABSTRACT-MACHINE-001-2b2471771a06a549.json`

16. Derivation 8: Fold circuit syntax and evaluation law

Claim: `SFT-COMP-FORM-CIRCUIT-001`

16.1 Scientific necessity and exact theorem

Fold circuit syntax and evaluation law is required to derive finite connected local transformations without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced finite connected local transformations kernel retains generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-ABSTRACT-MACHINE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

16.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for finite connected local transformations.

> *All finite generated connected-local-transformation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Resource size and depth comparisons are deferred to Complexity.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

16.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced finite connected local transformations kernel retains generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

16.4 Operational laws and consequences

- Carrier law: generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.

- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

16.5 Depth-independent closure

Base: The structural One supplies one canonical finite connected local transformations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite connected local transformations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

16.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-CIRCUIT-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-CIRCUIT-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-CIRCUIT-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Resource size and depth comparisons are deferred to Complexity.	`PASS`

16.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: circuit, Boolean circuit.

Resource size and depth comparisons are deferred to Complexity.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Resource size and depth comparisons are deferred to Complexity.

16.8 Exact evidence identities

- Source manifest: `sha256:1411276748c6c4373ac3ec63581aac086a1a3b2ad626d6c46aaa3b47a30d5a75`

- Complete-census certificate: `sha256:600387cd145201a86f224f4cb51476ca311cb2cca9535665ae50d39b7e4862c6`
- Derivation seal: `sha256:62e2b314384c853bb679b205fb39f47f76ee48f07eb26e225f0f2fee5861baf6`
- Independent implementation: `sha256:4d4a98f08cf35dab2175a66857e1f2764741107ab27f0809cc028907180c443b`
- Independent certificate: `sha256:b73405d93e9177c20dc3877407ce8bc3f169c481b8a298624f0e290625d87413`
- External validation: `sha256:d1e5487d81c976c6526694efaf7845bd437def19c61290a024f75abee8e5a7e7`
- Model-admitted engine receipt: `sha256:0a974615e0f53938a3c8551caabc041da5b7c241dfdc059febcb5d38269a0c94`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-CIRCUIT-001-0a974615e0f53938.json`

17. Derivation 9: Fold operational-process law

Claim: `SFT-COMP-FORM-OPERATIONAL-PROCESS-001`

17.1 Scientific necessity and exact theorem

Fold operational-process law is required to derive execution and observable process behaviour without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced execution and observable process behavior kernel retains initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

17.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for execution and observable process behaviour.

> *All generated finite execution and observable process behavior structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Does not import process algebra or distributed timing.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

17.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced execution and observable process behavior kernel retains initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

17.4 Operational laws and consequences

- Carrier law: initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

17.5 Depth-independent closure

Base: The structural One supplies one canonical execution and observable process behaviour form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated execution and observable process behaviour form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

17.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-OPERATIONAL-PROCESS-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-OPERATIONAL-PROCESS-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-OPERATIONAL-PROCESS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not import process algebra or distributed timing.	`PASS`

17.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: operational process, labelled transition system.

Does not import process algebra or distributed timing.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Does not import process algebra or distributed timing.

17.8 Exact evidence identities

- Source manifest: `sha256:ca32b606f98729a29a5a4a2c123efa575bd499b2522eb2b8b0eb1c73186eebf4`
- Complete-census certificate: `sha256:e5067d5b4f2f25d4b346d042203f2033e326cea7b653659447b60d7d47c5e93c`
- Derivation seal: `sha256:16f15c2211974316eb7f5595c532cc6e649fe07a22c56adf7a8673b7bc3466d3`

- Independent implementation: `sha256:edf9d2ca579edfbdcf5f372b21b3c37de52c998e8b92c5e4a136ab0d31b454ba`
- Independent certificate: `sha256:a37e121f9fca530b933f74e80c8877fd811ab4e64923192cc395984dfaf030a6`
- External validation: `sha256:b5d7bc8b74c9ed55eac3a59349e3a88148e1ef2411d26dcb672e6a97a0873579`
- Model-admitted engine receipt: `sha256:d3349d79836c6e8edf5199206fe0dc818a6210e9d3dc7f8000eb83f0b55ef282`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-OPERATIONAL-PROCESS-001-d3349d79836c6e8e.json`

18. Derivation 10: Fold process composition and decomposition law

Claim: `SFT-COMP-FORM-COMPOSITION-001`

18.1 Scientific necessity and exact theorem

Fold process composition and decomposition law is required to derive serial and parallel process assembly without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced serial and parallel process assembly kernel retains exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-OPERATIONAL-PROCESS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

18.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for serial and parallel process assembly.

> *All generated finite serial and parallel process assembly structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Concurrency laws requiring communication or shared causality remain downstream.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

18.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced serial and parallel process assembly kernel retains exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

18.4 Operational laws and consequences

- Carrier law: exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

18.5 Depth-independent closure

Base: The structural One supplies one canonical serial and parallel process assembly form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated serial and parallel process assembly form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

18.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-COMPOSITION-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-COMPOSITION-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-COMPOSITION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Concurrency laws requiring communication or shared causality remain downstream.	`PASS`

18.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: composition, decomposition.

Concurrency laws requiring communication or shared causality remain downstream.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Concurrency laws requiring communication or shared causality remain downstream.

18.8 Exact evidence identities

- Source manifest: `sha256:a5acf66751f0f1f687e326d1a5d57eb0d930f9341a51301ecb7778266222f3e1`
- Complete-census certificate: `sha256:828753dbe5d5d32a929bc631a6118bd4c2e9161986252e66ebb326b454d79a36`
- Derivation seal: `sha256:1a5b215a9ad08b3cf8665c8299fa6961b1a43c87847b1b705adfbc1419827716`
- Independent implementation: `sha256:01a1fd9217764d6beb2dfda7b127626b041b729945527f7cce6a4c4a7a22c756`
- Independent certificate: `sha256:11adf318b9899ccc822e4c9164ac86c61596df45c8b41bcb54d268785aa279a2`
- External validation: `sha256:91e36576fe2613746be520df07a60d9db80cd25dce24a78dd1b023d23c8c438b`

- Model-admitted engine receipt:
  `sha256:b0fea8f79a1c2c934c5b2dfaa37c9ba09e8b0f1416377175a645afe9f58738ad`
- Receipt path:
  `receipts/engine/model_admitted/SFT-COMP-FORM-COMPOSITION-001-b0fea8f79a1c2c93.json`

19. Derivation 11: Fold computational-model equivalence law

Claim: `SFT-COMP-FORM-MODEL-EQUIVALENCE-001`

19.1 Scientific necessity and exact theorem

Fold computational-model equivalence law is required to derive bidirectional execution-preserving translations without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced bidirectional execution-preserving translations kernel retains complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-COMPOSITION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

19.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for bidirectional execution-preserving translations.

> *All generated finite bidirectional execution-preserving translations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Equivalence applies only to the two registered generated model grammars.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

19.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced bidirectional execution-preserving translations kernel retains complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

19.4 Operational laws and consequences

- Carrier law: complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

19.5 Depth-independent closure

Base: The structural One supplies one canonical bidirectional execution-preserving translations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated bidirectional execution-preserving translations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

19.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-MODEL-EQUIVALENCE-001; result: `PASS`.

- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-MODEL-EQUIVALENCE-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-MODEL-EQUIVALENCE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Equivalence applies only to the two registered generated model grammars.	`PASS`

19.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: model equivalence, simulation.

Equivalence applies only to the two registered generated model grammars.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Equivalence applies only to the two registered generated model grammars.

19.8 Exact evidence identities

- Source manifest: `sha256:0162711d6244b48f48e65f584f7d4d7ebec34ca89f8ddd5803006fa81eb7c497`
- Complete-census certificate: `sha256:514c11ad3899b13152b2fbfa77adc36da6115e4c1452e7993b43d0261ab40488`
- Derivation seal: `sha256:3c0dad2425378918bacf1c277636db591165b0660c98f8c19d1f4ec68d7c1eb9`
- Independent implementation: `sha256:7f980a25b82bbe2793268f94eaf9f482120d4fc8ec8a889a6d345052c7e04f40`
- Independent certificate: `sha256:62db23d8401df6af4f79ceb122b3a85440ff45189a7bb09dc7d5e6722c6e99ce`
- External validation: `sha256:64e6d04fde05fcaba7e98b2cef7ca0a8c7f7de711e8a754d145b9f67e2377bc6`
- Model-admitted engine receipt: `sha256:911ff906b2d8407111a076f3abb06abe79334098a0e87faa54460f8453338139`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-MODEL-EQUIVALENCE-001-911ff906b2d84071.json`

20. Derivation 12: Fold universal-computation law

Claim: `SFT-COMP-FORM-UNIVERSALITY-001`

20.1 Scientific necessity and exact theorem

Fold universal-computation law is required to derive interpretation of generated machine descriptions without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced interpretation of generated machine descriptions kernel retains complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-MODEL-EQUIVALENCE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

20.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead.

Generation rule: Generate the literal product of the eight registered Formal Computation structural axes for interpretation of generated machine descriptions.

> *All generated finite interpretation of generated machine descriptions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Universality is exact for every description generated by the frozen grammar, not an unbounded completed totality.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`carrier`	`sampled-carrier`	sampled-carrier removes a required exact carrier condition.	`complete-generated-carrier`	complete-generated-carrier retains the complete registered carrier condition.
`identity`	`presentation-identity`	presentation-identity removes a required exact identity condition.	`canonical-held-identity`	canonical-held-identity retains the complete registered identity condition.
`relation`	`assumed-operation`	assumed-operation removes a required exact relation condition.	`source-bound-exact-relation`	source-bound-exact-relation retains the complete registered relation condition.
`trace`	`result-only`	result-only removes a required exact trace condition.	`complete-retained-trace`	complete-retained-trace retains the complete registered trace condition.
`boundary`	`implicit-terminal-boundary`	implicit-terminal-boundary removes a required exact boundary condition.	`explicit-initial-terminal-boundary`	explicit-initial-terminal-boundary retains the complete registered boundary condition.
`composition`	`opaque-assembly`	opaque-assembly removes a required exact composition condition.	`interface-preserving-composition`	interface-preserving-composition retains the complete registered composition condition.
`generality`	`fixed-examples`	fixed-examples removes a required exact generality condition.	`structural-successor`	structural-successor retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`addition`	`imported-machine-model`	imported-machine-model removes a required exact addition condition.	`no-extra-computational-premise`	no-extra-computational-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

20.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-carrier removes a required exact carrier condition.
64	presentation-identity removes a required exact identity condition.
32	assumed-operation removes a required exact relation condition.
16	result-only removes a required exact trace condition.
8	implicit-terminal-boundary removes a required exact boundary condition.
4	opaque-assembly removes a required exact composition condition.
2	fixed-examples removes a required exact generality condition.
1	imported-machine-model removes a required exact addition condition.

Shared claim-record clause `COMP-S003` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced interpretation of generated machine descriptions kernel retains complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

20.4 Operational laws and consequences

- Carrier law: complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

20.5 Depth-independent closure

Base: The structural One supplies one canonical interpretation of generated machine descriptions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated interpretation of generated machine descriptions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

20.6 Executed witnesses and adverse controls

- `formal_computation-witness-1` - formal_computation exact operational witness 1 for SFT-COMP-FORM-UNIVERSALITY-001; result: `PASS`.
- `formal_computation-witness-2` - formal_computation exact operational witness 2 for SFT-COMP-FORM-UNIVERSALITY-001; result: `PASS`.
- `formal_computation-witness-3` - formal_computation exact operational witness 3 for SFT-COMP-FORM-UNIVERSALITY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-carrier removes a required exact carrier condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Universality is exact for every description generated by the frozen grammar, not an unbounded completed totality.	`PASS`

20.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: universal computation, universal machine.

Universality is exact for every description generated by the frozen grammar, not an unbounded completed totality.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Universality is exact for every description generated by the frozen grammar, not an unbounded completed totality.

20.8 Exact evidence identities

- Source manifest: `sha256:6cbb5b4a2345df65875b39ced2bae1002a4efeb96a2aa4115a7faecf6aadf321`
- Complete-census certificate: `sha256:5d1074e1fc7f1c4b882429494e4239518885cc13ca5cd62ffb035974cfbc3caa`
- Derivation seal: `sha256:aa872db026edad77bae474d940bdead07caed584e3013fc04c986d0d927ad80d`
- Independent implementation: `sha256:f1df70199e93c004a3d9c9e2a3bf396a74626e742e81b0b8e213b3af7e9e8819`
- Independent certificate: `sha256:bf89e7bbbd33de8dc7caaf3f5bdf3d6979d628eacc66183a02411b407236e76b`
- External validation: `sha256:9901d8e8f04985a7b65da1a40504fa8f1df13e92f614953ef900fcd87ef4ace6`
- Model-admitted engine receipt: `sha256:83a401709b86991de5f38a49e7794379ea0c041fb2a406a245f900bf11bcb598`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-FORM-UNIVERSALITY-001-83a401709b86991d.json`

21. Derivation 13: Fold recognition and decision law

Claim: `SFT-COMP-CBL-RECOGNITION-DECISION-001`

21.1 Scientific necessity and exact theorem

Fold recognition and decision law is required to derive acceptance, rejection and continued execution without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced acceptance, rejection and continued execution kernel retains generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-UNIVERSALITY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

21.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger.

Generation rule: Generate the literal product of the eight registered Computability structural axes for acceptance, rejection and continued execution.

> *All generated finite acceptance, rejection and continued execution structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A recognizer need not terminate on every generated input; a decider must.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-descriptions`	sampled-descriptions removes a required exact domain condition.	`complete-generated-description-domain`	complete-generated-description-domain retains the complete registered domain condition.
`execution`	`verdict-without-run`	verdict-without-run removes a required exact execution condition.	`exact-recognition-execution-trace`	exact-recognition-execution-trace retains the complete registered execution condition.
`result`	`implicit-nontermination`	implicit-nontermination removes a required exact result condition.	`accept-reject-continue-classes`	accept-reject-continue-classes retains the complete registered result condition.
`self_reference`	`self-reference-excluded`	self-reference-excluded removes a required exact self_reference condition.	`generated-self-description-support`	generated-self-description-support retains the complete registered self_reference condition.
`totality`	`assumed-total-decider`	assumed-total-decider removes a required exact totality condition.	`explicit-totality-or-partiality-boundary`	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
`translation`	`answer-changing-map`	answer-changing-map removes a required exact translation condition.	`answer-preserving-reduction`	answer-preserving-reduction retains the complete registered translation condition.
`generality`	`fixed-depth-table`	fixed-depth-table removes a required exact generality condition.	`description-successor-certificate`	description-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-oracle`	hidden-oracle removes a required exact addition condition.	`no-undeclared-computational-power`	no-undeclared-computational-power retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

21.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

Shared claim-record clause `COMP-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced acceptance, rejection and continued execution kernel retains generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

21.4 Operational laws and consequences

- Carrier law: generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

21.5 Depth-independent closure

Base: The structural One supplies one canonical acceptance, rejection and continued execution form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated acceptance, rejection and continued execution form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

21.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-RECOGNITION-DECISION-001; result: `PASS`.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-RECOGNITION-DECISION-001; result: `PASS`.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-RECOGNITION-DECISION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A recognizer need not terminate on every generated input; a decider must.	`PASS`

21.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: recognizability, decidability.

A recognizer need not terminate on every generated input; a decider must.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A recognizer need not terminate on every generated input; a decider must.

21.8 Exact evidence identities

- Source manifest: `sha256:fcdd2db93deb173b9cd92f0ba025417ba3126423bffb0090d833fad544c5da89`
- Complete-census certificate: `sha256:1ac8edb72bda3ab3a65f70220c0981f627bf680a42946a7501995a1b54deb22b`
- Derivation seal: `sha256:11e94644591c3e7e51e1b0235bb9752f71fcd7ce64f457291bf83968e902a4a6`
- Independent implementation: `sha256:dc0f0486633f8e87e5f9e02b210b5376b2cf778691ca860a41eedcfa874929a6`
- Independent certificate: `sha256:0eb8c9783f7e87e434555864f84644fcd1c096bcc3bd1725540ffdd033fb0aa1`
- External validation: `sha256:9151b6040f1b47fae6bb0018232dc6b6399177add50d2f9a768f723f69bb6ebe`
- Model-admitted engine receipt: `sha256:c9c992b18e4408ecfd5bb004622b3d62d836752b387773eecb7dfc0e1daed9e8`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-RECOGNITION-DECISION-001-c9c992b18e4408ec.json`

22. Derivation 14: Fold halting boundary law

Claim: `SFT-COMP-CBL-HALTING-001`

22.1 Scientific necessity and exact theorem

Fold halting boundary law is required to derive terminal-state determination without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced terminal-state determination kernel retains encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-RECOGNITION-DECISION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

22.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction.

Generation rule: Generate the literal product of the eight registered Computability structural axes for terminal-state determination.

> *All generated finite terminal-state determination structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Does not decide halting for all generated universal-machine descriptions.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-descriptions`	sampled-descriptions removes a required exact domain condition.	`complete-generated-description-domain`	complete-generated-description-domain retains the complete registered domain condition.
`execution`	`verdict-without-run`	verdict-without-run removes a required exact execution condition.	`exact-recognition-execution-trace`	exact-recognition-execution-trace retains the complete registered execution condition.
`result`	`implicit-nontermination`	implicit-nontermination removes a required exact result condition.	`accept-reject-continue-classes`	accept-reject-continue-classes retains the complete registered result condition.
`self_reference`	`self-reference-excluded`	self-reference-excluded removes a required exact self_reference condition.	`generated-self-description-support`	generated-self-description-support retains the complete registered self_reference condition.
`totality`	`assumed-total-decider`	assumed-total-decider removes a required exact totality condition.	`explicit-totality-or-partiality-boundary`	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
`translation`	`answer-changing-map`	answer-changing-map removes a required exact translation condition.	`answer-preserving-reduction`	answer-preserving-reduction retains the complete registered translation condition.
`generality`	`fixed-depth-table`	fixed-depth-table removes a required exact generality condition.	`description-successor-certificate`	description-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-oracle`	hidden-oracle removes a required exact addition condition.	`no-undeclared-computational-power`	no-undeclared-computational-power retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

22.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.

Eliminated candidates	Exact first-failure reason
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

Shared claim-record clause `COMP-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced terminal-state determination kernel retains encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

22.4 Operational laws and consequences

- Carrier law: encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

22.5 Depth-independent closure

Base: The structural One supplies one canonical terminal-state determination form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated terminal-state determination form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

22.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-HALTING-001; result: `PASS`.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-HALTING-001; result: `PASS`.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-HALTING-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not decide halting for all generated universal-machine descriptions.	`PASS`

22.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: halting problem, diagonalization.

Does not decide halting for all generated universal-machine descriptions.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Does not decide halting for all generated universal-machine descriptions.

22.8 Exact evidence identities

- Source manifest: `sha256:66bd1cb6ca985f66e6ae136170730560e94ec8c59f80be5dbb0bcace9ea0b65c`
- Complete-census certificate: `sha256:cf77cceb2366e64454800a0363156968847e58931805d5f2ef68894baab70fd9`
- Derivation seal: `sha256:299c9b39f4e497fe084e28cb9d78834cef211cd11785de98b0fc7bb206f83d2d`
- Independent implementation: `sha256:e5acaf236971ed0d09b140257aaa940a41536fd2f5ee017ca337331933c9be54`
- Independent certificate: `sha256:b3be18262d2f70acfe69d15683428f479c2294731ad192d11d9a06a7d79cdf6d`
- External validation: `sha256:bb240ffe1dbdb05d04360ea66a0f4efd9284d5d1b72f80a8b72590e88dc77a02`
- Model-admitted engine receipt: `sha256:a33ae0f08db036bcde17a25eb3d6605dcfbcd3fa0492176faaf553110d2368e2`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-HALTING-001-a33ae0f08db036bc.json`

23. Derivation 15: Fold effective-enumeration law

Claim: `SFT-COMP-CBL-ENUMERATION-001`

23.1 Scientific necessity and exact theorem

Fold effective-enumeration law is required to derive generated description ordering without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced generated description ordering kernel retains complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-HALTING-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

23.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position.

Generation rule: Generate the literal product of the eight registered Computability structural axes for generated description ordering.

> *All generated finite generated description ordering structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No completed infinite enumeration is installed as an object.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-descriptions`	sampled-descriptions removes a required exact domain condition.	`complete-generated-description-domain`	complete-generated-description-domain retains the complete registered domain condition.
`execution`	`verdict-without-run`	verdict-without-run removes a required exact execution condition.	`exact-recognition-execution-trace`	exact-recognition-execution-trace retains the complete registered execution condition.
`result`	`implicit-nontermination`	implicit-nontermination removes a required exact result condition.	`accept-reject-continue-classes`	accept-reject-continue-classes retains the complete registered result condition.
`self_reference`	`self-reference-excluded`	self-reference-excluded removes a required exact self_reference condition.	`generated-self-description-support`	generated-self-description-support retains the complete registered self_reference condition.
`totality`	`assumed-total-decider`	assumed-total-decider removes a required exact totality condition.	`explicit-totality-or-partiality-boundary`	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
`translation`	`answer-changing-map`	answer-changing-map removes a required exact translation condition.	`answer-preserving-reduction`	answer-preserving-reduction retains the complete registered translation condition.
`generality`	`fixed-depth-table`	fixed-depth-table removes a required exact generality condition.	`description-successor-certificate`	description-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-oracle`	hidden-oracle removes a required exact addition condition.	`no-undeclared-computational-power`	no-undeclared-computational-power retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

23.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.

Eliminated candidates	Exact first-failure reason
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

Shared claim-record clause `COMP-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced generated description ordering kernel retains complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

23.4 Operational laws and consequences

- Carrier law: complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

23.5 Depth-independent closure

Base: The structural One supplies one canonical generated description ordering form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated description-ordering form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

23.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-ENUMERATION-001; result: `PASS`.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-ENUMERATION-001; result: `PASS`.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-ENUMERATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No completed infinite enumeration is installed as an object.	`PASS`

23.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: enumerability, effective enumeration.

No completed infinite enumeration is installed as an object.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No completed infinite enumeration is installed as an object.

23.8 Exact evidence identities

- Source manifest: `sha256:63973d4c123129f31559427f7455012727ae43b7f92ddc58dec264433f9fc9b1`
- Complete-census certificate: `sha256:70b88f63a8011f2ec1e30d67fd31e78efc4d5a1d401582c0ea91ac0f5bd05a82`
- Derivation seal: `sha256:6c0d27b9378dd92f4cdedc7f7d4dc90a02fe816f4c8fbeef2c826ae564712d08`
- Independent implementation: `sha256:c6a5dd8e67bf0218abf2a49489560529f1f066e15a1c597495aa6042686053cd`
- Independent certificate: `sha256:e2a65196b779a1bd41f25443155116ef6e7ffe265c9d0921e544927747b53717`
- External validation: `sha256:56a2658bc7bed4373c7763f34346566f995e7e95bcfcb87ede676a3ceb5e088c`
- Model-admitted engine receipt: `sha256:b4a3d735406f9ab57023c47ed0a12a7a221b278c75727cb0a7170dbdfec940f2`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-ENUMERATION-001-b4a3d735406f9ab5.json`

24. Derivation 16: Fold computability-reduction law

Claim: `SFT-COMP-CBL-REDUCTION-001`

24.1 Scientific necessity and exact theorem

Fold computability-reduction law is required to derive problem translation preserving answers without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced problem translation preserving answers kernel retains source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-ENUMERATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

24.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law.

Generation rule: Generate the literal product of the eight registered Computability structural axes for problem translation preserving answers.

> *All generated finite problem translation preserving answers structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Reduction claims require explicit encoders, decoders and generated domains.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-descriptions`	sampled-descriptions removes a required exact domain condition.	`complete-generated-description-domain`	complete-generated-description-domain retains the complete registered domain condition.
`execution`	`verdict-without-run`	verdict-without-run removes a required exact execution condition.	`exact-recognition-execution-trace`	exact-recognition-execution-trace retains the complete registered execution condition.
`result`	`implicit-nontermination`	implicit-nontermination removes a required exact result condition.	`accept-reject-continue-classes`	accept-reject-continue-classes retains the complete registered result condition.
`self_reference`	`self-reference-excluded`	self-reference-excluded removes a required exact self_reference condition.	`generated-self-description-support`	generated-self-description-support retains the complete registered self_reference condition.
`totality`	`assumed-total-decider`	assumed-total-decider removes a required exact totality condition.	`explicit-totality-or-partiality-boundary`	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
`translation`	`answer-changing-map`	answer-changing-map removes a required exact translation condition.	`answer-preserving-reduction`	answer-preserving-reduction retains the complete registered translation condition.
`generality`	`fixed-depth-table`	fixed-depth-table removes a required exact generality condition.	`description-successor-certificate`	description-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-oracle`	hidden-oracle removes a required exact addition condition.	`no-undeclared-computational-power`	no-undeclared-computational-power retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

24.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.

Eliminated candidates	Exact first-failure reason
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

Shared claim-record clause `COMP-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced problem translation preserving answers kernel retains source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

24.4 Operational laws and consequences

- Carrier law: source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

24.5 Depth-independent closure

Base: The structural One supplies one canonical problem translation preserving answers form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated problem translation preserving answers form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

24.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-REDUCTION-001; result: `PASS`.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-REDUCTION-001; result: `PASS`.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-REDUCTION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Reduction claims require explicit encoders, decoders and generated domains.	`PASS`

24.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: many-one reduction, computability reduction.

Reduction claims require explicit encoders, decoders and generated domains.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Reduction claims require explicit encoders, decoders and generated domains.

24.8 Exact evidence identities

- Source manifest: `sha256:7947d16c47683d213393a222dbf077c1313f6884b4fe89b8c84b69f488629669`
- Complete-census certificate: `sha256:2e6160c4db60b0b8f818a7ad8cc4f4f7782d13c3d9431e657d2b5d8b1e7ed56e`
- Derivation seal: `sha256:c3ae5424425de2ea0a730259170848da166eb040d9c9fe8f1d76d0254ea22811`
- Independent implementation: `sha256:173795ad853a56a81d7d98fd16dc14eccd38d80f3add1c8dc0a58720f6f43de5`
- Independent certificate: `sha256:c843b1c34313a579857158cca63c95281567d12d2cb77c0ec04f46ad8e07baf7`
- External validation: `sha256:1d8cac715551bdf0d1e71267a730cbf51e9175f803ebe560db4fefd7801a588e`
- Model-admitted engine receipt: `sha256:7d03f54baf0fc6d4bbe16f2ea73fac601ede53dceacbd55e2b032545196aaff2`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-REDUCTION-001-7d03f54baf0fc6d4.json`

25. Derivation 17: Fold degrees-of-computability law

Claim: `SFT-COMP-CBL-DEGREES-001`

25.1 Scientific necessity and exact theorem

Fold degrees-of-computability law is required to derive relative reduction classes without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced relative reduction classes kernel retains registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-REDUCTION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

25.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness.

Generation rule: Generate the literal product of the eight registered Computability structural axes for relative reduction classes.

> *All generated finite relative reduction classes structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Degrees are finite generated relation classes; no completed transfinite hierarchy is claimed.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-descriptions`	sampled-descriptions removes a required exact domain condition.	`complete-generated-description-domain`	complete-generated-description-domain retains the complete registered domain condition.
`execution`	`verdict-without-run`	verdict-without-run removes a required exact execution condition.	`exact-recognition-execution-trace`	exact-recognition-execution-trace retains the complete registered execution condition.
`result`	`implicit-nontermination`	implicit-nontermination removes a required exact result condition.	`accept-reject-continue-classes`	accept-reject-continue-classes retains the complete registered result condition.
`self_reference`	`self-reference-excluded`	self-reference-excluded removes a required exact self_reference condition.	`generated-self-description-support`	generated-self-description-support retains the complete registered self_reference condition.
`totality`	`assumed-total-decider`	assumed-total-decider removes a required exact totality condition.	`explicit-totality-or-partiality-boundary`	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
`translation`	`answer-changing-map`	answer-changing-map removes a required exact translation condition.	`answer-preserving-reduction`	answer-preserving-reduction retains the complete registered translation condition.
`generality`	`fixed-depth-table`	fixed-depth-table removes a required exact generality condition.	`description-successor-certificate`	description-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-oracle`	hidden-oracle removes a required exact addition condition.	`no-undeclared-computational-power`	no-undeclared-computational-power retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

25.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.

Eliminated candidates	Exact first-failure reason
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

Shared claim-record clause `COMP-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced relative reduction classes kernel retains registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

25.4 Operational laws and consequences

- Carrier law: registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

25.5 Depth-independent closure

Base: The structural One supplies one canonical relative reduction classes form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated relative reduction classes form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

25.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-DEGREES-001; result: `PASS`.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-DEGREES-001; result: `PASS`.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-DEGREES-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Degrees are finite generated relation classes; no completed transfinite hierarchy is claimed.	`PASS`

25.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: Turing degree, relative computability.

Degrees are finite generated relation classes; no completed transfinite hierarchy is claimed.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Degrees are finite generated relation classes; no completed transfinite hierarchy is claimed.

25.8 Exact evidence identities

- Source manifest: `sha256:e69d3de1e807f51f5435b2fd279dc63ebf903fc1a3abec1469dd6c567ad50189`
- Complete-census certificate: `sha256:7fdd77daa0ed7de987d15f92325c15b5a0f9034535436080d3ade6b19fc489bd`
- Derivation seal: `sha256:ad0d2f09c99201752cf963100d4b989df8bd6504cc4a12212f545eb45272c378`
- Independent implementation: `sha256:54e92220b00e34ab45dec83999c26efc88c4604686cf8bc98866a1143eabf7a3`
- Independent certificate: `sha256:4beaa5ffed228284c3bd0b27c642a6bceb0005d9de29c5b986fba873b1ac97d8`
- External validation: `sha256:d8e21565c305b5d29b57f36fd9d95aae62b6254c03faebc6c413c682b05b2e8f`
- Model-admitted engine receipt: `sha256:40189bc4fb7d083c3ffc7a7610e29e77cc5232bbde8514860c3afa328fe11a3e`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-DEGREES-001-40189bc4fb7d083c.json`

26. Derivation 18: Fold universal-machine execution law

Claim: `SFT-COMP-CBL-UNIVERSAL-MACHINE-001`

26.1 Scientific necessity and exact theorem

Fold universal-machine execution law is required to derive one machine executing encoded machines without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced one machine executing encoded machines kernel retains frozen machine grammar; prefix-free exact descriptions; encoded input; interpreter transition kernel; trace correspondence; halt and output preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-DEGREES-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

26.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: frozen machine grammar; prefix-free exact descriptions; encoded input; interpreter transition kernel; trace correspondence; halt and output preservation.

Generation rule: Generate the literal product of the eight registered Computability structural axes for one machine executing encoded machines.

> *All generated finite one machine executing encoded machines structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Extends Formal Universality into recognizability boundaries but not an omniscient halting oracle.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-descriptions`	sampled-descriptions removes a required exact domain condition.	`complete-generated-description-domain`	complete-generated-description-domain retains the complete registered domain condition.
`execution`	`verdict-without-run`	verdict-without-run removes a required exact execution condition.	`exact-recognition-execution-trace`	exact-recognition-execution-trace retains the complete registered execution condition.
`result`	`implicit-nontermination`	implicit-nontermination removes a required exact result condition.	`accept-reject-continue-classes`	accept-reject-continue-classes retains the complete registered result condition.
`self_reference`	`self-reference-excluded`	self-reference-excluded removes a required exact self_reference condition.	`generated-self-description-support`	generated-self-description-support retains the complete registered self_reference condition.
`totality`	`assumed-total-decider`	assumed-total-decider removes a required exact totality condition.	`explicit-totality-or-partiality-boundary`	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
`translation`	`answer-changing-map`	answer-changing-map removes a required exact translation condition.	`answer-preserving-reduction`	answer-preserving-reduction retains the complete registered translation condition.
`generality`	`fixed-depth-table`	fixed-depth-table removes a required exact generality condition.	`description-successor-certificate`	description-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-oracle`	hidden-oracle removes a required exact addition condition.	`no-undeclared-computational-power`	no-undeclared-computational-power retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

26.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

Shared claim-record clause `COMP-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

The forced one machine executing encoded machines kernel retains frozen machine grammar; prefix-free exact descriptions; encoded input; interpreter transition kernel; trace correspondence; halt and output preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

26.4 Operational laws and consequences

- Carrier law: frozen machine grammar; prefix-free exact descriptions; encoded input; interpreter transition kernel; trace correspondence; halt and output preservation.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

26.5 Depth-independent closure

Base: The structural One supplies one canonical one machine executing encoded machines form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated one machine executing encoded machines form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

26.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-UNIVERSAL-MACHINE-001; result: `PASS`.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-UNIVERSAL-MACHINE-001; result: `PASS`.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-UNIVERSAL-MACHINE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Extends Formal Universality into recognizability boundaries but not an omniscient halting oracle.	`PASS`

26.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: universal machine, interpreter.

Extends Formal Universality into recognizability boundaries but not an omniscient halting oracle.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Extends Formal Universality into recognizability boundaries but not an omniscient halting oracle.

26.8 Exact evidence identities

- Source manifest: `sha256:f1cddb7a0abca54bfc082e210aae0129fe23a0303fb9d8dd4e76735185692d7a`
- Complete-census certificate: `sha256:630ccc0d85a8e9228e341632fca4d29fb52eabd3792dc22a2fa8f6d4a386b652`
- Derivation seal: `sha256:bc727d9f76e920560ca4a2ff6fc8864fcdb4da986c10d8e3db80fe1df09d8320`
- Independent implementation: `sha256:f2890836bf1bba1f7dbc2c6641b34583b0906fcc25d4099cc5d993326b21ec85`
- Independent certificate: `sha256:d830526fc80fceff9278895cdb542cd9a72d6da659acaa0e60aa47d12841f058`
- External validation: `sha256:221ca0caa7445de51094f8fc76c69b2292b6118acf5bd19dc52f36acce24abe6`
- Model-admitted engine receipt: `sha256:63466fb169111c50266901bf7ac48255ced18ef727cd66d1f3b376b6beaa7633`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-UNIVERSAL-MACHINE-001-63466fb169111c50.json`

27. Derivation 19: Fold undecidable-problem law

Claim: `SFT-COMP-CBL-UNDECIDABILITY-001`

27.1 Scientific necessity and exact theorem

Fold undecidable-problem law is required to derive failure of total internal decision without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced failure of total internal decision kernel retains generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-UNIVERSAL-MACHINE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

27.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace.

Generation rule: Generate the literal product of the eight registered Computability structural axes for failure of total internal decision.

> *All generated finite failure of total internal decision structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. The theorem applies where the registered language supports the exact self-encoding and complement operation.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-descriptions`	sampled-descriptions removes a required exact domain condition.	`complete-generated-description-domain`	complete-generated-description-domain retains the complete registered domain condition.
`execution`	`verdict-without-run`	verdict-without-run removes a required exact execution condition.	`exact-recognition-execution-trace`	exact-recognition-execution-trace retains the complete registered execution condition.
`result`	`implicit-nontermination`	implicit-nontermination removes a required exact result condition.	`accept-reject-continue-classes`	accept-reject-continue-classes retains the complete registered result condition.
`self_reference`	`self-reference-excluded`	self-reference-excluded removes a required exact self_reference condition.	`generated-self-description-support`	generated-self-description-support retains the complete registered self_reference condition.
`totality`	`assumed-total-decider`	assumed-total-decider removes a required exact totality condition.	`explicit-totality-or-partiality-boundary`	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
`translation`	`answer-changing-map`	answer-changing-map removes a required exact translation condition.	`answer-preserving-reduction`	answer-preserving-reduction retains the complete registered translation condition.
`generality`	`fixed-depth-table`	fixed-depth-table removes a required exact generality condition.	`description-successor-certificate`	description-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-oracle`	hidden-oracle removes a required exact addition condition.	`no-undeclared-computational-power`	no-undeclared-computational-power retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

27.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

Shared claim-record clause `COMP-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced failure of total internal decision kernel retains generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

27.4 Operational laws and consequences

- Carrier law: generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

27.5 Depth-independent closure

Base: The structural One supplies one canonical failure of total internal decision form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated failure of total internal decision form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

27.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-UNDECIDABILITY-001; result: `PASS`.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-UNDECIDABILITY-001; result: `PASS`.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-UNDECIDABILITY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; The theorem applies where the registered language supports the exact self-encoding and complement operation.	`PASS`

27.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: undecidability, diagonal argument.

The theorem applies where the registered language supports the exact self-encoding and complement operation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- The theorem applies where the registered language supports the exact self-encoding and complement operation.

27.8 Exact evidence identities

- Source manifest: `sha256:d10fd9322aafb6432ebcb434fff6f1bd31e256c7c5ce93ae9a894b8bcf568786`
- Complete-census certificate: `sha256:f0655bf4421f5a52dfb30810d18c76896acfcaa3ef36d945c5672151a2d426da`
- Derivation seal: `sha256:42eefceee42f1ef84a8b765da3ecefbc91a79c5bfe7ce3cc6019e6ceddccf5d7`
- Independent implementation: `sha256:521c56670dedb8d993cca8984ca42bd99784d517bc12cb1989e071738a387845`
- Independent certificate: `sha256:8c3b3c911d7f33c654b756f4f59224e8614b0e9cdfcbff736daac8ce0bcf13f6`
- External validation: `sha256:b2820d28d6464e2afcfae7ee3eb56bb845b3a192556fb0cf0c7e6774d73bb5c0`
- Model-admitted engine receipt: `sha256:9de0276cb48ddbcea65e0a35f6f2deeac179a225ac85a8f93121e5e7ed97e692`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-UNDECIDABILITY-001-9de0276cb48ddbce.json`

28. Derivation 20: Fold incompleteness-boundary law

Claim: `SFT-COMP-CBL-INCOMPLETENESS-001`

28.1 Scientific necessity and exact theorem

Fold incompleteness-boundary law is required to derive internal proof and self-verification limits without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced internal proof and self-verification limits kernel retains effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-UNDECIDABILITY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

28.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle.

Generation rule: Generate the literal product of the eight registered Computability structural axes for internal proof and self-verification limits.

> *All generated finite internal proof and self-verification limits structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. This is a computational proof-system boundary, not an imported arithmetic incompleteness axiom.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-descriptions`	sampled-descriptions removes a required exact domain condition.	`complete-generated-description-domain`	complete-generated-description-domain retains the complete registered domain condition.
`execution`	`verdict-without-run`	verdict-without-run removes a required exact execution condition.	`exact-recognition-execution-trace`	exact-recognition-execution-trace retains the complete registered execution condition.
`result`	`implicit-nontermination`	implicit-nontermination removes a required exact result condition.	`accept-reject-continue-classes`	accept-reject-continue-classes retains the complete registered result condition.
`self_reference`	`self-reference-excluded`	self-reference-excluded removes a required exact self_reference condition.	`generated-self-description-support`	generated-self-description-support retains the complete registered self_reference condition.
`totality`	`assumed-total-decider`	assumed-total-decider removes a required exact totality condition.	`explicit-totality-or-partiality-boundary`	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
`translation`	`answer-changing-map`	answer-changing-map removes a required exact translation condition.	`answer-preserving-reduction`	answer-preserving-reduction retains the complete registered translation condition.
`generality`	`fixed-depth-table`	fixed-depth-table removes a required exact generality condition.	`description-successor-certificate`	description-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-oracle`	hidden-oracle removes a required exact addition condition.	`no-undeclared-computational-power`	no-undeclared-computational-power retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

28.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

Shared claim-record clause `COMP-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced internal proof and self-verification limits kernel retains effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

28.4 Operational laws and consequences

- Carrier law: effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.

- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

28.5 Depth-independent closure

Base: The structural One supplies one canonical internal proof and self-verification limits form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated internal proof and self-verification limits form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

28.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-INCOMPLETENESS-001; result: `PASS`.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-INCOMPLETENESS-001; result: `PASS`.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-INCOMPLETENESS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; This is a computational proof-system boundary, not an imported arithmetic incompleteness axiom.	`PASS`

28.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: incompleteness, self-reference.

This is a computational proof-system boundary, not an imported arithmetic incompleteness axiom.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- This is a computational proof-system boundary, not an imported arithmetic incompleteness axiom.

28.8 Exact evidence identities

- Source manifest: `sha256:563f132c7bc3d7b393606cd0cfb8678f07958ceaad483f330de156a08b2ad0eb`

- Complete-census certificate: `sha256:bece03b200d54ec76d2f512a5f16addc33681cf5c796038f1804ab24e733580e`
- Derivation seal: `sha256:39ccdf0743514fba3f9e01e41a67dfa4686805175ccd0fe687e6316aef358541`
- Independent implementation: `sha256:fce65fcba70bbdfe02de16c19f5f28328658fbf8d9c78050319b98a2baf774a3`
- Independent certificate: `sha256:f6f7786d6185c4e020b9031361db598d967b3772248ee3c7add2b0721457ab2b`
- External validation: `sha256:0ba02558215ddcb70efdb09565b963fcafeff462fd6b3410a05e0721472c967d`
- Model-admitted engine receipt: `sha256:f7fb969bc4fc339ff473e5ec2620e703695e6a12c3ab0b704594e9664e9ea277`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-INCOMPLETENESS-001-f7fb969bc4fc339f.json`

29. Derivation 21: Fold relative and oracle-computation law

Claim: `SFT-COMP-CBL-RELATIVE-ORACLE-001`

29.1 Scientific necessity and exact theorem

Fold relative and oracle-computation law is required to derive explicit external answer interfaces without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced explicit external answer interfaces kernel retains base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-INCOMPLETENESS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

29.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary.

Generation rule: Generate the literal product of the eight registered Computability structural axes for explicit external answer interfaces.

> *All generated finite explicit external answer interfaces structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. An oracle is a registered interface, never unexplained computational power.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-descriptions`	sampled-descriptions removes a required exact domain condition.	`complete-generated-description-domain`	complete-generated-description-domain retains the complete registered domain condition.
`execution`	`verdict-without-run`	verdict-without-run removes a required exact execution condition.	`exact-recognition-execution-trace`	exact-recognition-execution-trace retains the complete registered execution condition.
`result`	`implicit-nontermination`	implicit-nontermination removes a required exact result condition.	`accept-reject-continue-classes`	accept-reject-continue-classes retains the complete registered result condition.
`self_reference`	`self-reference-excluded`	self-reference-excluded removes a required exact self_reference condition.	`generated-self-description-support`	generated-self-description-support retains the complete registered self_reference condition.
`totality`	`assumed-total-decider`	assumed-total-decider removes a required exact totality condition.	`explicit-totality-or-partiality-boundary`	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
`translation`	`answer-changing-map`	answer-changing-map removes a required exact translation condition.	`answer-preserving-reduction`	answer-preserving-reduction retains the complete registered translation condition.
`generality`	`fixed-depth-table`	fixed-depth-table removes a required exact generality condition.	`description-successor-certificate`	description-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-oracle`	hidden-oracle removes a required exact addition condition.	`no-undeclared-computational-power`	no-undeclared-computational-power retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

29.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

Shared claim-record clause `COMP-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced explicit external answer interfaces kernel retains base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

29.4 Operational laws and consequences

- Carrier law: base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

29.5 Depth-independent closure

Base: The structural One supplies one canonical explicit external answer interfaces form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated explicit external answer interfaces form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

29.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-RELATIVE-ORACLE-001; result: `PASS`.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-RELATIVE-ORACLE-001; result: `PASS`.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-RELATIVE-ORACLE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; An oracle is a registered interface, never unexplained computational power.	`PASS`

29.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: oracle machine, relative computation.

An oracle is a registered interface, never unexplained computational power.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- An oracle is a registered interface, never unexplained computational power.

29.8 Exact evidence identities

- Source manifest: `sha256:12c67c86c538f90bc4536e45613aacaf1319d5f5ac02c0ae08396bca805a237e`
- Complete-census certificate: `sha256:85464270056da95e16f3985a90c6410a0652421d6d3a4e2d9e7e0a5ab78f9edd`
- Derivation seal: `sha256:6c056bd1d9d6e58d1c4cf9eb93bb767deeb145a363a6a7d85884a2696008114c`

- Independent implementation: `sha256:98e5f6212def0f26cd0cb6cfa352183ebdebf792aaa5184499fea5eb380abc64`
- Independent certificate: `sha256:f9757a676f3589d0cce885d22cb45b44377fd1720dcbb2953cee994b0c89911d`
- External validation: `sha256:718f8f589221365d114308744266476d1bca0bb9476be8e68a2ca1acde67b7be`
- Model-admitted engine receipt: `sha256:2a09b83377a56d06136df6922d9e9830c3b336e3487d3654d349c7f6f6746d01`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-RELATIVE-ORACLE-001-2a09b83377a56d06.json`

30. Derivation 22: Fold hypercomputation admissibility-limit law

Claim: `SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001`

30.1 Scientific necessity and exact theorem

Fold hypercomputation admissibility-limit law is required to derive claims exceeding generated universal computation without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced claims exceeding generated universal computation kernel retains claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-RELATIVE-ORACLE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

30.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition.

Generation rule: Generate the literal product of the eight registered Computability structural axes for claims exceeding generated universal computation.

> *All generated finite claims exceeding generated universal computation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No hypercomputation claim is admitted without a generated executable law and depth-independent certificate; finite demonstrations remain conditional.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-descriptions`	sampled-descriptions removes a required exact domain condition.	`complete-generated-description-domain`	complete-generated-description-domain retains the complete registered domain condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`execution`	`verdict-without-run`	verdict-without-run removes a required exact execution condition.	`exact-recognition-execution-trace`	exact-recognition-execution-trace retains the complete registered execution condition.
`result`	`implicit-nontermination`	implicit-nontermination removes a required exact result condition.	`accept-reject-continue-classes`	accept-reject-continue-classes retains the complete registered result condition.
`self_reference`	`self-reference-excluded`	self-reference-excluded removes a required exact self_reference condition.	`generated-self-description-support`	generated-self-description-support retains the complete registered self_reference condition.
`totality`	`assumed-total-decider`	assumed-total-decider removes a required exact totality condition.	`explicit-totality-or-partiality-boundary`	explicit-totality-or-partiality-boundary retains the complete registered totality condition.
`translation`	`answer-changing-map`	answer-changing-map removes a required exact translation condition.	`answer-preserving-reduction`	answer-preserving-reduction retains the complete registered translation condition.
`generality`	`fixed-depth-table`	fixed-depth-table removes a required exact generality condition.	`description-successor-certificate`	description-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-oracle`	hidden-oracle removes a required exact addition condition.	`no-undeclared-computational-power`	no-undeclared-computational-power retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

30.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-descriptions removes a required exact domain condition.
64	verdict-without-run removes a required exact execution condition.
32	implicit-nontermination removes a required exact result condition.
16	self-reference-excluded removes a required exact self_reference condition.
8	assumed-total-decider removes a required exact totality condition.
4	answer-changing-map removes a required exact translation condition.
2	fixed-depth-table removes a required exact generality condition.
1	hidden-oracle removes a required exact addition condition.

Shared claim-record clause `COMP-S007` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced claims exceeding generated universal computation kernel retains claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

30.4 Operational laws and consequences

- Carrier law: claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

30.5 Depth-independent closure

Base: The structural One supplies one canonical claims exceeding generated universal computation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated claims exceeding generated universal computation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

30.6 Executed witnesses and adverse controls

- `computability-witness-1` - computability exact operational witness 1 for SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001; result: `PASS`.
- `computability-witness-2` - computability exact operational witness 2 for SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001; result: `PASS`.
- `computability-witness-3` - computability exact operational witness 3 for SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-descriptions removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No hypercomputation claim is admitted without a generated executable law and depth-independent certificate; finite demonstrations remain conditional.	`PASS`

30.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: hypercomputation, computational limit.

No hypercomputation claim is admitted without a generated executable law and depth-independent certificate; finite demonstrations remain conditional.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No hypercomputation claim is admitted without a generated executable law and depth-independent certificate; finite demonstrations remain conditional.

30.8 Exact evidence identities

- Source manifest: `sha256:cb42d36a49aefee2c1b65da8c5e0a504792b30b9cfe777a69fc01b4f98608753`
- Complete-census certificate: `sha256:413dc020c84636e97a0bb1b2a86bc46cb24996c7a7092b8d0f47f361b5160a93`
- Derivation seal: `sha256:5b708f88bd4ff06b23706597a4ea720d6df25d109b06dd0aa7201a06267ad085`

- Independent implementation: `sha256:6b6764bb33684418d2d828cf9c1bc882739d517804a2ece25e8c1e56b79a6357`
- Independent certificate: `sha256:37c0063e48172aa654b2841935cde94ed2c44a732321a91de3cc86b2a5b58449`
- External validation: `sha256:e4a6b799d7ea881df152750b839cf35c871b97e84a80484817303e0fec561d43`
- Model-admitted engine receipt: `sha256:3d6f7b8fdd8b1ccaf30622dad922140dd6b441b19df9dd4b76ff17ee191439cc`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001-3d6f7b8fdd8b1cca.json`

31. Derivation 23: Fold input and problem-size law

Claim: `SFT-COMP-CPLX-INPUT-SIZE-001`

31.1 Scientific necessity and exact theorem

Fold input and problem-size law is required to derive exact generated input extent without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced exact generated input extent kernel retains canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

31.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for exact generated input extent.

> *All generated finite exact generated input extent structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Size is representation-relative unless an exact encoding correspondence is supplied.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

31.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced exact generated input extent kernel retains canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

31.4 Operational laws and consequences

- Carrier law: canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

31.5 Depth-independent closure

Base: The structural One supplies one canonical exact generated input extent form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact generated input extent form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

31.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-INPUT-SIZE-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-INPUT-SIZE-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-INPUT-SIZE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Size is representation-relative unless an exact encoding correspondence is supplied.	`PASS`

31.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: input size, problem size.

Size is representation-relative unless an exact encoding correspondence is supplied.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Size is representation-relative unless an exact encoding correspondence is supplied.

31.8 Exact evidence identities

- Source manifest: `sha256:c2cd5867ed1ede45da97082c15db4f5b544ab67e61633b2603322cd89f503b9b`
- Complete-census certificate: `sha256:a383f71b94033d5c6c7aafc4fc9ee43e526903df217ae1371330c894f2483a5b`
- Derivation seal: `sha256:59a9b0be31a5e733dce860eedfbe4bda67bde61c23ab06024202d6a115147f04`
- Independent implementation: `sha256:8b11f954a55f4184895929a2a1a8e8e2cc7edbf0510d5f079bec7017d7bda6ca`
- Independent certificate: `sha256:749367f01a05e86db5ae00d7d66abf0b6dcd40d036f06f166c7dcef695d99511`
- External validation: `sha256:9f28b7906f4b1e97bb7044a480308141fd074adb36c8a1d57e5bdd05b9ba2bd7`

- Model-admitted engine receipt: `sha256:3f002d11d6315fc6f32536022cefb2a3fad2bf857ea1761637481a827a8a2fc4`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-INPUT-SIZE-001-3f002d11d6315fc6.json`

32. Derivation 24: Fold time and space resource law

Claim: `SFT-COMP-CPLX-TIME-SPACE-001`

32.1 Scientific necessity and exact theorem

Fold time and space resource law is required to derive transition and retained-state resources without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced transition and retained-state resources kernel retains complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-INPUT-SIZE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

32.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for transition and retained-state resources.

> *All generated finite transition and retained-state resources structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Counts are exact host reports over generated traces, not foundational magnitudes.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

32.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced transition and retained-state resources kernel retains complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

32.4 Operational laws and consequences

- Carrier law: complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

32.5 Depth-independent closure

Base: The structural One supplies one canonical transition and retained-state resources form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated transition and retained-state resources form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

32.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-TIME-SPACE-001; result: `PASS`.

- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-TIME-SPACE-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-TIME-SPACE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Counts are exact host reports over generated traces, not foundational magnitudes.	`PASS`

32.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: time complexity, space complexity.

Counts are exact host reports over generated traces, not foundational magnitudes.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Counts are exact host reports over generated traces, not foundational magnitudes.

32.8 Exact evidence identities

- Source manifest: `sha256:f155e77d53f592a5b06eee8f93f6ad97642e0feaa0c05a7458ff978b5c5faced`
- Complete-census certificate: `sha256:6bc159a0c45385cc05dd89ba536edb90328cbedac03491c007940974c948238a`
- Derivation seal: `sha256:1fcc740897abe9b3adb6bdc54f8f5a7a4a1908d1da460d36b7751a0ba3fa0e36`
- Independent implementation: `sha256:55a51257da5a486414684d2eb845f65827c1a319fb2712f3619bf3da0868e96c`
- Independent certificate: `sha256:d69c776f3a72bdfe046e514a487a8e8365e1bfc47b33eb1827d7358db5dafb46`
- External validation: `sha256:8669c1478eabab8093f09aae4b52e18a24ecc93e744f3ae2099c67c15121a68d`
- Model-admitted engine receipt: `sha256:d6187e34cd91181461fa6614ebb82d710719cf6bebf4644d1bc08c81641df62b`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-TIME-SPACE-001-d6187e34cd911814.json`

33. Derivation 25: Fold circuit size and depth law

Claim: `SFT-COMP-CPLX-CIRCUIT-RESOURCE-001`

33.1 Scientific necessity and exact theorem

Fold circuit size and depth law is required to derive circuit structural resources without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced circuit structural resources kernel retains generated gates; wires; dependency order; longest causal layer trace; output support; shared-subcircuit identity; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-TIME-SPACE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

33.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated gates; wires; dependency order; longest causal layer trace; output support; shared-subcircuit identity.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for circuit structural resources.

> *All generated finite circuit structural resources structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Closes resource accounting, not arbitrary lower bounds.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

33.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced circuit structural resources kernel retains generated gates; wires; dependency order; longest causal layer trace; output support; shared-subcircuit identity; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

33.4 Operational laws and consequences

- Carrier law: generated gates; wires; dependency order; longest causal layer trace; output support; shared-subcircuit identity.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

33.5 Depth-independent closure

Base: The structural One supplies one canonical circuit structural resources form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated circuit structural resources form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

33.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-CIRCUIT-RESOURCE-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-CIRCUIT-RESOURCE-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-CIRCUIT-RESOURCE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closes resource accounting, not arbitrary lower bounds.	`PASS`

33.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: circuit size, circuit depth.

Closes resource accounting, not arbitrary lower bounds.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Closes resource accounting, not arbitrary lower bounds.

33.8 Exact evidence identities

- Source manifest: `sha256:846f4e21718ad950f5f8024aeb3a157f822c73a67e93ca085d618260f70a1f38`
- Complete-census certificate: `sha256:6c88b89eb6f1dfbe9db72ad1fbcecb9989ba7f92e6df6f89901f8fe56410691e`
- Derivation seal: `sha256:11d0d9927aa1c5cb64db2faeed8e29d4aad74135c81b3c68ea8ad080de297144`
- Independent implementation: `sha256:12e9bb6516941807a16f20797673f5421d0a4446a86e73aca97773d5d57fccfc`
- Independent certificate: `sha256:1a0022829a4bdd87943253b70fa5057822a6a456dbf6d8cf56f7a996826d3909`
- External validation: `sha256:f9af184469e9d277ee03491a44a563a9f131f5a86b3c7fd325ba362b0de85f57`
- Model-admitted engine receipt: `sha256:9f3e28733488ad63f954689caad30204963adbe8e2fa60f985be1140722116ae`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-CIRCUIT-RESOURCE-001-9f3e28733488ad63.json`

34. Derivation 26: Fold communication and query complexity law

Claim: `SFT-COMP-CPLX-COMMUNICATION-QUERY-001`

34.1 Scientific necessity and exact theorem

Fold communication and query complexity law is required to derive cross-interface and information-request resources without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced cross-interface and information-request resources kernel retains distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate; every operation is*

source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-CIRCUIT-RESOURCE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

34.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for cross-interface and information-request resources.

All generated finite cross-interface and information-request resources structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. The exact protocol and oracle identity must be registered.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

34.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.

Eliminated candidates	Exact first-failure reason
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced cross-interface and information-request resources kernel retains distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

34.4 Operational laws and consequences

- Carrier law: distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

34.5 Depth-independent closure

Base: The structural One supplies one canonical cross-interface and information-request resources form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated cross-interface and information-request resources form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

34.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-COMMUNICATION-QUERY-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-COMMUNICATION-QUERY-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-COMMUNICATION-QUERY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; The exact protocol and oracle identity must be registered.	`PASS`

34.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: communication complexity, query complexity.

The exact protocol and oracle identity must be registered.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- The exact protocol and oracle identity must be registered.

34.8 Exact evidence identities

- Source manifest: `sha256:c302cd798c3e0e45d7a29366ae64e227b6c5813fe121600d6e5e96f650f2d05a`
- Complete-census certificate: `sha256:f27a6346926b89a426e1b2639d1cbabe7aaaf222fa966b033932878d073eaef2`
- Derivation seal: `sha256:7d91bbd1b18c3aefb5ca8eb72d84964027c55fe09b733ffe688cf19e2491ac41`
- Independent implementation: `sha256:ca88ab267912a1c325ce19967fa6f9bb5104be65e508a42692808033bf18b3d7`
- Independent certificate: `sha256:b62d6baea8c3fb0502d436d71446deab6f9277fe93e8b49fb2845f1380457cb5`
- External validation: `sha256:10745814f262dafcab4969ed2534761d0e23085f68e6f31faa232b543b41d953`
- Model-admitted engine receipt: `sha256:c352ad5637226cc1d41ecbc8fb375f9593e7928d4157a97e8b7050ac18d19f3e`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-COMMUNICATION-QUERY-001-c352ad5637226cc1.json`

35. Derivation 27: Fold randomness-resource law

Claim: `SFT-COMP-CPLX-RANDOMNESS-001`

35.1 Scientific necessity and exact theorem

Fold randomness-resource law is required to derive scheduled unresolved branch support without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced scheduled unresolved branch support kernel retains complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-COMMUNICATION-QUERY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

35.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for scheduled unresolved branch support.

> *All generated finite scheduled unresolved branch support structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Randomness is not a causal premise; results quantify complete registered branch support or an explicitly measured external source.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

35.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced scheduled unresolved branch support kernel retains complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

35.4 Operational laws and consequences

- Carrier law: complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

35.5 Depth-independent closure

Base: The structural One supplies one canonical scheduled unresolved branch support form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated scheduled unresolved branch support form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

35.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-RANDOMNESS-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-RANDOMNESS-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-RANDOMNESS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Randomness is not a causal premise; results quantify complete registered branch support or an explicitly measured external source.	PASS

35.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: randomized complexity, random bits.

Randomness is not a causal premise; results quantify complete registered branch support or an explicitly measured external source.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Randomness is not a causal premise; results quantify complete registered branch support or an explicitly measured external source.

35.8 Exact evidence identities

- Source manifest: `sha256:0607afa19714526b32d36403953ce2de74281e7b9310eb134693acf7ed442306`
- Complete-census certificate: `sha256:adc6a5a3ebdc171d8e1332679a85020c371f2c5e4d12833b06ae098c96aed420`
- Derivation seal: `sha256:11553fb594b83f863eb99c1dcda12b7c55670a95e85ddeb507f0e96436cba33c`
- Independent implementation: `sha256:e86d848ef0f31cd3a4e10cbe82a0235e50943a06e0d9ffaff45e5744c6482459`
- Independent certificate: `sha256:dedc0a421764c9d7249870484673717c898c3cddf7ed33494cec492522a3dc34`
- External validation: `sha256:25f8a328196851e3b4fc33030d39baa1877734e8d287b0a9515d51206eeb02fa`
- Model-admitted engine receipt: `sha256:e4311dc1af067c40ad88072efc8c155d1647ad6b071c1003c48b0e597be64f8b`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-RANDOMNESS-001-e4311dc1af067c40.json`

36. Derivation 28: Fold reversibility-cost law

Claim: `SFT-COMP-CPLX-REVERSIBILITY-COST-001`

36.1 Scientific necessity and exact theorem

Fold reversibility-cost law is required to derive information retained for inverse execution without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced information retained for inverse execution kernel retains source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-RANDOMNESS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

36.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for information retained for inverse execution.

> *All generated finite information retained for inverse execution structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. This is the classical cost law; the universal reversible model belongs to the later quantum branch.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

36.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced information retained for inverse execution kernel retains source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

36.4 Operational laws and consequences

- Carrier law: source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

36.5 Depth-independent closure

Base: The structural One supplies one canonical information retained for inverse execution form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated information retained for inverse execution form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

36.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-REVERSIBILITY-COST-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-REVERSIBILITY-COST-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-REVERSIBILITY-COST-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; This is the classical cost law; the universal reversible model belongs to the later quantum branch.	`PASS`

36.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: reversible computation, Landauer cost.

This is the classical cost law; the universal reversible model belongs to the later quantum branch.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- This is the classical cost law; the universal reversible model belongs to the later quantum branch.

36.8 Exact evidence identities

- Source manifest: `sha256:43e6f36f3da6e8a178ac4fbdf14531b45ca6e55ac09ac5b4ec9a1526021b5903`
- Complete-census certificate: `sha256:2b9ccd74c371ca0b385c8062244e6b0c5a3d373db772dc2719405e107b85d5a2`
- Derivation seal: `sha256:8593ee37fadeb80b440d656e826431594044f76793053417dff646dc6da86bdd`
- Independent implementation: `sha256:62112db56cc1d6dde75ba41a7ba2e9b84a435cf17bc81262ad9d5e11c008d868`
- Independent certificate: `sha256:ad19f8f114159728b4ae9507a73e5c76f3cd94ca6d18e5586069be460d268278`
- External validation: `sha256:657b65250d381219f7c9577a80006d61e4e72c8b53388e955a68de6eceff7cce`
- Model-admitted engine receipt: `sha256:0a8552676c105fc6b3fdcd6aac90a7a68c428962c209460caacea2b81c496ed2`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-REVERSIBILITY-COST-001-0a8552676c105fc6.json`

37. Derivation 29: Fold parallel-complexity law

Claim: `SFT-COMP-CPLX-PARALLEL-001`

37.1 Scientific necessity and exact theorem

Fold parallel-complexity law is required to derive independent and causally layered work without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced independent and causally layered work kernel retains task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-REVERSIBILITY-COST-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

37.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for independent and causally layered work.

> *All generated finite independent and causally layered work structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No physical processor count or timing constant is imported.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

37.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced independent and causally layered work kernel retains task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

37.4 Operational laws and consequences

- Carrier law: task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

37.5 Depth-independent closure

Base: The structural One supplies one canonical independent and causally layered work form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated independent and causally layered work form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

37.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-PARALLEL-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-PARALLEL-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-PARALLEL-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No physical processor count or timing constant is imported.	`PASS`

37.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: parallel complexity, work and span.

No physical processor count or timing constant is imported.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No physical processor count or timing constant is imported.

37.8 Exact evidence identities

- Source manifest: `sha256:2a0ee9adf4736481f873cd67f7b4e348d44ebdc98940be3503dfe98fd403e4ba`
- Complete-census certificate: `sha256:810c108366d42966f12b55652d027a8ca3e89c684d4f81e136d9774e6b0c7066`
- Derivation seal: `sha256:15c1ba1323fe5baa1b8635a1bda7868db77477aa1f0c4981e4b11dc06af9f4ad`
- Independent implementation: `sha256:81ade383c13f9b93f5df4bc48531bcf5eaf6a2c6bb816dc89b3737ef6dc9ebae`
- Independent certificate: `sha256:dfbd1f51e697f60b15a455914c43b409255647b875c4a9882bb008ef789e72ea`
- External validation: `sha256:82c39a2ca55f4e9b10d7d4c7ff997b61bcd52472e2fca5ca2da470993b1b72fa`
- Model-admitted engine receipt: `sha256:d451551a66fdaf127bfd2e9738af1c7818753ea3dd2862a46db3108e32354424`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-PARALLEL-001-d451551a66fdaf12.json`

38. Derivation 30: Fold lower- and upper-bound law

Claim: `SFT-COMP-CPLX-BOUNDS-001`

38.1 Scientific necessity and exact theorem

Fold lower- and upper-bound law is required to derive resource comparison certificates without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced resource comparison certificates kernel retains problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-PARALLEL-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

38.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for resource comparison certificates.

> *All generated finite resource comparison certificates structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Arbitrary asymptotic lower bounds remain separate claims requiring their own generated families.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

38.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced resource comparison certificates kernel retains problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

38.4 Operational laws and consequences

- Carrier law: problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

38.5 Depth-independent closure

Base: The structural One supplies one canonical resource comparison certificates form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated resource comparison certificates form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

38.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-BOUNDS-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-BOUNDS-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-BOUNDS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Arbitrary asymptotic lower bounds remain separate claims requiring their own generated families.	`PASS`

38.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: lower bound, upper bound.

Arbitrary asymptotic lower bounds remain separate claims requiring their own generated families.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Arbitrary asymptotic lower bounds remain separate claims requiring their own generated families.

38.8 Exact evidence identities

- Source manifest: `sha256:93370860be77934c39962130d5c50209ea5eb1f0f631d4300fd00e9f9ddf41d9`
- Complete-census certificate: `sha256:930b712b9a8d08fa4fa577fc7fc83fee43ac2b8899c1c588cf8a58361f027c37`
- Derivation seal: `sha256:1cec4f2834aa22a4ecc619ed0c0541e1788ae7f620901e4c0c3065eb2ee7ac2d`
- Independent implementation: `sha256:7bd2df392101a75d5c8de34cd3952cbe8cbb50fffdae0927168f2afffbde223f`
- Independent certificate: `sha256:b347fc3d05e9cfd9931f0178ded39a23316344dd56773e09696d783ba933a55f`
- External validation: `sha256:f0bfb44bbb62a3b09c8cd4e1fe8751b4d016a368f0f3a516a00af53052f53b35`
- Model-admitted engine receipt: `sha256:7ba9b6310c1f0dd170c3c9b6abc0db4c257241df7af511faf3fb232b2d39db02`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-BOUNDS-001-7ba9b6310c1f0dd1.json`

39. Derivation 31: Fold reduction and completeness law

Claim: `SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001`

39.1 Scientific necessity and exact theorem

Fold reduction and completeness law is required to derive resource-preserving problem translations without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced resource-preserving problem translations kernel retains problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-BOUNDS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

39.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for resource-preserving problem translations.

> *All generated finite resource-preserving problem translations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Completeness is relative to the frozen problem class and resource law.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

39.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced resource-preserving problem translations kernel retains problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

39.4 Operational laws and consequences

- Carrier law: problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

39.5 Depth-independent closure

Base: The structural One supplies one canonical resource-preserving problem translations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated resource-preserving problem translations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

39.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Completeness is relative to the frozen problem class and resource law.	`PASS`

39.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: complexity reduction, complete problem.

Completeness is relative to the frozen problem class and resource law.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Completeness is relative to the frozen problem class and resource law.

39.8 Exact evidence identities

- Source manifest: `sha256:d4ea3f50576d38baaeb22d352a42da0dda94019c0bcc5eeb3aab586c6f8b6709`
- Complete-census certificate: `sha256:171884d544593303d4d7dbe208b8ff54569501ca5f9c64c74831f5ad091d5596`
- Derivation seal: `sha256:7ce5c8b6a21352a4cfdf7f441d39a955314633a9070682103521f83cfb41358d`
- Independent implementation: `sha256:12fc9feb0f395be2df932ad6211deaef3a7051d9638752248702b32398313bec`
- Independent certificate: `sha256:3710e881e46bb1cbd8b1788064009f6e420e795dcbdc2140690ed5e2bda71f46`
- External validation: `sha256:2adb304bdade25973af95ecd12fb91806996bbd575d4835b7a1dbde2357df1ea`
- Model-admitted engine receipt: `sha256:401b5e76d63dd3d31c2a38156dd3fab752e10b289cb7b0e38c9a71f0cf6f673a`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001-401b5e76d63dd3d3.json`

40. Derivation 32: Fold average- and worst-case law

Claim: `SFT-COMP-CPLX-AVERAGE-WORST-001`

40.1 Scientific necessity and exact theorem

Fold average- and worst-case law is required to derive complete instance-family resource profiles without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced complete instance-family resource profiles kernel retains generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

40.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for complete instance-family resource profiles.

All generated finite complete instance-family resource profiles structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Average claims require a complete exact support relation or registered empirical distribution.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` applies. The full, unchanged clause is preserved in the shared-clause appendix.

40.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` applies. The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced complete instance-family resource profiles kernel retains generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

40.4 Operational laws and consequences

- Carrier law: generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

40.5 Depth-independent closure

Base: The structural One supplies one canonical complete instance-family resource profiles form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated complete instance-family resource profiles form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

40.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-AVERAGE-WORST-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-AVERAGE-WORST-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-AVERAGE-WORST-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Average claims require a complete exact support relation or registered empirical distribution.	`PASS`

40.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: average-case complexity, worst-case complexity.

Average claims require a complete exact support relation or registered empirical distribution.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Average claims require a complete exact support relation or registered empirical distribution.

40.8 Exact evidence identities

- Source manifest: `sha256:70c3dc367ca77dbd939a780cf7faa396cd6ada546544332003b8cb8b7f65f17f`
- Complete-census certificate: `sha256:6be68eff0da24ae4189a24799092c315c3f7419fdc6b4b6daf67661d285d7109`
- Derivation seal: `sha256:9d6f8834fc4e6c4d7772ac100765066e8cd32a49ac2aca16328984c5567d3aa2`
- Independent implementation: `sha256:9fc79651d13c51167b280ff22f28bd599d3a56d69ba36f3b6aaf41f23a8f572e`
- Independent certificate: `sha256:59c844f91b8949eecae3fab5729be02bfc32aa99351bead60cb5f32e2c4c27a9`
- External validation: `sha256:9d98937e33feb40baa49f42b0a1e9d95c8cac0852d0c5ce6b89db55f6d8c0da4`
- Model-admitted engine receipt: `sha256:231135bfa6605c585af24c7a6f89d21b099f42462dc4b7a6c0820d761bce6bfb`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-AVERAGE-WORST-001-231135bfa6605c58.json`

41. Derivation 33: Fold approximation-complexity law

Claim: `SFT-COMP-CPLX-APPROXIMATION-001`

41.1 Scientific necessity and exact theorem

Fold approximation-complexity law is required to derive resource versus retained objective distinction without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced resource versus retained objective distinction kernel retains optimization domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-AVERAGE-WORST-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

41.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: optimisation domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for resource versus retained objective distinction.

> *All generated finite resource versus retained objective distinction structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No unregistered metric or irrational ratio is admitted.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

41.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced resource versus retained objective distinction kernel retains optimization domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

41.4 Operational laws and consequences

- Carrier law: optimisation domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

41.5 Depth-independent closure

Base: The structural One supplies one canonical resource versus retained objective distinction form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated resource versus retained objective distinction form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

41.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-APPROXIMATION-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-APPROXIMATION-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-APPROXIMATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No unregistered metric or irrational ratio is admitted.	`PASS`

41.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: approximation complexity, approximation ratio.

No unregistered metric or irrational ratio is admitted.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No unregistered metric or irrational ratio is admitted.

41.8 Exact evidence identities

- Source manifest: `sha256:da988e508999ce8992454d7b760a8e0c8cb256e20bc8bcb1c553ee2d478e0f6c`
- Complete-census certificate: `sha256:01715add870cff6f558b949706fbbf68b286c955c637a909e74cc0ef5f737968`
- Derivation seal: `sha256:661876c7efa17773960092329223bd1b8d3810c74167337614c4ca3861010c6f`
- Independent implementation: `sha256:49e8ad275c48b129eb12592995af3840143a8ca0dc7ebb66d8a7669f64daaf86`
- Independent certificate: `sha256:514e851668812207b41ed55e5cc45d02f101c37855b6c84347a0da2cceedff34`
- External validation: `sha256:71afb1d9276d8465dd2707358515b59d69b86fd671bca76be7680464eb4d47f3`
- Model-admitted engine receipt: `sha256:e8062d9de51f472dc26ca19cd2e838de843c9b1449d16c71962ad2c06e9f8357`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-APPROXIMATION-001-e8062d9de51f472d.json`

42. Derivation 34: Fold parameterized-complexity law

Claim: `SFT-COMP-CPLX-PARAMETERIZED-001`

42.1 Scientific necessity and exact theorem

Fold parameterized-complexity law is required to derive resource stratification by exact structural coordinates without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced resource stratification by exact structural coordinates kernel retains input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation; every operation is*

> *source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-APPROXIMATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

42.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for resource stratification by exact structural coordinates.

> *All generated finite resource stratification by exact structural coordinates structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Parameters must be forced structural coordinates, never fitted constants.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

42.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.

Eliminated candidates	Exact first-failure reason
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced resource stratification by exact structural coordinates kernel retains input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

42.4 Operational laws and consequences

- Carrier law: input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

42.5 Depth-independent closure

Base: The structural One supplies one canonical resource stratification by exact structural coordinates form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated resource stratification by exact structural coordinates form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

42.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-PARAMETERIZED-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-PARAMETERIZED-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-PARAMETERIZED-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Parameters must be forced structural coordinates, never fitted constants.	`PASS`

42.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: parameterized complexity, fixed-parameter tractability.

Parameters must be forced structural coordinates, never fitted constants.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Parameters must be forced structural coordinates, never fitted constants.

42.8 Exact evidence identities

- Source manifest: `sha256:a83039453f3937ef06d195c71447f74e854ce9c410a9860f00d5d2ef56c663bc`
- Complete-census certificate: `sha256:432c7d44a7c5f93e6b63ab3f1c3c8796a4bdc6fdab72f650eaba82086d16e5c7`
- Derivation seal: `sha256:f97fafb086546626bbe55c2f9544769c9b21ebf937e83b95a9c93439535d978a`
- Independent implementation: `sha256:d10266176661ea8f4c2625bac8f828473fd9a287f9e056a12ff5358d8e7c2de9`
- Independent certificate: `sha256:90751ccec65ad243aa877cde32c4ad9520033e0bcbc110e0b293ea452b879295`
- External validation: `sha256:0bd259151d1801258dce6352daa80fcef344a016412df453f479f07474c95eed`
- Model-admitted engine receipt: `sha256:1afcbc7d33437e51f057b4a63715fc3c5e41890025b3d093edb4fbeb15b370ac`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-PARAMETERIZED-001-1afcbc7d33437e51.json`

43. Derivation 35: Fold information and descriptive-complexity law

Claim: `SFT-COMP-CPLX-DESCRIPTIVE-001`

43.1 Scientific necessity and exact theorem

Fold information and descriptive-complexity law is required to derive shortest exact generated descriptions without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced shortest exact generated descriptions kernel retains frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-PARAMETERIZED-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

43.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger.

Generation rule: Generate the literal product of the eight registered Computational Complexity structural axes for shortest exact generated descriptions.

> *All generated finite shortest exact generated descriptions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Unrestricted shortest-description values remain uncomputable; finite censuses state their depth explicitly.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`input`	`presentation-length`	presentation-length removes a required exact input condition.	`canonical-encoding-support`	canonical-encoding-support retains the complete registered input condition.
`resource`	`unrecorded-cost`	unrecorded-cost removes a required exact resource condition.	`complete-resource-ledger`	complete-resource-ledger retains the complete registered resource condition.
`model`	`model-unspecified`	model-unspecified removes a required exact model condition.	`registered-computation-model`	registered-computation-model retains the complete registered model condition.
`comparison`	`unmatched-instances`	unmatched-instances removes a required exact comparison condition.	`common-family-comparison`	common-family-comparison retains the complete registered comparison condition.
`composition`	`cost-erasing-composition`	cost-erasing-composition removes a required exact composition condition.	`overhead-retaining-composition`	overhead-retaining-composition retains the complete registered composition condition.
`witness`	`claimed-bound`	claimed-bound removes a required exact witness condition.	`constructive-or-adversarial-bound-witness`	constructive-or-adversarial-bound-witness retains the complete registered witness condition.
`generality`	`benchmark-only`	benchmark-only removes a required exact generality condition.	`family-successor-certificate`	family-successor-certificate retains the complete registered generality condition.
`addition`	`imported-scale-or-rate`	imported-scale-or-rate removes a required exact addition condition.	`no-extra-resource-parameter`	no-extra-resource-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

43.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	presentation-length removes a required exact input condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-cost removes a required exact resource condition.
32	model-unspecified removes a required exact model condition.
16	unmatched-instances removes a required exact comparison condition.
8	cost-erasing-composition removes a required exact composition condition.
4	claimed-bound removes a required exact witness condition.
2	benchmark-only removes a required exact generality condition.
1	imported-scale-or-rate removes a required exact addition condition.

Shared claim-record clause `COMP-S008` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced shortest exact generated descriptions kernel retains frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

43.4 Operational laws and consequences

- Carrier law: frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

43.5 Depth-independent closure

Base: The structural One supplies one canonical shortest exact generated descriptions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated shortest exact generated descriptions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

43.6 Executed witnesses and adverse controls

- `complexity-witness-1` - complexity exact operational witness 1 for SFT-COMP-CPLX-DESCRIPTIVE-001; result: `PASS`.
- `complexity-witness-2` - complexity exact operational witness 2 for SFT-COMP-CPLX-DESCRIPTIVE-001; result: `PASS`.
- `complexity-witness-3` - complexity exact operational witness 3 for SFT-COMP-CPLX-DESCRIPTIVE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	presentation-length removes a required exact input condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Unrestricted shortest-description values remain uncomputable; finite censuses state their depth explicitly.	`PASS`

43.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: Kolmogorov complexity, descriptive complexity.

Unrestricted shortest-description values remain uncomputable; finite censuses state their depth explicitly.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Unrestricted shortest-description values remain uncomputable; finite censuses state their depth explicitly.

43.8 Exact evidence identities

- Source manifest: `sha256:87440f348bebb73171355c031df3baa6fd3c0a278a3bba811790e2f7384f2f76`
- Complete-census certificate: `sha256:7b0a69da1344d382e3afbfac17ebf150fa0aa40462fe8afa3be4a1b0ae6a4564`
- Derivation seal: `sha256:75c939154f2d150b36c52e8c27a9080fa0bb6df65b999fd5f9dd70dc6a71f6a4`
- Independent implementation: `sha256:70a8a83cc65b2f6a44ed0abde6ef8fe9211e60fbdd5176225e943725e32c6a7a`
- Independent certificate: `sha256:d7e0354f7e04bafa5d366e2c394282723134d2f980ce45bdaa624cd026b5b401`
- External validation: `sha256:4014842120d59fd7cb3078960451a7ac342f183f44b6ebd8a8d41d39d8bb2352`
- Model-admitted engine receipt: `sha256:bdbb14bfbdd53b05dd80b9b4fd0ea77e3f9eb303ab637ec6714029665a1f6eef`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-DESCRIPTIVE-001-bdbb14bfbdd53b05.json`

44. Derivation 36: Fold searching and ordering law

Claim: `SFT-COMP-ALG-SEARCH-ORDER-001`

44.1 Scientific necessity and exact theorem

Fold searching and ordering law is required to derive location and exact order construction without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced location and exact order construction kernel retains generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CPLX-DESCRIPTIVE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

44.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for location and exact order construction.

> *All generated finite location and exact order construction structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Only registered finite carriers and comparison relations are covered.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

44.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.

Eliminated candidates	Exact first-failure reason
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced location and exact order construction kernel retains generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

44.4 Operational laws and consequences

- Carrier law: generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

44.5 Depth-independent closure

Base: The structural One supplies one canonical location and exact order construction form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated location and exact order construction form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

44.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-SEARCH-ORDER-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-SEARCH-ORDER-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-SEARCH-ORDER-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only registered finite carriers and comparison relations are covered.	`PASS`

44.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: search, sorting.

Only registered finite carriers and comparison relations are covered.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Only registered finite carriers and comparison relations are covered.

44.8 Exact evidence identities

- Source manifest: `sha256:5763f896d0c066371fc1ff9577905144f07e46154d64f325dc543c7f904ba536`
- Complete-census certificate: `sha256:7a1636755ef57d0c54d8dfc12741169887017de61b65a9494dc13dbca4a02eb2`
- Derivation seal: `sha256:5410a5a2e280a6c31796b5735e3a1e7a936ed1b57d35fd24fe35123ea1a905b0`
- Independent implementation: `sha256:bc23f61ab78f015f932efa3d73e656a123c338d041692d13664cd89e5148ce46`
- Independent certificate: `sha256:b960ead246c2b465341e9c2a29216e0f80b39e26b0f908dcc6495d376b333e77`
- External validation: `sha256:db458ef04ab0525c0f379719e9d03e03c74effc48f16623c36ed8cca6bf8481e`
- Model-admitted engine receipt: `sha256:915d9a65bf080d49519f15583d8ffab68375f99e1af014be611cdc25c6626c78`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-SEARCH-ORDER-001-915d9a65bf080d49.json`

45. Derivation 37: Fold exact arithmetic-algorithm law

Claim: `SFT-COMP-ALG-ARITHMETIC-001`

45.1 Scientific necessity and exact theorem

Fold exact arithmetic-algorithm law is required to derive computation over exact positive parts without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced computation over exact positive parts kernel retains canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-SEARCH-ORDER-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

45.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for computation over exact positive parts.

> *All generated finite computation over exact positive parts structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No floating, negative, irrational or imaginary proof value is admitted.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

45.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.

Eliminated candidates	Exact first-failure reason
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced computation over exact positive parts kernel retains canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

45.4 Operational laws and consequences

- Carrier law: canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

45.5 Depth-independent closure

Base: The structural One supplies one canonical computation over exact positive parts form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated computation over exact positive parts form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

45.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-ARITHMETIC-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-ARITHMETIC-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-ARITHMETIC-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No floating, negative, irrational or imaginary proof value is admitted.	`PASS`

45.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: arithmetic algorithm, exact arithmetic.

No floating, negative, irrational or imaginary proof value is admitted.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No floating, negative, irrational or imaginary proof value is admitted.

45.8 Exact evidence identities

- Source manifest: `sha256:14502ce7be60ed58afaa4d0ce977254936c109bfc2a10c6b31042dd74b2b5add`
- Complete-census certificate: `sha256:fd5164391de2e2b5fdbfaecc8e9cfdb96d0595cce56a8b05faf9add5a870497f`
- Derivation seal: `sha256:c124e9fe09f662ef9e56992f8295e5145539549a2216e8b0d4e7ca2cbb9f5c49`
- Independent implementation: `sha256:fc377822b8b452cd195b9c076cac4feb03a72b5808dc38b3aee01b1ad69b1e63`
- Independent certificate: `sha256:0f9ad44ec485d994f209fa3832e1a9af89c20b5dc67c9f5e656db5992b1397a5`
- External validation: `sha256:9715086d06bc1f22f0398fe1cfcb9cea665f0a125bb5b497c4e8297f4826251c`
- Model-admitted engine receipt: `sha256:af254db83c3bc4c67b1c2daa5195c8f74a1a9e4fbcd08b159c677579b9c1ab52`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-ARITHMETIC-001-af254db83c3bc4c6.json`

46. Derivation 38: Fold string and sequence algorithm law

Claim: `SFT-COMP-ALG-STRINGS-SEQUENCES-001`

46.1 Scientific necessity and exact theorem

Fold string and sequence algorithm law is required to derive finite ordered symbol organisations without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced finite ordered symbol organizations kernel retains complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-ARITHMETIC-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

46.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for finite ordered symbol organisations.

> *All finite generated ordered-symbol-organisation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No ungenerated infinite stream is treated as complete.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

46.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.

Eliminated candidates	Exact first-failure reason
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced finite ordered symbol organizations kernel retains complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

46.4 Operational laws and consequences

- Carrier law: complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

46.5 Depth-independent closure

Base: The structural One supplies one canonical finite ordered symbol organisations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite ordered symbol organisations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

46.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-STRINGS-SEQUENCES-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-STRINGS-SEQUENCES-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-STRINGS-SEQUENCES-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No ungenerated infinite stream is treated as complete.	`PASS`

46.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: string algorithm, sequence algorithm.

No ungenerated infinite stream is treated as complete.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No ungenerated infinite stream is treated as complete.

46.8 Exact evidence identities

- Source manifest: `sha256:238638db306d480419c314ea9c5b665b7ec72f339f0f6fc4f7ddbd6be4752a55`
- Complete-census certificate: `sha256:68c63d4c5988cb3f4429f991d48810522bb1f62a5e0760a844869e9c73f74d31`
- Derivation seal: `sha256:8959ca63fab8c4543e86b8aa029aea4a48ca8f66670d4ec2312ab32eee0b2a89`
- Independent implementation: `sha256:20b535a46bd9170066a6f938dabfaa4ed83aedf5fb981eee653dcda6f8df5124`
- Independent certificate: `sha256:dc0b982cf407e60e815f8ec2d5045aaab3145ab9ccd3426d93a74d2531f68d84`
- External validation: `sha256:f6566f41be47c5ebd8b3fe42fe8bc341cd8a41f560280ba78f8bfd59001f8ad4`
- Model-admitted engine receipt: `sha256:5c2fb101d63e8d37ba076927e1de853261936a53c33400eba91c9adbd135caaf`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-STRINGS-SEQUENCES-001-5c2fb101d63e8d37.json`

47. Derivation 39: Fold tree and graph algorithm law

Claim: `SFT-COMP-ALG-TREES-GRAPHS-001`

47.1 Scientific necessity and exact theorem

Fold tree and graph algorithm law is required to derive finite relational traversal and structure without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced finite relational traversal and structure kernel retains generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination; every operation is source-bound, trace-complete,*

> *boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-STRINGS-SEQUENCES-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

47.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for finite relational traversal and structure.

> *All finite generated relational-traversal structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Applies to finite generated graphs; physical networks require empirical translation.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

47.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.

Eliminated candidates	Exact first-failure reason
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced finite relational traversal and structure kernel retains generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

47.4 Operational laws and consequences

- Carrier law: generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

47.5 Depth-independent closure

Base: The structural One supplies one canonical finite relational traversal and structure form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite relational traversal and structure form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

47.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-TREES-GRAPHS-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-TREES-GRAPHS-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-TREES-GRAPHS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Applies to finite generated graphs; physical networks require empirical translation.	`PASS`

47.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: graph algorithm, tree algorithm.

Applies to finite generated graphs; physical networks require empirical translation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Applies to finite generated graphs; physical networks require empirical translation.

47.8 Exact evidence identities

- Source manifest: `sha256:82e504863893dbb2b0f5068fd3e3e9fc4e68cbff929c6c8a053e94a86fc0b30e`
- Complete-census certificate: `sha256:f1630f72ede9480e5df1a74cc9afb4babf396c07f5dfea3373dba6cc38f1f609`
- Derivation seal: `sha256:0b0a095159e3c047d835c3e4bbedecc2efa8b4986fe1693c2e5793a33fad1c63`
- Independent implementation: `sha256:38f3cf31f1ade6c3a7438f948c812d8fc62eadcad6cca80e7bd97a55185fd25c`
- Independent certificate: `sha256:8379189b1b6c280a2b2311ca5d84cb7843e834a887f3115d401407767ac06270`
- External validation: `sha256:4deefa9e67a39f8f15daace40fc9734b41590d50fbc7ec7f2b6e2f58866b7f8d`
- Model-admitted engine receipt: `sha256:0c1d76188752d9f584b9bec6db855143f954a2f65b9ad57375bcd3e68809f604`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-TREES-GRAPHS-001-0c1d76188752d9f5.json`

48. Derivation 40: Fold algebraic and geometric algorithm law

Claim: `SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001`

48.1 Scientific necessity and exact theorem

Fold algebraic and geometric algorithm law is required to derive finite algebraic operations and incidence without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced finite algebraic operations and incidence kernel retains exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-TREES-GRAPHS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

48.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for finite algebraic operations and incidence.

> *All finite generated algebraic-operation-and-incidence structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Geometry remains finite and exact; no ungenerated continuum is imported.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

48.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

The forced finite algebraic operations and incidence kernel retains exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

48.4 Operational laws and consequences

- Carrier law: exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

48.5 Depth-independent closure

Base: The structural One supplies one canonical finite algebraic operations and incidence form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite algebraic operations and incidence form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

48.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Geometry remains finite and exact; no ungenerated continuum is imported.	`PASS`

48.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: algebraic algorithm, computational geometry.

Geometry remains finite and exact; no ungenerated continuum is imported.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Geometry remains finite and exact; no ungenerated continuum is imported.

48.8 Exact evidence identities

- Source manifest: `sha256:bb9b39ce14861c2ebdd176f20ae62d50cdab3842075ae516f848c6896d4fc45e`
- Complete-census certificate: `sha256:3aa7470cc7aec406c7c072cfcbf452b237eac34ca06d5ec7247a264dc0fb1308`
- Derivation seal: `sha256:81409fcc27f9a8511b396209692bbc2a28b900f3c0dfdce8c4c534c0f033d7ef`
- Independent implementation: `sha256:73d59f6f67dc89af760a385a65b33aad744e89a90b7c5dd722208a098432ea30`
- Independent certificate: `sha256:e06d8cce5e05c56e774a1101d444b554e16e3cb7f4ab4bd21d0c31f1cd207e0f`
- External validation: `sha256:779fd81207ac8c38ac5e261d9d80151e5868ca0d99cd774818697326c24817f3`
- Model-admitted engine receipt: `sha256:b2ba1fc787d153cd8673cf1166703cd608291c7136d7412bff9c9dc66f3de430`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001-b2ba1fc787d153cd.json`

49. Derivation 41: Fold dynamic-programming law

Claim: `SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001`

49.1 Scientific necessity and exact theorem

Fold dynamic-programming law is required to derive reuse of exact overlapping subproblem results without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced reuse of exact overlapping subproblem results kernel retains problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

49.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for reuse of exact overlapping subproblem results.

> *All generated finite reuse of exact overlapping subproblem results structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. The decomposition and optimal-substructure conditions must be proved for each problem.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

49.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced reuse of exact overlapping subproblem results kernel retains problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

49.4 Operational laws and consequences

- Carrier law: problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

49.5 Depth-independent closure

Base: The structural One supplies one canonical reuse of exact overlapping subproblem results form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated reuse of exact overlapping subproblem results form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

49.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; The decomposition and optimal-substructure conditions must be proved for each problem.	`PASS`

49.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: dynamic programming, memoization.

The decomposition and optimal-substructure conditions must be proved for each problem.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- The decomposition and optimal-substructure conditions must be proved for each problem.

49.8 Exact evidence identities

- Source manifest: `sha256:5b7f2b4a2bccef5a722bd45128cb0ec0077a5aed9db66a0b33d0dcad9757a330`
- Complete-census certificate: `sha256:d7c9bf091d95bc640e9e2108bff3c6d8e915545d309fc53a1db28f40fd3f09e7`
- Derivation seal: `sha256:cbb434e7fac471f725b58cafea9c8de8d29edc94e9641aba911b147392d433ce`
- Independent implementation: `sha256:4ff3d626485a1d771847ddf3c068086a7e991b4dbf6a53c47df08a9b20446fe9`
- Independent certificate: `sha256:2fc4b3935bb8d193969048a49d5f8e26392b5a087667239e58228cb18b23ad86`
- External validation: `sha256:132df1cbe4d7960c2524053d50748b28c616b110c46b1db58a2c82160b3a1bec`
- Model-admitted engine receipt: `sha256:f6b1c405c189bb4f86db10ec31ad2dc8adc8041e0833baeb6184ac1832ed2e17`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001-f6b1c405c189bb4f.json`

50. Derivation 42: Fold optimisation-algorithm law

Claim: `SFT-COMP-ALG-OPTIMIZATION-001`

50.1 Scientific necessity and exact theorem

Fold optimisation-algorithm law is required to derive selection of all exact optima without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced selection of all exact optima kernel retains complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

50.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for selection of all exact optima.

> *All generated finite selection of all exact optima structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Does not assume differentiability, convexity or a conventional metric.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

50.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced selection of all exact optima kernel retains complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

50.4 Operational laws and consequences

- Carrier law: complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.

- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

50.5 Depth-independent closure

Base: The structural One supplies one canonical selection of all exact optima form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated selection of all exact optima form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

50.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-OPTIMIZATION-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-OPTIMIZATION-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-OPTIMIZATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not assume differentiability, convexity or a conventional metric.	`PASS`

50.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: optimisation algorithm, exact optimum.

Does not assume differentiability, convexity or a conventional metric.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Does not assume differentiability, convexity or a conventional metric.

50.8 Exact evidence identities

- Source manifest: `sha256:85c97911077ffd40a3a67b5477c79766a87bba308799693b1a5fc6b8281028c7`

- Complete-census certificate: `sha256:95c08d8043fece309af5b1e542472b03b1d782d20b90c1378f1fd68612d60b72`
- Derivation seal: `sha256:4ee440a4de7a6270ba9d798a00174a5706c2974136615a61a24bfdc4651dea92`
- Independent implementation: `sha256:c9209269f93ffe1054bce5215f859ebea9fe5638dce655dbbb52047d94e8a1ad`
- Independent certificate: `sha256:5f293ae15ca747f138ce75beaf35f7a5deec7a65aa77d3056fa6eb4e44ded3ed`
- External validation: `sha256:df0821a0b5130ebbc42860538bee57f122cdb6a8e509201fb639d54edc33dba0`
- Model-admitted engine receipt: `sha256:74aac9551388975891fa29de8f8a7843eb1afcbc2a3df316784f34caf7ebd110`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-OPTIMIZATION-001-74aac95513889758.json`

51. Derivation 43: Fold randomized-algorithm law

Claim: `SFT-COMP-ALG-RANDOMIZED-001`

51.1 Scientific necessity and exact theorem

Fold randomized-algorithm law is required to derive complete scheduled branch execution without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced complete scheduled branch execution kernel retains deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-OPTIMIZATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

51.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for complete scheduled branch execution.

> *All generated finite complete scheduled branch execution structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Superdeterminism is preserved: apparent random choice is an unresolved or externally measured schedule, not an uncaused event.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

51.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced complete scheduled branch execution kernel retains deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

51.4 Operational laws and consequences

- Carrier law: deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

51.5 Depth-independent closure

Base: The structural One supplies one canonical complete scheduled branch execution form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated complete scheduled branch execution form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

51.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-RANDOMIZED-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-RANDOMIZED-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-RANDOMIZED-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Superdeterminism is preserved: apparent random choice is an unresolved or externally measured schedule, not an uncaused event.	`PASS`

51.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: randomized algorithm, probabilistic algorithm.

Superdeterminism is preserved: apparent random choice is an unresolved or externally measured schedule, not an uncaused event.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Superdeterminism is preserved: apparent random choice is an unresolved or externally measured schedule, not an uncaused event.

51.8 Exact evidence identities

- Source manifest: `sha256:71c76995d7b4206bee6895f9744a5edee06fd25a53e2cc9177e38bf9df238f3e`

- Complete-census certificate: `sha256:5325d143b0cf5951675567187534e15a9c168d778353a9f10c96c5c2b42e7dbc`
- Derivation seal: `sha256:660b6dd0cc83e1b9b5b002f08369fd5274b6f857c3c875579352bc279b6e5b12`
- Independent implementation: `sha256:dd42d329d28bec95bbb52eb2a0d11237935964122aa642def54ad716b7615c18`
- Independent certificate: `sha256:acba3f2adc7ce42c9c8aab1fe43243c633a3c380e20b5ddea1f722b7bcee8365`
- External validation: `sha256:3c5f4558ce2d237409b4c80bf00213d9dc8b3461ce506a0de72e7385baed043f`
- Model-admitted engine receipt: `sha256:1c9f175ed883e3267b8722b8481cc424959ae5980bab9a6cf81e691d0785e0ae`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-RANDOMIZED-001-1c9f175ed883e326.json`

52. Derivation 44: Fold parallel-algorithm law

Claim: `SFT-COMP-ALG-PARALLEL-001`

52.1 Scientific necessity and exact theorem

Fold parallel-algorithm law is required to derive causally independent algorithm steps without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced causally independent algorithm steps kernel retains dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-RANDOMIZED-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

52.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for causally independent algorithm steps.

> *All generated finite causally independent algorithm steps structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No physical speedup is claimed without an implementation experiment.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

52.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced causally independent algorithm steps kernel retains dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

52.4 Operational laws and consequences

- Carrier law: dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

52.5 Depth-independent closure

Base: The structural One supplies one canonical causally independent algorithm steps form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated causally independent algorithm steps form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

52.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-PARALLEL-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-PARALLEL-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-PARALLEL-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No physical speedup is claimed without an implementation experiment.	`PASS`

52.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: parallel algorithm, parallel schedule.

No physical speedup is claimed without an implementation experiment.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No physical speedup is claimed without an implementation experiment.

52.8 Exact evidence identities

- Source manifest: `sha256:163a71a7671bf378b29b06504ed879ddb632db4d2d558aff25cd0d7d912c4b31`
- Complete-census certificate: `sha256:f52a3bbccfc7ec823a8fb4c3bfc8d3a4bcb5ee879b1c21cf32eb1fbfb3a15583`
- Derivation seal: `sha256:568ed2eac60d7b9834407009080c157b70f5a31cb6a03762ca9660238d6a4216`

- Independent implementation: `sha256:1bb66536383966cbfc406201ad9e09aa506c9dc78b60816366d649a6b1499f78`
- Independent certificate: `sha256:9e0f1e409141fb2593e3015afd7188e5d260cb8dca6ccf3eb66894fa0736df9b`
- External validation: `sha256:6e9a823a975e4b99cd0abfbe7c2aa1a84ee3438a2f70578e37616c8e073cb9dc`
- Model-admitted engine receipt: `sha256:c16c0747c8994451d421ddb5bc14c4715d7cd4ea6c78588c3da0d318a66ca285`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-PARALLEL-001-c16c0747c8994451.json`

53. Derivation 45: Fold distributed-algorithm law

Claim: `SFT-COMP-ALG-DISTRIBUTED-001`

53.1 Scientific necessity and exact theorem

Fold distributed-algorithm law is required to derive local processes exchanging exact records without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced local processes exchanging exact records kernel retains process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-PARALLEL-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

53.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for local processes exchanging exact records.

> *All generated finite local processes exchanging exact records structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. System implementation and physical network claims are outside this formal theorem.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

53.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced local processes exchanging exact records kernel retains process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

53.4 Operational laws and consequences

- Carrier law: process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

53.5 Depth-independent closure

Base: The structural One supplies one canonical local processes exchanging exact records form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated local processes exchanging exact records form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

53.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-DISTRIBUTED-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-DISTRIBUTED-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-DISTRIBUTED-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; System implementation and physical network claims are outside this formal theorem.	`PASS`

53.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: distributed algorithm, message passing.

System implementation and physical network claims are outside this formal theorem.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- System implementation and physical network claims are outside this formal theorem.

53.8 Exact evidence identities

- Source manifest: `sha256:bbba34bfed0d86637ec45a9f7b4e3db903308295b1d2245cb63a1ab0654d34b8`
- Complete-census certificate: `sha256:33d933518ad295e9a5cd748482123a3d87c402b856733bf977aed5f5dd40ec12`
- Derivation seal: `sha256:8785aed4c9eaf6cd44caf7eea5d18b54c7830fd88b6afdbe1ccb3fc5b95ae2e5`
- Independent implementation: `sha256:abb34c0e7b42e093e771d0daec22d5b35f973b7c669013ad640a5bd3bbb8c3de`
- Independent certificate: `sha256:c91f515c7a13d50e8bbb9e8e41e6e074556f008800138ee68beddb8b79ce48d2`

- External validation: `sha256:f52cd6b95a6e1d6c10e77cc3690523ed9d0303cf0517c8efbb56d1b22b668361`
- Model-admitted engine receipt: `sha256:5dfc71b4941c7e3f629b7ee0ab6cd3ca1851222fbfaf68690fa67875ed1f7e60`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-DISTRIBUTED-001-5dfc71b4941c7e3f.json`

54. Derivation 46: Fold online and streaming algorithm law

Claim: `SFT-COMP-ALG-ONLINE-STREAMING-001`

54.1 Scientific necessity and exact theorem

Fold online and streaming algorithm law is required to derive prefix-bounded processing without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced prefix-bounded processing kernel retains generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-DISTRIBUTED-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

54.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for prefix-bounded processing.

> *All generated finite prefix-bounded processing structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A completed infinite stream is not admitted; every result names its generated prefix boundary.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

54.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced prefix-bounded processing kernel retains generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

54.4 Operational laws and consequences

- Carrier law: generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

54.5 Depth-independent closure

Base: The structural One supplies one canonical prefix-bounded processing form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated prefix-bounded processing form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

54.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-ONLINE-STREAMING-001; result: `PASS`.

- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-ONLINE-STREAMING-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-ONLINE-STREAMING-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A completed infinite stream is not admitted; every result names its generated prefix boundary.	`PASS`

54.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: online algorithm, streaming algorithm.

A completed infinite stream is not admitted; every result names its generated prefix boundary.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A completed infinite stream is not admitted; every result names its generated prefix boundary.

54.8 Exact evidence identities

- Source manifest: `sha256:a8fd89a9419fd5e03fa9bffce8c9e40f0a3501b9dc367b65d841b515b33ca401`
- Complete-census certificate: `sha256:e960b3f6b15556b07578faa2b545bc8fcb35a3343bb3903513766e52132058a6`
- Derivation seal: `sha256:aa17697469db35050d4793ba86d2bcbfbf2c6cb91c0ac3bc06c9a2b728449927`
- Independent implementation: `sha256:2645fa5246f2c6acd8fdb8ae2b5d7375a2da27904f8060ce8bb6f10ff936952a`
- Independent certificate: `sha256:86840c27e3c57fc8a9b6378bf020ea0b56e1e428a71d120792d65b38d5477d02`
- External validation: `sha256:1f9abd0ad082a83536f54eca31ebbd54ed32f9b7c82538cbadfedd5d28815fa9`
- Model-admitted engine receipt: `sha256:1978c4757d70afe4f9f64a74ddad4b9df88db850c160cb3840dbe18f2ada7a15`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-ONLINE-STREAMING-001-1978c4757d70afe4.json`

55. Derivation 47: Fold numerical-algorithm law

Claim: `SFT-COMP-ALG-NUMERICAL-001`

55.1 Scientific necessity and exact theorem

Fold numerical-algorithm law is required to derive exactly bounded approximation procedures without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced exactly bounded approximation procedures kernel retains exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-ONLINE-STREAMING-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

55.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for exactly bounded approximation procedures.

> *All generated finite exactly bounded approximation procedures structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Decimal or floating results are measured/implementation records and cannot establish the proof.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

55.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced exactly bounded approximation procedures kernel retains exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

55.4 Operational laws and consequences

- Carrier law: exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

55.5 Depth-independent closure

Base: The structural One supplies one canonical exactly bounded approximation procedures form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exactly bounded approximation procedures form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

55.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-NUMERICAL-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-NUMERICAL-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-NUMERICAL-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Decimal or floating results are measured/implementation records and cannot establish the proof.	`PASS`

55.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: numerical algorithm, numerical analysis.

Decimal or floating results are measured/implementation records and cannot establish the proof.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Decimal or floating results are measured/implementation records and cannot establish the proof.

55.8 Exact evidence identities

- Source manifest: `sha256:53aaa3f9f914f534a7317d16e99f8cdd13b5d8d5742a146500dc0e90c82859ff`
- Complete-census certificate: `sha256:d8e2c40d88ee24daadfdaf49061214c1c6e785ca77e839b2688d7032903c83bc`
- Derivation seal: `sha256:5f68377aa21130c232f506eb872b3c5ed5cb0eeff344de6f941f24c91fbd9476`
- Independent implementation: `sha256:197cdb1b49b090d0b84f1c28041860cb18342a7cb65b0a19e55697b32b3205a9`
- Independent certificate: `sha256:c9d66fcf52bfcea0b05cc05364626e91ff5b7c8b210f33ef2a03ba407de5fa90`
- External validation: `sha256:9ecc0471fe705aa388878b030820bfd0ba9801611de1780c7d0b7934ee91b598`
- Model-admitted engine receipt: `sha256:3f1c19281f4ac6c86c5bb13ae391ede42fa858c66e0571a4d6d66f27eeec098f`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-NUMERICAL-001-3f1c19281f4ac6c8.json`

56. Derivation 48: Fold symbolic-computation law

Claim: `SFT-COMP-ALG-SYMBOLIC-001`

56.1 Scientific necessity and exact theorem

Fold symbolic-computation law is required to derive canonical expression transformation without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced canonical expression transformation kernel retains generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-NUMERICAL-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

56.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for canonical expression transformation.

> *All generated finite canonical expression transformation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Only expressions within the frozen grammar are closed.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

56.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced canonical expression transformation kernel retains generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

56.4 Operational laws and consequences

- Carrier law: generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

56.5 Depth-independent closure

Base: The structural One supplies one canonical expression-transformation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated canonical expression transformation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

56.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-SYMBOLIC-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-SYMBOLIC-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-SYMBOLIC-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only expressions within the frozen grammar are closed.	`PASS`

56.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: symbolic computation, computer algebra.

Only expressions within the frozen grammar are closed.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Only expressions within the frozen grammar are closed.

56.8 Exact evidence identities

- Source manifest: `sha256:4f54e53491f964a781a06b806dd4c63e42bc5c4c58acf00842e360a73b9f62ab`
- Complete-census certificate: `sha256:586b45e768eea427658b4418273d997349f303b451609df1fd20e06dde4cf1c5`
- Derivation seal: `sha256:cfa87346438f6199df20a903922e69866cdac4918ac03bb3df563b888733db65`
- Independent implementation: `sha256:144a7d6137278c61bb5b2ec25919e1403d0967cab27d7f0974a48419b875dbdd`
- Independent certificate: `sha256:9e7f855770ac962b504ba0f8531b0052f95a8575a756a3756d5cfc11f0c6eafb`
- External validation: `sha256:8c88ba5ced2cdc6dc2cd67336f03d8554d6bb70fc68bacb4140adcc8a593bbbb`
- Model-admitted engine receipt: `sha256:3a84f78267c8d51b1e8c1ff0981ba89929b1c2b5231b5fedcb0d522b87f9fa8d`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-SYMBOLIC-001-3a84f78267c8d51b.json`

57. Derivation 49: Fold approximation-algorithm law

Claim: `SFT-COMP-ALG-APPROXIMATE-001`

57.1 Scientific necessity and exact theorem

Fold approximation-algorithm law is required to derive bounded resource solution with exact error ledger without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced bounded resource solution with exact error ledger kernel retains feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-SYMBOLIC-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

57.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace.

Generation rule: Generate the literal product of the eight registered Algorithms and Mathematical Data Structures structural axes for bounded resource solution with exact error ledger.

> *All generated finite bounded resource solution with exact error ledger structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No imported norm, epsilon parameter or irrational threshold is permitted.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`sampled-inputs`	sampled-inputs removes a required exact domain condition.	`complete-generated-input-domain`	complete-generated-input-domain retains the complete registered domain condition.
`precondition`	`implicit-precondition`	implicit-precondition removes a required exact precondition condition.	`exact-registered-precondition`	exact-registered-precondition retains the complete registered precondition condition.
`transition`	`answer-producing-heuristic`	answer-producing-heuristic removes a required exact transition condition.	`exact-step-relation`	exact-step-relation retains the complete registered transition condition.
`invariant`	`unchecked-intermediate-state`	unchecked-intermediate-state removes a required exact invariant condition.	`retained-invariant-ledger`	retained-invariant-ledger retains the complete registered invariant condition.
`output`	`result-without-certificate`	result-without-certificate removes a required exact output condition.	`reconstructible-output-certificate`	reconstructible-output-certificate retains the complete registered output condition.
`resource`	`unreported-resources`	unreported-resources removes a required exact resource condition.	`complete-time-space-trace`	complete-time-space-trace retains the complete registered resource condition.
`generality`	`example-program`	example-program removes a required exact generality condition.	`input-successor-certificate`	input-successor-certificate retains the complete registered generality condition.
`addition`	`imported-algorithm`	imported-algorithm removes a required exact addition condition.	`no-extra-answer-model`	no-extra-answer-model retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

57.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-inputs removes a required exact domain condition.

Eliminated candidates	Exact first-failure reason
64	implicit-precondition removes a required exact precondition condition.
32	answer-producing-heuristic removes a required exact transition condition.
16	unchecked-intermediate-state removes a required exact invariant condition.
8	result-without-certificate removes a required exact output condition.
4	unreported-resources removes a required exact resource condition.
2	example-program removes a required exact generality condition.
1	imported-algorithm removes a required exact addition condition.

Shared claim-record clause `COMP-S009` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced bounded resource solution with exact error ledger kernel retains feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

57.4 Operational laws and consequences

- Carrier law: feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

57.5 Depth-independent closure

Base: The structural One supplies one canonical bounded resource solution with exact error ledger form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated bounded resource solution with exact error ledger form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

57.6 Executed witnesses and adverse controls

- `algorithms-witness-1` - algorithms exact operational witness 1 for SFT-COMP-ALG-APPROXIMATE-001; result: `PASS`.
- `algorithms-witness-2` - algorithms exact operational witness 2 for SFT-COMP-ALG-APPROXIMATE-001; result: `PASS`.
- `algorithms-witness-3` - algorithms exact operational witness 3 for SFT-COMP-ALG-APPROXIMATE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-inputs removes a required exact domain condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No imported norm, epsilon parameter or irrational threshold is permitted.	`PASS`

57.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: approximation algorithm, error bound.

No imported norm, epsilon parameter or irrational threshold is permitted.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No imported norm, epsilon parameter or irrational threshold is permitted.

57.8 Exact evidence identities

- Source manifest: `sha256:35d1d30d54ffa205920e311090024fa5a26c8eed7dfbabaeccb58fd4bc892d46`
- Complete-census certificate: `sha256:cfe2c1b5daf58ab05a8b903443a2cb3e869c6c8653bda2faf2850c21cad2a658`
- Derivation seal: `sha256:e8803cab22548d6aaf21e549db85da47ff3586c6af729685dccc017a1a2f8ceb`
- Independent implementation: `sha256:ba63e01ccb420f64ca73dd0a592b1934c59848fc9bbae0045f153cb16e6b9e83`
- Independent certificate: `sha256:649e7983abd8098c4ef6eaefc7c1376f7709b4f78451cef1ead1f64036e2d541`
- External validation: `sha256:ac0157e48b5aa1fd22157b65b84f646284bf773de8fc411db161c5a918821e72`
- Model-admitted engine receipt: `sha256:52dfa606dc9c680d10c2c990b5df3ed29fe4f91be95690eb0d4880df5fd7dffc`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-ALG-APPROXIMATE-001-52dfa606dc9c680d.json`

58. Derivation 50: Fold program-syntax law

Claim: `SFT-COMP-SEM-SYNTAX-001`

58.1 Scientific necessity and exact theorem

Fold program-syntax law is required to derive generated program forms without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced generated program forms kernel retains complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-ALG-APPROXIMATE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

58.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for generated program forms.

All generated finite generated program forms structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Syntax alone assigns no execution meaning.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

58.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.

Eliminated candidates	Exact first-failure reason
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced generated program forms kernel retains complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

58.4 Operational laws and consequences

- Carrier law: complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

58.5 Depth-independent closure

Base: The structural One supplies one canonical generated program forms form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated program form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

58.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-SYNTAX-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-SYNTAX-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-SYNTAX-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Syntax alone assigns no execution meaning.	`PASS`

58.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: abstract syntax, program grammar.

Syntax alone assigns no execution meaning.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Syntax alone assigns no execution meaning.

58.8 Exact evidence identities

- Source manifest: `sha256:a87f61338ca3b9623a6adf94a9a13222ff3d1a2aa9b024687d9ae3d79397fc3c`
- Complete-census certificate: `sha256:3608ead17c98ddc04d8aefb2528e6adb2167bdf4bb5d252beaebf98e70983cc8`
- Derivation seal: `sha256:f044e21b0e7a95f6f56968665a58294e030e0bfef080756718c2aa9f35e7ff26`
- Independent implementation: `sha256:a107cd56990ec6701bea95ee4d4bfc2058d930bd3eabf27de0b2b32d9fdd0961`
- Independent certificate: `sha256:8b714790d85bdc3e7286ca82c336d9c1117fa2dc117dbd46ef229ef6d64a5adc`
- External validation: `sha256:81b83f0276bdde6805fb333b7889e1dbb969956453161217f2bb3cdf0d917c68`
- Model-admitted engine receipt: `sha256:0da324b3d1cd63a836a8494235259bffe143a912f6aace6745f55c1bcae10246`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-SYNTAX-001-0da324b3d1cd63a8.json`

59. Derivation 51: Fold binding and substitution law

Claim: `SFT-COMP-SEM-BINDING-SUBSTITUTION-001`

59.1 Scientific necessity and exact theorem

Fold binding and substitution law is required to derive name identity and replacement without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced name identity and replacement kernel retains held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-SYNTAX-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

59.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for name identity and replacement.

> *All generated finite name identity and replacement structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No presentation alias may replace canonical binding identity.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

59.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

The forced name identity and replacement kernel retains held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

59.4 Operational laws and consequences

- Carrier law: held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

59.5 Depth-independent closure

Base: The structural One supplies one canonical name identity and replacement form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated name identity and replacement form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

59.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-BINDING-SUBSTITUTION-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-BINDING-SUBSTITUTION-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-BINDING-SUBSTITUTION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No presentation alias may replace canonical binding identity.	`PASS`

59.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: binding, substitution.

No presentation alias may replace canonical binding identity.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No presentation alias may replace canonical binding identity.

59.8 Exact evidence identities

- Source manifest: `sha256:cef11ea2bef503bbbeb79fb136311d79b1422ba11e2f0632fae2aec3f0b20c12`
- Complete-census certificate: `sha256:aea1e5d04b24bf253e377c50f595dcb3d27b59d74b02887fc55adfa7b5f40eab`
- Derivation seal: `sha256:4d6312f18d849169c7052e425a5829af08525a4e25df7e057c9d99b787ba5e35`
- Independent implementation: `sha256:e83b481c5e02dad02da6129dda8424aaaac21723874d893cf02204356846ae4a`
- Independent certificate: `sha256:a757bc3b7aa1e6290b4b00eff8e3d2f07999383f3716d73e649dc90a05fcd9f8`
- External validation: `sha256:ba184fed5c0a038365c52d8da2a5658f865823719e1929cbbf4ed7bf90fd5f0e`
- Model-admitted engine receipt: `sha256:1200dc420f7ff8fcd37e5c3fd7e862fa3aa0d1466963d23d9fe3ce0a22ac103d`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-BINDING-SUBSTITUTION-001-1200dc420f7ff8fc.json`

60. Derivation 52: Fold evaluation law

Claim: `SFT-COMP-SEM-EVALUATION-001`

60.1 Scientific necessity and exact theorem

Fold evaluation law is required to derive program-state execution without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced program-state execution kernel retains syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-BINDING-SUBSTITUTION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

60.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for program-state execution.

> *All generated finite program-state execution structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Only exact positive parts and generated data forms are semantic values.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

60.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced program-state execution kernel retains syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

60.4 Operational laws and consequences

- Carrier law: syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

60.5 Depth-independent closure

Base: The structural One supplies one canonical program-state execution form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated program-state execution form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

60.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-EVALUATION-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-EVALUATION-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-EVALUATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only exact positive parts and generated data forms are semantic values.	`PASS`

60.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: evaluation, interpreter.

Only exact positive parts and generated data forms are semantic values.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- Only exact positive parts and generated data forms are semantic values.

60.8 Exact evidence identities

- Source manifest: `sha256:be036424b53d2ba9c6da1ff376a058174c88cecb90d87690bc310263e25b5b6f`
- Complete-census certificate: `sha256:42bea38f0ff3f867db7459bdae056bcea2ae33f442c26a51d1c8ef1285bdd443`
- Derivation seal: `sha256:3327352b243dc92f5a03be58dd7ef2f9509338a8551523c64daea50c75be5692`
- Independent implementation: `sha256:45c3b84d2fd37e2b113a0894dfca52199cb83b19f5637f9bf8f54b5081fd5200`
- Independent certificate: `sha256:f4bca7b32e999d76fc00c448137fa5dcfa056c2c270e7fb44b0bd9ce3ecfdb76`
- External validation: `sha256:d02ca6834776c37cadc0278024e7bd22283c70dfb96169cb5adbf74180d7dd10`
- Model-admitted engine receipt: `sha256:dfb202d42339a8c83ffbd8d1939ec3d6dd29819fb621b60f6dd559175315089a`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-EVALUATION-001-dfb202d42339a8c8.json`

61. Derivation 53: Fold operational and denotational correspondence law

Claim: `SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001`

61.1 Scientific necessity and exact theorem

Fold operational and denotational correspondence law is required to derive trace-to-meaning agreement without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced trace-to-meaning agreement kernel retains operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-EVALUATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

61.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for trace-to-meaning agreement.

> *All generated finite trace-to-meaning agreement structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Correspondence is proved only for the frozen language grammar.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

61.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced trace-to-meaning agreement kernel retains operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

61.4 Operational laws and consequences

- Carrier law: operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.

- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

61.5 Depth-independent closure

Base: The structural One supplies one canonical trace-to-meaning agreement form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated trace-to-meaning agreement form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

61.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Correspondence is proved only for the frozen language grammar.	`PASS`

61.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: operational semantics, denotational semantics.

Correspondence is proved only for the frozen language grammar.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Correspondence is proved only for the frozen language grammar.

61.8 Exact evidence identities

- Source manifest: `sha256:369034c6a53357958d53f61e96066a9b6539ffbcbe8e640dc752f258c243ebce`

- Complete-census certificate: `sha256:1e9121a5af21d756772b81910f2aa2b6b2d0e607baabf2e435df47fe084f217e`
- Derivation seal: `sha256:e8941a8b0de586b68b95ede9a238f45d9ffc7855a9c6492b52331ae4ced42991`
- Independent implementation: `sha256:5220d7142d967826226e5a0bf576cf87ab1f71af680374da69e82ace109ff839`
- Independent certificate: `sha256:dacb28b48850243cca84186f124d109878e9c4ecaf44f1cde9d151295f221fc1`
- External validation: `sha256:e408f1e0f0a65f69163d1db61238d7db90086204da7f256bbdda42edc721b778`
- Model-admitted engine receipt: `sha256:6db365c1b3c22883c276eec14485e6fb962528278f521654cba633fe6536e425`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001-6db365c1b3c22883.json`

62. Derivation 54: Fold type-theory law

Claim: `SFT-COMP-SEM-TYPE-001`

62.1 Scientific necessity and exact theorem

Fold type-theory law is required to derive structural admissibility of composition without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced structural admissibility of composition kernel retains generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

62.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for structural admissibility of composition.

> *All generated finite structural admissibility of composition structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No conventional type calculus is assumed beyond downstream comparison.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

62.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced structural admissibility of composition kernel retains generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

62.4 Operational laws and consequences

- Carrier law: generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

62.5 Depth-independent closure

Base: The structural One supplies one canonical structural admissibility of composition form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated structural admissibility of composition form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

62.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-TYPE-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-TYPE-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-TYPE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No conventional type calculus is assumed beyond downstream comparison.	`PASS`

62.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: type theory, type safety.

No conventional type calculus is assumed beyond downstream comparison.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No conventional type calculus is assumed beyond downstream comparison.

62.8 Exact evidence identities

- Source manifest: `sha256:9f4b1ebd3fca097f0b70cac79bf001afb8394ce01cd7ce64f858f2a700b2c253`
- Complete-census certificate: `sha256:560c422d0270c778a566861bdb33a0cd81e446427f702f7e3d688072c598abb3`
- Derivation seal: `sha256:fda760ee8e868f300216cd11a36d93a8e7b6845e8fcb70dcfd3802a003868364`
- Independent implementation: `sha256:241f76ae21ddba0448e9e66a4d78ddeb9f271415f6b99d065fe0d6f2fb296693`

- Independent certificate: `sha256:ba5a47bb0f58c6316a328e9317018057175150177e5259da5333957b760915a9`
- External validation: `sha256:8ade6d8bad00291ecb7005a195c0863d988e7381eb68b9a3f9127ef12b98f5cc`
- Model-admitted engine receipt: `sha256:f62ebd67f072c2e126b16d47d16cf1982426bc5c333be0c79d9148e43a505620`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-TYPE-001-f62ebd67f072c2e1.json`

63. Derivation 55: Fold program-equivalence law

Claim: `SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001`

63.1 Scientific necessity and exact theorem

Fold program-equivalence law is required to derive indistinguishable generated behaviour without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced indistinguishable generated behavior kernel retains two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-TYPE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

63.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for indistinguishable generated behaviour.

> *All generated finite indistinguishable generated behavior structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Equivalence is relative to the registered observation and input boundary.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

63.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced indistinguishable generated behavior kernel retains two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

63.4 Operational laws and consequences

- Carrier law: two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

63.5 Depth-independent closure

Base: The structural One supplies one canonical indistinguishable generated behaviour form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated indistinguishable generated behaviour form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

63.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Equivalence is relative to the registered observation and input boundary.	`PASS`

63.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: program equivalence, observational equivalence.

Equivalence is relative to the registered observation and input boundary.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Equivalence is relative to the registered observation and input boundary.

63.8 Exact evidence identities

- Source manifest: `sha256:a509788e7e52b8ebd47915633937dfa89a6c8badd67623c3055353489a9b0e16`
- Complete-census certificate: `sha256:6c456bc0b95c08f66914fee51db3916b28bfa3548775b435ae048d7eabc0c193`
- Derivation seal: `sha256:71c3f701a6c006e7d5b9d3ed693c33ac15f4b037304cb7e6db48e12d2c3f1939`
- Independent implementation: `sha256:eb20d8ce5ccc05057939be3429f6cc04f52154350f122608e3d34fd2013fe600`
- Independent certificate: `sha256:fca679a2f596aef6b1790e002eb6cab7145c49feb024f6f7f5e0fe3f2266e09b`
- External validation: `sha256:47b8bb4f01597ff56e8a874ee2ac589f2890d9ae8142f3421b559721bd462710`

- Model-admitted engine receipt: `sha256:ad7d0dba57a7c30f17d67ef1e94ba22b9fe9e126565b894d97f435d1f83e4a17`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001-ad7d0dba57a7c30f.json`

64. Derivation 56: Fold termination law

Claim: `SFT-COMP-SEM-TERMINATION-001`

64.1 Scientific necessity and exact theorem

Fold termination law is required to derive absence of continuing execution paths without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced absence of continuing execution paths kernel retains program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

64.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for absence of continuing execution paths.

> *All generated finite absence of continuing execution paths structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No total termination decider is claimed for the universal language.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

64.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced absence of continuing execution paths kernel retains program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

64.4 Operational laws and consequences

- Carrier law: program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

64.5 Depth-independent closure

Base: The structural One supplies one canonical absence of continuing execution paths form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated absence of continuing execution paths form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

64.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-TERMINATION-001; result: `PASS`.

- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-TERMINATION-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-TERMINATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No total termination decider is claimed for the universal language.	`PASS`

64.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: termination, well-foundedness.

No total termination decider is claimed for the universal language.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No total termination decider is claimed for the universal language.

64.8 Exact evidence identities

- Source manifest: `sha256:e19e68832904cb3077a97a63fefcd62a2d4c10e3ecf9957acf56bcd15e519a3e`
- Complete-census certificate: `sha256:5cd4884f6cc707074b03aaed07b0237e1b3cc0207fe43ed92df5f5b9f105aa0c`
- Derivation seal: `sha256:963c938eee4f90c5b8e76126431979d0bc0fa6cacf4ee4b0cd4eb1c781c4da6d`
- Independent implementation: `sha256:2b7e5bd0df6cb5e94e9d1b8b24d2de035186c83eb1772cd53cbc8bb1c215d59a`
- Independent certificate: `sha256:540dbe2314f20eed8acba59ca366bd2d6f526d4aaa19f1edb81f19d22babe6e9`
- External validation: `sha256:847873d743a601a413e64211aff7e4a9fac44da1ed5553a91c7f4ae11ae86b14`
- Model-admitted engine receipt: `sha256:dc2378dbe15b3cfe2521a462c2bae6a7cef270b027719a9345618a9f1971c752`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-TERMINATION-001-dc2378dbe15b3cfe.json`

65. Derivation 57: Fold program-correctness law

Claim: `SFT-COMP-SEM-CORRECTNESS-001`

65.1 Scientific necessity and exact theorem

Fold program-correctness law is required to derive agreement between execution and specification without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced agreement between execution and specification kernel retains precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-TERMINATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

65.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for agreement between execution and specification.

> *All generated finite agreement between execution and specification structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Correctness is conditional on the exact registered specification and input support.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

65.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced agreement between execution and specification kernel retains precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

65.4 Operational laws and consequences

- Carrier law: precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

65.5 Depth-independent closure

Base: The structural One supplies one canonical agreement between execution and specification form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated agreement between execution and specification form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

65.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-CORRECTNESS-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-CORRECTNESS-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-CORRECTNESS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Correctness is conditional on the exact registered specification and input support.	`PASS`

65.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: program correctness, Hoare logic.

Correctness is conditional on the exact registered specification and input support.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Correctness is conditional on the exact registered specification and input support.

65.8 Exact evidence identities

- Source manifest: `sha256:2572f0f43365ae13d16bd169aa23b6e6ee0387acacf3931936908e93803027b8`
- Complete-census certificate: `sha256:67da57cd8e8a450a2ccf8c727f6b25b4af134f170ed732c1699ebc665c181971`
- Derivation seal: `sha256:db3ea2b1eecab1439dbe15f31592764e3e6515c7c3290359f02037c991077b7a`
- Independent implementation: `sha256:5e849c094b2d3e312b9ebfc368174baf8c72a986093e0d9ba3c9a35ee6b2ed13`
- Independent certificate: `sha256:a2927c2c2496399b11d4660cef42aa88f2e01a212fc0a22dbe8f72c5f384d227`
- External validation: `sha256:e3b70acc247f6998bf64ebde81b82de51ed5f70cdb364abc73bb3475175da81c`
- Model-admitted engine receipt: `sha256:95af5c4c8a0cf03d91f3d1ceff4decef32ba66537923e85eebe51ca3335e78bf`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-CORRECTNESS-001-95af5c4c8a0cf03d.json`

66. Derivation 58: Fold formal-specification law

Claim: `SFT-COMP-SEM-SPECIFICATION-001`

66.1 Scientific necessity and exact theorem

Fold formal-specification law is required to derive exact admissible behaviour descriptions without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced exact admissible behavior descriptions kernel retains input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-CORRECTNESS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

66.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for exact admissible behaviour descriptions.

> *All generated finite exact admissible behavior descriptions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A prose intention cannot substitute for the generated specification relation.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

66.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced exact admissible behavior descriptions kernel retains input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

66.4 Operational laws and consequences

- Carrier law: input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

66.5 Depth-independent closure

Base: The structural One supplies one canonical exact admissible behaviour descriptions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact admissible behaviour descriptions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

66.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-SPECIFICATION-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-SPECIFICATION-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-SPECIFICATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A prose intention cannot substitute for the generated specification relation.	`PASS`

66.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: formal specification, refinement.

A prose intention cannot substitute for the generated specification relation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A prose intention cannot substitute for the generated specification relation.

66.8 Exact evidence identities

- Source manifest: `sha256:24c0ef32147af4486b7e3aba1a6d682302e7b69b6eababfe5ad6bc963c463439`
- Complete-census certificate: `sha256:dd53bfb73093fcfae8b8d87d11b94bea6cfab082b3ced8747d0b0fb8b97c18e6`
- Derivation seal: `sha256:29dbb881dca4124a901a109963c2678dc4d45236a921efb02d9432fd8bfbf3be`
- Independent implementation: `sha256:46650c558595b950805ed11f7bcac1b7f42cf1cb50c73235f9e4d9552e49286b`
- Independent certificate: `sha256:4638802ddd94e44f8e1c86e10ff4c8c015435545abfea2e7f1953c0f94ff3fe1`
- External validation: `sha256:cfdcebe686b9010ec4fb4002d01833f1b5a5b415fb651124e54ee97f489fd1a8`
- Model-admitted engine receipt: `sha256:4aeb9f85a830e501877b8cc19d41095d2bbfc96656ff92be8fe5a6655d351a51`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-SPECIFICATION-001-4aeb9f85a830e501.json`

67. Derivation 59: Fold program-transformation law

Claim: `SFT-COMP-SEM-TRANSFORMATION-001`

67.1 Scientific necessity and exact theorem

Fold program-transformation law is required to derive semantics-preserving program rewriting without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced semantics-preserving program rewriting kernel retains source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- SFT-COMP-SEM-SPECIFICATION-001
- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

67.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for semantics-preserving program rewriting.

> *All generated finite semantics-preserving program rewriting structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Optimization claims require separate resource evidence.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

67.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.

Eliminated candidates	Exact first-failure reason
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced semantics-preserving program rewriting kernel retains source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

67.4 Operational laws and consequences

- Carrier law: source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

67.5 Depth-independent closure

Base: The structural One supplies one canonical semantics-preserving program rewriting form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated semantics-preserving program rewriting form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

67.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-TRANSFORMATION-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-TRANSFORMATION-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-TRANSFORMATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Optimisation claims require separate resource evidence.	`PASS`

67.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: program transformation, refactoring.

Optimisation claims require separate resource evidence.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Optimisation claims require separate resource evidence.

67.8 Exact evidence identities

- Source manifest: `sha256:35b8d670b699a9d19bcb41ccb2460225e70490c7b379e7d97efbd245fa73e1a2`
- Complete-census certificate: `sha256:e8875557e249b5e24c2fdfbd042f7e590b54efeb479e14a29992bcece084f827`
- Derivation seal: `sha256:f7339e44653a24a42d3ca997d9b329bf80c25c3bcd78dd51a59d428980f32750`
- Independent implementation: `sha256:08c8b493d1011c559d7f1dc4158e7a0d78afcca76a1375879729e2286d25d92c`
- Independent certificate: `sha256:e3da2d11b82a871c6a2a4f41266025039358dc053ac8c838c32509ac3cda9c44`
- External validation: `sha256:2c73a1c68b6d4c7f142c1c7fb7f003619cd12c9edeef1a0b5854ec15ef0a2994`
- Model-admitted engine receipt: `sha256:cdc9731adba6f15b42e9eeba2e0c93d29bfeadf4ae281fbfed527d54918a7226`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-TRANSFORMATION-001-cdc9731adba6f15b.json`

68. Derivation 60: Fold semantics-preserving compilation law

Claim: `SFT-COMP-SEM-COMPILATION-001`

68.1 Scientific necessity and exact theorem

Fold semantics-preserving compilation law is required to derive translation between executable languages without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced translation between executable languages kernel retains source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger; every operation is*

> *source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-TRANSFORMATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

68.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for translation between executable languages.

> *All generated finite translation between executable languages structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Applies only to the two frozen language grammars and declared runtime interface.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

68.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.

Eliminated candidates	Exact first-failure reason
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced translation between executable languages kernel retains source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

68.4 Operational laws and consequences

- Carrier law: source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

68.5 Depth-independent closure

Base: The structural One supplies one canonical translation between executable languages form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated translation between executable languages form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

68.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-COMPILATION-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-COMPILATION-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-COMPILATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Applies only to the two frozen language grammars and declared runtime interface.	`PASS`

68.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: compiler correctness, compilation.

Applies only to the two frozen language grammars and declared runtime interface.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Applies only to the two frozen language grammars and declared runtime interface.

68.8 Exact evidence identities

- Source manifest: `sha256:ec2ac685a89a050f12b777bf76c8ab1682cbaa48ea97fe6bc5ce418cde0c9a95`
- Complete-census certificate: `sha256:ac9cfbc92b7ea2007942ef317956f5ab79527dcac6069b882950a91e0b3a6873`
- Derivation seal: `sha256:92e1e598f3b553e315bbfe123fe701956b1769053d33261670e6431995c95c7d`
- Independent implementation: `sha256:a8f2d406212e720b0ebb5c0f2d313a35b36a4dcb936a5d9625442bcf838a5cc3`
- Independent certificate: `sha256:8443c6e6dc2296d4e8e972d22a890e06d77964796b7a80f00632bdd47e9d49af`
- External validation: `sha256:30432a42c6a2a3e3bdc49bf3febd42443802c85172c313eb9da1244c13a0c98c`
- Model-admitted engine receipt: `sha256:1d1314650abf2f20bb952ce22c2d83de00e3bb90fa515fb5f25560c3e3bb96ae`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-COMPILATION-001-1d1314650abf2f20.json`

69. Derivation 61: Fold verification and proof-system law

Claim: `SFT-COMP-SEM-VERIFICATION-001`

69.1 Scientific necessity and exact theorem

Fold verification and proof-system law is required to derive machine-checkable program evidence without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced machine-checkable program evidence kernel retains specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-COMPILATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

69.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls.

Generation rule: Generate the literal product of the eight registered Semantics and Mathematical Programming Theory structural axes for machine-checkable program evidence.

All generated finite machine-checkable program evidence structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A passing test suite is not substituted for a proof over the complete declared boundary.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`syntax`	`partial-or-ambiguous-syntax`	partial-or-ambiguous-syntax removes a required exact syntax condition.	`complete-generated-syntax`	complete-generated-syntax retains the complete registered syntax condition.
`binding`	`presentation-name`	presentation-name removes a required exact binding condition.	`canonical-held-binding`	canonical-held-binding retains the complete registered binding condition.
`relation`	`informal-meaning`	informal-meaning removes a required exact relation condition.	`exact-semantic-relation`	exact-semantic-relation retains the complete registered relation condition.
`evaluation`	`result-only`	result-only removes a required exact evaluation condition.	`complete-evaluation-trace`	complete-evaluation-trace retains the complete registered evaluation condition.
`equivalence`	`visual-equivalence`	visual-equivalence removes a required exact equivalence condition.	`observation-bound-equivalence`	observation-bound-equivalence retains the complete registered equivalence condition.
`proof`	`test-only`	test-only removes a required exact proof condition.	`machine-checkable-proof-object`	machine-checkable-proof-object retains the complete registered proof condition.
`generality`	`sample-programs`	sample-programs removes a required exact generality condition.	`syntax-successor-certificate`	syntax-successor-certificate retains the complete registered generality condition.
`addition`	`imported-language-semantics`	imported-language-semantics removes a required exact addition condition.	`no-extra-semantic-premise`	no-extra-semantic-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

69.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	partial-or-ambiguous-syntax removes a required exact syntax condition.

Eliminated candidates	Exact first-failure reason
64	presentation-name removes a required exact binding condition.
32	informal-meaning removes a required exact relation condition.
16	result-only removes a required exact evaluation condition.
8	visual-equivalence removes a required exact equivalence condition.
4	test-only removes a required exact proof condition.
2	sample-programs removes a required exact generality condition.
1	imported-language-semantics removes a required exact addition condition.

Shared claim-record clause `COMP-S010` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced machine-checkable program evidence kernel retains specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

69.4 Operational laws and consequences

- Carrier law: specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

69.5 Depth-independent closure

Base: The structural One supplies one canonical machine-checkable program evidence form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated machine-checkable program evidence form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

69.6 Executed witnesses and adverse controls

- `semantics-witness-1` - semantics exact operational witness 1 for SFT-COMP-SEM-VERIFICATION-001; result: `PASS`.
- `semantics-witness-2` - semantics exact operational witness 2 for SFT-COMP-SEM-VERIFICATION-001; result: `PASS`.
- `semantics-witness-3` - semantics exact operational witness 3 for SFT-COMP-SEM-VERIFICATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	partial-or-ambiguous-syntax removes a required exact syntax condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A passing test suite is not substituted for a proof over the complete declared boundary.	`PASS`

69.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: program verification, proof system.

A passing test suite is not substituted for a proof over the complete declared boundary.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A passing test suite is not substituted for a proof over the complete declared boundary.

69.8 Exact evidence identities

- Source manifest: `sha256:56d7af3b754581d272d0847b611e8d23225a6d352b87c71a975c91011fd74501`
- Complete-census certificate: `sha256:5aba927de4346a6a9fc2d1ba3e7b0de80534383070f8fe6af6ef655861bb43e8`
- Derivation seal: `sha256:ab921149b3cdeea436b7570367e34b839a9a18bc9a4ef156f309283f8a7f543b`
- Independent implementation: `sha256:8321619907caf8c1ded379e15ff860599cc8f92598361ccf5c23a701ae40cf84`
- Independent certificate: `sha256:25deffb52997162e17bcd8ebc3d5aff42667672f858b9bedb610b358d40e8ef9`
- External validation: `sha256:54fcb3e9f8dca7638fa994ba1714cc738cbf8a5c01fabd5a568369008308019d`
- Model-admitted engine receipt: `sha256:e3312b01e5f796d522ced745f449997e595aade4f15b3b82b216840116f09f97`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEM-VERIFICATION-001-e3312b01e5f796d5.json`

70. Derivation 62: Fold concurrency law

Claim: `SFT-COMP-DIST-CONCURRENCY-001`

70.1 Scientific necessity and exact theorem

Fold concurrency law is required to derive multiple enabled process transitions without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced multiple enabled process transitions kernel retains process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEM-VERIFICATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

70.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for multiple enabled process transitions.

All generated finite multiple enabled process transitions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Concurrency is structural coexistence, not physical simultaneity.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

70.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced multiple enabled process transitions kernel retains process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

70.4 Operational laws and consequences

- Carrier law: process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

70.5 Depth-independent closure

Base: The structural One supplies one canonical multiple enabled process transitions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated multiple enabled process transitions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

70.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-CONCURRENCY-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-CONCURRENCY-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-CONCURRENCY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Concurrency is structural coexistence, not physical simultaneity.	`PASS`

70.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: concurrency, interleaving.

Concurrency is structural coexistence, not physical simultaneity.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Concurrency is structural coexistence, not physical simultaneity.

70.8 Exact evidence identities

- Source manifest: `sha256:6fe04cd19f195f77f3e6ea79e0fb949f368c1cea735186d9e4b72790574d9e26`
- Complete-census certificate: `sha256:d5e45de02fd60091d9e6fb785b2c767ad0258bcb3ffce22574bdb9b432ee46f5`
- Derivation seal: `sha256:3b3dce2d1c980ed0f261048b03453cdc320ccf71758f32588aead5dc2ed45f10`
- Independent implementation: `sha256:0413fd7e133b664c0ffb5febc57180d55a7208afe2afcc72d8bdf4014681f761`
- Independent certificate: `sha256:19b88fd27379a91ef6baa0cc1468a60dee4b9c725b5071c1c1ab0e7e9c6185b4`
- External validation: `sha256:c5f5d639dc53cf049646a3a76a5a158d50ab3a6cc14755448c935c7027b7af5a`
- Model-admitted engine receipt: `sha256:a3a2d3ef12a8166cc2117b17c3a470314146123a2c474c73ab363c2d5cf2e1c7`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-CONCURRENCY-001-a3a2d3ef12a8166c.json`

71. Derivation 63: Fold computational-causality law

Claim: `SFT-COMP-DIST-CAUSALITY-001`

71.1 Scientific necessity and exact theorem

Fold computational-causality law is required to derive event predecessor structure without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced event predecessor structure kernel retains generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-CONCURRENCY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

71.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for event predecessor structure.

> *All generated finite event predecessor structure structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Physical causal claims require a later empirical branch.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

71.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.

Eliminated candidates	Exact first-failure reason
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced event predecessor structure kernel retains generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

71.4 Operational laws and consequences

- Carrier law: generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

71.5 Depth-independent closure

Base: The structural One supplies one canonical event predecessor structure form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated event predecessor structure form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

71.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-CAUSALITY-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-CAUSALITY-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-CAUSALITY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Physical causal claims require a later empirical branch.	`PASS`

71.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: causality, happens-before.

Physical causal claims require a later empirical branch.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Physical causal claims require a later empirical branch.

71.8 Exact evidence identities

- Source manifest: `sha256:c04848152f5b1dcbb72a401209d05d24474b76270119aff5660ad1f13329e600`
- Complete-census certificate: `sha256:daa5fab5c8a0adf331ca7256f9f64699532ae48c60a03fb9ca1ec1653f5bd8ca`
- Derivation seal: `sha256:e9f316907a841ba2e2d3f4247665c19e61b4aea748aadf828a48d64bf0e35ed6`
- Independent implementation: `sha256:a48a7a468603ad934d8d4544a55963be61f283842e47b7b7334c6affdf5472ab`
- Independent certificate: `sha256:0d60e035ef2862fdeac2103dd4b7861b2b176d1acec4ffbfc4c6d8715d177925`
- External validation: `sha256:06d3b23bc054557f39397eccbc8efec57a1c3186c2c9d2aa54f5b2c87b6f3e2a`
- Model-admitted engine receipt: `sha256:902e87f46e187a03a813011d318a637eaab7c9fc3dedfc0fc2c59efcb0dffccf`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-CAUSALITY-001-902e87f46e187a03.json`

72. Derivation 64: Fold synchronization law

Claim: `SFT-COMP-DIST-SYNCHRONIZATION-001`

72.1 Scientific necessity and exact theorem

Fold synchronization law is required to derive jointly enabled process transitions without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced jointly enabled process transitions kernel retains participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state; every operation is source-bound, trace-complete,*

boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-CAUSALITY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

72.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for jointly enabled process transitions.

All generated finite jointly enabled process transitions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No timing or lock implementation is assumed.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

72.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.

Eliminated candidates	Exact first-failure reason
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced jointly enabled process transitions kernel retains participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

72.4 Operational laws and consequences

- Carrier law: participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

72.5 Depth-independent closure

Base: The structural One supplies one canonical jointly enabled process transitions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated jointly enabled process transitions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

72.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-SYNCHRONIZATION-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-SYNCHRONIZATION-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-SYNCHRONIZATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No timing or lock implementation is assumed.	`PASS`

72.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: synchronization, rendezvous.

No timing or lock implementation is assumed.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No timing or lock implementation is assumed.

72.8 Exact evidence identities

- Source manifest: `sha256:362b59789c63f76942ffb330df1b93fa6c8ef26cfd51deb8e47e395c5cdcde88`
- Complete-census certificate: `sha256:5e3eeb9ec25477629ac3d801457152c7bbb4c7bb9ed48c1177e725e90ffc7a4c`
- Derivation seal: `sha256:b400e103dedeca45bf97795420f164d6bc863dbd55be87c02e53e4948b9f9ded`
- Independent implementation: `sha256:28138e0387e7e6c371c9a6e9741e41627908bb7f65fdede735ff6f21d1255452`
- Independent certificate: `sha256:a23fe872cadd29b424590e9786b63c1a4924ea73bdce7e5d9b5dfc5b1ebe5602`
- External validation: `sha256:39a1d69fcde2915e256970362287e0796e1fbed21ddb9dbfd561c186892a4e53`
- Model-admitted engine receipt: `sha256:e1b5a01e71410f40f763377dcfc336f2d75bc664db00a510dbba8a075fa5aa0f`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-SYNCHRONIZATION-001-e1b5a01e71410f40.json`

73. Derivation 65: Fold communication law

Claim: `SFT-COMP-DIST-COMMUNICATION-001`

73.1 Scientific necessity and exact theorem

Fold communication law is required to derive exact record transfer without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced exact record transfer kernel retains sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-SYNCHRONIZATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`

- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

73.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for exact record transfer.

> *All generated finite exact record transfer structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Reliability, ordering and delay are separate declared channel properties.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
processes	anonymous-or-partial-processes	anonymous-or-partial-processes removes a required exact processes condition.	complete-held-process-identities	complete-held-process-identities retains the complete registered processes condition.
events	unrecorded-events	unrecorded-events removes a required exact events condition.	complete-local-event-ledgers	complete-local-event-ledgers retains the complete registered events condition.
causality	assumed-global-order	assumed-global-order removes a required exact causality condition.	exact-partial-causal-order	exact-partial-causal-order retains the complete registered causality condition.
communication	implicit-shared-state	implicit-shared-state removes a required exact communication condition.	source-bound-message-records	source-bound-message-records retains the complete registered communication condition.
faults	unspecified-faults	unspecified-faults removes a required exact faults condition.	declared-fault-support	declared-fault-support retains the complete registered faults condition.
knowledge	omniscient-process	omniscient-process removes a required exact knowledge condition.	observation-relative-local-knowledge	observation-relative-local-knowledge retains the complete registered knowledge condition.
generality	one-schedule	one-schedule removes a required exact generality condition.	schedule-successor-certificate	schedule-successor-certificate retains the complete registered generality condition.
addition	hidden-coordinator-or-clock	hidden-coordinator-or-clock removes a required exact addition condition.	no-extra-global-oracle	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

73.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced exact record transfer kernel retains sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

73.4 Operational laws and consequences

- Carrier law: sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

73.5 Depth-independent closure

Base: The structural One supplies one canonical exact record transfer form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact record transfer form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

73.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-COMMUNICATION-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-COMMUNICATION-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-COMMUNICATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Reliability, ordering and delay are separate declared channel properties.	`PASS`

73.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: communication, message passing.

Reliability, ordering and delay are separate declared channel properties.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum

- no free, fitted or learned parameter
- no application result or pretrained model
- Reliability, ordering and delay are separate declared channel properties.

73.8 Exact evidence identities

- Source manifest: `sha256:480c5500bb117e2400a88d8b7f127d73517f6b16d0cf2552d91df99ad87963cf`
- Complete-census certificate: `sha256:ea4f362cc4c5bc46a29be0fb05eaa5a874b7f38e3cc98d4b9e89566f819f6a3c`
- Derivation seal: `sha256:ee57d87e666ad6d34008a82623868fa0d37191217a64b946e92763d500ee9e6d`
- Independent implementation: `sha256:1a1371f4ca85d120e2b44065d766a0c078d3da9a064d413032cccfd29127353e`
- Independent certificate: `sha256:1cdbc1e3b2c636063f22f6c59325a1d8a2a801e8deb31abfde748e92975e6f28`
- External validation: `sha256:a691fbf7e762715853ece6b2497150169d72428cb03c329de72e725f6770be1a`
- Model-admitted engine receipt: `sha256:59defe90f59d3eef926714873a97cc9e2a4d9d14391e5d6d7458f51687b23423`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-COMMUNICATION-001-59defe90f59d3eef.json`

74. Derivation 66: Fold distributed partial-order law

Claim: `SFT-COMP-DIST-PARTIAL-ORDER-001`

74.1 Scientific necessity and exact theorem

Fold distributed partial-order law is required to derive causal event ordering without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced causal event ordering kernel retains event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-COMMUNICATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

74.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for causal event ordering.

> *All generated finite causal event ordering structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A total global order is not assumed where events are incomparable.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

74.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced causal event ordering kernel retains event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

74.4 Operational laws and consequences

- Carrier law: event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.

- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

74.5 Depth-independent closure

Base: The structural One supplies one canonical causal event ordering form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated causal event ordering form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

74.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-PARTIAL-ORDER-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-PARTIAL-ORDER-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-PARTIAL-ORDER-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A total global order is not assumed where events are incomparable.	`PASS`

74.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: partial order, event structure.

A total global order is not assumed where events are incomparable.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A total global order is not assumed where events are incomparable.

74.8 Exact evidence identities

- Source manifest: `sha256:56f3996a18b1f74c4eadbc772358e4ad5e142e16a21603e72d473c813e4c387b`

- Complete-census certificate: `sha256:e1c0c2bff2e7275889803fb9f574184af6a8c24f60e2922a73167f15a652c2cd`
- Derivation seal: `sha256:020e71262605b9bb2fcb82d912aae3c6d68941ae3f8bbf353eeacc051aba70ae`
- Independent implementation: `sha256:07732d983dfb8e2c1da409d19134eaceca8ace59feb78186dc37d4eea5df7580`
- Independent certificate: `sha256:9d32b2ce81cdbe022089e2b49fbcd2e85c900f05c78bb6928992ddf4a0fcf371`
- External validation: `sha256:d8a87eec1566bd3a23a3090815aa1a6dea8afa1268268100d1eefdd0e259da0e`
- Model-admitted engine receipt: `sha256:d04af273dac3a8a8cd3d64b1b7c72e3130c1e674320c6632b0c2b9e2a8225a0f`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-PARTIAL-ORDER-001-d04af273dac3a8a8.json`

75. Derivation 67: Fold consensus law

Claim: `SFT-COMP-DIST-CONSENSUS-001`

75.1 Scientific necessity and exact theorem

Fold consensus law is required to derive terminal common decision without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced terminal common decision kernel retains process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-PARTIAL-ORDER-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

75.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for terminal common decision.

> *All generated finite terminal common decision structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Every consensus claim must name its communication and fault model.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

75.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced terminal common decision kernel retains process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

75.4 Operational laws and consequences

- Carrier law: process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

75.5 Depth-independent closure

Base: The structural One supplies one canonical terminal common decision form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated terminal common decision form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

75.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-CONSENSUS-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-CONSENSUS-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-CONSENSUS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Every consensus claim must name its communication and fault model.	`PASS`

75.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: consensus, agreement.

Every consensus claim must name its communication and fault model.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Every consensus claim must name its communication and fault model.

75.8 Exact evidence identities

- Source manifest: `sha256:4a2c046f87b26d98e327093fd29479ef357aa0b24b055fd677bfac1aeef916ef`
- Complete-census certificate: `sha256:40ef6161a4b5eefd8d0ca925380a1fe9019676bd921170e9a57b16d9f4568424`
- Derivation seal: `sha256:cda13e15608d8e7b1a3f9aeb2859994b54270d7d838f6395ad8a193352f0c4c1`
- Independent implementation: `sha256:03b718ff935db905cfbecf0ddd1b5b7aef28447895c8ccf568ba3c02c8d43974`
- Independent certificate: `sha256:acdf2a8a619c4328635d5faceddde44915461068bd5fc53e010ce49aa41ff37e`

- External validation: `sha256:65fe96a9cb1b6d95a41f2d3bcbb7ef629252f95bf5546ed82d2e8c6582390160`
- Model-admitted engine receipt: `sha256:48ae9b96daea76ffeefb0c3b2b9869b3beea88dc03cfd1b5a90e6f250a4c4d1e`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-CONSENSUS-001-48ae9b96daea76ff.json`

76. Derivation 68: Fold agreement-impossibility law

Claim: `SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001`

76.1 Scientific necessity and exact theorem

Fold agreement-impossibility law is required to derive indistinguishable predecessor limits without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced indistinguishable predecessor limits kernel retains two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-CONSENSUS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

76.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for indistinguishable predecessor limits.

> *All generated finite indistinguishable predecessor limits structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Impossibility is relative to the exact asynchronous/fault grammar and does not transfer silently.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

76.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced indistinguishable predecessor limits kernel retains two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

76.4 Operational laws and consequences

- Carrier law: two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

76.5 Depth-independent closure

Base: The structural One supplies one canonical indistinguishable predecessor limits form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated indistinguishable predecessor limits form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

76.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Impossibility is relative to the exact asynchronous/fault grammar and does not transfer silently.	`PASS`

76.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: consensus impossibility, indistinguishability proof.

Impossibility is relative to the exact asynchronous/fault grammar and does not transfer silently.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Impossibility is relative to the exact asynchronous/fault grammar and does not transfer silently.

76.8 Exact evidence identities

- Source manifest: `sha256:63b144fb308a8aaf282bcbb96e4f355f58e2fea0b1da495cb1eaf4b051a24bed`
- Complete-census certificate: `sha256:f9a936ab5aa948101f178c2acee30fe81e5846c1da59d11e8784801be13add92`
- Derivation seal: `sha256:32e02c7b473ee4b9b2428c7bc89b1bbd83e07310152b5548182ce01d9ce41d3f`
- Independent implementation: `sha256:c4eeffda44abbf12bffefefad3c1ad32e374ec2e35f2ccdcdb232fabb851b421`
- Independent certificate: `sha256:f3b98f1d6b90feff2c25194bc0196b6076eed2df0983bf5d8a02c548c09e87be`
- External validation: `sha256:897f4e69e5287bc3147464d8e712bfc3087f0d8b8a2afbd2cab066599aff2ae4`

- Model-admitted engine receipt: `sha256:30b83eb403fb428502002c7b9d96f569a91195720f36d8e1675385e92919f48d`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001-30b83eb403fb4285.json`

77. Derivation 69: Fold replication and consistency law

Claim: `SFT-COMP-DIST-REPLICATION-CONSISTENCY-001`

77.1 Scientific necessity and exact theorem

Fold replication and consistency law is required to derive multiple retained state copies without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced multiple retained state copies kernel retains replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

77.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for multiple retained state copies.

> *All generated finite multiple retained state copies structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No implementation platform or wall-clock consistency is imported.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

77.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced multiple retained state copies kernel retains replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

77.4 Operational laws and consequences

- Carrier law: replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

77.5 Depth-independent closure

Base: The structural One supplies one canonical multiple retained state copies form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated multiple retained state copies form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

77.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-REPLICATION-CONSISTENCY-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-REPLICATION-CONSISTENCY-001; result: `PASS`.

- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-REPLICATION-CONSISTENCY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No implementation platform or wall-clock consistency is imported.	`PASS`

77.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: replication, consistency model.

No implementation platform or wall-clock consistency is imported.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No implementation platform or wall-clock consistency is imported.

77.8 Exact evidence identities

- Source manifest: `sha256:4801ddc5795d08b9ae79597d99aad0e2fb576dd02dd680e5954d31772fde1a83`
- Complete-census certificate: `sha256:ac7532dc862cb5d6b5d39c336f34ee22594ecb47adb1f9973758120913857a05`
- Derivation seal: `sha256:a72b53b74f5f1dd6ceed05797c12e0ffaf8eebe8751ff5e3169b4a40018038fd`
- Independent implementation: `sha256:cecc16c2c4b38824cda8a93e2a0e383b2e91e070970b18b35d91bc539315d6df`
- Independent certificate: `sha256:a94d015bc15181f18e941e198ccfde240b61e33c29edb0ae477fd31043f04d3b`
- External validation: `sha256:ad685d20578ad08d5ec909eb918daae661024ec45a05519097be78f761b0dd30`
- Model-admitted engine receipt: `sha256:79243bdec0bda3d90c090785f5a2b43444ff4a323b5d7b48f0a36f17c11dcf34`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-REPLICATION-CONSISTENCY-001-79243bdec0bda3d9.json`

78. Derivation 70: Fold computational fault-model law

Claim: `SFT-COMP-DIST-FAULT-MODEL-001`

78.1 Scientific necessity and exact theorem

Fold computational fault-model law is required to derive declared deviations from process rules without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced declared deviations from process rules kernel retains nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-REPLICATION-CONSISTENCY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

78.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for declared deviations from process rules.

> *All generated finite declared deviations from process rules structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Faults not generated by the model remain outside the guarantee.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

78.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced declared deviations from process rules kernel retains nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

78.4 Operational laws and consequences

- Carrier law: nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

78.5 Depth-independent closure

Base: The structural One supplies one canonical declared deviations from process rules form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated declared deviations from process rules form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

78.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-FAULT-MODEL-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-FAULT-MODEL-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-FAULT-MODEL-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	PASS
`tampered_source`	reject a changed official source identity	changed source identity differs	PASS
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	PASS
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Faults not generated by the model remain outside the guarantee.	PASS

78.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: fault model, fault tolerance.

Faults not generated by the model remain outside the guarantee.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Faults not generated by the model remain outside the guarantee.

78.8 Exact evidence identities

- Source manifest: `sha256:aed7d195a63919509b9ba1882ada8b2d58945898ae7e7281aff800ac5fdcc7e0`
- Complete-census certificate: `sha256:3a04bb762e3168e9c601da923ccf65aa71706910bac1a715a689a383c1dd7c1a`
- Derivation seal: `sha256:46ef8935bd002b706c9086252e41700af3efda873ca61ed9695b86d01b8bb10c`
- Independent implementation: `sha256:401a670fcf47677f6133385e134ba15e0c845145663b3a3fe267ee72f0cc8543`
- Independent certificate: `sha256:c56083388b4efc4858b69acdea2242163995edca605a8f532e908538453aa4ea`
- External validation: `sha256:b210256f53a86fd60631fbe0e6c7aa69c1164bd6d5fee88c3221aef797cd49a4`
- Model-admitted engine receipt: `sha256:812aad3547d57947e1e18963470f883725de08c4e299351306e308a1923bdf46`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-FAULT-MODEL-001-812aad3547d57947.json`

79. Derivation 71: Fold distributed-knowledge law

Claim: `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`

79.1 Scientific necessity and exact theorem

Fold distributed-knowledge law is required to derive information available to local processes without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced information available to local processes kernel retains global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-FAULT-MODEL-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

79.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for information available to local processes.

> *All generated finite information available to local processes structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Knowledge is observation-relative exact support, not an imported modal axiom.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

79.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced information available to local processes kernel retains global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

79.4 Operational laws and consequences

- Carrier law: global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

79.5 Depth-independent closure

Base: The structural One supplies one canonical information available to local processes form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated information available to local processes form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

79.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Knowledge is observation-relative exact support, not an imported modal axiom.	`PASS`

79.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: distributed knowledge, epistemic logic.

Knowledge is observation-relative exact support, not an imported modal axiom.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Knowledge is observation-relative exact support, not an imported modal axiom.

79.8 Exact evidence identities

- Source manifest: `sha256:d2ca5491f24d445df5d48630109f35c3d1e739ee4a899acfeacc4973f8b991f5`
- Complete-census certificate: `sha256:ee21a86f5706fbc5c100d2a5213995f180a31a18c5f39ba812086b9de0f3447e`
- Derivation seal: `sha256:dd0eef6ba8eba0f639d57fba8dc5e048b90dc3c83b4f4074b36d3de1fbc91254`
- Independent implementation: `sha256:65d80ba430b121e33988d04246aa04fcf58ee245cff28f35e0b2b095eeea4738`
- Independent certificate: `sha256:1c6d8edf0281e7a3cbfe30321cf9df2b4f87d8839ec1c901568d918f5011e8fc`
- External validation: `sha256:b5b38e7b5d6013a66581be1b1db819b1850351d585312e7b1d509786ca815e3c`
- Model-admitted engine receipt: `sha256:2e47ee8465ef950f9ad2d30eb331453e764abf2567a6fc83ed0ec99b9fceda41`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001-2e47ee8465ef950f.json`

80. Derivation 72: Fold locality law

Claim: `SFT-COMP-DIST-LOCALITY-001`

80.1 Scientific necessity and exact theorem

Fold locality law is required to derive bounded dependency neighborhoods without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced bounded dependency neighborhoods kernel retains network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

80.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for bounded dependency neighborhoods.

> *All generated finite bounded dependency neighborhoods structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Locality is relative to the registered network and round relation.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

80.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced bounded dependency neighborhoods kernel retains network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

80.4 Operational laws and consequences

- Carrier law: network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

80.5 Depth-independent closure

Base: The structural One supplies one canonical bounded dependency neighborhoods form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated bounded dependency neighborhoods form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

80.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-LOCALITY-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-LOCALITY-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-LOCALITY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Locality is relative to the registered network and round relation.	`PASS`

80.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: local computation, distributed locality.

Locality is relative to the registered network and round relation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Locality is relative to the registered network and round relation.

80.8 Exact evidence identities

- Source manifest: `sha256:b6ffc79d6147f815689d6f7de439921326a5c66486d30f4c4cf1810403dc681d`
- Complete-census certificate: `sha256:1af96f4623e9adf8de58fc51a08eb1f5fbf438ce2311b18767e099810f072429`
- Derivation seal: `sha256:52733a46f531bfb6cf359e04411633d64b8daa8d79e70c6bcbc2e114d7e98c69`
- Independent implementation: `sha256:f480454682af4bc3c9fcd5ff8cf7586bd356b42f0b30c571d68b11d485510478`
- Independent certificate: `sha256:7b519f5972fa40355453ff053370f91fa7cfe1e4c57bf309b3d21b93b0877892`
- External validation: `sha256:e049568757b31066d538d4633f44dc5ea8ec7822f55180090561079a7b29c8e8`
- Model-admitted engine receipt: `sha256:d75dcc0dc6937266743f829559bfe4b59dfe31fc0d2b186ccbd7979b13cba9fa`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-LOCALITY-001-d75dcc0dc6937266.json`

81. Derivation 73: Fold network-computation law

Claim: `SFT-COMP-DIST-NETWORK-COMPUTATION-001`

81.1 Scientific necessity and exact theorem

Fold network-computation law is required to derive computation over generated communication graphs without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced computation over generated communication graphs kernel retains network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-LOCALITY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

81.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers.

Generation rule: Generate the literal product of the eight registered Concurrent and Distributed Computation structural axes for computation over generated communication graphs.

All generated finite computation over generated communication graphs structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Operating systems, cloud platforms and physical networks are validation domains only.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`processes`	`anonymous-or-partial-processes`	anonymous-or-partial-processes removes a required exact processes condition.	`complete-held-process-identities`	complete-held-process-identities retains the complete registered processes condition.
`events`	`unrecorded-events`	unrecorded-events removes a required exact events condition.	`complete-local-event-ledgers`	complete-local-event-ledgers retains the complete registered events condition.
`causality`	`assumed-global-order`	assumed-global-order removes a required exact causality condition.	`exact-partial-causal-order`	exact-partial-causal-order retains the complete registered causality condition.
`communication`	`implicit-shared-state`	implicit-shared-state removes a required exact communication condition.	`source-bound-message-records`	source-bound-message-records retains the complete registered communication condition.
`faults`	`unspecified-faults`	unspecified-faults removes a required exact faults condition.	`declared-fault-support`	declared-fault-support retains the complete registered faults condition.
`knowledge`	`omniscient-process`	omniscient-process removes a required exact knowledge condition.	`observation-relative-local-knowledge`	observation-relative-local-knowledge retains the complete registered knowledge condition.
`generality`	`one-schedule`	one-schedule removes a required exact generality condition.	`schedule-successor-certificate`	schedule-successor-certificate retains the complete registered generality condition.
`addition`	`hidden-coordinator-or-clock`	hidden-coordinator-or-clock removes a required exact addition condition.	`no-extra-global-oracle`	no-extra-global-oracle retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

81.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	anonymous-or-partial-processes removes a required exact processes condition.

Eliminated candidates	Exact first-failure reason
64	unrecorded-events removes a required exact events condition.
32	assumed-global-order removes a required exact causality condition.
16	implicit-shared-state removes a required exact communication condition.
8	unspecified-faults removes a required exact faults condition.
4	omniscient-process removes a required exact knowledge condition.
2	one-schedule removes a required exact generality condition.
1	hidden-coordinator-or-clock removes a required exact addition condition.

Shared claim-record clause `COMP-S011` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced computation over generated communication graphs kernel retains network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

81.4 Operational laws and consequences

- Carrier law: network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

81.5 Depth-independent closure

Base: The structural One supplies one canonical computation over generated communication graphs form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated computation over generated communication graphs form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

81.6 Executed witnesses and adverse controls

- `distributed_computation-witness-1` - distributed_computation exact operational witness 1 for SFT-COMP-DIST-NETWORK-COMPUTATION-001; result: `PASS`.
- `distributed_computation-witness-2` - distributed_computation exact operational witness 2 for SFT-COMP-DIST-NETWORK-COMPUTATION-001; result: `PASS`.
- `distributed_computation-witness-3` - distributed_computation exact operational witness 3 for SFT-COMP-DIST-NETWORK-COMPUTATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	anonymous-or-partial-processes removes a required exact processes condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Operating systems, cloud platforms and physical networks are validation domains only.	`PASS`

81.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: network algorithm, distributed network.

Operating systems, cloud platforms and physical networks are validation domains only.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Operating systems, cloud platforms and physical networks are validation domains only.

81.8 Exact evidence identities

- Source manifest: `sha256:bde1392dce05056d4f751038f72b017f66a8616275d580e9c0b652193d5547d6`
- Complete-census certificate: `sha256:07b3b59bf2f17b9085383298005d3a0ff8a1030c811ceca62683ab91b00e5027`
- Derivation seal: `sha256:e5de17ce510c5dee17b619932baeff26f47fb39a45be46d95529ea1148b9cba6`
- Independent implementation: `sha256:40e6bd0f4a4cddad95e591f8e345388b6093afba2cdfd10fcd854215af618fce`
- Independent certificate: `sha256:f96f76e03ecacbfa6075d5fb3e6f25f8a0d814c7c994e2e0009644bf52d2e121`
- External validation: `sha256:0ff1e2f18949b5be3afa6a81b5dfb16479d840183d39010f58363b7a5433ae6c`
- Model-admitted engine receipt: `sha256:504861c17b0bc318c7dc6cfbfa43377ac42819e316735ab1b066b4dd11bb1951`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-DIST-NETWORK-COMPUTATION-001-504861c17b0bc318.json`

82. Derivation 74: Fold one-wayness law

Claim: `SFT-COMP-SEC-ONE-WAYNESS-001`

82.1 Scientific necessity and exact theorem

Fold one-wayness law is required to derive forward computation versus inverse search without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced forward computation versus inverse search kernel retains finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-DIST-NETWORK-COMPUTATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

82.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for forward computation versus inverse search.

> *All generated finite forward computation versus inverse search structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Unbounded finite inversion is not impossible; one-wayness is always resource-relative.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

82.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced forward computation versus inverse search kernel retains finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

82.4 Operational laws and consequences

- Carrier law: finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

82.5 Depth-independent closure

Base: The structural One supplies one canonical forward computation versus inverse search form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated forward computation versus inverse search form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

82.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-ONE-WAYNESS-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-ONE-WAYNESS-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-ONE-WAYNESS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Unbounded finite inversion is not impossible; one-wayness is always resource-relative.	`PASS`

82.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: one-way function, inversion hardness.

Unbounded finite inversion is not impossible; one-wayness is always resource-relative.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Unbounded finite inversion is not impossible; one-wayness is always resource-relative.

82.8 Exact evidence identities

- Source manifest: `sha256:6dd95b0fb4b12cd256da8c4cd50b0bf3e5a72be91dfc7c67216cfb4a09affb8a`
- Complete-census certificate: `sha256:43fb5c990476119271dd46fea841ed62af9eba3ed1b953da0cc908d649d04618`
- Derivation seal: `sha256:ddfec37a4c433a5afac7108f7c9efb94d851d621edc8c48b728cbcac61d14e40`
- Independent implementation: `sha256:17638c5f2b4650aa4143670864dcecd5cb1ff24031a8b29ddc170fb24655957b`
- Independent certificate: `sha256:e6bbfdf570e26533617e51b323bcb81bd4ba9cf746b583b62b02998fe61e35cd`
- External validation: `sha256:72270df40fff869ade128e4de351c353a0ad9fcac4088f77270223812c1bf960`
- Model-admitted engine receipt: `sha256:a4de8e0ec3e624835896831a4545adc63b5aa37cb7b2cecb26404bda0175eb62`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-ONE-WAYNESS-001-a4de8e0ec3e62483.json`

83. Derivation 75: Fold secrecy law

Claim: `SFT-COMP-SEC-SECRECY-001`

83.1 Scientific necessity and exact theorem

Fold secrecy law is required to derive observation-relative message indistinguishability without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced observation-relative message indistinguishability kernel retains message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-ONE-WAYNESS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

83.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for observation-relative message indistinguishability.

> *All generated finite observation-relative message indistinguishability structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Security requires exact adversary, resource and support definitions.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

83.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced observation-relative message indistinguishability kernel retains message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

83.4 Operational laws and consequences

- Carrier law: message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

83.5 Depth-independent closure

Base: The structural One supplies one canonical observation-relative message indistinguishability form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated observation-relative message indistinguishability form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

83.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-SECRECY-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-SECRECY-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-SECRECY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Security requires exact adversary, resource and support definitions.	`PASS`

83.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: secrecy, confidentiality.

Security requires exact adversary, resource and support definitions.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Security requires exact adversary, resource and support definitions.

83.8 Exact evidence identities

- Source manifest: `sha256:3ca8513b002c13e4a5f82ae192cdee76c0b56faf2a8190d0f762c5ab0b64d7f3`
- Complete-census certificate: `sha256:2d1544ac0c7261dedc525c5954e448b785f6c1c7f01785d091ce411d425d5be4`
- Derivation seal: `sha256:59ae9300995bd5a4e1ed134ea927bca7428ef8dcad953c194d96c85b25ec9aba`
- Independent implementation: `sha256:0be3b5dfc3878312f12f0414d1c476063532fd88932fe3be91f8b421f3fafdde`
- Independent certificate: `sha256:930976772bdb29fedd4075e465f45c4d9d4b626b8976852a9766ed754b60887e`
- External validation: `sha256:d5508adf2fe5b04b4e7189e2ca1a773a785d6cb98bffe668fdb6bea9758950c9`
- Model-admitted engine receipt: `sha256:73a1685ef3e59a0b4cf8b8b67a1590f8a3b94e1dd5ace2d7eff0a6bdd5183b74`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-SECRECY-001-73a1685ef3e59a0b.json`

84. Derivation 76: Fold integrity law

Claim: `SFT-COMP-SEC-INTEGRITY-001`

84.1 Scientific necessity and exact theorem

Fold integrity law is required to derive detection of unauthorized record change without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced detection of unauthorized record change kernel retains message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-SECRECY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

84.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for detection of unauthorized record change.

> *All generated finite detection of unauthorized record change structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Integrity does not imply secrecy or authorship.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

84.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced detection of unauthorized record change kernel retains message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

84.4 Operational laws and consequences

- Carrier law: message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

84.5 Depth-independent closure

Base: The structural One supplies one canonical detection of unauthorized record change form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated detection of unauthorized record change form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

84.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-INTEGRITY-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-INTEGRITY-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-INTEGRITY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Integrity does not imply secrecy or authorship.	`PASS`

84.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: integrity, tamper detection.

Integrity does not imply secrecy or authorship.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Integrity does not imply secrecy or authorship.

84.8 Exact evidence identities

- Source manifest: `sha256:2aeffa26e51414ad5f858b6994cd5efb7eed6ab1ecb17e9f4f4fa6d594c5171b`
- Complete-census certificate: `sha256:2559677b0e6803955ab30583cf2e86884658249ad059c01b2cc74d85420073c3`
- Derivation seal: `sha256:986a5cc25662f593e3440f93aa672ba5b6464b76a10caceee4f535d930938358`
- Independent implementation: `sha256:e50a46ec67178635ac4d6df9cde7330745719c7302ad7a655378325da782a6cb`
- Independent certificate: `sha256:3c39a2ab4febb4ae372aef6e5ab87cb65c2d9591e3bd22c49c7e123e49d63b76`
- External validation: `sha256:d85d50f86707d889c10663fa0bcdc985d2ad7fefafc363d25289c143515903b7`
- Model-admitted engine receipt: `sha256:54e959203614f553ff7ac956f549af5e057a240855b4175f237fb7363e080943`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-INTEGRITY-001-54e959203614f553.json`

85. Derivation 77: Fold authentication law

Claim: `SFT-COMP-SEC-AUTHENTICATION-001`

85.1 Scientific necessity and exact theorem

Fold authentication law is required to derive source-identity verification without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced source-identity verification kernel retains principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary; every operation is source-bound, trace-complete,*

> *boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-INTEGRITY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

85.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for source-identity verification.

> *All generated finite source-identity verification structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Identity claims are only as strong as the registered credential and adversary model.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

85.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.

Eliminated candidates	Exact first-failure reason
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced source-identity verification kernel retains principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

85.4 Operational laws and consequences

- Carrier law: principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

85.5 Depth-independent closure

Base: The structural One supplies one canonical source-identity verification form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated source-identity verification form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

85.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-AUTHENTICATION-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-AUTHENTICATION-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-AUTHENTICATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Identity claims are only as strong as the registered credential and adversary model.	`PASS`

85.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: authentication, challenge-response.

Identity claims are only as strong as the registered credential and adversary model.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Identity claims are only as strong as the registered credential and adversary model.

85.8 Exact evidence identities

- Source manifest: `sha256:0732acf36566e352c6679fee9ac8fe34183a7bcda30e89be6796be5fd15da9cf`
- Complete-census certificate: `sha256:5b6775e9c64e49a395b06f8d2261fa33fd08bc087f620e9877b112909970f3cd`
- Derivation seal: `sha256:19ab81842953854a3cb2d3354bb9670af318dac19fdbeaec14a93aeab64bfb82`
- Independent implementation: `sha256:8c06c672907c71606cb536667777f63d84e0add9a28cdaaf7d80c7e9206b3efb`
- Independent certificate: `sha256:30945f38ecef4be19638d405dbe7af6020ed8663aa260f2012d54eee4e231fe5`
- External validation: `sha256:cf6bfc6b8ab765bdc9f84ee432e122e03ea34764d788da1e97efcd71c6bfe5c1`
- Model-admitted engine receipt: `sha256:2c39303d9b2dd557650fe04f8acb2f132510ca3af8258dc10792a276796cb29c`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-AUTHENTICATION-001-2c39303d9b2dd557.json`

86. Derivation 78: Fold hashing law

Claim: `SFT-COMP-SEC-HASHING-001`

86.1 Scientific necessity and exact theorem

Fold hashing law is required to derive fixed-interface many-to-one record mapping without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced fixed-interface many-to-one record mapping kernel retains message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-AUTHENTICATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

86.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for fixed-interface many-to-one record mapping.

> *All generated finite fixed-interface many-to-one record mapping structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Collision resistance is resource-relative; collisions in finite larger domains are retained, not denied.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

86.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced fixed-interface many-to-one record mapping kernel retains message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

86.4 Operational laws and consequences

- Carrier law: message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

86.5 Depth-independent closure

Base: The structural One supplies one canonical fixed-interface many-to-one record mapping form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated fixed-interface many-to-one record mapping form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

86.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-HASHING-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-HASHING-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-HASHING-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Collision resistance is resource-relative; collisions in finite larger domains are retained, not denied.	`PASS`

86.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: hash function, collision resistance.

Collision resistance is resource-relative; collisions in finite larger domains are retained, not denied.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Collision resistance is resource-relative; collisions in finite larger domains are retained, not denied.

86.8 Exact evidence identities

- Source manifest: `sha256:3813266588daf1bad1e5cfb7490c5eeb712a7db71bdb3bfdea7e31f8b8f3a246`
- Complete-census certificate: `sha256:2e1ce8ae91707798d1b54be7dcec8e6464f43c4802f7f8b293f0e643b0e32f5b`
- Derivation seal: `sha256:01617a3c0bf0134dc0c4fa284d583d22153e56db32bc9ba49acc70880b1282d9`
- Independent implementation: `sha256:fd20652f052b8a1d5d18b9d12fdb3ee66fd12331f1075360660cff6aa0b45e23`
- Independent certificate: `sha256:3e486f817fe2b3d7bdee5c000d0aac1c12fc77145d7d9e5a0ee989dc62551e4e`
- External validation: `sha256:0436e07b628c57057b9ba0f7cb4a2a0051a444fd9275d02bbf0243ef7daee657`
- Model-admitted engine receipt: `sha256:30adbf79b53883ef1c3266d37eabc5389dc245bd879ff5be7c3c81df62767984`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-HASHING-001-30adbf79b53883ef.json`

87. Derivation 79: Fold commitment law

Claim: `SFT-COMP-SEC-COMMITMENT-001`

87.1 Scientific necessity and exact theorem

Fold commitment law is required to derive binding and hiding record phases without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced binding and hiding record phases kernel retains message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-HASHING-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

87.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for binding and hiding record phases.

> *All generated finite binding and hiding record phases structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Perfect hiding and perfect binding claims require compatible exact supports and may be mutually limited.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

87.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced binding and hiding record phases kernel retains message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

87.4 Operational laws and consequences

- Carrier law: message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

87.5 Depth-independent closure

Base: The structural One supplies one canonical binding and hiding record phases form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated binding and hiding record phases form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

87.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-COMMITMENT-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-COMMITMENT-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-COMMITMENT-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Perfect hiding and perfect binding claims require compatible exact supports and may be mutually limited.	`PASS`

87.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: commitment scheme, binding and hiding.

Perfect hiding and perfect binding claims require compatible exact supports and may be mutually limited.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- Perfect hiding and perfect binding claims require compatible exact supports and may be mutually limited.

87.8 Exact evidence identities

- Source manifest: `sha256:275b5aa4fbf532130c43d18e07f458a89f5a9125d26371830ea59649ff25696b`
- Complete-census certificate: `sha256:7c012f93d8e4a33696460bc6443c46923f978d2d539c74ee59bd3c44224f30e3`
- Derivation seal: `sha256:8d9f534c3964bff3c525e06a7a83a4dc6be41b3ec1ee7c57bcd60df2c6158686`
- Independent implementation: `sha256:784fc2f024c0611721529092f94ec3afc5d735b7d812c1d0b1805b7a2e27e078`
- Independent certificate: `sha256:7d125b12815107bbf048d8978fe647c04a510fe9c3f27c68e99cd5818c74015f`
- External validation: `sha256:33264f3ce62b46b936a0731aa0e0de8a6df818298318d342ac47e10b4816bd54`
- Model-admitted engine receipt: `sha256:402789a36ddf90549527f52c193fbb0d0065ed548efaeb8178b629a367c59e5f`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-COMMITMENT-001-402789a36ddf9054.json`

88. Derivation 80: Fold signature law

Claim: `SFT-COMP-SEC-SIGNATURE-001`

88.1 Scientific necessity and exact theorem

Fold signature law is required to derive publicly checkable source authorisation without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced publicly checkable source authorization kernel retains message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-COMMITMENT-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

88.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for publicly checkable source authorisation.

> *All generated finite publicly checkable source authorization structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No number-theoretic hardness assumption is imported; any computational claim states its resource boundary.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

88.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced publicly checkable source authorization kernel retains message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

88.4 Operational laws and consequences

- Carrier law: message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.

- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

88.5 Depth-independent closure

Base: The structural One supplies one canonical publicly checkable source authorisation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated publicly checkable source authorisation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

88.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-SIGNATURE-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-SIGNATURE-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-SIGNATURE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No number-theoretic hardness assumption is imported; any computational claim states its resource boundary.	`PASS`

88.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: digital signature, unforgeability.

No number-theoretic hardness assumption is imported; any computational claim states its resource boundary.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No number-theoretic hardness assumption is imported; any computational claim states its resource boundary.

88.8 Exact evidence identities

- Source manifest: `sha256:a051961ecca2e1bd80888f5d7d856ba23b3fb70a8a9848bdeb59b764f76771ca`
- Complete-census certificate: `sha256:8356a945a268513005817384c3da5f46b86ff408f57011cef2297a2e8dd0eb86`
- Derivation seal: `sha256:f6c665845c4cb731e7abc6523af0082873c08a343c7e4c7974205bc696fd63a9`
- Independent implementation: `sha256:9e1a001d6e4ab6299ffc425aaf2d7c3912bb1545278bc11e70e34efe3c45b0ef`
- Independent certificate: `sha256:ac4c20be5d8a6615688ca4fcd27541f4b7a0358fd7751b355dbf4822ffabce62`
- External validation: `sha256:bdd57b0d7eb4417ceebfbd4c9b257631e20f1f7bebf266b2a4d1be764c09dfb7`
- Model-admitted engine receipt: `sha256:1e80fcc3c81615ba2f131e83a91759abecf9a02f7a23b0a19b191a612d820bce`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-SIGNATURE-001-1e80fcc3c81615ba.json`

89. Derivation 81: Fold proof-of-knowledge law

Claim: `SFT-COMP-SEC-PROOF-KNOWLEDGE-001`

89.1 Scientific necessity and exact theorem

Fold proof-of-knowledge law is required to derive extractable witness-bearing interaction without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced extractable witness-bearing interaction kernel retains statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-SIGNATURE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

89.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for extractable witness-bearing interaction.

All generated finite extractable witness-bearing interaction structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Knowledge means existence of an exact extraction relation in the registered grammar.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

89.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

The forced extractable witness-bearing interaction kernel retains statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

89.4 Operational laws and consequences

- Carrier law: statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

89.5 Depth-independent closure

Base: The structural One supplies one canonical extractable witness-bearing interaction form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated extractable witness-bearing interaction form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

89.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-PROOF-KNOWLEDGE-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-PROOF-KNOWLEDGE-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-PROOF-KNOWLEDGE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Knowledge means existence of an exact extraction relation in the registered grammar.	`PASS`

89.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: proof of knowledge, witness extraction.

Knowledge means existence of an exact extraction relation in the registered grammar.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Knowledge means existence of an exact extraction relation in the registered grammar.

89.8 Exact evidence identities

- Source manifest: `sha256:d6129fbbe4e42e045f47e98d50fb7c69bc2b466dcb694ace1e513e217ba28b99`
- Complete-census certificate: `sha256:47f0d1f03b6c7d4ad9bcd6e3014e8bed0a2f28465bce0f86e9495b71a72cc451`

- Derivation seal: `sha256:096e1be819c3cbbc16da33d8aee4a9ec53e1f5d88153ea987802509c2c2bed13`
- Independent implementation: `sha256:f5004305f90756de4a02683a31cabda06491ba72784c1ce3036f03e625ea3523`
- Independent certificate: `sha256:240f016841b14142ad7671c8a07814cdaa2920e5aaf78f57fb3a092a895b5f68`
- External validation: `sha256:6651ed591513d864d58c47fc68f4c0a5b16042cc0776045deb5e71b793ed5227`
- Model-admitted engine receipt: `sha256:30375e18292a970fd81f642586376049976d61928fff9bff79bb0ee0692f0eb6`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-PROOF-KNOWLEDGE-001-30375e18292a970f.json`

90. Derivation 82: Fold zero-knowledge law

Claim: `SFT-COMP-SEC-ZERO-KNOWLEDGE-001`

90.1 Scientific necessity and exact theorem

Fold zero-knowledge law is required to derive verification without additional witness distinction without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced verification without additional witness distinction kernel retains statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-PROOF-KNOWLEDGE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

90.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for verification without additional witness distinction.

> *All generated finite verification without additional witness distinction structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Zero knowledge is relative to the registered verifier observation and simulator support.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

90.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced verification without additional witness distinction kernel retains statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

90.4 Operational laws and consequences

- Carrier law: statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

90.5 Depth-independent closure

Base: The structural One supplies one canonical verification without additional witness distinction form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated verification without additional witness distinction form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

90.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-ZERO-KNOWLEDGE-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-ZERO-KNOWLEDGE-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-ZERO-KNOWLEDGE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Zero knowledge is relative to the registered verifier observation and simulator support.	`PASS`

90.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: zero-knowledge proof, simulation.

Zero knowledge is relative to the registered verifier observation and simulator support.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Zero knowledge is relative to the registered verifier observation and simulator support.

90.8 Exact evidence identities

- Source manifest: `sha256:620bbad2881acb7c41f3224a48f16c5ae2c8aa8fde8fd162e9bb90f0268debc4`
- Complete-census certificate: `sha256:bf2e1d87dcd07838f14b6cec85d7487f0592f4bdd78710ae736cd2b8853b0274`
- Derivation seal: `sha256:5be0fa434efed7312395f86a9afb749199774bb3b538534566c3b6a783f37f8c`
- Independent implementation: `sha256:24b62399d900fdd7fe64843cd05cf1e6f3d765489dcecc3f0941281fa57dffb9`
- Independent certificate: `sha256:a730450b5720e8b9b620c379e228ea634d9317646f7b52e43cbaf4f97fc7720f`

- External validation: `sha256:c46233d93c746e6cf701fd5f16b696cafc5f0f2188fd8307c103102101cfd87b`
- Model-admitted engine receipt: `sha256:060ee3e0322b39395749298b995a9f4d67c148ade8eeba389c05906219d720ed`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-ZERO-KNOWLEDGE-001-060ee3e0322b3939.json`

91. Derivation 83: Fold multiparty-computation law

Claim: `SFT-COMP-SEC-MULTIPARTY-001`

91.1 Scientific necessity and exact theorem

Fold multiparty-computation law is required to derive joint function evaluation with local privacy without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced joint function evaluation with local privacy kernel retains party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-ZERO-KNOWLEDGE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

91.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for joint function evaluation with local privacy.

> *All generated finite joint function evaluation with local privacy structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Security requires an explicit corruption and communication model.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

91.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced joint function evaluation with local privacy kernel retains party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

91.4 Operational laws and consequences

- Carrier law: party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

91.5 Depth-independent closure

Base: The structural One supplies one canonical joint function evaluation with local privacy form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated joint function evaluation with local privacy form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

91.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-MULTIPARTY-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-MULTIPARTY-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-MULTIPARTY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Security requires an explicit corruption and communication model.	`PASS`

91.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: secure multiparty computation, MPC.

Security requires an explicit corruption and communication model.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Security requires an explicit corruption and communication model.

91.8 Exact evidence identities

- Source manifest: `sha256:f99a5d1d82db730ede50111e4923fbdf63981e58c2187b81d9e5e7a3a0398dc4`
- Complete-census certificate: `sha256:a2e3bc1c437dcf70d4a3becf015e03c0f5bd17d3efa7803076c8f1d2178fdd0b`
- Derivation seal: `sha256:e439ad17a32ce5c851815d3b14ab76da6da98235dfef93ce7e5a634a6a21d85e`
- Independent implementation: `sha256:540ddd1dc71c191068a84d648aecaa8c87dca286b22f8eba3a22a2525d5c5bc4`
- Independent certificate: `sha256:d2d42553baf0939f1eb080df51ff421d84598397699c3fe6debdfd601ddcc1b5`
- External validation: `sha256:1914804774000141c84f200cb7b301bd9c7c524f9c2825cda44fb861df601e2d`

- Model-admitted engine receipt:
  `sha256:211b27cbbbc88f0e82c4441ed0d47afe01703b5dd89892a3e35759025c5f29eb`
- Receipt path:
  `receipts/engine/model_admitted/SFT-COMP-SEC-MULTIPARTY-001-211b27cbbbc88f0e.json`

92. Derivation 84: Fold adversarial-computation law

Claim: `SFT-COMP-SEC-ADVERSARIAL-001`

92.1 Scientific necessity and exact theorem

Fold adversarial-computation law is required to derive computation against generated hostile strategies without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced computation against generated hostile strategies kernel retains system process; adversary grammar; resources; interaction schedules; security property; complete favorable and unfavorable rows; stopping rule; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-MULTIPARTY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

92.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: system process; adversary grammar; resources; interaction schedules; security property; complete favourable and unfavourable rows; stopping rule.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for computation against generated hostile strategies.

> *All generated finite computation against generated hostile strategies structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. An undefined adversary cannot support a security theorem.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

92.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced computation against generated hostile strategies kernel retains system process; adversary grammar; resources; interaction schedules; security property; complete favorable and unfavorable rows; stopping rule; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

92.4 Operational laws and consequences

- Carrier law: system process; adversary grammar; resources; interaction schedules; security property; complete favourable and unfavourable rows; stopping rule.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

92.5 Depth-independent closure

Base: The structural One supplies one canonical computation against generated hostile strategies form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated computation against generated hostile strategies form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

92.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-ADVERSARIAL-001; result: `PASS`.

- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-ADVERSARIAL-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-ADVERSARIAL-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; An undefined adversary cannot support a security theorem.	`PASS`

92.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: adversary model, security game.

An undefined adversary cannot support a security theorem.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- An undefined adversary cannot support a security theorem.

92.8 Exact evidence identities

- Source manifest: `sha256:be1520c84305b1e364a98fef71e3a03a2b35b42e3131c3722cfe77c1c24208b6`
- Complete-census certificate: `sha256:ff38ea69df3de315e9a0da0ac142dfec4255ba3459637f6ea73eaffb7c850e73`
- Derivation seal: `sha256:131a0d4c23230de9c1973b0660172a74c795f138add9a30cea8056abaa5f377f`
- Independent implementation: `sha256:667c35f6442a6883878e24e51c08894ee8f86106c7937bb824eca4df7a94d595`
- Independent certificate: `sha256:7b5cda64ba1d06a7f27e9723db24054715d8763d5e9b25d6a5087b88cbb06fe4`
- External validation: `sha256:bc5047e436959279dfd58647b6a20fae0e479e20fd3f6b94507402aacd616d4a`
- Model-admitted engine receipt: `sha256:feaa06a2fbc1430f8930075dada167efdfcb60df06ed5e21424557ce89b81b78`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-ADVERSARIAL-001-feaa06a2fbc1430f.json`

93. Derivation 85: Fold information-theoretic and computational-security law

Claim: `SFT-COMP-SEC-SECURITY-DEFINITION-001`

93.1 Scientific necessity and exact theorem

Fold information-theoretic and computational-security law is required to derive exact separation of unconditional and resource-bounded security without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced exact separation of unconditional and resource-bounded security kernel retains secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-ADVERSARIAL-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

93.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for exact separation of unconditional and resource-bounded security.

> *All generated finite exact separation of unconditional and resource-bounded security structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Information-theoretic and computational claims must never be conflated.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

93.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced exact separation of unconditional and resource-bounded security kernel retains secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

93.4 Operational laws and consequences

- Carrier law: secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

93.5 Depth-independent closure

Base: The structural One supplies one canonical exact separation of unconditional and resource-bounded security form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact separation of unconditional and resource-bounded security form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

93.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-SECURITY-DEFINITION-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-SECURITY-DEFINITION-001; result: `PASS`.

- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-SECURITY-DEFINITION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Information-theoretic and computational claims must never be conflated.	`PASS`

93.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: information-theoretic security, computational security.

Information-theoretic and computational claims must never be conflated.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Information-theoretic and computational claims must never be conflated.

93.8 Exact evidence identities

- Source manifest: `sha256:41a6b4267c2f2ae8babb09d61629ef19a5f4aad0cc3b512c5655628ea2e4d003`
- Complete-census certificate: `sha256:92b6f656b84f07e9eb480e65d3a87489c1916aca3bcfe79e44278a2cf78e846b`
- Derivation seal: `sha256:6b21cf6168cb2b3e44e495511de413b364806a7f4f525932ee7469ac1ba28d2a`
- Independent implementation: `sha256:c7d7f2d2ad3488af844772f361a82d489dd4775d704d8ac1f78a139c8736d496`
- Independent certificate: `sha256:fb830648518dacbf88f79aec34306fa7f19a7791d9a649d6e97483bfb7658a22`
- External validation: `sha256:8a7e1c759e576b62fa7c4db35de79ac8630f90be53a4d7b9d7faaf7ddf7af458`
- Model-admitted engine receipt: `sha256:e4af2a902c19997a8a81d6df7fe8ebd8246a2900aac642a6003b287d4bde7007`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-SECURITY-DEFINITION-001-e4af2a902c19997a.json`

94. Derivation 86: Fold post-quantum security-boundary law

Claim: `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`

94.1 Scientific necessity and exact theorem

Fold post-quantum security-boundary law is required to derive classical claims under an enlarged adversary interface without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced classical claims under an enlarged adversary interface kernel retains classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-SECURITY-DEFINITION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

94.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence.

Generation rule: Generate the literal product of the eight registered Cryptography and Computational Security structural axes for classical claims under an enlarged adversary interface.

> *All generated finite classical claims under an enlarged adversary interface structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No quantum operation is derived here; operational quantum adversaries wait for the Quantum Computation branch.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-messages-or-keys`	sampled-messages-or-keys removes a required exact support condition.	`complete-generated-challenge-support`	complete-generated-challenge-support retains the complete registered support condition.
`adversary`	`undefined-adversary`	undefined-adversary removes a required exact adversary condition.	`exact-adversary-process-grammar`	exact-adversary-process-grammar retains the complete registered adversary condition.
`interaction`	`outcome-only`	outcome-only removes a required exact interaction condition.	`complete-interaction-trace`	complete-interaction-trace retains the complete registered interaction condition.
`observation`	`implicit-leakage`	implicit-leakage removes a required exact observation condition.	`exact-adversary-observation`	exact-adversary-observation retains the complete registered observation condition.
`security`	`informal-security`	informal-security removes a required exact security condition.	`retained-closed-distinction-property`	retained-closed-distinction-property retains the complete registered security condition.
`resources`	`unbounded-claim`	unbounded-claim removes a required exact resources condition.	`declared-information-or-resource-bound`	declared-information-or-resource-bound retains the complete registered resources condition.
`generality`	`challenge-examples`	challenge-examples removes a required exact generality condition.	`support-successor-certificate`	support-successor-certificate retains the complete registered generality condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`addition`	`imported-hardness-assumption`	imported-hardness-assumption removes a required exact addition condition.	`no-extra-security-premise`	no-extra-security-premise retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

94.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-messages-or-keys removes a required exact support condition.
64	undefined-adversary removes a required exact adversary condition.
32	outcome-only removes a required exact interaction condition.
16	implicit-leakage removes a required exact observation condition.
8	informal-security removes a required exact security condition.
4	unbounded-claim removes a required exact resources condition.
2	challenge-examples removes a required exact generality condition.
1	imported-hardness-assumption removes a required exact addition condition.

Shared claim-record clause `COMP-S012` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced classical claims under an enlarged adversary interface kernel retains classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

94.4 Operational laws and consequences

- Carrier law: classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

94.5 Depth-independent closure

Base: The structural One supplies one canonical classical claims under an enlarged adversary interface form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated classical claims under an enlarged adversary interface form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

94.6 Executed witnesses and adverse controls

- `cryptography_security-witness-1` - cryptography_security exact operational witness 1 for SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001; result: `PASS`.
- `cryptography_security-witness-2` - cryptography_security exact operational witness 2 for SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001; result: `PASS`.
- `cryptography_security-witness-3` - cryptography_security exact operational witness 3 for SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-messages-or-keys removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No quantum operation is derived here; operational quantum adversaries wait for the Quantum Computation branch.	`PASS`

94.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: post-quantum cryptography, quantum adversary.

No quantum operation is derived here; operational quantum adversaries wait for the Quantum Computation branch.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No quantum operation is derived here; operational quantum adversaries wait for the Quantum Computation branch.

94.8 Exact evidence identities

- Source manifest: `sha256:db51baede6272149653383aac66a7f35aa42bc3af23f7a03f111bd71035fa022`
- Complete-census certificate: `sha256:473ff558f791c67050eed2c25afaa5e8f1f8fc2532903b6554acf8be82a63cc5`
- Derivation seal: `sha256:8134279bbf4e56299f7ac9f4a9473bcc5dd141f070698642f5fdac6b8cfc53e9`
- Independent implementation: `sha256:2dda32b30a46541bb6bb86c5ebe17d4883594404de13200d4bd086e1656e6ec1`
- Independent certificate: `sha256:a5a2e61fb892deaa58864f2562348e0e83b69771cb5249932797ab1d3f01e591`
- External validation: `sha256:76bac704135f577f50d1b819bfcb5395d19aa7490015fd3032691a1b592b23a6`
- Model-admitted engine receipt: `sha256:4e450e475eb74f6968064e4ba58530226f4a58f8a6c3fff12176306a4d028c4d`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001-4e450e475eb74f69.json`

95. Derivation 87: Fold inference law

Claim: `SFT-COMP-LEARN-INFERENCE-001`

95.1 Scientific necessity and exact theorem

Fold inference law is required to derive selection among generated hypotheses without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced selection among generated hypotheses kernel retains hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

95.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for selection among generated hypotheses.

> *All generated finite selection among generated hypotheses structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Inference cannot import a prior distribution or pretrained weight.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

95.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced selection among generated hypotheses kernel retains hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

95.4 Operational laws and consequences

- Carrier law: hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

95.5 Depth-independent closure

Base: The structural One supplies one canonical selection among generated hypotheses form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated selection among generated hypotheses form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

95.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-INFERENCE-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-INFERENCE-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-INFERENCE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Inference cannot import a prior distribution or pretrained weight.	`PASS`

95.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: inference, hypothesis selection.

Inference cannot import a prior distribution or pretrained weight.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Inference cannot import a prior distribution or pretrained weight.

95.8 Exact evidence identities

- Source manifest: `sha256:b205f5d2bcffba31b2a8517eb32f9478c8a4305e3dcf85ae9ed9082971a96036`
- Complete-census certificate: `sha256:b1bd2070f66348ce7b4214d62727122c435e9d6980474d4c5b391600df55da9c`
- Derivation seal: `sha256:86235ff5fef7bf72b8dce90c3aed7ffbd387d6ac9aec4bcf541f310352df4cb5`
- Independent implementation: `sha256:17c8d94beaa2ae8bd3ef6b1ee8007444129b98b82b09740194612e4434583399`
- Independent certificate: `sha256:bb6159b3a5f1ea8b49e4db61ce525e7b0c7f719296f9d966b1120bd369774de8`
- External validation: `sha256:4521c49947e66c92bb50422ef3747db8aadb7f6885d35a0b5cfd3fb40a8dee8d`
- Model-admitted engine receipt: `sha256:483c9c1cb618e3c57985fbbcda9d4c3450b6dc50874c3f86a5a09fd4a8e54160`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-INFERENCE-001-483c9c1cb618e3c5.json`

96. Derivation 88: Fold classification and prediction law

Claim: `SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001`

96.1 Scientific necessity and exact theorem

Fold classification and prediction law is required to derive sealed mapping from inputs to held labels without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced sealed mapping from inputs to held labels kernel retains input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-INFERENCE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

96.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for sealed mapping from inputs to held labels.

> *All generated finite sealed mapping from inputs to held labels structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Natural-data performance requires the separate empirical constitution.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

96.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced sealed mapping from inputs to held labels kernel retains input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

96.4 Operational laws and consequences

- Carrier law: input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

96.5 Depth-independent closure

Base: The structural One supplies one canonical sealed mapping from inputs to held labels form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated sealed mapping from inputs to held labels form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

96.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Natural-data performance requires the separate empirical constitution.	`PASS`

96.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: classification, prediction.

Natural-data performance requires the separate empirical constitution.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Natural-data performance requires the separate empirical constitution.

96.8 Exact evidence identities

- Source manifest: `sha256:c49f4676c62a1e29209b8955a86b21670b0e44f10a575b8eb3400fff94bc92d8`
- Complete-census certificate: `sha256:e2e8663b301ea772432ae0984bf1568ec622319f150a53914e7d6d9d6dc07214`
- Derivation seal: `sha256:69d028c417c15a65f8d335b120de039ccbebe7c4c70c8b316ef553a049d63c42`
- Independent implementation: `sha256:2e499ab68732d073e51c68bd9ccadff6a9dd19b2fb20ccba2853b66ade014b24`
- Independent certificate: `sha256:cf48193afa6527e13bf45359d46ac6145cca308770d6a52628430a328afcd0a4`
- External validation: `sha256:aeb1268d3d337b6596c4f33c8af49d6220db7e674a7247962608448O8d75f041`
- Model-admitted engine receipt: `sha256:2902c0a1035fc27073a7a1be9ad39806838e05a1ef7fb2cbf8e045f1705be080`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001-2902c0a1035fc270.json`

97. Derivation 89: Fold representation-learning boundary law

Claim: `SFT-COMP-LEARN-REPRESENTATION-001`

97.1 Scientific necessity and exact theorem

Fold representation-learning boundary law is required to derive generated encodings preserving declared distinctions without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced generated encodings preserving declared distinctions kernel retains source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

97.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for generated encodings preserving declared distinctions.

> *All generated finite generated encodings preserving declared distinctions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No opaque learned representation is admitted as a law without an exact transformation trace.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

97.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced generated encodings preserving declared distinctions kernel retains source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

97.4 Operational laws and consequences

- Carrier law: source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

97.5 Depth-independent closure

Base: The structural One supplies one canonical generated encodings preserving declared distinctions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated encoding that preserves declared distinctions adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

97.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-REPRESENTATION-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-REPRESENTATION-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-REPRESENTATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No opaque learned representation is admitted as a law without an exact transformation trace.	`PASS`

97.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: representation learning, feature representation.

No opaque learned representation is admitted as a law without an exact transformation trace.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No opaque learned representation is admitted as a law without an exact transformation trace.

97.8 Exact evidence identities

- Source manifest: `sha256:b137149be197d5b38e3b24cb5cb468d8ecf855fe881d1ee3a8dbcc617d7ab390`
- Complete-census certificate: `sha256:b55adedd6d5e18635cbcf6e75d74882a409d34581283f7b8117b0f845d921eca`
- Derivation seal: `sha256:9eaa78f002935ed855587d3be65f3de944d0a04472cd30757259663d777c07fe`
- Independent implementation: `sha256:acf5b1a98cf9f1c0bef4dcd2b233d8e23c0861dc2f8d2e18a2f03e859204ce54`
- Independent certificate: `sha256:a9d9123e1ce01feb9bba7752a76a7b7b8f72cab588a060badfcca338a18b8b2a`
- External validation: `sha256:3ae7189b33dfec76df2087ba9c398ab5d3aecdfac84eed0ab446e85e6db9c763`
- Model-admitted engine receipt: `sha256:aaf1bb06300eddd353f33d549cc9ee8ace48e5f131d530ff9bdb7976c1cd0061`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-REPRESENTATION-001-aaf1bb06300eddd3.json`

98. Derivation 90: Fold generalization law

Claim: `SFT-COMP-LEARN-GENERALIZATION-001`

98.1 Scientific necessity and exact theorem

Fold generalization law is required to derive extension from registered evidence to generated unseen support without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced extension from registered evidence to generated unseen support kernel retains hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-REPRESENTATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

98.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure condition.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for extension from registered evidence to generated unseen support.

> *All generated finite extension from registered evidence to generated unseen support structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Generalization is empirical unless a structural successor proof exhausts the target family.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

98.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced extension from registered evidence to generated unseen support kernel retains hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

98.4 Operational laws and consequences

- Carrier law: hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure condition.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

98.5 Depth-independent closure

Base: The structural One supplies one canonical extension from registered evidence to generated unseen support form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated extension from registered evidence to generated unseen support form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

98.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-GENERALIZATION-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-GENERALIZATION-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-GENERALIZATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Generalization is empirical unless a structural successor proof exhausts the target family.	`PASS`

98.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: generalization, out-of-sample prediction.

Generalization is empirical unless a structural successor proof exhausts the target family.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Generalization is empirical unless a structural successor proof exhausts the target family.

98.8 Exact evidence identities

- Source manifest: `sha256:8941fcdacbdb992fdffa7704d92aee58549b81bed0ed5f99bc7c358a1bb2340c`
- Complete-census certificate: `sha256:0a392f50c6824a6fab3cda881923c4aa02bb8438374791476c134a66dd0b7454`
- Derivation seal: `sha256:a9bc47b89cf7b5320b47b847a995c2a156143272e9219c67f0ea6f256ad0411e`
- Independent implementation: `sha256:eb08a6b81d8f850328b67873f31028e849338accc1bf929d01b1506aa763cb80`
- Independent certificate: `sha256:53095ce3818727b136a2ca6f4699d6f376ebc7219ea632a98129ffb4b4c18d2d`
- External validation: `sha256:73e4b939d6ae084174bf079d91dc2df84b6465f61b132169c263a04257f189f1`
- Model-admitted engine receipt: `sha256:c99db1b2646ffad5f54e060c0f1ae1b32a9d8ebedc719d36a6e3aec10abf446d`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-GENERALIZATION-001-c99db1b2646ffad5.json`

99. Derivation 91: Fold sample-complexity law

Claim: `SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001`

99.1 Scientific necessity and exact theorem

Fold sample-complexity law is required to derive evidence support required to close hypothesis distinctions without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced evidence support required to close hypothesis distinctions kernel retains hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-GENERALIZATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

99.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for evidence support required to close hypothesis distinctions.

> *All generated finite evidence support required to close hypothesis distinctions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No asymptotic formula is imported; each family requires a generated certificate.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

99.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced evidence support required to close hypothesis distinctions kernel retains hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

99.4 Operational laws and consequences

- Carrier law: hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

99.5 Depth-independent closure

Base: The structural One supplies one canonical evidence support required to close hypothesis distinctions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated evidence support required to close hypothesis distinctions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

99.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No asymptotic formula is imported; each family requires a generated certificate.	`PASS`

99.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: sample complexity, teaching dimension.

No asymptotic formula is imported; each family requires a generated certificate.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No asymptotic formula is imported; each family requires a generated certificate.

99.8 Exact evidence identities

- Source manifest: `sha256:a8c2a967ac606c5ee071bfab9ff0ea1676861940591718fa3a472127d3c28ddb`
- Complete-census certificate: `sha256:caa25ef318a02f773c9460e8ea08be4d447b317355cfc0b6285dff580c94c89b`
- Derivation seal: `sha256:9fddabe49ce98bba0f880bb1cff97015501842d31fe376af0da5c9096627a420`
- Independent implementation: `sha256:de4a1e8920b9a571c0477fbe9a05bec0fdf78829f65ce150603fcfa5c161ad9f`
- Independent certificate: `sha256:027768d0addb63d237cba6455555e463eb1f01c6b030c8a672fe5f335c660ea1`
- External validation: `sha256:10fed92e8c01ab8b517e587549341509dfe7349243814214346717 51f5b6b5e8`
- Model-admitted engine receipt: `sha256:d4b13522a727a382b31bddb83eda909f6706c4f7b9bc48633b8e1fd28c133092`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001-d4b13522a727a382.json`

100. Derivation 92: Fold optimisation-in-learning law

Claim: `SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001`

100.1 Scientific necessity and exact theorem

Fold optimisation-in-learning law is required to derive exact objective selection over hypotheses without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced exact objective selection over hypotheses kernel retains hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

100.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for exact objective selection over hypotheses.

> *All generated finite exact objective selection over hypotheses structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Gradient, differentiability and fitted parameter spaces are not assumed.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

100.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced exact objective selection over hypotheses kernel retains hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

100.4 Operational laws and consequences

- Carrier law: hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

100.5 Depth-independent closure

Base: The structural One supplies one canonical exact objective selection over hypotheses form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact objective selection over hypotheses form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

100.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Gradient, differentiability and fitted parameter spaces are not assumed.	`PASS`

100.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: learning optimisation, empirical risk.

Gradient, differentiability and fitted parameter spaces are not assumed.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Gradient, differentiability and fitted parameter spaces are not assumed.

100.8 Exact evidence identities

- Source manifest: `sha256:12dce3bb852954f394ecb4d67d026ccda30f3923c6851bfb0eb8c95930ed8698`
- Complete-census certificate: `sha256:3059d8e36f4a965707ebcc0a5a61f9030a6050f7ff64c49d0117e19826507bc8`
- Derivation seal: `sha256:db572a877ce20e6fa9c85144aa3abc6588031199d66e6f967b6d744fe908f9bc`
- Independent implementation: `sha256:a7e21f4888ff04407aee3540f61a11ac7d327f6d70bd97a48943ed830e819339`
- Independent certificate: `sha256:da8458ec44ab9e55677ace1d413bc4be984ddb2033182648add409771da500af`
- External validation: `sha256:086de46d3035bc9d85c7be6b357dcca7180725c96eb1502478ba96d1ce65d1fe`
- Model-admitted engine receipt: `sha256:543256016991fee945bc9d01d128355d0a9777a814dc7428ff9146ac93a8a398`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001-543256016991fee9.json`

101. Derivation 93: Fold computational-induction law

Claim: `SFT-COMP-LEARN-INDUCTION-001`

101.1 Scientific necessity and exact theorem

Fold computational-induction law is required to derive rule construction from finite evidence without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced rule construction from finite evidence kernel retains observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

101.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for rule construction from finite evidence.

> *All generated finite rule construction from finite evidence structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Finite agreement alone never proves unrestricted continuation.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

101.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced rule construction from finite evidence kernel retains observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

101.4 Operational laws and consequences

- Carrier law: observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

101.5 Depth-independent closure

Base: The structural One supplies one canonical rule construction from finite evidence form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated rule construction from finite evidence form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

101.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-INDUCTION-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-INDUCTION-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-INDUCTION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Finite agreement alone never proves unrestricted continuation.	`PASS`

101.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: induction, rule learning.

Finite agreement alone never proves unrestricted continuation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Finite agreement alone never proves unrestricted continuation.

101.8 Exact evidence identities

- Source manifest: `sha256:216c47067c713bcb4c79907da10663b2217e48cb311d6bf71983d40aa42baf90`
- Complete-census certificate: `sha256:565d863a25e8bdc9278071a9cc418c11c3fd46daece418426847aaba0820ebb4`
- Derivation seal: `sha256:3efc6894220eb6df93411e220a210816dfefa95a8195b2835ddc002134ed63ab`
- Independent implementation: `sha256:2a7ec40d95f8877c0c4985d214c3a60c199fec4ff536133fecb7776cf01da5c2`
- Independent certificate: `sha256:fb353e73890cb1f146d4612722f473d27a53ae6b55e398ede969035e4a2fc80c`
- External validation: `sha256:c1989a33f208cd8840a0743d971659c41fff7acfa4df1fe38b105401638050eb`
- Model-admitted engine receipt: `sha256:919a71359ffd05922865e02e03ef9013abb551bc0ccb5b0b95363aee5cd998ed`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-INDUCTION-001-919a71359ffd0592.json`

102. Derivation 94: Fold search and planning law

Claim: `SFT-COMP-LEARN-SEARCH-PLANNING-001`

102.1 Scientific necessity and exact theorem

Fold search and planning law is required to derive action-sequence construction toward exact goals without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced action-sequence construction toward exact goals kernel retains state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-INDUCTION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

102.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for action-sequence construction toward exact goals.

> *All generated finite action-sequence construction toward exact goals structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Only the generated environment model is covered.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

102.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.

Eliminated candidates	Exact first-failure reason
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced action-sequence construction toward exact goals kernel retains state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

102.4 Operational laws and consequences

- Carrier law: state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

102.5 Depth-independent closure

Base: The structural One supplies one canonical action-sequence construction toward exact goals form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated action-sequence construction toward exact goals form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

102.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-SEARCH-PLANNING-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-SEARCH-PLANNING-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-SEARCH-PLANNING-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only the generated environment model is covered.	`PASS`

102.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: planning, state-space search.

Only the generated environment model is covered.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Only the generated environment model is covered.

102.8 Exact evidence identities

- Source manifest: `sha256:fd1e4c6bce5884546b46d3b8bdb487c34e5f0d9fc144467cbbd255f53ec2627b`
- Complete-census certificate: `sha256:94004478e61e1ab8efbf1281203fc61c897cf44e38c8db4ad02b52365c7019ad`
- Derivation seal: `sha256:15ffc8f3e5def5649d23e98ea4fdc3424e78eaadf8b6d73c82bb60159dcfcf57`
- Independent implementation: `sha256:261e951d99ce2cda0b9fd6e33cada770b98f997f344ef9ab5d2983b8eae2f571`
- Independent certificate: `sha256:aa5ddf2ede9accea8029726d20fa5375af6ea31049796f13c29f611c8715ad5f`
- External validation: `sha256:4d3e3ee3ed6bcd4c5d13258a994ffbc068e989dfd5dd8c1472d0bdc2ac2979e6`
- Model-admitted engine receipt: `sha256:fae483c37bc6bdfa9b29553a47c1620a9038fb736fd8cf2af77fe96535e780c7`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-SEARCH-PLANNING-001-fae483c37bc6bdfa.json`

103. Derivation 95: Fold reinforcement-computation law

Claim: `SFT-COMP-LEARN-REINFORCEMENT-001`

103.1 Scientific necessity and exact theorem

Fold reinforcement-computation law is required to derive policy evaluation through generated interaction traces without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced policy evaluation through generated interaction traces kernel retains states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison; every*

> *operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-SEARCH-PLANNING-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

103.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for policy evaluation through generated interaction traces.

> *All generated finite policy evaluation through generated interaction traces structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No stochastic cause, fitted value function or external reward scale is imported.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

103.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.

Eliminated candidates	Exact first-failure reason
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced policy evaluation through generated interaction traces kernel retains states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

103.4 Operational laws and consequences

- Carrier law: states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

103.5 Depth-independent closure

Base: The structural One supplies one canonical policy evaluation through generated interaction traces form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated policy evaluation through generated interaction traces form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

103.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-REINFORCEMENT-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-REINFORCEMENT-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-REINFORCEMENT-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No stochastic cause, fitted value function or external reward scale is imported.	`PASS`

103.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: reinforcement learning, policy.

No stochastic cause, fitted value function or external reward scale is imported.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No stochastic cause, fitted value function or external reward scale is imported.

103.8 Exact evidence identities

- Source manifest: `sha256:6ddc0f26308bb7a994c49435c6b65a7e566854bd9a4777a0edb8505c18911acd`
- Complete-census certificate: `sha256:be3e82a9b47bfbde120ee0ed38771bbf9f844742494adc945dc3cb9b83edd28b`
- Derivation seal: `sha256:922810d79cb65be09158633bb008b07b816501fc5b47e20d5527bf4422f350fd`
- Independent implementation: `sha256:3e6c09be62e44db9928e0f51ac60f3c0a21d392640b02961341d29aa93b546bd`
- Independent certificate: `sha256:c3c5e2c21db8d0448d9c1f5fac9dfc1978a2e85db163404b0754b3a6be9d25e2`
- External validation: `sha256:656c0d9cc8004254483b20cb731012f582f72db6e28a241e4a7edc710c99ca83`
- Model-admitted engine receipt: `sha256:0cb38ecfea78976c1452c6d9974fff417f27c4cb05a6721698d9d8c1c4328a1a`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-REINFORCEMENT-001-0cb38ecfea78976c.json`

104. Derivation 96: Fold multi-agent computation law

Claim: `SFT-COMP-LEARN-MULTIAGENT-001`

104.1 Scientific necessity and exact theorem

Fold multi-agent computation law is required to derive interacting adaptive processes without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced interacting adaptive processes kernel retains agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-REINFORCEMENT-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

104.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for interacting adaptive processes.

> *All generated finite interacting adaptive processes structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Game-theoretic correspondence is downstream and requires exact preference relations.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

104.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced interacting adaptive processes kernel retains agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

104.4 Operational laws and consequences

- Carrier law: agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

104.5 Depth-independent closure

Base: The structural One supplies one canonical interacting adaptive processes form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated interacting adaptive processes form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

104.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-MULTIAGENT-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-MULTIAGENT-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-MULTIAGENT-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Game-theoretic correspondence is downstream and requires exact preference relations.	`PASS`

104.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: multi-agent systems, agent interaction.

Game-theoretic correspondence is downstream and requires exact preference relations.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Game-theoretic correspondence is downstream and requires exact preference relations.

104.8 Exact evidence identities

- Source manifest: `sha256:f62816b2440c54f007826324e7acdc51362bb01ebec449195d02bc2171ed0fb0`
- Complete-census certificate: `sha256:8377eb399e70932b70ca84cd87750e53ceb42a24143a5cb380483000859260 98`
- Derivation seal: `sha256:fb345343a53247874bd2912ed6f77073446b826010268bc6afc8fae76c14a47f`
- Independent implementation: `sha256:74bd7c749f6e5716fd7a91e3f9b6bc247a333b6ab385d4d268519368534aaf10`
- Independent certificate: `sha256:d32be77b1c6b6a3bb80e9210bedabe9bd0248c382462bc8404e6de95610ffbfc`
- External validation: `sha256:9b57258598dd95151a00eb00580d24a060d532ee02a7914cd1e6ac36a5b8b443`
- Model-admitted engine receipt: `sha256:9d95cd6493bea3f2e926a8beb970becf6593a0aad4c08b4ea2f999e6760202a2`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-MULTIAGENT-001-9d95cd6493bea3f2.json`

105. Derivation 97: Fold adaptation law

Claim: `SFT-COMP-LEARN-ADAPTATION-001`

105.1 Scientific necessity and exact theorem

Fold adaptation law is required to derive trace-bound change of computational rules or state without importing its conventional answer-producing model.

The engine-admitted theorem is:

The forced trace-bound change of computational rules or state kernel retains initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-MULTIAGENT-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

105.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for trace-bound change of computational rules or state.

All generated finite trace-bound change of computational rules or state structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Adaptation cannot silently alter the constitutional engine or claim law.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

105.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced trace-bound change of computational rules or state kernel retains initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

105.4 Operational laws and consequences

- Carrier law: initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

105.5 Depth-independent closure

Base: The structural One supplies one canonical trace-bound change of computational rules or state form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated trace-bound change of computational rules or state form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

105.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-ADAPTATION-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-ADAPTATION-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-ADAPTATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Adaptation cannot silently alter the constitutional engine or claim law.	PASS

105.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: adaptation, online learning.

Adaptation cannot silently alter the constitutional engine or claim law.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Adaptation cannot silently alter the constitutional engine or claim law.

105.8 Exact evidence identities

- Source manifest: `sha256:d86cf331ef2b3a2f54cc9d847ad323ae0a0637c68ba19ea2401be1abc075ea95`
- Complete-census certificate: `sha256:686d736a609a82cd61ffc1446cf5a08c81eeeb199a17e1300bb9908515862c0e`
- Derivation seal: `sha256:fbfe8663647aa11d4c95ce51175bb01a3c0ed27da3ae8148bf20d8313d646fd8`
- Independent implementation: `sha256:bd4698fccdf68c28d5cbeea2976f776b0f15dd2377b2ce00ca4714fa140b836a`
- Independent certificate: `sha256:1edd726dbaeaf10b6bdd6f7e816f7073bbaf8f3533e1013670f69d938117b268`
- External validation: `sha256:4de8725dbfb89714aadda6c0fc76ce9dbc2bb2534f36911908c3ddb74d32f174`
- Model-admitted engine receipt: `sha256:64ec3907b0e627e9e06f675e15a8ee10818781609c850b20948008691aa9d1cb`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-ADAPTATION-001-64ec3907b0e627e9.json`

106. Derivation 98: Fold computational limits-of-learning law

Claim: `SFT-COMP-LEARN-LEARNING-LIMITS-001`

106.1 Scientific necessity and exact theorem

Fold computational limits-of-learning law is required to derive indistinguishable evidence and unidentifiable hypotheses without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced indistinguishable evidence and unidentifiable hypotheses kernel retains hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-ADAPTATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

106.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for indistinguishable evidence and unidentifiable hypotheses.

> *All generated finite indistinguishable evidence and unidentifiable hypotheses structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Limits are relative to the exact evidence, hypothesis and target grammars.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

106.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced indistinguishable evidence and unidentifiable hypotheses kernel retains hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

106.4 Operational laws and consequences

- Carrier law: hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

106.5 Depth-independent closure

Base: The structural One supplies one canonical indistinguishable evidence and unidentifiable hypotheses form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated indistinguishable evidence and unidentifiable hypotheses form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

106.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-LEARNING-LIMITS-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-LEARNING-LIMITS-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-LEARNING-LIMITS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Limits are relative to the exact evidence, hypothesis and target grammars.	`PASS`

106.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: no-free-lunch, identifiability.

Limits are relative to the exact evidence, hypothesis and target grammars.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Limits are relative to the exact evidence, hypothesis and target grammars.

106.8 Exact evidence identities

- Source manifest: `sha256:55152f2e7369fb034b347faa7c8d385705344969ce55f80e29d0311b2912fda0`
- Complete-census certificate: `sha256:33009e7095b23df40e776554f188351b95bd11b85b677f3d070747836fc131ab`
- Derivation seal: `sha256:f5e5533666d646c7ef7bf627324974f71f2c66942cc4a1bfcf76e931fba31ea4`
- Independent implementation: `sha256:6f0b680771420d91f20a198e0be71505ba6bf4d1857ec9b562906dea65c7cadf`
- Independent certificate: `sha256:2ce5d5a80478e0d65c29ff54fe9ab22de89e9c7b86b381f3b2eee2fd2919ced4`
- External validation: `sha256:8c965863ce40503dee10d74d35ebe003bcbb75400e84133ab01119fc3d1c62fd`
- Model-admitted engine receipt: `sha256:969046a8a85fc0802d0269d6a4a46e706d278b71450e8b7c7dbd1188fe21bc90`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-LEARNING-LIMITS-001-969046a8a85fc080.json`

107. Derivation 99: Fold interpretability and verification law

Claim: `SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001`

107.1 Scientific necessity and exact theorem

Fold interpretability and verification law is required to derive traceable relation from inputs to decisions without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced traceable relation from inputs to decisions kernel retains model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-LEARNING-LIMITS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

107.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for traceable relation from inputs to decisions.

> *All generated finite traceable relation from inputs to decisions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A narrative explanation or score cannot replace the exact trace.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

107.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced traceable relation from inputs to decisions kernel retains model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

107.4 Operational laws and consequences

- Carrier law: model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

107.5 Depth-independent closure

Base: The structural One supplies one canonical traceable relation from inputs to decisions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated traceable relation from inputs to decisions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

107.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A narrative explanation or score cannot replace the exact trace.	`PASS`

107.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: interpretability, model verification.

A narrative explanation or score cannot replace the exact trace.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A narrative explanation or score cannot replace the exact trace.

107.8 Exact evidence identities

- Source manifest: `sha256:57953fb29c34d5ba8d75378b797de3cfeebe71f875ffa2c17dbfd0fde75d2ed4`
- Complete-census certificate: `sha256:b1f510b5bcd5dea8b61574693d523e057df0e57df49061a5cb052128bc70ad19`
- Derivation seal: `sha256:c8db64a84f070d98e4c50cdc3282e18181699e08907e6325c40ce7f8d0b3c710`
- Independent implementation: `sha256:56cf69782fa0d5db9f55475a297018e3351f9e65f24512b0a88548f3d1b67943`
- Independent certificate: `sha256:c9831ba4b85a410b2432b135d2d024421a92478a2ebe70c18e559d9e6ee863b6`
- External validation: `sha256:4983de7e26afc28ec869589a8285b7cac9c6a31bd71dfbc23b2f890b22223f45`
- Model-admitted engine receipt: `sha256:c0246290a461042d4797f427dfd27adaebf0954e5372091899b896b3418a1424`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001-c0246290a461042d.json`

108. Derivation 100: Fold classical-learning model law

Claim: `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`

108.1 Scientific necessity and exact theorem

Fold classical-learning model law is required to derive complete classical finite learning processes without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced complete classical finite learning processes kernel retains examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

108.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result.

Generation rule: Generate the literal product of the eight registered Learning and Intelligence Theory structural axes for complete classical finite learning processes.

> *All generated finite complete classical finite learning processes structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Quantum learning operations remain entirely outside this branch.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`support`	`sampled-hypotheses`	sampled-hypotheses removes a required exact support condition.	`complete-generated-hypothesis-support`	complete-generated-hypothesis-support retains the complete registered support condition.
`representation`	`opaque-representation`	opaque-representation removes a required exact representation condition.	`traceable-canonical-representation`	traceable-canonical-representation retains the complete registered representation condition.
`inference`	`pretrained-answer`	pretrained-answer removes a required exact inference condition.	`exact-evidence-to-hypothesis-relation`	exact-evidence-to-hypothesis-relation retains the complete registered inference condition.
`evaluation`	`development-score`	development-score removes a required exact evaluation condition.	`sealed-independent-evaluation`	sealed-independent-evaluation retains the complete registered evaluation condition.
`uncertainty`	`single-winner-erasure`	single-winner-erasure removes a required exact uncertainty condition.	`retained-alternative-ledger`	retained-alternative-ledger retains the complete registered uncertainty condition.
`resources`	`unreported-search`	unreported-search removes a required exact resources condition.	`complete-learning-resource-trace`	complete-learning-resource-trace retains the complete registered resources condition.
`generality`	`training-examples-only`	training-examples-only removes a required exact generality condition.	`structural-or-empirical-target-boundary`	structural-or-empirical-target-boundary retains the complete registered generality condition.
`addition`	`fitted-weight-or-prior`	fitted-weight-or-prior removes a required exact addition condition.	`no-extra-learning-parameter`	no-extra-learning-parameter retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

108.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	sampled-hypotheses removes a required exact support condition.
64	opaque-representation removes a required exact representation condition.
32	pretrained-answer removes a required exact inference condition.
16	development-score removes a required exact evaluation condition.
8	single-winner-erasure removes a required exact uncertainty condition.
4	unreported-search removes a required exact resources condition.
2	training-examples-only removes a required exact generality condition.
1	fitted-weight-or-prior removes a required exact addition condition.

Shared claim-record clause `COMP-S013` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced complete classical finite learning processes kernel retains examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

108.4 Operational laws and consequences

- Carrier law: examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

108.5 Depth-independent closure

Base: The structural One supplies one canonical complete classical finite learning processes form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated complete classical finite learning processes form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

108.6 Executed witnesses and adverse controls

- `learning_intelligence-witness-1` - learning_intelligence exact operational witness 1 for SFT-COMP-LEARN-CLASSICAL-LEARNING-001; result: `PASS`.
- `learning_intelligence-witness-2` - learning_intelligence exact operational witness 2 for SFT-COMP-LEARN-CLASSICAL-LEARNING-001; result: `PASS`.
- `learning_intelligence-witness-3` - learning_intelligence exact operational witness 3 for SFT-COMP-LEARN-CLASSICAL-LEARNING-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	sampled-hypotheses removes a required exact support condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Quantum learning operations remain entirely outside this branch.	`PASS`

108.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: classical learning, machine learning theory.

Quantum learning operations remain entirely outside this branch.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Quantum learning operations remain entirely outside this branch.

108.8 Exact evidence identities

- Source manifest: `sha256:84c7423aac0c5dd42a0da2038331ca17fc561cf85c8202db66054604dbf5a44e`
- Complete-census certificate: `sha256:de855f8a0d135b04ae3c2ba92de145a816d1607b24bf17c6bdd175b404fc2638`
- Derivation seal: `sha256:d0f21b1802e74a2e9b58544c4d87f8c3934cf28aa6628b44b863a14c4df9a29f`
- Independent implementation: `sha256:c009e53b2ed811511621e85d68a3f422e8467044881b0c670d98c74a365f57d0`
- Independent certificate: `sha256:17f40ce955c28ee371c3107d4e114ce125b9fd68dfe1ebdb760af5621159658e`
- External validation: `sha256:7f34c30e760456e4653d516838ec65c0f449d819f1a4fc2e2878ade0222383a2`
- Model-admitted engine receipt: `sha256:86cdfa5fb4a647883b4b1300ea87c83dddd01b8f34f96c3738fec0baa8f8a8ce`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-LEARN-CLASSICAL-LEARNING-001-86cdfa5fb4a64788.json`

109. Derivation 101: Fold exact and approximate calculation law

Claim: `SFT-COMP-SCI-EXACT-APPROXIMATE-001`

109.1 Scientific necessity and exact theorem

Fold exact and approximate calculation law is required to derive separation of proof values and bounded approximations without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced separation of proof values and bounded approximations kernel retains exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-LEARN-CLASSICAL-LEARNING-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

109.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for separation of proof values and bounded approximations.

> *All generated finite separation of proof values and bounded approximations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Approximate values cannot flow backward to select an exact law.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

109.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.

Eliminated candidates	Exact first-failure reason
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced separation of proof values and bounded approximations kernel retains exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

109.4 Operational laws and consequences

- Carrier law: exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

109.5 Depth-independent closure

Base: The structural One supplies one canonical separation of proof values and bounded approximations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated separation of proof values and bounded approximations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

109.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-EXACT-APPROXIMATE-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-EXACT-APPROXIMATE-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-EXACT-APPROXIMATE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Approximate values cannot flow backward to select an exact law.	`PASS`

109.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: exact computation, approximate computation.

Approximate values cannot flow backward to select an exact law.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Approximate values cannot flow backward to select an exact law.

109.8 Exact evidence identities

- Source manifest: `sha256:fb2986b63e83b23133a3823d055f0c3beb020e6b928274281935b3055ecfa40d`
- Complete-census certificate: `sha256:8ab424ccb156e7c2e467452c86467587b3e5cb0a0a1f61b5390ae0d8f6365401`
- Derivation seal: `sha256:8d29ccecdb3e151f4ca0c51c27e60e628a21c6471db567add4c18e7e8f8f53a3`
- Independent implementation: `sha256:5a5450ce9d27d00c9c63c225d46ecb4cb8ac2a8f713dc63ac331245ee27549c2`
- Independent certificate: `sha256:a873faf2411c2484210b6488ffa52a8d7f4f564c06d6caab2a38e3524b1304d1`
- External validation: `sha256:3dc1f3c153b7c6420f71d1f912eef13516f2952441df895cd2f246e0b8dc640a`
- Model-admitted engine receipt: `sha256:d04dba138ab4ed09a8fba0c3751fc8588a7ad3b34790ac1f536b44b013b1a905`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-EXACT-APPROXIMATE-001-d04dba138ab4ed09.json`

110. Derivation 102: Fold numerical-stability law

Claim: `SFT-COMP-SCI-STABILITY-001`

110.1 Scientific necessity and exact theorem

Fold numerical-stability law is required to derive output distinction under input or operation perturbation without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced output distinction under input or operation perturbation kernel retains exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-EXACT-APPROXIMATE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

110.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for output distinction under input or operation perturbation.

> *All generated finite output distinction under input or operation perturbation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No floating epsilon or norm is imported without derivation.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

110.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.

Eliminated candidates	Exact first-failure reason
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced output distinction under input or operation perturbation kernel retains exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

110.4 Operational laws and consequences

- Carrier law: exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

110.5 Depth-independent closure

Base: The structural One supplies one canonical output distinction under input or operation perturbation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated output distinction under input or operation perturbation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

110.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-STABILITY-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-STABILITY-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-STABILITY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No floating epsilon or norm is imported without derivation.	`PASS`

110.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: numerical stability, conditioning.

No floating epsilon or norm is imported without derivation.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No floating epsilon or norm is imported without derivation.

110.8 Exact evidence identities

- Source manifest: `sha256:d5cf72d09f2ddd68644ff7db62e2a034d814b2c271bb3a73e8348e6ee7825924`
- Complete-census certificate: `sha256:c936bf0f2a0fecc107d19c73caec2ed4c0aa05e8cb379920574a4cb536a8867a`
- Derivation seal: `sha256:a43ca827851cfe1c7db7b09f951670311b50ebcca5938c0c5d205e9e7c9a7f21`
- Independent implementation: `sha256:ed227d1ffe6b9cca5e1bd78b18ea37b96420ec366c45fc0886fd956d6f69dae0`
- Independent certificate: `sha256:f8a39efdb4e40d985b773a578ac4cbbe790f0491b94fdf2f9a1b2e5e5dce5d6f`
- External validation: `sha256:bd975ace29c4d474e3faf48d0b6399acf9f3e6cee4cd2f9f7178be691565b923`
- Model-admitted engine receipt: `sha256:f50a038a131659cc3156af54451f66c24d7562bb4284ab1ad6a71b6e97cc6980`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-STABILITY-001-f50a038a131659cc.json`

111. Derivation 103: Fold error-propagation law

Claim: `SFT-COMP-SCI-ERROR-PROPAGATION-001`

111.1 Scientific necessity and exact theorem

Fold error-propagation law is required to derive transport of exact approximation distinctions without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced transport of exact approximation distinctions kernel retains local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-STABILITY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

111.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for transport of exact approximation distinctions.

> *All generated finite transport of exact approximation distinctions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Signed arithmetic is represented by held orientation labels, not negative proof values.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

111.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.

Eliminated candidates	Exact first-failure reason
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced transport of exact approximation distinctions kernel retains local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

111.4 Operational laws and consequences

- Carrier law: local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

111.5 Depth-independent closure

Base: The structural One supplies one canonical transport of exact approximation distinctions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated transport of exact approximation distinctions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

111.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-ERROR-PROPAGATION-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-ERROR-PROPAGATION-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-ERROR-PROPAGATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Signed arithmetic is represented by held orientation labels, not negative proof values.	`PASS`

111.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: error propagation, uncertainty propagation.

Signed arithmetic is represented by held orientation labels, not negative proof values.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Signed arithmetic is represented by held orientation labels, not negative proof values.

111.8 Exact evidence identities

- Source manifest: `sha256:715d0ace314d6fe5a0307bef26e63634bcf52fb83416a5a01f6ccbb9cea1fa62`
- Complete-census certificate: `sha256:7a39d937f82bc63a6bb808c8deb7e81842764dfcfe753d000e76a9b4b29efc5a`
- Derivation seal: `sha256:3c1e26188805f325b58f26f59dd2b777a2b71a8294e318eb6a679564baab529d`
- Independent implementation: `sha256:ea298bac8ba3d6d280b08d831a3da7564bed420acc69dfff8630e933bfb791f0`
- Independent certificate: `sha256:8ff2ad3dbafe41f907f100ace8d23fb0c2b177a1e8720b8a1b6aec3c2c11e8b6`
- External validation: `sha256:10b7ab2f6ebc21135b80a940a993f81e0ab0a0120ca0db765d795507dc542d68`
- Model-admitted engine receipt: `sha256:083dc87700741205332c48c4a74aa5658b9f205141bcb39a13bb88288354230b`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-ERROR-PROPAGATION-001-083dc87700741205.json`

112. Derivation 104: Fold discretization law

Claim: `SFT-COMP-SCI-DISCRETIZATION-001`

112.1 Scientific necessity and exact theorem

Fold discretization law is required to derive finite generated representation of a declared model without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced finite generated representation of a declared model kernel retains source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-ERROR-PROPAGATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

112.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for finite generated representation of a declared model.

> *All finite generated-representation structures of a declared model whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. An ungenerated continuum is never the derivational premise.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

112.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.

Eliminated candidates	Exact first-failure reason
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced finite generated representation of a declared model kernel retains source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

112.4 Operational laws and consequences

- Carrier law: source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

112.5 Depth-independent closure

Base: The structural One supplies one canonical finite generated representation of a declared model form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated finite generated representation of a declared model form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

112.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-DISCRETIZATION-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-DISCRETIZATION-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-DISCRETIZATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; An ungenerated continuum is never the derivational premise.	`PASS`

112.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: discretization, mesh.

An ungenerated continuum is never the derivational premise.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- An ungenerated continuum is never the derivational premise.

112.8 Exact evidence identities

- Source manifest: `sha256:fb2079496ebb17f2dfb5cf163b7b360fee2dcbaaa0ad198198a4ee65af700d0e`
- Complete-census certificate: `sha256:57e7908787684a30402fccfb4ae3e6ac8e27c0f1f734dbf09ab2d77c10b6344e`
- Derivation seal: `sha256:65fec14eb6a66e240b026fbc893525962285ed8e6dae678f503d42de9b2da781`
- Independent implementation: `sha256:5c232f884af76629b5fe61f5338cdb2bd73cbb4a41d4b6890d01c357442e9d67`
- Independent certificate: `sha256:72050853d92aed0efab89da8ae0097f4801c8609a7dcfc27edc23fc7b8b59b91`
- External validation: `sha256:8a903c90d8dd50d73ddbba422eab2bfe78126a414ee94f93c2ab812b78d70b12`
- Model-admitted engine receipt: `sha256:bcf98ba0535b49f76382780ec0f1e22b9f184249bb3ccb596463f48d0532080c`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-DISCRETIZATION-001-bcf98ba0535b49f7.json`

113. Derivation 105: Fold convergence-certificate law

Claim: `SFT-COMP-SCI-CONVERGENCE-001`

113.1 Scientific necessity and exact theorem

Fold convergence-certificate law is required to derive successor refinement preserving and closing error distinctions without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced successor refinement preserving and closing error distinctions kernel retains refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-DISCRETIZATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

113.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for successor refinement preserving and closing error distinctions.

> *All generated finite successor refinement preserving and closing error distinctions structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Closure is depth-independent only when the successor certificate proves the registered descent property.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

113.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.

Eliminated candidates	Exact first-failure reason
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced successor refinement preserving and closing error distinctions kernel retains refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

113.4 Operational laws and consequences

- Carrier law: refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

113.5 Depth-independent closure

Base: The structural One supplies one canonical successor refinement preserving and closing error distinctions form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated successor refinement preserving and closing error distinctions form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

113.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-CONVERGENCE-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-CONVERGENCE-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-CONVERGENCE-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closure is depth-independent only when the successor certificate proves the registered descent property.	`PASS`

113.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: convergence, convergence proof.

Closure is depth-independent only when the successor certificate proves the registered descent property.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Closure is depth-independent only when the successor certificate proves the registered descent property.

113.8 Exact evidence identities

- Source manifest: `sha256:119a67a6a0e8a1a61cf2acd01d62f4a442ffa9bcc4efd6a8bc93b1ff439a6d42`
- Complete-census certificate: `sha256:a7f115d89adefcff54bae05486e6de093d175a54c2007af4913f5a12b5784d46`
- Derivation seal: `sha256:ed6247aa7d641db78d7d340bf3c5b3d14aa9fbab778506d4ba1aba8dd53d24ca`
- Independent implementation: `sha256:a1a4741ce2299d736fe56a7891feadc4e14e17b620a17b2356a08d5e6bb89598`
- Independent certificate: `sha256:75712ca92daab71c4ed80079d561ccaae73c8f058914420ba5740326759e65df`
- External validation: `sha256:ce2f9d06a52c46ff06ffb04608e4f3146f2c152ca69836601707192be0df483a`
- Model-admitted engine receipt: `sha256:a3ae509175198d515e8ae636ad4b669d942524409b96f97c4ce26a94a98c03b5`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-CONVERGENCE-001-a3ae509175198d51.json`

114. Derivation 106: Fold scientific symbolic-computation law

Claim: `SFT-COMP-SCI-SYMBOLIC-001`

114.1 Scientific necessity and exact theorem

Fold scientific symbolic-computation law is required to derive exact model-expression transformation without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced exact model-expression transformation kernel retains model syntax; exact rewrite rules; dimensional or type interfaces; normalization; identity proof; evaluator correspondence; result trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-CONVERGENCE-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

114.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: model syntax; exact rewrite rules; dimensional or type interfaces; normalization; identity proof; evaluator correspondence; result trace.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for exact model-expression transformation.

> *All generated finite exact model-expression transformation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Physical interpretation remains outside formal symbolic closure.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

114.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.

Eliminated candidates	Exact first-failure reason
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced exact model-expression transformation kernel retains model syntax; exact rewrite rules; dimensional or type interfaces; normalization; identity proof; evaluator correspondence; result trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

114.4 Operational laws and consequences

- Carrier law: model syntax; exact rewrite rules; dimensional or type interfaces; normalization; identity proof; evaluator correspondence; result trace.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

114.5 Depth-independent closure

Base: The structural One supplies one canonical exact model-expression transformation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact model-expression transformation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

114.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-SYMBOLIC-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-SYMBOLIC-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-SYMBOLIC-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`

Control	Expected	Observed	Result
boundary	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Physical interpretation remains outside formal symbolic closure.	PASS

114.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: symbolic scientific computing, computer algebra.

Physical interpretation remains outside formal symbolic closure.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Physical interpretation remains outside formal symbolic closure.

114.8 Exact evidence identities

- Source manifest: `sha256:8f3afcab73070c32caa7f7a244e2d3381cfe0c4c10361d97d213528b7a67cfa9`
- Complete-census certificate: `sha256:86c189a3ff38a24347c519207a6a20cdd6c35bed586bc3e768760803fea23466`
- Derivation seal: `sha256:ccdce6568fba17556bb00419d7c10978c1938ff74de2d20f214d126f993351ac`
- Independent implementation: `sha256:a47d9f806660a188331934953c9585f113b3b60beeea1e994b4878c5172007fc`
- Independent certificate: `sha256:f08802aacd47de59d3f84730e1e10748406d5e76f0e8b2d2ee36a2cd40dca71f`
- External validation: `sha256:a1d97bead16859a3ac163ad75406876848eb51cdbc35ae06549bfdf4827a341e`
- Model-admitted engine receipt: `sha256:484049210c99c3cf945baa6cf8f64f14beb56db6f50fe8ede3d322dee49c41b8`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-SYMBOLIC-001-484049210c99c3cf.json`

115. Derivation 107: Fold simulation law

Claim: `SFT-COMP-SCI-SIMULATION-001`

115.1 Scientific necessity and exact theorem

Fold simulation law is required to derive execution of a registered model relation without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced execution of a registered model relation kernel retains model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-SYMBOLIC-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

115.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for execution of a registered model relation.

> *All generated finite execution of a registered model relation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A simulation validates consequences of its model; it cannot establish the model as a natural law.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

115.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.

Eliminated candidates	Exact first-failure reason
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced execution of a registered model relation kernel retains model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

115.4 Operational laws and consequences

- Carrier law: model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

115.5 Depth-independent closure

Base: The structural One supplies one canonical execution of a registered model relation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated execution of a registered model relation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

115.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-SIMULATION-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-SIMULATION-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-SIMULATION-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A simulation validates consequences of its model; it cannot establish the model as a natural law.	`PASS`

115.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: simulation, computational model.

A simulation validates consequences of its model; it cannot establish the model as a natural law.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A simulation validates consequences of its model; it cannot establish the model as a natural law.

115.8 Exact evidence identities

- Source manifest: `sha256:c8e683c02a890d93d37979d9946e396fba5e64c5789dbdd9d1f08c65f97c7e45`
- Complete-census certificate: `sha256:433f57b352285b1a23fa05ce1bbfb89e9cc7a615c6e12252d08396d5d5e54c1e`
- Derivation seal: `sha256:b04cfb756fbbb395c383e325234af2577760c79c32252a1f80f6ba23a85a6cee`
- Independent implementation: `sha256:b93c4dfbe30b3dbd96c4865fc0ef9b9ad242cd0c0c0e1376087d48dbc6bb9712`
- Independent certificate: `sha256:1d19485b5b1604d13a29197016920d51da71b365201df597af35d2305911649e`
- External validation: `sha256:0f028f2611fc2b19447d3b2f57a2515829a4b95d8743d165903cb7ec50945962`
- Model-admitted engine receipt: `sha256:b06dd069154f18ea84aadfe5c47b070e0ec017113e01945e55690303f274e293`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-SIMULATION-001-b06dd069154f18ea.json`

116. Derivation 108: Fold computational-dynamics law

Claim: `SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001`

116.1 Scientific necessity and exact theorem

Fold computational-dynamics law is required to derive iterated exact finite state transformation without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced iterated exact finite state transformation kernel retains state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-SIMULATION-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-ENCODING-DECODING-001
- SFT-INFO-CONSERVATION-LOSS-001

116.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for iterated exact finite state transformation.

> *All generated finite iterated exact finite state transformation structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Physical time and continuous dynamics require later empirical correspondence.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

116.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced iterated exact finite state transformation kernel retains state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

116.4 Operational laws and consequences

- Carrier law: state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

116.5 Depth-independent closure

Base: The structural One supplies one canonical iterated exact finite state transformation form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated iterated exact finite state transformation form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

116.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Physical time and continuous dynamics require later empirical correspondence.	`PASS`

116.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: computational dynamics, dynamical simulation.

Physical time and continuous dynamics require later empirical correspondence.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Physical time and continuous dynamics require later empirical correspondence.

116.8 Exact evidence identities

- Source manifest: `sha256:fb8bacdb9b1b4ec965b144d3d65d72b86dd81919c721a370e797b55c99b236f1`
- Complete-census certificate: `sha256:2a564ff077594a99ced8cdb01d265feb58a6b2af6e0c87afa27d231badda5904`
- Derivation seal: `sha256:9c0a4479e8a6808361dfac8eaa8d54c89067ff7aad1cf3294f8ff62c9e35ec9c`
- Independent implementation: `sha256:e89c6e112f56650b644a04ff5b68e2ec9f48826e68f71975b1375d4beda1e1b3`
- Independent certificate: `sha256:c10c1dfce8f8e6f23c60190fb1800adee24259093c8d851aaf5fc5cd9f0b4160`
- External validation: `sha256:b4faf647cc85a39c4437bb902767c84e3556daf5bb17f7265166feb6b6881346`
- Model-admitted engine receipt: `sha256:9f40f7a193ca6e09bbc51fd61d7a526dc81c9224a6797bac0c0c57e8005ae0de`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001-9f40f7a193ca6e09.json`

117. Derivation 109: Fold inverse-problem law

Claim: `SFT-COMP-SCI-INVERSE-PROBLEM-001`

117.1 Scientific necessity and exact theorem

Fold inverse-problem law is required to derive reconstruction of sources from observations without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced reconstruction of sources from observations kernel retains source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

117.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for reconstruction of sources from observations.

> *All generated finite reconstruction of sources from observations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No fitted regularizer or prior may select among unresolved fibres unless separately registered.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

117.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced reconstruction of sources from observations kernel retains source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

117.4 Operational laws and consequences

- Carrier law: source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

117.5 Depth-independent closure

Base: The structural One supplies one canonical reconstruction of sources from observations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated reconstruction of sources from observations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

117.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-INVERSE-PROBLEM-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-INVERSE-PROBLEM-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-INVERSE-PROBLEM-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No fitted regularizer or prior may select among unresolved fibres unless separately registered.	`PASS`

117.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: inverse problem, identifiability.

No fitted regularizer or prior may select among unresolved fibres unless separately registered.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter

- no application result or pretrained model
- No fitted regularizer or prior may select among unresolved fibres unless separately registered.

117.8 Exact evidence identities

- Source manifest: `sha256:ebbe50d370311564302b91465ef3dcd85324bc6b44c2e4865abf130155e60ad0`
- Complete-census certificate: `sha256:69c171260b2d27e2ccd68e8868e52e07e63886eead5baf1257ac06a080cd05ed`
- Derivation seal: `sha256:9aedf19b7d236a4710da983fbd678c9f179bccb617188f080fa9540c02385310`
- Independent implementation: `sha256:631220d2e521292b1f4000a76e228380d7f613100c10076d8ebf18a09aac1162`
- Independent certificate: `sha256:6ac9ca23bb7c37bb14f2795f7e768b25108c651af84a744950cdccf4c7b67ee0`
- External validation: `sha256:7c54dcae327e865ff3d257363c532cb2016fdbf4cdf42f6b21375c9bd8c3112d`
- Model-admitted engine receipt: `sha256:41ca83901e2ba93484c943df39462f4196c046678f4d9cb17afafb5277cacdde`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-INVERSE-PROBLEM-001-41ca83901e2ba934.json`

118. Derivation 110: Fold computational-statistics law

Claim: `SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001`

118.1 Scientific necessity and exact theorem

Fold computational-statistics law is required to derive exact support summaries and inference algorithms without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced exact support summaries and inference algorithms kernel retains complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-INVERSE-PROBLEM-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

118.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for exact support summaries and inference algorithms.

> *All generated finite exact support summaries and inference algorithms structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Probability remains deterministic-support bookkeeping unless an external distribution is measured.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

118.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced exact support summaries and inference algorithms kernel retains complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

118.4 Operational laws and consequences

- Carrier law: complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

118.5 Depth-independent closure

Base: The structural One supplies one canonical exact support summaries and inference algorithms form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated exact support summaries and inference algorithms form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

118.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Probability remains deterministic-support bookkeeping unless an external distribution is measured.	`PASS`

118.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: computational statistics, statistical computing.

Probability remains deterministic-support bookkeeping unless an external distribution is measured.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Probability remains deterministic-support bookkeeping unless an external distribution is measured.

118.8 Exact evidence identities

- Source manifest: `sha256:89565510d8bf65cc4168e57422355f8a2cfeccdc09807867ac85716e37f2a1cb`
- Complete-census certificate: `sha256:a57c05e5723cfa9ddafce937b5a6c078fe7ada36e42f5a62ed58b3019d506b6a`

- Derivation seal: `sha256:acc45ba06d6b9dffc0e01169f023269669bdfea1c720e15db7c5feb630c26096`
- Independent implementation: `sha256:c6bb9adaf5e746b4ca7bed2d6a76b2f0183231070804981ed063b9f5eb8fa179`
- Independent certificate: `sha256:bdf581ea46da78d86afa92203678ff3422cc8bc86070705f8cd5e304c31c275a`
- External validation: `sha256:641f049a78a531c8dd32e5f49d3e31ed2788db1ba097dbbf3d0fd941eb2dc193`
- Model-admitted engine receipt: `sha256:af3c12e592f489613af6f00e0bed93b948f0ccc0a7f975ec89a55fce888ceffa`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001-af3c12e592f48961.json`

119. Derivation 111: Fold high-dimensional computation law

Claim: `SFT-COMP-SCI-HIGH-DIMENSIONAL-001`

119.1 Scientific necessity and exact theorem

Fold high-dimensional computation law is required to derive product-support computation with retained coordinate structure without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced product-support computation with retained coordinate structure kernel retains generated coordinate carriers; complete product support; sparse or factored relation; algorithm trace; resource ledger; lost-interaction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

119.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: generated coordinate carriers; complete product support; sparse or factored relation; algorithm trace; resource ledger; lost-interaction boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for product-support computation with retained coordinate structure.

> *All generated finite product-support computation with retained coordinate structure structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. Dimension is a generated coordinate trace, not a completed infinite vector space.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

119.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced product-support computation with retained coordinate structure kernel retains generated coordinate carriers; complete product support; sparse or factored relation; algorithm trace; resource ledger; lost-interaction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

119.4 Operational laws and consequences

- Carrier law: generated coordinate carriers; complete product support; sparse or factored relation; algorithm trace; resource ledger; lost-interaction boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

119.5 Depth-independent closure

Base: The structural One supplies one canonical product-support computation with retained coordinate structure form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated product-support computation with retained coordinate structure form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

119.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-HIGH-DIMENSIONAL-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-HIGH-DIMENSIONAL-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-HIGH-DIMENSIONAL-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Dimension is a generated coordinate trace, not a completed infinite vector space.	`PASS`

119.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: high-dimensional computing, curse of dimensionality.

Dimension is a generated coordinate trace, not a completed infinite vector space.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- Dimension is a generated coordinate trace, not a completed infinite vector space.

119.8 Exact evidence identities

- Source manifest: `sha256:e515d03a970bf5b51a7661b975738c624a4fc11224ae3597d6991b19c55eece3`
- Complete-census certificate: `sha256:8084ac3a48b17cb0a6a629d032ca3e3ba98dc2e4738b7ee73b9d6cec56af8921`
- Derivation seal: `sha256:ef184706f84eb5d737e955ed436ae8913d9b5cf2db1cafa541a94c0cc4165f79`
- Independent implementation: `sha256:66bba91ba3902bc2caf8dac70ef204e8cbad12d8f0f59634772f6f4f34d4cc04`
- Independent certificate: `sha256:db5a4afe687da48685c9b4f295bf78d0c71745c3db90405d81abab15632101f0`

- External validation: `sha256:4c20be1f568135bdfa36bb34a1ca13ebee60bb55f9d19c208a26bf65cb119b06`
- Model-admitted engine receipt: `sha256:08b4870cfa97a44e956720e0bf6ccfef2216790ad6ffcc0261c127f46625f84d`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-HIGH-DIMENSIONAL-001-08b4870cfa97a44e.json`

120. Derivation 112: Fold many-body computation law

Claim: `SFT-COMP-SCI-MANY-BODY-001`

120.1 Scientific necessity and exact theorem

Fold many-body computation law is required to derive composed local forms and interaction support without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced composed local forms and interaction support kernel retains body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-HIGH-DIMENSIONAL-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

120.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for composed local forms and interaction support.

> *All generated finite composed local forms and interaction support structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. No physical many-body law or quantum amplitude is imported.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

120.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced composed local forms and interaction support kernel retains body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

120.4 Operational laws and consequences

- Carrier law: body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

120.5 Depth-independent closure

Base: The structural One supplies one canonical composed local forms and interaction support form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated composed local forms and interaction support form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

120.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-MANY-BODY-001; result: `PASS`.

- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-MANY-BODY-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-MANY-BODY-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No physical many-body law or quantum amplitude is imported.	`PASS`

120.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: many-body computation, interaction system.

No physical many-body law or quantum amplitude is imported.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- No physical many-body law or quantum amplitude is imported.

120.8 Exact evidence identities

- Source manifest: `sha256:1836ffc81b8d6aedb59f67ef78d14e2a426b903858e85b7e47472d933b2c1e94`
- Complete-census certificate: `sha256:e8b0d3c3872d072d24e85200a62182589347d59df0a5cef00b81711e56d64c7e`
- Derivation seal: `sha256:9150974b48199453faca9a3ec42c5774fd3ea6dbd947b91fe3fcbb4a634421ab`
- Independent implementation: `sha256:1d2047ea1a1f1b8bf3b16ff8e2608a6d87b0df913c0a975134ffca4e7b842534`
- Independent certificate: `sha256:b478fd50145544b87303e4c72b0ed5f442a50fca78cdebbdf0fa26609d28b8d8`
- External validation: `sha256:3c8b8b90e8e91e724142f784b5c7d34340401fab6d2fabcdc2b6c40547f84649`
- Model-admitted engine receipt: `sha256:4e4db714accde19f571063b6d1c1c64c9f07650378a797107cb424bea54de4ca`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-MANY-BODY-001-4e4db714accde19f.json`

121. Derivation 113: Fold mathematical-modelling law

Claim: `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`

121.1 Scientific necessity and exact theorem

Fold mathematical-modelling law is required to derive explicit relation between formal structures and observations without importing its conventional answer-producing model.

The engine-admitted theorem is:

> *The forced explicit relation between formal structures and observations kernel retains phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-SCI-MANY-BODY-001`
- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-MATH-CATEGORY-TYPE-COMPOSITION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

121.2 Generated grammar and exact boundary

Admitted finite forms, exact relations, distinction ledgers and composition supply the carriers. Exhausting the eight group-specific axes forces the only structure retaining: phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records.

Generation rule: Generate the literal product of the eight registered Scientific Computation structural axes for explicit relation between formal structures and observations.

> *All generated finite explicit relation between formal structures and observations structures whose carriers, relations, traces and interfaces are composed from admitted Foundation, Mathematics and Information Science forms. A successful fit or opaque prediction does not force the model law.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`model`	`unstated-model`	unstated-model removes a required exact model condition.	`registered-exact-model-relation`	registered-exact-model-relation retains the complete registered model condition.
`inputs`	`sampled-or-fitted-inputs`	sampled-or-fitted-inputs removes a required exact inputs condition.	`complete-declared-input-support`	complete-declared-input-support retains the complete registered inputs condition.
`operation`	`opaque-calculation`	opaque-calculation removes a required exact operation condition.	`exact-executable-operation-trace`	exact-executable-operation-trace retains the complete registered operation condition.
`error`	`unsigned-error-scalar`	unsigned-error-scalar removes a required exact error condition.	`orientation-labelled-error-ledger`	orientation-labelled-error-ledger retains the complete registered error condition.
`closure`	`visual-convergence`	visual-convergence removes a required exact closure condition.	`base-successor-closure-certificate`	base-successor-closure-certificate retains the complete registered closure condition.
`validation`	`fit-only`	fit-only removes a required exact validation condition.	`sealed-falsifiable-comparison-boundary`	sealed-falsifiable-comparison-boundary retains the complete registered validation condition.
`generality`	`single-resolution`	single-resolution removes a required exact generality condition.	`generated-refinement-successor`	generated-refinement-successor retains the complete registered generality condition.
`addition`	`imported-equation-or-parameter`	imported-equation-or-parameter removes a required exact addition condition.	`no-extra-scientific-prior`	no-extra-scientific-prior retains the complete registered addition condition.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

121.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	unstated-model removes a required exact model condition.
64	sampled-or-fitted-inputs removes a required exact inputs condition.
32	opaque-calculation removes a required exact operation condition.
16	unsigned-error-scalar removes a required exact error condition.
8	visual-convergence removes a required exact closure condition.
4	fit-only removes a required exact validation condition.
2	single-resolution removes a required exact generality condition.
1	imported-equation-or-parameter removes a required exact addition condition.

Shared claim-record clause `COMP-S014` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

> *The forced explicit relation between formal structures and observations kernel retains phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

121.4 Operational laws and consequences

- Carrier law: phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records.
- Trace law: every accepted result reconstructs from the exact initial form and retained transition or relation rows.
- Boundary law: unregistered forms, powers, parameters, models and omitted rows halt the computation claim.
- Composition law: serial or parallel assembly preserves interfaces, component identities and lost/retained distinction ledgers.

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

121.5 Depth-independent closure

Base: The structural One supplies one canonical explicit relation between formal structures and observations form with its identity trace and no unrecorded predecessor.

Successor: Appending one generated explicit relation between formal structures and observations form adds its exact carrier row, relations, trace positions and new pair/interface distinctions while preserving every earlier record.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

121.6 Executed witnesses and adverse controls

- `scientific_computation-witness-1` - scientific_computation exact operational witness 1 for SFT-COMP-SCI-MATHEMATICAL-MODELLING-001; result: `PASS`.
- `scientific_computation-witness-2` - scientific_computation exact operational witness 2 for SFT-COMP-SCI-MATHEMATICAL-MODELLING-001; result: `PASS`.
- `scientific_computation-witness-3` - scientific_computation exact operational witness 3 for SFT-COMP-SCI-MATHEMATICAL-MODELLING-001; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	unstated-model removes a required exact model condition.	`PASS`

Control	Expected	Observed	Result
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A successful fit or opaque prediction does not force the model law.	`PASS`

121.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: mathematical modelling, scientific model.

A successful fit or opaque prediction does not force the model law.

The following imports remain prohibited at this boundary:

- no conventional machine, algorithm, language, security definition or learning model may select the law
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no completed infinity or ungenerated continuum
- no free, fitted or learned parameter
- no application result or pretrained model
- A successful fit or opaque prediction does not force the model law.

121.8 Exact evidence identities

- Source manifest: `sha256:e2da80d2e34c9e77c9c70e6fa18a7209bc39d59b1b69c4d7ab4f22c142f69eb4`
- Complete-census certificate: `sha256:c3bab5094675d4dc9b452804f521d6e149289070180ba9353b2590e635621d90`
- Derivation seal: `sha256:bf1cfe19e9f1107fc446eeb94708330d2a1f69e6593d9d1a18fcb2ef7938d7ec`
- Independent implementation: `sha256:5be00110e135b7023427d7d71b21f44efbe3ace7848a73e02c41a286eb0a78de`
- Independent certificate: `sha256:5599d7b3549eba1bb43cd7545c725f9fd7ddabf910f916c3ca1aeb0dc254e621`
- External validation: `sha256:852cfe82540a75b5e540d04049e107608bf2ddc7c663eca6626a69868ae516aa`
- Model-admitted engine receipt: `sha256:465c067c8c543effc5ba03cf8677a5caa2386f4695d1f3905b6e652060ba7ed4`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-SCI-MATHEMATICAL-MODELLING-001-465c067c8c543eff.json`

122. Derivation 114: Unrestricted native Fold Busy-Beaver law

Claim: `SFT-COMP-CBL-NATIVE-BUSY-BEAVER-002`

122.1 Scientific necessity and exact theorem

V2 Step 404 asserts an unrestricted theorem over the native Fold grammar; a bounded table or generic halting law does not close that same-strength value.

The engine-admitted theorem is:

> *For every supplied positive finite description depth k in the admitted native Fold closing-process grammar, every process halts within k lawful depth-lowering transitions and a depth-k source attains k; therefore BB_F(k)=k. Period-b recurrent processes carry an exact nonhalting certificate and are excluded from the closing maximum.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-HALTING-001`
- `SFT-COMP-CPLX-TIME-SPACE-001`
- `SFT-COMP-FORM-OPERATIONAL-PROCESS-001`

122.2 Generated grammar and exact boundary

The admitted machine has only the unique depth-lowering edge on a closing path. Complete word generation supplies the upper bound; any depth-k word supplies the equal lower witness; the label-prepending successor makes the equality depth-independent.

Generation rule: Generate the literal product of grammar, transition, halt, maximum, recurrence, successor, evidence and no-import coordinates.

> *Every positive finite native Fold description built only from generated fibre-label words, lawful depth-lowering Fold edges and separately registered recurrent period-b processes.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`grammar`	`external-transition-tables`	An external table imports a machine grammar.	`complete-native-fold-descriptions`	Every generated native word through the supplied depth occurs.
`transition`	`chosen-instruction-step`	A chosen instruction set changes the machine.	`one-lawful-depth-lowering-edge`	One Fold edge closes exactly one word position.
`halting`	`assumed-terminal`	A terminal label without a trace is not a halt certificate.	`exact-empty-one-terminal-trace`	The complete suffix trace reaches structural empty One.
`maximum`	`sampled-long-run`	Examples cannot establish the maximum over complete support.	`complete-upper-and-attaining-lower-witness`	All runs give the upper bound and a depth-k word attains it.
`recurrence`	`cycle-counted-as-halt`	A recurrent return has no terminal state.	`separate-exact-nonhalting-certificate`	The period-b return is classified outside the closing maximum.
`successor`	`depth-fourteen-table-only`	A long table is not a general certificate.	`prepend-one-label-successor`	Prepending one label adds exactly one lawful transition and one maximum unit.
`evidence`	`reported-number`	A number alone hides the process population.	`all-process-traces-and-attainment`	Every closing description retains its exact trace and the longest witness.
`addition`	`conventional-busy-beaver-import`	Conventional Turing tables would select a different grammar.	`no-extra-machine-premise`	The result is internal to the already-derived native Fold machine.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

122.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	An external table imports a machine grammar.
64	A chosen instruction set changes the machine.
32	A terminal label without a trace is not a halt certificate.
16	Examples cannot establish the maximum over complete support.
8	A recurrent return has no terminal state.
4	A long table is not a general certificate.
2	A number alone hides the process population.
1	Conventional Turing tables would select a different grammar.

The unique all-preserving member is `complete-native-fold-descriptions__one-lawful-depth-lowering-edge__exact-empty-one-terminal-trace__complete-upper-and-attaining-lower-witness__se` `parate-exact-nonhalting-certificate__prepend-one-label-successor__all-process-traces-and-attainment__no-extra-machine-premise`.

> *The unique complete kernel forces BB_F(k)=k for every supplied positive finite k in the admitted native Fold closing grammar, with recurrent processes separately certified nonhalting.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

122.4 Operational laws and consequences

- every closing edge lowers exact word depth once
- every native closing word of depth at most k halts within k
- a depth-k word attains k
- prepending one fibre label increments the maximum by one
- recurrent period-b processes never enter the closing maximum

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

122.5 Depth-independent closure

Base: At the first positive depth, either fibre label reaches structural empty One in one lawful edge, so BB_F(One)=One.

Successor: Prepending one generated fibre label to every depth-k word creates all depth-(k+1) words and adds exactly one mandatory closing edge; no shallower word exceeds that trace.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

122.6 Executed witnesses and adverse controls

- `depths-through-fourteen` - Complete native populations through depth fourteen attain their exact depth and no run exceeds it.; result: `PASS`.
- `first-attains` - The first depth-k word supplies an exact k-edge lower witness.; result: `PASS`.
- `recurrence-boundary` - A two-label recurrent alternation returns after its period without structural empty-One termination.; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	An external table imports a machine grammar.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no external Turing transition table; no numerical-zero description depth; no completed infinite description; no recurrent process counted as halting; no claim about conventional Busy Beaver	`PASS`

122.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: Busy Beaver, maximum halting time, native machine grammar.

This theorem is unrestricted over positive finite depth in the admitted native Fold process grammar. It makes no statement about arbitrary external Turing-machine tables.

The following imports remain prohibited at this boundary:

- no external Turing transition table
- no numerical-zero description depth
- no completed infinite description

- no recurrent process counted as halting
- no claim about conventional Busy Beaver

122.8 Exact evidence identities

- Source manifest: `sha256:997605ae50f25ccc162f4d3c9ef7ce97abf20f74d9503b48c94468ba96d18b6c`
- Complete-census certificate: `sha256:4ccb57932443a757c985bc418bf31d941d898d4b13bc0c740e1fb33d6c4fcb99`
- Derivation seal: `sha256:06a6a2865e474cc04ca2997bc2676bb1905e407384a36b51fb78789ae6b33da0`
- Independent implementation: `sha256:d24933201019bd380f9627f20b1b9f834afa2dd24d7f74ff4c0faf638fbbe112`
- Independent certificate: `sha256:13a7ea9a33905f153390064af3a4bb96e4c7d487f820299637e10adb09fa1c61`
- External validation: `sha256:ae2e4685c91d7f9963a3403ccf6aa90dc8023e0d0efba2914d75c36c100c7519`
- Model-admitted engine receipt: `sha256:92715d1d41f9021e01c27ac9962f7fa8c172b1c5237dd623bbab6da5bc1d143d`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CBL-NATIVE-BUSY-BEAVER-002-92715d1d41f9021e.json`

123. Derivation 115: Fold-P equals Fold-NP law

Claim: `SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002`

123.1 Scientific necessity and exact theorem

V2 Step 405 closes a native equality that the archived inventory explicitly left outside scope; equal-strength reconstruction requires the two containments and their boundary.

The engine-admitted theorem is:

> *Within the admitted native Fold process grammar, exact deterministic evaluation emits the complete lawful trace verified by the already-derived proof system; soundness makes every accepted certificate equal the unique evaluation. Both resources equal description depth, so P_F=NP_F throughout this grammar.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-CBL-RECOGNITION-DECISION-001`
- `SFT-COMP-CPLX-TIME-SPACE-001`
- `SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001`
- `SFT-COMP-SEM-VERIFICATION-001`

123.2 Generated grammar and exact boundary

The previously forced evaluator and proof system act on the same generated word and the same unique suffix edges. Evaluation emits the proof, and proof soundness admits no alternative result; their exact depth resources coincide.

Generation rule: Generate the literal product of domain, evaluator, certificate, verifier, two containments, successor and external-boundary coordinates.

> *Every generated positive finite native Fold word, its unique suffix-closing evaluation and certificates composed only from the registered trace proof system.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`domain`	`arbitrary-external-languages`	External languages import encodings and machines.	`complete-native-fold-word-family`	Every generated native word is included.

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`evaluator`	`answer-only-evaluation`	A verdict without its trace hides resource and soundness.	`unique-exact-closing-trace`	Evaluation retains every forced edge.
`certificate`	`unconstrained-advice`	Unconstrained advice imports answer power.	`compiled-lawful-evaluation-trace`	The certificate is exactly the source-bound transition trace.
`verifier`	`test-sampled-verifier`	Examples cannot prove soundness over generated support.	`edgewise-sound-complete-verifier`	The verifier checks source, every forced edge, terminal and length.
`p_to_np`	`assumed-first-containment`	A familiar containment name is not evidence.	`evaluator-emits-accepted-certificate`	Every exact evaluation supplies an accepted trace.
`np_to_p`	`nondeterministic-branch-import`	A foreign branch model changes the grammar.	`soundness-forces-unique-evaluation`	Every accepted trace equals the deterministic suffix run.
`successor`	`depth-seven-only`	Finite testing alone is not depth-independent.	`prepend-label-trace-successor`	One new label adds one evaluator and verifier edge.
`boundary`	`export-to-arbitrary-p-np`	External languages, encodings and polynomial conventions are absent.	`native-equality-only`	The equality remains exactly inside the admitted Fold grammar.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

123.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	External languages import encodings and machines.
64	A verdict without its trace hides resource and soundness.
32	Unconstrained advice imports answer power.
16	Examples cannot prove soundness over generated support.
8	A familiar containment name is not evidence.
4	A foreign branch model changes the grammar.
2	Finite testing alone is not depth-independent.
1	External languages, encodings and polynomial conventions are absent.

The unique all-preserving member is `complete-native-fold-word-family__unique-exact-closing-trace__compiled-lawful-evaluation-trace__edgewise-sound-complete-verifier__evaluator-emits-accepted-certificate__soundness-forces-unique-evaluation__prepend-label-trace-successor__native-equality-only`.

> *The sole all-preserving kernel forces P_F=NP_F with exact resource k for every supplied positive finite native description depth, without asserting conventional P=NP.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

123.4 Operational laws and consequences

- every deterministic evaluation emits its accepted exact trace
- every accepted exact trace equals the unique native evaluation
- evaluation and verification each traverse one edge per word position
- the two exact containments force native equality

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

123.5 Depth-independent closure

Base: At the first positive depth, the single edge is both the evaluation and its complete verified certificate.

Successor: Prepending either native label adds exactly one forced evaluation edge and one verifier check while preserving source, suffix and terminal equality.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

123.6 Executed witnesses and adverse controls

- `complete-depth-seven` - Every native source through depth seven has one accepted exact evaluation certificate.; result: `PASS`.
- `resource-equality` - Evaluation and verification lengths equal source depth through fourteen.; result: `PASS`.
- `tamper-rejection` - A changed first certificate state rejects for every depth through seven.; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	External languages import encodings and machines.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no arbitrary external language family; no nondeterministic machine import; no polynomial convention import; no conventional P-versus-NP conclusion; no hidden certificate advice	`PASS`

123.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: P versus NP, certificate verification, native complexity class.

This equality is confined to the admitted Fold word/process grammar. It does not decide conventional P versus NP for arbitrary languages, machines, encodings or Boolean gate bases.

The following imports remain prohibited at this boundary:

- no arbitrary external language family
- no nondeterministic machine import
- no polynomial convention import
- no conventional P-versus-NP conclusion
- no hidden certificate advice

123.8 Exact evidence identities

- Source manifest: `sha256:b9170896e049b2b60486de4a1d996676c7d1727624e35c870fdb7f755b4f9c82`
- Complete-census certificate: `sha256:9e1faaea7df0d45b834041b4ed840324d5ceb024eba2e6f1755cea212827ddae`
- Derivation seal: `sha256:b32e863be01454e0c156a13913b55a09303333f2a5946bbe412df7126f238b2c`
- Independent implementation: `sha256:3bb193e7add6ae94ba988cf174df28fa4e633efd664263e9ebb44e81654a61c3`
- Independent certificate: `sha256:567b4be1c67de0164587ff7af3849ac615a1cdf7fdf1dbe5d502b051f32a382a`
- External validation: `sha256:7540aff8437d360d4d7885a765194131a419b9bc9c46b096a5e6f52128a0c21f`
- Model-admitted engine receipt: `sha256:a6d19f95f567cd3d870b45330dbcb3b3387b5b3b47390c5ee1b92cbc5d2f6d74`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002-a6d19f95f567cd3d.json`

124. Derivation 116: Arbitrary admitted Fold-circuit lower-bound law

Claim: `SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002`

124.1 Scientific necessity and exact theorem

V2 Step 406 extends circuit resource counting to a necessary lower bound over every admitted circuit; a count of one implementation is not the same theorem.

The engine-admitted theorem is:

> *Across every circuit assembled from admitted Fold edges, closing a depth-k path requires at least k gates, complete source support requires width b^k, and complete layered coverage requires the sum of b^r edges for r from One through k. The registered circuit attains every bound.*

The derivation cites only these earlier model-admitted receipts:

- `SFT-COMP-FORM-CIRCUIT-001`
- `SFT-COMP-CPLX-CIRCUIT-RESOURCE-001`
- `SFT-COMP-CPLX-BOUNDS-001`

124.2 Generated grammar and exact boundary

The unique native edge from each word to its suffix cannot replace another source edge or close two dependent positions. Complete support therefore forces every input vertex and every layered edge; the constructed circuit meets the resulting bounds.

Generation rule: Generate the literal product of gate, path, width, size, subset exhaustion, attainment, successor and gate-basis-boundary coordinates.

> *Every circuit whose vertices are generated native Fold words and whose gates are lawful unique depth-r to depth-(r-1) Fold edges.*

Structural axis	Rejected coordinate	Exact rejection	Forced coordinate	Preservation supplied
`gate`	`rewired-or-multi-depth-gate`	A rewired or multi-depth edge is not a Fold transition.	`unique-lawful-one-depth-edge`	Each source has its forced suffix edge.
`path`	`reported-k-depth`	A reported implementation depth is not a lower bound.	`one-distinction-per-dependent-edge`	No lawful edge closes two dependent word positions.
`width`	`sampled-input-width`	Sampled sources omit generated inputs.	`all-b-to-k-source-words`	Every exact depth-k input requires its held source vertex.
`size`	`one-implementation-count`	Counting one circuit does not prove necessity.	`every-source-edge-required`	Each source word has one distinct forced outgoing edge.
`exhaustion`	`selected-subsets`	Selected omissions cannot prove arbitrary-circuit necessity.	`complete-forced-edge-subset-census`	Every subset at the declared census depth is classified.
`attainment`	`unattained-bound`	An unattained value need not be tight.	`registered-circuit-attains-all-bounds`	The complete forced-edge circuit meets path, width and size exactly.
`successor`	`finite-table-only`	A finite census does not generalise itself.	`add-next-complete-source-layer`	Depth successor adds b^(k+1) required source edges and one path edge.
`boundary`	`arbitrary-gate-basis-export`	An external gate can have different fan-in and semantics.	`admitted-fold-circuits-only`	The theorem stays within lawful Fold edges.

Shared claim-record clause `COMP-S001` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

124.3 Exhaustion, eliminations and unique survivor

Shared claim-record clause `COMP-S002` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

Eliminated candidates	Exact first-failure reason
128	A rewired or multi-depth edge is not a Fold transition.
64	A reported implementation depth is not a lower bound.
32	Sampled sources omit generated inputs.
16	Counting one circuit does not prove necessity.

Eliminated candidates	Exact first-failure reason
8	Selected omissions cannot prove arbitrary-circuit necessity.
4	An unattained value need not be tight.
2	A finite census does not generalise itself.
1	An external gate can have different fan-in and semantics.

The unique all-preserving member is `unique-lawful-one-depth-edge__one-distinction-per-dependent-edge__all-b-to-k-source-words__every-source-edge-required__complete-forced-edge-subset-census__registered-circuit-attains-all-bounds__add-next-complete-source-layer__admitted-fold-circuits-only`.

> *The unique kernel forces tight path k, width b^k and size sum(r=One..k)b^r lower bounds over every admitted Fold-edge circuit at every supplied positive finite depth.*

Shared claim-record clause `COMP-S004` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

124.4 Operational laws and consequences

- one lawful edge closes one dependent position
- every complete source word requires its own source vertex and outgoing edge
- all forced edges are necessary for complete layered coverage
- the complete registered circuit attains the three lower bounds

Shared claim-record clause `COMP-S005` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

124.5 Depth-independent closure

Base: At first positive depth there are b source words, one required edge from each, path depth One and width b.

Successor: Adding the next word position creates b^(k+1) new source words and forced edges, increases every closing path by one and preserves every prior layer.

Shared claim-record clause `COMP-S006` **applies.** The full, unchanged clause is preserved in the shared-clause appendix.

124.6 Executed witnesses and adverse controls

- `exact-resources` - Path, width and size match k, b^k and the complete edge sum through depth fourteen.; result: `PASS`.
- `subset-census` - Every subset of the fourteen forced edges through colour depth is exhausted and exactly the full set survives.; result: `PASS`.
- `attainment` - The complete forced-edge set contains every source edge once at every depth through fourteen.; result: `PASS`.

Control	Expected	Observed	Result
`false_premise`	reject the generated form lacking complete first-axis coverage	A rewired or multi-depth edge is not a Fold transition.	`PASS`
`tampered_source`	reject a changed official source identity	changed source identity differs	`PASS`
`tampered_artifact`	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities	`PASS`
`boundary`	halt imported models, parameters or excluded quantum machinery	no Boolean or external quantum gate basis; no rewired Fold edge; no sampled input support; no completed infinite circuit; no claim beyond admitted Fold circuits	`PASS`

124.7 Meaning, correspondence and limitations

The result closes the exact generated-finite kernel stated above. Conventional names enter only now, after the derivation seal, as correspondence labels: circuit lower bound, circuit depth, circuit width, circuit size.

The theorem covers every circuit in the admitted native Fold-edge grammar and does not assert lower bounds for external Boolean, arithmetic or quantum gate bases.

The following imports remain prohibited at this boundary:

- no Boolean or external quantum gate basis
- no rewired Fold edge
- no sampled input support
- no completed infinite circuit
- no claim beyond admitted Fold circuits

124.8 Exact evidence identities

- Source manifest: `sha256:068fbb0f57e3de411c7a1eafe20d2dfe573fb591359fcc5f9a79538da911b4ad`
- Complete-census certificate: `sha256:8634fe47e8c4a7a0cb69b91d5df2bad956c2cbc0edf8a0904af9696dd71c2f3d`
- Derivation seal: `sha256:99debdca97c8fab7140ac9aa63f6bb03ddccd72ed98eb8e9f306e1c5192a491c`
- Independent implementation: `sha256:9d8af6abadf3adf8437598d4b2f9a0eed3e98d6c59b97680e118de4cdba32702`
- Independent certificate: `sha256:b9f8d7a26d4cfe3e7a36f1d7baf36c71b581a97997fe0a86476c16be703d9ff5`
- External validation: `sha256:f66b8740ae0f7534381bb3ab6210a051b3a607b024f0cf9734ba4d4b8f1ae3d3`
- Model-admitted engine receipt: `sha256:18e485a78cc263bd16ceae80afee117f5971e549802d60b696645652b596074d`
- Receipt path: `receipts/engine/model_admitted/SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002-18e485a78cc263bd.json`

125. Branch synthesis and consequences

The 116 laws form one dependency chain from exact state change to models, computability, resources, algorithms, meaning, distributed knowledge, adversaries, learning and scientific calculation. The common invariant is complete provenance: every input belongs to a generated support, every operation names its relation, every execution retains a trace, every resource count names the representation, every loss names closed distinctions and every claimed generality supplies a successor certificate or an explicit finite boundary.

126. Complete V1/V2 reconstruction audit

The branch ownership audit reads all 763 V1/V2 census entries, decomposes the 70 computation-relevant source entries into 134 atomic obligations and closes 134 at same strength, with 0 open. It preserves 4 explicit corrections or reconciliations instead of hiding them behind a branch-level status.

Source	Entry	Atomic obligation	Disposition	V3 receipts
V1	`XII-4`	Every process on a complete finite generated carrier is decidable as terminal or recurrent by exhaustive state execution.	`closed`	`SFT-COMP-CBL-RECOGNITION-DECISION-001`, `SFT-COMP-CBL-HALTING-001`
V1	`XII-4`	The unrestricted total internal decider fails where generated self-description and held complement produce the exact contradiction trace.	`reconciled_proof_correction`	`SFT-COMP-CBL-UNDECIDABILITY-001`
V1	`XII-4`	Finite decidability does not license an undeclared oracle or completed-infinite computation.	`closed`	`SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001`

Source	Entry	Atomic obligation	Disposition	V3 receipts
V1	XII-5	The conventional P-versus-NP question remains outside the imported-language boundary, while the later native Fold comparison is separately forced.	reconciled_later_native_closure	SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002
V1	XIV-4	Every bounded generated sub-process has a complete exact simulation trace.	closed	SFT-COMP-SCI-SIMULATION-001, SFT-COMP-CBL-UNIVERSAL-MACHINE-001
V1	XIV-4	A complete self-containing one-to-one internal simulation is blocked by self-description and observation loss.	closed	SFT-COMP-CBL-INCOMPLETENESS-001, SFT-COMP-CBL-UNDECIDABILITY-001
V1	XIV-4	Nested simulations are admitted only as generated finite composed processes with explicit resource and interface ledgers.	closed	SFT-COMP-FORM-COMPOSITION-001, SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001
V1	XIV-8	Native Fold search follows its exact generated transition depth rather than importing a logarithmic or heuristic answer model.	reconciled_no_logarithm	SFT-COMP-ALG-SEARCH-ORDER-001, SFT-COMP-CPLX-TIME-SPACE-001
V1	XIV-8	A bounded process is guaranteed to halt or cycle and the class is decidable; it is not falsely guaranteed to halt.	reconciled_halt_or_cycle_correction	SFT-COMP-CBL-HALTING-001, SFT-COMP-CBL-RECOGNITION-DECISION-001
V1	XIV-8	Distributed integration requires exact synchronization, causal and consensus records rather than a hidden coordinator.	closed	SFT-COMP-DIST-SYNCHRONIZATION-001, SFT-COMP-DIST-CONSENSUS-001
V1	XIV-8	A registered invariant and complete proof trace provide an immediate correctness and verification boundary.	closed	SFT-COMP-SEM-CORRECTNESS-001, SFT-COMP-SEM-VERIFICATION-001
V1	X-3	A definite generated classical word can be read and copied with full source and output provenance.	closed	SFT-COMP-ALG-STRINGS-SEQUENCES-001, SFT-COMP-DIST-REPLICATION-CONSISTENCY-001
V1	X-3	The minimal internal replicator is a held template composed with a source-bound copy process; external process or file replication is not required.	closed	SFT-COMP-FORM-RECURSIVE-FUNCTION-001, SFT-COMP-FORM-OPERATIONAL-PROCESS-001, SFT-COMP-FORM-COMPOSITION-001
V1	C-1	A lawful simulation is an exact registered state-transition trace driven only by its sealed model relation.	closed	SFT-COMP-SCI-SIMULATION-001, SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
V1	C-2	A multi-stage simulation preserves the dependency order and proof identity of every stage.	closed	SFT-COMP-FORM-COMPOSITION-001, SFT-COMP-SCI-SIMULATION-001
V2	325	Same-strength reconstruction of Exact state and transition law: The forced generated configurations and change kernel retains complete state carrier; held action labels; source-bound transition rows; initial and terminal classes; retained execution traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-STATE-TRANSITION-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	327	Same-strength reconstruction of Fold input and problem-size law: The forced exact generated input extent kernel retains canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-INPUT-SIZE-001
V2	327	Same-strength reconstruction of Fold time and space resource law: The forced transition and retained-state resources kernel retains complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-TIME-SPACE-001
V2	330	Same-strength reconstruction of Fold operational-process law: The forced execution and observable process behavior kernel retains initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-OPERATIONAL-PROCESS-001
V2	330	Same-strength reconstruction of Fold process composition and decomposition law: The forced serial and parallel process assembly kernel retains exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-COMPOSITION-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	331	Same-strength reconstruction of Fold formal-language and grammar law: The forced generated symbol words and formation kernel retains complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-LANGUAGE-GRAMMAR-001
V2	332	Same-strength reconstruction of Fold finite automaton law: The forced finite recognition processes kernel retains finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-AUTOMATON-001
V2	333	Same-strength reconstruction of Fold rewriting-system law: The forced held-position replacement kernel retains generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-REWRITING-001
V2	334	Same-strength reconstruction of Fold recursion law: The forced structural finite recursion kernel retains structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-RECURSIVE-FUNCTION-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	334	Same-strength reconstruction of Fold binding and self-application calculus law: The forced binding, substitution and reduction kernel retains generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-LAMBDA-CALCULUS-001
V2	335	Same-strength reconstruction of Fold computational-model equivalence law: The forced bidirectional execution-preserving translations kernel retains complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-MODEL-EQUIVALENCE-001
V2	336	Same-strength reconstruction of Fold universal-computation law: The forced interpretation of generated machine descriptions kernel retains complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-UNIVERSALITY-001
V2	337	Same-strength reconstruction of Fold recognition and decision law: The forced acceptance, rejection and continued execution kernel retains generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CBL-RECOGNITION-DECISION-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	338	Same-strength reconstruction of Fold halting boundary law: The forced terminal-state determination kernel retains encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CBL-HALTING-001
V2	339	Same-strength reconstruction of Fold effective-enumeration law: The forced generated description ordering kernel retains complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CBL-ENUMERATION-001
V2	340	Same-strength reconstruction of Fold computability-reduction law: The forced problem translation preserving answers kernel retains source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CBL-REDUCTION-001
V2	341	Same-strength reconstruction of Fold undecidable-problem law: The forced failure of total internal decision kernel retains generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CBL-UNDECIDABILITY-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	342	Same-strength reconstruction of Fold relative and oracle-computation law: The forced explicit external answer interfaces kernel retains base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CBL-RELATIVE-ORACLE-001
V2	343	Same-strength reconstruction of Fold hypercomputation admissibility-limit law: The forced claims exceeding generated universal computation kernel retains claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001
V2	344	Same-strength reconstruction of Fold degrees-of-computability law: The forced relative reduction classes kernel retains registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CBL-DEGREES-001
V2	345	Same-strength reconstruction of Fold incompleteness-boundary law: The forced internal proof and self-verification limits kernel retains effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CBL-INCOMPLETENESS-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	346	Same-strength reconstruction of Fold circuit size and depth law: The forced circuit structural resources kernel retains generated gates; wires; dependency order; longest causal layer trace; output support; shared-subcircuit identity; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-CIRCUIT-RESOURCE-001
V2	347	Same-strength reconstruction of Fold communication and query complexity law: The forced cross-interface and information-request resources kernel retains distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-COMMUNICATION-QUERY-001
V2	348	Same-strength reconstruction of Fold randomness-resource law: The forced scheduled unresolved branch support kernel retains complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-RANDOMNESS-001
V2	349	Same-strength reconstruction of Fold reversibility-cost law: The forced information retained for inverse execution kernel retains source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-REVERSIBILITY-COST-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	350	Same-strength reconstruction of Fold parallel-complexity law: The forced independent and causally layered work kernel retains task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-PARALLEL-001
V2	352	Same-strength reconstruction of Fold lower- and upper-bound law: The forced resource comparison certificates kernel retains problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-BOUNDS-001
V2	352	Same-strength reconstruction of Fold reduction and completeness law: The forced resource-preserving problem translations kernel retains problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001
V2	353	Same-strength reconstruction of Fold average- and worst-case law: The forced complete instance-family resource profiles kernel retains generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-AVERAGE-WORST-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	354	Same-strength reconstruction of Fold approximation-complexity law: The forced resource versus retained objective distinction kernel retains optimization domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-APPROXIMATION-001
V2	355	Same-strength reconstruction of Fold parameterized-complexity law: The forced resource stratification by exact structural coordinates kernel retains input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-PARAMETERIZED-001
V2	356	Same-strength reconstruction of Fold information and descriptive-complexity law: The forced shortest exact generated descriptions kernel retains frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-CPLX-DESCRIPTIVE-001
V2	357	Same-strength reconstruction of Fold searching and ordering law: The forced location and exact order construction kernel retains generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-SEARCH-ORDER-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	358	Same-strength reconstruction of Fold exact arithmetic-algorithm law: The forced computation over exact positive parts kernel retains canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-ARITHMETIC-001
V2	359	Same-strength reconstruction of Fold string and sequence algorithm law: The forced finite ordered symbol organizations kernel retains complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-STRINGS-SEQUENCES-001
V2	360	Same-strength reconstruction of Fold tree and graph algorithm law: The forced finite relational traversal and structure kernel retains generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-TREES-GRAPHS-001
V2	361	Same-strength reconstruction of Fold algebraic and geometric algorithm law: The forced finite algebraic operations and incidence kernel retains exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	362	Same-strength reconstruction of Fold algebraic and geometric algorithm law: The forced finite algebraic operations and incidence kernel retains exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001
V2	363	Same-strength reconstruction of Fold dynamic-programming law: The forced reuse of exact overlapping subproblem results kernel retains problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001
V2	364	Same-strength reconstruction of Fold optimization-algorithm law: The forced selection of all exact optima kernel retains complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-OPTIMIZATION-001
V2	365	Same-strength reconstruction of Fold randomized-algorithm law: The forced complete scheduled branch execution kernel retains deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-RANDOMIZED-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	366	Same-strength reconstruction of Fold parallel-algorithm law: The forced causally independent algorithm steps kernel retains dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-PARALLEL-001
V2	367	Same-strength reconstruction of Fold distributed-algorithm law: The forced local processes exchanging exact records kernel retains process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-DISTRIBUTED-001
V2	368	Same-strength reconstruction of Fold online and streaming algorithm law: The forced prefix-bounded processing kernel retains generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-ONLINE-STREAMING-001
V2	369	Same-strength reconstruction of Fold numerical-algorithm law: The forced exactly bounded approximation procedures kernel retains exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-NUMERICAL-001
V2	370	Same-strength reconstruction of Fold symbolic-computation law: The forced canonical expression transformation kernel retains generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-SYMBOLIC-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	371	Same-strength reconstruction of Fold approximation-algorithm law: The forced bounded resource solution with exact error ledger kernel retains feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-ALG-APPROXIMATE-001
V2	373	Same-strength reconstruction of Fold program-syntax law: The forced generated program forms kernel retains complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-SYNTAX-001
V2	374	Same-strength reconstruction of Fold binding and substitution law: The forced name identity and replacement kernel retains held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-BINDING-SUBSTITUTION-001
V2	375	Same-strength reconstruction of Fold evaluation law: The forced program-state execution kernel retains syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-EVALUATION-001
V2	376	Same-strength reconstruction of Fold operational and denotational correspondence law: The forced trace-to-meaning agreement kernel retains operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	377	Same-strength reconstruction of Fold type-theory law: The forced structural admissibility of composition kernel retains generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-TYPE-001
V2	378	Same-strength reconstruction of Fold program-equivalence law: The forced indistinguishable generated behavior kernel retains two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001
V2	379	Same-strength reconstruction of Fold termination law: The forced absence of continuing execution paths kernel retains program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-TERMINATION-001
V2	379	Same-strength reconstruction of Fold program-correctness law: The forced agreement between execution and specification kernel retains precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-CORRECTNESS-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	380	Same-strength reconstruction of Fold formal-specification law: The forced exact admissible behavior descriptions kernel retains input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-SPECIFICATION-001
V2	381	Same-strength reconstruction of Fold program-transformation law: The forced semantics-preserving program rewriting kernel retains source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-TRANSFORMATION-001
V2	382	Same-strength reconstruction of Fold semantics-preserving compilation law: The forced translation between executable languages kernel retains source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-COMPILATION-001
V2	383	Same-strength reconstruction of Fold verification and proof-system law: The forced machine-checkable program evidence kernel retains specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEM-VERIFICATION-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	392	Same-strength reconstruction of Fold binding and self-application calculus law: The forced binding, substitution and reduction kernel retains generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-LAMBDA-CALCULUS-001
V2	392	Same-strength reconstruction of Fold circuit syntax and evaluation law: The forced finite connected local transformations kernel retains generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-FORM-CIRCUIT-001
V2	393	Same-strength reconstruction of Fold concurrency law: The forced multiple enabled process transitions kernel retains process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-CONCURRENCY-001
V2	393	Same-strength reconstruction of Fold computational-causality law: The forced event predecessor structure kernel retains generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-CAUSALITY-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	393	Same-strength reconstruction of Fold synchronization law: The forced jointly enabled process transitions kernel retains participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-SYNCHRONIZATION-001
V2	393	Same-strength reconstruction of Fold communication law: The forced exact record transfer kernel retains sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-COMMUNICATION-001
V2	393	Same-strength reconstruction of Fold distributed partial-order law: The forced causal event ordering kernel retains event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-PARTIAL-ORDER-001
V2	393	Same-strength reconstruction of Fold consensus law: The forced terminal common decision kernel retains process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-CONSENSUS-001
V2	393	Same-strength reconstruction of Fold agreement-impossibility law: The forced indistinguishable predecessor limits kernel retains two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	393	Same-strength reconstruction of Fold replication and consistency law: The forced multiple retained state copies kernel retains replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-REPLICATION-CONSISTENCY-001
V2	393	Same-strength reconstruction of Fold computational fault-model law: The forced declared deviations from process rules kernel retains nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-FAULT-MODEL-001
V2	393	Same-strength reconstruction of Fold distributed-knowledge law: The forced information available to local processes kernel retains global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001
V2	393	Same-strength reconstruction of Fold locality law: The forced bounded dependency neighborhoods kernel retains network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-LOCALITY-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	393	Same-strength reconstruction of Fold network-computation law: The forced computation over generated communication graphs kernel retains network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-DIST-NETWORK-COMPUTATION-001
V2	394	Same-strength reconstruction of Fold one-wayness law: The forced forward computation versus inverse search kernel retains finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-ONE-WAYNESS-001
V2	394	Same-strength reconstruction of Fold secrecy law: The forced observation-relative message indistinguishability kernel retains message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-SECRECY-001
V2	394	Same-strength reconstruction of Fold integrity law: The forced detection of unauthorized record change kernel retains message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-INTEGRITY-001
V2	394	Same-strength reconstruction of Fold authentication law: The forced source-identity verification kernel retains principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-AUTHENTICATION-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	394	Same-strength reconstruction of Fold hashing law: The forced fixed-interface many-to-one record mapping kernel retains message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-HASHING-001
V2	394	Same-strength reconstruction of Fold commitment law: The forced binding and hiding record phases kernel retains message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-COMMITMENT-001
V2	394	Same-strength reconstruction of Fold signature law: The forced publicly checkable source authorization kernel retains message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-SIGNATURE-001
V2	394	Same-strength reconstruction of Fold proof-of-knowledge law: The forced extractable witness-bearing interaction kernel retains statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-PROOF-KNOWLEDGE-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	394	Same-strength reconstruction of Fold zero-knowledge law: The forced verification without additional witness distinction kernel retains statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-ZERO-KNOWLEDGE-001
V2	394	Same-strength reconstruction of Fold multiparty-computation law: The forced joint function evaluation with local privacy kernel retains party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-MULTIPARTY-001
V2	394	Same-strength reconstruction of Fold adversarial-computation law: The forced computation against generated hostile strategies kernel retains system process; adversary grammar; resources; interaction schedules; security property; complete favorable and unfavorable rows; stopping rule; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-ADVERSARIAL-001
V2	394	Same-strength reconstruction of Fold information-theoretic and computational-security law: The forced exact separation of unconditional and resource-bounded security kernel retains secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-SECURITY-DEFINITION-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	394	Same-strength reconstruction of Fold post-quantum security-boundary law: The forced classical claims under an enlarged adversary interface kernel retains classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001
V2	395	Same-strength reconstruction of Fold inference law: The forced selection among generated hypotheses kernel retains hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-INFERENCE-001
V2	395	Same-strength reconstruction of Fold classification and prediction law: The forced sealed mapping from inputs to held labels kernel retains input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001
V2	395	Same-strength reconstruction of Fold representation-learning boundary law: The forced generated encodings preserving declared distinctions kernel retains source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-REPRESENTATION-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	395	Same-strength reconstruction of Fold generalization law: The forced extension from registered evidence to generated unseen support kernel retains hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-GENERALIZATION-001
V2	395	Same-strength reconstruction of Fold sample-complexity law: The forced evidence support required to close hypothesis distinctions kernel retains hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001
V2	395	Same-strength reconstruction of Fold optimization-in-learning law: The forced exact objective selection over hypotheses kernel retains hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001
V2	395	Same-strength reconstruction of Fold computational-induction law: The forced rule construction from finite evidence kernel retains observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-INDUCTION-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	395	Same-strength reconstruction of Fold search and planning law: The forced action-sequence construction toward exact goals kernel retains state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-SEARCH-PLANNING-001
V2	395	Same-strength reconstruction of Fold reinforcement-computation law: The forced policy evaluation through generated interaction traces kernel retains states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-REINFORCEMENT-001
V2	395	Same-strength reconstruction of Fold multi-agent computation law: The forced interacting adaptive processes kernel retains agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-MULTIAGENT-001
V2	395	Same-strength reconstruction of Fold adaptation law: The forced trace-bound change of computational rules or state kernel retains initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-ADAPTATION-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	395	Same-strength reconstruction of Fold computational limits-of-learning law: The forced indistinguishable evidence and unidentifiable hypotheses kernel retains hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-LEARNING-LIMITS-001
V2	395	Same-strength reconstruction of Fold interpretability and verification law: The forced traceable relation from inputs to decisions kernel retains model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001
V2	395	Same-strength reconstruction of Fold classical-learning model law: The forced complete classical finite learning processes kernel retains examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-LEARN-CLASSICAL-LEARNING-001
V2	396	Same-strength reconstruction of Fold exact and approximate calculation law: The forced separation of proof values and bounded approximations kernel retains exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-EXACT-APPROXIMATE-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	396	Same-strength reconstruction of Fold numerical-stability law: The forced output distinction under input or operation perturbation kernel retains exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-STABILITY-001
V2	396	Same-strength reconstruction of Fold error-propagation law: The forced transport of exact approximation distinctions kernel retains local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-ERROR-PROPAGATION-001
V2	396	Same-strength reconstruction of Fold discretization law: The forced finite generated representation of a declared model kernel retains source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-DISCRETIZATION-001
V2	396	Same-strength reconstruction of Fold convergence-certificate law: The forced successor refinement preserving and closing error distinctions kernel retains refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-CONVERGENCE-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	396	Same-strength reconstruction of Fold scientific symbolic-computation law: The forced exact model-expression transformation kernel retains model syntax; exact rewrite rules; dimensional or type interfaces; normalization; identity proof; evaluator correspondence; result trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-SYMBOLIC-001
V2	396	Same-strength reconstruction of Fold simulation law: The forced execution of a registered model relation kernel retains model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-SIMULATION-001
V2	396	Same-strength reconstruction of Fold computational-dynamics law: The forced iterated exact finite state transformation kernel retains state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001
V2	396	Same-strength reconstruction of Fold inverse-problem law: The forced reconstruction of sources from observations kernel retains source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-INVERSE-PROBLEM-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	396	Same-strength reconstruction of Fold computational-statistics law: The forced exact support summaries and inference algorithms kernel retains complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001
V2	396	Same-strength reconstruction of Fold high-dimensional computation law: The forced product-support computation with retained coordinate structure kernel retains generated coordinate carriers; complete product support; sparse or factored relation; algorithm trace; resource ledger; lost-interaction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-HIGH-DIMENSIONAL-001
V2	396	Same-strength reconstruction of Fold many-body computation law: The forced composed local forms and interaction support kernel retains body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-MANY-BODY-001
V2	396	Same-strength reconstruction of Fold mathematical-modelling law: The forced explicit relation between formal structures and observations kernel retains phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.	closed	SFT-COMP-SCI-MATHEMATICAL-MODELLING-001
V2	402	Prepending one native label forces the next complete state and description support.	closed	SFT-COMP-FORM-STATE-TRANSITION-001, SFT-COMP-CPLX-INPUT-SIZE-001
V2	402	The same label successor adds the next circuit layer with exact width, depth and size ledgers.	closed	SFT-COMP-CPLX-CIRCUIT-RESOURCE-001, SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002
V2	402	One new closing edge requires one additional held reverse label.	closed	SFT-COMP-CPLX-REVERSIBILITY-COST-001

Source	Entry	Atomic obligation	Disposition	V3 receipts
V2	404	For every supplied positive finite native description depth k, the complete closing-process maximum is BB_F(k)=k and recurrent processes are certified nonhalting.	closed	SFT-COMP-CBL-NATIVE-BUSY-BEAVER-002
V2	405	Exact evaluation emits the sound complete certificate and sound verification forces the unique evaluation, so P_F=NP_F inside the admitted native grammar.	closed	SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002
V2	406	Every admitted Fold-edge circuit requires path k, width b^k and complete layered edge sum; the registered circuit attains all bounds.	closed	SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002

The audit records four important corrections explicitly:

1. Unboundedness alone is not accepted as a proof of undecidability; the V3 result uses exact self-description and the held-complement diagonal contradiction.
2. A bounded deterministic process need not terminate; it must halt or enter a repeated configuration, and that finite class is decidable.
3. Earlier logarithmic/exponential efficiency prose is replaced by exact Fold description depth, support, transition, space and retained-information resources.
4. Native `P_F = NP_F` is a theorem of the admitted Fold grammar and is not silently exported as a verdict on conventional P versus NP.

127. Computation, open knowledge and institutional accountability

Computational science makes opaque authority mechanically testable. A proprietary oracle, undisclosed training corpus, result-only benchmark or inaccessible validator may be useful engineering, but it cannot establish a closed law because reviewers cannot reconstruct the premise-to-result route, enumerate alternatives or locate the exact failure boundary. This paper therefore publishes the grammars, candidate products, decision vectors, controls, validators, receipts and source rather than asking readers to trust a score or institutional label.

The institutional critique is evidential, not an allegation that every funded scientist or institution acts in bad faith. Reviews report associations between commercial sponsorship and favourable outcomes, influence on research agendas, limits in the evidence for grant-review reliability and fairness, and underpublication of null results. Those mechanisms show why capital, prestige, paywalls and consensus cannot substitute for an inspectable proof and data chain. Expertise remains valuable criticism; it is not an admission receipt.

Maria Smith developed this programme without conventional academic credentials, appointment or a funding gate. That fact does not prove any theorem; the public evidence must carry the entire scientific burden. It demonstrates what credential-first selection can exclude. The issue is not one outsider's biography but every mind lost when status or capital is treated as permission to create testable knowledge.

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128. Reproduction, criticism and falsification

Run `python3 -m sft verify-all` on macOS/Linux or `py -m sft verify-all` on Windows. Reviewers may invalidate a claim by producing an omitted coordinate inside its declared grammar, a second survivor, a simpler preserving form, an induction counterexample, an operational counterexample, a source mismatch or a failed adverse control. Hashes identify artifacts; they do not immunize the grammar from criticism.

129. Scope and next branch

The next dependency branch is Reversible and Quantum Computation. It may use these classical receipts but must independently derive branch support, phase, interference, entanglement, measurement, gates, quantum circuits, algorithms, communication, correction and limits without importing a conventional quantum formalism.

130. Complete-field execution - version 1.4

Version 1.3 froze and published the complete-field roadmap. Version 1.4 executes it. The 116 detailed foundational derivations above remain part of the paper and are not rewritten; this section is controlling for the current 369-obligation denominator and exposes every current claim, observation, boundary and receipt.

130.1 Complete family census

Order	Family	Scientific scope	Claims	Candidates	Controls	Status
1	`BASE`	foundational Fold machine and prior-corpus reconstruction	117	29,952	468	complete, receipt-backed, extension-open
2	`FORMX`	formal computation	22	5,632	88	complete, receipt-backed, extension-open
3	`CBLX`	computability	21	5,376	84	complete, receipt-backed, extension-open
4	`CPLXX`	computational complexity	33	8,448	132	complete, receipt-backed, extension-open
5	`ALGX`	algorithms and mathematical data structures	31	7,936	124	complete, receipt-backed, extension-open
6	`SEMX`	semantics and programming theory	25	6,400	100	complete, receipt-backed, extension-open
7	`DISTX`	concurrent and distributed computation	26	6,656	104	complete, receipt-backed, extension-open
8	`SECX`	cryptography and computational security	25	6,400	100	complete, receipt-backed, extension-open
9	`LEARNX`	learning and intelligence theory	26	6,656	104	complete, receipt-backed, extension-open
10	`SCIX`	scientific computation	25	6,400	100	complete, receipt-backed, extension-open
11	`VALID`	dated validation and reconciliation	12	3,072	48	complete, receipt-backed, extension-open
12	`HAND`	one-owner downstream handoffs	6	1,536	24	complete, receipt-backed, extension-open
Total	**12 families**	**complete frozen field census**	**369**	**94,464**	**1,476**	**369/369**

130.2 Admission constitution applied to every obligation

Each entry below states the scientific question, the forced law, the unique survivor, exhaustive enumeration, adverse controls, post-registration observation, source custody, excluded imports, dependency route and immutable receipt. Hashes identify the evidence; they do not replace the result. Machine-readable packages remain controlling and permit every row to be independently replayed.

130.3 Family `BASE` - 117/117 complete

Scope. Foundational fold machine and prior-corpus reconstruction. This family completely decides 29,952 candidates, retains 117 unique survivors and passes 468 adverse controls.

`SFT-COMP-OBL-BASE-001` - Exact state and transition law

- **Claim and ownership:** `SFT-COMP-FORM-STATE-TRANSITION-001`; Classical Computation family `BASE`; Exact state and transition law.
- **Forced law:** The forced generated configurations and change kernel retains complete state carrier; held action labels; source-bound transition rows; initial and terminal classes; retained execution traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced generated configurations and change kernel retains complete state carrier; held action labels;

source-bound transition rows; initial and terminal classes; retained execution traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not yet define a language, machine model or resource bound.
- **Evidence identities:** derivation `sha256:52d77e7c18c95f893f56bae7f7710e36ce0ab176b01d1f8b868be0ad6e0385f0`; independent implementation `sha256:c2917eb384de1fd43f363ae486fd3db8c6670b6199a9f0de76519dc719d78e62`; independent certificate `sha256:36071490b1ec6ae387124089b784f55bfdc63f7664cc91f3e45d45298b673299`; empirical/external `sha256:8df5268b3c4793c2dc19ad1082c591e928ab99d7aca1469d9d9870749143a0f9`; engine receipt `sha256:6ff1fdc0b1d041805addd6ef9350dee1fe38a41082072c9f7888d4a22975228f` at `receipts/engine/model_admitted/SFT-COMP-FORM-STATE-TRANSITION-001-6ff1fdc0b1d04180.json`.

`SFT-COMP-OBL-BASE-002` - Fold formal-language and grammar law

- **Claim and ownership:** `SFT-COMP-FORM-LANGUAGE-GRAMMAR-001`; Classical Computation family `BASE`; Fold formal-language and grammar law.
- **Forced law:** The forced generated symbol words and formation kernel retains complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced generated symbol words and formation kernel retains complete alphabet; empty-One word; finite successor words; named productions; derivation trees; exact membership certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-STATE-TRANSITION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closes generated finite languages and depth-independent word succession, not a completed infinite language object.
- **Evidence identities:** derivation `sha256:688db340733ac157427caf7da72b4bb5f97b86627b4a080ce828a3d5bd9adb4e`; independent implementation `sha256:70802d179f08a4a5a6b2f178c5bd11fa87d64689159b14694def28731b79e5c8`; independent certificate `sha256:8a40d9a22d068e806f0c2d49a9af923f6fa29101600069cb311d9f82d40d8cfd`; empirical/external `sha256:9ace6f1cbeddd3086e42f5b9dce75bf37c34fa29abb5ed8b97fd5bf9d476ac67`; engine receipt `sha256:9247a863bf580966b3db1091ef6024a0f4562506076136dce33acda2a6039d14` at `receipts/engine/model_admitted/SFT-COMP-FORM-LANGUAGE-GRAMMAR-001-9247a863bf580966.json`.

`SFT-COMP-OBL-BASE-003` - Fold finite automaton law

- **Claim and ownership:** `SFT-COMP-FORM-AUTOMATON-001`; Classical Computation family `BASE`; Fold finite automaton law.
- **Forced law:** The forced finite recognition processes kernel retains finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced finite recognition processes kernel retains finite states; complete input alphabet; source-bound transition relation; initial state; accepting class; complete run trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-LANGUAGE-GRAMMAR-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not yet assert equivalence with every other model or universality.
- **Evidence identities:** derivation `sha256:de24c317c1f621d5ec62ecb401e94a4fccfba4de95dabad9ab144a71748f731c`; independent implementation `sha256:0e08a4bae652b4772475643b97ad32e52bf14f55de8fefa0ff3218b5ef5e22e4`;

independent certificate
`sha256:c6d7e75b9dbd097a50865aa3fb476bc057d2a5a2b933b069a694f021a4cc7d27`;
empirical/external
`sha256:17f43954c36004983c2299b4c7468c5e5816b764b5c8735af7b0275b942a66d2`; engine receipt
`sha256:a28d36b12b07be8aa4f32af257b75d6da331b0cebfea99fda3d6240e9ccaf109` at
`receipts/engine/model_admitted/SFT-COMP-FORM-AUTOMATON-001-a28d36b12b07be8a.json`.

`SFT-COMP-OBL-BASE-004` - Fold rewriting-system law

- **Claim and ownership:** `SFT-COMP-FORM-REWRITING-001`; Classical Computation family `BASE`; Fold rewriting-system law.
- **Forced law:** The forced held-position replacement kernel retains generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced held-position replacement kernel retains generated word; exact pattern occurrence; retained rule identity; held replacement position; successor word; terminal, cyclic and branching trace classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-AUTOMATON-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only registered finite rules and generated words are admitted.
- **Evidence identities:** derivation
`sha256:7f1279c1b5b5d6b2e31d821616cdc40fbb8e89e782ad571f2cd65486a9569727`; independent implementation `sha256:1d03336bec2e9efb5b8a8338de71b228a4a517b8fee08c2146539375583d26b8`; independent certificate
`sha256:f5faf737b0c4341fe3c50b95d147053ef79cdb85a9ad4971d22b3304c7bb3466`;
empirical/external
`sha256:13b91a821e5ab3ea07326aed719fde2054ecd40561f9787f4d5d3b1218d24ef1`; engine receipt
`sha256:b1f6c54c1f8be568e0dbb809dc19f1f882840e53c693aa208db0c2410860defe` at
`receipts/engine/model_admitted/SFT-COMP-FORM-REWRITING-001-b1f6c54c1f8be568.json`.

`SFT-COMP-OBL-BASE-005` - Fold recursion law

- **Claim and ownership:** `SFT-COMP-FORM-RECURSIVE-FUNCTION-001`; Classical Computation family `BASE`; Fold recursion law.

- **Forced law:** The forced structural finite recursion kernel retains structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced structural finite recursion kernel retains structural-One base; generated successor trace; exact step relation; retained intermediate values; composition interface; terminal result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-REWRITING-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closes recursion over every generated finite successor trace, not completed transfinite recursion.
- **Evidence identities:** derivation `sha256:f0d1440d842607d5258c5accdcb83682ab3eaf0a58c3dceac901a2c9c730723d`; independent implementation `sha256:f0103e97dcbd277c3c09a157d04752e90b18f719eac050beb7d45f706b36c486`; independent certificate `sha256:9dc55f7ee3aa968c744de817d3bdef595ab14824948958aeb1d57417152fa061`; empirical/external `sha256:24365aa8a72572993447d2cafcf0abb11e9491ece06a19c1441ae242ceecc411`; engine receipt `sha256:291230ba67c75760c319e58fe0ae120c6c300a83e39d5f51c70de936c73e580c` at `receipts/engine/model_admitted/SFT-COMP-FORM-RECURSIVE-FUNCTION-001-291230ba67c75760.json`.

`SFT-COMP-OBL-BASE-006` - Fold binding and self-application calculus law

- **Claim and ownership:** `SFT-COMP-FORM-LAMBDA-CALCULUS-001`; Classical Computation family `BASE`; Fold binding and self-application calculus law.
- **Forced law:** The forced binding, substitution and reduction kernel retains generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced binding, substitution and reduction kernel retains generated term grammar; held binder identity; free and bound occurrence ledgers; capture-safe substitution; exact reduction; normal or cyclic trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-RECURSIVE-FUNCTION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No conventional lambda calculus is assumed; the result is the generated Fold term kernel.
- **Evidence identities:** derivation `sha256:2a809ddc1fcc1b0a016920d46311c76dcd0cd465af22930ae3b823a0ba7e14e3`; independent implementation `sha256:d745cfdbf195dec5c6646f1dfdc6039f85e6170cde0ce4f36f97627314a62ed6`; independent certificate `sha256:f89a69bbf74e6f348b37e64f672de3968639ca340f9d50e9c18b7a29a4a36c06`; empirical/external `sha256:6a8daaa4987a604def6e709ad88ba6c6c5a18065e3d9bc8ce61e12f537d000d6`; engine receipt `sha256:c5df90aeea7c0f8459bb98cdfa9343167723cc7b320af2c9fec60f7f5f5bee63` at `receipts/engine/model_admitted/SFT-COMP-FORM-LAMBDA-CALCULUS-001-c5df90aeea7c0f84.json`.

`SFT-COMP-OBL-BASE-007` - Fold abstract-machine law

- **Claim and ownership:** `SFT-COMP-FORM-ABSTRACT-MACHINE-001`; Classical Computation family `BASE`; Fold abstract-machine law.
- **Forced law:** The forced finite executable machine descriptions kernel retains description word; configuration carrier; transition kernel; generated store or tape; initial configuration; complete execution and halt trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced finite executable machine descriptions kernel retains description word; configuration carrier; transition kernel; generated store or tape; initial configuration; complete execution and halt trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-LAMBDA-CALCULUS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.

- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Machines outside the frozen description grammar remain outside the theorem.
- **Evidence identities:** derivation `sha256:1bf45a77a8defe269a94cc9471ce829b8b518f30acb09bcaab1a18d351cc508f`; independent implementation `sha256:b0c8e900076f04ce4e88db72f4300ddec7b44e871c574fda6828f04743b6437a`; independent certificate `sha256:3ce91feb3e76ea25d296bfcc777100313354b171985af741b72cbbe204dc2113`; empirical/external `sha256:0062195e4657ede78fadd3877e80e38bd845ffef5f981c6a577d68b4c6e04815`; engine receipt `sha256:2b2471771a06a549143d109bd88b40339b476c36ab65b278fd86a7b54adbdd1b` at `receipts/engine/model_admitted/SFT-COMP-FORM-ABSTRACT-MACHINE-001-2b2471771a06a549.json`.

`SFT-COMP-OBL-BASE-008` - Fold circuit syntax and evaluation law

- **Claim and ownership:** `SFT-COMP-FORM-CIRCUIT-001`; Classical Computation family `BASE`; Fold circuit syntax and evaluation law.
- **Forced law:** The forced finite connected local transformations kernel retains generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced finite connected local transformations kernel retains generated nodes and wires; typed interfaces; local transformations; feed-forward order or declared cycle; exact evaluation trace; output support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-ABSTRACT-MACHINE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Resource size and depth comparisons are deferred to Complexity.
- **Evidence identities:** derivation `sha256:62e2b314384c853bb679b205fb39f47f76ee48f07eb26e225f0f2fee5861baf6`; independent implementation `sha256:4d4a98f08cf35dab2175a66857e1f2764741107ab27f0809cc028907180c443b`; independent certificate `sha256:b73405d93e9177c20dc3877407ce8bc3f169c481b8a298624f0e290625d87413`;

empirical/external `sha256:d1e5487d81c976c6526694efaf7845bd437def19c61290a024f75abee8e5a7e7`; engine receipt `sha256:0a974615e0f53938a3c8551caabc041da5b7c241dfdc059febcb5d38269a0c94` at `receipts/engine/model_admitted/SFT-COMP-FORM-CIRCUIT-001-0a974615e0f53938.json`.

`SFT-COMP-OBL-BASE-009` - Fold operational-process law

- **Claim and ownership:** `SFT-COMP-FORM-OPERATIONAL-PROCESS-001`; Classical Computation family `BASE`; Fold operational-process law.
- **Forced law:** The forced execution and observable process behavior kernel retains initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced execution and observable process behavior kernel retains initial configuration; enabled transitions; retained choices; complete path trace; terminal, failed and cyclic classes; observation record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-CIRCUIT-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not import process algebra or distributed timing.
- **Evidence identities:** derivation `sha256:16f15c2211974316eb7f5595c532cc6e649fe07a22c56adf7a8673b7bc3466d3`; independent implementation `sha256:edf9d2ca579edfbdcf5f372b21b3c37de52c998e8b92c5e4a136ab0d31b454ba`; independent certificate `sha256:a37e121f9fca530b933f74e80c8877fd811ab4e64923192cc395984dfaf030a6`; empirical/external `sha256:b5d7bc8b74c9ed55eac3a59349e3a88148e1ef2411d26dcb672e6a97a0873579`; engine receipt `sha256:d3349d79836c6e8edf5199206fe0dc818a6210e9d3dc7f8000eb83f0b55ef282` at `receipts/engine/model_admitted/SFT-COMP-FORM-OPERATIONAL-PROCESS-001-d3349d79836c6e8e.json`.

`SFT-COMP-OBL-BASE-010` - Fold process composition and decomposition law

- **Claim and ownership:** `SFT-COMP-FORM-COMPOSITION-001`; Classical Computation family `BASE`; Fold process composition and decomposition law.
- **Forced law:** The forced serial and parallel process assembly kernel retains exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced serial and parallel process assembly kernel retains exact input and output interfaces; serial matching; independent parallel support; retained component identities; reconstruction witness; failure on mismatch; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-OPERATIONAL-PROCESS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Concurrency laws requiring communication or shared causality remain downstream.
- **Evidence identities:** derivation `sha256:1a5b215a9ad08b3cf8665c8299fa6961b1a43c87847b1b705adfbc1419827716`; independent implementation `sha256:01a1fd9217764d6beb2dfda7b127626b041b729945527f7cce6a4c4a7a22c756`; independent certificate `sha256:11adf318b9899ccc822e4c9164ac86c61596df45c8b41bcb54d268785aa279a2`; empirical/external `sha256:91e36576fe2613746be520df07a60d9db80cd25dce24a78dd1b023d23c8c438b`; engine receipt `sha256:b0fea8f79a1c2c934c5b2dfaa37c9ba09e8b0f1416377175a645afe9f58738ad` at `receipts/engine/model_admitted/SFT-COMP-FORM-COMPOSITION-001-b0fea8f79a1c2c93.json`.

`SFT-COMP-OBL-BASE-011` - Fold computational-model equivalence law

- **Claim and ownership:** `SFT-COMP-FORM-MODEL-EQUIVALENCE-001`; Classical Computation family `BASE`; Fold computational-model equivalence law.
- **Forced law:** The forced bidirectional execution-preserving translations kernel retains complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced bidirectional execution-preserving translations kernel retains complete source descriptions; total encoding; target descriptions; step or trace simulation; result preservation; reverse reconstruction; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-COMPOSITION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 ->

SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Equivalence applies only to the two registered generated model grammars.
- **Evidence identities:** derivation `sha256:3c0dad2425378918bacf1c277636db591165b0660c98f8c19d1f4ec68d7c1eb9`; independent implementation `sha256:7f980a25b82bbe2793268f94eaf9f482120d4fc8ec8a889a6d345052c7e04f40`; independent certificate `sha256:62db23d8401df6af4f79ceb122b3a85440ff45189a7bb09dc7d5e6722c6e99ce`; empirical/external `sha256:64e6d04fde05fcaba7e98b2cef7ca0a8c7f7de711e8a754d145b9f67e2377bc6`; engine receipt `sha256:911ff906b2d8407111a076f3abb06abe79334098a0e87faa54460f8453338139` at `receipts/engine/model_admitted/SFT-COMP-FORM-MODEL-EQUIVALENCE-001-911ff906b2d84071.json`.

`SFT-COMP-OBL-BASE-012` - Fold universal-computation law

- **Claim and ownership:** `SFT-COMP-FORM-UNIVERSALITY-001`; Classical Computation family `BASE`; Fold universal-computation law.
- **Forced law:** The forced interpretation of generated machine descriptions kernel retains complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`. Exact result: The forced interpretation of generated machine descriptions kernel retains complete frozen description grammar; encoded machine and input; one interpreter process; exact simulated trace; preserved terminal result; explicit resource overhead; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-MODEL-EQUIVALENCE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or

ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Universality is exact for every description generated by the frozen grammar, not an unbounded completed totality.

- **Evidence identities:** derivation `sha256:aa872db026edad77bae474d940bdead07caed584e3013fc04c986d0d927ad80d`; independent implementation `sha256:f1df70199e93c004a3d9c9e2a3bf396a74626e742e81b0b8e213b3af7e9e8819`; independent certificate `sha256:bf89e7bbbd33de8dc7caaf3f5bdf3d6979d628eacc66183a02411b407236e76b`; empirical/external `sha256:9901d8e8f04985a7b65da1a40504fa8f1df13e92f614953ef900fcd87ef4ace6`; engine receipt `sha256:83a401709b86991de5f38a49e7794379ea0c041fb2a406a245f900bf11bcb598` at `receipts/engine/model_admitted/SFT-COMP-FORM-UNIVERSALITY-001-83a401709b86991d.json`.

`SFT-COMP-OBL-BASE-013` - Fold recognition and decision law

- **Claim and ownership:** `SFT-COMP-CBL-RECOGNITION-DECISION-001`; Classical Computation family `BASE`; Fold recognition and decision law.
- **Forced law:** The forced acceptance, rejection and continued execution kernel retains generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`. Exact result: The forced acceptance, rejection and continued execution kernel retains generated input domain; partial recognizer traces; total decider traces; accepting and rejecting labels; explicit unresolved continuation; exact domain ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A recognizer need not terminate on every generated input; a decider must.
- **Evidence identities:** derivation `sha256:11e94644591c3e7e51e1b0235bb9752f71fcd7ce64f457291bf83968e902a4a6`; independent implementation `sha256:dc0f0486633f8e87e5f9e02b210b5376b2cf778691ca860a41eedcfa874929a6`; independent certificate `sha256:0eb8c9783f7e87e434555864f84644fcd1c096bcc3bd1725540ffdd033fb0aa1`; empirical/external `sha256:9151b6040f1b47fae6bb0018232dc6b6399177add50d2f9a768f723f69bb6ebe`; engine receipt `sha256:c9c992b18e4408ecfd5bb004622b3d62d836752b387773eecb7dfc0e1daed9e8` at `receipts/`

engine/model_admitted/SFT-COMP-CBL-RECOGNITION-DECISION-001-c9c992b18e4408ec.json.

SFT-COMP-OBL-BASE-014 - Fold halting boundary law

- **Claim and ownership:** SFT-COMP-CBL-HALTING-001; Classical Computation family BASE; Fold halting boundary law.
- **Forced law:** The forced terminal-state determination kernel retains encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power. Exact result: The forced terminal-state determination kernel retains encoded process and input; complete execution trace when available; terminal witness; self-description support; held complement action; contradiction certificate for total internal prediction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-CBL-RECOGNITION-DECISION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not decide halting for all generated universal-machine descriptions.
- **Evidence identities:** derivation sha256:299c9b39f4e497fe084e28cb9d78834cef211cd11785de98b0fc7bb206f83d2d; independent implementation sha256:e5acaf236971ed0d09b140257aaa940a41536fd2f5ee017ca337331933c9be54; independent certificate sha256:b3be18262d2f70acfe69d15683428f479c2294731ad192d11d9a06a7d79cdf6d; empirical/external sha256:bb240ffe1dbdb05d04360ea66a0f4efd9284d5d1b72f80a8b72590e88dc77a02; engine receipt sha256:a33ae0f08db036bcde17a25eb3d6605dcfbcd3fa0492176faaf553110d2368e2 at receipts/engine/model_admitted/SFT-COMP-CBL-HALTING-001-a33ae0f08db036bc.json.

SFT-COMP-OBL-BASE-015 - Fold effective-enumeration law

- **Claim and ownership:** SFT-COMP-CBL-ENUMERATION-001; Classical Computation family BASE; Fold effective-enumeration law.
- **Forced law:** The forced generated description ordering kernel retains complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Unique surviving structure:** `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`. Exact result: The forced generated description ordering kernel retains complete finite description support at each depth; canonical position labels; no duplicates; successor extension; recognizer linkage; reconstruction from position; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CBL-HALTING-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No completed infinite enumeration is installed as an object.
- **Evidence identities:** derivation `sha256:6c0d27b9378dd92f4cdedc7f7d4dc90a02fe816f4c8fbeef2c826ae564712d08`; independent implementation `sha256:c6a5dd8e67bf0218abf2a49489560529f1f066e15a1c597495aa6042686053cd`; independent certificate `sha256:e2a65196b779a1bd41f25443155116ef6e7ffe265c9d0921e544927747b53717`; empirical/external `sha256:56a2658bc7bed4373c7763f34346566f995e7e95bcfcb87ede676a3ceb5e088c`; engine receipt `sha256:b4a3d735406f9ab57023c47ed0a12a7a221b278c75727cb0a7170dbdfec940f2` at `receipts/engine/model_admitted/SFT-COMP-CBL-ENUMERATION-001-b4a3d735406f9ab5.json`.

`SFT-COMP-OBL-BASE-016` - Fold computability-reduction law

- **Claim and ownership:** `SFT-COMP-CBL-REDUCTION-001`; Classical Computation family `BASE`; Fold computability-reduction law.
- **Forced law:** The forced problem translation preserving answers kernel retains source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`. Exact result: The forced problem translation preserving answers kernel retains source domain; total encoding; target domain; target procedure; answer decoder; bidirectional correctness trace; composition law; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CBL-ENUMERATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 ->

SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Reduction claims require explicit encoders, decoders and generated domains.
- **Evidence identities:** derivation `sha256:c3ae5424425de2ea0a730259170848da166eb040d9c9fe8f1d76d0254ea22811`; independent implementation `sha256:173795ad853a56a81d7d98fd16dc14eccd38d80f3add1c8dc0a58720f6f43de5`; independent certificate `sha256:c843b1c34313a579857158cca63c95281567d12d2cb77c0ec04f46ad8e07baf7`; empirical/external `sha256:1d8cac715551bdf0d1e71267a730cbf51e9175f803ebe560db4fefd7801a588e`; engine receipt `sha256:7d03f54baf0fc6d4bbe16f2ea73fac601ede53dceacbd55e2b032545196aaff2` at `receipts/engine/model_admitted/SFT-COMP-CBL-REDUCTION-001-7d03f54baf0fc6d4.json`.

`SFT-COMP-OBL-BASE-017` - Fold degrees-of-computability law

- **Claim and ownership:** `SFT-COMP-CBL-DEGREES-001`; Classical Computation family `BASE`; Fold degrees-of-computability law.
- **Forced law:** The forced relative reduction classes kernel retains registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`. Exact result: The forced relative reduction classes kernel retains registered problems; mutual reduction relation; equivalence classes; strict one-way reachability; transitive closure; incomparability witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CBL-REDUCTION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or

ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Degrees are finite generated relation classes; no completed transfinite hierarchy is claimed.

- **Evidence identities:** derivation `sha256:ad0d2f09c99201752cf963100d4b989df8bd6504cc4a12212f545eb45272c378`; independent implementation `sha256:54e92220b00e34ab45dec83999c26efc88c4604686cf8bc98866a1143eabf7a3`; independent certificate `sha256:4beaa5ffed228284c3bd0b27c642a6bceb0005d9de29c5b986fba873b1ac97d8`; empirical/external `sha256:d8e21565c305b5d29b57f36fd9d95aae62b6254c03faebc6c413c682b05b2e8f`; engine receipt `sha256:40189bc4fb7d083c3ffc7a7610e29e77cc5232bbde8514860c3afa328fe11a3e` at `receipts/engine/model_admitted/SFT-COMP-CBL-DEGREES-001-40189bc4fb7d083c.json`.

`SFT-COMP-OBL-BASE-018` - Fold universal-machine execution law

- **Claim and ownership:** `SFT-COMP-CBL-UNIVERSAL-MACHINE-001`; Classical Computation family `BASE`; Fold universal-machine execution law.
- **Forced law:** The forced one machine executing encoded machines kernel retains frozen machine grammar; prefix-free exact descriptions; encoded input; interpreter transition kernel; trace correspondence; halt and output preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`. Exact result: The forced one machine executing encoded machines kernel retains frozen machine grammar; prefix-free exact descriptions; encoded input; interpreter transition kernel; trace correspondence; halt and output preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CBL-DEGREES-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Extends Formal Universality into recognizability boundaries but not an omniscient halting oracle.
- **Evidence identities:** derivation `sha256:bc727d9f76e920560ca4a2ff6fc8864fcdb4da986c10d8e3db80fe1df09d8320`; independent implementation `sha256:f2890836bf1bba1f7dbc2c6641b34583b0906fcc25d4099cc5d993326b21ec85`; independent certificate `sha256:d830526fc80fceff9278895cdb542cd9a72d6da659acaa0e60aa47d12841f058`; empirical/external `sha256:221ca0caa7445de51094f8fc76c69b2292b6118acf5bd19dc52f36acce24abe6`; engine receipt `sha256:63466fb169111c50266901bf7ac48255ced18ef727cd66d1f3b376b6beaa7633` at `receipts/engine/model_admitted/SFT-COMP-CBL-UNIVERSAL-MACHINE-001-63466fb169111c50.json`.

SFT-COMP-OBL-BASE-019 - Fold undecidable-problem law

- **Claim and ownership:** SFT-COMP-CBL-UNDECIDABILITY-001; Classical Computation family BASE; Fold undecidable-problem law.
- **Forced law:** The forced failure of total internal decision kernel retains generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`. Exact result: The forced failure of total internal decision kernel retains generated self-descriptions; assumed total decider; exact diagonal constructor; complementary verdict; self-application; contradiction trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CBL-UNIVERSAL-MACHINE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; The theorem applies where the registered language supports the exact self-encoding and complement operation.
- **Evidence identities:** derivation `sha256:42eefceee42f1ef84a8b765da3ecefbc91a79c5bfe7ce3cc6019e6ceddccf5d7`; independent implementation `sha256:521c56670dedb8d993cca8984ca42bd99784d517bc12cb1989e071738a387845`; independent certificate `sha256:8c3b3c911d7f33c654b756f4f59224e8614b0e9cdfcbff736daac8ce0bcf13f6`; empirical/external `sha256:b2820d28d6464e2afcfae7ee3eb56bb845b3a192556fb0cf0c7e6774d73bb5c0`; engine receipt `sha256:9de0276cb48ddbcea65e0a35f6f2deeac179a225ac85a8f93121e5e7ed97e692` at `receipts/engine/model_admitted/SFT-COMP-CBL-UNDECIDABILITY-001-9de0276cb48ddbce.json`.

SFT-COMP-OBL-BASE-020 - Fold incompleteness-boundary law

- **Claim and ownership:** SFT-COMP-CBL-INCOMPLETENESS-001; Classical Computation family BASE; Fold incompleteness-boundary law.
- **Forced law:** The forced internal proof and self-verification limits kernel retains effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-s`

uccessor-certificate__no-undeclared-computational-power. Exact result: The forced internal proof and self-verification limits kernel retains effective proof enumeration; consistency-preserving verifier; self-referential description; missing decision class; external trace record; no truth oracle; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-CBL-UNDECIDABILITY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; This is a computational proof-system boundary, not an imported arithmetic incompleteness axiom.
- **Evidence identities:** derivation sha256:39ccdf0743514fba3f9e01e41a67dfa4686805175ccd0fe687e6316aef358541; independent implementation sha256:fce65fcba70bbdfe02de16c19f5f28328658fbf8d9c78050319b98a2baf774a3; independent certificate sha256:f6f7786d6185c4e020b9031361db598d967b3772248ee3c7add2b0721457ab2b; empirical/external sha256:0ba02558215ddcb70efdb09565b963fcafeff462fd6b3410a05e0721472c967d; engine receipt sha256:f7fb969bc4fc339ff473e5ec2620e703695e6a12c3ab0b704594e9664e9ea277 at receipts/engine/model_admitted/SFT-COMP-CBL-INCOMPLETENESS-001-f7fb969bc4fc339f.json.

SFT-COMP-OBL-BASE-021 - Fold relative and oracle-computation law

- **Claim and ownership:** SFT-COMP-CBL-RELATIVE-ORACLE-001; Classical Computation family BASE; Fold relative and oracle-computation law.
- **Forced law:** The forced explicit external answer interfaces kernel retains base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power. Exact result: The forced explicit external answer interfaces kernel retains base process; named query alphabet; declared oracle relation; query and answer trace; relative recognition; oracle-identity ledger; nesting boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited

`Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; An oracle is a registered interface, never unexplained computational power.
- **Evidence identities:** derivation `sha256:6c056bd1d9d6e58d1c4cf9eb93bb767deeb145a363a6a7d85884a2696008114c`; independent implementation `sha256:98e5f6212def0f26cd0cb6cfa352183ebdebf792aaa5184499fea5eb380abc64`; independent certificate `sha256:f9757a676f3589d0cce885d22cb45b44377fd1720dcbb2953cee994b0c89911d`; empirical/external `sha256:718f8f589221365d114308744266476d1bca0bb9476be8e68a2ca1acde67b7be`; engine receipt `sha256:2a09b83377a56d06136df6922d9e9830c3b336e3487d3654d349c7f6f6746d01` at `receipts/engine/model_admitted/SFT-COMP-CBL-RELATIVE-ORACLE-001-2a09b83377a56d06.json`.

`SFT-COMP-OBL-BASE-022` - Fold hypercomputation admissibility-limit law

- **Claim and ownership:** `SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001`; Classical Computation family `BASE`; Fold hypercomputation admissibility-limit law.
- **Forced law:** The forced claims exceeding generated universal computation kernel retains claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`. Exact result: The forced claims exceeding generated universal computation kernel retains claimed input domain; operational transition law; finite evidence boundary; self-reference test; resource and oracle disclosures; falsification condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CBL-RELATIVE-ORACLE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No hypercomputation claim is admitted without a generated executable law and depth-independent certificate; finite

demonstrations remain conditional.

- **Evidence identities:** derivation `sha256:5b708f88bd4ff06b23706597a4ea720d6df25d109b06dd0aa7201a06267ad085`; independent implementation `sha256:6b6764bb33684418d2d828cf9c1bc882739d517804a2ece25e8c1e56b79a6357`; independent certificate `sha256:37c0063e48172aa654b2841935cde94ed2c44a732321a91de3cc86b2a5b58449`; empirical/external `sha256:e4a6b799d7ea881df152750b839cf35c871b97e84a80484817303e0fec561d43`; engine receipt `sha256:3d6f7b8fdd8b1ccaf30622dad922140dd6b441b19df9dd4b76ff17ee191439cc` at `receipts/engine/model_admitted/SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001-3d6f7b8fdd8b1cca.json`.

`SFT-COMP-OBL-BASE-023` - Fold input and problem-size law

- **Claim and ownership:** `SFT-COMP-CPLX-INPUT-SIZE-001`; Classical Computation family `BASE`; Fold input and problem-size law.
- **Forced law:** The forced exact generated input extent kernel retains canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced exact generated input extent kernel retains canonical input encoding; complete occupied positions; structural size trace; representation invariance ledger; instance and family boundaries; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Size is representation-relative unless an exact encoding correspondence is supplied.
- **Evidence identities:** derivation `sha256:59a9b0be31a5e733dce860eedfbe4bda67bde61c23ab06024202d6a115147f04`; independent implementation `sha256:8b11f954a55f4184895929a2a1a8e8e2cc7edbf0510d5f079bec7017d7bda6ca`; independent certificate `sha256:749367f01a05e86db5ae00d7d66abf0b6dcd40d036f06f166c7dcef695d99511`; empirical/external `sha256:9f28b7906f4b1e97bb7044a480308141fd074adb36c8a1d57e5bdd05b9ba2bd7`; engine receipt `sha256:3f002d11d6315fc6f32536022cefb2a3fad2bf857ea1761637481a827a8a2fc4` at `receipts/engine/model_admitted/SFT-COMP-CPLX-INPUT-SIZE-001-3f002d11d6315fc6.json`.

`SFT-COMP-OBL-BASE-024` - Fold time and space resource law

- **Claim and ownership:** `SFT-COMP-CPLX-TIME-SPACE-001`; Classical Computation family `BASE`; Fold time and space resource law.
- **Forced law:** The forced transition and retained-state resources kernel retains complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced transition and retained-state resources kernel retains complete execution transitions; live configuration support; retained history; peak support class; terminal trace; reversible record boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-INPUT-SIZE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Counts are exact host reports over generated traces, not foundational magnitudes.
- **Evidence identities:** derivation `sha256:1fcc740897abe9b3adb6bdc54f8f5a7a4a1908d1da460d36b7751a0ba3fa0e36`; independent implementation `sha256:55a51257da5a486414684d2eb845f65827c1a319fb2712f3619bf3da0868e96c`; independent certificate `sha256:d69c776f3a72bdfe046e514a487a8e8365e1bfc47b33eb1827d7358db5dafb46`; empirical/external `sha256:8669c1478eabab8093f09aae4b52e18a24ecc93e744f3ae2099c67c15121a68d`; engine receipt `sha256:d6187e34cd91181461fa6614ebb82d710719cf6bebf4644d1bc08c81641df62b` at `receipts/engine/model_admitted/SFT-COMP-CPLX-TIME-SPACE-001-d6187e34cd911814.json`.

`SFT-COMP-OBL-BASE-025` - Fold circuit size and depth law

- **Claim and ownership:** `SFT-COMP-CPLX-CIRCUIT-RESOURCE-001`; Classical Computation family `BASE`; Fold circuit size and depth law.
- **Forced law:** The forced circuit structural resources kernel retains generated gates; wires; dependency order; longest causal layer trace; output support; shared-subcircuit identity; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced circuit structural resources kernel retains generated gates; wires;

dependency order; longest causal layer trace; output support; shared-subcircuit identity; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-TIME-SPACE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closes resource accounting, not arbitrary lower bounds.
- **Evidence identities:** derivation `sha256:11d0d9927aa1c5cb64db2faeed8e29d4aad74135c81b3c68ea8ad080de297144`; independent implementation `sha256:12e9bb6516941807a16f20797673f5421d0a4446a86e73aca97773d5d57fccfc`; independent certificate `sha256:1a0022829a4bdd87943253b70fa5057822a6a456dbf6d8cf56f7a996826d3909`; empirical/external `sha256:f9af184469e9d277ee03491a44a563a9f131f5a86b3c7fd325ba362b0de85f57`; engine receipt `sha256:9f3e28733488ad63f954689caad30204963adbe8e2fa60f985be1140722116ae` at `receipts/engine/model_admitted/SFT-COMP-CPLX-CIRCUIT-RESOURCE-001-9f3e28733488ad63.json`.

`SFT-COMP-OBL-BASE-026` - Fold communication and query complexity law

- **Claim and ownership:** `SFT-COMP-CPLX-COMMUNICATION-QUERY-001`; Classical Computation family `BASE`; Fold communication and query complexity law.
- **Forced law:** The forced cross-interface and information-request resources kernel retains distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced cross-interface and information-request resources kernel retains distributed input support; sent records; received records; named query interface; answer trace; retained local knowledge; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-CIRCUIT-RESOURCE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; The exact protocol and oracle identity must be registered.
- **Evidence identities:** derivation `sha256:7d91bbd1b18c3aefb5ca8eb72d84964027c55fe09b733ffe688cf19e2491ac41`; independent implementation `sha256:ca88ab267912a1c325ce19967fa6f9bb5104be65e508a42692808033bf18b3d7`; independent certificate `sha256:b62d6baea8c3fb0502d436d71446deab6f9277fe93e8b49fb2845f1380457cb5`; empirical/external `sha256:10745814f262dafcab4969ed2534761d0e23085f68e6f31faa232b543b41d953`; engine receipt `sha256:c352ad5637226cc1d41ecbc8fb375f9593e7928d4157a97e8b7050ac18d19f3e` at `receipts/engine/model_admitted/SFT-COMP-CPLX-COMMUNICATION-QUERY-001-c352ad5637226cc1.json`.

`SFT-COMP-OBL-BASE-027` - Fold randomness-resource law

- **Claim and ownership:** `SFT-COMP-CPLX-RANDOMNESS-001`; Classical Computation family `BASE`; Fold randomness-resource law.
- **Forced law:** The forced scheduled unresolved branch support kernel retains complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced scheduled unresolved branch support kernel retains complete branch schedule; deterministic source process; held branch labels; execution per branch; output partition; schedule-description cost; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-COMMUNICATION-QUERY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Randomness is not a causal premise; results quantify complete registered branch support or an explicitly measured external source.
- **Evidence identities:** derivation `sha256:11553fb594b83f863eb99c1dcda12b7c55670a95e85ddeb507f0e96436cba33c`; independent

implementation `sha256:e86d848ef0f31cd3a4e10cbe82a0235e50943a06e0d9ffaff45e5744c6482459`; independent certificate `sha256:dedc0a421764c9d7249870484673717c898c3cddf7ed33494cec492522a3dc34`; empirical/external `sha256:25f8a328196851e3b4fc33030d39baa1877734e8d287b0a9515d51206eeb02fa`; engine receipt `sha256:e4311dc1af067c40ad88072efc8c155d1647ad6b071c1003c48b0e597be64f8b` at `receipts/engine/model_admitted/SFT-COMP-CPLX-RANDOMNESS-001-e4311dc1af067c40.json`.

`SFT-COMP-OBL-BASE-028` - Fold reversibility-cost law

- **Claim and ownership:** `SFT-COMP-CPLX-REVERSIBILITY-COST-001`; Classical Computation family `BASE`; Fold reversibility-cost law.
- **Forced law:** The forced information retained for inverse execution kernel retains source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced information retained for inverse execution kernel retains source configurations; transition fibres; merged-predecessor ledger; retained predecessor labels; inverse trace; erasure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-RANDOMNESS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; This is the classical cost law; the universal reversible model belongs to the later quantum branch.
- **Evidence identities:** derivation `sha256:8593ee37fadeb80b440d656e826431594044f76793053417dff646dc6da86bdd`; independent implementation `sha256:62112db56cc1d6dde75ba41a7ba2e9b84a435cf17bc81262ad9d5e11c008d868`; independent certificate `sha256:ad19f8f114159728b4ae9507a73e5c76f3cd94ca6d18e5586069be460d268278`; empirical/external `sha256:657b65250d381219f7c9577a80006d61e4e72c8b53388e955a68de6eceff7cce`; engine receipt `sha256:0a8552676c105fc6b3fdcd6aac90a7a68c428962c209460caacea2b81c496ed2` at `receipts/engine/model_admitted/SFT-COMP-CPLX-REVERSIBILITY-COST-001-0a8552676c105fc6.json`.

`SFT-COMP-OBL-BASE-029` - Fold parallel-complexity law

- **Claim and ownership:** `SFT-COMP-CPLX-PARALLEL-001`; Classical Computation family `BASE`; Fold parallel-complexity law.

- **Forced law:** The forced independent and causally layered work kernel retains task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced independent and causally layered work kernel retains task dependency graph; independent ready sets; layer schedule; work trace; span trace; synchronization rows; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No physical processor count or timing constant is imported.
- **Evidence identities:** derivation `sha256:15c1ba1323fe5baa1b8635a1bda7868db77477aa1f0c4981e4b11dc06af9f4ad`; independent implementation `sha256:81ade383c13f9b93f5df4bc48531bcf5eaf6a2c6bb816dc89b3737ef6dc9ebae`; independent certificate `sha256:dfbd1f51e697f60b15a455914c43b409255647b875c4a9882bb008ef789e72ea`; empirical/external `sha256:82c39a2ca55f4e9b10d7d4c7ff997b61bcd52472e2fca5ca2da470993b1b72fa`; engine receipt `sha256:d451551a66fdaf127bfd2e9738af1c7818753ea3dd2862a46db3108e32354424` at `receipts/engine/model_admitted/SFT-COMP-CPLX-PARALLEL-001-d451551a66fdaf12.json`.

`SFT-COMP-OBL-BASE-030` - Fold lower- and upper-bound law

- **Claim and ownership:** `SFT-COMP-CPLX-BOUNDS-001`; Classical Computation family `BASE`; Fold lower- and upper-bound law.
- **Forced law:** The forced resource comparison certificates kernel retains problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced resource comparison certificates kernel retains problem family; registered model; resource measure; constructive algorithm trace; adversarial indistinguishability family; lower witness; upper witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-PARALLEL-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Arbitrary asymptotic lower bounds remain separate claims requiring their own generated families.
- **Evidence identities:** derivation `sha256:1cec4f2834aa22a4ecc619ed0c0541e1788ae7f620901e4c0c3065eb2ee7ac2d`; independent implementation `sha256:7bd2df392101a75d5c8de34cd3952cbe8cbb50fffdae0927168f2afffbde223f`; independent certificate `sha256:b347fc3d05e9cfd9931f0178ded39a23316344dd56773e09696d783ba933a55f`; empirical/external `sha256:f0bfb44bbb62a3b09c8cd4e1fe8751b4d016a368f0f3a516a00af53052f53b35`; engine receipt `sha256:7ba9b6310c1f0dd170c3c9b6abc0db4c257241df7af511faf3fb232b2d39db02` at `receipts/engine/model_admitted/SFT-COMP-CPLX-BOUNDS-001-7ba9b6310c1f0dd1.json`.

`SFT-COMP-OBL-BASE-031` - Fold reduction and completeness law

- **Claim and ownership:** `SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001`; Classical Computation family `BASE`; Fold reduction and completeness law.
- **Forced law:** The forced resource-preserving problem translations kernel retains problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced resource-preserving problem translations kernel retains problem class inventory; complete source encodings; total reductions; answer preservation; resource-overhead ledger; hard and member certificates; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-BOUNDS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.

- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Completeness is relative to the frozen problem class and resource law.
- **Evidence identities:** derivation `sha256:7ce5c8b6a21352a4cfdf7f441d39a955314633a9070682103521f83cfb41358d`; independent implementation `sha256:12fc9feb0f395be2df932ad6211deaef3a7051d9638752248702b32398313bec`; independent certificate `sha256:3710e881e46bb1cbd8b1788064009f6e420e795dcbdc2140690ed5e2bda71f46`; empirical/external `sha256:2adb304bdade25973af95ecd12fb91806996bbd575d4835b7a1dbde2357df1ea`; engine receipt `sha256:401b5e76d63dd3d31c2a38156dd3fab752e10b289cb7b0e38c9a71f0cf6f673a` at `receipts/engine/model_admitted/SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001-401b5e76d63dd3d3.json`.

`SFT-COMP-OBL-BASE-032` **- Fold average- and worst-case law**

- **Claim and ownership:** `SFT-COMP-CPLX-AVERAGE-WORST-001`; Classical Computation family `BASE`; Fold average- and worst-case law.
- **Forced law:** The forced complete instance-family resource profiles kernel retains generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced complete instance-family resource profiles kernel retains generated instance support; exact resource per instance; greatest-resource class; exact positive support parts; distribution provenance; aggregate ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Average claims require a complete exact support relation or registered empirical distribution.
- **Evidence identities:** derivation `sha256:9d6f8834fc4e6c4d7772ac100765066e8cd32a49ac2aca16328984c5567d3aa2`; independent implementation `sha256:9fc79651d13c51167b280ff22f28bd599d3a56d69ba36f3b6aaf41f23a8f572e`; independent certificate

sha256:59c844f91b8949eecae3fab5729be02bfc32aa99351bead60cb5f32e2c4c27a9; empirical/external sha256:9d98937e33feb40baa49f42b0a1e9d95c8cac0852d0c5ce6b89db55f6d8c0da4; engine receipt sha256:231135bfa6605c585af24c7a6f89d21b099f42462dc4b7a6c0820d761bce6bfb at receipts/engine/model_admitted/SFT-COMP-CPLX-AVERAGE-WORST-001-231135bfa6605c58.json.

SFT-COMP-OBL-BASE-033 - Fold approximation-complexity law

- **Claim and ownership:** SFT-COMP-CPLX-APPROXIMATION-001; Classical Computation family BASE; Fold approximation-complexity law.
- **Forced law:** The forced resource versus retained objective distinction kernel retains optimization domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter. Exact result: The forced resource versus retained objective distinction kernel retains optimization domain; exact optimum class; candidate output; exact objective comparison; approximation ledger; resource trace; failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-CPLX-AVERAGE-WORST-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No unregistered metric or irrational ratio is admitted.
- **Evidence identities:** derivation sha256:661876c7efa17773960092329223bd1b8d3810c74167337614c4ca3861010c6f; independent implementation sha256:49e8ad275c48b129eb12592995af3840143a8ca0dc7ebb66d8a7669f64daaf86; independent certificate sha256:514e851668812207b41ed55e5cc45d02f101c37855b6c84347a0da2cceedff34; empirical/external sha256:71afb1d9276d8465dd2707358515b59d69b86fd671bca76be7680464eb4d47f3; engine receipt sha256:e8062d9de51f472dc26ca19cd2e838de843c9b1449d16c71962ad2c06e9f8357 at receipts/engine/model_admitted/SFT-COMP-CPLX-APPROXIMATION-001-e8062d9de51f472d.json.

SFT-COMP-OBL-BASE-034 - Fold parameterized-complexity law

- **Claim and ownership:** SFT-COMP-CPLX-PARAMETERIZED-001; Classical Computation family BASE; Fold parameterized-complexity law.
- **Forced law:** The forced resource stratification by exact structural coordinates kernel retains input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation; every operation is

source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced resource stratification by exact structural coordinates kernel retains input support; named structural parameter; parameter extraction proof; family partition; resource profiles; reduction preservation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-APPROXIMATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Parameters must be forced structural coordinates, never fitted constants.
- **Evidence identities:** derivation `sha256:f97fafb086546626bbe55c2f9544769c9b21ebf937e83b95a9c93439535d978a`; independent implementation `sha256:d10266176661ea8f4c2625bac8f828473fd9a287f9e056a12ff5358d8e7c2de9`; independent certificate `sha256:90751ccec65ad243aa877cde32c4ad9520033e0bcbc110e0b293ea452b879295`; empirical/external `sha256:0bd259151d1801258dce6352daa80fcef344a016412df453f479f07474c95eed`; engine receipt `sha256:1afcbc7d33437e51f057b4a63715fc3c5e41890025b3d093edb4fbeb15b370ac` at `receipts/engine/model_admitted/SFT-COMP-CPLX-PARAMETERIZED-001-1afcbc7d33437e51.json`.

`SFT-COMP-OBL-BASE-035` - Fold information and descriptive-complexity law

- **Claim and ownership:** `SFT-COMP-CPLX-DESCRIPTIVE-001`; Classical Computation family `BASE`; Fold information and descriptive-complexity law.
- **Forced law:** The forced shortest exact generated descriptions kernel retains frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`. Exact result: The forced shortest exact generated descriptions kernel retains frozen description grammar; complete descriptions through a declared depth; decoding relation; object reconstruction; minimal-description class; invariance ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-COMP-CPLX-PARAMETERIZED-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Unrestricted shortest-description values remain uncomputable; finite censuses state their depth explicitly.
- **Evidence identities:** derivation `sha256:75c939154f2d150b36c52e8c27a9080fa0bb6df65b999fd5f9dd70dc6a71f6a4`; independent implementation `sha256:70a8a83cc65b2f6a44ed0abde6ef8fe9211e60fbdd5176225e943725e32c6a7a`; independent certificate `sha256:d7e0354f7e04bafa5d366e2c394282723134d2f980ce45bdaa624cd026b5b401`; empirical/external `sha256:4014842120d59fd7cb3078960451a7ac342f183f44b6ebd8a8d41d39d8bb2352`; engine receipt `sha256:bdbb14bfbdd53b05dd80b9b4fd0ea77e3f9eb303ab637ec6714029665a1f6eef` at `receipts/engine/model_admitted/SFT-COMP-CPLX-DESCRIPTIVE-001-bdbb14bfbdd53b05.json`.

`SFT-COMP-OBL-BASE-036` - Fold searching and ordering law

- **Claim and ownership:** `SFT-COMP-ALG-SEARCH-ORDER-001`; Classical Computation family `BASE`; Fold searching and ordering law.
- **Forced law:** The forced location and exact order construction kernel retains generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`. Exact result: The forced location and exact order construction kernel retains generated carrier; exact equality; comparison relation; search trace; ordering invariant; retained duplicate classes; terminal certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-DESCRIPTIVE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.

- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only registered finite carriers and comparison relations are covered.
- **Evidence identities:** derivation `sha256:5410a5a2e280a6c31796b5735e3a1e7a936ed1b57d35fd24fe35123ea1a905b0`; independent implementation `sha256:bc23f61ab78f015f932efa3d73e656a123c338d041692d13664cd89e5148ce46`; independent certificate `sha256:b960ead246c2b465341e9c2a29216e0f80b39e26b0f908dcc6495d376b333e77`; empirical/external `sha256:db458ef04ab0525c0f379719e9d03e03c74effc48f16623c36ed8cca6bf8481e`; engine receipt `sha256:915d9a65bf080d49519f15583d8ffab68375f99e1af014be611cdc25c6626c78` at `receipts/engine/model_admitted/SFT-COMP-ALG-SEARCH-ORDER-001-915d9a65bf080d49.json`.

`SFT-COMP-OBL-BASE-037` - Fold exact arithmetic-algorithm law

- **Claim and ownership:** `SFT-COMP-ALG-ARITHMETIC-001`; Classical Computation family `BASE`; Fold exact arithmetic-algorithm law.
- **Forced law:** The forced computation over exact positive parts kernel retains canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`. Exact result: The forced computation over exact positive parts kernel retains canonical arithmetic inputs; structural operations; intermediate exact parts; normalization; termination trace; result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-ALG-SEARCH-ORDER-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No floating, negative, irrational or imaginary proof value is admitted.
- **Evidence identities:** derivation `sha256:c124e9fe09f662ef9e56992f8295e5145539549a2216e8b0d4e7ca2cbb9f5c49`; independent implementation `sha256:fc377822b8b452cd195b9c076cac4feb03a72b5808dc38b3aee01b1ad69b1e63`; independent certificate `sha256:0f9ad44ec485d994f209fa3832e1a9af89c20b5dc67c9f5e656db5992b1397a5`; empirical/external `sha256:9715086d06bc1f22f0398fe1cfcb9cea665f0a125bb5b497c4e8297f4826251c`; engine receipt

sha256:af254db83c3bc4c67b1c2daa5195c8f74a1a9e4fbcd08b159c677579b9c1ab52 at receipts/engine/model_admitted/SFT-COMP-ALG-ARITHMETIC-001-af254db83c3bc4c6.json.

SFT-COMP-OBL-BASE-038 - Fold string and sequence algorithm law

- **Claim and ownership:** SFT-COMP-ALG-STRINGS-SEQUENCES-001; Classical Computation family BASE; Fold string and sequence algorithm law.
- **Forced law:** The forced finite ordered symbol organizations kernel retains complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model. Exact result: The forced finite ordered symbol organizations kernel retains complete positions; held symbols; substring or subsequence relation; scan trace; match ledger; reconstruction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-ALG-ARITHMETIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No ungenerated infinite stream is treated as complete.
- **Evidence identities:** derivation sha256:8959ca63fab8c4543e86b8aa029aea4a48ca8f66670d4ec2312ab32eee0b2a89; independent implementation sha256:20b535a46bd9170066a6f938dabfaa4ed83aedf5fb981eee653dcda6f8df5124; independent certificate sha256:dc0b982cf407e60e815f8ec2d5045aaab3145ab9ccd3426d93a74d2531f68d84; empirical/external sha256:f6566f41be47c5ebd8b3fe42fe8bc341cd8a41f560280ba78f8bfd59001f8ad4; engine receipt sha256:5c2fb101d63e8d37ba076927e1de853261936a53c33400eba91c9adbd135caaf at receipts/engine/model_admitted/SFT-COMP-ALG-STRINGS-SEQUENCES-001-5c2fb101d63e8d37.json.

SFT-COMP-OBL-BASE-039 - Fold tree and graph algorithm law

- **Claim and ownership:** SFT-COMP-ALG-TREES-GRAPHS-001; Classical Computation family BASE; Fold tree and graph algorithm law.
- **Forced law:** The forced finite relational traversal and structure kernel retains generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model.

Exact result: The forced finite relational traversal and structure kernel retains generated vertices and edges; root or start identity; frontier support; visited ledger; path or component witness; termination; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-ALG-STRINGS-SEQUENCES-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Applies to finite generated graphs; physical networks require empirical translation.
- **Evidence identities:** derivation `sha256:0b0a095159e3c047d835c3e4bbedecc2efa8b4986fe1693c2e5793a33fad1c63`; independent implementation `sha256:38f3cf31f1ade6c3a7438f948c812d8fc62eadcad6cca80e7bd97a55185fd25c`; independent certificate `sha256:8379189b1b6c280a2b2311ca5d84cb7843e834a887f3115d401407767ac06270`; empirical/external `sha256:4deefa9e67a39f8f15daace40fc9734b41590d50fbc7ec7f2b6e2f58866b7f8d`; engine receipt `sha256:0c1d76188752d9f584b9bec6db855143f954a2f65b9ad57375bcd3e68809f604` at `receipts/engine/model_admitted/SFT-COMP-ALG-TREES-GRAPHS-001-0c1d76188752d9f5.json`.

`SFT-COMP-OBL-BASE-040` - Fold algebraic and geometric algorithm law

- **Claim and ownership:** `SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001`; Classical Computation family `BASE`; Fold algebraic and geometric algorithm law.
- **Forced law:** The forced finite algebraic operations and incidence kernel retains exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`. Exact result: The forced finite algebraic operations and incidence kernel retains exact carrier; closed operation tables or incidence relations; canonical coordinates; transformation trace; invariant ledger; result certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-ALG-TREES-GRAPHS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited`

`Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Geometry remains finite and exact; no ungenerated continuum is imported.
- **Evidence identities:** derivation `sha256:81409fcc27f9a8511b396209692bbc2a28b900f3c0dfdce8c4c534c0f033d7ef`; independent implementation `sha256:73d59f6f67dc89af760a385a65b33aad744e89a90b7c5dd722208a098432ea30`; independent certificate `sha256:e06d8cce5e05c56e774a1101d444b554e16e3cb7f4ab4bd21d0c31f1cd207e0f`; empirical/external `sha256:779fd81207ac8c38ac5e261d9d80151e5868ca0d99cd774818697326c24817f3`; engine receipt `sha256:b2ba1fc787d153cd8673cf1166703cd608291c7136d7412bff9c9dc66f3de430` at `receipts/engine/model_admitted/SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001-b2ba1fc787d153cd.json`.

`SFT-COMP-OBL-BASE-041` - Fold dynamic-programming law

- **Claim and ownership:** `SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001`; Classical Computation family `BASE`; Fold dynamic-programming law.
- **Forced law:** The forced reuse of exact overlapping subproblem results kernel retains problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`. Exact result: The forced reuse of exact overlapping subproblem results kernel retains problem decomposition graph; dependency order; subproblem identities; retained result table; composition recurrence; final reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; The decomposition and optimal-substructure conditions must be proved for each problem.

- **Evidence identities:** derivation `sha256:cbb434e7fac471f725b58cafea9c8de8d29edc94e9641aba911b147392d433ce`; independent implementation `sha256:4ff3d626485a1d771847ddf3c068086a7e991b4dbf6a53c47df08a9b20446fe9`; independent certificate `sha256:2fc4b3935bb8d193969048a49d5f8e26392b5a087667239e58228cb18b23ad86`; empirical/external `sha256:132df1cbe4d7960c2524053d50748b28c616b110c46b1db58a2c82160b3a1bec`; engine receipt `sha256:f6b1c405c189bb4f86db10ec31ad2dc8adc8041e0833baeb6184ac1832ed2e17` at `receipts/engine/model_admitted/SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001-f6b1c405c189bb4f.json`.

`SFT-COMP-OBL-BASE-042` - Fold optimization-algorithm law

- **Claim and ownership:** `SFT-COMP-ALG-OPTIMIZATION-001`; Classical Computation family `BASE`; Fold optimization-algorithm law.
- **Forced law:** The forced selection of all exact optima kernel retains complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`. Exact result: The forced selection of all exact optima kernel retains complete feasible support; exact comparison relation; candidate traversal; incumbent equivalence class; dominance eliminations; optimum certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Does not assume differentiability, convexity or a conventional metric.
- **Evidence identities:** derivation `sha256:4ee440a4de7a6270ba9d798a00174a5706c2974136615a61a24bfdc4651dea92`; independent implementation `sha256:c9209269f93ffe1054bce5215f859ebea9fe5638dce655dbbb52047d94e8a1ad`; independent certificate `sha256:5f293ae15ca747f138ce75beaf35f7a5deec7a65aa77d3056fa6eb4e44ded3ed`; empirical/external `sha256:df0821a0b5130ebbc42860538bee57f122cdb6a8e509201fb639d54edc33dba0`; engine receipt `sha256:74aac9551388975891fa29de8f8a7843eb1afcbc2a3df316784f34caf7ebd110` at `receipts/engine/model_admitted/SFT-COMP-ALG-OPTIMIZATION-001-74aac95513889758.json`.

SFT-COMP-OBL-BASE-043 - Fold randomized-algorithm law

- **Claim and ownership:** SFT-COMP-ALG-RANDOMIZED-001; Classical Computation family BASE; Fold randomized-algorithm law.
- **Forced law:** The forced complete scheduled branch execution kernel retains deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model. Exact result: The forced complete scheduled branch execution kernel retains deterministic algorithm kernel; registered schedule support; one trace per schedule; output partition; success support part; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-ALG-OPTIMIZATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Superdeterminism is preserved: apparent random choice is an unresolved or externally measured schedule, not an uncaused event.
- **Evidence identities:** derivation sha256:660b6dd0cc83e1b9b5b002f08369fd5274b6f857c3c875579352bc279b6e5b12; independent implementation sha256:dd42d329d28bec95bbb52eb2a0d11237935964122aa642def54ad716b7615c18; independent certificate sha256:acba3f2adc7ce42c9c8aab1fe43243c633a3c380e20b5ddea1f722b7bcee8365; empirical/external sha256:3c5f4558ce2d237409b4c80bf00213d9dc8b3461ce506a0de72e7385baed043f; engine receipt sha256:1c9f175ed883e3267b8722b8481cc424959ae5980bab9a6cf81e691d0785e0ae at receipts/engine/model_admitted/SFT-COMP-ALG-RANDOMIZED-001-1c9f175ed883e326.json.

SFT-COMP-OBL-BASE-044 - Fold parallel-algorithm law

- **Claim and ownership:** SFT-COMP-ALG-PARALLEL-001; Classical Computation family BASE; Fold parallel-algorithm law.
- **Forced law:** The forced causally independent algorithm steps kernel retains dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model.

Exact result: The forced causally independent algorithm steps kernel retains dependency graph; ready-set layers; local invariants; synchronization rows; combined trace; output reconstruction; work and span ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-ALG-RANDOMIZED-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No physical speedup is claimed without an implementation experiment.
- **Evidence identities:** derivation `sha256:568ed2eac60d7b9834407009080c157b70f5a31cb6a03762ca9660238d6a4216`; independent implementation `sha256:1bb66536383966cbfc406201ad9e09aa506c9dc78b60816366d649a6b1499f78`; independent certificate `sha256:9e0f1e409141fb2593e3015afd7188e5d260cb8dca6ccf3eb66894fa0736df9b`; empirical/external `sha256:6e9a823a975e4b99cd0abfbe7c2aa1a84ee3438a2f70578e37616c8e073cb9dc`; engine receipt `sha256:c16c0747c8994451d421ddb5bc14c4715d7cd4ea6c78588c3da0d318a66ca285` at `receipts/engine/model_admitted/SFT-COMP-ALG-PARALLEL-001-c16c0747c8994451.json`.

`SFT-COMP-OBL-BASE-045` - Fold distributed-algorithm law

- **Claim and ownership:** `SFT-COMP-ALG-DISTRIBUTED-001`; Classical Computation family `BASE`; Fold distributed-algorithm law.
- **Forced law:** The forced local processes exchanging exact records kernel retains process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`. Exact result: The forced local processes exchanging exact records kernel retains process identities; local states; message alphabet; send and receive rows; causal trace; fault boundary; global result reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-ALG-PARALLEL-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; System implementation and physical network claims are outside this formal theorem.
- **Evidence identities:** derivation `sha256:8785aed4c9eaf6cd44caf7eea5d18b54c7830fd88b6afdbe1ccb3fc5b95ae2e5`; independent implementation `sha256:abb34c0e7b42e093e771d0daec22d5b35f973b7c669013ad640a5bd3bbb8c3de`; independent certificate `sha256:c91f515c7a13d50e8bbb9e8e41e6e074556f008800138ee68beddb8b79ce48d2`; empirical/external `sha256:f52cd6b95a6e1d6c10e77cc3690523ed9d0303cf0517c8efbb56d1b22b668361`; engine receipt `sha256:5dfc71b4941c7e3f629b7ee0ab6cd3ca1851222fbfaf68690fa67875ed1f7e60` at `receipts/engine/model_admitted/SFT-COMP-ALG-DISTRIBUTED-001-5dfc71b4941c7e3f.json`.

`SFT-COMP-OBL-BASE-046` - Fold online and streaming algorithm law

- **Claim and ownership:** `SFT-COMP-ALG-ONLINE-STREAMING-001`; Classical Computation family `BASE`; Fold online and streaming algorithm law.
- **Forced law:** The forced prefix-bounded processing kernel retains generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`. Exact result: The forced prefix-bounded processing kernel retains generated input prefixes; retained state summary; update relation; output after each prefix; lost-distinction ledger; memory trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-ALG-DISTRIBUTED-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A completed infinite stream is not admitted; every result names its generated prefix boundary.
- **Evidence identities:** derivation `sha256:aa17697469db35050d4793ba86d2bcbfbf2c6cb91c0ac3bc06c9a2b728449927`; independent implementation `sha256:2645fa5246f2c6acd8fdb8ae2b5d7375a2da27904f8060ce8bb6f10ff936952a`; independent certificate

sha256:86840c27e3c57fc8a9b6378bf020ea0b56e1e428a71d120792d65b38d5477d02; empirical/external sha256:1f9abd0ad082a83536f54eca31ebbd54ed32f9b7c82538cbadfedd5d28815fa9; engine receipt sha256:1978c4757d70afe4f9f64a74ddad4b9df88db850c160cb3840dbe18f2ada7a15 at receipts/engine/model_admitted/SFT-COMP-ALG-ONLINE-STREAMING-001-1978c4757d70afe4.json.

SFT-COMP-OBL-BASE-047 - Fold numerical-algorithm law

- **Claim and ownership:** SFT-COMP-ALG-NUMERICAL-001; Classical Computation family BASE; Fold numerical-algorithm law.
- **Forced law:** The forced exactly bounded approximation procedures kernel retains exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model. Exact result: The forced exactly bounded approximation procedures kernel retains exact input parts; finite discretization; update trace; orientation-labelled error; stability witness; convergence certificate; output interval; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-ALG-ONLINE-STREAMING-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Decimal or floating results are measured/implementation records and cannot establish the proof.
- **Evidence identities:** derivation sha256:5f68377aa21130c232f506eb872b3c5ed5cb0eeff344de6f941f24c91fbd9476; independent implementation sha256:197cdb1b49b090d0b84f1c28041860cb18342a7cb65b0a19e55697b32b3205a9; independent certificate sha256:c9d66fcf52bfcea0b05cc05364626e91ff5b7c8b210f33ef2a03ba407de5fa90; empirical/external sha256:9ecc0471fe705aa388878b030820bfd0ba9801611de1780c7d0b7934ee91b598; engine receipt sha256:3f1c19281f4ac6c86c5bb13ae391ede42fa858c66e0571a4d6d66f27eeec098f at receipts/engine/model_admitted/SFT-COMP-ALG-NUMERICAL-001-3f1c19281f4ac6c8.json.

SFT-COMP-OBL-BASE-048 - Fold symbolic-computation law

- **Claim and ownership:** SFT-COMP-ALG-SYMBOLIC-001; Classical Computation family BASE; Fold symbolic-computation law.
- **Forced law:** The forced canonical expression transformation kernel retains generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression; every operation is source-bound,

trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Unique surviving structure:** `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`. Exact result: The forced canonical expression transformation kernel retains generated syntax tree; exact rewrite rules; held occurrence identity; normalization trace; equivalence proof; reconstructed expression; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-ALG-NUMERICAL-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only expressions within the frozen grammar are closed.
- **Evidence identities:** derivation `sha256:cfa87346438f6199df20a903922e69866cdac4918ac03bb3df563b888733db65`; independent implementation `sha256:144a7d6137278c61bb5b2ec25919e1403d0967cab27d7f0974a48419b875dbdd`; independent certificate `sha256:9e7f855770ac962b504ba0f8531b0052f95a8575a756a3756d5cfc11f0c6eafb`; empirical/external `sha256:8c88ba5ced2cdc6dc2cd67336f03d8554d6bb70fc68bacb4140adcc8a593bbbb`; engine receipt `sha256:3a84f78267c8d51b1e8c1ff0981ba89929b1c2b5231b5fedcb0d522b87f9fa8d` at `receipts/engine/model_admitted/SFT-COMP-ALG-SYMBOLIC-001-3a84f78267c8d51b.json`.

`SFT-COMP-OBL-BASE-049` - Fold approximation-algorithm law

- **Claim and ownership:** `SFT-COMP-ALG-APPROXIMATE-001`; Classical Computation family `BASE`; Fold approximation-algorithm law.
- **Forced law:** The forced bounded resource solution with exact error ledger kernel retains feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`. Exact result: The forced bounded resource solution with exact error ledger kernel retains feasible support; exact objective relation; approximate candidate; retained and lost objective distinctions; bound certificate; resource trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-COMP-ALG-SYMBOLIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No imported norm, epsilon parameter or irrational threshold is permitted.
- **Evidence identities:** derivation `sha256:e8803cab22548d6aaf21e549db85da47ff3586c6af729685dccc017a1a2f8ceb`; independent implementation `sha256:ba63e01ccb420f64ca73dd0a592b1934c59848fc9bbae0045f153cb16e6b9e83`; independent certificate `sha256:649e7983abd8098c4ef6eaefc7c1376f7709b4f78451cef1ead1f64036e2d541`; empirical/external `sha256:ac0157e48b5aa1fd22157b65b84f646284bf773de8fc411db161c5a918821e72`; engine receipt `sha256:52dfa606dc9c680d10c2c990b5df3ed29fe4f91be95690eb0d4880df5fd7dffc` at `receipts/engine/model_admitted/SFT-COMP-ALG-APPROXIMATE-001-52dfa606dc9c680d.json`.

`SFT-COMP-OBL-BASE-050` - Fold program-syntax law

- **Claim and ownership:** `SFT-COMP-SEM-SYNTAX-001`; Classical Computation family `BASE`; Fold program-syntax law.
- **Forced law:** The forced generated program forms kernel retains complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced generated program forms kernel retains complete token alphabet; formation productions; syntax trees; held constructor identities; membership derivation; malformed rejection; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-ALG-APPROXIMATE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Syntax alone assigns

no execution meaning.

- **Evidence identities:** derivation `sha256:f044e21b0e7a95f6f56968665a58294e030e0bfef080756718c2aa9f35e7ff26`; independent implementation `sha256:a107cd56990ec6701bea95ee4d4bfc2058d930bd3eabf27de0b2b32d9fdd0961`; independent certificate `sha256:8b714790d85bdc3e7286ca82c336d9c1117fa2dc117dbd46ef229ef6d64a5adc`; empirical/external `sha256:81b83f0276bdde6805fb333b7889e1dbb969956453161217f2bb3cdf0d917c68`; engine receipt `sha256:0da324b3d1cd63a836a8494235259bffe143a912f6aace6745f55c1bcae10246` at `receipts/engine/model_admitted/SFT-COMP-SEM-SYNTAX-001-0da324b3d1cd63a8.json`.

`SFT-COMP-OBL-BASE-051` - Fold binding and substitution law

- **Claim and ownership:** `SFT-COMP-SEM-BINDING-SUBSTITUTION-001`; Classical Computation family `BASE`; Fold binding and substitution law.
- **Forced law:** The forced name identity and replacement kernel retains held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced name identity and replacement kernel retains held binder identities; free and bound occurrence ledgers; scope relation; fresh-name construction; capture-safe substitution; reconstruction; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-SYNTAX-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No presentation alias may replace canonical binding identity.
- **Evidence identities:** derivation `sha256:4d6312f18d849169c7052e425a5829af08525a4e25df7e057c9d99b787ba5e35`; independent implementation `sha256:e83b481c5e02dad02da6129dda8424aaaac21723874d893cf02204356846ae4a`; independent certificate `sha256:a757bc3b7aa1e6290b4b00eff8e3d2f07999383f3716d73e649dc90a05fcd9f8`; empirical/external `sha256:ba184fed5c0a038365c52d8da2a5658f865823719e1929cbbf4ed7bf90fd5f0e`; engine receipt `sha256:1200dc420f7ff8fcd37e5c3fd7e862fa3aa0d1466963d23d9fe3ce0a22ac103d` at `receipts/engine/model_admitted/SFT-COMP-SEM-BINDING-SUBSTITUTION-001-1200dc420f7ff8fc.json`.

`SFT-COMP-OBL-BASE-052` - Fold evaluation law

- **Claim and ownership:** `SFT-COMP-SEM-EVALUATION-001`; Classical Computation family `BASE`; Fold evaluation law.
- **Forced law:** The forced program-state execution kernel retains syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced program-state execution kernel retains syntax tree; exact environment; evaluation relation; intermediate configurations; terminal value or failure; complete trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-BINDING-SUBSTITUTION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only exact positive parts and generated data forms are semantic values.
- **Evidence identities:** derivation `sha256:3327352b243dc92f5a03be58dd7ef2f9509338a8551523c64daea50c75be5692`; independent implementation `sha256:45c3b84d2fd37e2b113a0894dfca52199cb83b19f5637f9bf8f54b5081fd5200`; independent certificate `sha256:f4bca7b32e999d76fc00c448137fa5dcfa056c2c270e7fb44b0bd9ce3ecfdb76`; empirical/external `sha256:d02ca6834776c37cadc0278024e7bd22283c70dfb96169cb5adbf74180d7dd10`; engine receipt `sha256:dfb202d42339a8c83ffbd8d1939ec3d6dd29819fb621b60f6dd559175315089a` at `receipts/engine/model_admitted/SFT-COMP-SEM-EVALUATION-001-dfb202d42339a8c8.json`.

`SFT-COMP-OBL-BASE-053` - Fold operational and denotational correspondence law

- **Claim and ownership:** `SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001`; Classical Computation family `BASE`; Fold operational and denotational correspondence law.
- **Forced law:** The forced trace-to-meaning agreement kernel retains operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced trace-to-meaning agreement kernel retains operational traces; compositional meaning records; syntax constructors; result map; soundness direction; adequacy direction; counterexample boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported

answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-EVALUATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Correspondence is proved only for the frozen language grammar.
- **Evidence identities:** derivation `sha256:e8941a8b0de586b68b95ede9a238f45d9ffc7855a9c6492b52331ae4ced42991`; independent implementation `sha256:5220d7142d967826226e5a0bf576cf87ab1f71af680374da69e82ace109ff839`; independent certificate `sha256:dacb28b48850243cca84186f124d109878e9c4ecaf44f1cde9d151295f221fc1`; empirical/external `sha256:e408f1e0f0a65f69163d1db61238d7db90086204da7f256bbdda42edc721b778`; engine receipt `sha256:6db365c1b3c22883c276eec14485e6fb962528278f521654cba633fe6536e425` at `receipts/engine/model_admitted/SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001-6db365c1b3c22883.json`.

`SFT-COMP-OBL-BASE-054` - Fold type-theory law

- **Claim and ownership:** `SFT-COMP-SEM-TYPE-001`; Classical Computation family `BASE`; Fold type-theory law.
- **Forced law:** The forced structural admissibility of composition kernel retains generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced structural admissibility of composition kernel retains generated type forms; term forms; typing relation; exact interfaces; substitution preservation; evaluation preservation; rejected mismatch traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No conventional type calculus is assumed beyond downstream comparison.
- **Evidence identities:** derivation `sha256:fda760ee8e868f300216cd11a36d93a8e7b6845e8fcb70dcfd3802a003868364`; independent implementation `sha256:241f76ae21ddba0448e9e66a4d78ddeb9f271415f6b99d065fe0d6f2fb296693`; independent certificate `sha256:ba5a47bb0f58c6316a328e9317018057175150177e5259da5333957b760915a9`; empirical/external `sha256:8ade6d8bad00291ecb7005a195c0863d988e7381eb68b9a3f9127ef12b98f5cc`; engine receipt `sha256:f62ebd67f072c2e126b16d47d16cf1982426bc5c333be0c79d9148e43a505620` at `receipts/engine/model_admitted/SFT-COMP-SEM-TYPE-001-f62ebd67f072c2e1.json`.

`SFT-COMP-OBL-BASE-055` - Fold program-equivalence law

- **Claim and ownership:** `SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001`; Classical Computation family `BASE`; Fold program-equivalence law.
- **Forced law:** The forced indistinguishable generated behavior kernel retains two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced indistinguishable generated behavior kernel retains two program descriptions; common input support; complete traces; observation relation; retained/closed distinctions; equivalence certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-TYPE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Equivalence is relative to the registered observation and input boundary.
- **Evidence identities:** derivation `sha256:71c3f701a6c006e7d5b9d3ed693c33ac15f4b037304cb7e6db48e12d2c3f1939`; independent implementation `sha256:eb20d8ce5ccc05057939be3429f6cc04f52154350f122608e3d34fd2013fe600`;

independent certificate `sha256:fca679a2f596aef6b1790e002eb6cab7145c49feb024f6f7f5e0fe3f2266e09b`; empirical/external `sha256:47b8bb4f01597ff56e8a874ee2ac589f2890d9ae8142f3421b559721bd462710`; engine receipt `sha256:ad7d0dba57a7c30f17d67ef1e94ba22b9fe9e126565b894d97f435d1f83e4a17` at `receipts/engine/model_admitted/SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001-ad7d0dba57a7c30f.json`.

`SFT-COMP-OBL-BASE-056` - Fold termination law

- **Claim and ownership:** `SFT-COMP-SEM-TERMINATION-001`; Classical Computation family `BASE`; Fold termination law.
- **Forced law:** The forced absence of continuing execution paths kernel retains program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced absence of continuing execution paths kernel retains program and input support; transition relation; well-founded generated descent witness; complete traces; terminal coverage; cycle counterexample; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No total termination decider is claimed for the universal language.
- **Evidence identities:** derivation `sha256:963c938eee4f90c5b8e76126431979d0bc0fa6cacf4ee4b0cd4eb1c781c4da6d`; independent implementation `sha256:2b7e5bd0df6cb5e94e9d1b8b24d2de035186c83eb1772cd53cbc8bb1c215d59a`; independent certificate `sha256:540dbe2314f20eed8acba59ca366bd2d6f526d4aaa19f1edb81f19d22babe6e9`; empirical/external `sha256:847873d743a601a413e64211aff7e4a9fac44da1ed5553a91c7f4ae11ae86b14`; engine receipt `sha256:dc2378dbe15b3cfe2521a462c2bae6a7cef270b027719a9345618a9f1971c752` at `receipts/engine/model_admitted/SFT-COMP-SEM-TERMINATION-001-dc2378dbe15b3cfe.json`.

`SFT-COMP-OBL-BASE-057` - Fold program-correctness law

- **Claim and ownership:** `SFT-COMP-SEM-CORRECTNESS-001`; Classical Computation family `BASE`; Fold program-correctness law.

- **Forced law:** The forced agreement between execution and specification kernel retains precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced agreement between execution and specification kernel retains precondition support; program traces; postcondition relation; total or partial termination declaration; failure rows; proof certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-TERMINATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Correctness is conditional on the exact registered specification and input support.
- **Evidence identities:** derivation `sha256:db3ea2b1eecab1439dbe15f31592764e3e6515c7c3290359f02037c991077b7a`; independent implementation `sha256:5e849c094b2d3e312b9ebfc368174baf8c72a986093e0d9ba3c9a35ee6b2ed13`; independent certificate `sha256:a2927c2c2496399b11d4660cef42aa88f2e01a212fc0a22dbe8f72c5f384d227`; empirical/external `sha256:e3b70acc247f6998bf64ebde81b82de51ed5f70cdb364abc73bb3475175da81c`; engine receipt `sha256:95af5c4c8a0cf03d91f3d1ceff4decef32ba66537923e85eebe51ca3335e78bf` at `receipts/engine/model_admitted/SFT-COMP-SEM-CORRECTNESS-001-95af5c4c8a0cf03d.json`.

`SFT-COMP-OBL-BASE-058` - Fold formal-specification law

- **Claim and ownership:** `SFT-COMP-SEM-SPECIFICATION-001`; Classical Computation family `BASE`; Fold formal-specification law.
- **Forced law:** The forced exact admissible behavior descriptions kernel retains input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced exact admissible behavior descriptions kernel retains input and output carriers; relation of allowed rows; invariant support; liveness or terminal obligations; refinement boundary; falsification rows; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-CORRECTNESS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A prose intention cannot substitute for the generated specification relation.
- **Evidence identities:** derivation `sha256:29dbb881dca4124a901a109963c2678dc4d45236a921efb02d9432fd8bfbf3be`; independent implementation `sha256:46650c558595b950805ed11f7bcac1b7f42cf1cb50c73235f9e4d9552e49286b`; independent certificate `sha256:4638802ddd94e44f8e1c86e10ff4c8c015435545abfea2e7f1953c0f94ff3fe1`; empirical/external `sha256:cfdcebe686b9010ec4fb4002d01833f1b5a5b415fb651124e54ee97f489fd1a8`; engine receipt `sha256:4aeb9f85a830e501877b8cc19d41095d2bbfc96656ff92be8fe5a6655d351a51` at `receipts/engine/model_admitted/SFT-COMP-SEM-SPECIFICATION-001-4aeb9f85a830e501.json`.

`SFT-COMP-OBL-BASE-059` - Fold program-transformation law

- **Claim and ownership:** `SFT-COMP-SEM-TRANSFORMATION-001`; Classical Computation family `BASE`; Fold program-transformation law.
- **Forced law:** The forced semantics-preserving program rewriting kernel retains source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced semantics-preserving program rewriting kernel retains source syntax; transformation rules; target syntax; step trace; semantic correspondence; resource ledger; inverse or loss record; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-SPECIFICATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.

- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Optimization claims require separate resource evidence.
- **Evidence identities:** derivation `sha256:f7339e44653a24a42d3ca997d9b329bf80c25c3bcd78dd51a59d428980f32750`; independent implementation `sha256:08c8b493d1011c559d7f1dc4158e7a0d78afcca76a1375879729e2286d25d92c`; independent certificate `sha256:e3da2d11b82a871c6a2a4f41266025039358dc053ac8c838c32509ac3cda9c44`; empirical/external `sha256:2c73a1c68b6d4c7f142c1c7fb7f003619cd12c9edeef1a0b5854ec15ef0a2994`; engine receipt `sha256:cdc9731adba6f15b42e9eeba2e0c93d29bfeadf4ae281fbfed527d54918a7226` at `receipts/engine/model_admitted/SFT-COMP-SEM-TRANSFORMATION-001-cdc9731adba6f15b.json`.

`SFT-COMP-OBL-BASE-060` **- Fold semantics-preserving compilation law**

- **Claim and ownership:** `SFT-COMP-SEM-COMPILATION-001`; Classical Computation family `BASE`; Fold semantics-preserving compilation law.
- **Forced law:** The forced translation between executable languages kernel retains source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced translation between executable languages kernel retains source and target grammars; total compiler relation; source evaluation; target evaluation; result correspondence; failure rejection; overhead ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-TRANSFORMATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Applies only to the two frozen language grammars and declared runtime interface.
- **Evidence identities:** derivation `sha256:92e1e598f3b553e315bbfe123fe701956b1769053d33261670e6431995c95c7d`; independent implementation `sha256:a8f2d406212e720b0ebb5c0f2d313a35b36a4dcb936a5d9625442bcf838a5cc3`; independent certificate `sha256:8443c6e6dc2296d4e8e972d22a890e06d77964796b7a80f00632bdd47e9d49af`;

empirical/external `sha256:30432a42c6a2a3e3bdc49bf3febd42443802c85172c313eb9da1244c13a0c98c`; engine receipt `sha256:1d1314650abf2f20bb952ce22c2d83de00e3bb90fa515fb5f25560c3e3bb96ae` at `receipts/engine/model_admitted/SFT-COMP-SEM-COMPILATION-001-1d1314650abf2f20.json`.

`SFT-COMP-OBL-BASE-061` - Fold verification and proof-system law

- **Claim and ownership:** `SFT-COMP-SEM-VERIFICATION-001`; Classical Computation family `BASE`; Fold verification and proof-system law.
- **Forced law:** The forced machine-checkable program evidence kernel retains specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`. Exact result: The forced machine-checkable program evidence kernel retains specification; proof-object grammar; proof checker; source binding; program identity; correctness conclusion; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEM-COMPILATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A passing test suite is not substituted for a proof over the complete declared boundary.
- **Evidence identities:** derivation `sha256:ab921149b3cdeea436b7570367e34b839a9a18bc9a4ef156f309283f8a7f543b`; independent implementation `sha256:8321619907caf8c1ded379e15ff860599cc8f92598361ccf5c23a701ae40cf84`; independent certificate `sha256:25deffb52997162e17bcd8ebc3d5aff42667672f858b9bedb610b358d40e8ef9`; empirical/external `sha256:54fcb3e9f8dca7638fa994ba1714cc738cbf8a5c01fabd5a568369008308019d`; engine receipt `sha256:e3312b01e5f796d522ced745f449997e595aade4f15b3b82b216840116f09f97` at `receipts/engine/model_admitted/SFT-COMP-SEM-VERIFICATION-001-e3312b01e5f796d5.json`.

`SFT-COMP-OBL-BASE-062` - Fold concurrency law

- **Claim and ownership:** `SFT-COMP-DIST-CONCURRENCY-001`; Classical Computation family `BASE`; Fold concurrency law.
- **Forced law:** The forced multiple enabled process transitions kernel retains process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Unique surviving structure:** complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle. Exact result: The forced multiple enabled process transitions kernel retains process identities; local states; enabled actions; interleaving or independent-step support; complete schedules; observation traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-SEM-VERIFICATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Concurrency is structural coexistence, not physical simultaneity.
- **Evidence identities:** derivation sha256:3b3dce2d1c980ed0f261048b03453cdc320ccf71758f32588aead5dc2ed45f10; independent implementation sha256:0413fd7e133b664c0ffb5febc57180d55a7208afe2afcc72d8bdf4014681f761; independent certificate sha256:19b88fd27379a91ef6baa0cc1468a60dee4b9c725b5071c1c1ab0e7e9c6185b4; empirical/external sha256:c5f5d639dc53cf049646a3a76a5a158d50ab3a6cc14755448c935c7027b7af5a; engine receipt sha256:a3a2d3ef12a8166cc2117b17c3a470314146123a2c474c73ab363c2d5cf2e1c7 at receipts/engine/model_admitted/SFT-COMP-DIST-CONCURRENCY-001-a3a2d3ef12a8166c.json.

SFT-COMP-OBL-BASE-063 - Fold computational-causality law

- **Claim and ownership:** SFT-COMP-DIST-CAUSALITY-001; Classical Computation family BASE; Fold computational-causality law.
- **Forced law:** The forced event predecessor structure kernel retains generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle. Exact result: The forced event predecessor structure kernel retains generated events; process order; message edges; transitive closure; acyclicity; concurrent incomparability; retained provenance; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-DIST-CONCURRENCY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 ->

SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Physical causal claims require a later empirical branch.
- **Evidence identities:** derivation `sha256:e9f316907a841ba2e2d3f4247665c19e61b4aea748aadf828a48d64bf0e35ed6`; independent implementation `sha256:a48a7a468603ad934d8d4544a55963be61f283842e47b7b7334c6affdf5472ab`; independent certificate `sha256:0d60e035ef2862fdeac2103dd4b7861b2b176d1acec4ffbfc4c6d8715d177925`; empirical/external `sha256:06d3b23bc054557f39397eccbc8efec57a1c3186c2c9d2aa54f5b2c87b6f3e2a`; engine receipt `sha256:902e87f46e187a03a813011d318a637eaab7c9fc3dedfc0fc2c59efcb0dffccf` at `receipts/engine/model_admitted/SFT-COMP-DIST-CAUSALITY-001-902e87f46e187a03.json`.

`SFT-COMP-OBL-BASE-064` - Fold synchronization law

- **Claim and ownership:** `SFT-COMP-DIST-SYNCHRONIZATION-001`; Classical Computation family `BASE`; Fold synchronization law.
- **Forced law:** The forced jointly enabled process transitions kernel retains participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`. Exact result: The forced jointly enabled process transitions kernel retains participant identities; readiness states; rendezvous label; joint transition; blocked rows; causal trace; release state; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-DIST-CAUSALITY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No timing or lock

implementation is assumed.

- **Evidence identities:** derivation `sha256:b400e103dedeca45bf97795420f164d6bc863dbd55be87c02e53e4948b9f9ded`; independent implementation `sha256:28138e0387e7e6c371c9a6e9741e41627908bb7f65fdede735ff6f21d1255452`; independent certificate `sha256:a23fe872cadd29b424590e9786b63c1a4924ea73bdce7e5d9b5dfc5b1ebe5602`; empirical/external `sha256:39a1d69fcde2915e256970362287e0796e1fbed21ddb9dbfd561c186892a4e53`; engine receipt `sha256:e1b5a01e71410f40f763377dcfc336f2d75bc664db00a510dbba8a075fa5aa0f` at `receipts/engine/model_admitted/SFT-COMP-DIST-SYNCHRONIZATION-001-e1b5a01e71410f40.json`.

`SFT-COMP-OBL-BASE-065` - Fold communication law

- **Claim and ownership:** `SFT-COMP-DIST-COMMUNICATION-001`; Classical Computation family `BASE`; Fold communication law.
- **Forced law:** The forced exact record transfer kernel retains sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`. Exact result: The forced exact record transfer kernel retains sender and receiver; message identity; send event; channel row; receive event; retained/lost distinctions; delivery trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-DIST-SYNCHRONIZATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Reliability, ordering and delay are separate declared channel properties.
- **Evidence identities:** derivation `sha256:ee57d87e666ad6d34008a82623868fa0d37191217a64b946e92763d500ee9e6d`; independent implementation `sha256:1a1371f4ca85d120e2b44065d766a0c078d3da9a064d413032cccfd29127353e`; independent certificate `sha256:1cdbc1e3b2c636063f22f6c59325a1d8a2a801e8deb31abfde748e92975e6f28`; empirical/external `sha256:a691fbf7e762715853ece6b2497150169d72428cb03c329de72e725f6770be1a`; engine receipt `sha256:59defe90f59d3eef926714873a97cc9e2a4d9d14391e5d6d7458f51687b23423` at `receipts/engine/model_admitted/SFT-COMP-DIST-COMMUNICATION-001-59defe90f59d3eef.json`.

`SFT-COMP-OBL-BASE-066` - Fold distributed partial-order law

- **Claim and ownership:** `SFT-COMP-DIST-PARTIAL-ORDER-001`; Classical Computation family `BASE`; Fold distributed partial-order law.
- **Forced law:** The forced causal event ordering kernel retains event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`. Exact result: The forced causal event ordering kernel retains event carrier; reflexive identity ledger; antisymmetric strict precedence; transitive closure; incomparable pairs; linear-extension support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-DIST-COMMUNICATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A total global order is not assumed where events are incomparable.
- **Evidence identities:** derivation `sha256:020e71262605b9bb2fcb82d912aae3c6d68941ae3f8bbf353eeacc051aba70ae`; independent implementation `sha256:07732d983dfb8e2c1da409d19134eaceca8ace59feb78186dc37d4eea5df7580`; independent certificate `sha256:9d32b2ce81cdbe022089e2b49fbcd2e85c900f05c78bb6928992ddf4a0fcf371`; empirical/external `sha256:d8a87eec1566bd3a23a3090815aa1a6dea8afa1268268100d1eefdd0e259da0e`; engine receipt `sha256:d04af273dac3a8a8cd3d64b1b7c72e3130c1e674320c6632b0c2b9e2a8225a0f` at `receipts/engine/model_admitted/SFT-COMP-DIST-PARTIAL-ORDER-001-d04af273dac3a8a8.json`.

`SFT-COMP-OBL-BASE-067` - Fold consensus law

- **Claim and ownership:** `SFT-COMP-DIST-CONSENSUS-001`; Classical Computation family `BASE`; Fold consensus law.
- **Forced law:** The forced terminal common decision kernel retains process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`. Exact result: The forced terminal common decision kernel retains process inputs; proposal validity; message traces; decision states; agreement relation; termination declaration; fault boundary; every operation is

source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-DIST-PARTIAL-ORDER-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Every consensus claim must name its communication and fault model.
- **Evidence identities:** derivation `sha256:cda13e15608d8e7b1a3f9aeb2859994b54270d7d838f6395ad8a193352f0c4c1`; independent implementation `sha256:03b718ff935db905cfbecf0ddd1b5b7aef28447895c8ccf568ba3c02c8d43974`; independent certificate `sha256:acdf2a8a619c4328635d5faceddde44915461068bd5fc53e010ce49aa41ff37e`; empirical/external `sha256:65fe96a9cb1b6d95a41f2d3bcbb7ef629252f95bf5546ed82d2e8c6582390160`; engine receipt `sha256:48ae9b96daea76ffeefb0c3b2b9869b3beea88dc03cfd1b5a90e6f250a4c4d1e` at `receipts/engine/model_admitted/SFT-COMP-DIST-CONSENSUS-001-48ae9b96daea76ff.json`.

`SFT-COMP-OBL-BASE-068` - Fold agreement-impossibility law

- **Claim and ownership:** `SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001`; Classical Computation family `BASE`; Fold agreement-impossibility law.
- **Forced law:** The forced indistinguishable predecessor limits kernel retains two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`. Exact result: The forced indistinguishable predecessor limits kernel retains two executions; identical local observation; different required decisions; unrecorded predecessor distinction; scheduler support; contradiction witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-DIST-CONSENSUS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Impossibility is relative to the exact asynchronous/fault grammar and does not transfer silently.
- **Evidence identities:** derivation `sha256:32e02c7b473ee4b9b2428c7bc89b1bbd83e07310152b5548182ce01d9ce41d3f`; independent implementation `sha256:c4eeffda44abbf12bffefefad3c1ad32e374ec2e35f2ccdcdb232fabb851b421`; independent certificate `sha256:f3b98f1d6b90feff2c25194bc0196b6076eed2df0983bf5d8a02c548c09e87be`; empirical/external `sha256:897f4e69e5287bc3147464d8e712bfc3087f0d8b8a2afbd2cab066599aff2ae4`; engine receipt `sha256:30b83eb403fb428502002c7b9d96f569a91195720f36d8e1675385e92919f48d` at `receipts/engine/model_admitted/SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001-30b83eb403fb4285.json`.

`SFT-COMP-OBL-BASE-069` - Fold replication and consistency law

- **Claim and ownership:** `SFT-COMP-DIST-REPLICATION-CONSISTENCY-001`; Classical Computation family `BASE`; Fold replication and consistency law.
- **Forced law:** The forced multiple retained state copies kernel retains replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`. Exact result: The forced multiple retained state copies kernel retains replica identities; update events; propagation relation; conflict classes; consistency observation; convergence trace; lost-order ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No implementation platform or wall-clock consistency is imported.

- **Evidence identities:** derivation `sha256:a72b53b74f5f1dd6ceed05797c12e0ffaf8eebe8751ff5e3169b4a40018038fd`; independent implementation `sha256:cecc16c2c4b38824cda8a93e2a0e383b2e91e070970b18b35d91bc539315d6df`; independent certificate `sha256:a94d015bc15181f18e941e198ccfde240b61e33c29edb0ae477fd31043f04d3b`; empirical/external `sha256:ad685d20578ad08d5ec909eb918daae661024ec45a05519097be78f761b0dd30`; engine receipt `sha256:79243bdec0bda3d90c090785f5a2b43444ff4a323b5d7b48f0a36f17c11dcf34` at `receipts/engine/model_admitted/SFT-COMP-DIST-REPLICATION-CONSISTENCY-001-79243bdec0bda3d9.json`.

`SFT-COMP-OBL-BASE-070` - Fold computational fault-model law

- **Claim and ownership:** `SFT-COMP-DIST-FAULT-MODEL-001`; Classical Computation family `BASE`; Fold computational fault-model law.
- **Forced law:** The forced declared deviations from process rules kernel retains nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`. Exact result: The forced declared deviations from process rules kernel retains nominal transition relation; generated fault actions; affected components; detection observations; recovery relation; uncorrectable boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-DIST-REPLICATION-CONSISTENCY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Faults not generated by the model remain outside the guarantee.
- **Evidence identities:** derivation `sha256:46ef8935bd002b706c9086252e41700af3efda873ca61ed9695b86d01b8bb10c`; independent implementation `sha256:401a670fcf47677f6133385e134ba15e0c845145663b3a3fe267ee72f0cc8543`; independent certificate `sha256:c56083388b4efc4858b69acdea2242163995edca605a8f532e908538453aa4ea`; empirical/external `sha256:b210256f53a86fd60631fbe0e6c7aa69c1164bd6d5fee88c3221aef797cd49a4`; engine receipt `sha256:812aad3547d57947e1e18963470f883725de08c4e299351306e308a1923bdf46` at `receipts/engine/model_admitted/SFT-COMP-DIST-FAULT-MODEL-001-812aad3547d57947.json`.

SFT-COMP-OBL-BASE-071 - Fold distributed-knowledge law

- **Claim and ownership:** SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001; Classical Computation family BASE; Fold distributed-knowledge law.
- **Forced law:** The forced information available to local processes kernel retains global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle. Exact result: The forced information available to local processes kernel retains global execution support; local observation maps; indistinguishability classes; individual knowledge ledgers; common-knowledge iteration; decision boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-DIST-FAULT-MODEL-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Knowledge is observation-relative exact support, not an imported modal axiom.
- **Evidence identities:** derivation sha256:dd0eef6ba8eba0f639d57fba8dc5e048b90dc3c83b4f4074b36d3de1fbc91254; independent implementation sha256:65d80ba430b121e33988d04246aa04fcf58ee245cff28f35e0b2b095eeea4738; independent certificate sha256:1c6d8edf0281e7a3cbfe30321cf9df2b4f87d8839ec1c901568d918f5011e8fc; empirical/external sha256:b5b38e7b5d6013a66581be1b1db819b1850351d585312e7b1d509786ca815e3c; engine receipt sha256:2e47ee8465ef950f9ad2d30eb331453e764abf2567a6fc83ed0ec99b9fceda41 at receipts/engine/model_admitted/SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001-2e47ee8465ef950f.json.

SFT-COMP-OBL-BASE-072 - Fold locality law

- **Claim and ownership:** SFT-COMP-DIST-LOCALITY-001; Classical Computation family BASE; Fold locality law.
- **Forced law:** The forced bounded dependency neighborhoods kernel retains network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-g

lobal-oracle. Exact result: The forced bounded dependency neighborhoods kernel retains network graph; process neighborhood; local inputs; communication radius trace; locally indistinguishable instances; output constraint; lower witness; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Locality is relative to the registered network and round relation.
- **Evidence identities:** derivation `sha256:52733a46f531bfb6cf359e04411633d64b8daa8d79e70c6bcbc2e114d7e98c69`; independent implementation `sha256:f480454682af4bc3c9fcd5ff8cf7586bd356b42f0b30c571d68b11d485510478`; independent certificate `sha256:7b519f5972fa40355453ff053370f91fa7cfe1e4c57bf309b3d21b93b0877892`; empirical/external `sha256:e049568757b31066d538d4633f44dc5ea8ec7822f55180090561079a7b29c8e8`; engine receipt `sha256:d75dcc0dc6937266743f829559bfe4b59dfe31fc0d2b186ccbd7979b13cba9fa` at `receipts/engine/model_admitted/SFT-COMP-DIST-LOCALITY-001-d75dcc0dc6937266.json`.

`SFT-COMP-OBL-BASE-073` - Fold network-computation law

- **Claim and ownership:** `SFT-COMP-DIST-NETWORK-COMPUTATION-001`; Classical Computation family `BASE`; Fold network-computation law.
- **Forced law:** The forced computation over generated communication graphs kernel retains network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`. Exact result: The forced computation over generated communication graphs kernel retains network carrier; local machines; link channels; schedule support; message traces; aggregate result; resource and fault ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-DIST-LOCALITY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Operating systems, cloud platforms and physical networks are validation domains only.
- **Evidence identities:** derivation `sha256:e5de17ce510c5dee17b619932baeff26f47fb39a45be46d95529ea1148b9cba6`; independent implementation `sha256:40e6bd0f4a4cddad95e591f8e345388b6093afba2cdfd10fcd854215af618fce`; independent certificate `sha256:f96f76e03ecacbfa6075d5fb3e6f25f8a0d814c7c994e2e0009644bf52d2e121`; empirical/external `sha256:0ff1e2f18949b5be3afa6a81b5dfb16479d840183d39010f58363b7a5433ae6c`; engine receipt `sha256:504861c17b0bc318c7dc6cfbfa43377ac42819e316735ab1b066b4dd11bb1951` at `receipts/engine/model_admitted/SFT-COMP-DIST-NETWORK-COMPUTATION-001-504861c17b0bc318.json`.

`SFT-COMP-OBL-BASE-074` **- Fold one-wayness law**

- **Claim and ownership:** `SFT-COMP-SEC-ONE-WAYNESS-001`; Classical Computation family `BASE`; Fold one-wayness law.
- **Forced law:** The forced forward computation versus inverse search kernel retains finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced forward computation versus inverse search kernel retains finite source support; total forward map; image fibres; adversary process grammar; resource bound; exhaustive success ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-DIST-NETWORK-COMPUTATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Unbounded finite inversion is not impossible; one-wayness is always resource-relative.
- **Evidence identities:** derivation `sha256:ddfec37a4c433a5afac7108f7c9efb94d851d621edc8c48b728cbcac61d14e40`; independent

implementation `sha256:17638c5f2b4650aa4143670864dcecd5cb1ff24031a8b29ddc170fb24655957b`; independent certificate `sha256:e6bbfdf570e26533617e51b323bcb81bd4ba9cf746b583b62b02998fe61e35cd`; empirical/external `sha256:72270df40fff869ade128e4de351c353a0ad9fcac4088f77270223812c1bf960`; engine receipt `sha256:a4de8e0ec3e624835896831a4545adc63b5aa37cb7b2cecb26404bda0175eb62` at `receipts/engine/model_admitted/SFT-COMP-SEC-ONE-WAYNESS-001-a4de8e0ec3e62483.json`.

`SFT-COMP-OBL-BASE-075` - Fold secrecy law

- **Claim and ownership:** `SFT-COMP-SEC-SECRECY-001`; Classical Computation family `BASE`; Fold secrecy law.
- **Forced law:** The forced observation-relative message indistinguishability kernel retains message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced observation-relative message indistinguishability kernel retains message support; key or transformation support; ciphertext relation; adversary observation; closed message distinctions; disclosure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-ONE-WAYNESS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Security requires exact adversary, resource and support definitions.
- **Evidence identities:** derivation `sha256:59ae9300995bd5a4e1ed134ea927bca7428ef8dcad953c194d96c85b25ec9aba`; independent implementation `sha256:0be3b5dfc3878312f12f0414d1c476063532fd88932fe3be91f8b421f3fafdde`; independent certificate `sha256:930976772bdb29fedd4075e465f45c4d9d4b626b8976852a9766ed754b60887e`; empirical/external `sha256:d5508adf2fe5b04b4e7189e2ca1a773a785d6cb98bffe668fdb6bea9758950c9`; engine receipt `sha256:73a1685ef3e59a0b4cf8b8b67a1590f8a3b94e1dd5ace2d7eff0a6bdd5183b74` at `receipts/engine/model_admitted/SFT-COMP-SEC-SECRECY-001-73a1685ef3e59a0b.json`.

`SFT-COMP-OBL-BASE-076` - Fold integrity law

- **Claim and ownership:** `SFT-COMP-SEC-INTEGRITY-001`; Classical Computation family `BASE`; Fold integrity law.
- **Forced law:** The forced detection of unauthorized record change kernel retains message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces; every operation is source-bound,

trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced detection of unauthorized record change kernel retains message identity; protected record; allowed transformations; adversarial changes; verification relation; accepted and rejected traces; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-SECRECY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Integrity does not imply secrecy or authorship.
- **Evidence identities:** derivation `sha256:986a5cc25662f593e3440f93aa672ba5b6464b76a10caceee4f535d930938358`; independent implementation `sha256:e50a46ec67178635ac4d6df9cde7330745719c7302ad7a655378325da782a6cb`; independent certificate `sha256:3c39a2ab4febb4ae372aef6e5ab87cb65c2d9591e3bd22c49c7e123e49d63b76`; empirical/external `sha256:d85d50f86707d889c10663fa0bcdc985d2ad7fefafc363d25289c143515903b7`; engine receipt `sha256:54e959203614f553ff7ac956f549af5e057a240855b4175f237fb7363e080943` at `receipts/engine/model_admitted/SFT-COMP-SEC-INTEGRITY-001-54e959203614f553.json`.

`SFT-COMP-OBL-BASE-077` - Fold authentication law

- **Claim and ownership:** `SFT-COMP-SEC-AUTHENTICATION-001`; Classical Computation family `BASE`; Fold authentication law.
- **Forced law:** The forced source-identity verification kernel retains principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced source-identity verification kernel retains principal identities; credential support; challenge records; response relation; verifier trace; impersonation support; acceptance boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-COMP-SEC-INTEGRITY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Identity claims are only as strong as the registered credential and adversary model.
- **Evidence identities:** derivation `sha256:19ab81842953854a3cb2d3354bb9670af318dac19fdbeaec14a93aeab64bfb82`; independent implementation `sha256:8c06c672907c71606cb536667777f63d84e0add9a28cdaaf7d80c7e9206b3efb`; independent certificate `sha256:30945f38ecef4be19638d405dbe7af6020ed8663aa260f2012d54eee4e231fe5`; empirical/external `sha256:cf6bfc6b8ab765bdc9f84ee432e122e03ea34764d788da1e97efcd71c6bfe5c1`; engine receipt `sha256:2c39303d9b2dd557650fe04f8acb2f132510ca3af8258dc10792a276796cb29c` at `receipts/engine/model_admitted/SFT-COMP-SEC-AUTHENTICATION-001-2c39303d9b2dd557.json`.

`SFT-COMP-OBL-BASE-078` **- Fold hashing law**

- **Claim and ownership:** `SFT-COMP-SEC-HASHING-001`; Classical Computation family `BASE`; Fold hashing law.
- **Forced law:** The forced fixed-interface many-to-one record mapping kernel retains message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced fixed-interface many-to-one record mapping kernel retains message support; deterministic map; complete output support; collision fibres; preimage ledger; composition and domain separation; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-AUTHENTICATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or

ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Collision resistance is resource-relative; collisions in finite larger domains are retained, not denied.

- **Evidence identities:** derivation `sha256:01617a3c0bf0134dc0c4fa284d583d22153e56db32bc9ba49acc70880b1282d9`; independent implementation `sha256:fd20652f052b8a1d5d18b9d12fdb3ee66fd12331f1075360660cff6aa0b45e23`; independent certificate `sha256:3e486f817fe2b3d7bdee5c000d0aac1c12fc77145d7d9e5a0ee989dc62551e4e`; empirical/external `sha256:0436e07b628c57057b9ba0f7cb4a2a0051a444fd9275d02bbf0243ef7daee657`; engine receipt `sha256:30adbf79b53883ef1c3266d37eabc5389dc245bd879ff5be7c3c81df62767984` at `receipts/engine/model_admitted/SFT-COMP-SEC-HASHING-001-30adbf79b53883ef.json`.

`SFT-COMP-OBL-BASE-079` - Fold commitment law

- **Claim and ownership:** `SFT-COMP-SEC-COMMITMENT-001`; Classical Computation family `BASE`; Fold commitment law.
- **Forced law:** The forced binding and hiding record phases kernel retains message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced binding and hiding record phases kernel retains message support; opening support; commit relation; verification relation; hiding observation classes; binding collision ledger; phase trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-HASHING-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Perfect hiding and perfect binding claims require compatible exact supports and may be mutually limited.
- **Evidence identities:** derivation `sha256:8d9f534c3964bff3c525e06a7a83a4dc6be41b3ec1ee7c57bcd60df2c6158686`; independent implementation `sha256:784fc2f024c0611721529092f94ec3afc5d735b7d812c1d0b1805b7a2e27e078`; independent certificate `sha256:7d125b12815107bbf048d8978fe647c04a510fe9c3f27c68e99cd5818c74015f`; empirical/external `sha256:33264f3ce62b46b936a0731aa0e0de8a6df818298318d342ac47e10b4816bd54`; engine receipt `sha256:402789a36ddf90549527f52c193fbb0d0065ed548efaeb8178b629a367c59e5f` at `receipts/engine/model_admitted/SFT-COMP-SEC-COMMITMENT-001-402789a36ddf9054.json`.

`SFT-COMP-OBL-BASE-080` - Fold signature law

- **Claim and ownership:** `SFT-COMP-SEC-SIGNATURE-001`; Classical Computation family `BASE`; Fold signature law.
- **Forced law:** The forced publicly checkable source authorization kernel retains message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced publicly checkable source authorization kernel retains message support; signing relation; verification relation; key identities; accepted signatures; forgery adversary support; correctness trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-COMMITMENT-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No number-theoretic hardness assumption is imported; any computational claim states its resource boundary.
- **Evidence identities:** derivation `sha256:f6c665845c4cb731e7abc6523af0082873c08a343c7e4c7974205bc696fd63a9`; independent implementation `sha256:9e1a001d6e4ab6299ffc425aaf2d7c3912bb1545278bc11e70e34efe3c45b0ef`; independent certificate `sha256:ac4c20be5d8a6615688ca4fcd27541f4b7a0358fd7751b355dbf4822ffabce62`; empirical/external `sha256:bdd57b0d7eb4417ceebfbd4c9b257631e20f1f7bebf266b2a4d1be764c09dfb7`; engine receipt `sha256:1e80fcc3c81615ba2f131e83a91759abecf9a02f7a23b0a19b191a612d820bce` at `receipts/engine/model_admitted/SFT-COMP-SEC-SIGNATURE-001-1e80fcc3c81615ba.json`.

`SFT-COMP-OBL-BASE-081` - Fold proof-of-knowledge law

- **Claim and ownership:** `SFT-COMP-SEC-PROOF-KNOWLEDGE-001`; Classical Computation family `BASE`; Fold proof-of-knowledge law.
- **Forced law:** The forced extractable witness-bearing interaction kernel retains statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced extractable witness-bearing interaction

kernel retains statement support; witness relation; prover and verifier processes; transcript support; acceptance; extractor relation; adversary boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-SIGNATURE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Knowledge means existence of an exact extraction relation in the registered grammar.
- **Evidence identities:** derivation `sha256:096e1be819c3cbbc16da33d8aee4a9ec53e1f5d88153ea987802509c2c2bed13`; independent implementation `sha256:f5004305f90756de4a02683a31cabda06491ba72784c1ce3036f03e625ea3523`; independent certificate `sha256:240f016841b14142ad7671c8a07814cdaa2920e5aaf78f57fb3a092a895b5f68`; empirical/external `sha256:6651ed591513d864d58c47fc68f4c0a5b16042cc0776045deb5e71b793ed5227`; engine receipt `sha256:30375e18292a970fd81f642586376049976d61928fff9bff79bb0ee0692f0eb6` at `receipts/engine/model_admitted/SFT-COMP-SEC-PROOF-KNOWLEDGE-001-30375e18292a970f.json`.

`SFT-COMP-OBL-BASE-082` - Fold zero-knowledge law

- **Claim and ownership:** `SFT-COMP-SEC-ZERO-KNOWLEDGE-001`; Classical Computation family `BASE`; Fold zero-knowledge law.
- **Forced law:** The forced verification without additional witness distinction kernel retains statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced verification without additional witness distinction kernel retains statement and witness supports; real transcripts; simulator process; observation relation; acceptance correctness; retained/closed information ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-PROOF-KNOWLEDGE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Zero knowledge is relative to the registered verifier observation and simulator support.
- **Evidence identities:** derivation `sha256:5be0fa434efed7312395f86a9afb749199774bb3b538534566c3b6a783f37f8c`; independent implementation `sha256:24b62399d900fdd7fe64843cd05cf1e6f3d765489dcecc3f0941281fa57dffb9`; independent certificate `sha256:a730450b5720e8b9b620c379e228ea634d9317646f7b52e43cbaf4f97fc7720f`; empirical/external `sha256:c46233d93c746e6cf701fd5f16b696cafc5f0f2188fd8307c103102101cfd87b`; engine receipt `sha256:060ee3e0322b39395749298b995a9f4d67c148ade8eeba389c05906219d720ed` at `receipts/engine/model_admitted/SFT-COMP-SEC-ZERO-KNOWLEDGE-001-060ee3e0322b3939.json`.

`SFT-COMP-OBL-BASE-083` - Fold multiparty-computation law

- **Claim and ownership:** `SFT-COMP-SEC-MULTIPARTY-001`; Classical Computation family `BASE`; Fold multiparty-computation law.
- **Forced law:** The forced joint function evaluation with local privacy kernel retains party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced joint function evaluation with local privacy kernel retains party identities; private inputs; communication process; joint output relation; local views; simulation or leakage ledger; fault boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-ZERO-KNOWLEDGE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Security requires an explicit corruption and communication model.
- **Evidence identities:** derivation `sha256:e439ad17a32ce5c851815d3b14ab76da6da98235dfef93ce7e5a634a6a21d85e`; independent

implementation `sha256:540ddd1dc71c191068a84d648aecaa8c87dca286b22f8eba3a22a2525d5c5bc4`; independent certificate `sha256:d2d42553baf0939f1eb080df51ff421d84598397699c3fe6debdfd601ddcc1b5`; empirical/external `sha256:1914804774000141c84f200cb7b301bd9c7c524f9c2825cda44fb861df601e2d`; engine receipt `sha256:211b27cbbbc88f0e82c4441ed0d47afe01703b5dd89892a3e35759025c5f29eb` at `receipts/engine/model_admitted/SFT-COMP-SEC-MULTIPARTY-001-211b27cbbbc88f0e.json`.

`SFT-COMP-OBL-BASE-084` - Fold adversarial-computation law

- **Claim and ownership:** `SFT-COMP-SEC-ADVERSARIAL-001`; Classical Computation family `BASE`; Fold adversarial-computation law.
- **Forced law:** The forced computation against generated hostile strategies kernel retains system process; adversary grammar; resources; interaction schedules; security property; complete favorable and unfavorable rows; stopping rule; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced computation against generated hostile strategies kernel retains system process; adversary grammar; resources; interaction schedules; security property; complete favorable and unfavorable rows; stopping rule; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-MULTIPARTY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; An undefined adversary cannot support a security theorem.
- **Evidence identities:** derivation `sha256:131a0d4c23230de9c1973b0660172a74c795f138add9a30cea8056abaa5f377f`; independent implementation `sha256:667c35f6442a6883878e24e51c08894ee8f86106c7937bb824eca4df7a94d595`; independent certificate `sha256:7b5cda64ba1d06a7f27e9723db24054715d8763d5e9b25d6a5087b88cbb06fe4`; empirical/external `sha256:bc5047e436959279dfd58647b6a20fae0e479e20fd3f6b94507402aacd616d4a`; engine receipt `sha256:feaa06a2fbc1430f8930075dada167efdfcb60df06ed5e21424557ce89b81b78` at `receipts/engine/model_admitted/SFT-COMP-SEC-ADVERSARIAL-001-feaa06a2fbc1430f.json`.

`SFT-COMP-OBL-BASE-085` - Fold information-theoretic and computational-security law

- **Claim and ownership:** `SFT-COMP-SEC-SECURITY-DEFINITION-001`; Classical Computation family `BASE`; Fold information-theoretic and computational-security law.

- **Forced law:** The forced exact separation of unconditional and resource-bounded security kernel retains secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced exact separation of unconditional and resource-bounded security kernel retains secret support; observation classes; adversary support; resource law; exact success parts; perfect and bounded cases; reduction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-ADVERSARIAL-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Information-theoretic and computational claims must never be conflated.
- **Evidence identities:** derivation `sha256:6b21cf6168cb2b3e44e495511de413b364806a7f4f525932ee7469ac1ba28d2a`; independent implementation `sha256:c7d7f2d2ad3488af844772f361a82d489dd4775d704d8ac1f78a139c8736d496`; independent certificate `sha256:fb830648518dacbf88f79aec34306fa7f19a7791d9a649d6e97483bfb7658a22`; empirical/external `sha256:8a7e1c759e576b62fa7c4db35de79ac8630f90be53a4d7b9d7faaf7ddf7af458`; engine receipt `sha256:e4af2a902c19997a8a81d6df7fe8ebd8246a2900aac642a6003b287d4bde7007` at `receipts/engine/model_admitted/SFT-COMP-SEC-SECURITY-DEFINITION-001-e4af2a902c19997a.json`.

`SFT-COMP-OBL-BASE-086` - Fold post-quantum security-boundary law

- **Claim and ownership:** `SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001`; Classical Computation family `BASE`; Fold post-quantum security-boundary law.
- **Forced law:** The forced classical claims under an enlarged adversary interface kernel retains classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`. Exact result: The forced classical claims under an enlarged adversary interface kernel retains classical scheme; declared quantum-capable oracle interface; query and resource ledger; reduction dependencies; failure boundary; migration correspondence; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-SECURITY-DEFINITION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No quantum operation is derived here; operational quantum adversaries wait for the Quantum Computation branch.
- **Evidence identities:** derivation `sha256:8134279bbf4e56299f7ac9f4a9473bcc5dd141f070698642f5fdac6b8cfc53e9`; independent implementation `sha256:2dda32b30a46541bb6bb86c5ebe17d4883594404de13200d4bd086e1656e6ec1`; independent certificate `sha256:a5a2e61fb892deaa58864f2562348e0e83b69771cb5249932797ab1d3f01e591`; empirical/external `sha256:76bac704135f577f50d1b819bfcb5395d19aa7490015fd3032691a1b592b23a6`; engine receipt `sha256:4e450e475eb74f6968064e4ba58530226f4a58f8a6c3fff12176306a4d028c4d` at `receipts/engine/model_admitted/SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001-4e450e475eb74f69.json`.

`SFT-COMP-OBL-BASE-087` - Fold inference law

- **Claim and ownership:** `SFT-COMP-LEARN-INFERENCE-001`; Classical Computation family `BASE`; Fold inference law.
- **Forced law:** The forced selection among generated hypotheses kernel retains hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced selection among generated hypotheses kernel retains hypothesis support; evidence rows; compatibility relation; retained candidates; eliminated candidates; proof trace; uncertainty ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.

- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Inference cannot import a prior distribution or pretrained weight.
- **Evidence identities:** derivation `sha256:86235ff5fef7bf72b8dce90c3aed7ffbd387d6ac9aec4bcf541f310352df4cb5`; independent implementation `sha256:17c8d94beaa2ae8bd3ef6b1ee8007444129b98b82b09740194612e4434583399`; independent certificate `sha256:bb6159b3a5f1ea8b49e4db61ce525e7b0c7f719296f9d966b1120bd369774de8`; empirical/external `sha256:4521c49947e66c92bb50422ef3747db8aadb7f6885d35a0b5cfd3fb40a8dee8d`; engine receipt `sha256:483c9c1cb618e3c57985fbbcda9d4c3450b6dc50874c3f86a5a09fd4a8e54160` at `receipts/engine/model_admitted/SFT-COMP-LEARN-INFERENCE-001-483c9c1cb618e3c5.json`.

`SFT-COMP-OBL-BASE-088` - Fold classification and prediction law

- **Claim and ownership:** `SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001`; Classical Computation family `BASE`; Fold classification and prediction law.
- **Forced law:** The forced sealed mapping from inputs to held labels kernel retains input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced sealed mapping from inputs to held labels kernel retains input support; label support; hypothesis relation; training rows; sealed rule; target-custody boundary; prediction and falsification records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-LEARN-INFERENCE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Natural-data performance requires the separate empirical constitution.
- **Evidence identities:** derivation `sha256:69d028c417c15a65f8d335b120de039ccbebe7c4c70c8b316ef553a049d63c42`; independent implementation `sha256:2e499ab68732d073e51c68bd9ccadff6a9dd19b2fb20ccba2853b66ade014b24`; independent certificate `sha256:cf48193afa6527e13bf45359d46ac6145cca308770d6a52628430a328afcd0a4`;

empirical/external `sha256:aeb1268d3d337b6596c4f33c8af49d6220db7e674a724796260844808d75f041`; engine receipt `sha256:2902c0a1035fc27073a7a1be9ad39806838e05a1ef7fb2cbf8e045f1705be080` at `receipts/engine/model_admitted/SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001-2902c0a1035fc270.json`.

`SFT-COMP-OBL-BASE-089` - Fold representation-learning boundary law

- **Claim and ownership:** `SFT-COMP-LEARN-REPRESENTATION-001`; Classical Computation family `BASE`; Fold representation-learning boundary law.
- **Forced law:** The forced generated encodings preserving declared distinctions kernel retains source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced generated encodings preserving declared distinctions kernel retains source forms; representation grammar; encoding relation; decoding or task relation; retained and closed distinctions; complexity ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No opaque learned representation is admitted as a law without an exact transformation trace.
- **Evidence identities:** derivation `sha256:9eaa78f002935ed855587d3be65f3de944d0a04472cd30757259663d777c07fe`; independent implementation `sha256:acf5b1a98cf9f1c0bef4dcd2b233d8e23c0861dc2f8d2e18a2f03e859204ce54`; independent certificate `sha256:a9d9123e1ce01feb9bba7752a76a7b7b8f72cab588a060badfcca338a18b8b2a`; empirical/external `sha256:3ae7189b33dfec76df2087ba9c398ab5d3aecdfac84eed0ab446e85e6db9c763`; engine receipt `sha256:aaf1bb06300eddd353f33d549cc9ee8ace48e5f131d530ff9bdb7976c1cd0061` at `receipts/engine/model_admitted/SFT-COMP-LEARN-REPRESENTATION-001-aaf1bb06300eddd3.json`.

`SFT-COMP-OBL-BASE-090` - Fold generalization law

- **Claim and ownership:** `SFT-COMP-LEARN-GENERALIZATION-001`; Classical Computation family `BASE`; Fold generalization law.
- **Forced law:** The forced extension from registered evidence to generated unseen support kernel retains hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure

condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced extension from registered evidence to generated unseen support kernel retains hypothesis class; development rows; withheld support; structural invariance certificate; sealed predictor; complete target rows; failure condition; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-LEARN-REPRESENTATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Generalization is empirical unless a structural successor proof exhausts the target family.
- **Evidence identities:** derivation `sha256:a9bc47b89cf7b5320b47b847a995c2a156143272e9219c67f0ea6f256ad0411e`; independent implementation `sha256:eb08a6b81d8f850328b67873f31028e849338accc1bf929d01b1506aa763cb80`; independent certificate `sha256:53095ce3818727b136a2ca6f4699d6f376ebc7219ea632a98129ffb4b4c18d2d`; empirical/external `sha256:73e4b939d6ae084174bf079d91dc2df84b6465f61b132169c263a04257f189f1`; engine receipt `sha256:c99db1b2646ffad5f54e060c0f1ae1b32a9d8ebedc719d36a6e3aec10abf446d` at `receipts/engine/model_admitted/SFT-COMP-LEARN-GENERALIZATION-001-c99db1b2646ffad5.json`.

`SFT-COMP-OBL-BASE-091` - Fold sample-complexity law

- **Claim and ownership:** `SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001`; Classical Computation family `BASE`; Fold sample-complexity law.
- **Forced law:** The forced evidence support required to close hypothesis distinctions kernel retains hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced evidence support required to close hypothesis distinctions kernel retains hypothesis support; possible examples; distinction pairs; selected sample; retained ambiguity; complete teaching or witness set; resource ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-LEARN-GENERALIZATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No asymptotic formula is imported; each family requires a generated certificate.
- **Evidence identities:** derivation `sha256:9fddabe49ce98bba0f880bb1cff97015501842d31fe376af0da5c9096627a420`; independent implementation `sha256:de4a1e8920b9a571c0477fbe9a05bec0fdf78829f65ce150603fcfa5c161ad9f`; independent certificate `sha256:027768d0addb63d237cba6455555e463eb1f01c6b030c8a672fe5f335c660ea1`; empirical/external `sha256:10fed92e8c01ab8b517e587549341509dfe7349243814214346717 51f5b6b5e8`; engine receipt `sha256:d4b13522a727a382b31bddb83eda909f6706c4f7b9bc48633b8e1fd28c133092` at `receipts/engine/model_admitted/SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001-d4b13522a727a382.json`.

`SFT-COMP-OBL-BASE-092` - Fold optimization-in-learning law

- **Claim and ownership:** `SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001`; Classical Computation family `BASE`; Fold optimization-in-learning law.
- **Forced law:** The forced exact objective selection over hypotheses kernel retains hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced exact objective selection over hypotheses kernel retains hypothesis support; exact objective relation; evidence rows; complete search or certified elimination; tied optimum class; update trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.

- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Gradient, differentiability and fitted parameter spaces are not assumed.
- **Evidence identities:** derivation `sha256:db572a877ce20e6fa9c85144aa3abc6588031199d66e6f967b6d744fe908f9bc`; independent implementation `sha256:a7e21f4888ff04407aee3540f61a11ac7d327f6d70bd97a48943ed830e819339`; independent certificate `sha256:da8458ec44ab9e55677ace1d413bc4be984ddb2033182648add409771da500af`; empirical/external `sha256:086de46d3035bc9d85c7be6b357dcca7180725c96eb1502478ba96d1ce65d1fe`; engine receipt `sha256:543256016991fee945bc9d01d128355d0a9777a814dc7428ff9146ac93a8a398` at `receipts/engine/model_admitted/SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001-543256016991fee9.json`.

`SFT-COMP-OBL-BASE-093` **- Fold computational-induction law**

- **Claim and ownership:** `SFT-COMP-LEARN-INDUCTION-001`; Classical Computation family `BASE`; Fold computational-induction law.
- **Forced law:** The forced rule construction from finite evidence kernel retains observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced rule construction from finite evidence kernel retains observation rows; candidate rule grammar; consistency relation; minimality; successor test; retained alternatives; falsification boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Finite agreement alone never proves unrestricted continuation.
- **Evidence identities:** derivation `sha256:3efc6894220eb6df93411e220a210816dfefa95a8195b2835ddc002134ed63ab`; independent implementation `sha256:2a7ec40d95f8877c0c4985d214c3a60c199fec4ff536133fecb7776cf01da5c2`; independent certificate

`sha256:fb353e73890cb1f146d4612722f473d27a53ae6b55e398ede969035e4a2fc80c`; empirical/external `sha256:c1989a33f208cd8840a0743d971659c41fff7acfa4df1fe38b105401638050eb`; engine receipt `sha256:919a71359ffd05922865e02e03ef9013abb551bc0ccb5b0b95363aee5cd998ed` at `receipts/engine/model_admitted/SFT-COMP-LEARN-INDUCTION-001-919a71359ffd0592.json`.

`SFT-COMP-OBL-BASE-094` - Fold search and planning law

- **Claim and ownership:** `SFT-COMP-LEARN-SEARCH-PLANNING-001`; Classical Computation family `BASE`; Fold search and planning law.
- **Forced law:** The forced action-sequence construction toward exact goals kernel retains state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced action-sequence construction toward exact goals kernel retains state carrier; action transitions; initial state; goal class; path support; plan selection; failure and resource ledgers; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-LEARN-INDUCTION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Only the generated environment model is covered.
- **Evidence identities:** derivation `sha256:15ffc8f3e5def5649d23e98ea4fdc3424e78eaadf8b6d73c82bb60159dcfcf57`; independent implementation `sha256:261e951d99ce2cda0b9fd6e33cada770b98f997f344ef9ab5d2983b8eae2f571`; independent certificate `sha256:aa5ddf2ede9accea8029726d20fa5375af6ea31049796f13c29f611c8715ad5f`; empirical/external `sha256:4d3e3ee3ed6bcd4c5d13258a994ffbc068e989dfd5dd8c1472d0bdc2ac2979e6`; engine receipt `sha256:fae483c37bc6bdfa9b29553a47c1620a9038fb736fd8cf2af77fe96535e780c7` at `receipts/engine/model_admitted/SFT-COMP-LEARN-SEARCH-PLANNING-001-fae483c37bc6bdfa.json`.

`SFT-COMP-OBL-BASE-095` - Fold reinforcement-computation law

- **Claim and ownership:** `SFT-COMP-LEARN-REINFORCEMENT-001`; Classical Computation family `BASE`; Fold reinforcement-computation law.
- **Forced law:** The forced policy evaluation through generated interaction traces kernel retains states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison; every operation

is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced policy evaluation through generated interaction traces kernel retains states; actions; deterministic or scheduled transitions; exact outcome labels; policy support; complete trajectory ledger; policy comparison; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-LEARN-SEARCH-PLANNING-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No stochastic cause, fitted value function or external reward scale is imported.
- **Evidence identities:** derivation `sha256:922810d79cb65be09158633bb008b07b816501fc5b47e20d5527bf4422f350fd`; independent implementation `sha256:3e6c09be62e44db9928e0f51ac60f3c0a21d392640b02961341d29aa93b546bd`; independent certificate `sha256:c3c5e2c21db8d0448d9c1f5fac9dfc1978a2e85db163404b0754b3a6be9d25e2`; empirical/external `sha256:656c0d9cc8004254483b20cb731012f582f72db6e28a241e4a7edc710c99ca83`; engine receipt `sha256:0cb38ecfea78976c1452c6d9974fff417f27c4cb05a6721698d9d8c1c4328a1a` at `receipts/engine/model_admitted/SFT-COMP-LEARN-REINFORCEMENT-001-0cb38ecfea78976c.json`.

`SFT-COMP-OBL-BASE-096` - Fold multi-agent computation law

- **Claim and ownership:** `SFT-COMP-LEARN-MULTIAGENT-001`; Classical Computation family `BASE`; Fold multi-agent computation law.
- **Forced law:** The forced interacting adaptive processes kernel retains agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced interacting adaptive processes kernel retains agent identities; local observations; action supports; joint transition relation; communication; outcome classes; equilibrium or cycle ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-COMP-LEARN-REINFORCEMENT-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Game-theoretic correspondence is downstream and requires exact preference relations.
- **Evidence identities:** derivation `sha256:fb345343a53247874bd2912ed6f77073446b826010268bc6afc8fae76c14a47f`; independent implementation `sha256:74bd7c749f6e5716fd7a91e3f9b6bc247a333b6ab385d4d268519368534aaf10`; independent certificate `sha256:d32be77b1c6b6a3bb80e9210bedabe9bd0248c382462bc8404e6de95610ffbfc`; empirical/external `sha256:9b57258598dd95151a00eb00580d24a060d532ee02a7914cd1e6ac36a5b8b443`; engine receipt `sha256:9d95cd6493bea3f2e926a8beb970becf6593a0aad4c08b4ea2f999e6760202a2` at `receipts/engine/model_admitted/SFT-COMP-LEARN-MULTIAGENT-001-9d95cd6493bea3f2.json`.

`SFT-COMP-OBL-BASE-097` - Fold adaptation law

- **Claim and ownership:** `SFT-COMP-LEARN-ADAPTATION-001`; Classical Computation family `BASE`; Fold adaptation law.
- **Forced law:** The forced trace-bound change of computational rules or state kernel retains initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced trace-bound change of computational rules or state kernel retains initial process; observation history; admissible update grammar; retained rule identity; update trace; performance relation; rollback or failure boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-LEARN-MULTIAGENT-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.

- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Adaptation cannot silently alter the constitutional engine or claim law.
- **Evidence identities:** derivation `sha256:fbfe8663647aa11d4c95ce51175bb01a3c0ed27da3ae8148bf20d8313d646fd8`; independent implementation `sha256:bd4698fccdf68c28d5cbeea2976f776b0f15dd2377b2ce00ca4714fa140b836a`; independent certificate `sha256:1edd726dbaeaf10b6bdd6f7e816f7073bbaf8f3533e1013670f69d938117b268`; empirical/external `sha256:4de8725dbfb89714aadda6c0fc76ce9dbc2bb2534f36911908c3ddb74d32f174`; engine receipt `sha256:64ec3907b0e627e9e06f675e15a8ee10818781609c850b20948008691aa9d1cb` at `receipts/engine/model_admitted/SFT-COMP-LEARN-ADAPTATION-001-64ec3907b0e627e9.json`.

`SFT-COMP-OBL-BASE-098` - Fold computational limits-of-learning law

- **Claim and ownership:** `SFT-COMP-LEARN-LEARNING-LIMITS-001`; Classical Computation family `BASE`; Fold computational limits-of-learning law.
- **Forced law:** The forced indistinguishable evidence and unidentifiable hypotheses kernel retains hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`. Exact result: The forced indistinguishable evidence and unidentifiable hypotheses kernel retains hypothesis pair; identical observed rows; different unobserved requirements; learner process; forced same decision; counterexample support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-LEARN-ADAPTATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Limits are relative to the exact evidence, hypothesis and target grammars.
- **Evidence identities:** derivation `sha256:f5e5533666d646c7ef7bf627324974f71f2c66942cc4a1bfcf76e931fba31ea4`; independent implementation `sha256:6f0b680771420d91f20a198e0be71505ba6bf4d1857ec9b562906dea65c7cadf`; independent certificate `sha256:2ce5d5a80478e0d65c29ff54fe9ab22de89e9c7b86b381f3b2eee2fd2919ced4`; empirical/external

sha256:8c965863ce40503dee10d74d35ebe003bcbb75400e84133ab01119fc3d1c62fd; engine receipt sha256:969046a8a85fc0802d0269d6a4a46e706d278b71450e8b7c7dbd1188fe21bc90 at receipts/engine/model_admitted/SFT-COMP-LEARN-LEARNING-LIMITS-001-969046a8a85fc080.json.

SFT-COMP-OBL-BASE-099 - Fold interpretability and verification law

- **Claim and ownership:** SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001; Classical Computation family BASE; Fold interpretability and verification law.
- **Forced law:** The forced traceable relation from inputs to decisions kernel retains model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter. Exact result: The forced traceable relation from inputs to decisions kernel retains model description; exact execution trace; premise ledger; decision certificate; counterfactual support; verifier process; tamper controls; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-LEARN-LEARNING-LIMITS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A narrative explanation or score cannot replace the exact trace.
- **Evidence identities:** derivation sha256:c8db64a84f070d98e4c50cdc3282e18181699e08907e6325c40ce7f8d0b3c710; independent implementation sha256:56cf69782fa0d5db9f55475a297018e3351f9e65f24512b0a88548f3d1b67943; independent certificate sha256:c9831ba4b85a410b2432b135d2d024421a92478a2ebe70c18e559d9e6ee863b6; empirical/external sha256:4983de7e26afc28ec869589a8285b7cac9c6a31bd71dfbc23b2f890b22223f45; engine receipt sha256:c0246290a461042d4797f427dfd27adaebf0954e5372091899b896b3418a1424 at receipts/engine/model_admitted/SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001-c0246290a461042d.json.

SFT-COMP-OBL-BASE-100 - Fold classical-learning model law

- **Claim and ownership:** SFT-COMP-LEARN-CLASSICAL-LEARNING-001; Classical Computation family BASE; Fold classical-learning model law.
- **Forced law:** The forced complete classical finite learning processes kernel retains examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported

answer-producing premise.

- **Unique surviving structure:** complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter. Exact result: The forced complete classical finite learning processes kernel retains examples; hypotheses; inference or update process; exact support parts; sealed evaluation; resource ledger; reproducible result; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Quantum learning operations remain entirely outside this branch.
- **Evidence identities:** derivation sha256:d0f21b1802e74a2e9b58544c4d87f8c3934cf28aa6628b44b863a14c4df9a29f; independent implementation sha256:c009e53b2ed811511621e85d68a3f422e8467044881b0c670d98c74a365f57d0; independent certificate sha256:17f40ce955c28ee371c3107d4e114ce125b9fd68dfe1ebdb760af5621159658e; empirical/external sha256:7f34c30e760456e4653d516838ec65c0f449d819f1a4fc2e2878ade0222383a2; engine receipt sha256:86cdfa5fb4a647883b4b1300ea87c83dddd01b8f34f96c3738fec0baa8f8a8ce at receipts/engine/model_admitted/SFT-COMP-LEARN-CLASSICAL-LEARNING-001-86cdfa5fb4a64788.json.

SFT-COMP-OBL-BASE-101 - Fold exact and approximate calculation law

- **Claim and ownership:** SFT-COMP-SCI-EXACT-APPROXIMATE-001; Classical Computation family BASE; Fold exact and approximate calculation law.
- **Forced law:** The forced separation of proof values and bounded approximations kernel retains exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior. Exact result: The forced separation of proof values and bounded approximations kernel retains exact reference forms; approximation forms; orientation-labelled difference; retained precision provenance; output interval; reconstruction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.

- **Dependency and root trace:** SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Approximate values cannot flow backward to select an exact law.
- **Evidence identities:** derivation `sha256:8d29ccecdb3e151f4ca0c51c27e60e628a21c6471db567add4c18e7e8f8f53a3`; independent implementation `sha256:5a5450ce9d27d00c9c63c225d46ecb4cb8ac2a8f713dc63ac331245ee27549c2`; independent certificate `sha256:a873faf2411c2484210b6488ffa52a8d7f4f564c06d6caab2a38e3524b1304d1`; empirical/external `sha256:3dc1f3c153b7c6420f71d1f912eef13516f2952441df895cd2f246e0b8dc640a`; engine receipt `sha256:d04dba138ab4ed09a8fba0c3751fc8588a7ad3b34790ac1f536b44b013b1a905` at `receipts/engine/model_admitted/SFT-COMP-SCI-EXACT-APPROXIMATE-001-d04dba138ab4ed09.json`.

`SFT-COMP-OBL-BASE-102` - Fold numerical-stability law

- **Claim and ownership:** `SFT-COMP-SCI-STABILITY-001`; Classical Computation family `BASE`; Fold numerical-stability law.
- **Forced law:** The forced output distinction under input or operation perturbation kernel retains exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`. Exact result: The forced output distinction under input or operation perturbation kernel retains exact input support; registered perturbations; algorithm traces; exact output differences; amplification relation; stable and unstable classes; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SCI-EXACT-APPROXIMATE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.

- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No floating epsilon or norm is imported without derivation.
- **Evidence identities:** derivation `sha256:a43ca827851cfe1c7db7b09f951670311b50ebcca5938c0c5d205e9e7c9a7f21`; independent implementation `sha256:ed227d1ffe6b9cca5e1bd78b18ea37b96420ec366c45fc0886fd956d6f69dae0`; independent certificate `sha256:f8a39efdb4e40d985b773a578ac4cbbe790f0491b94fdf2f9a1b2e5e5dce5d6f`; empirical/external `sha256:bd975ace29c4d474e3faf48d0b6399acf9f3e6cee4cd2f9f7178be691565b923`; engine receipt `sha256:f50a038a131659cc3156af54451f66c24d7562bb4284ab1ad6a71b6e97cc6980` at `receipts/engine/model_admitted/SFT-COMP-SCI-STABILITY-001-f50a038a131659cc.json`.

`SFT-COMP-OBL-BASE-103` - Fold error-propagation law

- **Claim and ownership:** `SFT-COMP-SCI-ERROR-PROPAGATION-001`; Classical Computation family `BASE`; Fold error-propagation law.
- **Forced law:** The forced transport of exact approximation distinctions kernel retains local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`. Exact result: The forced transport of exact approximation distinctions kernel retains local error ledgers; dependency graph; orientation labels; composition rules; accumulated bound; cancellation/merging provenance; terminal ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SCI-STABILITY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Signed arithmetic is represented by held orientation labels, not negative proof values.
- **Evidence identities:** derivation `sha256:3c1e26188805f325b58f26f59dd2b777a2b71a8294e318eb6a679564baab529d`; independent implementation `sha256:ea298bac8ba3d6d280b08d831a3da7564bed420acc69dfff8630e933bfb791f0`; independent certificate `sha256:8ff2ad3dbafe41f907f100ace8d23fb0c2b177a1e8720b8a1b6aec3c2c11e8b6`; empirical/external `sha256:10b7ab2f6ebc21135b80a940a993f81e0ab0a0120ca0db765d795507dc542d68`; engine receipt `sha256:083dc87700741205332c48c4a74aa5658b9f205141bcb39a13bb88288354230b` at `receipts/`

engine/model_admitted/SFT-COMP-SCI-ERROR-PROPAGATION-001-083dc87700741205.json.

SFT-COMP-OBL-BASE-104 - Fold discretization law

- **Claim and ownership:** SFT-COMP-SCI-DISCRETIZATION-001; Classical Computation family BASE; Fold discretization law.
- **Forced law:** The forced finite generated representation of a declared model kernel retains source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior. Exact result: The forced finite generated representation of a declared model kernel retains source relation; generated cells; coverage map; retained and closed distinctions; local operators; refinement successor; boundary ledger; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-SCI-ERROR-PROPAGATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; An ungenerated continuum is never the derivational premise.
- **Evidence identities:** derivation sha256:65fec14eb6a66e240b026fbc893525962285ed8e6dae678f503d42de9b2da781; independent implementation sha256:5c232f884af76629b5fe61f5338cdb2bd73cbb4a41d4b6890d01c357442e9d67; independent certificate sha256:72050853d92aed0efab89da8ae0097f4801c8609a7dcfc27edc23fc7b8b59b91; empirical/external sha256:8a903c90d8dd50d73ddbba422eab2bfe78126a414ee94f93c2ab812b78d70b12; engine receipt sha256:bcf98ba0535b49f76382780ec0f1e22b9f184249bb3ccb596463f48d0532080c at receipts/engine/model_admitted/SFT-COMP-SCI-DISCRETIZATION-001-bcf98ba0535b49f7.json.

SFT-COMP-OBL-BASE-105 - Fold convergence-certificate law

- **Claim and ownership:** SFT-COMP-SCI-CONVERGENCE-001; Classical Computation family BASE; Fold convergence-certificate law.
- **Forced law:** The forced successor refinement preserving and closing error distinctions kernel retains refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-succ

`essor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`. Exact result: The forced successor refinement preserving and closing error distinctions kernel retains refinement trace; exact outputs; comparison relation; nested error classes; structural base; successor descent; target boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SCI-DISCRETIZATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Closure is depth-independent only when the successor certificate proves the registered descent property.
- **Evidence identities:** derivation `sha256:ed6247aa7d641db78d7d340bf3c5b3d14aa9fbab778506d4ba1aba8dd53d24ca`; independent implementation `sha256:a1a4741ce2299d736fe56a7891feadc4e14e17b620a17b2356a08d5e6bb89598`; independent certificate `sha256:75712ca92daab71c4ed80079d561ccaae73c8f058914420ba5740326759e65df`; empirical/external `sha256:ce2f9d06a52c46ff06ffb04608e4f3146f2c152ca69836601707192be0df483a`; engine receipt `sha256:a3ae509175198d515e8ae636ad4b669d942524409b96f97c4ce26a94a98c03b5` at `receipts/engine/model_admitted/SFT-COMP-SCI-CONVERGENCE-001-a3ae509175198d51.json`.

`SFT-COMP-OBL-BASE-106` - Fold scientific symbolic-computation law

- **Claim and ownership:** `SFT-COMP-SCI-SYMBOLIC-001`; Classical Computation family `BASE`; Fold scientific symbolic-computation law.
- **Forced law:** The forced exact model-expression transformation kernel retains model syntax; exact rewrite rules; dimensional or type interfaces; normalization; identity proof; evaluator correspondence; result trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`. Exact result: The forced exact model-expression transformation kernel retains model syntax; exact rewrite rules; dimensional or type interfaces; normalization; identity proof; evaluator correspondence; result trace; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SCI-CONVERGENCE-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 ->

SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Physical interpretation remains outside formal symbolic closure.
- **Evidence identities:** derivation `sha256:ccdce6568fba17556bb00419d7c10978c1938ff74de2d20f214d126f993351ac`; independent implementation `sha256:a47d9f806660a188331934953c9585f113b3b60beeea1e994b4878c5172007fc`; independent certificate `sha256:f08802aacd47de59d3f84730e1e10748406d5e76f0e8b2d2ee36a2cd40dca71f`; empirical/external `sha256:a1d97bead16859a3ac163ad75406876848eb51cdbc35ae06549bfdf4827a341e`; engine receipt `sha256:484049210c99c3cf945baa6cf8f64f14beb56db6f50fe8ede3d322dee49c41b8` at `receipts/engine/model_admitted/SFT-COMP-SCI-SYMBOLIC-001-484049210c99c3cf.json`.

`SFT-COMP-OBL-BASE-107` - Fold simulation law

- **Claim and ownership:** `SFT-COMP-SCI-SIMULATION-001`; Classical Computation family `BASE`; Fold simulation law.
- **Forced law:** The forced execution of a registered model relation kernel retains model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`. Exact result: The forced execution of a registered model relation kernel retains model state carrier; transition law; initial support; schedule; complete trace; observation map; comparison boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SCI-SYMBOLIC-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A simulation validates consequences of its model; it cannot establish the model as a natural law.

- **Evidence identities:** derivation `sha256:b04cfb756fbbb395c383e325234af2577760c79c32252a1f80f6ba23a85a6cee`; independent implementation `sha256:b93c4dfbe30b3dbd96c4865fc0ef9b9ad242cd0c0c0e1376087d48dbc6bb9712`; independent certificate `sha256:1d19485b5b1604d13a29197016920d51da71b365201df597af35d2305911649e`; empirical/external `sha256:0f028f2611fc2b19447d3b2f57a2515829a4b95d8743d165903cb7ec50945962`; engine receipt `sha256:b06dd069154f18ea84aadfe5c47b070e0ec017113e01945e55690303f274e293` at `receipts/engine/model_admitted/SFT-COMP-SCI-SIMULATION-001-b06dd069154f18ea.json`.

`SFT-COMP-OBL-BASE-108` - Fold computational-dynamics law

- **Claim and ownership:** `SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001`; Classical Computation family `BASE`; Fold computational-dynamics law.
- **Forced law:** The forced iterated exact finite state transformation kernel retains state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`. Exact result: The forced iterated exact finite state transformation kernel retains state support; update relation; successor time labels; orbit traces; terminal and cycle classes; invariant ledger; perturbation support; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SCI-SIMULATION-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Physical time and continuous dynamics require later empirical correspondence.
- **Evidence identities:** derivation `sha256:9c0a4479e8a6808361dfac8eaa8d54c89067ff7aad1cf3294f8ff62c9e35ec9c`; independent implementation `sha256:e89c6e112f56650b644a04ff5b68e2ec9f48826e68f71975b1375d4beda1e1b3`; independent certificate `sha256:c10c1dfce8f8e6f23c60190fb1800adee24259093c8d851aaf5fc5cd9f0b4160`; empirical/external `sha256:b4faf647cc85a39c4437bb902767c84e3556daf5bb17f7265166feb6b6881346`; engine receipt `sha256:9f40f7a193ca6e09bbc51fd61d7a526dc81c9224a6797bac0c0c57e8005ae0de` at `receipts/engine/model_admitted/SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001-9f40f7a193ca6e09.json`.

SFT-COMP-OBL-BASE-109 - Fold inverse-problem law

- **Claim and ownership:** SFT-COMP-SCI-INVERSE-PROBLEM-001; Classical Computation family BASE; Fold inverse-problem law.
- **Forced law:** The forced reconstruction of sources from observations kernel retains source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior. Exact result: The forced reconstruction of sources from observations kernel retains source support; forward relation; observation; predecessor fibres; identifiable and ambiguous classes; retained prior exclusions; reconstruction certificate; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No fitted regularizer or prior may select among unresolved fibres unless separately registered.
- **Evidence identities:** derivation sha256:9aedf19b7d236a4710da983fbd678c9f179bccb617188f080fa9540c02385310; independent implementation sha256:631220d2e521292b1f4000a76e228380d7f613100c10076d8ebf18a09aac1162; independent certificate sha256:6ac9ca23bb7c37bb14f2795f7e768b25108c651af84a744950cdccf4c7b67ee0; empirical/external sha256:7c54dcae327e865ff3d257363c532cb2016fdbf4cdf42f6b21375c9bd8c3112d; engine receipt sha256:41ca83901e2ba93484c943df39462f4196c046678f4d9cb17afafb5277cacdde at receipts/engine/model_admitted/SFT-COMP-SCI-INVERSE-PROBLEM-001-41ca83901e2ba934.json.

SFT-COMP-OBL-BASE-110 - Fold computational-statistics law

- **Claim and ownership:** SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001; Classical Computation family BASE; Fold computational-statistics law.
- **Forced law:** The forced exact support summaries and inference algorithms kernel retains complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refin

`ement-successor__no-extra-scientific-prior`. Exact result: The forced exact support summaries and inference algorithms kernel retains complete sample support; exact count relations; statistic map; fibres; algorithm trace; uncertainty ledger; empirical boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SCI-INVERSE-PROBLEM-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Probability remains deterministic-support bookkeeping unless an external distribution is measured.
- **Evidence identities:** derivation `sha256:acc45ba06d6b9dffc0e01169f023269669bdfea1c720e15db7c5feb630c26096`; independent implementation `sha256:c6bb9adaf5e746b4ca7bed2d6a76b2f0183231070804981ed063b9f5eb8fa179`; independent certificate `sha256:bdf581ea46da78d86afa92203678ff3422cc8bc86070705f8cd5e304c31c275a`; empirical/external `sha256:641f049a78a531c8dd32e5f49d3e31ed2788db1ba097dbbf3d0fd941eb2dc193`; engine receipt `sha256:af3c12e592f489613af6f00e0bed93b948f0ccc0a7f975ec89a55fce888ceffa` at `receipts/engine/model_admitted/SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001-af3c12e592f48961.json`.

`SFT-COMP-OBL-BASE-111` - Fold high-dimensional computation law

- **Claim and ownership:** `SFT-COMP-SCI-HIGH-DIMENSIONAL-001`; Classical Computation family `BASE`; Fold high-dimensional computation law.
- **Forced law:** The forced product-support computation with retained coordinate structure kernel retains generated coordinate carriers; complete product support; sparse or factored relation; algorithm trace; resource ledger; lost-interaction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`. Exact result: The forced product-support computation with retained coordinate structure kernel retains generated coordinate carriers; complete product support; sparse or factored relation; algorithm trace; resource ledger; lost-interaction boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 ->

SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; Dimension is a generated coordinate trace, not a completed infinite vector space.
- **Evidence identities:** derivation `sha256:ef184706f84eb5d737e955ed436ae8913d9b5cf2db1cafa541a94c0cc4165f79`; independent implementation `sha256:66bba91ba3902bc2caf8dac70ef204e8cbad12d8f0f59634772f6f4f34d4cc04`; independent certificate `sha256:db5a4afe687da48685c9b4f295bf78d0c71745c3db90405d81abab15632101f0`; empirical/external `sha256:4c20be1f568135bdfa36bb34a1ca13ebee60bb55f9d19c208a26bf65cb119b06`; engine receipt `sha256:08b4870cfa97a44e956720e0bf6ccfef2216790ad6ffcc0261c127f46625f84d` at `receipts/engine/model_admitted/SFT-COMP-SCI-HIGH-DIMENSIONAL-001-08b4870cfa97a44e.json`.

`SFT-COMP-OBL-BASE-112` - Fold many-body computation law

- **Claim and ownership:** `SFT-COMP-SCI-MANY-BODY-001`; Classical Computation family `BASE`; Fold many-body computation law.
- **Forced law:** The forced composed local forms and interaction support kernel retains body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`. Exact result: The forced composed local forms and interaction support kernel retains body identities; local states; interaction graph; joint support; update relation; factorization and correlation ledgers; resource boundary; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SCI-HIGH-DIMENSIONAL-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; No physical many-body law or quantum amplitude is imported.

- **Evidence identities:** derivation `sha256:9150974b48199453faca9a3ec42c5774fd3ea6dbd947b91fe3fcbb4a634421ab`; independent implementation `sha256:1d2047ea1a1f1b8bf3b16ff8e2608a6d87b0df913c0a975134ffca4e7b842534`; independent certificate `sha256:b478fd50145544b87303e4c72b0ed5f442a50fca78cdebbdf0fa26609d28b8d8`; empirical/external `sha256:3c8b8b90e8e91e724142f784b5c7d34340401fab6d2fabcdc2b6c40547f84649`; engine receipt `sha256:4e4db714accde19f571063b6d1c1c64c9f07650378a797107cb424bea54de4ca` at `receipts/engine/model_admitted/SFT-COMP-SCI-MANY-BODY-001-4e4db714accde19f.json`.

`SFT-COMP-OBL-BASE-113` - Fold mathematical-modelling law

- **Claim and ownership:** `SFT-COMP-SCI-MATHEMATICAL-MODELLING-001`; Classical Computation family `BASE`; Fold mathematical-modelling law.
- **Forced law:** The forced explicit relation between formal structures and observations kernel retains phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Unique surviving structure:** `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`. Exact result: The forced explicit relation between formal structures and observations kernel retains phenomenon disclosure; model grammar; derivation or constitutional provenance; parameter prohibition; sealed predictions; external measurements; falsification and revision records; every operation is source-bound, trace-complete, boundary-explicit, composition-preserving, successor-general and contains no imported answer-producing premise.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-SCI-MANY-BODY-001 -> SFT-FOUNDATION-FORM-ENFORCEMENT-001 -> SFT-MATH-DISCRETE-001 -> SFT-MATH-GRAPH-NETWORK-001 -> SFT-MATH-LOGIC-PROOF-001 -> SFT-MATH-CATEGORY-TYPE-COMPOSITION-001 -> SFT-INFO-ENCODING-DECODING-001 -> SFT-INFO-CONSERVATION-LOSS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no conventional machine, algorithm, language, security definition or learning model may select the law; no numerical zero, negative, irrational, imaginary or floating proof quantity; no completed infinity or ungenerated continuum; no free, fitted or learned parameter; no application result or pretrained model; A successful fit or opaque prediction does not force the model law.
- **Evidence identities:** derivation `sha256:bf1cfe19e9f1107fc446eeb94708330d2a1f69e6593d9d1a18fcb2ef7938d7ec`; independent implementation `sha256:5be00110e135b7023427d7d71b21f44efbe3ace7848a73e02c41a286eb0a78de`; independent certificate `sha256:5599d7b3549eba1bb43cd7545c725f9fd7ddabf910f916c3ca1aeb0dc254e621`; empirical/external `sha256:852cfe82540a75b5e540d04049e107608bf2ddc7c663eca6626a69868ae516aa`; engine receipt `sha256:465c067c8c543effc5ba03cf8677a5caa2386f4695d1f3905b6e652060ba7ed4` at `receipts/engine/model_admitted/SFT-COMP-SCI-MATHEMATICAL-MODELLING-001-465c067c8c543eff.json`.

SFT-COMP-OBL-BASE-114 - Unrestricted native Fold Busy-Beaver law

- **Claim and ownership:** SFT-COMP-CBL-NATIVE-BUSY-BEAVER-002; Classical Computation family BASE; Unrestricted native Fold Busy-Beaver law.
- **Forced law:** For every supplied positive finite description depth k in the admitted native Fold closing-process grammar, every process halts within k lawful depth-lowering transitions and a depth-k source attains k; therefore BB_F(k)=k. Period-b recurrent processes carry an exact nonhalting certificate and are excluded from the closing maximum.
- **Unique surviving structure:** complete-native-fold-descriptions__one-lawful-depth-lowering-edge__exact-empty-one-terminal-trace__complete-upper-and-attaining-lower-witness__separate-exact-nonhalting-certificate__prepend-one-label-successor__all-process-traces-and-attainment__no-extra-machine-premise. Exact result: The unique complete kernel forces BB_F(k)=k for every supplied positive finite k in the admitted native Fold closing grammar, with recurrent processes separately certified nonhalting.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-CBL-HALTING-001 -> SFT-COMP-CPLX-TIME-SPACE-001 -> SFT-COMP-FORM-OPERATIONAL-PROCESS-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: independently_replicated.
- **Prohibited imports and boundary:** no external Turing transition table; no numerical-zero description depth; no completed infinite description; no recurrent process counted as halting; no claim about conventional Busy Beaver.
- **Evidence identities:** derivation sha256:06a6a2865e474cc04ca2997bc2676bb1905e407384a36b51fb78789ae6b33da0; independent implementation sha256:d24933201019bd380f9627f20b1b9f834afa2dd24d7f74ff4c0faf638fbbe112; independent certificate sha256:13a7ea9a33905f153390064af3a4bb96e4c7d487f820299637e10adb09fa1c61; empirical/external None; engine receipt sha256:92715d1d41f9021e01c27ac9962f7fa8c172b1c5237dd623bbab6da5bc1d143d at receipts/engine/model_admitted/SFT-COMP-CBL-NATIVE-BUSY-BEAVER-002-92715d1d41f9021e.json.

SFT-COMP-OBL-BASE-115 - Fold-P equals Fold-NP law

- **Claim and ownership:** SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002; Classical Computation family BASE; Fold-P equals Fold-NP law.
- **Forced law:** Within the admitted native Fold process grammar, exact deterministic evaluation emits the complete lawful trace verified by the already-derived proof system; soundness makes every accepted certificate equal the unique evaluation. Both resources equal description depth, so P_F=NP_F throughout this grammar.
- **Unique surviving structure:** complete-native-fold-word-family__unique-exact-closing-trace__compiled-lawful-evaluation-trace__edgewise-sound-complete-verifier__evaluator-emits-accepted-certificate__soundness-forces-unique-evaluation__prepend-label-trace-successor__native-equality-only. Exact result: The sole all-preserving kernel forces P_F=NP_F with exact resource k for every supplied positive finite native description depth, without asserting conventional P=NP.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-COMP-CBL-RECOGNITION-DECISION-001 -> SFT-COMP-CPLX-TIME-SPACE-001 -> SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001 -> SFT-COMP-SEM-VERIFICATION-001. Registered root theorem: reached through the cited Foundation/Mathematics/Information receipts. Imported axioms: 0; free parameters: 0.

- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no arbitrary external language family; no nondeterministic machine import; no polynomial convention import; no conventional P-versus-NP conclusion; no hidden certificate advice.
- **Evidence identities:** derivation `sha256:b32e863be01454e0c156a13913b55a09303333f2a5946bbe412df7126f238b2c`; independent implementation `sha256:3bb193e7add6ae94ba988cf174df28fa4e633efd664263e9ebb44e81654a61c3`; independent certificate `sha256:567b4be1c67de0164587ff7af3849ac615a1cdf7fdf1dbe5d502b051f32a382a`; empirical/external `None`; engine receipt `sha256:a6d19f95f567cd3d870b45330dbcb3b3387b5b3b47390c5ee1b92cbc5d2f6d74` at `receipts/engine/model_admitted/SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002-a6d19f95f567cd3d.json`.

`SFT-COMP-OBL-BASE-116` - Arbitrary admitted Fold-circuit lower-bound law

- **Claim and ownership:** `SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002`; Classical Computation family `BASE`; Arbitrary admitted Fold-circuit lower-bound law.
- **Forced law:** Across every circuit assembled from admitted Fold edges, closing a depth-k path requires at least k gates, complete source support requires width b^k, and complete layered coverage requires the sum of b^r edges for r from One through k. The registered circuit attains every bound.
- **Unique surviving structure:** `unique-lawful-one-depth-edge__one-distinction-per-dependent-edge__all-b-to-k-source-words__every-source-edge-required__complete-forced-edge-subset-census__registered-circuit-attains-all-bounds__add-next-complete-source-layer__admitted-fold-circuits-only`. Exact result: The unique kernel forces tight path k, width b^k and size sum(r=One..k)b^r lower bounds over every admitted Fold-edge circuit at every supplied positive finite depth.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-CPLX-CIRCUIT-RESOURCE-001 -> SFT-COMP-CPLX-BOUNDS-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no Boolean or external quantum gate basis; no rewired Fold edge; no sampled input support; no completed infinite circuit; no claim beyond admitted Fold circuits.
- **Evidence identities:** derivation `sha256:99debdca97c8fab7140ac9aa63f6bb03ddccd72ed98eb8e9f306e1c5192a491c`; independent implementation `sha256:9d8af6abadf3adf8437598d4b2f9a0eed3e98d6c59b97680e118de4cdba32702`; independent certificate `sha256:b9f8d7a26d4cfe3e7a36f1d7baf36c71b581a97997fe0a86476c16be703d9ff5`; empirical/external `None`; engine receipt `sha256:18e485a78cc263bd16ceae80afee117f5971e549802d60b696645652b596074d` at `receipts/engine/model_admitted/SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002-18e485a78cc263bd.json`.

`SFT-COMP-OBL-BASE-117` - Conditional conventional decision-family transport law

- **Claim and ownership:** `SFT-COMP-CPLX-CONVENTIONAL-TRANSLATION-003`; Classical Computation family `BASE`; Conditional conventional decision-family transport law.

- **Forced law:** A conventionally presented decision family inherits the admitted native Fold P_F=NP_F equality only when a registered total encoding covers every declared instance, preserves verdicts, deterministic traces and sound-complete certificates in both directions, and retains explicit polynomial size and execution bounds on a common input-size carrier. The theorem executes one complete external word family by an exact bijective translation; without those conditions no conclusion about arbitrary conventional P versus NP is admitted.
- **Unique surviving structure:** `complete-registered-external-family__total-bijective-source-bound-encoding__exact-bidirectional-verdict-preservation__complete-stepwise-trace-translation__sound-complete-bidirectional-certificate-map__explicit-common-size-polynomial-overhead__prepend-label-translation-successor__conditional-family-transport-only`. Exact result: The sole all-preserving kernel transports P_F=NP_F to the complete registered external two-label decision family with identity size and trace overhead, and proves that arbitrary conventional P-versus-NP export is inadmissible without the same complete translation certificate.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002 -> SFT-COMP-FORM-MODEL-EQUIVALENCE-001 -> SFT-COMP-CBL-REDUCTION-001 -> SFT-COMP-CPLX-INPUT-SIZE-001 -> SFT-COMP-CPLX-BOUNDS-001 -> SFT-COMP-SEM-VERIFICATION-001. Registered root theorem: `reached through the cited Foundation/Mathematics/Information receipts`. Imported axioms: `0`; free parameters: `0`.
- **Observation and independent reconstruction:** the preserved foundational package carries its admitted external-validation and implementation-distinct certificate identities; no later target was used to rewrite its historical registration.
- **Sources and custody:** receipt-bound exact computational observation corpus. External status: `independently_replicated`.
- **Prohibited imports and boundary:** no arbitrary conventional P-versus-NP conclusion; no partial, sampled or answer-changing encoding; no unmatched input-size or resource convention; no hidden advice, oracle or nondeterministic machine import; no completed infinite language object.
- **Evidence identities:** derivation `sha256:7f92ee63abcd8366fb3866ae1ad3ea2f4180a18d8ae91f6163c90b8657c34d4c`; independent implementation `sha256:3a883efc1dfa2f583c3fa86b0aaef724f4455ef58402f6ea8859d9a3a9e38d1c`; independent certificate `sha256:e21a55586caa5777b578cac35f5c93dc423389304495d51ca98cbd7f4fac212e`; empirical/external `None`; engine receipt `sha256:2b389d7cdae40491b9aeb9808b617aa660969daf44b047b15d29c481f0ac5419` at `receipts/engine/model_admitted/SFT-COMP-CPLX-CONVENTIONAL-TRANSLATION-003-2b389d7cdae40491.json`.

130.4 Family `FORMX` - 22/22 complete

Scope. Formal computation. This family completely decides 5,632 candidates, retains 22 unique survivors and passes 88 adverse controls.

`SFT-COMP-OBL-FORMX-001` - Configuration identity and canonical state encoding

- **Claim and ownership:** `SFT-COMP-FORMX-CONFIGURATION-IDENTITY-001`; Classical Computation family `FORMX`; Configuration identity and canonical state encoding.
- **Forced law:** A formal configuration is the canonical product of one retained process state, one complete generated storage word and one retained position or focus; equality requires equality of every coordinate.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__canonical-configuration-carrier__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-001 uniquely retains canonical-configuration-carrier, complete execution custody, root forcing, post-registry observation and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-STATE-TRANSITION-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-001: configuration_round_trip; exact execution: {"coordinate_count":3,"focus_retained":true,"state_retained":true,"storage_retained":true}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:6997e905dd2ba9bb6482853f115df4c4f5f74aaadb28534912ac5c4f9bac1c04`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:23e22ff7efe44ca86372a48129d46019ebca037f5a335fbcff159a2b3db85313`; engine receipt `sha256:05ef72b504a615bf9f8e04128b538c827f594e0e7f55391e473c57ce8463b86e` at `receipts/engine/model_admitted/SFT-COMP-FORMX-CONFIGURATION-IDENTITY-001-05ef72b504a615bf.json`.

`SFT-COMP-OBL-FORMX-002` - Partial and total transition relations

- **Claim and ownership:** `SFT-COMP-FORMX-PARTIAL-TOTAL-TRANSITION-002`; Classical Computation family `FORMX`; Partial and total transition relations.
- **Forced law:** A transition relation is total on its declared domain exactly when every generated source has a retained image; an absent row is a partial boundary and multiple rows retain branching rather than selecting one.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__source-complete-transition-relation__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-002 uniquely retains source-complete-transition-relation, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-STATE-TRANSITION-001 -> SFT-COMP-FORMX-CONFIGURATION-IDENTITY-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-002: partial_total_transition_distinction; exact execution: {"absent_source_row_retained_as_partial":true,"complete_source_rows":2,"multiple_images_not_silently_selected":true}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status:

empirically_tested_and_independently_replicated.

- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation sha256:2442c09fd0e34a3feefd1a2ada378966a431033a692797fcb65d6e2b52d6d172; independent implementation sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e; independent certificate sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896; empirical/external sha256:f8839ac7d68176e28a186b977ca17142dc523d21f83d47ed75818bcd206c9286; engine receipt sha256:922ed7d7b1ff3af08a6442d7d3a0d47a4c9b0c76ea5089c3e6ba56ab3c41fd75 at receipts/engine/model_admitted/SFT-COMP-FORMX-PARTIAL-TOTAL-TRANSITION-002-922ed7d7b1ff3af0.json.

SFT-COMP-OBL-FORMX-003 - Acceptance, rejection and nontermination distinction

- **Claim and ownership:** SFT-COMP-FORMX-TERMINAL-OUTCOME-DISTINCTION-003; Classical Computation family FORMX; Acceptance, rejection and nontermination distinction.
- **Forced law:** Acceptance and rejection are separately labelled terminal classes; nontermination is a retained recurrent or open execution trace and is never relabelled as either terminal outcome.
- **Unique surviving structure:** complete-canonical-carrier__source-bound-complete-transition__three-way-terminal-recurrence-ledger__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary. Exact result: FORMX-003 uniquely retains three-way-terminal-recurrence-ledger, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-OPERATIONAL-PROCESS-001 -> SFT-COMP-FORMX-PARTIAL-TOTAL-TRANSITION-002. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** FORMX-003: terminal_outcome_partition; exact execution: {"open_boundary_separate":true,"recurrent_trace_separate":true,"terminal_labels":["accept","reject"]}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation sha256:da1d1f50a86b2759fc9dc965b19c8e024e88f90c54b294f84dd382f81ee0e58b; independent implementation sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e; independent certificate sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896; empirical/external sha256:d0562364a01e337291ec47682457882e9a2009828f4733563816654073bd0f9; engine receipt

`sha256:ddfc25a2da658167388daf5077efe788462d3216876e360cbedcd199258b34e7` at `receipts/engine/model_admitted/SFT-COMP-FORMX-TERMINAL-OUTCOME-DISTINCTION-003-ddfc25a2da658167.json`.

`SFT-COMP-OBL-FORMX-004` - Language union, intersection and complement correspondence

- **Claim and ownership:** `SFT-COMP-FORMX-LANGUAGE-BOOLEAN-CORRESPONDENCE-004`; Classical Computation family `FORMX`; Language union, intersection and complement correspondence.
- **Forced law:** Union retains every word in either complete language, intersection retains exactly shared words, and relative complement retains source words absent from the declared comparison support.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__complete-support-language-operations__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-004 uniquely retains complete-support-language-operations, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-LANGUAGE-GRAMMAR-001 -> SFT-COMP-FORMX-TERMINAL-OUTCOME-DISTINCTION-003. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-004: language_operations; exact execution: {"intersection_support":1,"left_support":2,"relative_complement_support":1,"right_support":2,"union_support":3}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:c5fdcf5d769d82bf9a57c3de3fa8c820f5635df8e89d8e8db56718adf8f7603c`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:19beb0c460dc1646a14f9db2c87e4825d79c949c3326eef9baa16192cbf6412f`; engine receipt `sha256:a52baf651920e7ff284f84abe21503b55b70d132d5feb1a6f0954f9264f1f17c` at `receipts/engine/model_admitted/SFT-COMP-FORMX-LANGUAGE-BOOLEAN-CORRESPONDENCE-004-a52baf651920e7ff.json`.

`SFT-COMP-OBL-FORMX-005` - Concatenation, iteration and generated-language closure

- **Claim and ownership:** `SFT-COMP-FORMX-CONCATENATION-ITERATION-005`; Classical Computation family `FORMX`; Concatenation, iteration and generated-language closure.
- **Forced law:** Language concatenation is the complete ordered joining of every left word with every right word; finite iteration begins with the empty One word and applies that product once per retained successor.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__ordered-concatenation-successor-closure__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor`

`-or-explicit-boundary`. Exact result: FORMX-005 uniquely retains ordered-concatenation-successor-closure, complete execution custody, root forcing, post-registry observation and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-LANGUAGE-GRAMMAR-001 -> SFT-COMP-FORMX-LANGUAGE-BOOLEAN-CORRESPONDENCE-004. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-005: concatenation_iteration; exact execution: {"alphabet_size":2,"empty_one_word_retained":true,"ordered_word_support":4,"successor_depth":2}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:00b6b49c2e459c8340d3de2a368fd15b798311b05388be7e9d443203fe993560`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:30f62a799ee6930ec9c90c50a24547991ba43cbce3e21bf70305c5d5a3efdfcb`; engine receipt `sha256:308dd8c15433530c5782863b61e213beba2cc3dff8538fdf053a598284655a58` at `receipts/engine/model_admitted/SFT-COMP-FORMX-CONCATENATION-ITERATION-005-308dd8c15433530c.json`.

`SFT-COMP-OBL-FORMX-006` - Grammar derivation trees and ambiguity custody

- **Claim and ownership:** `SFT-COMP-FORMX-DERIVATION-TREE-AMBIGUITY-006`; Classical Computation family `FORMX`; Grammar derivation trees and ambiguity custody.
- **Forced law:** A grammar derivation is a complete source-bound rewrite tree; ambiguity exists exactly when one generated terminal word retains multiple distinct derivation trees, all of which remain open.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__complete-derivation-tree-ledger__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-006 uniquely retains complete-derivation-tree-ledger, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-LANGUAGE-GRAMMAR-001 -> SFT-COMP-FORMX-CONCATENATION-ITERATION-005. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-006: derivation_tree_ambiguity; exact execution: {"all_trees_retained":true,"complete_parse_tree_count":2,"terminal_word":"aaa"}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation

execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:7906ecba57d4ac08f2b95bc17ab56e37f8f7295f045faa090b6327d787b3825a`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:1daf2e81e494741fae4b0650627ba92504fffcd6464a4172bb9e5ef67a02029f`; engine receipt `sha256:e4b80cdfe27e9bb0379164532369b43b3dd83978a0c0f6ec53a506ba30235886` at `receipts/engine/model_admitted/SFT-COMP-FORMX-DERIVATION-TREE-AMBIGUITY-006-e4b80cdfe27e9bb0.json`.

`SFT-COMP-OBL-FORMX-007` - Parsing, recognition and generation correspondence

- **Claim and ownership:** `SFT-COMP-FORMX-PARSE-RECOGNIZE-GENERATE-007`; Classical Computation family `FORMX`; Parsing, recognition and generation correspondence.
- **Forced law:** Parsing, recognition and generation coincide only when they range over the same complete grammar boundary and preserve a witness tree or rejecting trace for every generated word.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__parse-recognize-generate-equivalence__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-007 uniquely retains parse-recognize-generate-equivalence, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-AUTOMATON-001 -> SFT-COMP-FORMX-DERIVATION-TREE-AMBIGUITY-006. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-007: parse_recognize_generate; exact execution: {"all_generated_words_parsed":true,"generated_widths_checked":[1,2,3,4],"rejecting_trace_retained":true}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:7f99b3df1a2b802d811d2acfbe0f8a007a1f0e284f26c597696ada7535811250`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`;

independent certificate
`sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`;
empirical/external
`sha256:203ef8c75f1084254e0e8fd8581fbc12a214b5e4525c4fc4bf8a562aa040d817`; engine receipt `sha256:3941b75acdf1ab76fcb29f63ca1351cc1aa20b2f9a497b368325c4621e595f28` at `receipts/engine/model_admitted/SFT-COMP-FORMX-PARSE-RECOGNIZE-GENERATE-007-3941b75acdf1ab76.json`.

`SFT-COMP-OBL-FORMX-008` - Automaton product, quotient and minimization

- **Claim and ownership:** `SFT-COMP-FORMX-AUTOMATON-PRODUCT-QUOTIENT-008`; Classical Computation family `FORMX`; Automaton product, quotient and minimization.
- **Forced law:** Automaton product retains both component states; quotienting merges exactly states with identical complete future observations, and minimization is the least such observation-preserving quotient.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__observation-equivalent-state-quotient__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-008 uniquely retains observation-equivalent-state-quotient, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-AUTOMATON-001 -> SFT-COMP-FORMX-PARSE-RECOGNIZE-GENERATE-007. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-008: automaton_product_quotient; exact execution: {"future_observation_classes":2,"generated_words_checked":15,"parity_observation_preserved":true}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:7ae6dcfbe34ee6eb5be2913ece66c2951ef7829e498145277ca2db0c9d6eb1fa`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:e055c590eabf4dc5c59c918d057e5e49d6546e015c8739a0e0dcc1c9c63731e4`; engine receipt `sha256:72a37de3f8ee4eba738462f193e7b0134ede3e14cf36dd962891a8eece3b7422` at `receipts/engine/model_admitted/SFT-COMP-FORMX-AUTOMATON-PRODUCT-QUOTIENT-008-72a37de3f8ee4eba.json`.

`SFT-COMP-OBL-FORMX-009` - Finite-state transduction and output custody

- **Claim and ownership:** `SFT-COMP-FORMX-FINITE-TRANSDUCTION-009`; Classical Computation family `FORMX`; Finite-state transduction and output custody.

- **Forced law:** A finite transducer retains for every source-state and input-label pair both its successor state and complete emitted word; composition preserves both state and output traces.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__state-and-output-transduction__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-009 uniquely retains state-and-output-transduction, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-AUTOMATON-001 -> SFT-COMP-FORMX-AUTOMATON-PRODUCT-QUOTIENT-008. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-009: finite_transduction; exact execution: {"output_word":["x","y","x"],"source_word":["a","b","a"],"state_trace_retained":true}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:3d64c7fc3f094661173e9e068075e9f9bbf0753b9ee792c8933333bc7566a450`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:56ccb534080b9f719b4dd74a98a026ecbb763b139082cc14852acdae9ee24fb3`; engine receipt `sha256:972b0967ec11b9f859dcc1f1d76b24d285d9a0c00dfef003f3b8ed26bed6a29f` at `receipts/engine/model_admitted/SFT-COMP-FORMX-FINITE-TRANSDUCTION-009-972b0967ec11b9f8.json`.

`SFT-COMP-OBL-FORMX-010` - Stack, queue and register storage correspondence

- **Claim and ownership:** `SFT-COMP-FORMX-STORAGE-MACHINE-CORRESPONDENCE-010`; Classical Computation family `FORMX`; Stack, queue and register storage correspondence.
- **Forced law:** Stack, queue and register machines are distinguished by their retained storage access relation; translations are admitted only when every operation, stored label and execution order is reconstructible.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__typed-storage-operation-ledger__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-010 uniquely retains typed-storage-operation-ledger, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-ABSTRACT-MACHINE-001 -> SFT-COMP-FORMX-FINITE-TRANSDUCTION-009. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** FORMX-010: storage_machine_correspondence; exact execution: {"operations_typed":true,"queue_remainder":["b","c"],"source_word":["a","b","c"],"stack_output":["c","b","a"]}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:67969cf0209aae48ca1666d1dd8efa118a940dd803f4de75efe26f45a0652d59`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:498c9862d071c4893ac31a709f357d03a6edd94e0d0b17382bec056636eeea13`; engine receipt `sha256:42d1ee0d0b70f6af59c575cc6499f17220e2c1a4d753a5709f71185e3a9d5085` at `receipts/engine/model_admitted/SFT-COMP-FORMX-STORAGE-MACHINE-CORRESPONDENCE-010-42d1ee0d0b70f6af.json`.

`SFT-COMP-OBL-FORMX-011` - Rewriting termination and normal forms

- **Claim and ownership:** `SFT-COMP-FORMX-REWRITE-NORMAL-FORM-011`; Classical Computation family `FORMX`; Rewriting termination and normal forms.
- **Forced law:** A rewrite system terminates on its declared grammar when every lawful step descends a registered finite order; a normal form is exactly a reachable form with no lawful successor.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__well-founded-rewrite-normalization__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-011 uniquely retains well-founded-rewrite-normalization, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-REWRITING-001 -> SFT-COMP-FORMX-STORAGE-MACHINE-CORRESPONDENCE-010. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-011: rewrite_normal_form; exact execution: {"normal_form":"bbaa","source_word":"abab","terminal_forms":1,"well_founded_descent":true}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.

- **Evidence identities:** derivation `sha256:9b06e9d9e5a7d5cb45e7fdd92fc03b1592f3c4a7cf0e1225de8eb52ef4e17a65`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:a9d6372b7e5f1d5c95b40c6d0be9354c30262047027bc374197466d87790065c`; engine receipt `sha256:ec6b6dd611fa6125e4580ecdf3dcc50e2aba6c1c1bcad91d697787575e1a302b` at `receipts/engine/model_admitted/SFT-COMP-FORMX-REWRITE-NORMAL-FORM-011-ec6b6dd611fa6125.json`.

`SFT-COMP-OBL-FORMX-012` - Rewriting confluence and critical-pair custody

- **Claim and ownership:** `SFT-COMP-FORMX-REWRITE-CONFLUENCE-012`; Classical Computation family `FORMX`; Rewriting confluence and critical-pair custody.
- **Forced law:** Confluence is forced when every pair of divergent lawful rewrites has a retained common descendant; complete critical-pair enumeration supplies the finite local certificate.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__complete-critical-pair-joinability__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-012 uniquely retains complete-critical-pair-joinability, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-REWRITING-001 -> SFT-COMP-FORMX-REWRITE-NORMAL-FORM-011. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-012: critical_pair_confluence; exact execution: {"all_critical_pairs_join":true,"common_normal_forms":["a"],"divergent_first_steps":2,"source_word":"aaa"}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:09a078f72ebe703d3df9809b5d28f934ae2a2cdc9ab1106374160d171677ccfe`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:1d8b76c4ea92db573ef82d198d691f3f97c6ebdbdc1864a66bff5c54a043967e`; engine receipt `sha256:75151477eacbcf657be6c606b04e8059925e185c6fbc60ee5afc449cdfac047f` at `receipts/engine/model_admitted/SFT-COMP-FORMX-REWRITE-CONFLUENCE-012-75151477eacbcf65.json`.

`SFT-COMP-OBL-FORMX-013` - Recursive composition and finite iteration

- **Claim and ownership:** `SFT-COMP-FORMX-RECURSIVE-COMPOSITION-013`; Classical Computation family `FORMX`; Recursive composition and finite iteration.

- **Forced law:** A recursive computation retains one base value and one source-bound successor action; composition substitutes complete output traces and finite iteration applies the action once per generated control label.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__base-successor-recursive-composition__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-013 uniquely retains base-successor-recursive-composition, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-RECURSIVE-FUNCTION-001 -> SFT-COMP-FORMX-REWRITE-CONFLUENCE-012. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-013: recursive_composition; exact execution: {"control_word":["a","b","c"],"output_word":["x","x","x"],"successor_applications":3}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:2813aff0271908dda95e9e4191886e15e0f0dd31cf5584a9a1156992ecc0783a`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:e377d5f816ce4f154f88ae28d9fac34fdd46d102c363c68787783f14781ce32c`; engine receipt `sha256:29fbc24de44568a20cd2c2f495e5cfef22f7f15a3ac6d061cf2392226eb8ccf2` at `receipts/engine/model_admitted/SFT-COMP-FORMX-RECURSIVE-COMPOSITION-013-29fbc24de44568a2.json`.

`SFT-COMP-OBL-FORMX-014` - Primitive recursion and minimization correspondence

- **Claim and ownership:** `SFT-COMP-FORMX-PRIMITIVE-RECURSION-MINIMIZATION-014`; Classical Computation family `FORMX`; Primitive recursion and minimization correspondence.
- **Forced law:** Primitive recursion is base-plus-successor construction over a generated word; minimization returns the first retained prefix whose terminal label satisfies the declared predicate or an explicit absent-result boundary.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__generated-prefix-least-witness__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-014 uniquely retains generated-prefix-least-witness, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-RECURSIVE-FUNCTION-001 -> SFT-COMP-FORMX-RECURSIVE-COMPOSITION-013. Registered root theorem:

SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Post-registration observation:** FORMX-014: primitive_recursion_minimization; exact execution: {"base_word":["s"],"control_width":2,"least_witness_prefix":["a","a","b"],"recursive_output":["s","x","x"]}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation sha256:f2026c81bb602b401977e6e2aa6507ca439d53ba3d8fd10601f04a0ba19e99d0; independent implementation sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e; independent certificate sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896; empirical/external sha256:f0378f9661ad85393c3b1a7c55f595a77664089a20c79fdc6fb5c6949157d895; engine receipt sha256:6f820a230c272c19e12c63a5a3c787ffa2dd423065d162799a908692d62175df at receipts/engine/model_admitted/SFT-COMP-FORMX-PRIMITIVE-RECURSION-MINIMIZATION-014-6f820a230c272c19.json.

SFT-COMP-OBL-FORMX-015 - Lambda reduction, capture avoidance and normal forms

- **Claim and ownership:** SFT-COMP-FORMX-LAMBDA-CAPTURE-NORMAL-015; Classical Computation family FORMX; Lambda reduction, capture avoidance and normal forms.
- **Forced law:** Lambda-like substitution replaces only free occurrences, renames a conflicting binder before descent and retains alpha-equivalent structure; normality is absence of a registered reduction redex.
- **Unique surviving structure:** complete-canonical-carrier__source-bound-complete-transition__capture-avoiding-bound-substitution__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary. Exact result: FORMX-015 uniquely retains capture-avoiding-bound-substitution, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-LAMBDA-CALCULUS-001 -> SFT-COMP-FORMX-PRIMITIVE-RECURSION-MINIMIZATION-014. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** FORMX-015: lambda_capture_avoidance; exact execution: {"capture_avoided":true,"fresh_binder":"u","replacement":"y","result":"lambda u . y","source":"lambda y . x"}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary,

floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.

- **Evidence identities:** derivation `sha256:ec8af15d2f213a52ebae20431944b4356cfb7b85aaf131a3841e71bd103d39b0`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:7cb1637888dae42c8cefdc667658ee3cd6d9c5d9ec61a4a713a824c20207795b`; engine receipt `sha256:6bf29af9584f6bf9beddd7af94026b30bf783ccc3d75161d3be77f3d724c6747` at `receipts/engine/model_admitted/SFT-COMP-FORMX-LAMBDA-CAPTURE-NORMAL-015-6bf29af9584f6bf9.json`.

`SFT-COMP-OBL-FORMX-016` - Abstract-machine simulation invariants

- **Claim and ownership:** `SFT-COMP-FORMX-MACHINE-SIMULATION-INVARIANT-016`; Classical Computation family `FORMX`; Abstract-machine simulation invariants.
- **Forced law:** One abstract machine simulates another only through a total configuration relation preserving initial form, terminal outcome and each source step by a finite nonempty target trace with explicit overhead.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__stepwise-configuration-simulation__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-016 uniquely retains stepwise-configuration-simulation, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-MODEL-EQUIVALENCE-001 -> SFT-COMP-FORMX-LAMBDA-CAPTURE-NORMAL-015. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-016: machine_simulation; exact execution: {"source_output":["c","b","a"],"source_word":["a","b","c"],"stack_trace_steps":6,"target_output":["c","b","a"]}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:7157d6e0a06a6919d6793931d1fd9a0e3a6f931ec5279e35ca58b91cc7f710d9`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:337e246d6aafbf694b7072ca39372c3c348edb35a12bfe6a72a7e085c148cc06`; engine receipt `sha256:74b76cc1c1ac9a63fae07f6b56264188473a100804e25e1077bbbd4fadfb6cd1` at `receipts/engine/model_admitted/SFT-COMP-FORMX-MACHINE-SIMULATION-INVARIANT-016-74b76cc1c1ac9a63.json`.

SFT-COMP-OBL-FORMX-017 - Circuit fan-in, fan-out and acyclic evaluation

- **Claim and ownership:** SFT-COMP-FORMX-CIRCUIT-ACYCLIC-EVALUATION-017; Classical Computation family FORMX; Circuit fan-in, fan-out and acyclic evaluation.
- **Forced law:** A formal circuit is a finite acyclic gate relation with declared inputs and outputs; evaluation follows dependency order and retains every intermediate held label and fan relation.
- **Unique surviving structure:** complete-canonical-carrier__source-bound-complete-transition__topological-gate-evaluation__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary. Exact result: FORMX-017 uniquely retains topological-gate-evaluation, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-CIRCUIT-001 -> SFT-COMP-FORMX-MACHINE-SIMULATION-INVARIANT-016. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** FORMX-017: circuit_evaluation; exact execution: {"all_intermediates_retained":true,"gate_count":2,"input_labels":["held","changed"],"terminal_label":"changed"}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation sha256:4095617e4bc561022646962b9bcb505286c61083ab99f35ad49c5df2193d3bff; independent implementation sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e; independent certificate sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896; empirical/external sha256:06b3cbb2bb684d74d339898ed7474c49b961e403fa8b744710a7a7045f244601; engine receipt sha256:bf7008c723dd9079f6efcb7ff27fea97bdd4b94e1d6f483d6ae158e85e650218 at receipts/engine/model_admitted/SFT-COMP-FORMX-CIRCUIT-ACYCLIC-EVALUATION-017-bf7008c723dd9079.json.

SFT-COMP-OBL-FORMX-018 - Sequential and combinational process correspondence

- **Claim and ownership:** SFT-COMP-FORMX-SEQUENTIAL-COMBINATIONAL-018; Classical Computation family FORMX; Sequential and combinational process correspondence.
- **Forced law:** A combinational process depends only on its declared input support; a sequential process additionally retains state, and finite unrolling produces an equivalent acyclic circuit only with every state record preserved.
- **Unique surviving structure:** complete-canonical-carrier__source-bound-complete-transition__state-retained-sequential-unrolling__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary. Exact result: FORMX-018 uniquely retains state-retained-sequential-unrolling, complete execution custody, root forcing, post-registry observation and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-OPERATIONAL-PROCESS-001 -> SFT-COMP-FORMX-CIRCUIT-ACYCLIC-EVALUATION-017. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-018: sequential_unrolling; exact execution: {"combinational_output":"held","finite_unrolling_preserves_output":true,"sequential_state_trace":[["a"],["a","b"]]}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:c641bb0a889a8e7cd59629a87a2fe32c90e288197b6da53743b8aa9878ff4978`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:1f0b339cd9aa9f06213e63c819d46cd4c7722fd434ad39001d8827c5e1c3fedc`; engine receipt `sha256:6909c6914657670ae1b1930ed093622de2d1fc793a5aa15131d20456cb1b832b` at `receipts/engine/model_admitted/SFT-COMP-FORMX-SEQUENTIAL-COMBINATIONAL-018-6909c6914657670a.json`.

`SFT-COMP-OBL-FORMX-019` - Process algebra composition and observational equivalence

- **Claim and ownership:** `SFT-COMP-FORMX-PROCESS-ALGEBRA-EQUIVALENCE-019`; Classical Computation family `FORMX`; Process algebra composition and observational equivalence.
- **Forced law:** Parallel process composition generates every interleaving preserving each local order; observational equivalence requires equality of complete declared observation traces, not one selected schedule.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__complete-interleaving-trace-equivalence__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-019 uniquely retains complete-interleaving-trace-equivalence, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-COMPOSITION-001 -> SFT-COMP-FORMX-SEQUENTIAL-COMBINATIONAL-018. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-019: process_interleaving; exact execution: {"all_local_orders_preserved":true,"complete_interleaving_count":6,"left_steps":2,"right_steps":2}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:8969629765ae06755febb4b1a78a495bfb8e962ceec25b2cc8e19940e79b11a3`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:93c42da1bf974d163876ec61466b5dc11b355f1d152c0276c5e7a1dc31057fb0`; engine receipt `sha256:cd1f9cd0c479e29f67359615d2436e9c7846ce517ecf9cb66d1f67d54515d4d5` at `receipts/engine/model_admitted/SFT-COMP-FORMX-PROCESS-ALGEBRA-EQUIVALENCE-019-cd1f9cd0c479e29f.json`.

`SFT-COMP-OBL-FORMX-020` - Universal interpreter and self-interpretation

- **Claim and ownership:** `SFT-COMP-FORMX-UNIVERSAL-SELF-INTERPRETATION-020`; Classical Computation family `FORMX`; Universal interpreter and self-interpretation.
- **Forced law:** A universal Fold interpreter accepts a generated process description as data, executes only its registered operations and emits the same terminal value and trace as direct execution; its own description is admitted by the same grammar.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__description-driven-universal-interpretation__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-020 uniquely retains description-driven-universal-interpretation, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-FORMX-PROCESS-ALGEBRA-EQUIVALENCE-019. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-020: universal_interpretation; exact execution: {"instruction_count":4,"self_description_admitted":true,"terminal_word":["b"],"trace_complete":true}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:60e737b74ffd5bd383326749292c78973205eeec195f63886086f4e2d1d1931f`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`;

independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:1316cf9e31d9072fcc484c5ae4724482d94284de5de2e930b0657a5d718aa727`; engine receipt `sha256:a02e52a528eced7ce952aed8e063a18fd4ad825c971d29a0036cac34f6787811` at `receipts/engine/model_admitted/SFT-COMP-FORMX-UNIVERSAL-SELF-INTERPRETATION-020-a02e52a528eced7c.json`.

`SFT-COMP-OBL-FORMX-021` - Effective model translation and overhead custody

- **Claim and ownership:** `SFT-COMP-FORMX-MODEL-TRANSLATION-OVERHEAD-021`; Classical Computation family `FORMX`; Effective model translation and overhead custody.
- **Forced law:** Computational models are equivalent only when total translations preserve inputs, outcomes and complete traces in both directions while retaining exact description and execution overhead.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__bidirectional-trace-preserving-translation__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-021 uniquely retains bidirectional-trace-preserving-translation, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-MODEL-EQUIVALENCE-001 -> SFT-COMP-FORMX-UNIVERSAL-SELF-INTERPRETATION-020. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-021: model_translation_overhead; exact execution: {"reverse_reconstructible":true,"source_output":["a","b"],"source_steps":2,"translated_output":["a","b"],"translated_steps":2}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:c66d4ecbf94ea7af5bed6a8fed53d160f2379a9ecca0bb8a552bae5e0537aeb4`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:a157e135fa03e1bc70c94badb0fa1f3d23cc03033d83f3bb7fc63020513585b9`; engine receipt `sha256:cc88c1b0901080bea719456844010d90fb80ac69a08d521bb8db23813935a933` at `receipts/engine/model_admitted/SFT-COMP-FORMX-MODEL-TRANSLATION-OVERHEAD-021-cc88c1b0901080be.json`.

`SFT-COMP-OBL-FORMX-022` - Formal-computation completeness certificate

- **Claim and ownership:** `SFT-COMP-FORMX-COMPLETENESS-022`; Classical Computation family `FORMX`; Formal-computation completeness certificate.

- **Forced law:** Formal-computation completeness is the one-to-one reconciliation of all twenty-two frozen extension obligations with unique survivors, adverse controls, exact observations, independent reconstructions and untouched-engine receipts.
- **Unique surviving structure:** `complete-canonical-carrier__source-bound-complete-transition__twenty-two-obligation-no-omission-ledger__complete-step-trace__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__finite-successor-or-explicit-boundary`. Exact result: FORMX-022 uniquely retains twenty-two-obligation-no-omission-ledger, complete execution custody, root forcing, post-registry observation and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORM-UNIVERSALITY-001 -> SFT-COMP-FORMX-MODEL-TRANSLATION-OVERHEAD-021. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** FORMX-022: formal_computation_completeness; exact execution: {"duplicate_owners":0,"independent_execution_rows":22,"omitted_owners":0,"registered_obligations":22}; complete formal-computation observation custody: 22 records; all rows preserved; complete finite state, transition, grammar, trace, circuit and translation execution supplies the observation; no imported machine axiom, hidden branch or target-selected answer enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-FORMAL-COMPUTATION-OBSERVER, SFT-V1-V2-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported machine model, theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, selected branch, omitted parse, unregistered oracle or stochastic cause; no failed route retires an obligation or changes the protected engine or verification authority.
- **Evidence identities:** derivation `sha256:7c893e2e01a97c6356662334187dd61edbb38794f492be4828493c3f88916d66`; independent implementation `sha256:1598b3c44a4bfd83f98db696bd8f10ec0347e5527364d39d42b67ceb651c2e8e`; independent certificate `sha256:08d995f6dd39e2405efc5ced7e08791c17dfcd390517afbd6a4f5cc80c05f896`; empirical/external `sha256:6a33433d3d07ab3c57aad65c3f9c62c1f3b25864dd0428a8b6d216b2545db567`; engine receipt `sha256:4cdcde71060d5990ee806c8c9ea98ae13a031e61e0eef641c695b7b854feece3` at `receipts/engine/model_admitted/SFT-COMP-FORMX-COMPLETENESS-022-4cdcde71060d5990.json`.

130.5 Family `CBLX` - 21/21 complete

Scope. Computability. This family completely decides 5,376 candidates, retains 21 unique survivors and passes 84 adverse controls.

`SFT-COMP-OBL-CBLX-001` - Decidable and recognizable language closure

- **Claim and ownership:** `SFT-COMP-CBLX-DECIDABLE-RECOGNIZABLE-CLOSURE-001`; Classical Computation family `CBLX`; Decidable and recognizable language closure.
- **Forced law:** Decidable supports remain decidable under complete finite union, intersection and relative complement; recognizable supports remain recognizable under operations whose dovetailed traces retain every accepting witness.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__decision-and-recognition-closure__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-001 uniquely retains decision-and-recognition-closure, complete self-application custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-RECOGNITION-DECISION-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-001: decision_closure; exact execution: {"domain_size":4,"intersection_size":1,"relative_complement_size":1,"union_size":3}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:8c844930e00087b77ad1dfb5110a9768be7e158a3112c3ead0aa15e14ed793d4`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:b321a0a023c7c030e5158e76bf51778b14c548b37b5bed6da480db2064f5df9b`; engine receipt `sha256:ebb5093c994b260f89fe43d90812062a281979b5491777b7f4384eaf7160afed` at `receipts/engine/model_admitted/SFT-COMP-CBLX-DECIDABLE-RECOGNIZABLE-CLOSURE-001-ebb5093c994b260f.json`.

`SFT-COMP-OBL-CBLX-002` - Recognizable and co-recognizable decision boundary

- **Claim and ownership:** `SFT-COMP-CBLX-CO-RECOGNIZABLE-BOUNDARY-002`; Classical Computation family `CBLX`; Recognizable and co-recognizable decision boundary.
- **Forced law:** A support is decidable when one recognizer accepts its members and a second recognizer accepts its declared complement, because complete dovetailing forces exactly one terminal witness for every generated input.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__paired-recognizer-decision__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-002 uniquely retains paired-recognizer-decision, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-RECOGNITION-DECISION-001 -> SFT-COMP-CBLX-DECIDABLE-RECOGNIZABLE-CLOSURE-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-002: paired_recognizer_decision; exact execution: {"complement_support":2,"domain_covered":true,"member_support":2,"overlap":0}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:3f538dfa93bf94eba75c85a56db13cfa0afe875e4b9d5ba68168acc9c3514f52`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:b574264b3a347333f042a6e8b43c985a868cb80b2c78776fcb140e09fe93aa11`; engine receipt `sha256:4d2ac779b57b7ba851730b976f8c756d98bd79d99a9a2be70690d5b9ff59b4b7` at `receipts/engine/model_admitted/SFT-COMP-CBLX-CO-RECOGNIZABLE-BOUNDARY-002-4d2ac779b57b7ba8.json`.

`SFT-COMP-OBL-CBLX-003` - Effective enumeration and dovetailing

- **Claim and ownership:** `SFT-COMP-CBLX-DOVETAIL-ENUMERATION-003`; Classical Computation family `CBLX`; Effective enumeration and dovetailing.
- **Forced law:** Effective dovetailing executes one further retained step of each generated process in successor order, so no finite accepting or emitting trace is permanently omitted.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__fair-prefix-dovetail-enumeration__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-003 uniquely retains fair-prefix-dovetail-enumeration, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-ENUMERATION-001 -> SFT-COMP-CBLX-CO-RECOGNIZABLE-BOUNDARY-002. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-003: fair_dovetail; exact execution: {"emissions_omitted":0,"finite_emissions":["a","b","c"],"processes":3}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:3c1df33b9fdf67dab75c51ed4eb7f65db302c0fd4b8dfa4222c7cfb924ee5cd4`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate

`sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:aaa1e929f9f895dd2ec61c0f5f318eaafbaa08f5e4e62c0c01a3a63daa2a0e42`; engine receipt `sha256:206e232b7ac03edf9a031fa59b6bf1c02b2beec9f2e0a4912e9298449f8d2cc0` at `receipts/engine/model_admitted/SFT-COMP-CBLX-DOVETAIL-ENUMERATION-003-206e232b7ac03edf.json`.

`SFT-COMP-OBL-CBLX-004` - Diagonal language construction

- **Claim and ownership:** `SFT-COMP-CBLX-DIAGONAL-LANGUAGE-004`; Classical Computation family `CBLX`; Diagonal language construction.
- **Forced law:** For a complete generated machine-description table, the diagonal language assigns each description the opposite terminal verdict from that machine on its own description; therefore no listed row decides the diagonal support.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__self-row-verdict-complement__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-004 uniquely retains self-row-verdict-complement, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-UNDECIDABILITY-001 -> SFT-COMP-CBLX-DOVETAIL-ENUMERATION-003. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-004: diagonal_language; exact execution: {"machine_rows":3,"opposite_verdicts":3,"self_rows_compared":3}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:c71f177203be4beaf75d75429eff9f69470801e7238d84a60a42ad22b4650617`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:f24cc57c72b0627d40044da40211d28f5bfb20f4525c74fec1c5cbf3482f3274`; engine receipt `sha256:85e2e72ac9a0953cad077e8dfb8b5ff44a5f8d42e51bbc15e2988a0498c598c1` at `receipts/engine/model_admitted/SFT-COMP-CBLX-DIAGONAL-LANGUAGE-004-85e2e72ac9a0953c.json`.

`SFT-COMP-OBL-CBLX-005` - Self-reference and fixed-point construction

- **Claim and ownership:** `SFT-COMP-CBLX-SELF-REFERENCE-FIXED-POINT-005`; Classical Computation family `CBLX`; Self-reference and fixed-point construction.
- **Forced law:** A generated description can retain its own canonical code as input; composing the quoting constructor with any admitted transformation yields a process whose execution is observationally identical to applying that transformation to its own description.

- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__retained-description-fixed-point__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-005 uniquely retains retained-description-fixed-point, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-UNIVERSAL-MACHINE-001 -> SFT-COMP-CBLX-DIAGONAL-LANGUAGE-004. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-005: fixed_point; exact execution: {"description_retained":true,"fixed_behavior_equal":true,"self_input_retained":true}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f6ce3837ecc2f60095e077ea7d07a4f07ee5d9b93956dd4e3ff4c1033d7d642d`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:34f4c20b87461ebd5fad0a76ff162b7deaafada7acf7cf3be6e7ed98f42dfbfb`; engine receipt `sha256:8bff973be333e45d57e4f8197e626478b13d105eecab15216b4ebaa9bb80ce77` at `receipts/engine/model_admitted/SFT-COMP-CBLX-SELF-REFERENCE-FIXED-POINT-005-8bff973be333e45d.json`.

`SFT-COMP-OBL-CBLX-006` - Recursion-theorem correspondence

- **Claim and ownership:** `SFT-COMP-CBLX-RECURSION-THEOREM-006`; Classical Computation family `CBLX`; Recursion-theorem correspondence.
- **Forced law:** Every total admitted description transformation has a generated self-referential process whose behavior equals the transformed behavior of its own description, with the quotation and execution traces retained.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__description-transform-fixed-point__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-006 uniquely retains description-transform-fixed-point, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-UNIVERSAL-MACHINE-001 -> SFT-COMP-CBLX-SELF-REFERENCE-FIXED-POINT-005. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** CBLX-006: recursion_theorem; exact execution: {"description_transform_total":true,"fixed_behavior_equal":true,"quotation_trace_retained":true}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f4e5505242f43df90773d5304cee06b4651a24fe43fec32e8ab96542fcde9f87`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:ed05bca4e70a3af0a8c420d20c36681dcb4e7fab3ca228a67d76885487294f05`; engine receipt `sha256:b4ffa3ec2dbc99836f568bbb592e0358ffad889d575770044821893ad2e320db` at `receipts/engine/model_admitted/SFT-COMP-CBLX-RECURSION-THEOREM-006-b4ffa3ec2dbc9983.json`.

`SFT-COMP-OBL-CBLX-007` - Semantic-property undecidability boundary

- **Claim and ownership:** `SFT-COMP-CBLX-SEMANTIC-PROPERTY-UNDECIDABILITY-007`; Classical Computation family `CBLX`; Semantic-property undecidability boundary.
- **Forced law:** No total internal process decides every nontrivial property determined solely by admitted process behavior: composing such a verdict with the opposite behavior constructs the exact self-negating contradiction.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__nontrivial-behavior-diagonal-boundary__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-007 uniquely retains nontrivial-behavior-diagonal-boundary, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-UNDECIDABILITY-001 -> SFT-COMP-CBLX-RECURSION-THEOREM-006. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-007: semantic_property_boundary; exact execution: {"self_negating_fixed_verdicts":0,"terminal_verdicts":["accept","reject"]}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:13f9b34e60772c01a2d0e737dde2815c2365ca617cd82315079d9ab2b4dd0fe7`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:c0dee0e4ebb7c10f389f4effa530b1ac4042e7810a1789eea63f93cf90b32f33`; engine receipt `sha256:8a7e29f98636fb2ab938f2a2c3e2cec40b4846594d743e1df9d4a955aee2b85e` at `receipts/engine/model_admitted/SFT-COMP-CBLX-SEMANTIC-PROPERTY-UNDECIDABILITY-007-8a7e29f98636fb2a.json`.

`SFT-COMP-OBL-CBLX-008` - Many-one reduction and composition

- **Claim and ownership:** `SFT-COMP-CBLX-MANY-ONE-REDUCTION-008`; Classical Computation family `CBLX`; Many-one reduction and composition.
- **Forced law:** A many-one Fold reduction is one total source-to-target map preserving and reflecting the terminal verdict for every generated instance; reductions compose by exact map composition and trace concatenation.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__total-verdict-preserving-map__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-008 uniquely retains total-verdict-preserving-map, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-REDUCTION-001 -> SFT-COMP-CBLX-SEMANTIC-PROPERTY-UNDECIDABILITY-007. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-008: many_one_reduction; exact execution: {"mapped_instances":2,"source_instances":2,"verdicts_preserved":2}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:273ccc39a0f7c8f92cd12dbaed6ae7e8c5464d2ebe04b355c5756519b912599f`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:2ce3af1a8b9d6b4bc6510375fc6c9a3ec6ca1f5473c03d36fb83f03fdd22d4b2`; engine receipt `sha256:f3a170cd132c5f82e27e3c0f8a65ab640585bc024e8cd8228ac6cfbc03ebbf04` at `receipts/engine/model_admitted/SFT-COMP-CBLX-MANY-ONE-REDUCTION-008-f3a170cd132c5f82.json`.

`SFT-COMP-OBL-CBLX-009` - Turing-style reduction correspondence

- **Claim and ownership:** `SFT-COMP-CBLX-TURING-REDUCTION-009`; Classical Computation family `CBLX`; Turing-style reduction correspondence.

- **Forced law:** An adaptive oracle reduction retains every query, answer and next-query dependency; its result is admissible only relative to that complete answer record and declared oracle resource.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__adaptive-recorded-query-reduction__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-009 uniquely retains adaptive-recorded-query-reduction, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-RELATIVE-ORACLE-001 -> SFT-COMP-CBLX-MANY-ONE-REDUCTION-008. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-009: adaptive_oracle_reduction; exact execution: {"answers":["accept","reject"],"order_retained":true,"queries":["q1","q2"],"query_count":2}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:40aa7674593d661bb36c88360bf79062ac0f9a442cf81587b590c8988b81b815`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:237e9de95bc1e7a0caea2d5d31d178cf66d42a09e2eafef935788b4dac8a647a`; engine receipt `sha256:c88f6e8bf37f767383f7a32619430ca628605d38daeb13bbab36a3e89907853f` at `receipts/engine/model_admitted/SFT-COMP-CBLX-TURING-REDUCTION-009-c88f6e8bf37f7673.json`.

`SFT-COMP-OBL-CBLX-010` - Enumeration reducibility correspondence

- **Claim and ownership:** `SFT-COMP-CBLX-ENUMERATION-REDUCIBILITY-010`; Classical Computation family `CBLX`; Enumeration reducibility correspondence.
- **Forced law:** Enumeration reducibility is witnessed by a generated finite family of positive query sets whose complete appearance in the oracle enumeration forces each emitted source item, without negative or absent-answer inference.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__finite-positive-query-witness__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-010 uniquely retains finite-positive-query-witness, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-ENUMERATION-001 -> SFT-COMP-CBLX-TURING-REDUCTION-009. Registered root theorem:

SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Post-registration observation:** CBLX-010: enumeration_reducibility; exact execution: {"all_positive_queries_observed":true,"negative_queries_used":0,"positive_query_sets":2}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:30dac79ea968771bdbf3bb46296cac3945978bf9a7451eae94de9b9c7512651f; independent implementation sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d; independent certificate sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19; empirical/external sha256:495e2bf6f7d08b7a4da90803e276a22ea9a9fd2b9a2e6036a4c68be0a278229c; engine receipt sha256:c8e6aed32024c375f07ad947ed27a740caffc906239b1ad675ccf91947dbddb1 at receipts/engine/model_admitted/SFT-COMP-CBLX-ENUMERATION-REDUCIBILITY-010-c8e6aed32024c375.json.

SFT-COMP-OBL-CBLX-011 - Computability degree ordering

- **Claim and ownership:** SFT-COMP-CBLX-DEGREE-ORDER-011; Classical Computation family CBLX; Computability degree ordering.
- **Forced law:** Computability degrees are equivalence classes under mutual registered reducibility; the induced relation is reflexive, transitive and antisymmetric on classes while incomparable classes remain distinct.
- **Unique surviving structure:** complete-generated-domain__complete-trace-execution__mutual-reducibility-equivalence-order__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit. Exact result: CBLX-011 uniquely retains mutual-reducibility-equivalence-order, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-DEGREES-001 -> SFT-COMP-CBLX-ENUMERATION-REDUCIBILITY-010. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** CBLX-011: degree_order; exact execution: {"degree_classes":3,"explicit_reductions":2,"reflexive_rows":3,"transitive_reduction":["A","C"]}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause;

bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:008f4238b82981916ae819a3c8dbe908a4e13849ba52bf349f9e7e1e770bfbf8`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:bedd0d7138b475d55c120bcb6555f451f4796a3c0ab56b4a1a04ff8218e26ffa`; engine receipt `sha256:6fcea0a78e3759e302f40a2b8500b0d22bfb3907c01d044b3f3eed3e960163e6` at `receipts/engine/model_admitted/SFT-COMP-CBLX-DEGREE-ORDER-011-6fcea0a78e3759e3.json`.

`SFT-COMP-OBL-CBLX-012` - Jump and relative-computation succession

- **Claim and ownership:** `SFT-COMP-CBLX-JUMP-RELATIVE-SUCCESSION-012`; Classical Computation family `CBLX`; Jump and relative-computation succession.
- **Forced law:** The jump of a declared oracle support is the diagonal self-halting support of machines relative to that oracle; it is recognizable with the next answer record but not decided by any machine in the prior complete enumeration.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__relative-self-halting-diagonal__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-012 uniquely retains relative-self-halting-diagonal, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-DEGREES-001 -> SFT-COMP-CBLX-DEGREE-ORDER-011. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-012: relative_jump; exact execution: {"diagonal_rows":3,"machine_rows":3,"prior_self_verdicts_all_opposed":true}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:29edd10bbabd2d7e9b70b41f6deec56f1985c5e3c5c6a04d8bb02fbcf6e0d637`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:b59d2ddc73d3ac6c95f172fae2d991afce6adba17488b8a399b66cb9bd8467ed`; engine receipt `sha256:1726d2afb081664fda9c8468d6c156e0c0efa030eb635901f804df7c4fb0b247` at `receipts/engine/model_admitted/SFT-COMP-CBLX-JUMP-RELATIVE-SUCCESSION-012-1726d2afb081664f.json`.

SFT-COMP-OBL-CBLX-013 - Oracle answer-record custody

- **Claim and ownership:** SFT-COMP-CBLX-ORACLE-ANSWER-CUSTODY-013; Classical Computation family CBLX; Oracle answer-record custody.
- **Forced law:** An oracle computation is relative computation, not an unexplained stronger process: every answer must retain query identity, oracle identity, order and downstream use so the result can be reconstructed relative to exactly that source.
- **Unique surviving structure:** complete-generated-domain__complete-trace-execution__query-answer-provenance-ledger__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit. Exact result: CBLX-013 uniquely retains query-answer-provenance-ledger, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-RELATIVE-ORACLE-001 -> SFT-COMP-CBLX-JUMP-RELATIVE-SUCCESSION-012. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** CBLX-013: oracle_custody; exact execution: {"answer_identity_retained":true,"answer_order_retained":true,"oracle_identity_retained":true,"query_identity_retained":true}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:20fd2d4df647c0d2f89a4a90f74296fc2b34f8db6ef30f48694ce218e4efd76c; independent implementation sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d; independent certificate sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19; empirical/external sha256:0df1446aad16ed5dea6369a829819b4a1ffcf833b68169af590546e7fba31401; engine receipt sha256:08b45db8aa6523cfeee683c6307838e61ade9ff4c294538785a3b787e0f12b72 at receipts/engine/model_admitted/SFT-COMP-CBLX-ORACLE-ANSWER-CUSTODY-013-08b45db8aa6523cf.json.

SFT-COMP-OBL-CBLX-014 - Arithmetical-hierarchy correspondence boundary

- **Claim and ownership:** SFT-COMP-CBLX-ARITHMETICAL-HIERARCHY-BOUNDARY-014; Classical Computation family CBLX; Arithmetical-hierarchy correspondence boundary.
- **Forced law:** Quantifier hierarchy corresponds to alternating complete existential and universal searches over generated finite prefixes; no completed infinite truth object or unrecorded oracle is admitted by the correspondence.
- **Unique surviving structure:** complete-generated-domain__complete-trace-execution__finite-quantifier-alternation-ledger__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit. Exact result: CBLX-014 uniquely retains finite-quantifier-alternation-ledger, complete self-application custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-RELATIVE-ORACLE-001 -> SFT-COMP-CBLX-ORACLE-ANSWER-CUSTODY-013. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-014: quantifier_hierarchy; exact execution: {"existential_result":true,"finite_domain_size":2,"universal_existential_result":true,"universal_result":false}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:3a211f040f43012ef24ddef676a95a354960685cc9ffe95ef2f5462802715edc`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:f43ba42944f6b2714f61764baad02dfdfce6e779d55517d33ba5ef7a0ddf887f`; engine receipt `sha256:f105b1030b22d9a7ca3016a29975066fb44d6f07510a91c845664a0cf3148d2c` at `receipts/engine/model_admitted/SFT-COMP-CBLX-ARITHMETICAL-HIERARCHY-BOUNDARY-014-f105b1030b22d9a7.json`.

`SFT-COMP-OBL-CBLX-015` - Post-correspondence finite witness boundary

- **Claim and ownership:** `SFT-COMP-CBLX-POST-CORRESPONDENCE-WITNESS-015`; Classical Computation family `CBLX`; Post-correspondence finite witness boundary.
- **Forced law:** A Post-style correspondence instance is recognized by complete enumeration of finite tile-index words and exact equality of their two concatenated label traces; absence at a finite depth is not unrestricted nonexistence.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__complete-tile-sequence-witness__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-015 uniquely retains complete-tile-sequence-witness, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-ENUMERATION-001 -> SFT-COMP-CBLX-ARITHMETICAL-HIERARCHY-BOUNDARY-014. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-015: post_correspondence; exact execution: {"least_witness_indices":[0],"matched_word":["a"],"search_depth":2,"tile_count":3}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:71722cca087363f116a7a5d830ac4b97462953db5d0f25e2b3f12a9085b72a54`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:4e548a32534cba1702d54b9d74e18cc016bd7cc3ddc9c3c958f50c7e2d9a9e89`; engine receipt `sha256:0a5997a26e1f1fd9b3d9b1faa90a7fb37a02ad74d4ee65590f090b322072a02b` at `receipts/engine/model_admitted/SFT-COMP-CBLX-POST-CORRESPONDENCE-WITNESS-015-0a5997a26e1f1fd9.json`.

`SFT-COMP-OBL-CBLX-016` - Entscheidungsproblem correspondence boundary

- **Claim and ownership:** `SFT-COMP-CBLX-ENTSCHEIDUNGSPROBLEM-BOUNDARY-016`; Classical Computation family `CBLX`; Entscheidungsproblem correspondence boundary.
- **Forced law:** Every finite registered formula prefix may be exhaustively decided when its proof search closes, but no total internal procedure decides every generated statement once self-application can form its opposite verdict.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__generated-prefix-decision-self-reference-limit__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-016 uniquely retains generated-prefix-decision-self-reference-limit, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-UNDECIDABILITY-001 -> SFT-COMP-CBLX-POST-CORRESPONDENCE-WITNESS-015. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-016: entscheidungsproblem; exact execution: {"generated_prefix_decisions_retained":true,"total_self_negating_decider_exists":false}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:8e4a5549835145675ad60aff9db0e1ed62e7785ebfeee506c32c7c2e01e39cd4`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`;

independent certificate
`sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`;
empirical/external
`sha256:eb7d2e4d7f84a685c6586fbac0939232d5c65f656aeda70c5611cd57e159bda1`; engine receipt
`sha256:185ee2b6e356466324cbe70cc81198cbbfdfa4588e6ce4fc49e976ae53cf9936` at `receipts/engine/model_admitted/SFT-COMP-CBLX-ENTSCHEIDUNGSPROBLEM-BOUNDARY-016-185ee2b6e3564663.json`.

`SFT-COMP-OBL-CBLX-017` - Incompleteness and internal consistency boundary

- **Claim and ownership:** `SFT-COMP-CBLX-INCOMPLETENESS-CONSISTENCY-BOUNDARY-017`; Classical Computation family `CBLX`; Incompleteness and internal consistency boundary.
- **Forced law:** A sufficiently expressive sound generated proof process cannot close every statement about its own executions or certify its complete consistency from only its internal proof records; the missing self-distinction requires an explicitly stronger external record.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__self-verification-missing-distinction__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-017 uniquely retains self-verification-missing-distinction, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-INCOMPLETENESS-001 -> SFT-COMP-CBLX-ENTSCHEIDUNGSPROBLEM-BOUNDARY-016. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-017: incompleteness; exact execution: {"complete_internal_consistency_certificate_present":false,"external_record_required":true,"internal_proof_rows":2}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:4e210b416e18a384fcc3326a99b952320cb5ddf50180687374099aed3e00e6f9`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:c0acc266d9ae310889d651403f729dfa0afda6ffa6903bda764efb63ff1652cd`; engine receipt `sha256:d21509d08b502f830693c6d19eb030d8c3e71c0c3762ea15e4b1a45dc070ad7c` at `receipts/engine/model_admitted/SFT-COMP-CBLX-INCOMPLETENESS-CONSISTENCY-BOUNDARY-017-d21509d08b502f83.json`.

`SFT-COMP-OBL-CBLX-018` - Busy-Beaver domination theorem

- **Claim and ownership:** `SFT-COMP-CBLX-BUSY-BEAVER-DOMINATION-018`; Classical Computation family `CBLX`; Busy-Beaver domination theorem.

- **Forced law:** For each positive finite description depth in the admitted closing-process grammar, complete enumeration has a maximum halting runtime attained by a generated description; any total function represented within a strictly smaller depth-bound is eventually exceeded within that grammar.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__closing-grammar-runtime-maximum__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-018 uniquely retains closing-grammar-runtime-maximum, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-NATIVE-BUSY-BEAVER-002 -> SFT-COMP-CBLX-INCOMPLETENESS-CONSISTENCY-BOUNDARY-017. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-018: busy_beaver_domination; exact execution: {"depths_checked":[1,2,3,4,5,6,7,8],"maximum_runtime_equals_depth":true,"smaller_bound_dominated":true}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:926bd48a06fcf7cf6fd451bfb3ea70a429113592bd6358361a8f8eb2e911d55a`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:4b21d977ce71f0b8a92a0c5124f842e2f17cc786e431690b1309bf2636d2562f`; engine receipt `sha256:bd7d42e6dcdc3d369578cb0b8bb334b024147eb7fa95557ea28f504c3d620e46` at `receipts/engine/model_admitted/SFT-COMP-CBLX-BUSY-BEAVER-DOMINATION-018-bd7d42e6dcdc3d36.json`.

`SFT-COMP-OBL-CBLX-019` - Finite Busy-Beaver census protocol

- **Claim and ownership:** `SFT-COMP-CBLX-BUSY-BEAVER-FINITE-CENSUS-019`; Classical Computation family `CBLX`; Finite Busy-Beaver census protocol.
- **Forced law:** A finite Busy-Beaver result is exact only when every description through the declared depth is generated, every halting or recurrence trace is certified, all maxima and ties are retained, and the finite boundary is explicit.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__complete-depth-bounded-runtime-census__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-019 uniquely retains complete-depth-bounded-runtime-census, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-NATIVE-BUSY-BEAVER-002 -> SFT-COMP-CBLX-BUSY-BEAVER-DOMINATION-018. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-019: finite_busy_beaver_census; exact execution: {"all_ties_retained":true,"alphabet_size":2,"depths_exhausted":[1,2,3,4,5,6,7],"maxima":[1,2,3,4,5,6,7]}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:6eaa8969f306a9c6264b9752ca67f5c332053784398705c7cff3b766ea1e68f7`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:c663cccd64d79e63b42074e3c425260a1d29cbbdddaf47fb02fce0ea3fff1af0`; engine receipt `sha256:c119219f70bdd4b936a398e1ba5106f21274910e7fe24d6fe10e7b60b5b1374d` at `receipts/engine/model_admitted/SFT-COMP-CBLX-BUSY-BEAVER-FINITE-CENSUS-019-c119219f70bdd4b9.json`.

`SFT-COMP-OBL-CBLX-020` - Hypercomputation admissibility and physical-record boundary

- **Claim and ownership:** `SFT-COMP-CBLX-HYPERCOMPUTATION-ADMISSIBILITY-020`; Classical Computation family `CBLX`; Hypercomputation admissibility and physical-record boundary.
- **Forced law:** A claimed computation beyond the admitted universal model is admissible only with a generated operational trace or a declared relative oracle record; an answer without either is not a computation and cannot enter the model.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__trace-or-relative-oracle-admissibility__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-020 uniquely retains trace-or-relative-oracle-admissibility, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001 -> SFT-COMP-CBLX-BUSY-BEAVER-FINITE-CENSUS-019. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-020: hypercomputation_admissibility; exact execution: {"operational_trace_or_oracle_record_required":true,"physical_realization_handoff":true,"unrecorded_answers_admitted":false}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:5a13fb2b9b7c8cab6457b92792dc715eb2616ba991ba8fc0a504277a5f0da1b1`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external `sha256:2667d3690c3c158aea5a93a4888fad60f72214bd0f63ed9a99f46de102c9d9bf`; engine receipt `sha256:8fd953b4892eb44c0a1d20601b2121c553b0852b558c490af8f007412444ea6a` at `receipts/engine/model_admitted/SFT-COMP-CBLX-HYPERCOMPUTATION-ADMISSIBILITY-020-8fd953b4892eb44c.json`.

`SFT-COMP-OBL-CBLX-021` - Computability completeness and no-omission certificate

- **Claim and ownership:** `SFT-COMP-CBLX-COMPLETENESS-021`; Classical Computation family `CBLX`; Computability completeness and no-omission certificate.
- **Forced law:** Computability completeness is the one-to-one reconciliation of all twenty-one frozen obligations with unique survivors, adverse controls, exact executions, independent reconstructions and untouched-engine receipts.
- **Unique surviving structure:** `complete-generated-domain__complete-trace-execution__twenty-one-obligation-no-omission-ledger__retained-self-description__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-execution__depth-certificate-or-explicit-limit`. Exact result: CBLX-021 uniquely retains twenty-one-obligation-no-omission-ledger, complete self-application custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022 -> SFT-COMP-CBL-HYPERCOMPUTATION-LIMIT-001 -> SFT-COMP-CBLX-HYPERCOMPUTATION-ADMISSIBILITY-020. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CBLX-021: computability_completeness; exact execution: {"duplicate_owners":0,"independent_execution_rows":21,"omitted_owners":0,"registered_obligations":21}; complete family observation custody: 21 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPUTABILITY-OBSERVER, SFT-V1-V2-COMPUTABILITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported computability theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden transition, unregistered oracle, selected branch or stochastic cause; bounded exhaustion is not relabelled as an unrestricted negative result; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:e6bc2b724268bd815249409d3cbccdb46a5b7d79a10c049455c77a276de967ab`; independent implementation `sha256:5d90ece0c089f0e988e580b97664eb4fdc32993946977032d83065ee8e3b252d`; independent certificate `sha256:e97c3251847432d9279134100b82410f154e465c361392ae0759e0be9641bb19`; empirical/external

`sha256:038f97865c21143dd29d7eb133e2357f3e48c12b19043f2c2962604183c3fdef`; engine receipt `sha256:e96c59bfea8b5a95027908eb23c03f3da2a172e044b6bf3181060f30c39cd028` at `receipts/engine/model_admitted/SFT-COMP-CBLX-COMPLETENESS-021-e96c59bfea8b5a95.json`.

130.6 Family `CPLXX` - 33/33 complete

Scope. Computational complexity. This family completely decides 8,448 candidates, retains 33 unique survivors and passes 132 adverse controls.

`SFT-COMP-OBL-CPLXX-001` - Canonical input-length and instance-family carrier

- **Claim and ownership:** `SFT-COMP-CPLXX-INPUT-LENGTH-CARRIER-001`; Classical Computation family `CPLXX`; Canonical input-length and instance-family carrier.
- **Forced law:** Input size is the retained count of generated symbol positions in one canonical encoding; problem size additionally records the complete instance family and encoding translation boundary.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__canonical-generated-input-size__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-001 uniquely retains canonical-generated-input-size, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-INPUT-SIZE-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-001: canonical_input_length; exact execution: {"all_encodings_same_depth":true,"complete_instance_count":8,"word_depth":3}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:2ac0af4aee93a68df91b12f330ec6c7740c79e4b5fba2eaf6b2dd4a84232abb1`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:75bc2368ea47f2e673966a22f5a9c014abc989e545d52953942a242e2736bde2`; engine receipt `sha256:5ae2c108a4e22c1175fb4feccf1870d4925263cea2921a47c49d05495cdc39c7` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-INPUT-LENGTH-CARRIER-001-5ae2c108a4e22c11.json`.

`SFT-COMP-OBL-CPLXX-002` - Deterministic time-class correspondence

- **Claim and ownership:** `SFT-COMP-CPLXX-DETERMINISTIC-TIME-CLASS-002`; Classical Computation family `CPLXX`; Deterministic time-class correspondence.

- **Forced law:** Deterministic time is the retained transition count of the unique execution trace; a time class is the complete generated family whose traces satisfy one registered successor resource law.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__exact-transition-trace-time__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-002 uniquely retains exact-transition-trace-time, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-TIME-SPACE-001 -> SFT-COMP-CPLXX-INPUT-LENGTH-CARRIER-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-002: deterministic_time; exact execution: {"instance_count":8,"transition_count_each":3,"unique_trace_each":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:8c9a5fdebbc3ee7deb50796eb551f2a407ef88e1dae5ca852e9e333aae7b0e23`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:60066c2a29268cf244d42abd566dfffa23cbc11f9bbc1345f459e5ffa384a871`; engine receipt `sha256:43c83ffdd9b8eba6895e93c55ee3419c0e4df5d2579780f891183601e72ace07` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-DETERMINISTIC-TIME-CLASS-002-43c83ffdd9b8eba6.json`.

`SFT-COMP-OBL-CPLXX-003` - Deterministic space-class correspondence

- **Claim and ownership:** `SFT-COMP-CPLXX-DETERMINISTIC-SPACE-CLASS-003`; Classical Computation family `CPLXX`; Deterministic space-class correspondence.
- **Forced law:** Deterministic space is the greatest retained live configuration support along the unique trace, with reusable storage distinguished from accumulated provenance records.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__maximum-retained-configuration-space__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-003 uniquely retains maximum-retained-configuration-space, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 ->

SFT-COMP-CPLX-TIME-SPACE-001 -> SFT-COMP-CPLXX-DETERMINISTIC-TIME-CLASS-002. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** CPLXX-003: deterministic_space; exact execution: {"instance_count":8,"maximum_live_word_width":3,"provenance_record_separate":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:e441078ed286849f9961c8ef62391ff2c032fc418438172ac361b7ce1c8b49fe`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:099160916a2475e2ea999c16840e0a375e50512d490b3913997150167de4e647`; engine receipt `sha256:ac351aee769a0e2311437f55a34476ea31426c47e721e4f979e1f05583ac56c2` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-DETERMINISTIC-SPACE-CLASS-003-ac351aee769a0e23.json`.

`SFT-COMP-OBL-CPLXX-004` - Time and space hierarchy succession

- **Claim and ownership:** `SFT-COMP-CPLXX-HIERARCHY-SUCCESSION-004`; Classical Computation family `CPLXX`; Time and space hierarchy succession.
- **Forced law:** A resource hierarchy separates when each successor family contains an explicit process requiring the new retained depth or support and every smaller registered bound is exhausted.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__strict-resource-successor-hierarchy__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-004 uniquely retains strict-resource-successor-hierarchy, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-BOUNDS-001 -> SFT-COMP-CPLXX-DETERMINISTIC-SPACE-CLASS-003. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-004: resource_hierarchy; exact execution: {"attained_resources":[1,2,3,4,5,6],"depths":[1,2,3,4,5,6],"strict_successors":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or

completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:aefda3539de01a4b049e18c1e34718fd0ebdbe0107d9ec1641805a41b030c037`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:b6a0ddc432f504826efa2a5dea2fda901619250a3aa3b22a9ac3a4de483654cd`; engine receipt `sha256:3fb3a7123665c18a4b00b674f5e1c043a0ef171d2fee060c77efe4af8df2a191` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-HIERARCHY-SUCCESSION-004-3fb3a7123665c18a.json`.

`SFT-COMP-OBL-CPLXX-005` - Nondeterministic support as complete deterministic branch support

- **Claim and ownership:** `SFT-COMP-CPLXX-NONDETERMINISTIC-SUPPORT-005`; Classical Computation family `CPLXX`; Nondeterministic support as complete deterministic branch support.
- **Forced law:** Nondeterministic execution is the complete generated support of deterministic labelled branches; no branch occurs stochastically and acceptance retains at least one accepting trace plus the full support ledger.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__complete-deterministic-branch-product__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-005 uniquely retains complete-deterministic-branch-product, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-RANDOMNESS-001 -> SFT-COMP-CPLXX-HIERARCHY-SUCCESSION-004. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-005: nondeterministic_support; exact execution: {"branch_labels":2,"complete_branch_count":16,"depth":4,"stochastic_cause_used":false}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:99410f3ace7c99d6ee397c05148dfc65d99364ed103034be07069a0572809645`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:62199344d68b1f0eb65ab694bb1b263c5eecd07628c31261acb34c9754c8a16c`; engine receipt `sha256:5132f99ae3fb750cf58830a8c5466356ac4b0ebfcfe46ddeb5569a031a32e17f` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-NONDETERMINISTIC-SUPPORT-005-5132f99ae3fb750c`

`.json`.

`SFT-COMP-OBL-CPLXX-006` - Certificate length and verification resource

- **Claim and ownership:** `SFT-COMP-CPLXX-CERTIFICATE-VERIFICATION-RESOURCE-006`; Classical Computation family `CPLXX`; Certificate length and verification resource.
- **Forced law:** A certificate is a retained finite witness whose verifier reconstructs the declared verdict; its length and verification resources are counted on the same canonical input carrier.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__trace-bound-certificate-resource__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-006 uniquely retains trace-bound-certificate-resource, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-SEM-VERIFICATION-001 -> SFT-COMP-CPLXX-NONDETERMINISTIC-SUPPORT-005. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-006: certificate_resource; exact execution: {"all_verdicts_reconstructed":true,"certificate_length":3,"instances":8,"verification_steps":3}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:2a8d0e36f661ed38875b39d0fc6ee6a4b50f046b83b2f8cbceb6dc0d74a919b1`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:1b9873e185a22f1544926f6c261f3317d737b645e3d5e899bc25ea08c213e3a6`; engine receipt `sha256:e6b15c6b24274b3854feef82be148d5ba275b206a669b447c5d3806a0d2c217d` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-CERTIFICATE-VERIFICATION-RESOURCE-006-e6b15c6b24274b38.json`.

`SFT-COMP-OBL-CPLXX-007` - Fold-P and Fold-NP scoped equality boundary

- **Claim and ownership:** `SFT-COMP-CPLXX-FOLD-P-NP-SCOPE-007`; Classical Computation family `CPLXX`; Fold-P and Fold-NP scoped equality boundary.
- **Forced law:** Within the admitted native closing-process grammar, deterministic evaluation emits the unique complete trace accepted by the verifier, so deterministic execution and certificate verification share the same positive finite depth resource.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__native-trace-certificate-equality__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-007 uniquely retains native-trace-certificate-equality, complete resource custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002 -> SFT-COMP-CPLXX-CERTIFICATE-VERIFICATION-RESOURCE-006. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-007: native_fold_p_np; exact execution: {"certificate_resource":3,"deterministic_resource":3,"native_depth":3,"scope":"admitted-native-closing-grammar"}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:fc04198aa221d14821dbe28a5d5e991b7523381d85fe836cb88cb5d4b80d6996`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:4273dd4bae73fc4d7111fbd4b106f89364aa487992cab9d9da7be156ec3739dc`; engine receipt `sha256:defaca0bdd112782bf55a8c0ef3c716526c6a9f975da569aa1240237e354aba7` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-FOLD-P-NP-SCOPE-007-defaca0bdd112782.json`.

`SFT-COMP-OBL-CPLXX-008` - Conventional P-versus-NP transport boundary

- **Claim and ownership:** `SFT-COMP-CPLXX-CONVENTIONAL-P-NP-TRANSPORT-008`; Classical Computation family `CPLXX`; Conventional P-versus-NP transport boundary.
- **Forced law:** A conventional decision family inherits a native complexity relation only through a total bidirectional encoding preserving every instance, verdict, trace, certificate and explicit polynomial resource on one common size carrier.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__total-bidirectional-resource-transport__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-008 uniquely retains total-bidirectional-resource-transport, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-CONVENTIONAL-TRANSLATION-003 -> SFT-COMP-CPLXX-FOLD-P-NP-SCOPE-007. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-008: conventional_transport; exact execution: {"arbitrary_export":false,"bijection":true,"external_instances":8,"native_instances":8,"verdict_trace_certificate_preserved":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:78a4576dd6e2f8fb7ae01803fbe6c44e23e1afa503f54821f266b757661afc05`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:7871b0a766166db977b86758b52043e69df77a1ec5553d643a4eb3da5a2d1227`; engine receipt `sha256:399f5778bedbf1152360ec18540b8785170ff1b25e8e195311b4c2389d78660b` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-CONVENTIONAL-P-NP-TRANSPORT-008-399f5778bedbf115.json`.

`SFT-COMP-OBL-CPLXX-009` - Co-decision and complement-class correspondence

- **Claim and ownership:** `SFT-COMP-CPLXX-COMPLEMENT-CLASS-009`; Classical Computation family `CPLXX`; Co-decision and complement-class correspondence.
- **Forced law:** A complement class exchanges the separately retained accepting and rejecting terminal labels on the same complete input support; nontermination is not silently converted into either verdict.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__terminal-verdict-complement__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-009 uniquely retains terminal-verdict-complement, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001 -> SFT-COMP-CPLXX-CONVENTIONAL-P-NP-TRANSPORT-008. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-009: complement_class; exact execution: {"nontermination_relabelled":false,"terminal_classes":["accept","reject"],"verdicts_exchanged":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:b23dc4f06416a06fd15cc4693fb615d499388e0bd7d5aa5930f65d2ed65864f0`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`;

independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:f01cda4b66347604072e5954f91176479442aa0ffc146c14120d0a92b2a31a67`; engine receipt `sha256:5fbe91c1b06981bdbcd5df244115e4ee81c0c8f4f69bc918fcfc64379b7a3297` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-COMPLEMENT-CLASS-009-5fbe91c1b06981bd.json`.

`SFT-COMP-OBL-CPLXX-010` - Polynomial-space and alternating-process correspondence

- **Claim and ownership:** `SFT-COMP-CPLXX-ALTERNATION-SPACE-010`; Classical Computation family `CPLXX`; Polynomial-space and alternating-process correspondence.
- **Forced law:** Alternating execution retains existential and universal branch labels; depth-first evaluation uses one active path plus branch records, establishing correspondence only at the registered depth and encoding boundary.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__depth-first-alternation-space__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-010 uniquely retains depth-first-alternation-space, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-TIME-SPACE-001 -> SFT-COMP-CPLXX-COMPLEMENT-CLASS-009. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-010: alternating_space; exact execution: {"existential_nodes":1,"one_active_path_retained":true,"root_verdict":"accept","terminal_nodes":3,"universal_nodes":1}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:62bfa23cd93783472e0a94299dd051d781ac195f4c64e5e3e059224ca09acaa7`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:f8da44bdf1b280e1b4073b60cd9391fb3504bb9685d291dbdb314dd169e0a21e`; engine receipt `sha256:035fff1d6218f0c2b5706f5f40cce634be22c2aeed95879b60f6a62ff6860d31` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-ALTERNATION-SPACE-010-035fff1d6218f0c2.json`.

`SFT-COMP-OBL-CPLXX-011` - Exponential-resource succession

- **Claim and ownership:** `SFT-COMP-CPLXX-EXPONENTIAL-SUCCESSION-011`; Classical Computation family `CPLXX`; Exponential-resource succession.
- **Forced law:** Each generated branch label multiplies complete support by the fibre count, forcing exponential support growth while path depth remains the retained successor count.

- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__complete-branching-support-succession__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-011 uniquely retains complete-branching-support-succession, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-BOUNDS-001 -> SFT-COMP-CPLXX-ALTERNATION-SPACE-010. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-011: exponential_support; exact execution: {"depths":[1,2,3,4,5,6],"support":[2,4,8,16,32,64]}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:b75f25ae30877521f7ed9eaf33be2487dbf874be7d97932239fde5bb6276636b`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:e24e1b8ec97d15c76f81b1a5c76463bff30117edf19e6cdba95d272c16cf0f4f`; engine receipt `sha256:3f4a2e70c6a97c4e31201af7ff72e59896208a1c817f044476fddde20560550f` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-EXPONENTIAL-SUCCESSION-011-3f4a2e70c6a97c4e.json`.

`SFT-COMP-OBL-CPLXX-012` - Circuit family uniformity and nonuniformity

- **Claim and ownership:** `SFT-COMP-CPLXX-CIRCUIT-UNIFORMITY-012`; Classical Computation family `CPLXX`; Circuit family uniformity and nonuniformity.
- **Forced law:** A circuit family is uniform when one admitted generator constructs every depth-indexed member with a retained generation-resource trace; otherwise each member requires a separately retained advice description.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__generated-family-uniformity-record__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-012 uniquely retains generated-family-uniformity-record, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-CIRCUIT-RESOURCE-001 -> SFT-COMP-CPLXX-EXPONENTIAL-SUCCESSION-011. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** CPLXX-012: circuit_uniformity; exact execution: {"family_depths":[1,2,3,4,5,6],"nonuniform_advice_rows":6,"uniform_generator_present":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:08c0580086e6ed525da8fc6ea0ba9e8fd82283f7f6c6fe7ef2af70f0fe446813`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:4d7ca5bc0fd6d51698bca5dd8e87c2fdf188c21345aacb8ce398524c75dba676`; engine receipt `sha256:71a76ad6406f38c2c57786270aaafe84c2fd99f8599be3617079832aace2ce3d` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-CIRCUIT-UNIFORMITY-012-71a76ad6406f38c2.json`.

`SFT-COMP-OBL-CPLXX-013` - Circuit size, depth and width trade ledger

- **Claim and ownership:** `SFT-COMP-CPLXX-CIRCUIT-SIZE-DEPTH-WIDTH-013`; Classical Computation family `CPLXX`; Circuit size, depth and width trade ledger.
- **Forced law:** Circuit complexity retains size, depth and maximum width as separate exact coordinates; no coordinate is inferred from another without an explicit gate-basis and support theorem.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__circuit-resource-vector__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-013 uniquely retains circuit-resource-vector, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-CIRCUIT-RESOURCE-001 -> SFT-COMP-CPLXX-CIRCUIT-UNIFORMITY-012. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-013: circuit_resource_vector; exact execution: {"coordinates_separate":true,"depth":4,"size":30,"width":16}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:2881fc882fd766ea611ab3bcec61bd8d013917ab8c99f445e8e19cf5759843e2`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:39195fde89b15c88266c62380f9469d89d20fa884b100fe1f4b2a14890010a03`; engine receipt `sha256:04a764d2868070278f9b3a0373ad1df827f0e450645b5152c9e5a518687d2907` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-CIRCUIT-SIZE-DEPTH-WIDTH-013-04a764d286807027.json`.

`SFT-COMP-OBL-CPLXX-014` - Formula, branching-program and circuit correspondence

- **Claim and ownership:** `SFT-COMP-CPLXX-FORMULA-BRANCHING-CIRCUIT-014`; Classical Computation family `CPLXX`; Formula, branching-program and circuit correspondence.
- **Forced law:** Formulae, branching programs and circuits correspond only through translations preserving input support, output labels, shared-subcomputation records and exact resource overhead.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__trace-preserving-graph-model-translation__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-014 uniquely retains trace-preserving-graph-model-translation, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-FORM-MODEL-EQUIVALENCE-001 -> SFT-COMP-CPLXX-CIRCUIT-SIZE-DEPTH-WIDTH-013. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-014: graph_model_translation; exact execution: {"branching_path":3,"circuit_path":3,"formula_path":3,"input_depth":3,"trace_preserved":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:fb739f72fe515304727948e3826d6def9abfc91340c26269f3a80a613e7b196f`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:5d3791cffd87d92eb7a4c45e27c0fd785797df4450db6ccf8a6987ca5b31a129`; engine receipt `sha256:230d2589acde78139b2b2a6e6447467107881a8d69715d4712418379a94955d2` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-FORMULA-BRANCHING-CIRCUIT-014-230d2589acde7813.json`.

SFT-COMP-OBL-CPLXX-015 - Parallel depth and processor work

- **Claim and ownership:** SFT-COMP-CPLXX-PARALLEL-WORK-DEPTH-015; Classical Computation family CPLXX; Parallel depth and processor work.
- **Forced law:** Parallel complexity retains total work and causal layer depth separately; a speedup is admitted only with the complete processor, communication and synchronization ledger.
- **Unique surviving structure:** canonical-complete-input__time-space-depth-width-record-ledger__layered-work-depth-ledger__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport. Exact result: CPLXX-015 uniquely retains layered-work-depth-ledger, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-PARALLEL-001 -> SFT-COMP-CPLXX-FORMULA-BRANCHING-CIRCUIT-014. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** CPLXX-015: parallel_work_depth; exact execution: {"depth":3,"input_width":8,"layer_widths":[8,4,2,1],"work_and_depth_separate":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:8f20b454b8f3f35ee92043c0060d131e6fcb662264b6fe015cc6461844637aca; independent implementation sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773; independent certificate sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25; empirical/external sha256:671ad2a4e8a4bd6951add7f21c95eada7e5056c65301b84f94e84fbe4d813cb8; engine receipt sha256:00b10208411ce8ad75203482b23b25e53712e75dc77c14c7f6e317fd5a92af50 at receipts/engine/model_admitted/SFT-COMP-CPLXX-PARALLEL-WORK-DEPTH-015-00b10208411ce8ad.json.

SFT-COMP-OBL-CPLXX-016 - Communication rounds and transmitted distinctions

- **Claim and ownership:** SFT-COMP-CPLXX-COMMUNICATION-ROUNDS-DISTINCTIONS-016; Classical Computation family CPLXX; Communication rounds and transmitted distinctions.
- **Forced law:** Communication complexity is the retained message-label and round count required to separate all input pairs that demand different outputs under the declared party partition.
- **Unique surviving structure:** canonical-complete-input__time-space-depth-width-record-ledger__transcript-distinguishability-resource__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport. Exact result: CPLXX-016 uniquely retains transcript-distinguishability-resource, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-COMMUNICATION-QUERY-001 -> SFT-COMP-CPLXX-PARALLEL-WORK-DEPTH-015. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-016: communication_distinctions; exact execution: {"different_class_size":2,"input_pairs":4,"same_class_size":2,"transcript_classes":2}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:c5b19da918cee0b6c84b7d3699e8ab2131ddbb3940dabea51f317b272d06d0fd`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:b8f3b07489ee8f8526cd158707494716f830a44447690db336ae151e732488db`; engine receipt `sha256:cebe1722bcc156d61c89a210e3b7739a3fb128ebbd5de85689f030d3ba5edfc7` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-COMMUNICATION-ROUNDS-DISTINCTIONS-016-cebe1722bcc156d6.json`.

`SFT-COMP-OBL-CPLXX-017` - Decision-tree and query lower bounds

- **Claim and ownership:** `SFT-COMP-CPLXX-DECISION-TREE-QUERY-LOWER-017`; Classical Computation family `CPLXX`; Decision-tree and query lower bounds.
- **Forced law:** A query lower bound follows when fewer queried distinctions leave at least two generated inputs with different required outputs in one observation class; complete leaves certify attainment.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__leaf-distinguishability-query-bound__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-017 uniquely retains leaf-distinguishability-query-bound, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-COMMUNICATION-QUERY-001 -> SFT-COMP-CPLXX-COMMUNICATION-ROUNDS-DISTINCTIONS-016. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-017: query_lower_bound; exact execution: {"complete_tree_nodes":31,"input_depth":4,"required_leaves":16,"unresolved_pair_below_depth":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:775f8f7eb0e8f57356000c8f72bde546d214ecce99fb1719f5d09ff6539a9c09`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:752262d09b8bd10a8f16dddb69d3564503c0970ab637465ec38d1fecb7d88c9c`; engine receipt `sha256:a1d051b66c6793bfafabd4cd98c8b0dba995eef687a39ba84afe1685bb914e99` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-DECISION-TREE-QUERY-LOWER-017-a1d051b66c6793bf.json`.

`SFT-COMP-OBL-CPLXX-018` - Randomized resource as unresolved deterministic support

- **Claim and ownership:** `SFT-COMP-CPLXX-RANDOMIZED-RESOURCE-018`; Classical Computation family `CPLXX`; Randomized resource as unresolved deterministic support.

- **Forced law:** Randomized complexity is the exact resource ledger over complete deterministic branch support and exact outcome parts; no stochastic cause or ungenerated probability enters.

- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__exact-branch-ledger-randomness__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-018 uniquely retains exact-branch-ledger-randomness, complete resource custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-RANDOMNESS-001 -> SFT-COMP-CPLXX-DECISION-TREE-QUERY-LOWER-017. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** CPLXX-018: randomized_support; exact execution: {"accept_part":"3/4","branches":4,"reject_part":"1/4","stochastic_cause_used":false}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:303864c1461b22305617083bad312af10121df7cdcf23b4f1d716c9b71c9c0d3`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:b77443811d1fa4fd0248a0f762842d13839edb285d372e5051901a731deff7ad`; engine receipt `sha256:9823096535f3dea363d02f3535e87e7eccb23aab1c16245f30706a7654ea70b6` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-RANDOMIZED-RESOURCE-018-9823096535f3dea3.json`.

`SFT-COMP-OBL-CPLXX-019` - Derandomization correspondence boundary

- **Claim and ownership:** `SFT-COMP-CPLXX-DERANDOMIZATION-BOUNDARY-019`; Classical Computation family `CPLXX`; Derandomization correspondence boundary.
- **Forced law:** A deterministic representative replaces complete branch support without loss only when every branch has the same declared outcome or a separately forced hitting structure retains every relevant distinction.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__branch-invariant-canonical-selection__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-019 uniquely retains branch-invariant-canonical-selection, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-RANDOMNESS-001 -> SFT-COMP-CPLXX-RANDOMIZED-RESOURCE-018. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-019: derandomization_boundary; exact execution: {"branch_invariant_outcomes":1,"branch_invariant_support":4,"mixed_support_outcomes":2,"unforced_selection_rejected":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:967b7622e972a69ec9c9c9fe568eb461e5a961690efd0bd4d8c943615175282a`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:c43ff44831220b5b9ac8f9c92e5ab9e20c49f4af52a23dbffd7afe0e385f9a80`; engine receipt `sha256:5e8a3d58ed0a745b7e330fade666a9ec718b0cea5b04b0fb0823a4de27b9349f` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-DERANDOMIZATION-BOUNDARY-019-5e8a3d58ed0a745b.json`.

`SFT-COMP-OBL-CPLXX-020` - Counting-complexity support correspondence

- **Claim and ownership:** `SFT-COMP-CPLXX-COUNTING-SUPPORT-020`; Classical Computation family `CPLXX`; Counting-complexity support correspondence.
- **Forced law:** Counting complexity retains the exact number and part of generated branches satisfying the verifier; it does not replace the branch census with an analytic or stochastic approximation.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__exact-accepting-branch-count__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-020 uniquely retains exact-accepting-branch-count,

complete resource custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-DESCRIPTIVE-001 -> SFT-COMP-CPLXX-DERANDOMIZATION-BOUNDARY-019. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-020: counting_support; exact execution: {"accepting_branches":3,"branches":4,"exact_census_retained":true,"rejecting_branches":1}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:8f6309008b1bb6722cd6cd2915cf42e860a31fc05ffc8963c1e1161d7ab5bff4`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:8b9fcfb1b4428395a1ae5a53b9229e0acbe41df6a0be7aaea77f3ba549456e6d`; engine receipt `sha256:762a065484a88a128480137c509c3e33043aa846355c018cb7a908ba3b4de072` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-COUNTING-SUPPORT-020-762a065484a88a12.json`.

`SFT-COMP-OBL-CPLXX-021` - Reduction closure and complete-problem construction

- **Claim and ownership:** `SFT-COMP-CPLXX-REDUCTION-COMPLETE-PROBLEM-021`; Classical Computation family `CPLXX`; Reduction closure and complete-problem construction.
- **Forced law:** A problem is complete for a declared family only when it belongs to that family and every registered member has a total verdict-preserving reduction with explicit compositional resource overhead.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__resource-preserving-reduction-closure__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-021 uniquely retains resource-preserving-reduction-closure, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-REDUCTION-COMPLETENESS-001 -> SFT-COMP-CPLXX-COUNTING-SUPPORT-020. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-021: reduction_completeness; exact execution: {"composition_preserves_verdict":true,"intermediate_instances":2,"source_instances":2,"terminal_instances":2}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:16879828dbcb296a335f93d01f53482b8d07c2ed2464e87e8812d135908e8524`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:f00131d6f02c65dd67fb7a68bfcdd78bcc5c80428921e99e16fd00954fc98c64`; engine receipt `sha256:04bcd8150ae9201d3922fff124d4e4dea5f83bc08a7d5429afd1407c660e37eb` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-REDUCTION-COMPLETE-PROBLEM-021-04bcd8150ae9201d.json`.

`SFT-COMP-OBL-CPLXX-022` - Upper-bound witness and exact resource certificate

- **Claim and ownership:** `SFT-COMP-CPLXX-UPPER-BOUND-CERTIFICATE-022`; Classical Computation family `CPLXX`; Upper-bound witness and exact resource certificate.
- **Forced law:** An upper bound requires an admitted algorithm, complete execution trace and exact resource ledger for every generated instance under the declared encoding.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__attained-execution-upper-bound__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-022 uniquely retains attained-execution-upper-bound, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-BOUNDS-001 -> SFT-COMP-CPLXX-REDUCTION-COMPLETE-PROBLEM-021. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-022: upper_bound; exact execution: {"bound_attained":true,"execution_steps":4,"input_depth":4}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:038bf7219a196088a5677bce05da1748a92fcc10463ce71c8a3e9cc0806e306f`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate

`sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:5cadf94ef273f7520d9b5833b300e3a147d5ed0c271433e7ec4e9da8421c08a3`; engine receipt `sha256:1d1976900019513aa47be3e617a48201c6ff957e658eab69ed821ecc723725c6` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-UPPER-BOUND-CERTIFICATE-022-1d1976900019513a.json`.

`SFT-COMP-OBL-CPLXX-023` - Lower-bound adversary and indistinguishability certificate

- **Claim and ownership:** `SFT-COMP-CPLXX-LOWER-BOUND-ADVERSARY-023`; Classical Computation family `CPLXX`; Lower-bound adversary and indistinguishability certificate.
- **Forced law:** A lower bound requires complete elimination of smaller-resource processes or an adversary retaining two differently labelled instances indistinguishable under every smaller observation trace.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__unresolved-input-pair-lower-bound__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-023 uniquely retains unresolved-input-pair-lower-bound, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-BOUNDS-001 -> SFT-COMP-CPLXX-UPPER-BOUND-CERTIFICATE-022. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-023: lower_bound; exact execution: {"indistinguishable_pairs_with_different_suffix":8,"input_depth":4,"observed_prefix_depth":3,"smaller_observation_insufficient":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:d649cb87aeac36329148d357874d986d7cfe547c0494cba7682f05c009b2cd39`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:bf543d2907905094062af07b01db34d193b602e9c65c8813cb6e3938bfe3766b`; engine receipt `sha256:15e9760aaf52bc4208057d4c02b56d46178cc25973c384ebcc0e64cdf1bb2f7a` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-LOWER-BOUND-ADVERSARY-023-15e9760aaf52bc42.json`.

`SFT-COMP-OBL-CPLXX-024` - Arbitrary circuit lower-bound theorem boundary

- **Claim and ownership:** `SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024`; Classical Computation family `CPLXX`; Arbitrary circuit lower-bound theorem boundary.
- **Forced law:** Across circuits assembled from admitted Fold edges, closing depth requires that many gates, complete support requires fibre-count-to-depth width, and complete layers require the exact sum of generated edges; other gate bases require

separate transport.

- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__native-fold-edge-circuit-lower-bound__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-024 uniquely retains native-fold-edge-circuit-lower-bound, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002 -> SFT-COMP-CPLXX-LOWER-BOUND-ADVERSARY-023. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-024: native_circuit_lower_bound; exact execution: {"depths":[1,2,3,4,5,6,7],"path_bounds":[1,2,3,4,5,6,7],"size_at_depth_seven":254,"width_at_depth_seven":128}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:b7c86f9906ec59e02fe474bfaff0be9fa65360938af087a98ed17ffcada36264`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:9734c300cac8a83ce9463d53e3d1604afb111cbc7a9ae165c0758187a4aa5461`; engine receipt `sha256:7390b05427a5d0b2b977e869671412ea085fb74eb8030be9906b89529d4e11b4` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024-7390b05427a5d0b2.json`.

`SFT-COMP-OBL-CPLXX-025` - Worst-case, average-case and distribution custody

- **Claim and ownership:** `SFT-COMP-CPLXX-WORST-AVERAGE-DISTRIBUTION-025`; Classical Computation family `CPLXX`; Worst-case, average-case and distribution custody.
- **Forced law:** Worst case is the greatest retained resource over the complete declared instance support; average case is the exact part-weighted sum under a registered support ledger whose provenance cannot be fitted to outcomes.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__complete-instance-resource-distribution__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-025 uniquely retains complete-instance-resource-distribution, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 ->

SFT-COMP-CPLX-AVERAGE-WORST-001 -> SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** CPLXX-025: worst_average; exact execution: {"distribution_rows":4,"exact_average":"2/1","resource_vector":[1,2,4,1],"worst":4}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:36308be3a078d219cb0864876ec40f98cc0bdb89d975093e3ba7f76afb5c1af9`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:abc2f5fb398e9b49d970fd3e04542e531e477bda407bdeeddb21556d7c0f996b`; engine receipt `sha256:dae9c818e1f164594ebc834ddcfac60c53394c27287ca7c0adf723c35f6148c6` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-WORST-AVERAGE-DISTRIBUTION-025-dae9c818e1f16459.json`.

`SFT-COMP-OBL-CPLXX-026` - Approximation ratio as exact-part relation

- **Claim and ownership:** `SFT-COMP-CPLXX-APPROXIMATION-RATIO-026`; Classical Computation family `CPLXX`; Approximation ratio as exact-part relation.
- **Forced law:** Approximation quality is an exact part relating the candidate value to the independently forced optimum at a declared orientation; floating estimates and hidden optimum oracles are inadmissible.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__exact-candidate-optimum-part__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-026 uniquely retains exact-candidate-optimum-part, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-APPROXIMATION-001 -> SFT-COMP-CPLXX-WORST-AVERAGE-DISTRIBUTION-025. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-026: approximation_ratio; exact execution: {"candidate":6,"floating_used":false,"optimum":4,"oriented_exact_part":"3/2"}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or

completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:e1cee72617315c2b34a2ba1a9926c6cbf2b2cde640fa47f52463b64c01852e5a`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:cc4b8d30efab9c3c3c1614da2fd4037c4f61950c5e62a0c3aa68ec937ae3805a`; engine receipt `sha256:001f2200d972eeb636174b144c096dd66c7dcec9602bae3a596559e3b61c73f4` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-APPROXIMATION-RATIO-026-001f2200d972eeb6.json`.

`SFT-COMP-OBL-CPLXX-027` - Parameterized size and fixed-parameter resource

- **Claim and ownership:** `SFT-COMP-CPLXX-PARAMETERIZED-SIZE-027`; Classical Computation family `CPLXX`; Parameterized size and fixed-parameter resource.
- **Forced law:** Parameterized complexity retains input size and parameter as distinct generated coordinates; fixed-parameter tractability requires a parameter-only carrier composed with an explicit size-resource law.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__input-parameter-resource-pair__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-027 uniquely retains input-parameter-resource-pair, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-PARAMETERIZED-001 -> SFT-COMP-CPLXX-APPROXIMATION-RATIO-026. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-027: parameterized_resource; exact execution: {"coordinates_separate":true,"input_depth":3,"parameters":[1,2],"resource_bound_preserved":true}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:7fb18df0f67b5400778f5bba5dde55615c4e8b81ea15c4b1255144d0ef87de68`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:2679b2b6b6def18f6194226d2868b1d850e6adbe4834e6ec5e9f2a2c95245e0c`; engine receipt `sha256:704b2a3563f7cfe8c1bea6e61f126a9e4ed800e110eff891c6d9411f4cbd7b3b` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-PARAMETERIZED-SIZE-027-704b2a3563f7cfe8.json`.

`SFT-COMP-OBL-CPLXX-028` - Kernelization correspondence boundary

- **Claim and ownership:** `SFT-COMP-CPLXX-KERNELIZATION-BOUNDARY-028`; Classical Computation family `CPLXX`; Kernelization correspondence boundary.
- **Forced law:** Kernelization is a total transformation to an equivalent instance bounded by the retained parameter, together with a reverse verdict or witness reconstruction and exact preprocessing resource.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__equivalent-bounded-kernel-reconstruction__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-028 uniquely retains equivalent-bounded-kernel-reconstruction, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-PARAMETERIZED-001 -> SFT-COMP-CPLXX-PARAMETERIZED-SIZE-027. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-028: kernelization; exact execution: {"kernel":["a","b"],"parameter":2,"reverse_locations":{"a":[0,1,3],"b":[2]},"source":["a","a","b","a"]}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:fafdb98f491635cbd120711b71d4a57c5069d5b07fda8ef14a677c2ab8b06d31`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:fdf9439b45f113274162697fdc42f887bec4b9d5c27e7b615702e6cdd1ef8688`; engine receipt `sha256:87638f73e2058c88517b04585491f76378c6294e71582e710f0532b3d368b616` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-KERNELIZATION-BOUNDARY-028-87638f73e2058c88.json`.

`SFT-COMP-OBL-CPLXX-029` - Amortized and aggregate resource accounting

- **Claim and ownership:** `SFT-COMP-CPLXX-AMORTIZED-RESOURCE-029`; Classical Computation family `CPLXX`; Amortized and aggregate resource accounting.
- **Forced law:** Amortized complexity retains every operation cost and prefix total; redistributed credits are held records, never negative values, and the exact aggregate divided across the complete sequence supplies the average.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__prefix-complete-aggregate-ledger__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-029 uniquely retains prefix-complete-aggregate-ledger, complete resource custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-AVERAGE-WORST-001 -> SFT-COMP-CPLXX-KERNELIZATION-BOUNDARY-028. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-029: amortized_resource; exact execution: {"aggregate":6,"exact_average":"3/2","negative_credit_used":false,"operation_costs":[1,1,3,1],"prefix_totals":[1,2,5,6]}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f8a4c28c4ff2dd2fe5d9995d0fef4c69fd88132e80bd0a7780f463bdd4117bff`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:151962230238aba69da946a9faa24b05d8a2a34929f2d3152b61d772b55bbb3a`; engine receipt `sha256:4225a625872ee296077521ee036f4f1f376ce1111528dff9ab57c9f3af12306c` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-AMORTIZED-RESOURCE-029-4225a625872ee296.json`.

`SFT-COMP-OBL-CPLXX-030` - Online competitive-resource correspondence

- **Claim and ownership:** `SFT-COMP-CPLXX-ONLINE-COMPETITIVE-030`; Classical Computation family `CPLXX`; Online competitive-resource correspondence.
- **Forced law:** An online resource guarantee compares each irrevocable prefix decision with an independently generated full-information optimum using an exact oriented part and a complete adversarial input support.
- **Unique surviving structure:** `canonical-complete-input__time-space-depth-width-record-ledger__prefix-decision-offline-comparison__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport`. Exact result: CPLXX-030 uniquely retains prefix-decision-offline-comparison, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-APPROXIMATION-001 -> SFT-COMP-CPLXX-AMORTIZED-RESOURCE-029. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-030: online_competitive; exact execution: {"complete_prefixes_retained":true,"exact_ratio":"3/2","offline_optimum":4,"online_cost":6}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status:

empirically_tested_and_independently_replicated.

- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation sha256:89f04a3c62511db2bc5e1031005d694cf2c0bea9042a2c5b478416aed9772784; independent implementation sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773; independent certificate sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25; empirical/external sha256:9050bb1f7a89c274e913ce23c57a4cd7bd092740bbbb081e944556f5f477ac30; engine receipt sha256:279968d86a5e21e52f32c08a227c4b5569f59713faf0c0d715687743d51e891e at receipts/engine/model_admitted/SFT-COMP-CPLXX-ONLINE-COMPETITIVE-030-279968d86a5e21e5.json.

SFT-COMP-OBL-CPLXX-031 - Description and program-size complexity

- **Claim and ownership:** SFT-COMP-CPLXX-DESCRIPTION-PROGRAM-SIZE-031; Classical Computation family CPLXX; Description and program-size complexity.

- **Forced law:** Description complexity is the least generated program-word length producing the retained object under one fixed universal grammar; changing grammar requires an explicit translation constant record.

- **Unique surviving structure:** canonical-complete-input__time-space-depth-width-record-ledger__fixed-grammar-least-description__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport. Exact result: CPLXX-031 uniquely retains fixed-grammar-least-description, complete resource custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-DESCRIPTIVE-001 -> SFT-COMP-CPLXX-ONLINE-COMPETITIVE-030. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Post-registration observation:** CPLXX-031: description_complexity; exact execution: {"compressed_description_length":2,"grammar_fixed":true,"least_registered_length":2,"literal_description_length":3}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.

- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation sha256:2bc051758230b92aaf31c7c43fd4c40d3dd084b87f7ff6cad62b35b8fc830245; independent implementation sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773; independent certificate sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25; empirical/external sha256:2ef832431b45bdc5cca2f644a0896d1d540a941dee75badffa54ea822ac1b00c; engine receipt sha256:0c967d21b5fb44f18bcca78966604644c5be20ee266a00611cd12d28f6ea6b08 at receipts/engine/model_admitted/SFT-COMP-CPLXX-DESCRIPTION-PROGRAM-SIZE-031-0c967d21b5fb44f1.json.

SFT-COMP-OBL-CPLXX-032 - Reversible time-space-record tradeoff

- **Claim and ownership:** SFT-COMP-CPLXX-REVERSIBLE-TRADEOFF-032; Classical Computation family CPLXX; Reversible time-space-record tradeoff.
- **Forced law:** Reversible simulation retains every predecessor distinction closed by an irreversible step; time, live space and reverse-record size form one exact resource ledger.
- **Unique surviving structure:** canonical-complete-input__time-space-depth-width-record-ledger__predecessor-record-reversible-resource__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport. Exact result: CPLXX-032 uniquely retains predecessor-record-reversible-resource, complete resource custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-REVERSIBILITY-COST-001 -> SFT-COMP-CPLXX-DESCRIPTION-PROGRAM-SIZE-031. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** CPLXX-032: reversible_tradeoff; exact execution: {"lost_predecessor_labels":0,"restored":["a","b","c"],"reverse_record":["c","b","a"],"source":["a","b","c"]}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:98c42be21f6cae046e4990c64497c59228a262d457f34ba0a1a580b98b41160f; independent implementation sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773; independent certificate sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25; empirical/external sha256:659d2182836230298638f82b02d5a1c6b21d3baee1c3df95f0cbdcd31b83ba80; engine receipt sha256:d26bf5549335ce6b922488c1533e1f67572b1f0e333c00c4f8d683f8e4312fb6 at receipts/engine/model_admitted/SFT-COMP-CPLXX-REVERSIBLE-TRADEOFF-032-d26bf5549335ce6b.json.

SFT-COMP-OBL-CPLXX-033 - Complexity-family completeness certificate

- **Claim and ownership:** SFT-COMP-CPLXX-COMPLETENESS-033; Classical Computation family CPLXX; Complexity-family completeness certificate.
- **Forced law:** Complexity completeness is the one-to-one reconciliation of all thirty-three frozen obligations with unique survivors, controls, exact observations, independent reconstructions and untouched-engine receipts.
- **Unique surviving structure:** canonical-complete-input__time-space-depth-width-record-ledger__thirty-three-obligation-no-omission-ledger__complete-generated-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-resource-execution__depth-certificate-or-explicit-transport. Exact result: CPLXX-033 uniquely retains

thirty-three-obligation-no-omission-ledger, complete resource custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-CPLX-BOUNDS-001 -> SFT-COMP-CPLXX-REVERSIBLE-TRADEOFF-032. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** CPLXX-033: complexity_completeness; exact execution: {"duplicate_owners":0,"independent_execution_rows":33,"omitted_owners":0,"registered_obligations":33}; complete family observation custody: 33 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-COMPLEXITY-OBSERVER, SFT-V1-V2-COMPLEXITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported complexity-class answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden advice, unregistered oracle, sampled branch support or stochastic cause; native Fold results are not exported to arbitrary conventional families without the registered transport certificate; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:91bf9f0e9660d470c8ca169e9e52e027d9312f6f557a28fd928d23af2cba4e9b`; independent implementation `sha256:11eee89779db64305b6ab24d37614c81d2ccaf230db16f664a9e435b4141a773`; independent certificate `sha256:d4f168a579a960f2ea80dd493bc97263a28c9c79d75861a64b8aa2d6a9d71f25`; empirical/external `sha256:db64b100bedef1d7b80aa0aa51a0342bec80e8de192ebdee0312417d6d868e10`; engine receipt `sha256:7f5c0426902fa647237c31c60bd398fb98183edfebb74e11f14a62c3e916ad48` at `receipts/engine/model_admitted/SFT-COMP-CPLXX-COMPLETENESS-033-7f5c0426902fa647.json`.

130.7 Family `ALGX` - 31/31 complete

Scope. Algorithms and mathematical data structures. This family completely decides 7,936 candidates, retains 31 unique survivors and passes 124 adverse controls.

`SFT-COMP-OBL-ALGX-001` - Algorithm specification, invariant and termination certificate

- **Claim and ownership:** `SFT-COMP-ALGX-SPECIFICATION-INVARIANT-001`; Classical Computation family `ALGX`; Algorithm specification, invariant and termination certificate.
- **Forced law:** An algorithm is a total source-bound transition process over a complete mathematical input organization; correctness retains its invariant at every step, termination retains a finite descent certificate, and the terminal result satisfies the registered specification.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__invariant-trace-termination-certificate__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-001 uniquely retains invariant-trace-termination-certificate, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-SEARCH-ORDER-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** ALGX-001: algorithm_certificate; exact execution: {"input":["a","b","c"],"invariant_retained":true,"observed_positions":[0,1],"target":"b","terminal_position":1}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:850e825687af8af971ec438ccc7132aedf101541dbb295603c7f92dc613b8c75`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:6403fc4a27537395993281cb7b9763fe10b402b7a6b063c3f7555ad924582512`; engine receipt `sha256:70eb3cb7e25655fecad3c36d1930fe5102cb6c821dd7dba96c7cbbbbc0740367` at `receipts/engine/model_admitted/SFT-COMP-ALGX-SPECIFICATION-INVARIANT-001-70eb3cb7e25655fe.json`.

`SFT-COMP-OBL-ALGX-002` - Exact linear and ordered search

- **Claim and ownership:** `SFT-COMP-ALGX-SEARCH-002`; Classical Computation family `ALGX`; Exact linear and ordered search.
- **Forced law:** Linear search observes each retained position in order until the target is found or complete support is exhausted; ordered search may close a whole interval only after the comparison relation proves that interval cannot contain the target.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__complete-search-observation-trace__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-002 uniquely retains complete-search-observation-trace, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-SEARCH-ORDER-001 -> SFT-COMP-ALGX-SPECIFICATION-INVARIANT-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-002: search; exact execution: {"absent_result_supported":true,"linear_terminal_position":3,"ordered_terminal_position":2}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or

changes protected authority.

- **Evidence identities:** derivation `sha256:4449cb8f05d58323da64a28eb892d915bd18b190dfb8529b9c651769a06b2841`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:311bf41c3bade2daedfa60ec1fd74b971f1f2149549d198cd5a731d8e5c7746f`; engine receipt `sha256:80f98e73bea6d1180666a9966985a84433df743beb73f2f0ffd687ffe3faf374` at `receipts/engine/model_admitted/SFT-COMP-ALGX-SEARCH-002-80f98e73bea6d118.json`.

`SFT-COMP-OBL-ALGX-003` - Comparison sorting and permutation custody

- **Claim and ownership:** `SFT-COMP-ALGX-COMPARISON-SORT-003`; Classical Computation family `ALGX`; Comparison sorting and permutation custody.
- **Forced law:** Comparison sorting returns a nondecreasing permutation of the complete input multiset; every duplication and original identity is retained, and its comparison trace certifies both order and permutation custody.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__stable-permutation-preserving-order__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-003 uniquely retains stable-permutation-preserving-order, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-SEARCH-ORDER-001 -> SFT-COMP-ALGX-SEARCH-002. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-003: comparison_sort; exact execution: {"input":[3,1,2,1],"multiset_preserved":true,"output":[1,1,2,3],"stable_identity_retained":true}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:ca305d19037c8c9d88abf91e960afa60b123840e82591504afd9251a6a022016`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:279eaf75a41c2c4a8f6f52588bfaf4a905a156240c22fe66e55f8e9405dadea2`; engine receipt `sha256:6b19d5a7cb429e8eb756b6995520a73de408a1e0f55aa48709a80bb75e1cb2af` at `receipts/engine/model_admitted/SFT-COMP-ALGX-COMPARISON-SORT-003-6b19d5a7cb429e8e.json`.

`SFT-COMP-OBL-ALGX-004` - Noncomparison ordering correspondence

- **Claim and ownership:** `SFT-COMP-ALGX-NONCOMPARISON-ORDER-004`; Classical Computation family `ALGX`; Noncomparison ordering correspondence.
- **Forced law:** Noncomparison ordering is admissible only when a complete finite key alphabet supplies canonical buckets; scanning those buckets in alphabet order emits every source item exactly once without importing a comparison result.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__alphabet-bucket-order__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-004 uniquely retains alphabet-bucket-order, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-SEARCH-ORDER-001 -> SFT-COMP-ALGX-COMPARISON-SORT-003. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-004: bucket_order; exact execution: {"alphabet":["a","b","c"],"input":["c","a","b","a"],"output":["a","a","b","c"]}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f0d5a368fa30298b5a8f83e900669a9619c815723059d521d8881b4d47d29902`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:83d3ef5232d7d5c5e696161f026a1bc385542f88277a5b0d3261cdffe5e15ddc`; engine receipt `sha256:3e8e751e67ac2709bf1ae0719c70012f0eca7b19576a1a07e9df8333c02423a9` at `receipts/engine/model_admitted/SFT-COMP-ALGX-NONCOMPARISON-ORDER-004-3e8e751e67ac2709.json`.

`SFT-COMP-OBL-ALGX-005` - Integer addition, multiplication and division algorithms

- **Claim and ownership:** `SFT-COMP-ALGX-INTEGER-ARITHMETIC-005`; Classical Computation family `ALGX`; Integer addition, multiplication and division algorithms.
- **Forced law:** Exact addition is disjoint word junction, multiplication is complete pair formation, and division is repeated complete-part extraction with an explicitly retained unmatched remainder; an absent divisor halts.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__exact-word-arithmetic__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-005 uniquely retains exact-word-arithmetic, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-ARITHMETIC-001 -> SFT-COMP-ALGX-NONCOMPARISON-ORDER-004. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-005: arithmetic; exact execution: {"division_quotient":2,"division_remainder":1,"left_width":2,"product_cells":6,"right_width":3,"sum_width":5}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:d59bf5415d796647bfbd71de89fc431e054be491c5dbd5bf46a47ae11d82248d`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:4bcd9b10ea77456825b0f5b8b7697d25d37552daab5763bc26944374efd42eb7`; engine receipt `sha256:1986b80f0a0c3f213bb6f96367d88010f554e9c5979caf3346e831b2ce505ddb` at `receipts/engine/model_admitted/SFT-COMP-ALGX-INTEGER-ARITHMETIC-005-1986b80f0a0c3f21.json`.

`SFT-COMP-OBL-ALGX-006` - Greatest-common-part and modular arithmetic algorithms

- **Claim and ownership:** `SFT-COMP-ALGX-GCD-MODULAR-006`; Classical Computation family `ALGX`; Greatest-common-part and modular arithmetic algorithms.
- **Forced law:** The greatest common part is forced by repeated exact remainder descent, and modular arithmetic retains the quotient class and exact remainder without signed, floating or continuum values.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__remainder-descent-modular-arithmetic__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-006 uniquely retains remainder-descent-modular-arithmetic, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-ARITHMETIC-001 -> SFT-COMP-ALGX-INTEGER-ARITHMETIC-005. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-006: gcd_modular; exact execution: {"greatest_common_part":6,"inputs":[18,12],"modular_power_remainder":4}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:35d18a1adbbb824abed346ee8a0540315042bddef05280e6b48e09f9efdad465`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:e978260f677e303f4ebcc17c193ae8261767ec3e34ddaa5fae25ea9c6eb1667b`; engine receipt `sha256:a25b98e0c5b5fdef7ac59ce929ae72bab0b879c7b2267d323f469780e0ca1ca1` at `receipts/engine/model_admitted/SFT-COMP-ALGX-GCD-MODULAR-006-a25b98e0c5b5fdef.json`.

`SFT-COMP-OBL-ALGX-007` - Exact rational-part arithmetic algorithms

- **Claim and ownership:** `SFT-COMP-ALGX-RATIONAL-ARITHMETIC-007`; Classical Computation family `ALGX`; Exact rational-part arithmetic algorithms.

- **Forced law:** Rational-part arithmetic uses exact common refinement for addition and complete product refinement for multiplication; every result remains a finite part of a generated whole with canonical reduction.

- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__exact-common-refinement-rational-arithmetic__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-007 uniquely retains exact-common-refinement-rational-arithmetic, complete correctness custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-ARITHMETIC-001 -> SFT-COMP-ALGX-GCD-MODULAR-006. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** ALGX-007: rational_arithmetic; exact execution: {"floating_used":false,"product":"1/2","sum":"1/2"}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:6cf58d55daf9014728209f70159e154f25db5f466f5522ad27c231246472af4a`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:5d41f07b4ddfa8a66b278cfafdcd52179537b2087f6aacf0ccc5a134d1a58042`; engine receipt `sha256:3529fb809c8a267c1c8aa95c98e8a7dc7a1c5c6c250a73585686a98cb8df073a` at `receipts/engine/model_admitted/SFT-COMP-ALGX-RATIONAL-ARITHMETIC-007-3529fb809c8a267c.json`.

`SFT-COMP-OBL-ALGX-008` - String matching and finite-pattern search

- **Claim and ownership:** `SFT-COMP-ALGX-STRING-MATCH-008`; Classical Computation family `ALGX`; String matching and finite-pattern search.
- **Forced law:** Finite-pattern search compares the complete pattern at every lawful text position and returns the exact ordered position ledger, including overlapping matches and an explicit absent-result record.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__complete-pattern-position-ledger__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-008 uniquely retains complete-pattern-position-ledger, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-STRINGS-SEQUENCES-001 -> SFT-COMP-ALGX-RATIONAL-ARITHMETIC-007. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-008: string_matching; exact execution: {"match_positions":[0,2],"overlaps_retained":true,"pattern":"aba","text":"ababa"}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:b16d90c40dd20bda742acc5c9a21b68f164871e79fb1f6c7941f082a5ba054ee`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:1a9da697ff951b6b9e26ac37fb21f48547eb19439fb4e572f2f9155f8a45fb0b`; engine receipt `sha256:875c43ecbc48ec320bac5b9772905896bc33dc395fd02ad2747cfd7cd5a41121` at `receipts/engine/model_admitted/SFT-COMP-ALGX-STRING-MATCH-008-875c43ecbc48ec32.json`.

`SFT-COMP-OBL-ALGX-009` - Sequence alignment and edit structure

- **Claim and ownership:** `SFT-COMP-ALGX-SEQUENCE-EDIT-009`; Classical Computation family `ALGX`; Sequence alignment and edit structure.
- **Forced law:** Sequence edit distance is the least complete trace of retained insert, remove and substitute operations; dynamic table cells enumerate all lawful predecessor operations and retain every tie.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__edit-witness-dynamic-table__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-009 uniquely retains edit-witness-dynamic-table, complete correctness custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-STRINGS-SEQUENCES-001 -> SFT-COMP-ALGX-STRING-MATCH-008. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-009: sequence_edit; exact execution: {"least_edit_count":1,"source":"abc","target":"ac","witness_operation":"remove-b"}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:2bbd986786db89a51ab0d514493299b2747012d0acd0ab2f3ad432bb74fb46c8`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:b667a7d307ab3a13f708443922c15d2d13e4a050b30d4142fc507899c9b48dbb`; engine receipt `sha256:907e84359f8e9d7abcfafdfed6694683d4869b3022986be0eb269effdd22416a` at `receipts/engine/model_admitted/SFT-COMP-ALGX-SEQUENCE-EDIT-009-907e84359f8e9d7a.json`.

`SFT-COMP-OBL-ALGX-010` - Tree traversal, balancing and search organization

- **Claim and ownership:** `SFT-COMP-ALGX-TREE-ORGANIZATION-010`; Classical Computation family `ALGX`; Tree traversal, balancing and search organization.
- **Forced law:** A tree algorithm retains root, child order and parent custody; traversal visits every reachable node once, search follows the registered order, and balancing is admissible only with preserved inorder identity and explicit rotations.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__rooted-tree-operation-ledger__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-010 uniquely retains rooted-tree-operation-ledger, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-TREES-GRAPHS-001 -> SFT-COMP-ALGX-SEQUENCE-EDIT-009. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-010: tree_traversal; exact execution: {"duplicates":0,"nodes_visited":4,"preorder":["r","a","c","b"],"root":"r"}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status:

`empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:549077a10723b78711592a40f394d3cdafbebaa4049fb00571efa428c92dc279`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:8cc21f81ccbcc7e52bce0509cedc5c7401fbb9c46061c0a1aa172b2594867b96`; engine receipt `sha256:cb77a18fe66ea6934eae0804e816160510d73199c16f178833816fd04314557a` at `receipts/engine/model_admitted/SFT-COMP-ALGX-TREE-ORGANIZATION-010-cb77a18fe66ea693.json`.

`SFT-COMP-OBL-ALGX-011` - Graph traversal and reachability

- **Claim and ownership:** `SFT-COMP-ALGX-GRAPH-REACHABILITY-011`; Classical Computation family `ALGX`; Graph traversal and reachability.
- **Forced law:** Graph reachability is the least successor closure of a declared source under complete adjacency; traversal retains frontier order, predecessor witnesses and every reached vertex exactly once.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__frontier-complete-reachability__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-011 uniquely retains frontier-complete-reachability, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-TREES-GRAPHS-001 -> SFT-COMP-ALGX-TREE-ORGANIZATION-010. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-011: graph_reachability; exact execution: {"reach_order":["a","b","c","d"],"reached_vertices":4,"source":"a"}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:bd94108413c22be6e70975c450644b2442bd5f5558eee1a764726d9ae11abab4`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:7d72c9b8e99ac85a752b1f4add8f8408b4aac50c9f65207ae005ff3cb880cc01`; engine receipt

`sha256:091ce9e2862a174871a807be845136ed822cf7af7ad94d0a9d1bd4ad9e93a6a6` at `receipts/engine/model_admitted/SFT-COMP-ALGX-GRAPH-REACHABILITY-011-091ce9e2862a1748.json`.

`SFT-COMP-OBL-ALGX-012` - Shortest-path and path-composition algorithms

- **Claim and ownership:** `SFT-COMP-ALGX-SHORTEST-PATH-012`; Classical Computation family `ALGX`; Shortest-path and path-composition algorithms.
- **Forced law:** Shortest-path computation composes exact nonnegative edge parts, retains the least discovered cost and reconstructible path for every reached vertex, and rejects any unregistered signed or hidden edge value.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__path-composition-least-cost-ledger__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-012 uniquely retains path-composition-least-cost-ledger, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-TREES-GRAPHS-001 -> SFT-COMP-ALGX-GRAPH-REACHABILITY-011. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-012: shortest_path; exact execution: {"composed_cost":3,"direct_cost":5,"selected_path":["a","b","c"],"source":"a","target":"c"}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:ec75a39e9ab1512bf9559c6ebf82b0eb17ba2db09d43271d45033a18166a28d9`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:af0599d6a5c20e1ac8fd5b31f41781dc67d78a22fc6d0309db8232feb1bf8f6f`; engine receipt `sha256:bf61c594bae20ae2c96e2d397d9a9300b716bf1e8f163c8903cd58414aa98260` at `receipts/engine/model_admitted/SFT-COMP-ALGX-SHORTEST-PATH-012-bf61c594bae20ae2.json`.

`SFT-COMP-OBL-ALGX-013` - Spanning-tree and connectivity algorithms

- **Claim and ownership:** `SFT-COMP-ALGX-SPANNING-CONNECTIVITY-013`; Classical Computation family `ALGX`; Spanning-tree and connectivity algorithms.
- **Forced law:** A spanning tree is a cycle-free edge subset reaching every vertex in the declared connected component; its predecessor edge for each nonroot vertex supplies the exact connectivity witness.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__cycle-free-connectivity-witness__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-013 uniquely retains cycle-free-connectivity-witness, complete correctness custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-TREES-GRAPHS-001 -> SFT-COMP-ALGX-SHORTEST-PATH-012. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-013: spanning_tree; exact execution: {"all_reached":true,"cycle_free":true,"tree_edges":[["a","b"],["a","c"],["b","d"]],"vertices":4}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:85df435054ed10f376d29abe852af6ad4eeb8f4329b73ba872354a90817b4b76`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:7758457a47abffc71b8132e9a97be1e7b22efd747a615bc654df81c252416614`; engine receipt `sha256:70706956bf7f0dc9ab038a422c8f117b34019e51033f2387240679e4de1b1030` at `receipts/engine/model_admitted/SFT-COMP-ALGX-SPANNING-CONNECTIVITY-013-70706956bf7f0dc9.json`.

`SFT-COMP-OBL-ALGX-014` - Network flow and cut algorithms

- **Claim and ownership:** `SFT-COMP-ALGX-FLOW-CUT-014`; Classical Computation family `ALGX`; Network flow and cut algorithms.
- **Forced law:** A network flow retains per-edge capacity and transported parts, conserves every internal carrier, and is maximal only when an exact source-sink cut with equal capacity is independently reconstructed.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__flow-conservation-cut-equality__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-014 uniquely retains flow-conservation-cut-equality, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-TREES-GRAPHS-001 -> SFT-COMP-ALGX-SPANNING-CONNECTIVITY-013. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-014: flow_cut; exact execution: {"cut_capacities":[3,4,3],"equality":true,"least_cut":3,"path_flows":[2,1],"total_flow":3}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:d220350cacdca5c406087541eb53e47a9ccb3fa06c76f8c5c99b18b3214312ce`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:b6c895bfeb39461bcafd6b59ced8b98dd321f09847f2a86894f00da86d2d770d`; engine receipt `sha256:950fb2aad5b407823e69b4870424de88307396fea4001321eefaa069c040ea85` at `receipts/engine/model_admitted/SFT-COMP-ALGX-FLOW-CUT-014-950fb2aad5b40782.json`.

`SFT-COMP-OBL-ALGX-015` - Matching and assignment algorithms

- **Claim and ownership:** `SFT-COMP-ALGX-MATCHING-ASSIGNMENT-015`; Classical Computation family `ALGX`; Matching and assignment algorithms.
- **Forced law:** A matching is an injective retained relation between generated vertex families; maximum or minimum assignment requires complete feasible-pair enumeration or a separately forced optimality certificate.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__injective-pairing-optimum__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-015 uniquely retains injective-pairing-optimum, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-TREES-GRAPHS-001 -> SFT-COMP-ALGX-FLOW-CUT-014. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-015: matching; exact execution: {"complete_matchings":1,"left":["a","b"],"matched_pairs":[["a","x"],["b","y"]],"right":["x","y"]}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:ae5815a7ecf30d508c8049093b8405dfa6970e0919b16a5fc4d8767e06cb6d9a`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`;

empirical/external `sha256:97761872219d9496aaf3d66bd210272b774c77b6d8248284bcbab8a859a4afda`; engine receipt `sha256:f508ce7800234e1a4491a4a3276d4ef851a3df3a9b58b1fb13337321b1b6cfa2` at `receipts/engine/model_admitted/SFT-COMP-ALGX-MATCHING-ASSIGNMENT-015-f508ce7800234e1a.json`.

`SFT-COMP-OBL-ALGX-016` - Algebraic elimination and exact linear solving

- **Claim and ownership:** `SFT-COMP-ALGX-ALGEBRAIC-SOLVE-016`; Classical Computation family `ALGX`; Algebraic elimination and exact linear solving.
- **Forced law:** Exact algebraic solving enumerates or eliminates over generated exact parts while preserving every equation; a unique solution is admitted only when it satisfies the full system and all alternatives are eliminated.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__exact-equation-solution-census__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-016 uniquely retains exact-equation-solution-census, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001 -> SFT-COMP-ALGX-MATCHING-ASSIGNMENT-015. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-016: exact_linear_solving; exact execution: {"equations":["x+y=2","x+2y=3"],"positive_candidates_per_coordinate":3,"unique_solution":{"x":"1/1","y":"1/1"}}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:5496c25f4cc439f61b44d9e3394770851e0ec8cf859cad1413b28056a06f38a3`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:dfa51c5f11d881c866ec46fc9050aa71a452ec7593f7f74f0ea47ae4456f82ab`; engine receipt `sha256:f960a59398cee625bef0c860554d09baa09d7c1e5e0077072467f753ab1c70ec` at `receipts/engine/model_admitted/SFT-COMP-ALGX-ALGEBRAIC-SOLVE-016-f960a59398cee625.json`.

`SFT-COMP-OBL-ALGX-017` - Polynomial and symbolic algebra algorithms

- **Claim and ownership:** `SFT-COMP-ALGX-POLYNOMIAL-SYMBOLIC-017`; Classical Computation family `ALGX`; Polynomial and symbolic algebra algorithms.
- **Forced law:** Symbolic polynomial computation retains canonical exponent-to-coefficient records; addition merges equal exponents and multiplication generates every cross-term before canonical collection.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__canonical-polynomial-coefficient-map__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-ce`

rtificate-or-explicit-handoff. Exact result: ALGX-017 uniquely retains canonical-polynomial-coefficient-map, complete correctness custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001 -> SFT-COMP-ALGX-ALGEBRAIC-SOLVE-016. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** ALGX-017: symbolic_polynomial; exact execution: {"left":"x+1","product_coefficients":{"degree-0":1,"degree-1":2,"degree-2":1},"right":"x+1"}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:9af39151f06239689720912cf9a2e6528a44ed8fc46bcfe430028015aa1bc231; independent implementation sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606; independent certificate sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa; empirical/external sha256:ce0cce447cfda29f92be72c380999e781e6dead32acc4b57aebaf966e0fb0349; engine receipt sha256:0af11b5c6ee29033e717f8237d168a0a6d1bf0840b960685207197e6057d9c29 at receipts/engine/model_admitted/SFT-COMP-ALGX-POLYNOMIAL-SYMBOLIC-017-0af11b5c6ee29033.json.

SFT-COMP-OBL-ALGX-018 - Computational geometry orientation and intersection

- **Claim and ownership:** SFT-COMP-ALGX-GEOMETRIC-ORIENTATION-018; Classical Computation family ALGX; Computational geometry orientation and intersection.
- **Forced law:** Geometric orientation compares the two exact positive determinant sums and returns held left, right or aligned labels; intersection retains endpoint order and never imports signed continuum coordinates.
- **Unique surviving structure:** complete-generated-input__invariant-trace-process__held-orientation-intersection-ledger__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff. Exact result: ALGX-018 uniquely retains held-orientation-intersection-ledger, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001 -> SFT-COMP-ALGX-POLYNOMIAL-SYMBOLIC-017. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** ALGX-018: geometry_orientation; exact execution: {"aligned_example":"aligned","left_example":"left","signed_scalar_used":false}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden

branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:40635821577bed3dc78272ddeea6472748893a0ba8b1453b050011d330fdefa3`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:b4ab5f8a6bce55e2e687c43bf1cfa4338f84b9e8a727258c2949536c7b209219`; engine receipt `sha256:1746097e1dfcee054d9d781b7703c49209c85135683d56487f9a60a7f1ea74d1` at `receipts/engine/model_admitted/SFT-COMP-ALGX-GEOMETRIC-ORIENTATION-018-1746097e1dfcee05.json`.

`SFT-COMP-OBL-ALGX-019` - Convex-hull and spatial-order algorithms

- **Claim and ownership:** `SFT-COMP-ALGX-CONVEX-HULL-019`; Classical Computation family `ALGX`; Convex-hull and spatial-order algorithms.
- **Forced law:** A convex boundary consists exactly of directed point pairs whose remaining generated support lies on at most one held side; spatial ordering retains every boundary tie and interior exclusion witness.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__supporting-edge-spatial-boundary__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-019 uniquely retains supporting-edge-spatial-boundary, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-ALGEBRAIC-GEOMETRIC-001 -> SFT-COMP-ALGX-GEOMETRIC-ORIENTATION-018. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-019: convex_hull; exact execution: {"directed_boundary_edges":8,"interior_points":1,"points":5,"undirected_boundary_edges":4}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:30297903c1c00446b058690b99db09e0bda872774e5d89689c7170e372c2af0e`; independent

implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:ab74ee88972820e36fb78f681e9ae769c36f0f320c7be95278dc682f3f6d32e2`; engine receipt `sha256:f35c33680cf31cc79c4ff6d89224667c03f3c9d49f95f82cff95528a526c0b5a` at `receipts/engine/model_admitted/SFT-COMP-ALGX-CONVEX-HULL-019-f35c33680cf31cc7.json`.

`SFT-COMP-OBL-ALGX-020` - Dynamic-programming optimal-substructure law

- **Claim and ownership:** `SFT-COMP-ALGX-DYNAMIC-PROGRAMMING-020`; Classical Computation family `ALGX`; Dynamic-programming optimal-substructure law.
- **Forced law:** Dynamic programming is forced when subproblems recur with identical canonical identity: each is solved once, every dependency is retained, and the terminal table composes the exact optimal or counting result.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__overlap-reconciled-subproblem-table__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-020 uniquely retains overlap-reconciled-subproblem-table, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-DYNAMIC-PROGRAMMING-001 -> SFT-COMP-ALGX-CONVEX-HULL-019. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-020: dynamic_programming; exact execution: {"exact_tiling_count":13,"subproblem_rows":7,"tiling_width":6}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:0bf55791706f42f4a6e844eec7a1f9574379fab1ae40bff098472cff11508fc6`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:d76497dcf31290e32fc4c909af5e3c93238de209b60379c800b6901378d7426b`; engine receipt `sha256:83155df4a73403027cf9d4e77931b849978ec7215c40bd8e13c5a5bb0fa2641d` at `receipts/engine/model_admitted/SFT-COMP-ALGX-DYNAMIC-PROGRAMMING-020-83155df4a7340302.json`.

`SFT-COMP-OBL-ALGX-021` - Greedy-choice admissibility boundary

- **Claim and ownership:** `SFT-COMP-ALGX-GREEDY-BOUNDARY-021`; Classical Computation family `ALGX`; Greedy-choice admissibility boundary.
- **Forced law:** A greedy choice is admissible only when a complete exchange certificate transforms every optimum into one containing the chosen element without worsening the exact objective; otherwise the greedy route remains unclosed.

- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__exchange-certified-greedy-choice__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-021 uniquely retains exchange-certified-greedy-choice, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-OPTIMIZATION-001 -> SFT-COMP-ALGX-DYNAMIC-PROGRAMMING-020. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-021: greedy_boundary; exact execution: {"exchange_checked":true,"intervals":4,"selected_count":3,"selected_intervals":[[1,3],[3,4],[4,6]]}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:3401a250ebcdf55fd21a077793cdff3d6b370534513a23cdfb5e10b925806865`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:d288e4b81541ff4bc6c7fbf3e362fed33676bcab1d9fc3d8eddb59ae22a9e50c`; engine receipt `sha256:b651accbad18cf0e4312bc77e789e9057dd7a0ea2740cd1baa5bdf31683c9411` at `receipts/engine/model_admitted/SFT-COMP-ALGX-GREEDY-BOUNDARY-021-b651accbad18cf0e.json`.

`SFT-COMP-OBL-ALGX-022` - Combinatorial optimization and branch-and-bound

- **Claim and ownership:** `SFT-COMP-ALGX-BRANCH-BOUND-022`; Classical Computation family `ALGX`; Combinatorial optimization and branch-and-bound.
- **Forced law:** Branch-and-bound generates every feasible choice branch unless an exact retained bound proves no descendant can improve the incumbent; optimality is the surviving best value plus complete pruning certificates.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__complete-feasible-subset-bound__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-022 uniquely retains complete-feasible-subset-bound, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-OPTIMIZATION-001 -> SFT-COMP-ALGX-GREEDY-BOUNDARY-021. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** ALGX-022: branch_and_bound; exact execution: {"all_feasible_subsets_retained":true,"capacity":5,"items":3,"optimal_value":7}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:109e3d64e179ec2e1ce18f028c3f2ed8ea5b9a50c94b9adbf966a1f635e47c33`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:aa27f870fc5aeddce26ba255bff0685bc14fd351fd788c77efff4fa264b7feaf`; engine receipt `sha256:5a5d109a8c5ef478bf8b0a1ede52e69c6a5042f4e5d39225908211589eaffd8e` at `receipts/engine/model_admitted/SFT-COMP-ALGX-BRANCH-BOUND-022-5a5d109a8c5ef478.json`.

`SFT-COMP-OBL-ALGX-023` - Randomized algorithm complete-support execution

- **Claim and ownership:** `SFT-COMP-ALGX-RANDOMIZED-SUPPORT-023`; Classical Computation family `ALGX`; Randomized algorithm complete-support execution.
- **Forced law:** A randomized algorithm is executed as the complete deterministic support of its generated branch labels; outcomes, resources and exact parts are retained branchwise and no stochastic cause selects a path.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__complete-deterministic-random-support__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-023 uniquely retains complete-deterministic-random-support, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-RANDOMIZED-001 -> SFT-COMP-ALGX-BRANCH-BOUND-022. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-023: randomized_support; exact execution: {"branch_labels":2,"complete_branches":8,"depth":3,"stochastic_cause_used":false}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:8906438f3b5cd8eb3eb6f2b8330c667decf0228039179248171252deb8367842`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:0a07f61a693c43b5e4d88a5dcd2746b72c52028cf0aa0f4778b73131e97f0da4`; engine receipt `sha256:70966a9f35449066307818f149845c0eb686985d4328883283390b80996857d0` at `receipts/engine/model_admitted/SFT-COMP-ALGX-RANDOMIZED-SUPPORT-023-70966a9f35449066.json`.

`SFT-COMP-OBL-ALGX-024` - Parallel work-depth algorithm law

- **Claim and ownership:** `SFT-COMP-ALGX-PARALLEL-WORK-DEPTH-024`; Classical Computation family `ALGX`; Parallel work-depth algorithm law.
- **Forced law:** A parallel algorithm retains causal layers, operations per layer, total work and communication separately; correctness equals the sequential specification only when every dependency and merge is preserved.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__parallel-layer-work-depth__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-024 uniquely retains parallel-layer-work-depth, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-PARALLEL-001 -> SFT-COMP-ALGX-RANDOMIZED-SUPPORT-023. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-024: parallel_algorithm; exact execution: {"input_width":8,"layer_widths":[8,4,2,1],"work_depth_separate":true}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:88a6a33c37d80f97ebb112cf9f2b509e6db6cd62783acecc156c556683797173`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:582e89aa6e1c7b2edf789e50b8983f3cf6c851a6139af1c47d42ae0de0770d6b`; engine receipt `sha256:2926e2fa0084479f138e46819e36fe57faaec786a5001ca610c8f7ff91530213` at `receipts/engine/model_admitted/SFT-COMP-ALGX-PARALLEL-WORK-DEPTH-024-2926e2fa0084479f.json`.

`SFT-COMP-OBL-ALGX-025` - Distributed local-state algorithm law

- **Claim and ownership:** `SFT-COMP-ALGX-DISTRIBUTED-LOCAL-025`; Classical Computation family `ALGX`; Distributed local-state algorithm law.

- **Forced law:** A distributed algorithm advances from exact local state and received messages; each round retains sender, receiver, payload and causal order, and a global conclusion requires reconstruction from all declared local traces.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__round-indexed-local-state__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-025 uniquely retains round-indexed-local-state, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-DISTRIBUTED-001 -> SFT-COMP-ALGX-PARALLEL-WORK-DEPTH-024. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-025: distributed_algorithm; exact execution: {"round_support_sizes":[1,2,3,4],"source":"a","terminal_knowledge":["a","b","c","d"]}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:694f34a87c4e11a175c251363e29d387640f6c77affe47885fefde4b2860a049`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:ca2c5e253d62fd67d3f2beaef35db9d883aa6da4f7a2ee57fb298fa26582cddb`; engine receipt `sha256:176b072112610b2f4d138a97978cab490064602572f1c46e26515e2affd6840c` at `receipts/engine/model_admitted/SFT-COMP-ALGX-DISTRIBUTED-LOCAL-025-176b072112610b2f.json`.

`SFT-COMP-OBL-ALGX-026` - Online decision and competitive ledger

- **Claim and ownership:** `SFT-COMP-ALGX-ONLINE-COMPETITIVE-026`; Classical Computation family `ALGX`; Online decision and competitive ledger.
- **Forced law:** An online algorithm uses only the retained prefix available at each irrevocable decision; its competitive guarantee is an exact oriented part against an independently generated full-input optimum over complete adversarial support.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__prefix-only-competitive-ledger__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-026 uniquely retains prefix-only-competitive-ledger, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 ->

SFT-COMP-ALG-ONLINE-STREAMING-001 -> SFT-COMP-ALGX-DISTRIBUTED-LOCAL-025. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** ALGX-026: online_algorithm; exact execution: {"exact_competitive_part":"3/2","future_access_used":false,"offline_optimum":4,"online_cost":6}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:54f504196a635bde96cee1233a0933ff81fb66be9da0343af048c2b139280640`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:62b1bcd724a4e073faa20ae99f86effaf99de84bc969acfb1afe8abfe2e8b746`; engine receipt `sha256:4b79f08cea3b5df1eef3bd3e01ec15dd28a182b8a4eb92ee4d15834f30044489` at `receipts/engine/model_admitted/SFT-COMP-ALGX-ONLINE-COMPETITIVE-026-4b79f08cea3b5df1.json`.

`SFT-COMP-OBL-ALGX-027` - Streaming memory and approximation ledger

- **Claim and ownership:** `SFT-COMP-ALGX-STREAMING-MEMORY-027`; Classical Computation family `ALGX`; Streaming memory and approximation ledger.
- **Forced law:** A streaming algorithm updates one bounded retained summary per input item; exact or approximate answers require an explicit memory ledger, merge law, error enclosure and complete stream-order boundary.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__bounded-memory-stream-summary__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-027 uniquely retains bounded-memory-stream-summary, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-ONLINE-STREAMING-001 -> SFT-COMP-ALGX-ONLINE-COMPETITIVE-026. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-027: streaming_algorithm; exact execution: {"distinct_support":["a","b","c"],"stream":["a","b","a","c"],"summary_width":3}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library

implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:f1aea1e080fc5d3c7970613f4f32aee62474d7ee82e13d479f94aad00d84ac8b`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:0a8c78b273e1b549a01821b382bbf65e2cd94c8fb86e68dcf6e71c10a241a71e`; engine receipt `sha256:fd34848982a108037ca3be62f1205b7081a01c391a1d075c7ba5f9b60cef754d` at `receipts/engine/model_admitted/SFT-COMP-ALGX-STREAMING-MEMORY-027-fd34848982a10803.json`.

`SFT-COMP-OBL-ALGX-028` - Numerical iteration with exact error custody

- **Claim and ownership:** `SFT-COMP-ALGX-NUMERICAL-ERROR-028`; Classical Computation family `ALGX`; Numerical iteration with exact error custody.
- **Forced law:** Numerical iteration remains admissible only over exact rational parts or symbolic enclosures; every step retains truncation, residual and propagated-error records and halts when the registered exact enclosure condition is met.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__exact-rational-error-enclosure__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-028 uniquely retains exact-rational-error-enclosure, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-NUMERICAL-001 -> SFT-COMP-ALGX-STREAMING-MEMORY-027. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-028: numerical_iteration; exact execution: {"all_values_rational":true,"floating_used":false,"limit":"1/1","start":"3/1","steps":3,"terminal":"5/4"}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:aac9ef30631a840229d0e396818a355daf2cd1f9d69f474486adbc12c9f6aee9`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:d1a70a28d2aec9b58503ceb2c59bc75215453a7c11da42a3ac1119e85e0dfd53`; engine receipt `sha256:9e090d95bf38a8e080707a935673ec0c7edea4c3c95ceef7a0c47ab2d5eec6e8` at `receipts/engine/model_admitted/SFT-COMP-ALGX-NUMERICAL-ERROR-028-9e090d95bf38a8e0.json`.

SFT-COMP-OBL-ALGX-029 - Symbolic simplification and equivalence custody

- **Claim and ownership:** SFT-COMP-ALGX-SYMBOLIC-SIMPLIFY-029; Classical Computation family ALGX; Symbolic simplification and equivalence custody.
- **Forced law:** Symbolic simplification is a terminating provenance-retaining rewrite to an equivalent canonical expression; every rule preserves denotation and confluence retains one normal form or every unresolved alternative.
- **Unique surviving structure:** complete-generated-input__invariant-trace-process__terminating-equivalence-rewrite__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff. Exact result: ALGX-029 uniquely retains terminating-equivalence-rewrite, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-SYMBOLIC-001 -> SFT-COMP-ALGX-NUMERICAL-ERROR-028. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** ALGX-029: symbolic_simplification; exact execution: {"equivalence_preserved":true,"normal_form":"a","source":"join(a,a)","terminates":true}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:f7273d3c3fedbcd8dc67809505f91682eb51eb02d66da348f76f36d3f43953dd; independent implementation sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606; independent certificate sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa; empirical/external sha256:92ec2a27a012ade5104407983f522388ebad994f6065a0eff34f8b1823e88d57; engine receipt sha256:4eedc31ec7fa86ec449aa4d61561831d7647943182e18d8ab437f45329f94a78 at receipts/engine/model_admitted/SFT-COMP-ALGX-SYMBOLIC-SIMPLIFY-029-4eedc31ec7fa86ec.json.

SFT-COMP-OBL-ALGX-030 - Approximation scheme and guarantee certificate

- **Claim and ownership:** SFT-COMP-ALGX-APPROXIMATION-SCHEME-030; Classical Computation family ALGX; Approximation scheme and guarantee certificate.
- **Forced law:** An approximation scheme generates a result for each retained accuracy part, certifies its exact relation to an independently forced optimum and states its resource dependence without floating tolerances or fitted thresholds.
- **Unique surviving structure:** complete-generated-input__invariant-trace-process__exact-guarantee-part__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff. Exact result: ALGX-030 uniquely retains exact-guarantee-part, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-APPROXIMATE-001 -> SFT-COMP-ALGX-SYMBOLIC-SIMPLIFY-029. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-030: approximation_scheme; exact execution: {"candidate_guarantee":"7/8","floating_tolerance_used":false,"guarantee_met":true,"registered_requirement":"3/4"}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:e9facf7f5970b8c318cebdb1c518df764faed802f79974783638e8ea1f175a06`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:609f80c4bf6b9defef09a01b62e75779d0fa18e1918b80c9e8c18b30d4448503`; engine receipt `sha256:e837401c4a85055e82fed3fc6ef60c3702d6cdf2856848f2e43da01229125e26` at `receipts/engine/model_admitted/SFT-COMP-ALGX-APPROXIMATION-SCHEME-030-e837401c4a85055e.json`.

`SFT-COMP-OBL-ALGX-031` - Algorithm and data-organization completeness certificate

- **Claim and ownership:** `SFT-COMP-ALGX-COMPLETENESS-031`; Classical Computation family `ALGX`; Algorithm and data-organization completeness certificate.
- **Forced law:** Algorithm and data-organization completeness is the one-to-one reconciliation of all thirty-one frozen obligations with unique survivors, adverse controls, exact executions, independent reconstructions and untouched-engine receipts.
- **Unique surviving structure:** `complete-generated-input__invariant-trace-process__thirty-one-obligation-no-omission-ledger__complete-correctness-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-algorithm-execution__successor-certificate-or-explicit-handoff`. Exact result: ALGX-031 uniquely retains thirty-one-obligation-no-omission-ledger, complete correctness custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-ALG-APPROXIMATE-001 -> SFT-COMP-ALGX-APPROXIMATION-SCHEME-030. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** ALGX-031: algorithm_completeness; exact execution: {"duplicate_owners":0,"independent_execution_rows":31,"omitted_owners":0,"registered_obligations":31}; complete family observation custody: 31 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-ALGORITHM-OBSERVER, SFT-V1-V2-ALGORITHM-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported algorithm theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden branch, sampled input support, unregistered oracle or stochastic cause; no library implementation or favorable benchmark substitutes for the mathematical algorithm; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:00f8af8ab2a9e4a2d4fc0779d54733485a01c4ecaf28955b79d9b42bec7af2ca`; independent implementation `sha256:5ec8e7e6cf4fd8dfcab8f08b3fd1701c9cc59695532a7b75ec8027302c82a606`; independent certificate `sha256:c4369279bafa18ea994005f652442e6f97d5f6e6e277de9f50b5af9f0105abaa`; empirical/external `sha256:cf9ab04b5cfb665ed92d5ea7411bb033e41666d80e7c3dafa501060d83205760`; engine receipt `sha256:183e2469566102a9df47110d8430f40aa28e1917d6650f6f8649da02614c1bdc` at `receipts/engine/model_admitted/SFT-COMP-ALGX-COMPLETENESS-031-183e2469566102a9.json`.

130.8 Family `SEMX` - 25/25 complete

Scope. Semantics and programming theory. This family completely decides 6,400 candidates, retains 25 unique survivors and passes 100 adverse controls.

`SFT-COMP-OBL-SEMX-001` - Abstract syntax trees and well-formed program terms

- **Claim and ownership:** `SFT-COMP-SEMX-AST-WELL-FORMED-001`; Classical Computation family `SEMX`; Abstract syntax trees and well-formed program terms.

- **Forced law:** Program syntax is the complete generated tree of constructor labels and child positions; a term is well formed exactly when every constructor has its declared arity and every name occurrence is lawful at its scope.

- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__canonical-well-formed-syntax__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-001 uniquely retains canonical-well-formed-syntax, complete semantic custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-SYNTAX-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SEMX-001: ast_well_formed; exact execution: {"closed_term":"let x=a in join(x,b)","constructor_rows":4,"unbound_name_rejected":true}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:7c123fb7e934d7adaa70902961fa73927f5afed37177b2579255ae5e55b76b41`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:57d6caf550bb68fc3703919fad130a2138352a2a42dc3afe7225154895d80f40`; engine receipt `sha256:f9937dfa65924f7a8ea2f1189be92d03892a5554d26f201352894a02b11761d6` at `receipts/engine/model_admitted/SFT-COMP-SEMX-AST-WELL-FORMED-001-f9937dfa65924f7a.json`.

`SFT-COMP-OBL-SEMX-002` - Free, bound and scoped-name distinction

- **Claim and ownership:** `SFT-COMP-SEMX-FREE-BOUND-SCOPE-002`; Classical Computation family `SEMX`; Free, bound and scoped-name distinction.
- **Forced law:** A name occurrence is bound only by its nearest retained binder path, free when no such path exists, and ill scoped when a closed program contains it without a binder record.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__free-bound-scope-ledger__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-002 uniquely retains free-bound-scope-ledger, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-BINDING-SUBSTITUTION-001 -> SFT-COMP-SEMX-AST-WELL-FORMED-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-002: free_bound_scope; exact execution: {"bound_name":"x","free_support":["y"],"nearest_binder_retained":true}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:57f8d3d7ce1ee525c5d306094e45bfcba670649cf72b406ef8ed7df766d0cf4a`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:1fc37590f469c4f6138e42812f3cb1c5904fca8fd85937cbed386c8f110f2588`; engine receipt `sha256:5b0d956dd758d1728213b2b4e68eb1751d10c3a30c72f4e150359b9532f6a705` at `receipts/engine/model_admitted/SFT-COMP-SEMX-FREE-BOUND-SCOPE-002-5b0d956dd758d172.json`.

`SFT-COMP-OBL-SEMX-003` - Alpha-equivalence and lawful renaming

- **Claim and ownership:** `SFT-COMP-SEMX-ALPHA-RENAMING-003`; Classical Computation family `SEMX`; Alpha-equivalence and lawful renaming.
- **Forced law:** Alpha-equivalence is the bijective renaming of a binder and exactly its bound occurrences to a fresh label; free names, constructor structure and evaluation remain unchanged.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__alpha-renaming-equivalence__independently-checkable-judgment__literal-c`

omplete-product__there-is-no-nothing-lineage__post-registry-exact-program-executio n__declared-language-or-explicit-translation. Exact result: SEMX-003 uniquely retains alpha-renaming-equivalence, complete semantic custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-BINDING-SUBSTITUTION-001 -> SFT-COMP-SEMX-FREE-BOUND-SCOPE-002. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** SEMX-003: alpha_equivalence; exact execution: {"bound_occurrences_renamed":1,"free_names_changed":false,"fresh_binder":"u","source_binder":"x"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:8b3959ae01fc89072d63b5c9c0c9971a64d18b65821b124a5e117ba135953cc3; independent implementation sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407; independent certificate sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70; empirical/external sha256:6a7af669837d80ad9cd1400b5669b8a0591b2570ecd8b00b24c7d9d6cfbb85e1; engine receipt sha256:516639864237c32835cffdf9bb17be08545ed86aeb5fd7573bc622c18976c4dc at receipts/engine/model_admitted/SFT-COMP-SEMX-ALPHA-RENAMING-003-516639864237c328.json.

SFT-COMP-OBL-SEMX-004 - Capture-avoiding substitution

- **Claim and ownership:** SFT-COMP-SEMX-CAPTURE-AVOIDING-SUBSTITUTION-004; Classical Computation family SEMX; Capture-avoiding substitution.
- **Forced law:** Substitution replaces exactly free occurrences and renames any conflicting binder before descent, thereby preserving name identity without capture or silent aliasing.
- **Unique surviving structure:** complete-well-formed-term__complete-operational-and-compositio nal-trace__capture-avoiding-substitution__independently-checkable-judgment__litera l-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execu tion__declared-language-or-explicit-translation. Exact result: SEMX-004 uniquely retains capture-avoiding-substitution, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-BINDING-SUBSTITUTION-001 -> SFT-COMP-SEMX-ALPHA-RENAMING-003. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** SEMX-004: capture_avoiding_substitution; exact execution: {"capture_occurred":false,"conflicting_binder_renamed":"u","replacement_free_name":"y","substituted_name":"x"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the

observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:4f4da96079db3299e63d3453c05d789601fecb2a5ee9e7091ce3c92e283dd289`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:7180f8bbceff49f28db4ab463478e494afd93148dca903b35edbeb6699eeba1a`; engine receipt `sha256:1536ffef25ca51f5f5e4daec28011fbaad98577f6578320f2119256446fc8bf1` at `receipts/engine/model_admitted/SFT-COMP-SEMX-CAPTURE-AVOIDING-SUBSTITUTION-004-1536ffef25ca51f5.json`.

`SFT-COMP-OBL-SEMX-005` - Small-step operational evaluation

- **Claim and ownership:** `SFT-COMP-SEMX-SMALL-STEP-005`; Classical Computation family `SEMX`; Small-step operational evaluation.
- **Forced law:** Small-step semantics is the source-bound relation selecting the unique declared evaluation context and one lawful redex contraction while retaining every intermediate term.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__one-step-contextual-reduction__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-005 uniquely retains one-step-contextual-reduction, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-EVALUATION-001 -> SFT-COMP-SEMX-CAPTURE-AVOIDING-SUBSTITUTION-004. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-005: small_step; exact execution: {"intermediates_retained":2,"source":"join(a,b)","terminal_word":["a","b"],"transition_count":1}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:2ad408eb79749c2b73185ecc7997f72abd82c7b612ec77cee7b4f2f41e157adf`; independent

implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:d888b9fb8da43cb6c424c6968a4b266917594eb8713240e40484643d6ed14ea0`; engine receipt `sha256:8fa06f93a05cc64a2b93e882b1f177e1edc2082db91afcdfc1587480765623b9` at `receipts/engine/model_admitted/SFT-COMP-SEMX-SMALL-STEP-005-8fa06f93a05cc64a.json`.

`SFT-COMP-OBL-SEMX-006` - Big-step operational evaluation

- **Claim and ownership:** `SFT-COMP-SEMX-BIG-STEP-006`; Classical Computation family `SEMX`; Big-step operational evaluation.
- **Forced law:** Big-step semantics is the complete derivation from one program and environment to its terminal value, with every premise result retained in the derivation tree.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__whole-derivation-evaluation__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-006 uniquely retains whole-derivation-evaluation, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-EVALUATION-001 -> SFT-COMP-SEMX-SMALL-STEP-005. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-006: big_step; exact execution: {"environment_binding_retained":true,"source":"let x=a in join(x,b)","terminal_word":["a","b"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:4493756a7839a733b0eb9aa0f45750cabd493b988293f8b5e76e8b44a7bb10b5`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:09e662278be976bfa4e04ff9a04e5ba0c4315791fae911673da6b136565af64c`; engine receipt `sha256:17707d9d4ffe36a0157f4aacbd46f2e07283520bddecb0a0e16995d0e82d693d` at `receipts/engine/model_admitted/SFT-COMP-SEMX-BIG-STEP-006-17707d9d4ffe36a0.json`.

`SFT-COMP-OBL-SEMX-007` - Evaluation-context composition

- **Claim and ownership:** `SFT-COMP-SEMX-EVALUATION-CONTEXT-007`; Classical Computation family `SEMX`; Evaluation-context composition.
- **Forced law:** An evaluation context is a generated term with one retained hole; filling and context composition preserve that unique position and determine the next lawful reduction boundary.

- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__single-hole-context-composition__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-007 uniquely retains single-hole-context-composition, complete semantic custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-EVALUATION-001 -> SFT-COMP-SEMX-BIG-STEP-006. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SEMX-007: evaluation_context; exact execution: {"context":"join(hole,b)","filled_term":"join(a,b)","hole_count":1,"terminal_word":["a","b"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:8c6215febf6545a05a1ad8188782f82ea593d0a165ae1fe105536b20657cdefd`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:d0e36232dad19275d8a71df5a45630ae6203bd1948402c7479ff792034234538`; engine receipt `sha256:c5461e1574d94896c0637f5ac18f3603e2f437bb732f81cf6f35a1b219092ec1` at `receipts/engine/model_admitted/SFT-COMP-SEMX-EVALUATION-CONTEXT-007-c5461e1574d94896.json`.

`SFT-COMP-OBL-SEMX-008` - Denotational meaning as compositional Fold map

- **Claim and ownership:** `SFT-COMP-SEMX-DENOTATIONAL-COMPOSITION-008`; Classical Computation family `SEMX`; Denotational meaning as compositional Fold map.

- **Forced law:** Denotational meaning is a compositional Fold map assigning each constructor the exact composition of its child meanings and retaining the environment interpretation of names.

- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__compositional-fold-denotation__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-008 uniquely retains compositional-fold-denotation, complete semantic custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001 -> SFT-COMP-SEMX-EVALUATION-CONTEXT-007. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SEMX-008: denotational_composition; exact execution: {"composed_meaning":["a","b"],"left_meaning":["a"],"right_meaning":["b"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:9150998b0dae970bea9014c2c29bdc730320a4805a2fe3fadb802e24284c2378`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:b9ab89188a616e88e8ead62b4cbb939889b377ad8da38511f5efb2623d582279`; engine receipt `sha256:161a065120e382997cd400a6e7ece0719d31ad8c41185c32349db9a11e6d3231` at `receipts/engine/model_admitted/SFT-COMP-SEMX-DENOTATIONAL-COMPOSITION-008-161a065120e38299.json`.

`SFT-COMP-OBL-SEMX-009` - Operational-denotational adequacy

- **Claim and ownership:** `SFT-COMP-SEMX-OPERATIONAL-DENOTATIONAL-ADEQUACY-009`; Classical Computation family `SEMX`; Operational-denotational adequacy.
- **Forced law:** Operational-denotational adequacy holds when every terminating operational trace yields the denotation and every denotational terminal value has a reconstructible operational derivation within the declared language.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__operational-denotational-adequacy__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-009 uniquely retains operational-denotational-adequacy, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-OPERATIONAL-DENOTATIONAL-001 -> SFT-COMP-SEMX-DENOTATIONAL-COMPOSITION-008. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-009: operational_denotational_adequacy; exact execution: {"denotational_value":["a","b"],"directions_checked":2,"operational_value":["a","b"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route

retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:24d3e8092562f0ae6982821b490252d73aaa3d5c7e9b3a14eaf09f679b6672f3`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:3690d5aacd053b4413bbca5877ade9cbe5cd8a5788ef20e150948a26d42dd247`; engine receipt `sha256:16c5aa07ecd2b339aad20d57f2e7c2746b1870afe4ee1053c5b010dd0be3037c` at `receipts/engine/model_admitted/SFT-COMP-SEMX-OPERATIONAL-DENOTATIONAL-ADEQUACY-009-16c5aa07ecd2b339.json`.

`SFT-COMP-OBL-SEMX-010` - Full-abstraction correspondence boundary

- **Claim and ownership:** `SFT-COMP-SEMX-FULL-ABSTRACTION-BOUNDARY-010`; Classical Computation family `SEMX`; Full-abstraction correspondence boundary.
- **Forced law:** Full abstraction is admitted only for a complete registered context grammar: two terms are equivalent exactly when their denotations agree precisely when every generated context produces the same observation.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__declared-context-full-abstraction__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-010 uniquely retains declared-context-full-abstraction, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001 -> SFT-COMP-SEMX-OPERATIONAL-DENOTATIONAL-ADEQUACY-009. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-010: full_abstraction_boundary; exact execution: {"export_unrestricted":false,"observations_equal":true,"registered_contexts":["hole","join(hole,a)"],"terms":["join(a,b)","ab"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:22ed0843c2b038f4bbbbda6f8daa47586ca12855d16122ba9b4e771755b1b2e4`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:bfd6971e0c5b12a218e3a45d60472bf125b6689c719dfeecb19ef8afd8e43e82`; engine receipt `sha256:486d14064cfb1261819ac89b7f8de2c7d0acbb564fc8024e7345d6169acd0354` at `receipts/engine/model_admitted/SFT-COMP-SEMX-FULL-ABSTRACTION-BOUNDARY-010-486d14064cfb1261.json`.

SFT-COMP-OBL-SEMX-011 - Type formation, introduction and elimination

- **Claim and ownership:** SFT-COMP-SEMX-TYPE-RULES-011; Classical Computation family SEMX; Type formation, introduction and elimination.
- **Forced law:** A type system is generated by explicit formation, introduction and elimination relations; every judgment retains its term, context, type and complete premise tree.
- **Unique surviving structure:** complete-well-formed-term__complete-operational-and-compositional-trace__type-formation-introduction-elimination__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation. Exact result: SEMX-011 uniquely retains type-formation-introduction-elimination, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-TYPE-001 -> SFT-COMP-SEMX-FULL-ABSTRACTION-BOUNDARY-010. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** SEMX-011: type_rules; exact execution: {"inferred_type":"word","premise_tree_retained":true,"term":"let x=a in join(x,b)"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:12bc3e5c36209566469115b55e903e74c60c736b503053dd60826b1a3b246ba5; independent implementation sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407; independent certificate sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70; empirical/external sha256:deb66664898b934f2b9aebcf3023c7f825204fa640659ac862a6bd5cc3599509; engine receipt sha256:65b9f5ff0aab1c7f63eeb73273e56a2b50b130833e33fbe0b69f7c3f10e8e77d at receipts/engine/model_admitted/SFT-COMP-SEMX-TYPE-RULES-011-65b9f5ff0aab1c7f.json.

SFT-COMP-OBL-SEMX-012 - Type checking and type inference correspondence

- **Claim and ownership:** SFT-COMP-SEMX-TYPE-INFERENCE-012; Classical Computation family SEMX; Type checking and type inference correspondence.
- **Forced law:** Type checking verifies a supplied type judgment, while inference generates the unique type or all retained alternatives from syntax-directed rules; unresolved or conflicting constraints halt.
- **Unique surviving structure:** complete-well-formed-term__complete-operational-and-compositional-trace__syntax-directed-type-judgment__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation. Exact result: SEMX-012 uniquely retains syntax-directed-type-judgment, complete semantic custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-TYPE-001 -> SFT-COMP-SEMX-TYPE-RULES-011. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-012: type_inference; exact execution: {"branch_types_equal":true,"inferred_type":"word","term":"same(a,a,b,a)"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:0b6c159ff6b2108ec0c9791710586a2b0fece361ed80640ed048f75ec6c99248`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:8501b7df69370accca8617b4888b912b1c61a06b001a9302fcf52fbc2503d186`; engine receipt `sha256:8fe747263cf1a16065934e787e315a030bb431bae39d32b3bc2d3b960a520636` at `receipts/engine/model_admitted/SFT-COMP-SEMX-TYPE-INFERENCE-012-8fe747263cf1a160.json`.

`SFT-COMP-OBL-SEMX-013` - Polymorphism and parametricity boundary

- **Claim and ownership:** `SFT-COMP-SEMX-POLYMORPHISM-PARAMETRICITY-013`; Classical Computation family `SEMX`; Polymorphism and parametricity boundary.
- **Forced law:** Parametric behavior is forced only when one term acts uniformly over every generated type fibre and cannot inspect an unprovided representation distinction.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__representation-independent-parametricity__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-013 uniquely retains representation-independent-parametricity, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-TYPE-001 -> SFT-COMP-SEMX-TYPE-INFERENCE-012. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-013: parametric_identity; exact execution: {"identity_results_preserved":3,"representation_inspection":false,"type_fibres":["word","vector","pair"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:bff71fd15c9bf7378f7397d478c092056bfed2cab049f4873a7bc8ba5af03301`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:f21e4647e6217ec02c6f766fb3d2223a570bd509406ba77c093f6c4cf6d2894d`; engine receipt `sha256:35f9a63bdc416323e4b251126fb3b9883d382222321e1f892790ef7e23e61a65` at `receipts/engine/model_admitted/SFT-COMP-SEMX-POLYMORPHISM-PARAMETRICITY-013-35f9a63bdc416323.json`.

`SFT-COMP-OBL-SEMX-014` - Dependent evidence and proposition-as-type correspondence

- **Claim and ownership:** `SFT-COMP-SEMX-DEPENDENT-EVIDENCE-014`; Classical Computation family `SEMX`; Dependent evidence and proposition-as-type correspondence.
- **Forced law:** Dependent evidence retains an index in the type carrier and a witness whose generated structure matches that index; propositions correspond to types only at this explicit evidence boundary.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__type-indexed-evidence-carrier__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-014 uniquely retains type-indexed-evidence-carrier, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-TYPE-001 -> SFT-COMP-SEMX-POLYMORPHISM-PARAMETRICITY-013. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-014: dependent_evidence; exact execution: {"mismatched_index_rejected":true,"retained_index":3,"witness_width":3}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:23560c2af2f43dbc35adc499328f8ae7d5893f1a16eeff012b31a4a167e46783`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate

sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70; empirical/external sha256:c2da9f60fe90372b5594645b01f089ed217bf3368a4e809206fcc2d1a44feab1; engine receipt sha256:6c4c0457213d3fbca0f5c60c4a7bc54a65a214c0395d37e159509bd2799d9549 at receipts/engine/model_admitted/SFT-COMP-SEMX-DEPENDENT-EVIDENCE-014-6c4c0457213d3fbc.json.

SFT-COMP-OBL-SEMX-015 - State, effect and exception semantics

- **Claim and ownership:** SFT-COMP-SEMX-STATE-EFFECT-EXCEPTION-015; Classical Computation family SEMX; State, effect and exception semantics.
- **Forced law:** State, effects and exceptions are retained operational records: state reads and writes preserve store provenance, effects remain ordered, and an exception terminates only its declared continuation.
- **Unique surviving structure:** complete-well-formed-term__complete-operational-and-compositional-trace__state-effect-exception-trace__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation. Exact result: SEMX-015 uniquely retains state-effect-exception-trace, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-EVALUATION-001 -> SFT-COMP-SEMX-DEPENDENT-EVIDENCE-014. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** SEMX-015: state_effect_exception; exact execution: {"effects_after_exception":0,"effects_executed":["set-x","get-x","raise-halt"],"terminal":"exception-halt"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:5b9c1eb59580909d2629fa609445f2736ed9d6e636e3db24c7632a35f6febd55; independent implementation sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407; independent certificate sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70; empirical/external sha256:a266c9c4c1aa9cfd4f1fe6b045f3f04b0b427d2ec6a74822f6987800bd94a220; engine receipt sha256:6be8516daeee0e7564b5cc97f2730f07bd9b66d5233bc3a71773f5cc55e90dee at receipts/engine/model_admitted/SFT-COMP-SEMX-STATE-EFFECT-EXCEPTION-015-6be8516daeee0e75.json.

SFT-COMP-OBL-SEMX-016 - Contextual equivalence and bisimulation

- **Claim and ownership:** SFT-COMP-SEMX-CONTEXTUAL-BISIMULATION-016; Classical Computation family SEMX; Contextual equivalence and bisimulation.
- **Forced law:** Contextual equivalence requires identical observations in every generated context, while bisimulation additionally retains a matching successor relation after every lawful step.

- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__all-context-observation-equivalence__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-016 uniquely retains all-context-observation-equivalence, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-PROGRAM-EQUIVALENCE-001 -> SFT-COMP-SEMX-STATE-EFFECT-EXCEPTION-015. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-016: contextual_equivalence; exact execution: {"all_observations_equal":true,"optimized":"ab","registered_context_count":2,"source":"join(a,b)"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:0aa1e27eae76b1bee207c4fe57b0e10dfcea58f92bc0584bf37a4916a9065482`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:a6d1893cf1b94b300844dc977e5589a1470a0ee623d580052f935a921ad6ac11`; engine receipt `sha256:ec7102f78869885bf468157071f968a44265cd0d503a331ba5d9ae5b57cd5a23` at `receipts/engine/model_admitted/SFT-COMP-SEMX-CONTEXTUAL-BISIMULATION-016-ec7102f78869885b.json`.

`SFT-COMP-OBL-SEMX-017` - Termination measure and well-founded descent

- **Claim and ownership:** `SFT-COMP-SEMX-TERMINATION-DESCENT-017`; Classical Computation family `SEMX`; Termination measure and well-founded descent.
- **Forced law:** Termination is forced by a well-founded generated measure that strictly descends on every transition and reaches a declared terminal class without a fitted bound.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__well-founded-term-descent__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-017 uniquely retains well-founded-term-descent, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-TERMINATION-001 -> SFT-COMP-SEMX-CONTEXTUAL-BISIMULATION-016. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SEMX-017: termination_measure; exact execution: {"source_size":3,"strict_descent":true,"successor_size":1}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:c28e6765840b5fb6ee072dad2602cad228a7c2d624d3a8c973237394a556e7a3`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:566c372860d8bc89ac984eadcbb7a219e61f853c2b89fd1ff4893afbe9508802`; engine receipt `sha256:e8a0d80e975829d3f611d06b45ea68622cebe30115c9047aaca1a5c563a9ef71` at `receipts/engine/model_admitted/SFT-COMP-SEMX-TERMINATION-DESCENT-017-e8a0d80e975829d3.json`.

`SFT-COMP-OBL-SEMX-018` - Partial correctness and total correctness

- **Claim and ownership:** `SFT-COMP-SEMX-PARTIAL-TOTAL-CORRECTNESS-018`; Classical Computation family `SEMX`; Partial correctness and total correctness.
- **Forced law:** Partial correctness requires every terminating trace from a precondition to satisfy the postcondition; total correctness additionally requires the registered termination certificate for every generated input.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__pre-post-termination-correctness__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-018 uniquely retains pre-post-termination-correctness, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-CORRECTNESS-001 -> SFT-COMP-SEMX-TERMINATION-DESCENT-017. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-018: partial_total_correctness; exact execution: {"generated_states":2,"postcondition_rows":2,"precondition_rows":2,"termination_certificate":"single append transition"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:2b0b429ef9884bae31d28583d01c2a5aeec7c31101f72a40744f606daba466e7`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:0d9f90c7eb0f46ddf7c3365537759c0620e2c2bd411bd85cdf3f2cb790aee784`; engine receipt `sha256:4d14a78649a97b9fc51447970158d5ba0414783efffb003e633f0fc50ede15e0` at `receipts/engine/model_admitted/SFT-COMP-SEMX-PARTIAL-TOTAL-CORRECTNESS-018-4d14a78649a97b9f.json`.

`SFT-COMP-OBL-SEMX-019` - Assertion logic and invariant preservation

- **Claim and ownership:** `SFT-COMP-SEMX-ASSERTION-INVARIANT-019`; Classical Computation family `SEMX`; Assertion logic and invariant preservation.
- **Forced law:** Assertion logic is admissible when each command preserves its declared invariant and the composed proof reconstructs every intermediate assertion from exact state transitions.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__invariant-preserving-assertion-logic__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-019 uniquely retains invariant-preserving-assertion-logic, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-CORRECTNESS-001 -> SFT-COMP-SEMX-PARTIAL-TOTAL-CORRECTNESS-018. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-019: assertion_invariant; exact execution: {"generated_states":2,"initial_minimum_width":1,"invariant_preserved":true,"terminal_minimum_width":2}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:ef2cb642f44cc81b50dbe05215e673e68e4c9992ff761f68c0614bc8ceb31a35`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:06d67fc701e703cbbe9eb71e08fe96f615fc1c3beaa4307516661ac344a0584f`; engine receipt `sha256:2c8b3d8de65efb5076cfc6383c4370d169baf0bef3101bf2f35183ed89712b35` at `receipts/engine/model_admitted/SFT-COMP-SEMX-ASSERTION-INVARIANT-019-2c8b3d8de65efb50.json`.

`SFT-COMP-OBL-SEMX-020` - Formal specification and refinement ordering

- **Claim and ownership:** `SFT-COMP-SEMX-SPECIFICATION-REFINEMENT-020`; Classical Computation family `SEMX`; Formal specification and refinement ordering.

- **Forced law:** A specification is a complete input-output-trace relation; refinement is inclusion of observable behaviors with every implementation result and trace permitted by the source specification.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__behavior-subset-refinement-order__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-020 uniquely retains behavior-subset-refinement-order, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-SPECIFICATION-001 -> SFT-COMP-SEMX-ASSERTION-INVARIANT-019. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-020: specification_refinement; exact execution: {"behavior_inclusion":true,"generated_inputs":2,"source_outputs":["a","ba"],"target_outputs":["a","ba"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:485435dd1fbd9777dc17f39b54e92aaabb81b0c7e1c7872355a8e95312a56ad3`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:d75245b89bbbad762947ad77175d1c8ff8065d8a70fe3b54dba6e2dd6573ebbd`; engine receipt `sha256:bfe321ce3fca7ca7a2b0fb078b9947bfb2cfbb2fd249305a14faedf32ea1d19b` at `receipts/engine/model_admitted/SFT-COMP-SEMX-SPECIFICATION-REFINEMENT-020-bfe321ce3fca7ca7.json`.

`SFT-COMP-OBL-SEMX-021` - Program transformation and optimization correctness

- **Claim and ownership:** `SFT-COMP-SEMX-PROGRAM-TRANSFORMATION-021`; Classical Computation family `SEMX`; Program transformation and optimization correctness.
- **Forced law:** Program transformation is correct only when the transformed term preserves the declared observation for every generated input and retains a simulation or equivalence certificate.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__semantics-preserving-transformation__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-021 uniquely retains semantics-preserving-transformation, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 ->

SFT-COMP-SEM-TRANSFORMATION-001 -> SFT-COMP-SEMX-SPECIFICATION-REFINEMENT-020. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SEMX-021: program_transformation; exact execution: {"source":"join(a,b)","source_value":["a","b"],"transformed":"ab","transformed_value":["a","b"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:240f62971b77453ae604879bf5ba76c7b10d9d240cb8475911519376d13c5441`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:8bd0fb84a96e7a643a5e296c3e3aff6d4a53a7664f034c1ab1ff6a931502e583`; engine receipt `sha256:365414c6f3faeec0c984d8eb78ce99d23a57a2075629a0f07af8da53d10e1371` at `receipts/engine/model_admitted/SFT-COMP-SEMX-PROGRAM-TRANSFORMATION-021-365414c6f3faeec0.json`.

`SFT-COMP-OBL-SEMX-022` - Compiler pass simulation and semantic preservation

- **Claim and ownership:** `SFT-COMP-SEMX-COMPILER-SIMULATION-022`; Classical Computation family `SEMX`; Compiler pass simulation and semantic preservation.
- **Forced law:** Compilation preserves semantics when each source step is matched by a finite target trace, terminal values correspond, and exact code-size and execution overhead are retained.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__step-simulating-compiler-correctness__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-022 uniquely retains step-simulating-compiler-correctness, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-COMPILATION-001 -> SFT-COMP-SEMX-PROGRAM-TRANSFORMATION-021. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-022: compiler_simulation; exact execution: {"source_steps":1,"source_value":["a","b"],"target_instructions":3,"target_value":["a","b"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or

completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:791851ece2e089c39403af0aa9df832bf977379bdda57777c87e5b9716584d42`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:9bb0747916a3e9ff5f96014485a4afc10e628826801305638d17185cb3ff2be7`; engine receipt `sha256:eefea6935ebb19346833ee549a4249a0089c2108f42145393e9e7274fac357ba` at `receipts/engine/model_admitted/SFT-COMP-SEMX-COMPILER-SIMULATION-022-eefea6935ebb1934.json`.

`SFT-COMP-OBL-SEMX-023` - Intermediate-language composition

- **Claim and ownership:** `SFT-COMP-SEMX-INTERMEDIATE-COMPOSITION-023`; Classical Computation family `SEMX`; Intermediate-language composition.
- **Forced law:** Intermediate-language composition is lawful when adjacent translations share an identical boundary representation and their simulation relations compose without losing source states or target traces.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__composed-intermediate-simulation__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-023 uniquely retains composed-intermediate-simulation, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-COMPILATION-001 -> SFT-COMP-SEMX-COMPILER-SIMULATION-022. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-023: intermediate_composition; exact execution: {"interfaces_identical":true,"source":"join(join(a,b),a)","target_instructions":5,"terminal_value":["a","b","a"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f664f1955df893e2cec400f14149f00ec3378c35b424454cba5b0d4f44c90774`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:e093c9db689043e3877e8946a0d474fc739dceb40a86aa8f528eb729cc9a0ef6`; engine receipt `sha256:e1791bc5734b7e9811dbc79d78284476ec6de29fc3d46b8e7103fb17ca508e81` at `receipts/engine/model_admitted/SFT-COMP-SEMX-INTERMEDIATE-COMPOSITION-023-e1791bc5734b7e98.json`.

`SFT-COMP-OBL-SEMX-024` - Proof-carrying program and certificate checking

- **Claim and ownership:** `SFT-COMP-SEMX-PROOF-CARRYING-024`; Classical Computation family `SEMX`; Proof-carrying program and certificate checking.
- **Forced law:** A proof-carrying program contains a finite certificate whose independent checker reconstructs well-formedness, type, specification and resource claims without trusting the producer.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__proof-carrying-certificate-check__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-024 uniquely retains proof-carrying-certificate-check, complete semantic custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-VERIFICATION-001 -> SFT-COMP-SEMX-INTERMEDIATE-COMPOSITION-023. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-024: proof_carrying_program; exact execution: {"checked_type":"word","checked_value":["a","b"],"producer_trusted":false,"term_well_formed":true}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f5c4f8be279ef05ac22b50efb7763609bd42a0d7eaeaffb058d9c5baf45c68ea`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:2b10af9c353218f040e8147acdc6869cd2d9e0003b862c1b73ce047f6eb151cd`; engine receipt `sha256:d29d272caf8b9e28b5704cae106e38047a590116668c8b145af76e825b062abb` at `receipts/engine/model_admitted/SFT-COMP-SEMX-PROOF-CARRYING-024-d29d272caf8b9e28.json`.

`SFT-COMP-OBL-SEMX-025` - Semantics and programming-theory completeness certificate

- **Claim and ownership:** `SFT-COMP-SEMX-COMPLETENESS-025`; Classical Computation family `SEMX`; Semantics and programming-theory completeness certificate.
- **Forced law:** Semantics and programming-theory completeness is the one-to-one reconciliation of all twenty-five frozen obligations with unique survivors, controls, exact executions, independent reconstructions and untouched-engine receipts.
- **Unique surviving structure:** `complete-well-formed-term__complete-operational-and-compositional-trace__twenty-five-obligation-no-omission-ledger__independently-checkable-judgment__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-program-execution__declared-language-or-explicit-translation`. Exact result: SEMX-025 uniquely retains twenty-five-obligation-no-omission-ledger, complete semantic custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-SEM-VERIFICATION-001 -> SFT-COMP-SEMX-PROOF-CARRYING-024. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SEMX-025: semantics_completeness; exact execution: {"duplicate_owners":0,"independent_execution_rows":25,"omitted_owners":0,"registered_obligations":25}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SEMANTICS-OBSERVER, SFT-V1-V2-SEMANTICS-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported language semantics or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden environment, unbound name, selected evaluation path or trusted opaque compiler; no software-development process or implementation benchmark substitutes for program science; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:7fec01187f28765a219fd1960dd915e7139a709eec74fa7ca62f50992909e603`; independent implementation `sha256:201b7cd1670ef37f74a292e8e815dba6e7288548b8450e9144ae4b12eaedc407`; independent certificate `sha256:be52097d5157091a7900ba4cb748d161c53ca7fca422df015bb1802ffbbfbe70`; empirical/external `sha256:248e510787c5165205c7dd9573ad01f03a4ddee1b402b497461b5821d6f6c2c3`; engine receipt `sha256:4b394dd0722839dc2179f6647fc7d4fbbb1acbf1b9e87c15398211141dbaf0a1` at `receipts/engine/model_admitted/SFT-COMP-SEMX-COMPLETENESS-025-4b394dd0722839dc.json`.

130.9 Family `DISTX` - 26/26 complete

Scope. Concurrent and distributed computation. This family completely decides 6,656 candidates, retains 26 unique survivors and passes 104 adverse controls.

`SFT-COMP-OBL-DISTX-001` - Event identity and local process order

- **Claim and ownership:** `SFT-COMP-DISTX-EVENT-LOCAL-ORDER-001`; Classical Computation family `DISTX`; Event identity and local process order.
- **Forced law:** A distributed event is a unique process-bound transition record; each process supplies a total local successor order while events on distinct processes remain unordered until a retained communication or synchronization edge relates them.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__event-identity-local-order__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-001 uniquely retains event-identity-local-order, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-CAUSALITY-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-001: event_local_order; exact execution: {"events":["p1","p2","p3"],"local_edges":[["p1","p2"],["p2","p3"]],"process":"p"}; complete family observation custody:

26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:66b95da2c077b097c23d372348160a71fc592598ede85e64f9124db1d50c99e1`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:ed75b5963d389276a3ae91f95ce68131f9830c70f03700c6d596ba15fc36a17e`; engine receipt `sha256:76821b4e759f6be4268ebf825735416f5c56ea3661b07098fc945a054c9bbf9f` at `receipts/engine/model_admitted/SFT-COMP-DISTX-EVENT-LOCAL-ORDER-001-76821b4e759f6be4.json`.

`SFT-COMP-OBL-DISTX-002` - Concurrent interleaving and partial-order equivalence

- **Claim and ownership:** `SFT-COMP-DISTX-INTERLEAVING-PARTIAL-ORDER-002`; Classical Computation family `DISTX`; Concurrent interleaving and partial-order equivalence.
- **Forced law:** Concurrent executions are equivalent exactly when their linear traces differ only by exchanges of independent events and reconstruct the same event partial order, local states and observations.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__partial-order-interleaving-class__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-002 uniquely retains partial-order-interleaving-class, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-PARTIAL-ORDER-001 -> SFT-COMP-DISTX-EVENT-LOCAL-ORDER-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-002: partial_order_interleavings; exact execution: {"events":["a","b","c"],"lawful_linearizations":[["a","b","c"],["b","a","c"]],"required_edges":[["a","c"],["b","c"]]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:89292bc670ae42aab81378dc8eaad33004a1d5e6e2945d8ced63c109f809f739`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:6083e09c9116b762d790250fe4b4413be7d3f30a449a5cfc1f76f6bba731dde7`; engine receipt `sha256:f5564192b19f9e0afa64ca8318f15064287fff61073b66b6de88d25bdae00522` at `receipts/engine/model_admitted/SFT-COMP-DISTX-INTERLEAVING-PARTIAL-ORDER-002-f5564192b19f9e0a.json`.

`SFT-COMP-OBL-DISTX-003` **- Happens-before causality relation**

- **Claim and ownership:** `SFT-COMP-DISTX-HAPPENS-BEFORE-003`; Classical Computation family `DISTX`; Happens-before causality relation.
- **Forced law:** Happens-before is the least irreflexive transitive relation containing every local successor edge and every retained send-to-receive edge; no wall-clock order or unrecorded influence is inserted.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__transitive-happens-before-closure__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-003 uniquely retains transitive-happens-before-closure, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-CAUSALITY-001 -> SFT-COMP-DISTX-INTERLEAVING-PARTIAL-ORDER-002. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-003: happens_before; exact execution: {"closure_edges":3,"forced_transitive_edge":["a","c"],"primitive_edges":[["a","b"],["b","c"]]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:269b20477bd097586be8c7ceadf34a2c8933e7a2ad0415291e67439b65211c57`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:af948d5c905e16ae7366201a89bc97ee7c1d00390815eb874b79d075e9a016be`; engine receipt `sha256:bc4bfbd5fddd12999046bce7d97cbec501389e934168b445b4bd61e565808337` at `receipts/engine/model_admitted/SFT-COMP-DISTX-HAPPENS-BEFORE-003-bc4bfbd5fddd1299.json`.

`SFT-COMP-OBL-DISTX-004` - Message send, receipt and channel custody

- **Claim and ownership:** `SFT-COMP-DISTX-MESSAGE-CUSTODY-004`; Classical Computation family `DISTX`; Message send, receipt and channel custody.
- **Forced law:** A received message is lawful only when its identity, sender, receiver and payload reconstruct a prior send and a declared channel path; loss, duplication and reordering remain explicit channel outcomes.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__send-receive-channel-ledger__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-004 uniquely retains send-receive-channel-ledger, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-COMMUNICATION-001 -> SFT-COMP-DISTX-HAPPENS-BEFORE-003. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-004: message_custody; exact execution: {"message":"m","orphan_receipts":0,"receipt_count":1,"receiver":"q","send_count":1,"sender":"p"}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:a8ac49ed7d775e29335a36e0870d513fed4e1fb914c1b56a4b7598fc709d0422`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:e388bd27b7faa8725aaae60032767aecce3c0fe3fc2deeda3b716ef1c288ab2d`; engine receipt `sha256:bebaca914d3d511b36c4991d8bc13c1a3567e0565b4200f13bb77887fba08337` at `receipts/engine/model_admitted/SFT-COMP-DISTX-MESSAGE-CUSTODY-004-bebaca914d3d511b.json`.

`SFT-COMP-OBL-DISTX-005` - Synchronous and asynchronous execution boundary

- **Claim and ownership:** `SFT-COMP-DISTX-SYNCHRONY-BOUNDARY-005`; Classical Computation family `DISTX`; Synchronous and asynchronous execution boundary.
- **Forced law:** Synchronous execution retains a send-receive acknowledgement boundary before continuation, whereas asynchronous execution retains a queue and permits unrelated transitions before delivery; neither timing class is inferred without a registered bound.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__declared-synchrony-delivery-boundary__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-005 uniquely retains declared-synchrony-delivery-boundary, complete distributed trace custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-SYNCHRONIZATION-001 -> SFT-COMP-DISTX-MESSAGE-CUSTODY-004. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-005: synchrony_boundary; exact execution: {"asynchronous_queue_width":1,"synchronous_events":["send","receive","ack"],"timing_inferred":false}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:725cca26779254ce635aef583d0e9923b0ded8b5b8f7e382fe8d8057d699dd7a`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:1500668c57e7ae49306f848144abf01b7ff4aab767d068d8489e1edf760c5926`; engine receipt `sha256:7b9b75463296e04a5c938f15e118920fe44e0e82a2629bec0e207050c7042907` at `receipts/engine/model_admitted/SFT-COMP-DISTX-SYNCHRONY-BOUNDARY-005-7b9b75463296e04a.json`.

`SFT-COMP-OBL-DISTX-006` **- Mutual exclusion and critical-section safety**

- **Claim and ownership:** `SFT-COMP-DISTX-MUTUAL-EXCLUSION-006`; Classical Computation family `DISTX`; Mutual exclusion and critical-section safety.
- **Forced law:** Mutual exclusion is the invariant that at most one generated process holds the unique critical-section token in every reachable state; entry without the token or overlapping ownership is eliminated.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__single-holder-critical-section-invariant__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-006 uniquely retains single-holder-critical-section-invariant, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-SYNCHRONIZATION-001 -> SFT-COMP-DISTX-SYNCHRONY-BOUNDARY-005. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-006: mutual_exclusion; exact execution: {"lawful_schedule_max_holders":1,"overlap_control_rejected":true,"token_identity_retained":true}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:5f3045013312c566d6507a7d77f1da479ae3ecddd5e1c5aaa66a6edaad0a4036`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:e39d74362b02b4295bb14fb3c30abe929d99165360bcd205807125d91de3f7cd`; engine receipt `sha256:aef98b4628abf0ac53c11479eca1b2cd2f60e55ba3bfcde93274a4322a1268d2` at `receipts/engine/model_admitted/SFT-COMP-DISTX-MUTUAL-EXCLUSION-006-aef98b4628abf0ac.json`.

`SFT-COMP-OBL-DISTX-007` - Deadlock, livelock and progress distinction

- **Claim and ownership:** `SFT-COMP-DISTX-PROGRESS-CLASSES-007`; Classical Computation family `DISTX`; Deadlock, livelock and progress distinction.
- **Forced law:** Deadlock is a nonterminal state with no enabled transition, livelock is a recurrent nonterminal transition class without declared progress, and progress reaches a registered terminal or advances its well-founded obligation.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__deadlock-livelock-progress-trichotomy__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-007 uniquely retains deadlock-livelock-progress-trichotomy, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-SYNCHRONIZATION-001 -> SFT-COMP-DISTX-MUTUAL-EXCLUSION-006. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-007: progress_classes; exact execution: {"distinct_classes":3,"nonterminal_cycle_retained":true,"terminal_classes":["deadlock","livelock","progress"]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:5bcaff1440c15e4fcb69ac31e7f0d2a2e154da36c9008d669690bbf41dde05e9`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate

sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054; empirical/external sha256:40fa09e9e841da4c1fd6859896c1887cd0b85d20208436cca868cbe91154fbc6; engine receipt sha256:93b28be26d36fadb84da23cb02720beb12ef3d7cbc379520470b5d32c989ad69 at receipts/engine/model_admitted/SFT-COMP-DISTX-PROGRESS-CLASSES-007-93b28be26d36fadb.json.

SFT-COMP-OBL-DISTX-008 - Barrier, semaphore and rendezvous correspondence

- **Claim and ownership:** SFT-COMP-DISTX-COORDINATION-PRIMITIVES-008; Classical Computation family DISTX; Barrier, semaphore and rendezvous correspondence.
- **Forced law:** Barriers, semaphores and rendezvous are distinct token-and-event organizations: a barrier releases after every registered arrival, a semaphore transfers bounded permits, and a rendezvous joins one matching send and receipt.
- **Unique surviving structure:** complete-process-bound-events__complete-message-synchronization-ledger__coordination-token-correspondence__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary. Exact result: DISTX-008 uniquely retains coordination-token-correspondence, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-SYNCHRONIZATION-001 -> SFT-COMP-DISTX-PROGRESS-CLASSES-007. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** DISTX-008: coordination_primitives; exact execution: {"barrier_release_rows":1,"participants":2,"rendezvous_pair_retained":true,"semaphore_permit_retained":true}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:9fc34d84e35e5bb69804a20e3d819436f27f5614a9490db55ed91761b7a573da; independent implementation sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5; independent certificate sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054; empirical/external sha256:ae2fcd47a3f65c4059fb0765ef638d41aa104a96926230cea12df5389dcb9f87; engine receipt sha256:ee454fbd5f51cc371c0ad9f39e4dd9c2b2e2abb061cb25929175ed1f2aa95675 at receipts/engine/model_admitted/SFT-COMP-DISTX-COORDINATION-PRIMITIVES-008-ee454fbd5f51cc37.json.

SFT-COMP-OBL-DISTX-009 - Logical clock and causal timestamp correspondence

- **Claim and ownership:** SFT-COMP-DISTX-LOGICAL-CLOCK-009; Classical Computation family DISTX; Logical clock and causal timestamp correspondence.
- **Forced law:** A logical clock is the least positive event label increasing on every local edge and strictly increasing across every send-receive edge; equal or incomparable labels never create causality not present in the trace.

- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__causality-preserving-logical-clock__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-009 uniquely retains causality-preserving-logical-clock, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-CAUSALITY-001 -> SFT-COMP-DISTX-COORDINATION-PRIMITIVES-008. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-009: logical_clock; exact execution: {"causal_inequality":"send-before-receipt","receipt_clock":2,"send_clock":1,"wall_clock_used":false}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f3038b3e1c19b8c989cdd90ad20f4de18d7ec02a96bd1570535c8629feff566a`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:9e110aff4199e50fb4cc0ee889248e07d2757a64c940d0d3b1e0c1c13fd00d93`; engine receipt `sha256:d0dabe6250f5edd299efba51b0c34b7cded09c3dca59d4cf714dc6ce77cea2f2` at `receipts/engine/model_admitted/SFT-COMP-DISTX-LOGICAL-CLOCK-009-d0dabe6250f5edd2.json`.

`SFT-COMP-OBL-DISTX-010` - Broadcast, multicast and point-to-point communication

- **Claim and ownership:** `SFT-COMP-DISTX-DELIVERY-MODES-010`; Classical Computation family `DISTX`; Broadcast, multicast and point-to-point communication.
- **Forced law:** Point-to-point, multicast and broadcast differ only by their registered recipient support; every delivery retains one sender, one payload and each exact receiver without silently expanding the network.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__recipient-set-delivery-relation__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-010 uniquely retains recipient-set-delivery-relation, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-COMMUNICATION-001 -> SFT-COMP-DISTX-LOGICAL-CLOCK-009. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** DISTX-010: delivery_modes; exact execution: {"broadcast_deliveries":3,"multicast_deliveries":2,"point_deliveries":1,"registered_receivers":3}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:6367fddcc4b68f4236a6c00fe7e820fc8c26166f3e7bc5723d78fb326da5e539`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:e6a1b43e57d773abf4ba7d37ed8fe44a997158d73c6b7c43ca3b46b6182376b2`; engine receipt `sha256:b153e59c46ccbcf5f7606d96f5aeffadca07d5598b94d04ff135f191e569f958` at `receipts/engine/model_admitted/SFT-COMP-DISTX-DELIVERY-MODES-010-b153e59c46ccbcf5.json`.

`SFT-COMP-OBL-DISTX-011` - Failure-free consensus construction

- **Claim and ownership:** `SFT-COMP-DISTX-FAILURE-FREE-CONSENSUS-011`; Classical Computation family `DISTX`; Failure-free consensus construction.
- **Forced law:** Failure-free consensus is a terminating complete exchange whose deterministic decision map gives every participant one identical value drawn from the generated input support while retaining validity and agreement.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__failure-free-agreement-decision__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-011 uniquely retains failure-free-agreement-decision, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-CONSENSUS-001 -> SFT-COMP-DISTX-DELIVERY-MODES-010. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-011: failure_free_consensus; exact execution: {"decision":"a","inputs":["b","a","b"],"participant_decisions":["a","a","a"],"validity":true}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:f420843a10b44e961eb3c3ac4c654bafec10791e0570caf3fd335f15f37ee083`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:8d17cd95d32b4aa0b6546d21b9851a4c9da2dcdec8e9d6e251e9bb780336803e`; engine receipt `sha256:b520393f78b6a4187857256f726510749d181ab03a10081a5a8a9a010e80ba3f` at `receipts/engine/model_admitted/SFT-COMP-DISTX-FAILURE-FREE-CONSENSUS-011-b520393f78b6a418.json`.

`SFT-COMP-OBL-DISTX-012` - Crash-fault consensus boundary

- **Claim and ownership:** `SFT-COMP-DISTX-CRASH-CONSENSUS-BOUNDARY-012`; Classical Computation family `DISTX`; Crash-fault consensus boundary.
- **Forced law:** Crash-fault consensus requires an explicit failure and timing model; whenever a crash erases the only distinguishing message, processes with identical retained views cannot be required to decide differently, and any stronger guarantee needs an added lawful detector or synchrony record.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__crash-indistinguishability-boundary__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-012 uniquely retains crash-indistinguishability-boundary, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001 -> SFT-COMP-DISTX-FAILURE-FREE-CONSENSUS-011. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-012: crash_fault_boundary; exact execution: {"hidden_remote_states_distinct":true,"retained_local_views_equal":true,"unconditional_total_decision_admitted":false}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:0ca4160f0edef7bbd158bd8cd38510e5b885b4e63b97ea71996f9c5b7f17af4a`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:6f5d3b6905d4ae9a3fae56048af8ef119a69a13fc3481b14b3d14d7744c80dd7`; engine receipt `sha256:00ad944609968e9b642476482beb8fc9c52e1659ff1476ef053557f40842e589` at `receipts/engine/model_admitted/SFT-COMP-DISTX-CRASH-CONSENSUS-BOUNDARY-012-00ad944609968e9b.json`.

`SFT-COMP-OBL-DISTX-013` - Byzantine-fault agreement boundary

- **Claim and ownership:** `SFT-COMP-DISTX-BYZANTINE-AGREEMENT-BOUNDARY-013`; Classical Computation family `DISTX`; Byzantine-fault agreement boundary.
- **Forced law:** Byzantine agreement requires quorums whose every pair retains enough intersection to include a nonadversarial witness under the declared fault support; the exact participant, fault and quorum counts remain part of the theorem boundary.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__adversarial-quorum-intersection-boundary__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-013 uniquely retains adversarial-quorum-intersection-boundary, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-FAULT-MODEL-001 -> SFT-COMP-DISTX-CRASH-CONSENSUS-BOUNDARY-012. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-013: byzantine_quorum; exact execution: {"declared_fault_support":1,"least_pair_intersection":2,"nonadversarial_intersection_forced":true,"participants":4,"quorum_width":3}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:b4f7058f2bb1dcda2a1daf92e3346a719bc63e2248ef04a05e0f1ebb25ad08c2`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:467cd9fec1bd467f3ec6d6aa92ff7515e26c331a3d6f5d03c5301e05182378fb`; engine receipt `sha256:2f881eb13ee8addfbe479e2e9aa490fbcd3cb907db78475b936864d860671384` at `receipts/engine/model_admitted/SFT-COMP-DISTX-BYZANTINE-AGREEMENT-BOUNDARY-013-2f881eb13ee8addf.json`.

`SFT-COMP-OBL-DISTX-014` - Hidden-predecessor agreement impossibility

- **Claim and ownership:** `SFT-COMP-DISTX-HIDDEN-PREDECESSOR-014`; Classical Computation family `DISTX`; Hidden-predecessor agreement impossibility.
- **Forced law:** If distinct predecessor configurations merge to one observation without a retained reverse label, no process seeing only that image can identify which predecessor occurred; agreement predicates depending on the hidden distinction cannot be universally decided there.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__unrecorded-predecessor-impossibility__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-d`

istributed-execution__declared-fault-time-topology-boundary. Exact result: DISTX-014 uniquely retains unrecorded-predecessor-impossibility, complete distributed trace custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-AGREEMENT-IMPOSSIBILITY-001 -> SFT-COMP-DISTX-BYZANTINE-AGREEMENT-BOUNDARY-013. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** DISTX-014: hidden_predecessor; exact execution: {"predecessor_count":2,"predecessor_identifiable":false,"reverse_label_retained":false,"terminal_images":1}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:e72f626684f2cd8bb2f6b8d77e90e16d1ceca86f20a94ef35fb5ab9678d43f13; independent implementation sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5; independent certificate sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054; empirical/external sha256:f7c92a9281bfa5d8594c860aa2e2709ca1b23b1fa92b5ad2db7e1fec44845dcc; engine receipt sha256:153ace35214c8e9bd4b9e7c4613122f017673b3050e2da32c95b3986e6079571 at receipts/engine/model_admitted/SFT-COMP-DISTX-HIDDEN-PREDECESSOR-014-153ace35214c8e9b.json.

SFT-COMP-OBL-DISTX-015 - Failure detector and synchrony-assumption custody

- **Claim and ownership:** SFT-COMP-DISTX-FAILURE-DETECTOR-CUSTODY-015; Classical Computation family DISTX; Failure detector and synchrony-assumption custody.
- **Forced law:** A failure detector is a process producing suspicion records under an explicit completeness, accuracy and timing contract; it cannot be treated as knowledge or imported synchrony when its evidence is absent.
- **Unique surviving structure:** complete-process-bound-events__complete-message-synchronization-ledger__detector-assumption-ledger__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary. Exact result: DISTX-015 uniquely retains detector-assumption-ledger, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-FAULT-MODEL-001 -> SFT-COMP-DISTX-HIDDEN-PREDECESSOR-014. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** DISTX-015: failure_detector_custody; exact execution: {"basis":"missed-declared-round", "knowledge_claimed":false,"suspected_process":"q","synchrony_contract":"bounded-round"}; complete family

observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:a0b7c05f1893e315b25541fc3ef521ff6c1ccd87813b212bb0c2d1ebceff6684`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:82c043c7c10bb0b937e4258e0c363ac36d61160333ebf05412a8916e2ccc67d4`; engine receipt `sha256:5e0d11563372ced650fc8c8be5e5021399eafa4801a3a6db67e363ea2ba9e2c4` at `receipts/engine/model_admitted/SFT-COMP-DISTX-FAILURE-DETECTOR-CUSTODY-015-5e0d11563372ced6.json`.

`SFT-COMP-OBL-DISTX-016` - Replication and state-machine correspondence

- **Claim and ownership:** `SFT-COMP-DISTX-REPLICATION-STATE-MACHINE-016`; Classical Computation family `DISTX`; Replication and state-machine correspondence.
- **Forced law:** State-machine replication applies one identical totally ordered command trace to deterministic replica transitions; identical initial states and commands force identical states, while divergence retains the first differing input, order or transition.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__deterministic-replicated-state-trace__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-016 uniquely retains deterministic-replicated-state-trace, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-REPLICATION-CONSISTENCY-001 -> SFT-COMP-DISTX-FAILURE-DETECTOR-CUSTODY-015. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-016: replication; exact execution: {"command_trace":["a","b"],"deterministic_transition":true,"distinct_terminal_states":1,"replicas":3}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an

obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:8fbd5062f742b4f8268db7c80786518962c02e9271194178f3b64a807584bf4a`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:12f67a0cd5077d443edc61e09286296ffcf28740ca83191f852576d356381a3a`; engine receipt `sha256:d0f84ac77f0430c92fab8269d9956ef954f2a3b77d910d37d43a2cad9c05637d` at `receipts/engine/model_admitted/SFT-COMP-DISTX-REPLICATION-STATE-MACHINE-016-d0f84ac77f0430c9.json`.

`SFT-COMP-OBL-DISTX-017` - Linearizable and sequential consistency boundary

- **Claim and ownership:** `SFT-COMP-DISTX-LINEARIZABLE-SEQUENTIAL-017`; Classical Computation family `DISTX`; Linearizable and sequential consistency boundary.
- **Forced law:** Linearizability preserves real-time precedence and admits one sequential explanation, while sequential consistency preserves each process order without adding an unobserved real-time edge; every read must be justified by its retained write history.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__real-time-and-program-order-consistency-boundary__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-017 uniquely retains real-time-and-program-order-consistency-boundary, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-REPLICATION-CONSISTENCY-001 -> SFT-COMP-DISTX-REPLICATION-STATE-MACHINE-016. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-017: consistency_boundary; exact execution: {"linearizable_history":true,"program_order_edges_retained":true,"read_mismatch_control":true,"real_time_edges_retained":true}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:6229c56c36d73fcf5829073c1dc6c56e7274ce1d7c7da7b54aea803c40336065`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:994a0ae05f311886b9c0ccb905137f78a94d2ba0fe737fc44a345ef01c13e4c0`; engine receipt `sha256:9d8774d36c1fe34228d6f3254bc4ba189698bf729cdcba04cbf128c288495a31` at `receipts/`

`engine/model_admitted/SFT-COMP-DISTX-LINEARIZABLE-SEQUENTIAL-017-9d8774d36c1fe342.json`.

`SFT-COMP-OBL-DISTX-018` - Causal and eventual consistency boundary

- **Claim and ownership:** `SFT-COMP-DISTX-CAUSAL-EVENTUAL-018`; Classical Computation family `DISTX`; Causal and eventual consistency boundary.
- **Forced law:** Causal consistency preserves happens-before at every replica; eventual consistency additionally requires every permanently delivered update to appear in a converged state after communication resumes, without promising an unregistered convergence time.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__causal-order-eventual-convergence-boundary__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-018 uniquely retains causal-order-eventual-convergence-boundary, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-REPLICATION-CONSISTENCY-001 -> SFT-COMP-DISTX-LINEARIZABLE-SEQUENTIAL-017. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-018: causal_eventual; exact execution: {"causal_edges_preserved":2,"merge_after_updates":true,"unregistered_convergence_time":false,"updates":["write-a","write-b"]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:b9d4d9295ee4103381e555016365336b089441f2e2a22835e317f191e4196d62`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:8126d9c034d1809b329168f7e636cc382e7e33717a5a6cdeac61427721ff3a7e`; engine receipt `sha256:58eb9e40384fe05b3259221b8d7c63ea92c4d75e6f3de0ce7a7cf6570f693171` at `receipts/engine/model_admitted/SFT-COMP-DISTX-CAUSAL-EVENTUAL-018-58eb9e40384fe05b.json`.

`SFT-COMP-OBL-DISTX-019` - Quorum intersection and replicated decision

- **Claim and ownership:** `SFT-COMP-DISTX-QUORUM-INTERSECTION-019`; Classical Computation family `DISTX`; Quorum intersection and replicated decision.
- **Forced law:** A replicated decision is quorum-safe exactly when every two decision quorums intersect in the required retained witness support; disjoint admissible quorums cannot force one value without an additional relation.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__intersecting-replicated-quorum__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distrib`

`uted-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-019 uniquely retains intersecting-replicated-quorum, complete distributed trace custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-REPLICATION-CONSISTENCY-001 -> SFT-COMP-DISTX-CAUSAL-EVENTUAL-018. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-019: quorum_intersection; exact execution: {"least_intersection":1,"participants":3,"quorum_count":3,"quorum_width":2}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:7b281dbd629d4cbdb2cce94a1f508d60eb42136399019355af01d47c63bb62d7`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:dc985ce694832a4f3f8e4a569e972a61339a88055e9572ed81194c56087c5258`; engine receipt `sha256:b77fa2a83844e81f064af60c6422096ea7fb6d981e05b74d6750decad8f08043` at `receipts/engine/model_admitted/SFT-COMP-DISTX-QUORUM-INTERSECTION-019-b77fa2a83844e81f.json`.

`SFT-COMP-OBL-DISTX-020` - Distributed transaction atomicity boundary

- **Claim and ownership:** `SFT-COMP-DISTX-TRANSACTION-ATOMICITY-020`; Classical Computation family `DISTX`; Distributed transaction atomicity boundary.
- **Forced law:** A distributed transaction commits only when every registered participant vote and durable decision record supports commit; any retained rejection or missing required evidence selects abort, and partial external effects violate atomicity.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__all-or-abort-transaction-boundary__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-020 uniquely retains all-or-abort-transaction-boundary, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-REPLICATION-CONSISTENCY-001 -> SFT-COMP-DISTX-QUORUM-INTERSECTION-019. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-020: transaction_atomicity; exact execution: {"all_yes_decision":"commit","partial_effect_admitted":false,"retained_no_decision":"abort"}; complete family

observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:1267648dd0134e09124755439c27b1c9fbccfdd6dddba4b3903ec7d1d2010d6c`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:186b84d2b250187ab35bb9319752cbd3d348859f41bf2b19de6f18e9bb365ac8`; engine receipt `sha256:bdb5547fa3b213665530242badc2aa35da2addce3bbc50c82b4f3481cebf1da0` at `receipts/engine/model_admitted/SFT-COMP-DISTX-TRANSACTION-ATOMICITY-020-bdb5547fa3b21366.json`.

`SFT-COMP-OBL-DISTX-021` - Local, shared and common knowledge distinction

- **Claim and ownership:** `SFT-COMP-DISTX-DISTRIBUTED-KNOWLEDGE-021`; Classical Computation family `DISTX`; Local, shared and common knowledge distinction.
- **Forced law:** Local knowledge is the distinctions retained by one process, shared knowledge is their generated union, and common knowledge is the recursively supported intersection only when every required communication level is explicitly retained.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__local-shared-common-knowledge-ledger__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-021 uniquely retains local-shared-common-knowledge-ledger, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-DISTRIBUTED-KNOWLEDGE-001 -> SFT-COMP-DISTX-TRANSACTION-ATOMICITY-020. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-021: distributed_knowledge; exact execution: {"common_support":["b"],"local_views":[["a","b"],["b","c"]],"shared_support":["a","b","c"]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:acd54323e7d94e83c929d194bc7abe6144b9a0aa889441817b9ca8b22850b2a5`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:2ad4dee5507e6ba6a8137ae193b6dacfad15be8683971394cb15cc30366aa82c`; engine receipt `sha256:5b97b6c393021b68652cacb90e23057bfc5d5957368f8b3437361108f9176f92` at `receipts/engine/model_admitted/SFT-COMP-DISTX-DISTRIBUTED-KNOWLEDGE-021-5b97b6c393021b68.json`.

`SFT-COMP-OBL-DISTX-022` - Locality radius and information-propagation lower bound

- **Claim and ownership:** `SFT-COMP-DISTX-LOCALITY-RADIUS-022`; Classical Computation family `DISTX`; Locality radius and information-propagation lower bound.
- **Forced law:** After a finite number of local communication rounds, a process can depend only on states within that exact graph radius; any function distinguishing equal-radius views has a propagation lower bound beyond those rounds.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__radius-bounded-information-propagation__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-022 uniquely retains radius-bounded-information-propagation, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-LOCALITY-001 -> SFT-COMP-DISTX-DISTRIBUTED-KNOWLEDGE-021. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-022: locality_radius; exact execution: {"instant_remote_access":false,"round_layers":[["a"],["b"],["c"],["d"]],"rounds_to_d":3,"source":"a"}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:d85006f5f4f4f59cdc1c12b4e28f6c8ca7893615126fde657f9133ff6467f0ad`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:9ef41d6b6f0438e964020a6ef7a5c7617e5c10ef3a943e6d586799ee3af1016d`; engine receipt `sha256:458a83a0584cbfddaee4adffb898f55f52a346e81a1c6c2311b44753dd8215a6` at `receipts/engine/model_admitted/SFT-COMP-DISTX-LOCALITY-RADIUS-022-458a83a0584cbfdd.json`.

SFT-COMP-OBL-DISTX-023 - Network topology and distributed computability

- **Claim and ownership:** SFT-COMP-DISTX-NETWORK-TOPOLOGY-023; Classical Computation family DISTX; Network topology and distributed computability.
- **Forced law:** Distributed computability is conditioned by the exact communication graph: information and decisions can cross only generated reachability paths, and disconnected components cannot jointly decide a predicate requiring an unshared distinction.
- **Unique surviving structure:** complete-process-bound-events__complete-message-synchronization-ledger__topology-conditioned-computability__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary. Exact result: DISTX-023 uniquely retains topology-conditioned-computability, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-NETWORK-COMPUTATION-001 -> SFT-COMP-DISTX-LOCALITY-RADIUS-022. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** DISTX-023: network_topology; exact execution: {"path_required":true,"reachable":["a","b","c","d"],"reachable_count":4,"source":"a"}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:fa303a2e567c3dd7b4ff5649655b722a54e0bfa3b6d05211c2d5eab0dda37dbd; independent implementation sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5; independent certificate sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054; empirical/external sha256:282907b1c9ae21f9845a7f995d8a88da14e07b47886d6356f915a05b594c7e78; engine receipt sha256:803d514dbacbe8457e1d1240c0dc3118c833396f23e26839b8ff2b0f8524c233 at receipts/engine/model_admitted/SFT-COMP-DISTX-NETWORK-TOPOLOGY-023-803d514dbacbe845.json.

SFT-COMP-OBL-DISTX-024 - Dynamic-network and partition custody

- **Claim and ownership:** SFT-COMP-DISTX-PARTITION-CUSTODY-024; Classical Computation family DISTX; Dynamic-network and partition custody.
- **Forced law:** A dynamic network is a time-indexed edge relation; every partition, restoration, queued message and unavailable path remains in the trace, and no delivery or global-state inference crosses an absent edge.
- **Unique surviving structure:** complete-process-bound-events__complete-message-synchronization-ledger__time-indexed-partition-ledger__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary. Exact result: DISTX-024 uniquely retains time-indexed-partition-ledger, complete distributed trace custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-NETWORK-COMPUTATION-001 -> SFT-COMP-DISTX-NETWORK-TOPOLOGY-023. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-024: partition_custody; exact execution: {"components":[["a","b"],["c","d"]],"cross_partition_delivery":false,"source_a_reachable":["a","b"]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:7a864d49934bc4a9fa0ae7cd05669abb2ff40cbcedf024aef0760924fbfb432a`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:f7ac8de707a5da1ba61ed86bc6dd954ba65ab641518fdec131be3b7ef71c11c2`; engine receipt `sha256:21aac73eb668978af9f4749daadd622ae1338e322f1360292fd438914ecc9399` at `receipts/engine/model_admitted/SFT-COMP-DISTX-PARTITION-CUSTODY-024-21aac73eb668978a.json`.

`SFT-COMP-OBL-DISTX-025` - Distributed safety, liveness and fairness certificate

- **Claim and ownership:** `SFT-COMP-DISTX-SAFETY-LIVENESS-FAIRNESS-025`; Classical Computation family `DISTX`; Distributed safety, liveness and fairness certificate.
- **Forced law:** Distributed correctness keeps safety, liveness and fairness separate: safety excludes every bad reachable state, liveness supplies progress under declared assumptions, and fairness records which continuously enabled actions must occur.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__safety-liveness-fairness-certificate__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-025 uniquely retains safety-liveness-fairness-certificate, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-NETWORK-COMPUTATION-001 -> SFT-COMP-DISTX-PARTITION-CUSTODY-024. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-025: safety_liveness_fairness; exact execution: {"continuously_enabled_actions_served":2,"safety_control_passed":true,"terminal_progress_reached":true}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:7dae93810681777b7721d8fc03e66b6131e2dbf7a1ca2bb2ee202a11460046e1`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`; independent certificate `sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`; empirical/external `sha256:fe63b74bb0dff1607ff38eaf93993bf8d7f5a73db7b3a479d1b475df293d0488`; engine receipt `sha256:ed7ecefdab3d3a01655579c0a95aa84c9c1a9e24c198f7788224933d4d00e6ce` at `receipts/engine/model_admitted/SFT-COMP-DISTX-SAFETY-LIVENESS-FAIRNESS-025-ed7ecefdab3d3a01.json`.

`SFT-COMP-OBL-DISTX-026` - Concurrent and distributed completeness certificate

- **Claim and ownership:** `SFT-COMP-DISTX-COMPLETENESS-026`; Classical Computation family `DISTX`; Concurrent and distributed completeness certificate.
- **Forced law:** Concurrent and distributed completeness is the one-to-one reconciliation of all twenty-six frozen obligations with unique survivors, adverse controls, exact executions, independent reconstructions and untouched-engine receipts.
- **Unique surviving structure:** `complete-process-bound-events__complete-message-synchronization-ledger__twenty-six-obligation-no-omission-ledger__complete-adversarial-trace-certificate__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-distributed-execution__declared-fault-time-topology-boundary`. Exact result: DISTX-026 uniquely retains twenty-six-obligation-no-omission-ledger, complete distributed trace custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-DIST-NETWORK-COMPUTATION-001 -> SFT-COMP-DISTX-SAFETY-LIVENESS-FAIRNESS-025. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** DISTX-026: distributed_completeness; exact execution: {"duplicate_owners":0,"independent_execution_rows":26,"omitted_owners":0,"registered_obligations":26}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-DISTRIBUTED-OBSERVER, SFT-V1-V2-DISTRIBUTED-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported distributed theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden message, process state, scheduler choice, timing bound, detector or failure; no platform implementation or favorable execution substitutes for the complete distributed law; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:7ac6853a62b1ea37648acec0f01c653f57b77e64e7a47108e5e25e7f7b476d85`; independent implementation `sha256:e45bd5140887e6c6beb1659af71e184bfb11c008fd25aa9bc1f28fa8ed05c3f5`;

independent certificate
`sha256:af5dbfb518c594ad7338d96edb356484fe1344f57294f3591f6b912827ba7054`;
empirical/external
`sha256:72fbe364b5abaca047dfb075442d340cb145d4f8d079e3d1cb5cdec09017be02`; engine receipt
`sha256:65e0b337b2ee8af57d45b7377189789b050396602af45f818603f7368ff7b49d` at `receipts/engine/model_admitted/SFT-COMP-DISTX-COMPLETENESS-026-65e0b337b2ee8af5.json`.

130.10 Family `SECX` - 25/25 complete

Scope. Cryptography and computational security. This family completely decides 6,400 candidates, retains 25 unique survivors and passes 100 adverse controls.

`SFT-COMP-OBL-SECX-001` - Adversary, view, resource and success-event definition

- **Claim and ownership:** `SFT-COMP-SECX-ADVERSARY-DEFINITION-001`; Classical Computation family `SECX`; Adversary, view, resource and success-event definition.
- **Forced law:** A security claim begins with one explicit adversarial process, its exact initial information, permitted queries, finite resources and success event; changing any coordinate creates a different claim rather than an implicit stronger result.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__adversary-view-resource-success-ledger__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-001 uniquely retains adversary-view-resource-success-ledger, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-ADVERSARIAL-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-001: adversary_definition; exact execution: {"implicit_capabilities":0,"public_items":1,"queries":1,"resource_classes":1,"success_event":"invert"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:df40f6821972dd52173f43190a057c9a9c657d0fea6b1b3caaf03a4c10d74c94`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:1ccb327af2825a75f61066cdc63f6ef8e01c2edf2488fbe54c5203213e516dce`; engine receipt `sha256:6957c2467a28c4a498771da9800f20de2e76ba62bc4af7ae2930ca2544cc84af` at `receipts/engine/model_admitted/SFT-COMP-SECX-ADVERSARY-DEFINITION-001-6957c2467a28c4a4.json`.

SFT-COMP-OBL-SECX-002 - Information-theoretic secrecy

- **Claim and ownership:** SFT-COMP-SECX-INFORMATION-THEORETIC-SECRECY-002; Classical Computation family SECX; Information-theoretic secrecy.
- **Forced law:** Information-theoretic secrecy holds exactly when every protected message induces the same complete adversarial observation support with equal retained branch multiplicities, independent of computational limits.
- **Unique surviving structure:** complete-adversary-view__exact-resource-success-ledger__equal-ciphertext-support-secrecy__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary. Exact result: SECX-002 uniquely retains equal-ciphertext-support-secrecy, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-SECRECY-001 -> SFT-COMP-SECX-ADVERSARY-DEFINITION-001. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** SECX-002: information_theoretic_secrecy; exact execution: {"ciphertext_support_each":["left","right"],"messages":["left","right"],"support_multiplicities_equal":true}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:ad612e8039c895732118ba4a4ebf7e642e7775b93e48cd69c2e6e691a23ce1fb; independent implementation sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c; independent certificate sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51; empirical/external sha256:613a53844ba5a37661d862d94771774593f0ff062ca6553ccd221beb6ab8d0ca; engine receipt sha256:a557544e4ebeb032beb64533484d8f25f320ddb2af9b4ebd0f1fba4244b2e625 at receipts/engine/model_admitted/SFT-COMP-SECX-INFORMATION-THEORETIC-SECRECY-002-a557544e4ebeb032.json.

SFT-COMP-OBL-SECX-003 - Computational indistinguishability correspondence

- **Claim and ownership:** SFT-COMP-SECX-INDISTINGUISHABILITY-003; Classical Computation family SECX; Computational indistinguishability correspondence.
- **Forced law:** Computational indistinguishability is indexed by a generated observer and resource ledger: two ensembles correspond only when every observer in that complete class has the admitted exact success relation.
- **Unique surviving structure:** complete-adversary-view__exact-resource-success-ledger__resource-indexed-view-indistinguishability__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary. Exact result: SECX-003 uniquely retains resource-indexed-view-indistinguishability, complete adversarial custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-SECURITY-DEFINITION-001 -> SFT-COMP-SECX-INFORMATION-THEORETIC-SECRECY-002. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-003: computational_indistinguishability; exact execution: {"complete_support_equal":true,"left_view":["left","right"],"observer_class_explicit":true,"right_view":["right","left"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:2b33524eb262b8f402414e077f9f21a50fa2cd4a236994f91de0b0fccbeb6391`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:ef13ed226897f97618b5121ce93be8adbb7f63b60d74f042996869abb06fe77d`; engine receipt `sha256:e8d08965120d7c93451b16e0358e627b3928dd6b4b6bfe3bcc53e0605ebc5a7f` at `receipts/engine/model_admitted/SFT-COMP-SECX-INDISTINGUISHABILITY-003-e8d08965120d7c93.json`.

`SFT-COMP-OBL-SECX-004` - One-way transformation and inversion resource

- **Claim and ownership:** `SFT-COMP-SECX-ONE-WAY-RESOURCE-004`; Classical Computation family `SECX`; One-way transformation and inversion resource.
- **Forced law:** A transformation is one-way only relative to a registered input support, forward process, inversion process and resource bound; unrestricted finite enumeration remains an explicit inversion route and cannot be hidden.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__bounded-inversion-resource-relation__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-004 uniquely retains bounded-inversion-resource-relation, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-ONE-WAYNESS-001 -> SFT-COMP-SECX-INDISTINGUISHABILITY-003. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-004: one_way_resource; exact execution: {"complete_enumeration_inversion":"c","limited_inversion":"absent-from-declared-query-support","limited_queries":2,"mapping_width":3}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status:

empirically_tested_and_independently_replicated.

- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:97a03118e4d444dabf8079152adb3dd59349acd3756a18207a129a2526ed2ce3; independent implementation sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c; independent certificate sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51; empirical/external sha256:7fbd60c0fa1d0782352ac90c4c7b8130249207a3607b915091e137c5e47c1e5a; engine receipt sha256:87d821ee1643ce93d8ecadca3a74dd3c2d6f140724fadc609e64d3ab6cf00421 at receipts/engine/model_admitted/SFT-COMP-SECX-ONE-WAY-RESOURCE-004-87d821ee1643ce93.json.

SFT-COMP-OBL-SECX-005 - Hard-core distinction correspondence boundary

- **Claim and ownership:** SFT-COMP-SECX-HARD-CORE-BOUNDARY-005; Classical Computation family SECX; Hard-core distinction correspondence boundary.
- **Forced law:** A hard-core distinction is admissible only when its predictor and advantage are separately generated from the one-way relation; finite examples establish the correspondence boundary, not unrestricted hardness.
- **Unique surviving structure:** complete-adversary-view__exact-resource-success-ledger__predictor-advantage-boundary__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary. Exact result: SECX-005 uniquely retains predictor-advantage-boundary, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-ONE-WAYNESS-001 -> SFT-COMP-SECX-ONE-WAY-RESOURCE-004. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** SECX-005: hard_core_boundary; exact execution: {"retained_terminal_distinctions":["left","right"],"seed_support":2,"unrestricted_hardness_claimed":false}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:3a84ccc55889cd744f866d9e910f8faf86014faf32bd66151eb29b291934efc9; independent implementation sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c; independent certificate sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51; empirical/external sha256:c95d49b4d57a1e68bbbea79938ced4f2795f6ac7c0e9f0ed38a561d848eddeeb; engine receipt

sha256:793a3b0d423f60b0a952227cdeb402fc097b7c15cbd05db708e09b8b7d473cca at receipts/engine/model_admitted/SFT-COMP-SECX-HARD-CORE-BOUNDARY-005-793a3b0d423f60b0.json.

SFT-COMP-OBL-SECX-006 - Pseudorandom generator support correspondence

- **Claim and ownership:** SFT-COMP-SECX-PSEUDORANDOM-GENERATOR-006; Classical Computation family SECX; Pseudorandom generator support correspondence.
- **Forced law:** A pseudorandom generator expands a smaller seed support into a registered output support whose observer relation is compared with complete uniform support under an exact resource class; support omission is never called randomness.
- **Unique surviving structure:** complete-adversary-view__exact-resource-success-ledger__seed-to-expanded-support-relation__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary. Exact result: SECX-006 uniquely retains seed-to-expanded-support-relation, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-SECURITY-DEFINITION-001 -> SFT-COMP-SECX-HARD-CORE-BOUNDARY-005. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** SECX-006: pseudorandom_generator; exact execution: {"complete_three-label-support":8,"generated_output_support":2,"seed_support":2,"unbounded_distinguishable":true}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:4f76eb93696a8c656a78346400c52ccee29050306786d23d4d33692b17d3710c; independent implementation sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c; independent certificate sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51; empirical/external sha256:b4ce2456d491af3abda828602bca2165f490dc4ac37d3b996aa94522fbe40e47; engine receipt sha256:872f5e886b09ea01822d81f7a77e293855d6ec2af77c105767fe2ed83862d68b at receipts/engine/model_admitted/SFT-COMP-SECX-PSEUDORANDOM-GENERATOR-006-872f5e886b09ea01.json.

SFT-COMP-OBL-SECX-007 - Pseudorandom function correspondence

- **Claim and ownership:** SFT-COMP-SECX-PSEUDORANDOM-FUNCTION-007; Classical Computation family SECX; Pseudorandom function correspondence.
- **Forced law:** A pseudorandom function family is a keyed generated set of total input-output maps whose complete query transcripts are compared against the registered function support for a declared observer budget.
- **Unique surviving structure:** complete-adversary-view__exact-resource-success-ledger__keyed-function-family-relation__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__decla

`red-scheme-adversary-handoff-boundary`. Exact result: SECX-007 uniquely retains keyed-function-family-relation, complete adversarial custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-SECURITY-DEFINITION-001 -> SFT-COMP-SECX-PSEUDORANDOM-GENERATOR-006. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-007: pseudorandom_function; exact execution: {"input_support":2,"keys":2,"left_key_map":["left","right"],"right_key_map":["right","left"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:3cbdd4337c0390132d1e037f4b997884fd78ef28b1b0ed3c5a5930e7cff9d8bf`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:8bb8b54481ef357ba60ce0e3232c0720c3a8ff425d88e5dd0c3948a9b763dade`; engine receipt `sha256:98ac85b848c75f8c79c39911e0b42703a00750355874025823925980 7377fb99` at `receipts/engine/model_admitted/SFT-COMP-SECX-PSEUDORANDOM-FUNCTION-007-98ac85b848c75f8c.json`.

`SFT-COMP-OBL-SECX-008` - Symmetric encryption correctness and secrecy

- **Claim and ownership:** `SFT-COMP-SECX-SYMMETRIC-ENCRYPTION-008`; Classical Computation family `SECX`; Symmetric encryption correctness and secrecy.
- **Forced law:** Symmetric encryption is correct when decryption reverses every key-message encryption trace and secret only when complete ciphertext views meet the selected information-theoretic or resource-indexed definition.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__decrypt-encrypt-identity-and-view-secrecy__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-008 uniquely retains decrypt-encrypt-identity-and-view-secrecy, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-SECRECY-001 -> SFT-COMP-SECX-PSEUDORANDOM-FUNCTION-007. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-008: symmetric_encryption; exact execution: {"ciphertext_support_equal":true,"correctness_rows":4,"failed_correctness_rows":0,"keys":2,"messages":2}; complete

family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:abe612ab88e92aba73a963ec16d1c22efa63bbbb1e3e84078fff37a45e1f7cd6`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:cd319aa3df2fcbe57cb12ec27bfcd65f4fd6c19a646b1d44ac08deff61c276b9`; engine receipt `sha256:b0926efc08f26bf1e09a141d9624734281fab28b8636ade2aa54d4e915501d0f` at `receipts/engine/model_admitted/SFT-COMP-SECX-SYMMETRIC-ENCRYPTION-008-b0926efc08f26bf1.json`.

`SFT-COMP-OBL-SECX-009` - Public-key encryption correspondence boundary

- **Claim and ownership:** `SFT-COMP-SECX-PUBLIC-KEY-BOUNDARY-009`; Classical Computation family `SECX`; Public-key encryption correspondence boundary.
- **Forced law:** Public-key encryption requires separate public forward and secret reverse structures; any claimed secrecy retains key generation, adversary resources, correctness and the finite-enumeration boundary.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__trapdoor-and-enumeration-boundary__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-009 uniquely retains trapdoor-and-enumeration-boundary, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-SECRECY-001 -> SFT-COMP-SECX-SYMMETRIC-ENCRYPTION-008. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-009: public_key_boundary; exact execution: {"finite_unbounded_enumeration_recovers":true,"forward_table_public":true,"reverse_table_trapdoor":true,"unrestricted_secrecy_claimed":false}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:676d048e0f1269d69cfb792bf08c60607f5a39e10ba4d4324c618f9e6d69e700`; independent

implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:bae8202dcabf57403fd8f23a762e58410145ec0d565f17cea32386b5922a5b1c`; engine receipt `sha256:3e6e390f842a737be687676f55f75f62aa41e64e1ac96fed8dd091f57555224b` at `receipts/engine/model_admitted/SFT-COMP-SECX-PUBLIC-KEY-BOUNDARY-009-3e6e390f842a737b.json`.

`SFT-COMP-OBL-SECX-010` - Message authentication and integrity

- **Claim and ownership:** `SFT-COMP-SECX-MESSAGE-AUTHENTICATION-010`; Classical Computation family `SECX`; Message authentication and integrity.
- **Forced law:** Message authentication accepts every honestly keyed message-tag pair and rejects every generated unauthorized alteration within the declared adversary support; integrity never follows from secrecy alone.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__keyed-integrity-verification__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-010 uniquely retains keyed-integrity-verification, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-INTEGRITY-001 -> SFT-COMP-SECX-PUBLIC-KEY-BOUNDARY-009. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-010: authentication_integrity; exact execution: {"altered_message_rejected":true,"honest_pair_accepted":true,"keyed_tag_retained":true}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:638a6e41bd4323864782eff305bc5d3053c1dff65e32acd8a9ebf458f941cff5`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:4c3f255a3fdfab7c31626873ebaf3d72d1b0259f1a9f23a2415a192b3f072ab0`; engine receipt `sha256:69ea57ab41bd6a76e29c1e4e313f8300f57a15955d3e240a36cbbb8152b61541` at `receipts/engine/model_admitted/SFT-COMP-SECX-MESSAGE-AUTHENTICATION-010-69ea57ab41bd6a76.json`.

`SFT-COMP-OBL-SECX-011` - Entity authentication and freshness

- **Claim and ownership:** `SFT-COMP-SECX-ENTITY-FRESHNESS-011`; Classical Computation family `SECX`; Entity authentication and freshness.

- **Forced law:** Entity authentication binds identity to a fresh retained challenge and verified response; replayed, absent or duplicated freshness records cannot establish a new session.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__fresh-challenge-authentication__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-011 uniquely retains fresh-challenge-authentication, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-AUTHENTICATION-001 -> SFT-COMP-SECX-MESSAGE-AUTHENTICATION-010. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-011: entity_freshness; exact execution: {"fresh_challenges":["fresh-a","fresh-b"],"replay_as_new_session":false,"responses_distinct":true}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:ccfe7c95d05a3a211fb47e02efdc3bb059f67c5923de96934afacfa25ded21fc`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:56c2bdf4cf00c5f669110eaadb5fcc71ae880dccfa46f33260366f058ca764c2`; engine receipt `sha256:b862ae4e306a3daa5dd406e536d746bca83c8baed622b1588314a631be2e1ed5` at `receipts/engine/model_admitted/SFT-COMP-SECX-ENTITY-FRESHNESS-011-b862ae4e306a3daa.json`.

`SFT-COMP-OBL-SECX-012` - Hash compression, preimage and collision properties

- **Claim and ownership:** `SFT-COMP-SECX-HASH-PROPERTIES-012`; Classical Computation family `SECX`; Hash compression, preimage and collision properties.
- **Forced law:** A hash is a total compression map with exact domain and image support; preimages and collisions are structural relations, while resistance is separately indexed by the adversary and resource grammar.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__compression-preimage-collision-ledger__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-012 uniquely retains compression-preimage-collision-ledger, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 ->

SFT-COMP-SEC-HASHING-001 -> SFT-COMP-SECX-ENTITY-FRESHNESS-011. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SECX-012: hash_properties; exact execution: {"compression":true,"domain_width":4,"image_width":2,"retained_collision":[["left","left"],["right","right"]]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:79d91c3d8f9cf15a820f3d052c25748e71452be6b1f6356ffae5f9dd005cb6bf`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:1712a9c30a81d5abbe5cebb5602326ece7c9cef16b40f4aabb278ee73f42bd39`; engine receipt `sha256:dc7cab9f930db38e057aabacb09e435d632a748a2dfedc42509ad578f952dc2a` at `receipts/engine/model_admitted/SFT-COMP-SECX-HASH-PROPERTIES-012-dc7cab9f930db38e.json`.

`SFT-COMP-OBL-SECX-013` - Commitment hiding and binding

- **Claim and ownership:** `SFT-COMP-SECX-COMMITMENT-013`; Classical Computation family `SECX`; Commitment hiding and binding.
- **Forced law:** A commitment retains a public record and later opening; hiding concerns pre-opening views and binding concerns the absence of two accepted distinct openings within the complete declared support.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__hiding-binding-opening-ledger__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-013 uniquely retains hiding-binding-opening-ledger, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-COMMITMENT-001 -> SFT-COMP-SECX-HASH-PROPERTIES-012. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-013: commitment; exact execution: {"binding_control_detected":true,"commitment_records":2,"opening_correctness":true,"openings_per_record":2,"simultaneous_unrestricted_hiding_binding_claimed":false}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success

event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:0c4aaac39d78db773ec583f9ff3e07d9e55d245ca6c91bc83ee10f4a27ea84fc`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:373708a9e4d39d698c945d32b0bcfe06e10b4ef6cee1f679519f40a4804cb0b7`; engine receipt `sha256:59d06a56168b9c2987a390594bd50b788a193ee0abcf90e34b98e8ada70ee4f2` at `receipts/engine/model_admitted/SFT-COMP-SECX-COMMITMENT-013-59d06a56168b9c29.json`.

`SFT-COMP-OBL-SECX-014` - Digital signature correctness and unforgeability

- **Claim and ownership:** `SFT-COMP-SECX-DIGITAL-SIGNATURE-014`; Classical Computation family `SECX`; Digital signature correctness and unforgeability.
- **Forced law:** A signature scheme retains key generation, signing and public verification correctness; unforgeability is only the exact absence of a successful fresh-message forgery in the registered adversarial resource class.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__verification-and-forgery-resource-ledger__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-014 uniquely retains verification-and-forgery-resource-ledger, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-SIGNATURE-001 -> SFT-COMP-SECX-COMMITMENT-013. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-014: digital_signature; exact execution: {"altered_message_rejected":true,"honest_signature_accepted":true,"unrestricted_unforgeability_claimed":false}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:df7263e901760d3ea84b22e6e6471562e3e175908084fb60f4dd436c0e70171a`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:328e887dd18be28de7cb1fe90d7b8a3c67e28a70e24261c875b295482757ccc0`; engine receipt `sha256:56ce0a88da8b990e4037fd6e10df1bfdc63a80e31ac264e2c85b6f548d221637` at `receipts/engine/model_admitted/SFT-COMP-SECX-DIGITAL-SIGNATURE-014-56ce0a88da8b990e.json`.

`SFT-COMP-OBL-SECX-015` - Key establishment and authenticated exchange

- **Claim and ownership:** `SFT-COMP-SECX-KEY-ESTABLISHMENT-015`; Classical Computation family `SECX`; Key establishment and authenticated exchange.
- **Forced law:** Key establishment yields matching retained session keys from authenticated local traces; transcript exposure, freshness, identity and active-adversary capabilities remain explicit inputs to the security relation.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__authenticated-shared-key-transcript__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-015 uniquely retains authenticated-shared-key-transcript, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-AUTHENTICATION-001 -> SFT-COMP-SECX-DIGITAL-SIGNATURE-014. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-015: key_establishment; exact execution: {"authentication_boundary_retained":true,"shared_key_parts":["left","right"],"transcript_items":2}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f35e68248b40be2df8a90c267219854eb659c71b522171fe91a099ab47a58017`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:07f6c51637fc972e9ae8d1126a415df4fd5da2890a1ac684e5df54a590dc3ff5`; engine receipt `sha256:2a2c4a0983fbe18243f9d266346d4dc2bd45ec0329b4e8087e32136c70132003` at `receipts/engine/model_admitted/SFT-COMP-SECX-KEY-ESTABLISHMENT-015-2a2c4a0983fbe182.json`.

`SFT-COMP-OBL-SECX-016` - Secret sharing and reconstruction threshold

- **Claim and ownership:** `SFT-COMP-SECX-SECRET-SHARING-016`; Classical Computation family `SECX`; Secret sharing and reconstruction threshold.
- **Forced law:** Secret sharing distributes exact labelled shares so every authorized subset reconstructs one secret and every unauthorized subset retains the declared ambiguity; threshold and participant support are structural coordinates.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__threshold-share-reconstruction__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-016 uniquely retains threshold-share-reconstruction, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-MULTIPARTY-001 -> SFT-COMP-SECX-KEY-ESTABLISHMENT-015. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-016: secret_sharing; exact execution: {"authorized_subsets":3,"shares":3,"threshold":2,"unauthorized_single_share":"insufficient-shares"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:79cd9e9f08db17525c44b8afd4c2f3828872c0f08c654aab4baffaa375d02b0b`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:75a016d8810a19000d7b3ad6a85f6d0b40173948e0f3512aba73be4605927231`; engine receipt `sha256:275ed3e54bd78277b771634df6f26e4c7018887f3d8758f1f1cbd1e81367cfc6` at `receipts/engine/model_admitted/SFT-COMP-SECX-SECRET-SHARING-016-275ed3e54bd78277.json`.

`SFT-COMP-OBL-SECX-017` - Proof of knowledge and extractor boundary

- **Claim and ownership:** `SFT-COMP-SECX-PROOF-KNOWLEDGE-017`; Classical Computation family `SECX`; Proof of knowledge and extractor boundary.
- **Forced law:** A proof of knowledge accepts only with a relation witness and an extractor that reconstructs one witness from the complete registered response structure; one favorable transcript is insufficient.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__dual-challenge-extraction-boundary__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-017 uniquely retains dual-challenge-extraction-boundary, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-PROOF-KNOWLEDGE-001 -> SFT-COMP-SECX-SECRET-SHARING-016. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-017: proof_of_knowledge; exact execution: {"challenge_support":2,"dual_response_extraction":"secret","single_transcript_extraction":"insufficient-transcripts"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:b7bba99e2187c94592e204ec4d85f8598ff6948fb450c956de14c179d6fa743e`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:3d66c3fc920be68dea024e49204bee00f48ddef598f2a1dba6ff43b411351692`; engine receipt `sha256:17692a70e5ab4bb87c83eb3865dc7a1480a811181f6f411fc09d822145fabcb6` at `receipts/engine/model_admitted/SFT-COMP-SECX-PROOF-KNOWLEDGE-017-17692a70e5ab4bb8.json`.

`SFT-COMP-OBL-SECX-018` - Zero-knowledge simulator correspondence

- **Claim and ownership:** `SFT-COMP-SECX-ZERO-KNOWLEDGE-018`; Classical Computation family `SECX`; Zero-knowledge simulator correspondence.

- **Forced law:** Zero knowledge requires equality of the complete registered verifier-view support produced by the real interaction and an independent simulator without witness access, under the exact observer boundary.

- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__real-simulated-view-support-equality__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-018 uniquely retains real-simulated-view-support-equality, complete adversarial custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-ZERO-KNOWLEDGE-001 -> SFT-COMP-SECX-PROOF-KNOWLEDGE-017. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SECX-018: zero_knowledge; exact execution: {"complete_support_equal":true,"real_view_rows":2,"simulated_view_rows":2,"simulator_witness_access":false}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:9b7d2135ba1320dc91dd5c3b1dbe155192ef7af91cc7c2913334e2feeb9d5e66`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:1845a3ef62ac8dd91ab2ce03d47beeaef1671858f546dcb91da85fa8c24d4fbb`; engine receipt `sha256:72ff687bf0660893b61147ef1df53895108c5d13593f9750a525d3a4e1ac2823` at `receipts/`

engine/model_admitted/SFT-COMP-SECX-ZERO-KNOWLEDGE-018-72ff687bf0660893.json.

SFT-COMP-OBL-SECX-019 - Secure multiparty computation and view custody

- **Claim and ownership:** SFT-COMP-SECX-MULTIPARTY-COMPUTATION-019; Classical Computation family SECX; Secure multiparty computation and view custody.
- **Forced law:** Secure multiparty computation preserves the declared function output while each corrupted coalition view corresponds to an independently generated ideal view; inputs, outputs, leakage and corruption timing remain retained.
- **Unique surviving structure:** complete-adversary-view__exact-resource-success-ledger__function-output-local-view-custody__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary. Exact result: SECX-019 uniquely retains function-output-local-view-custody, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-MULTIPARTY-001 -> SFT-COMP-SECX-ZERO-KNOWLEDGE-018. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** SECX-019: secure_multiparty; exact execution: {"function_output":["left","right"],"leakage_boundary_retained":true,"local_views":2,"participants":2}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:1a383058ad816007016b60f4b6aadf311db4bb91051509e80c3f6cec63e6df78; independent implementation sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c; independent certificate sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51; empirical/external sha256:b5178d3d5fa18931e40c867f86e062d85ddc3aa2eb6ca72853e0e2b97f9de021; engine receipt sha256:437eeb9f9fe1ad076ed626e291603c03ab8736d01870a13a5b2682cc91ebabbf at receipts/engine/model_admitted/SFT-COMP-SECX-MULTIPARTY-COMPUTATION-019-437eeb9f9fe1ad07.json.

SFT-COMP-OBL-SECX-020 - Oblivious-transfer correspondence boundary

- **Claim and ownership:** SFT-COMP-SECX-OBLIVIOUS-TRANSFER-020; Classical Computation family SECX; Oblivious-transfer correspondence boundary.
- **Forced law:** Oblivious transfer delivers exactly the selected message, hides the receiver choice from the sender view and hides unselected messages from the receiver view only within its registered information and resource boundary.
- **Unique surviving structure:** complete-adversary-view__exact-resource-success-ledger__choice-private-message-transfer__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary. Exact result: SECX-020 uniquely retains choice-private-message-transfer, complete adversarial custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-MULTIPARTY-001 -> SFT-COMP-SECX-MULTIPARTY-COMPUTATION-019. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-020: oblivious_transfer; exact execution: {"choice":"right","messages":["a","b"],"selected":"b","sender_choice_view":"choice-hidden"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:381ea6783e820681e22258e4b02701c9baf50e9df735e703c400aec2a44e392d`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:801dd5ef04aebf0eafaf5412c3185f3fabbba3614bcfb6d896c1e5f8cfb778a8`; engine receipt `sha256:b986014291018533189215l1a3a508edd8f4a45e6d58c90ffa1690f2075125fa` at `receipts/engine/model_admitted/SFT-COMP-SECX-OBLIVIOUS-TRANSFER-020-b986014291018533.json`.

`SFT-COMP-OBL-SECX-021` - Adaptive, concurrent and composable adversaries

- **Claim and ownership:** `SFT-COMP-SECX-COMPOSABLE-ADVERSARY-021`; Classical Computation family `SECX`; Adaptive, concurrent and composable adversaries.
- **Forced law:** Adaptive, concurrent and composable security is a separate environment-indexed relation over complete interleavings and corruption times; standalone success does not silently transport to composition.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__environment-indexed-adversary-composition__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-021 uniquely retains environment-indexed-adversary-composition, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-ADVERSARIAL-001 -> SFT-COMP-SECX-OBLIVIOUS-TRANSFER-020. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-021: composable_adversary; exact execution: {"complete_environment_rows":8,"corruption_classes":2,"session_classes":2,"timing_classes":2}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:5acaa5e756f07f2d6d47eb9ce1c459429b8cd821ff53aa4d404aaff33f1ed40b`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:b51aa2e0055c2ac3d221713b72b72991b3fcadfb22bb20e4a79b9de5a7050b67`; engine receipt `sha256:1e5ae793371e0436a70e4d5c5e3587485ce8a5006e0daa7c52e317df53de537f` at `receipts/engine/model_admitted/SFT-COMP-SECX-COMPOSABLE-ADVERSARY-021-1e5ae793371e0436.json`.

`SFT-COMP-OBL-SECX-022` - Side-channel information ownership handoff

- **Claim and ownership:** `SFT-COMP-SECX-SIDE-CHANNEL-HANDOFF-022`; Classical Computation family `SECX`; Side-channel information ownership handoff.
- **Forced law:** Timing, power, memory, electromagnetic and physical leakage belong to an explicit implementation observation channel; algorithmic security hands these measurements to Engineering Translation without erasing them.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__implementation-leakage-handoff__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-022 uniquely retains implementation-leakage-handoff, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-SECURITY-DEFINITION-001 -> SFT-COMP-SECX-COMPOSABLE-ADVERSARY-021. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-022: side_channel_handoff; exact execution: {"algorithmic_channels":["ciphertext"],"implementation_channels":["time","power"],"owner":"engineering-translation"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:49e71ba1d1c2f5cb4a702427cf71baef619681b077dd9d14212a1be26878f3ad`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`;

empirical/external `sha256:7ac4252895f1f0cca76f833c69018918905235c73f519816ecdd3b0b6d23e6fb`; engine receipt `sha256:b3fc57d1251004cff61d2d91ea4a1bd3acc53d7f3bdaa7c53352bd0014db3b88` at `receipts/engine/model_admitted/SFT-COMP-SECX-SIDE-CHANNEL-HANDOFF-022-b3fc57d1251004cf.json`.

`SFT-COMP-OBL-SECX-023` - Post-quantum security reduction boundary

- **Claim and ownership:** `SFT-COMP-SECX-POST-QUANTUM-BOUNDARY-023`; Classical Computation family `SECX`; Post-quantum security reduction boundary.
- **Forced law:** Post-quantum security requires the same claim under a separately registered quantum adversary, query and resource grammar plus a semantics-preserving reduction; classical resistance alone cannot select it.
- **Unique surviving structure:** `complete-adversary-view__exact-resource-success-ledger__quantum-resource-reduction-boundary__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary`. Exact result: SECX-023 uniquely retains quantum-resource-reduction-boundary, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-SECX-SIDE-CHANNEL-HANDOFF-022. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SECX-023: post_quantum_boundary; exact execution: {"classical_resource_registry":"separate","quantum_resource_registry":"separate","silent_transport":false}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:6590ed1cf41716dd5d8a833caaa7ee6cf46e86c21ceaefee19befac60850aae0`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:a5310da72ebf10a53aa70414d0dc750a5b65dc30e2c9da91855045a4d59f539e`; engine receipt `sha256:0e17c394d4e2a6f268636f8a1d2f84f595e45d040bfa265be76f329372427d8b` at `receipts/engine/model_admitted/SFT-COMP-SECX-POST-QUANTUM-BOUNDARY-023-0e17c394d4e2a6f2.json`.

`SFT-COMP-OBL-SECX-024` - Quantum cryptography ownership handoff

- **Claim and ownership:** `SFT-COMP-SECX-QUANTUM-CRYPTO-HANDOFF-024`; Classical Computation family `SECX`; Quantum cryptography ownership handoff.
- **Forced law:** Protocols requiring quantum states, measurements or channels hand those operations to Quantum Computation while this branch retains adversary, secrecy, authentication and compositional security definitions.

- **Unique surviving structure:** complete-adversary-view__exact-resource-success-ledger__quantum-channel-ownership-handoff__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary. Exact result: SECX-024 uniquely retains quantum-channel-ownership-handoff, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-SECX-POST-QUANTUM-BOUNDARY-023. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** SECX-024: quantum_cryptography_handoff; exact execution: {"classical_security_owner":"computation-security","handoff_explicit":true,"quantum_channel_owner":"quantum-computation"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:f3b87c399d43fa8fd59d73eca6dffabc1f255c4c1a33f6f0b50c6b454b79efb0; independent implementation sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c; independent certificate sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51; empirical/external sha256:a4fd90c5871b087863cd571bdf3fdd42a914dd01809fb2a6d748e4eebe67b95d; engine receipt sha256:8fc26fb47557c5f1d8702eccf9c6248883f3ad5964470c14b623b062b7681879 at receipts/engine/model_admitted/SFT-COMP-SECX-QUANTUM-CRYPTO-HANDOFF-024-8fc26fb47557c5f1.json.

SFT-COMP-OBL-SECX-025 - Cryptographic security completeness certificate

- **Claim and ownership:** SFT-COMP-SECX-COMPLETENESS-025; Classical Computation family SECX; Cryptographic security completeness certificate.
- **Forced law:** Cryptographic security completeness is the one-to-one reconciliation of all twenty-five frozen obligations with unique survivors, adverse controls, exact executions, independent reconstructions and untouched-engine receipts.
- **Unique surviving structure:** complete-adversary-view__exact-resource-success-ledger__twenty-five-obligation-no-omission-ledger__complete-adversarial-control-support__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-security-execution__declared-scheme-adversary-handoff-boundary. Exact result: SECX-025 uniquely retains twenty-five-obligation-no-omission-ledger, complete adversarial custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-SEC-POST-QUANTUM-BOUNDARY-001 -> SFT-COMP-SECX-QUANTUM-CRYPTO-HANDOFF-024. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.

- **Post-registration observation:** SECX-025: security_completeness; exact execution: {"duplicate_owners":0,"independent_execution_rows":25,"omitted_owners":0,"registered_obligations":25}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SECURITY-OBSERVER, SFT-V1-V2-SECURITY-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported cryptographic theorem answer or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden key, oracle, leakage channel, adversary action, resource or success event; no toy execution silently exports to unrestricted practical security; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:acb0f24f97a9a0b2ed5c35fa493541c2bfa89db7a3b536b7d40e55bda8bc52b9`; independent implementation `sha256:7d6f4ebcc70b4afa8ffd8a6a1cffe1718e7d2ffdf8108065949d697e359afe8c`; independent certificate `sha256:392b8d5f2eeba07b41605eeab03b69ec5f8be1ba6d6730513678e45893955c51`; empirical/external `sha256:897b1fd981692575443971ee2d8e41c9202bb6744dda655e3f14163f7c6aaf05`; engine receipt `sha256:0aeaaa54bd0851429dd8bcac639036b5ae8eb44720318c61f708172f8cd8fd77` at `receipts/engine/model_admitted/SFT-COMP-SECX-COMPLETENESS-025-0aeaaa54bd085142.json`.

130.11 Family `LEARNX` - 26/26 complete

Scope. Learning and intelligence theory. This family completely decides 6,656 candidates, retains 26 unique survivors and passes 104 adverse controls.

`SFT-COMP-OBL-LEARNX-001` - Learning problem, example and target identity

- **Claim and ownership:** `SFT-COMP-LEARNX-PROBLEM-IDENTITY-001`; Classical Computation family `LEARNX`; Learning problem, example and target identity.
- **Forced law:** A learning problem retains the identity of every example, observable feature, target, admissible prediction and evaluation support before any hypothesis is selected.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__example-target-identity-ledger__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-001 uniquely retains example-target-identity-ledger, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-INFERENCE-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-001: learning_identity; exact execution: {"examples":4,"feature_width":2,"target_labels":["left","right"],"unique_example_ids":4}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or

completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:b785c506d3682d0a20ea40138430dc168d42be5f1f7c2b48c95356cede63505c`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:af6a325882196f10da414d9d26bd8ecd9ea28f40204d30615903139874867f72`; engine receipt `sha256:30941a547292c1e8f50330e38964be8887913739e45194cd982be8c1350260bd` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-PROBLEM-IDENTITY-001-30941a547292c1e8.json`.

`SFT-COMP-OBL-LEARNX-002` - Hypothesis family generation and representation

- **Claim and ownership:** `SFT-COMP-LEARNX-HYPOTHESIS-FAMILY-002`; Classical Computation family `LEARNX`; Hypothesis family generation and representation.
- **Forced law:** A hypothesis family is the complete generated set of representable input-output processes under one declared grammar; architecture names or opaque pretrained priors cannot replace enumeration.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__complete-hypothesis-family__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-002 uniquely retains complete-hypothesis-family, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-REPRESENTATION-001 -> SFT-COMP-LEARNX-PROBLEM-IDENTITY-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-002: hypothesis_family; exact execution: {"feature_positions":2,"generated_hypotheses":4,"label_orientations":2,"opaque_models":0}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:e7198b5c48023dd81c01cc951123bec3df286b9a75c7f50ba8d8d561c226e944`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:68f1920a0a446160410810bc0f37b1dd71c97243dbf5a5d01f519c99090b4d7d`; engine receipt `sha256:63236200da2e56d3f92710237df821370f395e31dde8e57dbc271929b52bff40` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-HYPOTHESIS-FAMILY-002-63236200da2e56d3.json`.

`SFT-COMP-OBL-LEARNX-003` - Exact loss and risk as parts of a finite whole

- **Claim and ownership:** `SFT-COMP-LEARNX-LOSS-RISK-003`; Classical Computation family `LEARNX`; Exact loss and risk as parts of a finite whole.
- **Forced law:** Loss counts exact retained disagreement events and risk is that count as a reduced part of the complete evaluation whole; floating surrogates and hidden weighting are excluded.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__exact-loss-risk-part__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-003 uniquely retains exact-loss-risk-part, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001 -> SFT-COMP-LEARNX-HYPOTHESIS-FAMILY-002. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-003: exact_loss_risk; exact execution: {"alternate_mistakes":2,"alternate_risk":"1/2","best_mistakes":0,"best_risk":"0/1 artifact absence","floating_used":false}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:d9460d794dd217f9796941e3935e641143e66c981680112fc3a7277319e6c62e`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:c7aa413b4604f06bae204f995c8daa268d0b97e272d04dd54c99c4402e0ed275`; engine receipt `sha256:b5dfccf83410a4048f4d2db964d69d3b5037e947fa5d89854ca28a976734c638` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-LOSS-RISK-003-b5dfccf83410a404.json`.

`SFT-COMP-OBL-LEARNX-004` - Training, validation and held-out observation custody

- **Claim and ownership:** `SFT-COMP-LEARNX-HELD-OUT-CUSTODY-004`; Classical Computation family `LEARNX`; Training, validation and held-out observation custody.
- **Forced law:** Training, validation and held-out supports are disjoint identity ledgers frozen before their respective observations; no target row migrates between roles after results are opened.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__disjoint-training-validation-test-custody__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-004 uniquely retains disjoint-training-validation-test-custody, complete learning custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-GENERALIZATION-001 -> SFT-COMP-LEARNX-LOSS-RISK-003. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-004: held_out_custody; exact execution: {"identities_retained":true,"overlap_rows":0,"test_rows":1,"training_rows":2,"validation_rows":1}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:ec96ca15bf2eb069b34210183260d451539d12d9901e3f846f06c0eaff9bf849`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:7b53a89c8a496daf91891a2463e2cf0364adbc815978566da06b7815a7f2ee8c`; engine receipt `sha256:86f6702af9187302c91d9167566ea3883f74368356094acb54e237bcda81e5f7` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-HELD-OUT-CUSTODY-004-86f6702af9187302.json`.

`SFT-COMP-OBL-LEARNX-005` - Empirical-risk minimization boundary

- **Claim and ownership:** `SFT-COMP-LEARNX-EMPIRICAL-RISK-005`; Classical Computation family `LEARNX`; Empirical-risk minimization boundary.
- **Forced law:** Empirical-risk minimization returns every hypothesis with least exact training risk after complete family evaluation; ties remain retained and training success does not itself establish generalization.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__complete-empirical-minimizer-set__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-005 uniquely retains complete-empirical-minimizer-set, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001 -> SFT-COMP-LEARNX-HELD-OUT-CUSTODY-004. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-005: empirical_risk; exact execution: {"hypotheses_evaluated":4,"minimizer":"feature-1-left","ties_hidden":false,"unique_minimizers":1}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status:

`empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:5a0c1f9d1f8dd701fec6392268317186e2556a75bd3dd5f36cb6be0fa2f471e1`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:52e3c06ea96e16989fd67c376f18c0626850e7e7981a5645f93f301744e10505`; engine receipt `sha256:040b6d1498580d30954a204f39c0f494a466455ae6128162db761d1da6bc7b1d` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-EMPIRICAL-RISK-005-040b6d1498580d30.json`.

`SFT-COMP-OBL-LEARNX-006` - Generalization as unseen-support preservation

- **Claim and ownership:** `SFT-COMP-LEARNX-GENERALIZATION-006`; Classical Computation family `LEARNX`; Generalization as unseen-support preservation.
- **Forced law:** Generalization is the preservation of the registered target relation on unseen generated support, measured separately from training performance with every favorable and adverse row retained.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__unseen-support-preservation__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-006 uniquely retains unseen-support-preservation, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-GENERALIZATION-001 -> SFT-COMP-LEARNX-EMPIRICAL-RISK-005. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-006: generalization; exact execution: {"failed_rows":0,"preserved_rows":2,"training_reuse":false,"unseen_rows":2}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:a1776abf7cc68cbcf2dee23f2c3fd051f9a8d05a2662a66f93b976e3455b80ce`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:066ca1f29418f69355cccc4a4b4954e50b20cf91bd447e59682d9ea86aa08741`; engine receipt

sha256:f38a67360079b5bb74f153bc527433fce719d4e9f5ba10ee6005185353503b91 at receipts/engine/model_admitted/SFT-COMP-LEARNX-GENERALIZATION-006-f38a67360079b5bb.json.

SFT-COMP-OBL-LEARNX-007 - Sample complexity from distinguishability count

- **Claim and ownership:** SFT-COMP-LEARNX-SAMPLE-COMPLEXITY-007; Classical Computation family LEARNX; Sample complexity from distinguishability count.
- **Forced law:** Sample complexity is indexed by the number of target distinctions the family and observer must resolve under the declared success relation; it is not imported as a distribution-free scalar.
- **Unique surviving structure:** complete-example-target-ledger__complete-hypothesis-update-process__distinguishability-indexed-sample-boundary__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary. Exact result: LEARNX-007 uniquely retains distinguishability-indexed-sample-boundary, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001 -> SFT-COMP-LEARNX-GENERALIZATION-006. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** LEARNX-007: sample_complexity; exact execution: {"complete_labelings":4,"distinguished_points":2,"support_complete":true}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:cc9bad6ef44a437d1ad23e6d2c06f8c27e929310c3ed7c964c41347578c8aad3; independent implementation sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677; independent certificate sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b; empirical/external sha256:0dacefba30a53ae1cb08f3503e57b1ffb2844e204e592fb1a21dd540ca6647e9; engine receipt sha256:2d12207bb8fe7ee9ea656193d8c07276f66114c6f37ae47e05cc2884ce18c370 at receipts/engine/model_admitted/SFT-COMP-LEARNX-SAMPLE-COMPLEXITY-007-2d12207bb8fe7ee9.json.

SFT-COMP-OBL-LEARNX-008 - Capacity and shattering correspondence boundary

- **Claim and ownership:** SFT-COMP-LEARNX-CAPACITY-SHATTERING-008; Classical Computation family LEARNX; Capacity and shattering correspondence boundary.
- **Forced law:** Capacity is the exact set of target labelings realized on generated points; shattering holds only when every labeling is present, and absent labelings remain explicit counterexamples.
- **Unique surviving structure:** complete-example-target-ledger__complete-hypothesis-update-process__realized-labeling-capacity__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary. Exact result: LEARNX-008 uniquely retains

realized-labeling-capacity, complete learning custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-SAMPLE-COMPLEXITY-001 -> SFT-COMP-LEARNX-SAMPLE-COMPLEXITY-007. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-008: capacity_shattering; exact execution: {"one_point_labelings_realized":2,"one_point_shattered":true,"three_point_labelings_required":8,"three_point_shattered":false}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:c57ef775180b1122779eb7d48e46886ce892da802c2ac08858ac57bfed9ae176`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:e2693e58ec2c8b986fdd7909e72617531f5102991ad2607b619dc320b253ef72`; engine receipt `sha256:d94094145ec4c004883e62028162c8ebf8fb252fe79ba1a84893b8c1b7e97022` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-CAPACITY-SHATTERING-008-d94094145ec4c004.json`.

`SFT-COMP-OBL-LEARNX-009` - Probably-approximately-correct correspondence

- **Claim and ownership:** `SFT-COMP-LEARNX-PAC-CORRESPONDENCE-009`; Classical Computation family `LEARNX`; Probably-approximately-correct correspondence.
- **Forced law:** Probably-approximately-correct language corresponds to exact finite branch support: success and error are reduced parts of the complete registered whole, never stochastic causes or floating tolerances.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__finite-support-success-error-correspondence__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-009 uniquely retains finite-support-success-error-correspondence, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-GENERALIZATION-001 -> SFT-COMP-LEARNX-CAPACITY-SHATTERING-008. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-009: pac_correspondence; exact execution: {"complete_branches":4,"stochastic_cause_used":false,"success_branches":3,"success_part":"3/4"}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported

theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:5daae7807dbfd01f13ef60fa3add1e3adddb44c9b9b21e808e9b1220f01a66f4`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:7ba7d8cac6158b2738e11d3ad5e3d1351c2c869d05c11c1a6c6bbc2110dbb545`; engine receipt `sha256:5f357fcfb279edda673850eee251e29431fdbcc347b3d7c103b9ad8552e2c4f8` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-PAC-CORRESPONDENCE-009-5f357fcfb279edda.json`.

`SFT-COMP-OBL-LEARNX-010` - Classification and decision-boundary computation

- **Claim and ownership:** `SFT-COMP-LEARNX-CLASSIFICATION-010`; Classical Computation family `LEARNX`; Classification and decision-boundary computation.
- **Forced law:** Classification applies one generated decision relation to every input and retains its boundary, label, trace, correct rows and errors; an output label without this custody is not a learned law.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__decision-boundary-classifier__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-010 uniquely retains decision-boundary-classifier, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001 -> SFT-COMP-LEARNX-PAC-CORRESPONDENCE-009. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-010: classification; exact execution: {"errors":0,"predictions":["left","right","left","right"],"targets":["left","right","left","right"]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:a90eaad953ca68dcb437062401c6d7732c6e8d9664d7e84cbf15b484f258fb27`; independent

implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:d0d1459fc8ac328dde4ff3a7706a2e172101eb90a22477ca3f9c3d01abdb39d7`; engine receipt `sha256:9c60131c72dfa432787498f3fa9a2ed2cf9a1bfe8a4812b3fc3ce0d3bebb3d32` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-CLASSIFICATION-010-9c60131c72dfa432.json`.

`SFT-COMP-OBL-LEARNX-011` - Regression as exact representative relation

- **Claim and ownership:** `SFT-COMP-LEARNX-REGRESSION-011`; Classical Computation family `LEARNX`; Regression as exact representative relation.
- **Forced law:** Regression selects an exact representative relation over finite rational-part targets and retains residuals, ordering, loss and enclosure; continuum or fitted real-valued assumptions are unnecessary.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__exact-representative-regression__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-011 uniquely retains exact-representative-regression, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001 -> SFT-COMP-LEARNX-CLASSIFICATION-010. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-011: regression; exact execution: {"continuum_used":false,"exact_representative":"1/1","targets":["1/2","3/2"]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:ef6f4cc6e0a87eb87a4226d0fd39bf89de802863392bb2e6a5acbc7abe970755`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:b594541e40c2d01d1313fe32b51cfba741b1034221578354d0901cd768b5e7c0`; engine receipt `sha256:5d56e6c97dd7a707dffb1ecdd466cb1fdbbbf9db0fb47306b7366e2a8ca573f4` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-REGRESSION-011-5d56e6c97dd7a707.json`.

`SFT-COMP-OBL-LEARNX-012` - Feature selection and sufficient representation

- **Claim and ownership:** `SFT-COMP-LEARNX-FEATURE-SELECTION-012`; Classical Computation family `LEARNX`; Feature selection and sufficient representation.
- **Forced law:** A feature support is sufficient only when no two generated examples equal on retained features yet require distinct targets; selection preserves every eliminated distinction and failure witness.

- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__target-sufficient-feature-support__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-012 uniquely retains target-sufficient-feature-support, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-REPRESENTATION-001 -> SFT-COMP-LEARNX-REGRESSION-011. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-012: feature_selection; exact execution: {"counterexamples_retained":true,"feature_one_sufficient":true,"feature_two_sufficient":false}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:11c8bfe89a530815c77babfdb0adf91ac52babda1ca7ad56a830cefe564ff8eb`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:d39220367c3c54952d79bb9f76e1fa99c62b9085913a5cac5beccb19d12f9a7b`; engine receipt `sha256:a77a3b0d20e2c5f1d27b6705513297c345fdd06ff436b7447d345f3a0368dc43` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-FEATURE-SELECTION-012-a77a3b0d20e2c5f1.json`.

`SFT-COMP-OBL-LEARNX-013` - Unsupervised partition and clustering custody

- **Claim and ownership:** `SFT-COMP-LEARNX-CLUSTERING-013`; Classical Computation family `LEARNX`; Unsupervised partition and clustering custody.
- **Forced law:** Unsupervised clustering enumerates the complete declared partition support and evaluates one exact structural objective; labels applied after inspection cannot select the partition.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__complete-partition-objective-ledger__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-013 uniquely retains complete-partition-objective-ledger, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-INDUCTION-001 -> SFT-COMP-LEARNX-FEATURE-SELECTION-012. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** LEARNX-013: clustering; exact execution: {"cluster_labels":2,"complete_assignment_support":8,"items":3,"posthoc_labels_select_partition":false}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f8fc273e19b72a00ada24564b323ef238309589579c44545f84f761670f2fc29`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:acfc071230e14d7f7dcc567975431f442058f9f99caf2c809f0394c632199231`; engine receipt `sha256:38c359fe762df8857cf434575c13253c812005e5743f2f71787bb382fd96ab5f` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-CLUSTERING-013-38c359fe762df885.json`.

`SFT-COMP-OBL-LEARNX-014` - Generative-support reconstruction boundary

- **Claim and ownership:** `SFT-COMP-LEARNX-GENERATIVE-SUPPORT-014`; Classical Computation family `LEARNX`; Generative-support reconstruction boundary.
- **Forced law:** A generative learner is compared by the exact support and branch multiplicities it reconstructs from registered seeds; plausible samples cannot substitute for complete support custody.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__seed-to-generated-support-reconstruction__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-014 uniquely retains seed-to-generated-support-reconstruction, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-INDUCTION-001 -> SFT-COMP-LEARNX-CLUSTERING-013. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-014: generative_support; exact execution: {"generated_support":[["left","held"],["right","held"]],"missing_outputs":0,"seeds":["left","right"]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:de55e2a352cb4228e9057226860530ec17523799fc0d0b7c450a7bfaedd66bd4`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:0bf913ee5635e41ee9ca437d05ea56d2acd9b7ef5581f122996dc5d20821e50f`; engine receipt `sha256:124a2783afd7773c5811dbc7f1d6aa9e3f4d28e41282a04046bf138c4841ce27` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-GENERATIVE-SUPPORT-014-124a2783afd7773c.json`.

`SFT-COMP-OBL-LEARNX-015` - Deterministic-support Bayesian correspondence

- **Claim and ownership:** `SFT-COMP-LEARNX-BAYESIAN-CORRESPONDENCE-015`; Classical Computation family `LEARNX`; Deterministic-support Bayesian correspondence.
- **Forced law:** Bayesian correspondence is deterministic branch reweighting: exact prior parts multiply exact likelihood parts and renormalize over complete support without introducing stochastic causation.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__exact-branch-weight-posterior-correspondence__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-015 uniquely retains exact-branch-weight-posterior-correspondence, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-INFERENCE-001 -> SFT-COMP-LEARNX-GENERATIVE-SUPPORT-014. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-015: bayesian_correspondence; exact execution: {"likelihood":{"left":"3/4","right":"1/4"},"posterior":{"left":"3/4","right":"1/4"},"prior":{"left":"1/2","right":"1/2"},"stochastic_cause_used":false}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:04132ad1658e3093c5dfd55be868ebba53105ad1e6de6f838fa6842b1f0ac01f`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:fd5e045705c0a74a456cab80117c1faf5effe6dabd221f13236a1b84a5956a7f`; engine receipt `sha256:734446248f1f30ac3c7982aad99b1ca661aec54b44e4e35618edda3f95a1317f` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-BAYESIAN-CORRESPONDENCE-015-734446248f1f30ac.json`.

SFT-COMP-OBL-LEARNX-016 - Learning optimization and convergence certificate

- **Claim and ownership:** SFT-COMP-LEARNX-OPTIMIZATION-CONVERGENCE-016; Classical Computation family LEARNX; Learning optimization and convergence certificate.
- **Forced law:** Learning optimization converges only with a retained well-founded objective or enclosure that strictly descends on every update and terminates at the declared fixed or minimal class.
- **Unique surviving structure:** complete-example-target-ledger__complete-hypothesis-update-process__strict-descent-learning-convergence__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary. Exact result: LEARNX-016 uniquely retains strict-descent-learning-convergence, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001 -> SFT-COMP-LEARNX-BAYESIAN-CORRESPONDENCE-015. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** LEARNX-016: optimization_convergence; exact execution: {"initial_measure":4,"strict_descent":true,"terminal_measure":1,"trace":[4,3,2,1]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:e4d5d8a8143c6011bfd8dfae515effab026aef55638c5b134900c2f2c3a3f9a0; independent implementation sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677; independent certificate sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b; empirical/external sha256:2ef62e1b25a697ef59d6b8a02fa5f6066d7fd6a9d8ddf908b8728d576d1719b4; engine receipt sha256:1292904052c2a8fbe2059f7313cf8970e0b9cd690875d0540c004f2fd09d213c at receipts/engine/model_admitted/SFT-COMP-LEARNX-OPTIMIZATION-CONVERGENCE-016-1292904052c2a8fb.json.

SFT-COMP-OBL-LEARNX-017 - Online learning and regret as exact ledger

- **Claim and ownership:** SFT-COMP-LEARNX-ONLINE-REGRET-017; Classical Computation family LEARNX; Online learning and regret as exact ledger.
- **Forced law:** Online learning observes only the retained prefix before each decision; regret is the exact accumulated loss ledger against a separately generated comparator over the same complete sequence.
- **Unique surviving structure:** complete-example-target-ledger__complete-hypothesis-update-process__prefix-loss-comparator-regret-ledger__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary. Exact result: LEARNX-017 uniquely retains prefix-loss-comparator-regret-ledger, complete learning custody, root forcing, post-registry execution and no extra

rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-ADAPTATION-001 -> SFT-COMP-LEARNX-OPTIMIZATION-CONVERGENCE-016. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-017: online_regret; exact execution: {"comparator_loss":1,"exact_regret":0,"future_access_used":false,"learner_loss":1,"rounds":3}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:6dfa3a71fb4d7234bea762f5d15792d036b4233718e6767463dcfcf35eeabda4`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:b52882a555c6d0ef98bd149a4be0f5efaf585a3cd0c5520162d2c29d5f84e3c0`; engine receipt `sha256:11fb4503637f00d22cf13372a640746d1a4f65f3c3ba28b0f088da202c25a7a7` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-ONLINE-REGRET-017-11fb4503637f00d2.json`.

`SFT-COMP-OBL-LEARNX-018` - Concept drift and adaptation

- **Claim and ownership:** `SFT-COMP-LEARNX-CONCEPT-DRIFT-018`; Classical Computation family `LEARNX`; Concept drift and adaptation.
- **Forced law:** Concept drift is a time-indexed change in the target or support relation; adaptation is evaluated on post-change identities without rewriting prior observations or silently moving the boundary.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__time-indexed-target-drift-ledger__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-018 uniquely retains time-indexed-target-drift-ledger, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-ADAPTATION-001 -> SFT-COMP-LEARNX-ONLINE-REGRET-017. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-018: concept_drift; exact execution: {"after_target":"right","before_target":"left","change_retained":true,"feature":"held","prior_rows_rewritten":false}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:942b351eaad464aba88a3ced5e516ab6326cb40ca06e247776e63bde789d1fb4`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:052179d2a6983302f155bc883f75a725e22fe07c65aa14f9b7a3478f54d8b874`; engine receipt `sha256:fc6dfbeaf677c3065f6d89f7cc8492d02cee3caba9f1efcdd58b20081e662e2d` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-CONCEPT-DRIFT-018-fc6dfbeaf677c306.json`.

`SFT-COMP-OBL-LEARNX-019` - Search, planning and heuristic admissibility

- **Claim and ownership:** `SFT-COMP-LEARNX-SEARCH-PLANNING-019`; Classical Computation family `LEARNX`; Search, planning and heuristic admissibility.
- **Forced law:** Search and planning retain states, actions, successors, path resources and terminal goals; a heuristic is admissible only when its exact estimate never exceeds the independently generated remaining cost.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__admissible-estimate-search-trace__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-019 uniquely retains admissible-estimate-search-trace, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-SEARCH-PLANNING-001 -> SFT-COMP-LEARNX-CONCEPT-DRIFT-018. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-019: search_planning; exact execution: {"direct_cost":3,"heuristic_admissible":true,"selected_cost":2,"selected_path":["a","b","c"],"source":"a","target":"c"}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:875e4896a335f8c209df245dd081d0101af9adb141cb224363b88d80405afa35`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`;

empirical/external `sha256:a01532fde8acebcd21cf3d6877690308bd790c66b8ed46ce3522724d99ab6da6`; engine receipt `sha256:2501fb3af3240a93b1a9efcb75d755aba0c4894e8133b38e3786315faf2f1894` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-SEARCH-PLANNING-019-2501fb3af3240a93.json`.

`SFT-COMP-OBL-LEARNX-020` - Reinforcement state, action and return custody

- **Claim and ownership:** `SFT-COMP-LEARNX-REINFORCEMENT-020`; Classical Computation family `LEARNX`; Reinforcement state, action and return custody.
- **Forced law:** Reinforcement learning retains every state, action, transition, observation and exact finite return; branch selection and exploration are complete deterministic supports rather than stochastic causes.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__state-action-transition-return-ledger__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-020 uniquely retains state-action-transition-return-ledger, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-REINFORCEMENT-001 -> SFT-COMP-LEARNX-SEARCH-PLANNING-019. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-020: reinforcement_return; exact execution: {"branch_trace_retained":true,"finite_return":"1/1","reward_parts":["1/2","1/3","1/6"]}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:9d39bf09ed1c202cd06ab9b19c89494db5e53a564ffb1089ae16d939b20afbda`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:95668a381d70a2cc917a60f5cec6056d2d45f6bb45a9a3d9de9efd4033e9833f`; engine receipt `sha256:4576edd5a65dde5aaa6103cf3b03ee6256f1986320bd47e08df66a7a5957c1fd` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-REINFORCEMENT-020-4576edd5a65dde5a.json`.

`SFT-COMP-OBL-LEARNX-021` - Multi-agent learning and strategic observation

- **Claim and ownership:** `SFT-COMP-LEARNX-MULTI-AGENT-021`; Classical Computation family `LEARNX`; Multi-agent learning and strategic observation.
- **Forced law:** Multi-agent learning retains joint actions, local views, communication and outcome relations for every participant; strategic claims require complete opponent and information boundaries.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__joint-policy-strategic-view-ledger__complete-held-out-adverse-ledger__litera`

`l-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-021 uniquely retains joint-policy-strategic-view-ledger, complete learning custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-MULTIAGENT-001 -> SFT-COMP-LEARNX-REINFORCEMENT-020. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-021: multi_agent; exact execution: {"actions_each":2,"agents":2,"joint_action_rows":4,"strategic_views_retained":true}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:51ec537fb518a4c6c3a495274f509b725c39b4eeee888caa3687409db405e07e`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:763f314251c8bab9e32d65206f73151a0032fdf8fb10f9a6f31d82e7ccea1961`; engine receipt `sha256:d21f568f31b6f1df324828ca3fe3984c72b10fad3d23a645793ade7f8dd84107` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-MULTI-AGENT-021-d21f568f31b6f1df.json`.

`SFT-COMP-OBL-LEARNX-022` - Interpretability as reconstructible decision trace

- **Claim and ownership:** `SFT-COMP-LEARNX-INTERPRETABILITY-022`; Classical Computation family `LEARNX`; Interpretability as reconstructible decision trace.
- **Forced law:** Interpretability is an independently reconstructible path from retained input distinctions through rules or state transitions to the output, not a post-hoc narrative detached from execution.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__reconstructible-decision-trace__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-022 uniquely retains reconstructible-decision-trace, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001 -> SFT-COMP-LEARNX-MULTI-AGENT-021. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-022: interpretability; exact execution: {"posthoc_only":false,"trace":[["observe","safe"],["rule","safe-to-act"],["act","act"]],"trace_steps":3}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported

theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:c6b4032d429c8b064eac6d01cf9ece32edae10d9b360717bfbf2d263a1992edc`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:3e05a3056ca4ea2c1fdd9108d8b59288772c42a20d8c5217f95fc44fdcceb9ad`; engine receipt `sha256:998e6fbeda643e6dc34b8196fb58636dca1f9fae64365f5b2c1ccdff55afc857` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-INTERPRETABILITY-022-998e6fbeda643e6d.json`.

`SFT-COMP-OBL-LEARNX-023` - Robustness, distribution shift and adversarial examples

- **Claim and ownership:** `SFT-COMP-LEARNX-ROBUSTNESS-SHIFT-023`; Classical Computation family `LEARNX`; Robustness, distribution shift and adversarial examples.
- **Forced law:** Robustness holds only over a registered perturbation neighborhood whose every point preserves the required observation; distribution shifts and adversarial rows remain separately identified.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__registered-neighborhood-shift-certificate__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-023 uniquely retains registered-neighborhood-shift-certificate, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-LEARNING-LIMITS-001 -> SFT-COMP-LEARNX-INTERPRETABILITY-022. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-023: robustness_shift; exact execution: {"adverse_coordinate_control_detected":true,"neighborhood_registered":true,"stable_neighborhoods":2,"stable_rows":4}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:459806a8fde323bb910734296659fe968167872343fcb8c5af0b0ef10e4a1884`; independent

implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:1ba1fff4e22119895a511cf92bdd7cc1d12da345f0a6f1424ca8caa93a635d01`; engine receipt `sha256:3c099fc6e9523688b49828cd95ff8278058f88cae65fad8b4d1cbb79de7d5d6a` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-ROBUSTNESS-SHIFT-023-3c099fc6e9523688.json`.

`SFT-COMP-OBL-LEARNX-024` - Verification of learned-process behavior

- **Claim and ownership:** `SFT-COMP-LEARNX-LEARNED-VERIFICATION-024`; Classical Computation family `LEARNX`; Verification of learned-process behavior.
- **Forced law:** A learned process is verified by exhaustive generated support or a separately forced inductive certificate proving its behavior, resources and halting boundary; sampled accuracy is insufficient.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__complete-support-behavior-verification__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-024 uniquely retains complete-support-behavior-verification, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-INTERPRETABILITY-VERIFICATION-001 -> SFT-COMP-LEARNX-ROBUSTNESS-SHIFT-023. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-024: learned_verification; exact execution: {"property_failures":0,"sample_only":false,"support_rows":4,"verified_rows":4}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:56d422626fc3e777fb33fa0864420a82324793a9d3f7446ca5c56d4ea5eb65f0`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:28a19f283e4d7af71895b6d86387f86f833946ab1b4a547d76629b6263d38538`; engine receipt `sha256:7a75ae59b3b58abd147e008ff6b88867576b3f49ab7a498a0fd3f7204e74d48f` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-LEARNED-VERIFICATION-024-7a75ae59b3b58abd.json`.

`SFT-COMP-OBL-LEARNX-025` - No-free-lunch and identifiability limits

- **Claim and ownership:** `SFT-COMP-LEARNX-IDENTIFIABILITY-LIMIT-025`; Classical Computation family `LEARNX`; No-free-lunch and identifiability limits.

- **Forced law:** When two target laws have identical retained training views but differ on unseen support, no learner restricted to that view can identify both; every stronger claim requires a lawful added distinction.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__observational-identifiability-boundary__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-025 uniquely retains observational-identifiability-boundary, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-LEARNING-LIMITS-001 -> SFT-COMP-LEARNX-LEARNED-VERIFICATION-024. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-025: identifiability_limit; exact execution: {"simultaneous_identification_possible":false,"training_views_equal":true,"unseen_targets_distinct":true}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:1ecb34ec3b322793a133ad3dd4efb5780314efd2a5ee5f1a90bdd85c8dfda8d5`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:dc9051245794e71b8a29a2f8cbfb75a04fa4e35450a85becb006114246198fcf`; engine receipt `sha256:b803edc846a67c4a5bf8a750e73ba026174dbed38c67c54f4ddd1b2aa62295eb` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-IDENTIFIABILITY-LIMIT-025-b803edc846a67c4a.json`.

`SFT-COMP-OBL-LEARNX-026` - Computational learning completeness certificate

- **Claim and ownership:** `SFT-COMP-LEARNX-COMPLETENESS-026`; Classical Computation family `LEARNX`; Computational learning completeness certificate.
- **Forced law:** Computational learning completeness is the one-to-one reconciliation of all twenty-six frozen obligations with unique survivors, adverse controls, exact executions, independent reconstructions and untouched-engine receipts.
- **Unique surviving structure:** `complete-example-target-ledger__complete-hypothesis-update-process__twenty-six-obligation-no-omission-ledger__complete-held-out-adverse-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-learning-execution__declared-support-shift-application-boundary`. Exact result: LEARNX-026 uniquely retains twenty-six-obligation-no-omission-ledger, complete learning custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-LEARN-CLASSICAL-LEARNING-001 -> SFT-COMP-LEARNX-IDENTIFIABILITY-LIMIT-025. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** LEARNX-026: learning_completeness; exact execution: {"duplicate_owners":0,"independent_execution_rows":26,"omitted_owners":0,"registered_obligations":26}; complete family observation custody: 26 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-LEARNING-OBSERVER, SFT-V1-V2-LEARNING-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, pretrained consensus model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden training row, target label, branch, randomness, hyperparameter or oracle; no favorable benchmark or opaque prediction substitutes for complete learning evidence; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:d19cc728ca3f86290b5ef87872c64cf893fe46c16d8cce45764f2c4bb08289e4`; independent implementation `sha256:c0740adbe44dcb74ae632777ff9bffd5d1c4e7bd6013f839ce6317db3df30677`; independent certificate `sha256:14852197c8d1d03dc3395a170d104f6d0c02f48f635f93572a431505cf417c6b`; empirical/external `sha256:23a8fbeecb46663f90a2b16ca7f57d2e3b5ada3536eb3dc88150c4bbb05dd260`; engine receipt `sha256:91c835ed6de921461bb462de45296841fa7d509444e1ab83d85d4ea66ca74b75` at `receipts/engine/model_admitted/SFT-COMP-LEARNX-COMPLETENESS-026-91c835ed6de92146.json`.

130.12 Family `SCIX` - 25/25 complete

Scope. Scientific computation. This family completely decides 6,400 candidates, retains 25 unique survivors and passes 100 adverse controls.

`SFT-COMP-OBL-SCIX-001` - Exact and approximate result distinction

- **Claim and ownership:** `SFT-COMP-SCIX-EXACT-APPROXIMATE-001`; Classical Computation family `SCIX`; Exact and approximate result distinction.
- **Forced law:** An exact result is one canonical finite Fold expression; an approximate result is an explicit rational enclosure containing it, with representation, residual and stopping evidence retained.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__exact-value-or-explicit-enclosure__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-001 uniquely retains exact-value-or-explicit-enclosure, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-EXACT-APPROXIMATE-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-001: exact_approximate; exact execution: {"approximation":"33/100","contained":true,"enclosure":["33/100","34/100"],"exact":"1/3"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:18a41d561708e3336f8ad40bcf2f6348d64e06b0d68892e8872b175fdacb898a`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:891c71ae3a07ecb332f794a764756407d160d24ddcf41bca3271d565e2c6cefd`; engine receipt `sha256:84970622b0cf5e5e93dd36403e37249069a6538b6e55a8ed6d31d8f2081afdbd` at `receipts/engine/model_admitted/SFT-COMP-SCIX-EXACT-APPROXIMATE-001-84970622b0cf5e5e.json`.

`SFT-COMP-OBL-SCIX-002` - Finite-precision representation correspondence

- **Claim and ownership:** `SFT-COMP-SCIX-FINITE-PRECISION-002`; Classical Computation family `SCIX`; Finite-precision representation correspondence.
- **Forced law:** Finite precision is a generated rational grid with exact spacing, range and tie rule; it corresponds to a value only through a retained rounding or enclosure map.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__finite-rational-grid-correspondence__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-002 uniquely retains finite-rational-grid-correspondence, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-EXACT-APPROXIMATE-001 -> SFT-COMP-SCIX-EXACT-APPROXIMATE-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-002: finite_precision; exact execution: {"grid_denominator":4,"nearest":"3/2","source":"7/5","tie_rule":"least-exact-grid-part"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:5d4d7f932872a6221947fdd35d25445e6e1fe89a0fdf675b92f1661244dfcca5`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`;

empirical/external `sha256:20911c35ae01447d647864fac515171f0a133a1421bac5f45c29a80ab0b22ebb`; engine receipt `sha256:aecf5eb5e4d25c707b0d79cbbaccd39602cebff9958623fc2498ec4fa7fb309c` at `receipts/engine/model_admitted/SFT-COMP-SCIX-FINITE-PRECISION-002-aecf5eb5e4d25c70.json`.

`SFT-COMP-OBL-SCIX-003` - Rounding and truncation error ledger

- **Claim and ownership:** `SFT-COMP-SCIX-ROUNDING-ERROR-003`; Classical Computation family `SCIX`; Rounding and truncation error ledger.
- **Forced law:** Rounding and truncation retain the exact source, represented result, oriented separation and propagation position; no floating artifact is silently promoted to an exact value.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__rounding-truncation-separation-ledger__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-003 uniquely retains rounding-truncation-separation-ledger, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-ERROR-PROPAGATION-001 -> SFT-COMP-SCIX-FINITE-PRECISION-002. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-003: rounding_truncation; exact execution: {"exact":"7/5","floating_used":false,"represented":"3/2","separation":"1/10"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:e443dcdf46ebcca08d2e56ce9582c23af70f6cae6328274eb2e99f48e9033ec0`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:a6c942f8a2297e96473b3ef481e9c6cb6f6c243ae2956e2d1ce4335860438106`; engine receipt `sha256:d157fdac7fee80f80dedd8cc48a478392527392a70fb91f2ff9a4e44962bfc77` at `receipts/engine/model_admitted/SFT-COMP-SCIX-ROUNDING-ERROR-003-d157fdac7fee80f8.json`.

`SFT-COMP-OBL-SCIX-004` - Forward and backward error

- **Claim and ownership:** `SFT-COMP-SCIX-FORWARD-BACKWARD-ERROR-004`; Classical Computation family `SCIX`; Forward and backward error.
- **Forced law:** Forward error compares computed and target outputs, while backward error reconstructs the least admitted input relation that makes the computed output exact; both retain their support boundary.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__forward-backward-error-pair__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-exe`

`cution__declared-model-mesh-data-boundary`. Exact result: SCIX-004 uniquely retains forward-backward-error-pair, complete numerical custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-ERROR-PROPAGATION-001 -> SFT-COMP-SCIX-ROUNDING-ERROR-003. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-004: forward_backward; exact execution: {"both_ledgers_retained":true,"forward_separation":"1/10","reconstructed_exact_input":"7/5"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:02c45b2085218766bf230f6b5439d3a3ff4dbf6737c85c0e6736cdecafd69adb`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:4ba27ef9d18bd0f6ce6f0f54cdc2b014f7c80dc9f2873196be5fafd8ba27617d`; engine receipt `sha256:1c3ddd7a6308e1ae90df969f0866502e89ab06df231c3df724838f4fa6fb0b98` at `receipts/engine/model_admitted/SFT-COMP-SCIX-FORWARD-BACKWARD-ERROR-004-1c3ddd7a6308e1ae.json`.

`SFT-COMP-OBL-SCIX-005` - Conditioning and sensitivity

- **Claim and ownership:** `SFT-COMP-SCIX-CONDITIONING-005`; Classical Computation family `SCIX`; Conditioning and sensitivity.
- **Forced law:** Conditioning is the exact output-distinction amplification relative to an input distinction over a registered support; sensitivity is never inferred from one favorable perturbation.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__input-output-sensitivity-ratio__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-005 uniquely retains input-output-sensitivity-ratio, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-STABILITY-001 -> SFT-COMP-SCIX-FORWARD-BACKWARD-ERROR-004. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-005: conditioning; exact execution: {"amplification":"2/1","input_separation":"1/1","output_separation":"2/1"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden

branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:97629a8324b72cb8ba7e71530bb2b90fae1ab02fc93e198fc24e5b070347813f`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:6b508a4852862ce780c813f909c8ba322629ec92295d0d9388f33706118cc3e4`; engine receipt `sha256:5d9c3952e1472f918727bf655f82a7779a118e79b6ed9e15af68d9210ee707d4` at `receipts/engine/model_admitted/SFT-COMP-SCIX-CONDITIONING-005-5d9c3952e1472f91.json`.

`SFT-COMP-OBL-SCIX-006` - Numerical stability under composition

- **Claim and ownership:** `SFT-COMP-SCIX-STABILITY-COMPOSITION-006`; Classical Computation family `SCIX`; Numerical stability under composition.
- **Forced law:** A composed numerical process is stable only when each transition transports an exact enclosure and the terminal enclosure contains every generated perturbation branch.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__composed-error-enclosure__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-006 uniquely retains composed-error-enclosure, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-STABILITY-001 -> SFT-COMP-SCIX-CONDITIONING-005. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-006: stability_composition; exact execution: {"all_step_errors_retained":true,"steps":2,"terminal_enclosure":["27/10","33/10"]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:ef1d4884dc17f0f31e3bdb06eb1b7ef1c58a68cf15bd73acbadbaf140955fd3a`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`;

independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:7b5b839cdf7cbe810d9a37e16383b65c3930f7dd66dfe5bc19afb99c12323f7c`; engine receipt `sha256:638203d938543911fa1934c6e2812bf7f9633778432e5212c5cde289feb18aab` at `receipts/engine/model_admitted/SFT-COMP-SCIX-STABILITY-COMPOSITION-006-638203d938543911.json`.

`SFT-COMP-OBL-SCIX-007` - Convergence order and stopping certificate

- **Claim and ownership:** `SFT-COMP-SCIX-CONVERGENCE-STOPPING-007`; Classical Computation family `SCIX`; Convergence order and stopping certificate.
- **Forced law:** Convergence retains the exact error sequence and reduction ratios; stopping requires a registered residual or enclosure certificate rather than a fitted floating tolerance.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__exact-convergence-ratio-stopping__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-007 uniquely retains exact-convergence-ratio-stopping, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-CONVERGENCE-001 -> SFT-COMP-SCIX-STABILITY-COMPOSITION-006. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-007: convergence; exact execution: {"errors":["1/2","1/4","1/8"],"reduction_ratios":["2/1","2/1"],"stopping_certificate":"registered-error-sequence"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:badc6ed0a6cc1ea004e4ec45a10cd8bd4095a141ffe08b7e3635178ce021a2c0`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:637e39906ff4ec84f7be5203a9b6a4d5c72532d8cd30fe6a53b7a9ded5a23c76`; engine receipt `sha256:3186d61ac593644f8382867ff73dccf9751b34d3ad8e988b4965a40a0c8c37af` at `receipts/engine/model_admitted/SFT-COMP-SCIX-CONVERGENCE-STOPPING-007-3186d61ac593644f.json`.

`SFT-COMP-OBL-SCIX-008` - Discretization and consistency

- **Claim and ownership:** `SFT-COMP-SCIX-DISCRETIZATION-008`; Classical Computation family `SCIX`; Discretization and consistency.
- **Forced law:** Discretization replaces a declared finite domain with a generated mesh; consistency compares each discrete relation with the source relation under exact refinement custody.

- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__mesh-consistency-ledger__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-008 uniquely retains mesh-consistency-ledger, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-DISCRETIZATION-001 -> SFT-COMP-SCIX-CONVERGENCE-STOPPING-007. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-008: discretization; exact execution: {"interval":["1/1","2/1"],"mesh":["1/1","5/4","3/2","7/4","2/1"],"pieces":4}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:abf795c492f51d917a7f743707e75aad6a7d731852feca7a4e2fd124954e17da`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:6d4627275231decb7dee073d4dfbc6e99e96c0060efd37f0c5612db5e7d4c5a3`; engine receipt `sha256:7e2ac4ca4d88351b3194eb9ec9e4e63713811f381a1b381aac4935b29c33a253` at `receipts/engine/model_admitted/SFT-COMP-SCIX-DISCRETIZATION-008-7e2ac4ca4d88351b.json`.

`SFT-COMP-OBL-SCIX-009` - Interpolation and approximation custody

- **Claim and ownership:** `SFT-COMP-SCIX-INTERPOLATION-009`; Classical Computation family `SCIX`; Interpolation and approximation custody.
- **Forced law:** Interpolation retains nodes, basis, evaluation point, interpolant and exact residual enclosure; approximation quality is stated only on the registered domain support.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__interpolant-residual-custody__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-009 uniquely retains interpolant-residual-custody, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-DISCRETIZATION-001 -> SFT-COMP-SCIX-DISCRETIZATION-008. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SCIX-009: interpolation; exact execution: {"interpolated":"4/1","nodes":[["1/1","2/1"],["3/1","6/1"]],"point":"2/1"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:04e31b6e208c8fc66be1ba6d80fb992988346f7011c955780e705aaef99a8f14`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:b2c52fcb73591af3414a5168803f07e2cb5149e4b1463bd02117f957ac6ea02b`; engine receipt `sha256:706080777f8f25f8e0b0c0ba13d6e64676bb3d6c323339b294e0e5782d922e7c` at `receipts/engine/model_admitted/SFT-COMP-SCIX-INTERPOLATION-009-706080777f8f25f8.json`.

`SFT-COMP-OBL-SCIX-010` - Quadrature and exact residual bounds

- **Claim and ownership:** `SFT-COMP-SCIX-QUADRATURE-010`; Classical Computation family `SCIX`; Quadrature and exact residual bounds.
- **Forced law:** Quadrature is an exact weighted sum over generated nodes whose residual is separately enclosed or evaluated against a symbolic integral on the declared function class.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__quadrature-residual-enclosure__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-010 uniquely retains quadrature-residual-enclosure, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-EXACT-APPROXIMATE-001 -> SFT-COMP-SCIX-INTERPOLATION-009. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-010: quadrature; exact execution: {"function":"square","nodes":["1/1","2/1","3/1"],"residual_boundary_retained":true,"trapezoid_sum":"9/1"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:97902775d5f2fb9150465c553e0d8c8e013ea50af391de14ea34556d1fb941d5`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:3045f4d4a56839e3f7af89288fc269deb173930c27d854166181d03d0cce1953`; engine receipt `sha256:f17f245b625c883f6361ef370b156ff9f0e6e0b5824b415391fc03e571bbc3ab` at `receipts/engine/model_admitted/SFT-COMP-SCIX-QUADRATURE-010-f17f245b625c883f.json`.

`SFT-COMP-OBL-SCIX-011` - Root-finding and interval custody

- **Claim and ownership:** `SFT-COMP-SCIX-ROOT-INTERVAL-011`; Classical Computation family `SCIX`; Root-finding and interval custody.
- **Forced law:** Root finding preserves a nested interval or exact residual whose endpoints retain opposite ordered relation labels; a point is exact only when its substituted relation closes.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__nested-root-interval__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-011 uniquely retains nested-root-interval, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-CONVERGENCE-001 -> SFT-COMP-SCIX-QUADRATURE-010. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-011: root_interval; exact execution: {"exact_root":"2/1","initial_interval":["1/1","3/1"],"substitution_closes":true,"target_square":"4/1"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:b75fbbd6575fe9f011d0e62265a7026716e8d839a94d4a213599202833dd676b`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:0424e2b6ffae256202dbce90f91e794d5b048129157f595b805c15f71e62443a`; engine receipt `sha256:80eefe844eff96b435c08072e5ff6bd0908a04f0f6268a0b0892507ccb232265` at `receipts/engine/model_admitted/SFT-COMP-SCIX-ROOT-INTERVAL-011-80eefe844eff96b4.json`.

`SFT-COMP-OBL-SCIX-012` - Exact and approximate linear-system solving

- **Claim and ownership:** `SFT-COMP-SCIX-LINEAR-SYSTEM-012`; Classical Computation family `SCIX`; Exact and approximate linear-system solving.

- **Forced law:** A linear-system solution retains every equation, exact elimination step and residual; approximate solving additionally retains conditioning and componentwise enclosures.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__residual-certified-linear-solution__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-012 uniquely retains residual-certified-linear-solution, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-STABILITY-001 -> SFT-COMP-SCIX-ROOT-INTERVAL-011. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-012: linear_system; exact execution: {"equations":["x+y=3","x+2y=5"],"residuals":["absence","absence"],"solution":{"x":"1/1","y":"2/1"}}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:dbe958c63d3a976ce9851ce8b14787895b735f5d069c71131637c254834fc5ae`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:8c869435bea10ada62000ca619b2cb4030472e6926bafe455f6bf25344516982`; engine receipt `sha256:4a587a26b1637905f09ca4076c568e13b070f1c4a59588f1a4ef6ea37c474400` at `receipts/engine/model_admitted/SFT-COMP-SCIX-LINEAR-SYSTEM-012-4a587a26b1637905.json`.

`SFT-COMP-OBL-SCIX-013` - Eigenvalue and mode computation boundary

- **Claim and ownership:** `SFT-COMP-SCIX-EIGEN-MODES-013`; Classical Computation family `SCIX`; Eigenvalue and mode computation boundary.
- **Forced law:** Mode computation retains the declared operator, candidate eigenvalue, vector and exact residual; spectral claims do not export beyond the finite operator or enclosure.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__mode-residual-boundary__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-013 uniquely retains mode-residual-boundary, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001 -> SFT-COMP-SCIX-LINEAR-SYSTEM-012. Registered root

theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SCIX-013: eigen_modes; exact execution: {"modes":[{"value":"2/1","vector":["1/1","absence"]},{"value":"3/1","vector":["absence","1/1"]}],"operator":"diagonal(2,3)","residuals_close":true}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f61bb29b5045eaa1f5610afc4cc81ec6d29af8050517fdd6a0bdefe4e66d756e`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:6744007cd5dbef495a524c4f2bbe7b1e5dae6a9996c4e2ecb6f4e8e5b05a7f6f`; engine receipt `sha256:faab21d93c3569d70ab00eb9c910efcc1702b38fa9824b4ceb083072c08a5126` at `receipts/engine/model_admitted/SFT-COMP-SCIX-EIGEN-MODES-013-faab21d93c3569d7.json`.

`SFT-COMP-OBL-SCIX-014` - Ordinary differential-system discretization

- **Claim and ownership:** `SFT-COMP-SCIX-ORDINARY-SYSTEM-014`; Classical Computation family `SCIX`; Ordinary differential-system discretization.
- **Forced law:** Ordinary differential computation retains the initial state, update relation, time mesh, local residual and propagated enclosure at every finite step.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__time-step-consistency-trace__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-014 uniquely retains time-step-consistency-trace, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-COMPUTATIONAL-DYNAMICS-001 -> SFT-COMP-SCIX-EIGEN-MODES-013. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-014: ordinary_system; exact execution: {"derivative":"y","initial":"1/1","step":"1/2","terminal":"3/2","trace_retained":true}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes

protected authority.

- **Evidence identities:** derivation `sha256:dc759175d7a479703c3934ab0c5294c604a40b8d0991e66347d72baaf22789f7`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:bd0113801f996b913562b0c2f55bcf00be5c824f30dbbd1b39f55cf2ed99cbb5`; engine receipt `sha256:4093d5dc1741e934fe74530d37058a6b2503c72675ab0264508d4e282c18eb91` at `receipts/engine/model_admitted/SFT-COMP-SCIX-ORDINARY-SYSTEM-014-4093d5dc1741e934.json`.

`SFT-COMP-OBL-SCIX-015` - Partial differential-system discretization

- **Claim and ownership:** `SFT-COMP-SCIX-PARTIAL-SYSTEM-015`; Classical Computation family `SCIX`; Partial differential-system discretization.
- **Forced law:** Partial differential computation retains the space-time mesh, boundary and initial records, stencil relation, stability evidence and refinement comparison.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__space-time-grid-consistency__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-015 uniquely retains space-time-grid-consistency, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-DISCRETIZATION-001 -> SFT-COMP-SCIX-ORDINARY-SYSTEM-014. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-015: partial_system; exact execution: {"boundary_retained":true,"smoothed_interior":[2,3,4],"source_row":[1,2,3,4,5]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:f52aba9aa9a44db7010490f193fcea607bf61287ce899b3084d57e50c0538b90`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:d27111e4be1b64c823d073287a5fe7e503dfbb72d3a7dd9508057703d170dcd0`; engine receipt `sha256:49cc1bb64ed0ec8d967548abe8668ce6cffe2780f1b94dfac4f8c8093eda694e` at `receipts/engine/model_admitted/SFT-COMP-SCIX-PARTIAL-SYSTEM-015-49cc1bb64ed0ec8d.json`.

`SFT-COMP-OBL-SCIX-016` - Stochastic-simulation deterministic-support correspondence

- **Claim and ownership:** `SFT-COMP-SCIX-STOCHASTIC-SUPPORT-016`; Classical Computation family `SCIX`; Stochastic-simulation deterministic-support correspondence.
- **Forced law:** Stochastic simulation corresponds to the complete deterministic support of generated branch labels, with every path, weight and outcome retained and no stochastic cause imported.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__complete-deterministic-path-support__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-016 uniquely retains complete-deterministic-path-support, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-COMP-SCIX-PARTIAL-SYSTEM-015. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-016: stochastic_support; exact execution: {"branch_labels":2,"complete_paths":8,"depth":3,"stochastic_cause_used":false}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:11693622f433ecaf5c59d1e9ad01b8098a639f08a471e8b57f11c1ca8338afe4`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:172ea748ac050380da9e146276996308eae725cdccf665c1991ef966ddb63e54`; engine receipt `sha256:f0aff985707147ecf0c4774347e4c55889b03a67ad65468bce7db9f4c34afd39` at `receipts/engine/model_admitted/SFT-COMP-SCIX-STOCHASTIC-SUPPORT-016-f0aff985707147ec.json`.

`SFT-COMP-OBL-SCIX-017` - Monte-Carlo support and sampling correspondence

- **Claim and ownership:** `SFT-COMP-SCIX-MONTE-CARLO-017`; Classical Computation family `SCIX`; Monte-Carlo support and sampling correspondence.
- **Forced law:** Monte Carlo corresponds to a declared subset or complete generated support with exact branch averages and sampling custody; a sampled estimate never becomes an exact whole without enclosure evidence.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__support-average-sampling-ledger__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-017 uniquely retains support-average-sampling-ledger, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-COMP-SCIX-STOCHASTIC-SUPPORT-016. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-017: monte_carlo_support; exact execution: {"exact_average":"5/2","sample_promoted_to_whole":false,"support_values":[1,2,3,4],"support_width":4}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:69c2c22c44d84c92cba2be88ae657ea2225dffd3d1e2008703cedb80510ffdb5`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:8773125e10a8426078bf7c9d4b95cb9997f58646c4efe31f311510a60e4ed14f`; engine receipt `sha256:e9a9aaa784128e934df6eabda95521eeec9f1e144ded58198f6bffa4cfcf2dbf` at `receipts/engine/model_admitted/SFT-COMP-SCIX-MONTE-CARLO-017-e9a9aaa784128e93.json`.

`SFT-COMP-OBL-SCIX-018` - Inverse-problem identifiability and regularization boundary

- **Claim and ownership:** `SFT-COMP-SCIX-INVERSE-IDENTIFIABILITY-018`; Classical Computation family `SCIX`; Inverse-problem identifiability and regularization boundary.
- **Forced law:** An inverse problem is identifiable exactly when each observed image has one admissible parameter predecessor; regularization selects only through an independently forced structural relation.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__forward-map-identifiability-fibres__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-018 uniquely retains forward-map-identifiability-fibres, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-INVERSE-PROBLEM-001 -> SFT-COMP-SCIX-MONTE-CARLO-017. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-018: inverse_identifiability; exact execution: {"identifiable":false,"image_same_predecessors":["a","b"],"parameters":["a","b","c"],"regularizer_selected":false}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:dd1f1b31396856e09d4ec40a14677de4168717fea43d2af4a17f21677263b241`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:6146f410ba419fc0f295a8eff3e809cfd8e808f83cecc11da84055f4c38dafc7`; engine receipt `sha256:49b61f2bdba7a6f227f6f40403032d511b57cec32fc7e3b893dc78f33257c96e` at `receipts/engine/model_admitted/SFT-COMP-SCIX-INVERSE-IDENTIFIABILITY-018-49b61f2bdba7a6f2.json`.

`SFT-COMP-OBL-SCIX-019` - Computational statistics and estimator custody

- **Claim and ownership:** `SFT-COMP-SCIX-COMPUTATIONAL-STATISTICS-019`; Classical Computation family `SCIX`; Computational statistics and estimator custody.

- **Forced law:** A computational estimator retains its complete data support, exact statistic, sampling relation, bias or residual ledger and every adverse row.

- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__estimator-support-ledger__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-019 uniquely retains estimator-support-ledger, complete numerical custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-COMPUTATIONAL-STATISTICS-001 -> SFT-COMP-SCIX-INVERSE-IDENTIFIABILITY-018. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SCIX-019: computational_statistics; exact execution: {"complete_support_retained":true,"mean":"2/1","median":"2/1","values":[1,3,2]}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:3c558a9c423347549411a7a05446ceb88a83dbbfb5395c9b96b87d664c351037`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:cb78a2fa1c4b0f02c0552a959a159a3b0bb687224239f2a0a9f5246f43f3521c`; engine receipt sha256:47c01bce3a142e2488bc94eb8b8f54c449b81498274d47c131e534f8cffeec45 at receipts/engine/model_admitted/SFT-COMP-SCIX-COMPUTATIONAL-STATISTICS-019-47c01bce3a142e24.json.

SFT-COMP-OBL-SCIX-020 - High-dimensional and sparse computation

- **Claim and ownership:** SFT-COMP-SCIX-SPARSE-COMPUTATION-020; Classical Computation family SCIX; High-dimensional and sparse computation.
- **Forced law:** High-dimensional computation retains coordinate identities; sparse computation acts only on explicitly stored nonempty entries while preserving absent-coordinate semantics and complexity.
- **Unique surviving structure:** exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__sparse-index-operation-ledger__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary. Exact result: SCIX-020 uniquely retains sparse-index-operation-ledger, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-HIGH-DIMENSIONAL-001 -> SFT-COMP-SCIX-COMPUTATIONAL-STATISTICS-019. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** SCIX-020: sparse_computation; exact execution: {"absent_coordinates_retained":true,"result":[8,"absence",15],"stored_entries":2,"width":3}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation sha256:4f829a4261873533e67b76e4590a9c8cbf3b26ce9db3f894627a6dedc7135f10; independent implementation sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c; independent certificate sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337; empirical/external sha256:b62aa05880f45d8998a0abbdeb951db535042603d0d8ac847ba8816e7252c6a2; engine receipt sha256:f1ed9b4cf0352258f2949bafe28ffa8e75ae80ff092eb776589ba891d9602f48 at receipts/engine/model_admitted/SFT-COMP-SCIX-SPARSE-COMPUTATION-020-f1ed9b4cf0352258.json.

SFT-COMP-OBL-SCIX-021 - Many-body state-space organization

- **Claim and ownership:** SFT-COMP-SCIX-MANY-BODY-021; Classical Computation family SCIX; Many-body state-space organization.
- **Forced law:** Many-body computation organizes the complete product of retained component states and every interaction edge; reductions require exact symmetry or compression certificates.
- **Unique surviving structure:** exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__product-state-many-body-organization__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary. Exact result: SCIX-021 uniquely retains

product-state-many-body-organization, complete numerical custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-MANY-BODY-001 -> SFT-COMP-SCIX-SPARSE-COMPUTATION-020. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-021: many_body; exact execution: {"bodies":4,"component_states":2,"interactions_hidden":false,"product_states":16}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:5162786e12585e5cf3ea75c5ffd98d319f49fe288da49e9cfadfa3b92166281d`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:d4dcc73ee5dcb189d3e4a0e6422481635cf0ad9f7ef23345f533e2c2caee8109`; engine receipt `sha256:ce3a4d103911ceb9bba216ba594faba7ace61018364ea674e458de6f7dc0312a` at `receipts/engine/model_admitted/SFT-COMP-SCIX-MANY-BODY-021-ce3a4d103911ceb9.json`.

`SFT-COMP-OBL-SCIX-022` - Symbolic-numeric correspondence

- **Claim and ownership:** `SFT-COMP-SCIX-SYMBOLIC-NUMERIC-022`; Classical Computation family `SCIX`; Symbolic-numeric correspondence.
- **Forced law:** Symbolic-numeric correspondence evaluates one canonical symbolic expression into exact rational values or enclosures and retains every rewrite, substitution and residual.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__symbolic-exact-evaluation-correspondence__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-022 uniquely retains symbolic-exact-evaluation-correspondence, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-SYMBOLIC-001 -> SFT-COMP-SCIX-MANY-BODY-021. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-022: symbolic_numeric; exact execution: {"exact_output":"9/1","expression":"1+2x+x^2","input":"2/1","rewrite_trace_retained":true}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:d12e9f5a28611e009146c95e52f37f4cb64f0ce96a9de8614198881b741062ba`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:13ee6fab062c8d303109c21506077aa4fbdb1d84e20ae9c0e4943c49eea226ac`; engine receipt `sha256:260edf1246e3aaaae8bd9da08ab57d55079f15eaf82639c7bff7060efd496431` at `receipts/engine/model_admitted/SFT-COMP-SCIX-SYMBOLIC-NUMERIC-022-260edf1246e3aaaa.json`.

`SFT-COMP-OBL-SCIX-023` - Simulation verification and validation

- **Claim and ownership:** `SFT-COMP-SCIX-SIMULATION-VALIDATION-023`; Classical Computation family `SCIX`; Simulation verification and validation.

- **Forced law:** Simulation verification compares implementation with its declared equations, while validation compares registered consequences with external observations opened after prediction sealing; neither substitutes for the other.

- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__code-equation-data-validation-ledger__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-023 uniquely retains code-equation-data-validation-ledger, complete numerical custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-SIMULATION-001 -> SFT-COMP-SCIX-SYMBOLIC-NUMERIC-022. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SCIX-023: simulation_validation; exact execution: {"equation":"two-times-input","equation_output":"6/1","external_validation_separate":true,"implementation_output":"6/1","input":"3/1"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:73d89c259b76b092690b3e4283a9e40e4149f14137b976416e269149aa9e1d70`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`;

empirical/external `sha256:64af4cfd409a2f6de92f2506e90c77e4c0f0293b6104b66b2780da71581fcce5`; engine receipt `sha256:faa8d98dc849ef1b84b040db3cf804cf34f67132c6f52a8ecd576c8514f867fb` at `receipts/engine/model_admitted/SFT-COMP-SCIX-SIMULATION-VALIDATION-023-faa8d98dc849ef1b.json`.

`SFT-COMP-OBL-SCIX-024` - Reproducible mathematical-model provenance

- **Claim and ownership:** `SFT-COMP-SCIX-MODEL-PROVENANCE-024`; Classical Computation family `SCIX`; Reproducible mathematical-model provenance.
- **Forced law:** A reproducible mathematical model retains source identities, equations, assumptions, inputs, transformations, outputs, uncertainties, software environment and immutable result receipts.
- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__source-model-input-output-provenance__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-024 uniquely retains source-model-input-output-provenance, complete numerical custody, root forcing, post-registry execution and no extra rule.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-SCIX-SIMULATION-VALIDATION-023. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** SCIX-024: model_provenance; exact execution: {"equation":"two-times-input","inputs":["3/1"],"output":"6/1","receipt_bound":true,"source":"registered"}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.
- **Evidence identities:** derivation `sha256:389aafa725818666025c970cf1bc84989ae1282f15ef4714b8a17c843c787817`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:e3db368c328b15050279e198a1e59aab178bc37bee1a45061a69011060d30fd0`; engine receipt `sha256:a4c13c0e5c6508aeaf516ae056755ea8ef564d38bd7ccf810da767e3f35f7cd8` at `receipts/engine/model_admitted/SFT-COMP-SCIX-MODEL-PROVENANCE-024-a4c13c0e5c6508ae.json`.

`SFT-COMP-OBL-SCIX-025` - Scientific-computation completeness certificate

- **Claim and ownership:** `SFT-COMP-SCIX-COMPLETENESS-025`; Classical Computation family `SCIX`; Scientific-computation completeness certificate.
- **Forced law:** Scientific-computation completeness is the one-to-one reconciliation of all twenty-five frozen obligations with unique survivors, adverse controls, exact executions, independent reconstructions and untouched-engine receipts.

- **Unique surviving structure:** `exact-rational-or-symbolic-enclosure__complete-trace-error-ledger__twenty-five-obligation-no-omission-ledger__complete-verification-validation-ledger__literal-complete-product__there-is-no-nothing-lineage__post-registry-exact-scientific-execution__declared-model-mesh-data-boundary`. Exact result: SCIX-025 uniquely retains twenty-five-obligation-no-omission-ledger, complete numerical custody, root forcing, post-registry execution and no extra rule.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-SCI-MATHEMATICAL-MODELLING-001 -> SFT-COMP-SCIX-MODEL-PROVENANCE-024. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** SCIX-025: scientific_completeness; exact execution: {"duplicate_owners":0,"independent_execution_rows":25,"omitted_owners":0,"registered_obligations":25}; complete family observation custody: 25 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-INDEPENDENT-EXACT-SCIENTIFIC-OBSERVER, SFT-V1-V2-SCIENTIFIC-COMPUTATION-OBSERVATION-CORPUS. External status: `empirically_tested_and_independently_replicated`.

- **Prohibited imports and boundary:** no axiom, imported numerical model or target outcome selects the survivor; host absence and artifact counters are not admitted numerical-zero objects; no negative, irrational, imaginary, floating or completed-infinite proof scalar; no hidden precision, tolerance, sample, mesh, approximation, input or external target; no favorable simulation substitutes for verification and external validation; no failed route retires an obligation or changes protected authority.

- **Evidence identities:** derivation `sha256:e19df6ae76233eeb8ec090621ba1fd1cb125b8b2b8f7f10b087f1b3221f65137`; independent implementation `sha256:29674c26dbdb34d8f35470b6c848411d8475d527a60899fa7678e7b3084ffa1c`; independent certificate `sha256:68d3a7243817854d47ac5bebe15dbe0fade25f86ee13077c84e19a83f9acf337`; empirical/external `sha256:91cce41f69c065874f8a485c64e8803390d5122666715221d855c2d71912f4c3`; engine receipt `sha256:e90843fef110814cfe879738e175945b56260eed2950bf4aa9382c209593a515` at `receipts/engine/model_admitted/SFT-COMP-SCIX-COMPLETENESS-025-e90843fef110814c.json`.

130.13 Family `VALID` - 12/12 complete

Scope. Dated validation and reconciliation. This family completely decides 3,072 candidates, retains 12 unique survivors and passes 48 adverse controls.

`SFT-COMP-OBL-VALID-001` - Formal-computation complete validation vector

- **Claim and ownership:** `SFT-COMP-VALID-FORMAL-VECTOR-001`; Classical Computation family `VALID`; Formal-computation complete validation vector.

- **Forced law:** Formal-computation complete validation vector is the exact one-to-one reconciliation of its frozen obligations, current untouched-engine receipts, complete candidate enumerations, unique survivors, four adverse controls per claim, post-registry observations, implementation-distinct reconstructions and explicit scope boundaries.

- **Unique surviving structure:** `complete-frozen-obligation-map__exact-current-receipt-replay__formal-family-receipt-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary`. Exact result: VALID-001 uniquely retains formal-family-receipt-vector, exact evidence custody, root lineage and no extra admission.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.

- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-FORMX-COMPLETENESS-022. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** VALID-001: formx_validation; exact execution: {"candidate_rows":5632,"claims":22,"family":"FORMX","independent_reconstructions":22,"passed_controls":88,"post_registry_observations":22,"receipt_replay":"22/22","unique_survivors":22}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation `sha256:75751b3cad4a737f1cf034bffbbf351a10e34163f4be3048a8d0fc9c355b189a`; independent implementation `sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0`; independent certificate `sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; empirical/external `sha256:cc9b0c6cd51d3170257ef0927ec4b12de2e044b902bd40afa408058195333b9b`; engine receipt `sha256:2214e4f5c89ae7ad15ccc694fe5f0e288eea4e2f7c29197a1dc0b7504fb2e511` at `receipts/engine/model_admitted/SFT-COMP-VALID-FORMAL-VECTOR-001-2214e4f5c89ae7ad.json`.

`SFT-COMP-OBL-VALID-002` - Computability complete validation vector

- **Claim and ownership:** `SFT-COMP-VALID-COMPUTABILITY-VECTOR-002`; Classical Computation family `VALID`; Computability complete validation vector.
- **Forced law:** Computability complete validation vector is the exact one-to-one reconciliation of its frozen obligations, current untouched-engine receipts, complete candidate enumerations, unique survivors, four adverse controls per claim, post-registry observations, implementation-distinct reconstructions and explicit scope boundaries.
- **Unique surviving structure:** `complete-frozen-obligation-map__exact-current-receipt-replay__computability-family-receipt-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary`. Exact result: VALID-002 uniquely retains computability-family-receipt-vector, exact evidence custody, root lineage and no extra admission.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CBLX-COMPLETENESS-021 -> SFT-COMP-VALID-FORMAL-VECTOR-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** VALID-002: cblx_validation; exact execution: {"candidate_rows":5376,"claims":21,"family":"CBLX","independent_reconstructions":21,"passed_controls":84,"post_registry_observations":21,"receipt_replay":"21/21","unique_survivors":21}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no

bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.

- **Evidence identities:** derivation `sha256:c09d70e57df4f246d9937a94bcb09b4e692ae78409a4b08146a75d0d497faee3`; independent implementation `sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0`; independent certificate `sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; empirical/external `sha256:eb2af18ebead733d763555efdd071b55bed22214244a3de7287e6c9c3e0aa55a`; engine receipt `sha256:169f52f6943eb9820aa3ed78670f24269c81664551b0e692c02a121db4db7e51` at `receipts/engine/model_admitted/SFT-COMP-VALID-COMPUTABILITY-VECTOR-002-169f52f6943eb982.json`.

`SFT-COMP-OBL-VALID-003` - Complexity complete validation vector

- **Claim and ownership:** `SFT-COMP-VALID-COMPLEXITY-VECTOR-003`; Classical Computation family `VALID`; Complexity complete validation vector.
- **Forced law:** Complexity complete validation vector is the exact one-to-one reconciliation of its frozen obligations, current untouched-engine receipts, complete candidate enumerations, unique survivors, four adverse controls per claim, post-registry observations, implementation-distinct reconstructions and explicit scope boundaries.
- **Unique surviving structure:** `complete-frozen-obligation-map__exact-current-receipt-replay__complexity-family-receipt-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary`. Exact result: VALID-003 uniquely retains complexity-family-receipt-vector, exact evidence custody, root lineage and no extra admission.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-VALID-COMPUTABILITY-VECTOR-002. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** VALID-003: cplxx_validation; exact execution: {"candidate_rows":8448,"claims":33,"family":"CPLXX","independent_reconstructions":33,"passed_controls":132,"post_registry_observations":33,"receipt_replay":"33/33","unique_survivors":33}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation `sha256:fb1ca642c1b04ec315249b0cee4c17fd674bf68fb5c8975b665273c815119137`; independent implementation `sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0`; independent certificate `sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; empirical/external `sha256:a2a03e293495bb0d1f5febb6661881e0231ce7f10ff6b7327694fe11ba2677d8`; engine receipt `sha256:8296b2a8bce1d2a4b4a6134a4d87a2558a45fb6b232501d5cc3db6cb931c93c6` at `receipts/engine/model_admitted/SFT-COMP-VALID-COMPLEXITY-VECTOR-003-8296b2a8bce1d2a4.json`.

SFT-COMP-OBL-VALID-004 - Algorithms and data-organization complete validation vector

- **Claim and ownership:** `SFT-COMP-VALID-ALGORITHM-VECTOR-004`; Classical Computation family `VALID`; Algorithms and data-organization complete validation vector.
- **Forced law:** Algorithms and data-organization complete validation vector is the exact one-to-one reconciliation of its frozen obligations, current untouched-engine receipts, complete candidate enumerations, unique survivors, four adverse controls per claim, post-registry observations, implementation-distinct reconstructions and explicit scope boundaries.
- **Unique surviving structure:** `complete-frozen-obligation-map__exact-current-receipt-replay__algorithm-family-receipt-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary`. Exact result: VALID-004 uniquely retains algorithm-family-receipt-vector, exact evidence custody, root lineage and no extra admission.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-ALGX-COMPLETENESS-031 -> SFT-COMP-VALID-COMPLEXITY-VECTOR-003. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** VALID-004: algx_validation; exact execution: {"candidate_rows":7936,"claims":31,"family":"ALGX","independent_reconstructions":31,"passed_controls":124,"post_registry_observations":31,"receipt_replay":"31/31","unique_survivors":31}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation `sha256:dcb133f675d87981b198f391f79750b241a00b893791cacaf6e47641311a83f7`; independent implementation `sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0`; independent certificate `sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; empirical/external `sha256:967d40cc917d5c0a810c303046ff5d05edf9f1b0fc13c4bb9e813668bf06d2e3`; engine receipt `sha256:69e96ff30bdc4259556241a51ccf3e61ebd2166a10ae860bd94c73e46e0362b2` at `receipts/engine/model_admitted/SFT-COMP-VALID-ALGORITHM-VECTOR-004-69e96ff30bdc4259.json`.

SFT-COMP-OBL-VALID-005 - Semantics and programming-theory complete validation vector

- **Claim and ownership:** `SFT-COMP-VALID-SEMANTICS-VECTOR-005`; Classical Computation family `VALID`; Semantics and programming-theory complete validation vector.
- **Forced law:** Semantics and programming-theory complete validation vector is the exact one-to-one reconciliation of its frozen obligations, current untouched-engine receipts, complete candidate enumerations, unique survivors, four adverse controls per claim, post-registry observations, implementation-distinct reconstructions and explicit scope boundaries.
- **Unique surviving structure:** `complete-frozen-obligation-map__exact-current-receipt-replay__semantics-family-receipt-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary`. Exact result: VALID-005 uniquely retains semantics-family-receipt-vector, exact evidence custody, root lineage and no extra admission.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SEMX-COMPLETENESS-025 -> SFT-COMP-VALID-ALGORITHM-VECTOR-004. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** VALID-005: semx_validation; exact execution: {"candidate_rows":6400,"claims":25,"family":"SEMX","independent_reconstructions":25,"passed_controls":100,"post_registry_observations":25,"receipt_replay":"25/25","unique_survivors":25}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation `sha256:3cdd449a1176f39adfd53e30175ae32982b6ae5fb62083eae567bd9174266eb1`; independent implementation `sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0`; independent certificate `sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; empirical/external `sha256:22c2751e2f3090a0bf0cb51893fff8e85a0cc75c05bb4f579ff40982bf05e050`; engine receipt `sha256:e30690b6087cf1eab319a1420f9bc902159ad30f77cf8e14182205d501dd605f` at `receipts/engine/model_admitted/SFT-COMP-VALID-SEMANTICS-VECTOR-005-e30690b6087cf1ea.json`.

`SFT-COMP-OBL-VALID-006` - Concurrency and distributed complete validation vector

- **Claim and ownership:** `SFT-COMP-VALID-DISTRIBUTED-VECTOR-006`; Classical Computation family `VALID`; Concurrency and distributed complete validation vector.
- **Forced law:** Concurrency and distributed complete validation vector is the exact one-to-one reconciliation of its frozen obligations, current untouched-engine receipts, complete candidate enumerations, unique survivors, four adverse controls per claim, post-registry observations, implementation-distinct reconstructions and explicit scope boundaries.
- **Unique surviving structure:** `complete-frozen-obligation-map__exact-current-receipt-replay__distributed-family-receipt-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary`. Exact result: VALID-006 uniquely retains distributed-family-receipt-vector, exact evidence custody, root lineage and no extra admission.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-DISTX-COMPLETENESS-026 -> SFT-COMP-VALID-SEMANTICS-VECTOR-005. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** VALID-006: distx_validation; exact execution: {"candidate_rows":6656,"claims":26,"family":"DISTX","independent_reconstructions":26,"passed_controls":104,"post_registry_observations":26,"receipt_replay":"26/26","unique_survivors":26}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status:

empirically_tested_and_independently_replicated.

- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation sha256:6fccf2e6e4429b05980c05180d23c3967953717e9bd14d87250b652709d568b8; independent implementation sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0; independent certificate sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; empirical/external sha256:3583b222e0cd9eaddab95cdd681f696fa83190e38944b639acb929ea32b3453b; engine receipt sha256:833de968a9cb7abd35b2a08a889d570e055976acf6d75d01424b6a10651a1c2e at receipts/engine/model_admitted/SFT-COMP-VALID-DISTRIBUTED-VECTOR-006-833de968a9cb7abd.json.

SFT-COMP-OBL-VALID-007 - Cryptography and security complete validation vector

- **Claim and ownership:** SFT-COMP-VALID-SECURITY-VECTOR-007; Classical Computation family VALID; Cryptography and security complete validation vector.
- **Forced law:** Cryptography and security complete validation vector is the exact one-to-one reconciliation of its frozen obligations, current untouched-engine receipts, complete candidate enumerations, unique survivors, four adverse controls per claim, post-registry observations, implementation-distinct reconstructions and explicit scope boundaries.
- **Unique surviving structure:** complete-frozen-obligation-map__exact-current-receipt-replay__security-family-receipt-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary. Exact result: VALID-007 uniquely retains security-family-receipt-vector, exact evidence custody, root lineage and no extra admission.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SECX-COMPLETENESS-025 -> SFT-COMP-VALID-DISTRIBUTED-VECTOR-006. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** VALID-007: secx_validation; exact execution: {"candidate_rows":6400,"claims":25,"family":"SECX","independent_reconstructions":25,"passed_controls":100,"post_registry_observations":25,"receipt_replay":"25/25","unique_survivors":25}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation sha256:3941365e6e0f23bbc862da858eae6f4485b782864e9a6109913ce9ea43705fdd; independent implementation sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0; independent certificate sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; empirical/external sha256:14b9f1801163a8ffd26c32da9bcb23882590e82b79164d85f462bbeef9e139a4; engine receipt sha256:199feee7fae46ae0cb3af0dea58d10e1b061b49c76c6979472c5403c78397e30 at receipts/

engine/model_admitted/SFT-COMP-VALID-SECURITY-VECTOR-007-199feee7fae46ae0.json.

SFT-COMP-OBL-VALID-008 - Learning and intelligence complete validation vector

- **Claim and ownership:** SFT-COMP-VALID-LEARNING-VECTOR-008; Classical Computation family VALID; Learning and intelligence complete validation vector.
- **Forced law:** Learning and intelligence complete validation vector is the exact one-to-one reconciliation of its frozen obligations, current untouched-engine receipts, complete candidate enumerations, unique survivors, four adverse controls per claim, post-registry observations, implementation-distinct reconstructions and explicit scope boundaries.
- **Unique surviving structure:** complete-frozen-obligation-map__exact-current-receipt-replay__learning-family-receipt-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary. Exact result: VALID-008 uniquely retains learning-family-receipt-vector, exact evidence custody, root lineage and no extra admission.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is depth_independent.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-LEARNX-COMPLETENESS-026 -> SFT-COMP-VALID-SECURITY-VECTOR-007. Registered root theorem: SFT-ROOT-THERE-IS-NO-NOTHING. Imported axioms: 0; free parameters: 0.
- **Post-registration observation:** VALID-008: learnx_validation; exact execution: {"candidate_rows":6656,"claims":26,"family":"LEARNX","independent_reconstructions":26,"passed_controls":104,"post_registry_observations":26,"receipt_replay":"26/26","unique_survivors":26}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: empirically_tested_and_independently_replicated.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation sha256:221accb19c2b0555473e1cb49578aa9ca28e44c49b352cab954068612f127dbc; independent implementation sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0; independent certificate sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17; empirical/external sha256:89558225818f3ed3a6aa231550f1d0e484eef94e9fd7a22f4ae8672d511b518a; engine receipt sha256:8f85c9c55cb5f772ddcb17e7a64f963a52f699a0c604e972db1911ced7d94b4c at receipts/engine/model_admitted/SFT-COMP-VALID-LEARNING-VECTOR-008-8f85c9c55cb5f772.json.

SFT-COMP-OBL-VALID-009 - Scientific-computation complete validation vector

- **Claim and ownership:** SFT-COMP-VALID-SCIENTIFIC-VECTOR-009; Classical Computation family VALID; Scientific-computation complete validation vector.
- **Forced law:** Scientific-computation complete validation vector is the exact one-to-one reconciliation of its frozen obligations, current untouched-engine receipts, complete candidate enumerations, unique survivors, four adverse controls per claim, post-registry observations, implementation-distinct reconstructions and explicit scope boundaries.
- **Unique surviving structure:** complete-frozen-obligation-map__exact-current-receipt-replay__scientific-family-receipt-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary. Exact result: VALID-009 uniquely retains scientific-family-receipt-vector, exact evidence custody, root lineage and no extra admission.

- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SCIX-COMPLETENESS-025 -> SFT-COMP-VALID-LEARNING-VECTOR-008. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** VALID-009: scix_validation; exact execution: {"candidate_rows":6400,"claims":25,"family":"SCIX","independent_reconstructions":25,"passed_controls":100,"post_registry_observations":25,"receipt_replay":"25/25","unique_survivors":25}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation `sha256:a38989125e9d47c3af1342579e16cdca019c4b16a6e9ddf242fd20abf6539c67`; independent implementation `sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0`; independent certificate `sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; empirical/external `sha256:ef4606f3927d38eda781b340cfcf8cdf9bb71693df58ac2fbea406aca771299a`; engine receipt `sha256:16de2551ef53859b6dd85b7fa5d2785b7c0830248e156b09a4f196edfaa120bd` at `receipts/engine/model_admitted/SFT-COMP-VALID-SCIENTIFIC-VECTOR-009-16de2551ef53859b.json`.

`SFT-COMP-OBL-VALID-010` - Famous-problem theorem and finite-census boundary vector

- **Claim and ownership:** `SFT-COMP-VALID-FAMOUS-BOUNDARY-010`; Classical Computation family `VALID`; Famous-problem theorem and finite-census boundary vector.
- **Forced law:** The famous-problem vector separates depth-independent theorems, bounded exhaustive censuses and unrestricted open boundaries; no finite Busy-Beaver, Fold-P/Fold-NP, circuit or hypercomputation result silently exports beyond its registered grammar.
- **Unique surviving structure:** `complete-frozen-obligation-map__exact-current-receipt-replay__theorem-finite-frontier-boundary-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary`. Exact result: VALID-010 uniquely retains theorem-finite-frontier-boundary-vector, exact evidence custody, root lineage and no extra admission.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-CPLXX-COMPLETENESS-033 -> SFT-COMP-VALID-SCIENTIFIC-VECTOR-009. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** VALID-010: famous_problem_boundary; exact execution: {"registered_boundary_claims":["SFT-COMP-CBLX-DIAGONAL-LANGUAGE-004","SFT-COMP-CBLX-BUSY-BEAVER-FINITE-CENSUS-019","SFT-COMP-CPLXX-FOLD-P-NP-SCOPE-007","SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024"],"silent_export":false,"theorem_finite_unrestricted_classes_separate":true}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters

- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation `sha256:fd555425d4c3abb345eabb7291f92f5f3cdfb2380732865df00caeb971f7a8e2`; independent implementation `sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0`; independent certificate `sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; empirical/external `sha256:c5d5661daafb0f186db6ae3a157a48238b26fe508f4aa94850e5b2b705d4bf4f`; engine receipt `sha256:21cc1ef42523def5e87ac5eee6b194d094ba54cda2ebe2e4dae2191c39e13c5a` at `receipts/engine/model_admitted/SFT-COMP-VALID-FAMOUS-BOUNDARY-010-21cc1ef42523def5.json`.

`SFT-COMP-OBL-VALID-011` - Adverse, absent, unresolved and ownership-boundary vector

- **Claim and ownership:** `SFT-COMP-VALID-ADVERSE-BOUNDARY-011`; Classical Computation family `VALID`; Adverse, absent, unresolved and ownership-boundary vector.
- **Forced law:** The adverse vector preserves every favorable, adverse, absent, unavailable, unresolved and downstream-owned row without deleting a failed route, selecting on outcomes or changing the protected authority.
- **Unique surviving structure:** `complete-frozen-obligation-map__exact-current-receipt-replay__complete-adverse-ownership-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary`. Exact result: VALID-011 uniquely retains complete-adverse-ownership-vector, exact evidence custody, root lineage and no extra admission.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SCIX-COMPLETENESS-025 -> SFT-COMP-VALID-FAMOUS-BOUNDARY-010. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** VALID-011: adverse_boundary_vector; exact execution: {"adverse_absent_unavailable_and_handoff_rows_preserved":true,"extension_claims":234,"failed_route_retired":false,"passed_control_rows":936}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation `sha256:0a4d8478f82077b313ed0c3099946673935cf8df6dc1e944a1475f8936dde808`; independent implementation `sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0`; independent certificate `sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; empirical/external `sha256:6e2200f6f4b1a4a58bb01f5c039c29bf4aa912921da70f507ec30c4a5221d6df`; engine receipt

`sha256:044dea4ce05cb1dd1c9d2a14a6c2b1bc0a5825e71e8a6e952016e49e5a197948` at `receipts/engine/model_admitted/SFT-COMP-VALID-ADVERSE-BOUNDARY-011-044dea4ce05cb1dd.json`.

`SFT-COMP-OBL-VALID-012` - Classical Computational Science empirical and formal Grand Lock

- **Claim and ownership:** `SFT-COMP-VALID-GRAND-LOCK-012`; Classical Computation family `VALID`; Classical Computational Science empirical and formal Grand Lock.
- **Forced law:** The Classical Computational Science Grand Lock is the exact reconciliation of BASE and all nine completed extension families through the frozen census, receipts, controls, observations, independent certificates and root lineage, with only the separately registered HAND family remaining open at this boundary.
- **Unique surviving structure:** `complete-frozen-obligation-map__exact-current-receipt-replay__classical-computation-grand-lock-vector__all-controls-and-independent-certificates__literal-complete-product__there-is-no-nothing-lineage__complete-post-registry-observation-ledger__explicit-theorem-census-handoff-boundary`. Exact result: VALID-012 uniquely retains classical-computation-grand-lock-vector, exact evidence custody, root lineage and no extra admission.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-SCIX-COMPLETENESS-025 -> SFT-COMP-VALID-ADVERSE-BOUNDARY-011. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** VALID-012: computation_grand_lock; exact execution: {"base_claims":117,"current_receipt_rows":351,"extension_claims":234,"extension_passed_controls":936,"extension_unique_survivors":234,"generated_extension_candidates":59904,"implementation_distinct_reconstructions":234,"open_handoff_rows":6,"open_validation_rows":12,"post_registry_observations":234,"protected_authority_changed":false,"reconciliation_identity":"sha256:b481c02042105d7235af18f1391dd8e7f3471a7a944ca03c6dc168efc701248d"}; complete family observation custody: 12 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V10, SFT-V3-CURRENT-ENGINE-RECEIPTS, SFT-V3-INDEPENDENT-CERTIFICATES. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, target outcome or publication status selects the validation result; no missing, duplicate, stale or noncurrent receipt is accepted; no failed control, reconstruction or observation row is suppressed; no bounded result is exported as an unrestricted theorem; no downstream application substitutes for its owning law; no protected engine or verification-authority edit is permitted.
- **Evidence identities:** derivation `sha256:8d10e7321a4f5c4a28d3b34cb4a7d3c2da8904967473711b934ad515ab6ca009`; independent implementation `sha256:101ed366fcbc360634847e6f724d88f61824020203ce10dda10ab5aca85eb3e0`; independent certificate `sha256:01bdad497ce9ca162a119883efe4fb92f71b4217e582718bc690142e417bda17`; empirical/external `sha256:e0c457df222704646310cef7e75b366ac76d52bea7e213f09af12646e2ea81fc`; engine receipt `sha256:31489be2c8fb8943fe78df5391ce2b191dcb3d8cc7b5281a88478addb67e8a50` at `receipts/engine/model_admitted/SFT-COMP-VALID-GRAND-LOCK-012-31489be2c8fb8943.json`.

130.14 Family `HAND` - 6/6 complete

Scope. One-owner downstream handoffs. This family completely decides 1,536 candidates, retains 6 unique survivors and passes 24 adverse controls.

`SFT-COMP-OBL-HAND-001` - Classical Computation one-owner downstream handoff

- **Claim and ownership:** `SFT-COMP-HAND-ONE-OWNER-001`; Classical Computation family `HAND`; Classical Computation one-owner downstream handoff.

- **Forced law:** Every downstream obligation has exactly one current owner and one explicit input-output boundary; duplicated ownership, orphaned work and application-selected laws halt the handoff.
- **Unique surviving structure:** `one-declared-owner__complete-input-output-interface__one-obligation-one-owner-map__immutable-prior-evidence-custody__literal-complete-product__there-is-no-nothing-lineage__dated-complete-open-extension__explicit-domain-quantum-engineering-boundary`. Exact result: HAND-001 uniquely retains one-obligation-one-owner-map, one-owner custody, immutable prior evidence, open extension and no extra scope.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-VALID-GRAND-LOCK-012. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** HAND-001: one_owner; exact execution: {"duplicate_subjects":0,"orphan_subjects":0,"owner_rows":[{"owner":"quantum-computation","subject":"quantum states, gates, channels and fault tolerance"},{"owner":"engineering-translation","subject":"software, hardware, platforms and side channels"},{"owner":"owning-domain-science","subject":"domain measurements"},{"owner":"frontier-application-rebuild","subject":"Unison AI, Fold Chess, Fold Go and Fold Protein experiments"}]}; complete family observation custody: 6 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V11, SFT-V3-BRANCH-OWNERSHIP-TREE, SFT-V3-OPEN-EXTENSION-POLICY. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, application result or publication state selects a handoff; no obligation has two owners or no owner; no empirical claim bypasses sealed target custody; no classical claim silently absorbs quantum or engineering operations; no lawful extension rewrites a prior receipt; no protected authority edit is permitted.
- **Evidence identities:** derivation `sha256:61b5e8d6fcd377edd9e2b34463cc7e01adbee1f844566174d6da957ced2c2bf0`; independent implementation `sha256:5463cf22bd991837912a99c38ad38c24fd06eaf5a9eaa3d409f6159241660ea4`; independent certificate `sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f`; empirical/external `sha256:7c235d8839cbba3b4beb934de13bbbfb89d7eb2c9261b7f2bac1e7e04fa85c52`; engine receipt `sha256:c66f7306bf18e9aadf4ca9b63c96087ea0db83585443c133b11d9c840a8bcbf8` at `receipts/engine/model_admitted/SFT-COMP-HAND-ONE-OWNER-001-c66f7306bf18e9aa.json`.

`SFT-COMP-OBL-HAND-002` - Classical Computation measurement-boundary handoff

- **Claim and ownership:** `SFT-COMP-HAND-MEASUREMENT-BOUNDARY-002`; Classical Computation family `HAND`; Classical Computation measurement-boundary handoff.
- **Forced law:** Classical Computation owns computational form, trace and resource laws, while measured consequences remain with the relevant physical, chemical, biological, medical, Earth or astronomical branch and return only through sealed comparison.
- **Unique surviving structure:** `one-declared-owner__complete-input-output-interface__domain-measurement-owner-map__immutable-prior-evidence-custody__literal-complete-product__there-is-no-nothing-lineage__dated-complete-open-extension__explicit-domain-quantum-engineering-boundary`. Exact result: HAND-002 uniquely retains domain-measurement-owner-map, one-owner custody, immutable prior evidence, open extension and no extra scope.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-VALID-GRAND-LOCK-012 ->

SFT-COMP-HAND-ONE-OWNER-001. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** HAND-002: measurement_boundary; exact execution: {"computational_owner":"classical-computation","measurement_owner":"owning-domain-science","return_route":"sealed-external-comparison"}; complete family observation custody: 6 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V11, SFT-V3-BRANCH-OWNERSHIP-TREE, SFT-V3-OPEN-EXTENSION-POLICY. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, application result or publication state selects a handoff; no obligation has two owners or no owner; no empirical claim bypasses sealed target custody; no classical claim silently absorbs quantum or engineering operations; no lawful extension rewrites a prior receipt; no protected authority edit is permitted.
- **Evidence identities:** derivation `sha256:13c0a7c14e20df28a1de5868bbee234c76c038bfaa2e725938309d88571b3f4d`; independent implementation `sha256:5463cf22bd991837912a99c38ad38c24fd06eaf5a9eaa3d409f6159241660ea4`; independent certificate `sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f`; empirical/external `sha256:b5785c091a994563fdd81c71750f2b566cf2a723617b6d7433b5af5a65c3b81d`; engine receipt `sha256:ca36797c3c58ac3c2f9572975fcf54bbed302d0826471c4c5a3a8863ae205ad3` at `receipts/engine/model_admitted/SFT-COMP-HAND-MEASUREMENT-BOUNDARY-002-ca36797c3c58ac3c.json`.

`SFT-COMP-OBL-HAND-003` - Classical Computation formal-to-empirical handoff

- **Claim and ownership:** `SFT-COMP-HAND-FORMAL-EMPIRICAL-003`; Classical Computation family `HAND`; Classical Computation formal-to-empirical handoff.
- **Forced law:** A formal result becomes empirically tested only through a value-free target registry, sealed prediction, independently custodied observation, complete comparison and preserved favorable, adverse, absent and unavailable rows.
- **Unique surviving structure:** `one-declared-owner__complete-input-output-interface__formal-prediction-observation-comparison-chain__immutable-prior-evidence-custody__literal-complete-product__there-is-no-nothing-lineage__dated-complete-open-extension__explicit-domain-quantum-engineering-boundary`. Exact result: HAND-003 uniquely retains formal-prediction-observation-comparison-chain, one-owner custody, immutable prior evidence, open extension and no extra scope.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-VALID-GRAND-LOCK-012 -> SFT-COMP-HAND-MEASUREMENT-BOUNDARY-002. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** HAND-003: formal_empirical; exact execution: {"ordered_stages":["formal-law","value-free-registry","sealed-prediction","custodied-observation","complete-comparison"],"suppressed_outcome_rows":0}; complete family observation custody: 6 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V11, SFT-V3-BRANCH-OWNERSHIP-TREE, SFT-V3-OPEN-EXTENSION-POLICY. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, application result or publication state selects a handoff; no obligation has two owners or no owner; no empirical claim bypasses sealed target custody; no classical claim silently absorbs quantum or engineering operations; no lawful extension rewrites a prior receipt; no protected authority edit is permitted.

- **Evidence identities:** derivation `sha256:ee417057a893a8bece9c3d2c95a83cf6907a127cad2953ac07b491ae526e1fd6`; independent implementation `sha256:5463cf22bd991837912a99c38ad38c24fd06eaf5a9eaa3d409f6159241660ea4`; independent certificate `sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f`; empirical/external `sha256:f286f7a0972b42ad57923fbb22eb026db765a736f8fd1218417cbe5398133cb8`; engine receipt `sha256:e5436a3b978b62a59992e52c92021f48eb9fc768a9e67e9b109806a94f78afdd` at `receipts/engine/model_admitted/SFT-COMP-HAND-FORMAL-EMPIRICAL-003-e5436a3b978b62a5.json`.

`SFT-COMP-OBL-HAND-004` - Classical-to-quantum operational handoff

- **Claim and ownership:** `SFT-COMP-HAND-CLASSICAL-QUANTUM-004`; Classical Computation family `HAND`; Classical-to-quantum operational handoff.
- **Forced law:** Classical state, process and information traces hand reversible and quantum operations to Quantum Computation through an explicit semantics-preserving correspondence; quantum states, gates, measurements, channels and fault tolerance remain quantum-owned.
- **Unique surviving structure:** `one-declared-owner__complete-input-output-interface__classical-quantum-operation-boundary__immutable-prior-evidence-custody__literal-complete-product__there-is-no-nothing-lineage__dated-complete-open-extension__explicit-domain-quantum-engineering-boundary`. Exact result: HAND-004 uniquely retains classical-quantum-operation-boundary, one-owner custody, immutable prior evidence, open extension and no extra scope.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-VALID-GRAND-LOCK-012 -> SFT-COMP-HAND-FORMAL-EMPIRICAL-003. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** HAND-004: classical_quantum; exact execution: {"classical_owner":"classical-computation","quantum_owner":"quantum-computation","silent_transfer":false,"transferred_records":["state-composition-interface","trace-interface","resource-interface"]}; complete family observation custody: 6 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V11, SFT-V3-BRANCH-OWNERSHIP-TREE, SFT-V3-OPEN-EXTENSION-POLICY. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, application result or publication state selects a handoff; no obligation has two owners or no owner; no empirical claim bypasses sealed target custody; no classical claim silently absorbs quantum or engineering operations; no lawful extension rewrites a prior receipt; no protected authority edit is permitted.
- **Evidence identities:** derivation `sha256:323d7798fdd015d54554c847b1e8a7a41237d8cdab096bae8894d9f1add7db34`; independent implementation `sha256:5463cf22bd991837912a99c38ad38c24fd06eaf5a9eaa3d409f6159241660ea4`; independent certificate `sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f`; empirical/external `sha256:16d421437f3cdf3d19ed11b4bd343cd0d9f1285d885613d4888e6805f4fc5b97`; engine receipt `sha256:2ce0d9f2abe0141ce4f7283069e50825792dea85fa3904e2ff0c8bf75629c596` at `receipts/engine/model_admitted/SFT-COMP-HAND-CLASSICAL-QUANTUM-004-2ce0d9f2abe0141c.json`.

`SFT-COMP-OBL-HAND-005` - Classical Computation open-extension handoff

- **Claim and ownership:** `SFT-COMP-HAND-OPEN-EXTENSION-005`; Classical Computation family `HAND`; Classical Computation open-extension handoff.

- **Forced law:** Branch completion is dated completion against the frozen census, never permanent closure: lawful additions receive new obligations and receipts without invalidating or rewriting prior identities.
- **Unique surviving structure:** `one-declared-owner__complete-input-output-interface__dated-completion-open-extension-ledger__immutable-prior-evidence-custody__literal-complete-product__there-is-no-nothing-lineage__dated-complete-open-extension__explicit-domain-quantum-engineering-boundary`. Exact result: HAND-005 uniquely retains dated-completion-open-extension-ledger, one-owner custody, immutable prior evidence, open extension and no extra scope.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-VALID-GRAND-LOCK-012 -> SFT-COMP-HAND-CLASSICAL-QUANTUM-004. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`. Imported axioms: `0`; free parameters: `0`.
- **Post-registration observation:** HAND-005: open_extension; exact execution: {"extension_policy":"new obligations and receipts without rewriting prior identities","permanent_lock":false,"status":"dated-complete-against-frozen-census"}; complete family observation custody: 6 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V11, SFT-V3-BRANCH-OWNERSHIP-TREE, SFT-V3-OPEN-EXTENSION-POLICY. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, application result or publication state selects a handoff; no obligation has two owners or no owner; no empirical claim bypasses sealed target custody; no classical claim silently absorbs quantum or engineering operations; no lawful extension rewrites a prior receipt; no protected authority edit is permitted.
- **Evidence identities:** derivation `sha256:7986afeb68ae6d7a6f6947c4fb6fad37b84c74589eed4439e27365e078cf511d`; independent implementation `sha256:5463cf22bd991837912a99c38ad38c24fd06eaf5a9eaa3d409f6159241660ea4`; independent certificate `sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f`; empirical/external `sha256:2c82eb312a57a2bab9f79b3e2fb75a993719c1003e28e803940a5cb6f3d9788d`; engine receipt `sha256:78da65a6b8b65d6d3c3462075f93d3300df2ff87cb6f7925dadf56e254d3e434` at `receipts/engine/model_admitted/SFT-COMP-HAND-OPEN-EXTENSION-005-78da65a6b8b65d6d.json`.

`SFT-COMP-OBL-HAND-006` - Classical Computation cross-branch completeness certificate

- **Claim and ownership:** `SFT-COMP-HAND-CROSS-BRANCH-COMPLETENESS-006`; Classical Computation family `HAND`; Classical Computation cross-branch completeness certificate.
- **Forced law:** The Classical Computation cross-branch certificate reconciles all 369 frozen obligations, assigns every downstream interface once, preserves open extension and completes the branch without permitting applications to select its laws.
- **Unique surviving structure:** `one-declared-owner__complete-input-output-interface__six-handoff-no-omission-certificate__immutable-prior-evidence-custody__literal-complete-product__there-is-no-nothing-lineage__dated-complete-open-extension__explicit-domain-quantum-engineering-boundary`. Exact result: HAND-006 uniquely retains six-handoff-no-omission-certificate, one-owner custody, immutable prior evidence, open extension and no extra scope.
- **Exhaustion and falsification:** 256 candidates were generated and decided; exactly one survived; 4 of 4 controls passed; closure is `depth_independent`.
- **Dependency and root trace:** SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 -> SFT-COMP-VALID-GRAND-LOCK-012 -> SFT-COMP-HAND-OPEN-EXTENSION-005. Registered root theorem: `SFT-ROOT-THERE-IS-NO-NOTHING`.

Imported axioms: `0`; free parameters: `0`.

- **Post-registration observation:** HAND-006: cross_branch_completeness; exact execution: {"duplicate_owners":0,"frozen_obligations":369,"handoff_obligations":6,"omitted_owners":0,"owner_rows":4,"post_handoff_expected":369,"pre_handoff_closed":363,"predecessor_reconciliation_identity":"sha256:6db45c0e72dee5cc697d67dcc2e1e2df15b475d3c2bf6b6a6c8a57c2aa18b290","protected_authority_changed":false}; complete family observation custody: 6 records; all rows preserved; complete generated execution supplies the observation; no imported theorem answer, hidden branch or target-selected survivor enters
- **Sources and custody:** SFT-V3-COMPUTATION-RECONCILIATION-V11, SFT-V3-BRANCH-OWNERSHIP-TREE, SFT-V3-OPEN-EXTENSION-POLICY. External status: `empirically_tested_and_independently_replicated`.
- **Prohibited imports and boundary:** no axiom, application result or publication state selects a handoff; no obligation has two owners or no owner; no empirical claim bypasses sealed target custody; no classical claim silently absorbs quantum or engineering operations; no lawful extension rewrites a prior receipt; no protected authority edit is permitted.
- **Evidence identities:** derivation `sha256:f26a47c054d86aedd23deae81cb6043d5ff68f2940a275a7ad91644c97d0d2e3`; independent implementation `sha256:5463cf22bd991837912a99c38ad38c24fd06eaf5a9eaa3d409f6159241660ea4`; independent certificate `sha256:14aa427629e75135c5645f40e2f50246fa9ff6641dc466de204baa168496891f`; empirical/external `sha256:360500b9ecf6f4e82fa5282e4150f82ac00e3036115f5f24a9acf7f05d631463`; engine receipt `sha256:c035c034f2623e06068a000dff4a6211a9ace07869fb7fcf48d3350ee119f2ca` at `receipts/engine/model_admitted/SFT-COMP-HAND-CROSS-BRANCH-COMPLETENESS-006-c035c034f2623e06.json`.

131. What empirical means in this computational branch

Empirical means grounded in observation or experience rather than accepted solely by theory or authority. In computation the directly observable objects are generated inputs, machine states, transitions, traces, resource ledgers, outputs, tamper responses and independent executions. The field-wide extension therefore seals its grammars and target identities first, runs exact executions afterward, preserves every row and compares the result with an implementation-distinct reconstruction. Where a claim concerns physical devices, natural data or application performance, that measurement remains with the owning downstream branch. This boundary does not make computation non-empirical; it prevents a formal execution from pretending to have measured hardware or nature.

The branch admits no result-only benchmark. A computation result is acceptable only with its initial configuration, generated description, complete transition trace, dependencies, resource accounting, controls, source identities, receipt and reproducible execution. Apparent random outcomes are observed deterministic schedules over complete registered support. Security and learning results retain their exact finite adversary and hypothesis grammars and cannot be inflated into practical claims beyond those supports.

132. Dated completion, falsification and extension

The branch is complete to census `sha256:411959c9505e6e09ec57e03923572f249cb5402996dca17911b3e1616e81b6ea` and reconciliation `sha256:84e436aa27ed392c4c2a5cab77833952a027f43e878151627e958e64be3ee30e`: 369 closed, structural absence of an open registered row, and no omitted census member. This is not permanent closure. A reviewer may challenge any claim with an omitted coordinate, second survivor, simpler preserving form, counterexample to the successor certificate, mismatched source, unreproducible trace, failed control or broken receipt. A lawful new problem enters as a versioned obligation without rewriting prior evidence.

The twelve VALID claims form the dated Classical Computation validation Grand Lock: they replay and partition the 351 predecessor receipts, retain theorem/finite-census/unrestricted distinctions and preserve adverse and ownership rows. The six HAND claims then assign quantum operations, physical measurements, software/hardware engineering and application experiments exactly once. The name Grand Lock denotes this dated validation layer; it does not erase open scientific criticism or prevent a lawful later extension.

133. Complete-field conclusion

Classical Computation is complete to its frozen 29 July 2026 census: **369/369 obligations, 12/12 families, 94,464 candidate decisions, 369 unique survivors, 1,476 passed adverse controls, 369 independent reconstructions and no open registered obligation**. The protected engine remains `sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a` and the protected verification authority remains `sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8`.

The scientific contribution is a unified computational object, not a list of renamed conventional subjects. State, symbol, language, machine, algorithm, resource, program meaning, causal process, adversary, learner and simulation are successive exact organisations of retained Fold distinctions. Every familiar historical correspondence is downstream of the forced SFT carrier. This permits comparison with Turing, Church, Gödel, Shannon, von Neumann, Landauer and Bennett while refusing to let prestige or precedent choose the law.

The dated validation Grand Lock preserves the adverse and ownership rows attached to all 351 predecessor receipts; it does not turn an implementation trace into a measurement of hardware or nature. No correction or later observation is used to rewrite an earlier receipt. The six handoffs keep quantum operations, physical measurement, platform engineering and application performance with their owning branches. Classical Computation is therefore closed only to this frozen field census and remains open to a registered counterexample, correction or new computational problem.

The public contribution is equally concrete. Every registration, alternative, elimination, survivor, control, observation, reconstruction and receipt is available for inspection. Maria Smith's lack of credentialed access is not evidence for the mathematics; it is evidence of what credential and capital gates can exclude. Ernos Labs therefore invites unrestricted criticism while reserving its standards designation for work that preserves the public empirical constitution, complete evidence, open licensing, retained authorship and unchanged admission route.

Data and code availability

The open repository is `MettaMazza/ernos-labs-sft-platform`. The live model-admitted index is `census/claims.json`; the Classical Computation field boundary is `census/computation_discipline_obligations.json`. Every represented claim has its own package under `claims/<claim-id>/` and its immutable model-admitted receipt under `receipts/engine/model_admitted/`.

The publication evidence map is `publications/successors/computation/evidence_map_v1_4.json` and the candidate manifest is `publications/successors/computation/manifest_v1_4.json`. Registered external records and source captures are retained under `experiments/external_sources/computation/` together with their source identities, transport outcomes and hashes. Files too large for ordinary Git remain checksum-bound and are distributed through the applicable existing Zenodo record rather than omitted or replaced.

Build, render, replay and verification programs are retained under `tools/`, with branch implementations under `sft/`, independent reconstructions under `generated/` and tests under `tests/`. The final approved Markdown, rendered PDF, evidence map, manifest, metadata and checksums must agree before deposit. No successful test is treated as empirical confirmation unless the claim-specific evidence protocol separately authorises that status.

Shared claim-record clauses

The clauses below previously appeared verbatim in multiple claim records. They are preserved once here to reduce mechanical repetition. Each in-text clause reference applies this exact wording at that claim location; no scientific statement, status, condition or qualification has been removed.

`COMP-S001` - applied at 116 claim locations

The literal product contains **256** candidates. The census preserves 256 candidate identities and the elimination receipt preserves 256 decisions. There are **1** survivors and **255** eliminations.

`COMP-S002` - applied at 116 claim locations

Every rejected candidate fails at the first non-preserving coordinate in registered axis order. The following ledger groups all eliminations by their exact first failure; the counts sum to the complete rejected support.

`COMP-S003` - applied at 12 claim locations

The unique all-preserving member is `complete-generated-carrier__canonical-held-identity__source-bound-exact-relation__complete-retained-trace__explicit-initial-terminal-boundary__interface-preserving-composition__structural-successor__no-extra-computational-premise`.

`COMP-S004` - applied at 116 claim locations

Minimality follows because changing any one of the eight admitted coordinates replaces it with the generated failure coordinate and removes its named condition. Named-shape uniqueness follows because the complete product contains one and only one tuple composed entirely of admitted coordinates.

`COMP-S005` - applied at 116 claim locations

These are construction laws, not renamed conventional equations. They specify exact carriers, relations, traces, interfaces and boundary failures. Host integers count generated artifacts but never become SFT proof quantities.

`COMP-S006` - applied at 116 claim locations

The certificate therefore does not infer unrestricted completion from a depth table. It proves that every fresh generated finite successor extends the same classification while preserving prior identities and adding all new relations. No completed infinity is created.

`COMP-S007` - applied at 10 claim locations

The unique all-preserving member is `complete-generated-description-domain__exact-recognition-execution-trace__accept-reject-continue-classes__generated-self-description-support__explicit-totality-or-partiality-boundary__answer-preserving-reduction__description-successor-certificate__no-undeclared-computational-power`.

`COMP-S008` - applied at 13 claim locations

The unique all-preserving member is `canonical-encoding-support__complete-resource-ledger__registered-computation-model__common-family-comparison__overhead-retaining-composition__constructive-or-adversarial-bound-witness__family-successor-certificate__no-extra-resource-parameter`.

`COMP-S009` - applied at 14 claim locations

The unique all-preserving member is `complete-generated-input-domain__exact-registered-precondition__exact-step-relation__retained-invariant-ledger__reconstructible-output-certificate__complete-time-space-trace__input-successor-certificate__no-extra-answer-model`.

`COMP-S010` - applied at 12 claim locations

The unique all-preserving member is `complete-generated-syntax__canonical-held-binding__exact-semantic-relation__complete-evaluation-trace__observation-bound-equivalence__machine-checkable-proof-object__syntax-successor-certificate__no-extra-semantic-premise`.

`COMP-S011` - applied at 12 claim locations

The unique all-preserving member is `complete-held-process-identities__complete-local-event-ledgers__exact-partial-causal-order__source-bound-message-records__declared-fault-support__observation-relative-local-knowledge__schedule-successor-certificate__no-extra-global-oracle`.

`COMP-S012` - applied at 13 claim locations

The unique all-preserving member is `complete-generated-challenge-support__exact-adversary-process-grammar__complete-interaction-trace__exact-adversary-observation__retained-closed-distinction-property__declared-information-or-resource-bound__support-successor-certificate__no-extra-security-premise`.

`COMP-S013` - applied at 14 claim locations

The unique all-preserving member is `complete-generated-hypothesis-support__traceable-canonical-representation__exact-evidence-to-hypothesis-relation__sealed-independent-evaluation__retained-alternative-ledger__complete-learning-resource-trace__structural-or-empirical-target-boundary__no-extra-learning-parameter`.

`COMP-S014` - applied at 13 claim locations

The unique all-preserving member is `registered-exact-model-relation__complete-declared-input-support__exact-executable-operation-trace__orientation-labelled-error-ledger__base-successor-closure-certificate__sealed-falsifiable-comparison-boundary__generated-refinement-successor__no-extra-scientific-prior`.

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Complete current-census successor supplement

The authoritative current scope of this paper is 375 live model-admitted claims (`computation` 375). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

The preceding manuscript did not contain 6 later-admitted claim section(s). They are included below in dependency order so the successor covers the complete current branch surface.

Exact SFT causal attention-transformer assembly

Claim ID: `SFT-COMP-AIX-CAUSAL-ATTENTION-TRANSFORMER-001`

Exact statement: An SFT causal attention-transformer is the finite composition of exact source-bound token and position carriers, query-key-value projections, predecessor-only attention partitions, identity-retaining multihead value contractions, exact normalization, residual and oriented complement-gate blocks, and a complete output-support projection. Future positions are structural absence rather than numerical masks; every intermediate part, rounding or enclosure and causal edge is retained; a one-position base and append-position/layer successor preserve all prior causal outputs; no imported transformer runtime, pretrained weight or target result selects the architecture.

Dependency route: `SFT-COMP-FORM-STATE-TRANSITION-001` -> `SFT-COMP-FORM-COMPOSITION-001` -> `SFT-COMP-ALG-STRINGS-SEQUENCES-001` -> `SFT-COMP-SCIX-FINITE-PRECISION-002` -> `SFT-MATH-GRAPH-DIRECTED-CAUSAL-REACHABILITY-008` -> `SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001` -> `SFT-MATH-LINEAR-MAP-COMPOSITION-002` -> `SFT-MATH-LINEAR-INNER-PRODUCT-METRIC-008` -> `SFT-MATH-LINEAR-TENSOR-CONTRACTION-012` -> `SFT-MATH-PROB-FINITE-DISTRIBUTION-006` -> `SFT-MATH-NUM-EXACT-REPRESENTATION-ROUNDING-001`

Candidate generator: Generate the literal product of token support, projection, causality, partition, head assembly, block trace, successor and import-boundary coordinates before any UnisonFold result is opened.

Grammar boundary: Every finite exact causal self-attention network assembled from generated token/position supports, declared layer/head/coordinate carriers, exact oriented rational or registered finite-grid parameters, structurally predecessor-only score supports, complete attention partitions, exact block traces and complete output support.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `complete-ordered-token-position-support__exact-source-bound-q-k-v-projections__structural-predecessor-only-support__complete-exact-causal-attention-partition__identity-retaining-multihead-contraction__exact-normalize-residual-gate-block-trace__append-position-and-layer-successor__no-imported-runtime-weight-or-target-selector`

Exact result: The unique exact SFT causal attention-transformer retains complete token-position support, source-bound Q/K/V projections, structural predecessor-only causality, a complete exact attention partition, identity-retaining multihead contraction, exact normalization/residual/gate traces, append-position and append-layer successors, and no imported runtime, weights or target selector.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:a19ceb388637ad5cf05ab67f93eb8bec21ca6765358bedae3a4d326e0248887b` at `receipts/engine/model_admitted/SFT-COMP-AIX-CAUSAL-ATTENTION-TRANSFORMER-001-a19ceb388637ad5c.json`

Control	Passed	Expected	Observed
false_premise	True	reject the generated form lacking complete first-axis coverage	A sampled context cannot establish the architecture carrier.
tampered_source	True	reject a changed official source identity	changed source identity differs
tampered_artifact	True	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities
boundary	True	halt imported models, parameters or excluded quantum machinery	no numerical negative infinity or numerical mask value; no floating, irrational, imaginary or completed-infinite proof scalar; no hidden attention row, dropped predecessor, collapsed head or unretained rounding; no pretrained parameter, external transformer runtime or target answer as an architectural premise; no conversational-quality, optimization, physical-speed or quantum-computation conclusion

Exact source-bound teacher-observation learning

Claim ID: `SFT-COMP-AIX-EXACT-TEACHER-LEARNING-002`

Exact statement: An external pretrained teacher may supply development observations to an SFT-native learner only after the architecture, teacher and corpus identities, exact bit-to-rational observation map, complete output partition, exact disagreement objective, generated update grammar, data roles and stop rule are frozen. Teacher bits become exact source-bound parts without floating arithmetic; no teacher parameter, tokenizer implementation or runtime enters the student. Every unchanged and declared oriented successor is scored by the same exact student execution, only strict objective descent may change state, tied successors and all adverse routes are retained, persisted artifacts bind complete parent and evidence lineage and run without the teacher, and unseen conversational generalization remains a separate sealed evaluation.

Dependency route: `SFT-COMP-AIX-CAUSAL-ATTENTION-TRANSFORMER-001` -> `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001` -> `SFT-COMP-HAND-MEASUREMENT-BOUNDARY-002` -> `SFT-COMP-LEARN-CLASSICAL-LEARNING-001` -> `SFT-COMP-LEARN-REPRESENTATION-001` -> `SFT-COMP-LEARN-ADAPTATION-001` -> `SFT-COMP-LEARNX-HYPOTHESIS-FAMILY-002` -> `SFT-COMP-LEARNX-LOSS-RISK-003` -> `SFT-COMP-LEARNX-HELD-OUT-CUSTODY-004` -> `SFT-COMP-LEARNX-GENERALIZATION-006` -> `SFT-COMP-LEARNX-OPTIMIZATION-CONVERGENCE-016` -> `SFT-COMP-LEARNX-LEARNED-VERIFICATION-024`

Candidate generator: Generate the literal product of teacher identity, record encoding, output support, data custody, objective, update, selection/persistence and unseen-boundary coordinates before the new exact-learning execution is opened.

Grammar boundary: Every finite exact teacher-observation learning execution over a previously admitted SFT causal attention-transformer, a frozen external teacher/corpus identity, captured finite output records, a complete declared output grouping, a generated exact parameter/update carrier and separately frozen evaluation supports.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `frozen-source-bound-external-teacher__exact-bit-record-to-rational-parts__complete-declared-teacher-output-partition__frozen-disjoint-data-role-ledgers__exact-complete-disagreement-ledger__generated-exact-update-candidate-grammar__strict-descent-tie-adverse-lineage-ledger__teacher-free-artifact-and-sealed-unseen-boundary`

Exact result: The unique exact teacher-observation learner retains a frozen source-bound external teacher, exact bit-record-to-rational parts, complete teacher output support, disjoint data-role custody, a complete exact disagreement ledger, generated exact update candidates, strict-descent/tie/adverse/persistence lineage and a teacher-free artifact with a separate sealed unseen boundary.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:89bc370a1f554bbc3c9d17df43b1b144232988caf817a86154860f3e0fb61d95` at `receipts/engine/model_admitted/SFT-COMP-AIX-EXACT-TEACHER-LEARNING-002-89bc370a1f554bbc.json`

Control	Passed	Expected	Observed
false_premise	True	reject the generated form lacking complete first-axis coverage	A mutable or runtime teacher can silently select later states.
tampered_source	True	reject a changed official source identity	changed source identity differs
tampered_artifact	True	reject missing, duplicated or additional survivors	literal product contains one survivor and unique identities
boundary	True	halt imported models, parameters or excluded quantum machinery	no teacher weight, tokenizer implementation, hidden state, gradient, optimizer state or runtime dependency inside the student; no floating arithmetic, floating surrogate, hidden weighting or unregistered tolerance in the learning decision; no target-selected update grammar, erased tie, discarded failure or overwritten checkpoint lineage; no development prompt, teacher observation or training score may enter the sealed unseen support; no training agreement alone establishes generalized conversation, physical efficiency or quantum advantage

SFT reconstruction of the permanent rational-formula lower bound

Claim ID: `SFT-COMP-OAI26-PERMANENT-FORMULA-001`

Exact statement: For every generated n at least thirty-two and every admissible canonical encoding of a valid rational formula over structurally paired exact complex coordinates whose evaluation equals the encoded permanent polynomial in the encoded fraction ring, the source lower bound on variable leaves holds as an exact-real enclosure inequality.

Dependency route: `SFT-MATH-SYMB-CANONICAL-EXPRESSION-001` -> `SFT-COMP-CPLXX-FORMULA-BRANCHING-CIRCUIT-014` ->

`SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024` -> `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: canonical rational-formula syntax x validity/evaluation traces x permanent expression support x leaf-resource certificates x size-successor induction. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-root-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: For every generated n at least thirty-two and every admissible canonical encoding of a valid rational formula over structurally paired exact complex coordinates whose evaluation equals the encoded permanent polynomial in the encoded fraction ring, the source lower bound on variable leaves holds as an exact-real enclosure inequality.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:990358ae43cdbd32dc7deb9d83cafee45862e9d6d700df510a4d4b43fa929771` at `receipts/engine/model_admitted/SFT-COMP-OAI26-PERMANENT-FORMULA-001-990358ae43cdbd32.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT reconstruction of GapCVP400 NP-hardness

Claim ID: `SFT-COMP-OAI26-GAPCVP400-002`

Exact statement: For every admissible generated bit language carrying an exact SFT IsNP witness, there exists a total generated bit-list reduction to the exact encoded gapCVP400 promise, a polynomial-resource machine certificate, and complete per-input yes-preservation and no-preservation proofs; the four PromiseReduction fields are retained.

Dependency route: `SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006` -> `SFT-MATH-GEOM-EUCLIDEAN-DISTANCE-002` -> `SFT-COMP-CPLXX-REDUCTION-COMPLETE-PROBLEM-021` -> `SFT-COMP-CPLXX-APPROXIMATION-RATIO-026`

Candidate generator: Compose, without an outcome axis, the admitted candidate generators of: generated bit languages x NP witness machines x total reduction maps x exact lattice instances x promise verdict/resource traces. Generate theorem-specific witnesses, arbitrary-input proof steps, eliminations and certificates.

Grammar boundary: Every SFT-admissible encoding of every source binder and every original hypothesis/conjunct. Universal natural scope requires base-and-successor closure; existential scope requires the complete registered witness grammar; a rejected carrier is not a mathematical negation.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-source__exact-quantifier-conjunct-order__exact-native-correspondence__preexisting-branch-grammar-composition__complete-prior-receipt-trace__complete-r` `oot-to-result-chain__complete-executable-certificates__sft-root-bound-forward-derivation`

Exact result: PROVED: For every admissible generated bit language carrying an exact SFT IsNP witness, there exists a total generated bit-list reduction to the exact encoded gapCVP400 promise, a polynomial-resource machine certificate, and complete per-input yes-preservation and no-preservation proofs; the four PromiseReduction fields are retained.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:2216e2b7da1eb97c1e1bc89820433bb2e8fa3986c9680f8c788546d24d734afd` at `receipts/engine/model_admitted/SFT-COMP-OAI26-GAPCVP400-002-2216e2b7da1eb97c.json`

Control	Passed	Expected	Observed
false_premise	True	reject a changed source target	A changed or paraphrased target is not the registered proposition.
tampered_source	True	reject a changed source identity	changed source identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional survivor	the complete route product contains exactly one all-admitted derivation
boundary	True	reject carrier denial finite examples imported proof authority or incomplete chains	exact correspondence, full proof graph, arbitrary-input closure and executable certificates all passed

SFT disproof of source-artifact validity: PermanentFormulaLowerBound.permanent_rational_formula_logarithmic_lower_bound

Claim ID: `SFT-COMP-OAI26-PERMANENT-FORMULA-VALIDITY-001`

Exact statement: Not SFTValid(exact frozen artifact PermanentFormulaLowerBound.permanent_rational_formula_logarithmic_lower_bound): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires complex fraction-ring formulas with subtraction/division and a completed real logarithmic resource scalar.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-FOUNDATION-EXACT-OPERATIONS-001` -> `SFT-MATH-SYMB-CANONICAL-EXPRESSION-001` -> `SFT-COMP-CPLXX-FORMULA-BRANCHING-CIRCUIT-014` -> `SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact PermanentFormulaLowerBound.permanent_rational_formula_logarithmic_lower_bound)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact PermanentFormulaLowerBound.permanent_rational_formula_logarithmic_lower_bound). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:cb63b42d62cde6a79d1fb9478726ca6da5f2073afc8dc4ee5a4622d2459badae` at `receipts/engine/model_admitted/SFT-COMP-OAI26-PERMANENT-FORMULA-VALIDITY-001-cb63b42d62cde6a7.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

SFT disproof of source-artifact validity: GapCVP.Comparator.gapCVP400IsNPHard

Claim ID: `SFT-COMP-OAI26-GAPCVP400-VALIDITY-002`

Exact statement: Not SFTValid(exact frozen artifact GapCVP.Comparator.gapCVP400IsNPHard): assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires the completed family of all bit languages and conventional reductions into signed integer lattices with a real gap factor.

Dependency route: `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001` -> `SFT-MATH-LOGIC-PROOF-001` -> `SFT-FOUNDATION-EXACT-OPERATIONS-001` -> `SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006` -> `SFT-MATH-GEOM-EUCLIDEAN-DISTANCE-002` -> `SFT-COMP-CPLXX-REDUCTION-COMPLETE-PROBLEM-021` -> `SFT-COMP-CPLXX-APPROXIMATION-RATIO-026`

Candidate generator: Compose exact source custody, exact quotation, exposed axiom/carrier evidence, pre-existing admission laws, a complete contradiction chain, executable checks and a strict native/source nontransfer boundary; no verdict coordinate.

Grammar boundary: SFTValid(exact frozen artifact GapCVP.Comparator.gapCVP400IsNPHard)

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-frozen-artifact__exact-statement-and-quantifier-quotation__nonempty-source-axiom-vector-exposed__necessary-source-component-exposed__preexisting-admission-and-domain-laws__assume-valid-and-derive-actual-contradiction__complete-executable-evidence__native-reconstruction-kept-distinct`

Exact result: DISPROVED: SFTValid(exact frozen artifact GapCVP.Comparator.gapCVP400IsNPHard). SFT validity forces an empty axiom vector and admitted carriers; the exact source exposes three axioms and the theorem-specific excluded component. The assumption therefore entails a contradiction.

Closure: `depth_independent`; minimality `True`; named-shape uniqueness `True`

External status: `independently_replicated`

Engine receipt: `sha256:f5b14c6e98feafce610f973db57677ef55bb2acde6ad4ea2cbb398f779eeeb87` at `receipts/engine/model_admitted/SFT-COMP-OAI26-GAPCVP400-VALIDITY-002-f5b14c6e98feafce.json`

Control	Passed	Expected	Observed
false_premise	True	reject the false premise that the frozen source axiom vector is empty	the exact three-entry source vector is nonempty
tampered_source	True	reject any changed source identity	the changed identity differs from the executed source manifest
tampered_artifact	True	reject a missing duplicate or additional complete contradiction route	the complete route product has one all-evidence survivor
boundary	True	reject transfer from the SFT-native reconstruction to the external artifact	the contradiction proves only the exact source-validity negation and records the reconstruction as distinct

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.4.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

This paper reports the complete-field Classical Computation branch of the third clean-room reconstruction of Smithian Fold Theory (SFT), complete to its frozen dated census and explicitly open to lawful extension. The census contains **369 obligations**

in 12 dependency-ordered families. Every obligation has an untouched-engine model-admission receipt, a complete 256-member candidate census, exactly one survivor, four adverse controls, a depth-independent finite-successor or explicit boundary certificate, an implementation-distinct reconstruction and an observation record with registered custody. Together the branch preserves **94,464 generated candidates, 369 unique survivors and 1,476 passed controls**. The final reconciliation identity is `sha256:84e436aa27ed392c4c2a5cab77833952a027f43e878151627e958e64be3ee30e`.

The reconstruction spans formal computation; computability; computational complexity; algorithms and mathematical data structures; semantics and programming theory; concurrent and distributed computation; cryptography and computational security; learning and intelligence theory; and scientific computation. It derives a native SFT tape machine, exact languages and automata, rewriting and recursion, model equivalence, universality, halting and incompleteness boundaries, exact resource laws, deterministic-support randomized computation, semantics-preserving compilation, consensus and impossibility boundaries, explicit adversary laws, traceable learning and exact scientific calculation. Its native famous-problem results remain precisely typed: `BB_F(k)=k`, `P_F=NP_F`, and the depth-`k` full-edge circuit lower bound hold inside the admitted Fold grammar and are not silently exported to conventional external grammars.

The proof domain imports no conventional machine, axiom, free or fitted parameter, semantic numerical zero, negative proof magnitude, irrational or imaginary proof scalar, floating proof quantity, completed infinity, ontic randomness, cryptographic hardness assumption, pretrained model, benchmark answer or application result. Displayed `0` is structural absence and schedule support is deterministic. Quantum operations, physical hardware, software platforms and the later Unison AI, Fold Chess, Fold Go and Fold Protein rebuilds remain explicit downstream handoffs.

Version 1.4 preserves the complete version 1.3 foundation and its public scientific argument, then executes the full-field roadmap claim by claim. No protected engine or verification-authority source was modified. Every later extension must enter as a new registered question and pass the same public protocol.

Keywords: Smithian Fold Theory; complete-field classical computation; Fold machine; computability; halting; complexity; algorithms; semantics; distributed computation; cryptography; learning theory; scientific computing; superdeterminism; computational proof; open science.

Scope and ownership boundary

This paper owns the `computation` surface comprising 375 current claims. Ownership identifies responsibility for derivation and falsification; it does not prevent criticism or lawful cross-branch use. Application performance, consensus, credentials and Lean acceptance cannot transfer ownership or select a law.

Mathematical, computational and empirical constitutions

The mathematical constitution retains exact generated carriers, relations, candidate grammars, decisions, survivors and closure boundaries. The computational constitution retains executable identities, complete traces, controls, independent reconstructions and receipts. The empirical constitution retains source registration, transport, pre-seal and post-seal chronology, custody, measurement or observation, uncertainty, favourable and adverse rows, missingness, unresolved records and falsification conditions. None of these evidence classes substitutes for another.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Successor machine-identity appendix

- Preceding source: `publications/successors/computation/AFTER_TURING_THE_FOLD_MACHINE_PAPER_001_V1_4.md`
- Preceding source SHA-256: `sha256:e1406b1c91db69b88d92107c4fc96d017fef968299583ef97eec217f60b1e54f`
- Lean report: `generated/lean4_validation/reports/whole_model_validation.json`

- Lean report SHA-256: `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`
- Ordered census SHA-256: `sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71`
- Execution manifest SHA-256: `sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41`
- Protected engine seal: `sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a`
- Verification-authority seal: `sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8`
- Publication authorised: `false`
- Remote actions performed: `false`

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned `PASS`. Its immutable report identity for this successor is `sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf`.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **375** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar: `unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter. Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors. `#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted gate` only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.5.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 375 current claim(s) within `computation` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.